

Understanding the Effect of Ionic Liquids and Deep Eutectic Solvents on Extraction Processes

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Abstract

Ionic liquids (ILs) and deep eutectic solvents (DESs) are being actively explored as alternative to conventional solvents due to their non-volatility and reduced flammability, with interest in the use of these solvents emerging for a variety of applications. Moreover, IL and DES properties can be tuned by ion/component selection and their composition, allowing them to be designed as promising solvents for these applications, including specific extraction processes such as biomass extraction. This project explores the development of DESs and ILs for extraction processes from two different perspectives. The first is the design of novel DESs with significantly improved hydrogen bond donating or accepting capacity towards being able to perform extraction processes more efficiently. The other was to undertake a systematic exploration of the extraction of key components of the *Pinus radiata* bark to understand the influence of IL and DES solvent effects on these outcomes.

To improve extraction processes, the ability to design novel DESs with better hydrogen bond donating or accepting capability than existing DESs was explored. The development of strong hydrogen-bond donating solvents were explored using guanidinium salts as a constituent along with 20 different hydrogen bond donor combinations. The polarity of DESs was explored using the solvatochromic dye set developed by Kamlet and Taft to measure the hydrogen bonding strength of the DESs. Out of all DESs, the guanidinium chloride: lactic acid mixture was found to possess the best hydrogen bond donating ability (α). This was comparable to, if not slightly improved, compared to existing strong hydrogen bond donating solvents such as 1,1,1,3,3,3-hexafluoro-2-propanol. Attempts to develop strong hydrogen-bond accepting solvents were also made by exploring potential solvate ILs with strongly hydrogen bond accepting anions and examining the concept of reverse DESs where the salt was added with a strong hydrogen bond acceptor rather than a hydrogen bond donor. These attempts were not successful due to the formation of mainly high melting mixtures rather than room temperature liquids.

Moreover, the new DESs that were prepared, have the potential to be more cost-effective and environmentally friendly than ILs with related guanidinium cations, which have recently been shown to be excellent solvents for the purification of a variety of difficult-to-solubilize pharmaceuticals. The solvents that could be successfully prepared were used to examine the solubilization of active pharmaceutical ingredients such as fenofibrate and the extraction of key components from *Pinus radiata* bark. For the fenofibrate solubilisation, the DES based on

guanidinium chloride and levulinic acid displayed the highest solubility although this was significantly less than could be obtained using the related IL solvent systems.

Pinus radiata is one of the most extensively used woods in New Zealand. The bark of *Pinus radiata* consists of various classes of substances such as suberin, lignin, cellulose, polyphenols, and extractives but is currently used for largely low value applications and is considered to be predominantly a waste product. Many of the compounds present in the bark demonstrate bioactivity such as antifungal, anti-tumor, anti-carcinogenic, anti-diabetic and anti-inflammatory properties and the structural biopolymers are of interest as potential sources of bio renewable materials and biofuels.

Given the potential importance of components of *Pinus radiata* bark, one of the goals of this work was to understand how the structure of ILs and DESs affects the extraction and recovery of essential components from *Pinus radiata* bark. As ILs can alter their properties based on the structure of the ions selected, various types of ILs were developed by selecting a range of cations and anions of systematically differing hydrogen-bonding abilities to model their effects on the extraction of key bark components. The length of the alkyl chains on the IL ions was also modified to provide insight into the effect of the hydrophobic component of the IL on the extraction capacity and selectivity. Overall, 15 ILs, along with room temperature DESs developed such as guanidinium chloride: lactic acid and guanidinium chloride: glycerol, were used. Additionally, triethylammonium hydrogensulfate ([TEA][HSO_4]) was also examined for bark extraction due to it being reported as an effective solvent for extracting lignin from biomass sources.

ILs were found to be more effective at removing aromatics and waxes than sugars/polysaccharides as demonstrated by the solid-state NMR and pyrolysis GCMS results. This was even true for dimethylphosphate ($[Me_2PO_4]^-$) based ILs which have been described as being effective for the solubilisation of carbohydrates due to strong hydrogen bond accepting ability of the anion, which was attributed to the competition with multiple components of the biomass. $[C_nC_1pyrr]^+$ cations facilitate the preferential extraction of the hydrophobic components when combined with a strong hydrogen bond accepting anion ($[Me_2PO_4]^-$). In contrast, $[C_nC_1im]^+$ cation based ILs were more efficient at extracting lignin and tannin fractions. Moreover, the dissolution of the carbohydrate and lignin or tannin components depend on the alkyl chain length and these components decreases with increasing non-polar domain of the cation. However, the extraction of hydrophobic compounds such as waxes and

suberinic material tended to increase with increasing alkyl chain length. In terms of DESs, guanidinium chloride: lactic acid mixture was found to be effective in extracting a significant amount of carbohydrate than other components present in the bark.

The recovery and the isolation of the extracts was examined. When employing non-volatile solvents like ILs and DESs, this is an area that is sometimes underexplored, yet it is critical to any application of these solvents. The recovery of components from the bark extracts was explored using antisolvent addition. It was found that the use of water as an antisolvent enabled selective precipitation of some bark fractions compared to the use of other antisolvents that were ineffective hydrogen bond donors.

The $[Me_2PO_4]^-$ IL extracts led to higher yields of carbohydrates than could be recovered from $[N(CN)_2]^-$ ILs or the DES with the addition of water as an antisolvent. Moreover, the highest proportion of carbohydrate was recovered from $[C_4C_1im][Me_2PO_4]$ due to the presence of the imidazolium cation in combination with the strong hydrogen bond acceptor anion. Regarding the isolation of suberins/waxes as hydrophobic components, $[C_4C_1pyrr][Me_2PO_4]$ contributed to extracting and enabling the recovery of the highest proportion of suberin/waxes due to hydrophobicity of the cation. The long chain aliphatic $[Me_2PO_4]^-$ ILs only gave mixed fractions and limited selectivity of the components was observed due to their improved solubility on addition of water to these ILs.

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Declaration

I declare that the work for this thesis entitled ‘Understanding the effect of Ionic liquids and Deep eutectic solvents on the extraction processes’ has been carried out by me, except:

- Beatrix Theobald performed the Pyrolysis GC-MS experiment
- Laura Raymond and Stefan Hill performed ^{13}C solid-state NMR experiment
- Some of the viscosity measurements performed by Roland Brown.

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List of Abbreviations

DCM	dichloromethane
[$C_nC_1$im]	1-alkyl-3-methylimidazolium
[$C_nC_1$pyrr]	1-alkyl-3-methylpyrrolidinium
DMSO-d6	deuterated dimethyl sulfoxide
DMSO	dimethylsulfoxide
°C	degrees Celsius
mg	milligram
mPa s	millipascal second
h	hour
eq	equivalent
K	Kelvin
mL	millilitre
H-bonding	hydrogen-bonding
ILs	ionic liquids
DESs	deep eutectic solvents
m.p.	melting point
Ppts	precipitates
NMR	nuclear magnetic resonance
PILs	protic ionic liquids
AILs	aprotic ionic liquids
TSILS	task specific ionic liquids
wt	% weight percent
TEA	triethylammonium
HBA	hydrogen bond acceptor
HBD	hydrogen bond donor
SILs	solvate ionic liquids

RDESs	reverse deep eutectic solvents
HFIP	1,1,1,3,3,3-hexafluoro-2-propanol
LA	lactic acid
LevA	levulinic acid
Acet	acetamide
Gly	glycerol
GuCl	guanidinium chloride
ChCl	choline chloride
FT-IR	fourier transformation infrared
IR	infrared
HRMS	high performance mass spectroscopy
Py-GC-MS	pyrolysis gas chromatography- mass spectroscopy
HPLC-UV	high performance liquid chromatography ultraviolet visible
TMA-OH-py-GC-MS	tetramethylammonium hydroxide pyrolysis gas chromatography mass spectroscopy
HS-SPME-GCMS	head space solid phase microextraction gas chromatography mass spectroscopy
TPC	total phenolic content
GAE	gallic acid equivalent
K-T parameters	kamlet-taft parameters
API	active pharmaceuticals ingredients

1. Introduction

1.1. Green Chemistry

Sheldon presents a working definition of green chemistry as follows: "*green chemistry efficiently utilises (preferably renewable) raw materials, eliminates waste and avoids the use of toxic and/or hazardous reagents and solvents in the manufacture and application of chemical products*."[1] Green chemistry has been described as a process that reduces harm rather than one that causes no harm.[2] The primary aim of green chemistry is to prevent hazards at their source by minimising or eliminating the use of hazardous chemical feedstocks, reagents, solvents and products.[3]

The use of green solvents to replace conventionally used solvents is a vast topic of research and development. This is because solvents are a large contributor to waste produced from chemical processes and introduce significant hazards given conventional solvents are often flammable, volatile and can be toxic.[4] It has been proposed that green solvents must meet 12 criteria to be viable and desirable alternatives. These criteria include their affordability, recyclability, technical grade, synthesis, toxicity, biodegradability, performance, stability, flammability, easy to store, and renewability.[5] The "greenness" of a solvent depends on the application that it being used for and how it compares to other alternatives. For example, water is often considered to be a green solvent, but it is very energy intensive to distil, reducing the greenness of water for processes where it needs to be evaporated.[6]

Researchers have created and explored novel solvents and solvent systems based on green chemistry concepts to attempt to eliminate some of the inherent risk connected with existing solvents.[7]

The following are some of the novel solvent classes that have been investigated as alternative green solvents:

- Aqueous solvents, including aqueous biphasic mixtures
- Supercritical or dense phase fluids,
- Ionic liquids (ILs),
- Deep eutectic solvents (DESs),
- Switchable solvents

- Fluorous solvents, DESs and ILs will be discussed in greater detail in Section 1.4 and Section 1.5.

1.2. Solvents

Solvents are the dominant substance in solutions which are liquid or in solid phases comprising more than one substance.[8] One accepted definition of a solvent is "Any substance that dissolves another substance so that the resulting mixture is a homogenous solution".[7] Solvents are linked by their function as inert fluids and solid solutions that dissolve another substance (the solute). Since gases lack intermolecular interactions, they are not able to act as solvents.[9]

The largest use of solvents for chemical processes is in the solution phase. In traditional chemical processes, solvents are used for dissolving reactants, extracting and washing products, separating mixtures and cleaning reaction apparatus.[10] The primary purpose of these liquid solvents in chemical synthesis is to facilitate mass transfer and to control chemical reactions in terms of reaction rate, yields, and selectivity due to solvation effects that influence relative stability of reactants and key intermediates. While solvents often do this by dissolving the reactants and intermediates to form homogenous solutions, this requires that the subsequent product has to be separated from the solvent through methods that are often energy intensive or generate substantial further solvent waste.[8] The separation, purification and recycling of solvents are critical criteria that have direct impact on both the industrial viability and environmental impact of a process.[5] In this context, low boiling point organic solvents are often favoured because they allow for easier product isolation and recovery/recycling via low-temperature distillation, which prevents thermally sensitive compounds from degrading. At the same time, organic solvents have a number of environmental and safety issues, including flammability and volatility.[5] Therefore, although the invention of different organic solvents has led to in some remarkable advances in chemistry, the legacy of such solvents has resulted in a number of environmental and health concerns.

Various other cleaner solvents have been evaluated as alternatives.[1] Water was one of the first substances explored as an alternative solvent to replace organic solvents because it has many advantages, including being non-hazardous, high abundance and low cost. However, it is important to note that water has some drawbacks as well, including, a relatively temperature-restricted liquid state range, a high heat capacity,[4] and that it is not effective at dissolving many organic compounds.[10] The compatibility of the solvent with the solute also needs to be considered in selecting an appropriate solvent as some solutes can react with particular

solvents.[9] This is particularly an issue with the use of water as a solvent for chemical reactions. Therefore, many other alternative and benign solvents have been explored for new and safe media with the goal of increasing the long-term sustainability and safety of chemical processes. Some of the solvents such as ILs and DESs have been discussed in detailed in section 1.4 and 1.5.

1.3. Solvent Polarity: Betaine dye and Kamlet-Taft scale

To understand the effect of a solvent on solutes for reaction or extraction processes, it is important to understand the nature of solvent-solute interactions in the system. Factors such as solubility, relative thermodynamic stability, and reaction rates are all affected by solvent-solute interactions, and solvent selection governed by this understanding can allow thermodynamic and kinetic control of a chemical reaction or be used to tailor improved selectivity for an extraction process. The nature of solvent-solute interactions is often considered in terms of solvent polarity. Boerhaave stated in 1965 that "the polarity of a solvent is determined by its solvation capability (or solvation power) for reactants and activated complexes as well as for molecules in their ground and excited states".[11] As a result, this is dependent on the activity of all possible intermolecular forces between solvent and solute molecules, both specific and nonspecific. Coulomb interactions between ions, directional interactions between dipoles, inductive, dispersion, hydrogen-bonding, and charge transfer forces, as well as solvophobic interactions, are examples of these intermolecular forces. Only those interactions that result in definite chemical changes in the solute molecules, such as protonation, oxidation, reduction, complex formation, or other chemical processes, are eliminated. In the meantime, 'IUPAC Recommendation 1994' for terms used in physical organic chemistry has been updated to include this more realistic definition of solvent polarity.[12]

Since solvent polarity affects the nature of solvent-solute interactions and therefore influences solvent effects, the measurement of parameters reflecting solvent polarity has been widely explored to try and predict the nature of these effects. An early and widely used approach to modelling these solvent-solute interactions was through the use of macroscopic solvent parameters due to the simplicity of their measurement. These parameters mainly involve physical solvent properties such as relative permittivity or refractive index.[13]

These macroscopic physical property measures are often insufficient as they do not take into account specific solvent-solute interactions such as hydrogen-bonding, which often play a

dominant role in solvent-solute interactions. Therefore, no one macroscopic physical property could possibly account for the variety of solvent-solute interactions on the molecular level.[13]

The lack of simple theoretical expressions for calculating solvent effects, as well as, the inadequacy of defining "solvent polarity" in terms of single physical constants, has prompted attempts to introduce empirical solvent polarity scales based on solvent sensitive reference processes. This could be done by assuming some particular reaction rate, equilibrium, or spectral absorption as a suitable model, selecting an appropriate reference process that is sufficiently solvent-sensitive, and assume that this process has all possible solute/solvent interactions that are also present in related solvent-influenced processes. This polarity scale should therefore give an empirical measure of the solvation capability of a particular solvent for the given reference process. An appropriately chosen reference process could also be considered as a probe of the solvation shell of the standard solute – a probe that sums up a wide variety of possible intermolecular interactions such as ion-dipole, dipole-induced dipole, hydrogen-bonding interactions, etc. The most common empirical solvent polarity scales, which have been suggested as methods of measuring solvent polarity, will be discussed in the next sections 1.3.1, 1.3.2 and 1.3.3.

1.3.1. E_T^N Scale of Solvent Polarity

As discussed in section 1.2, the relative permittivity of a solvent, which is a macroscopic parameter, may not be sufficient to account for a variety of solvent effects that are dependent on specific solvent-solute interactions because this parameter is a macroscopic representation of the solvent that ignores microscopic interactions. Therefore, to compensate for the weakness of the macroscopic parameter approach, probe molecules that are sensitive to the nature of solvent-solute interactions have been employed. The most commonly used of these probes takes advantage of a physical process called solvatochromism.

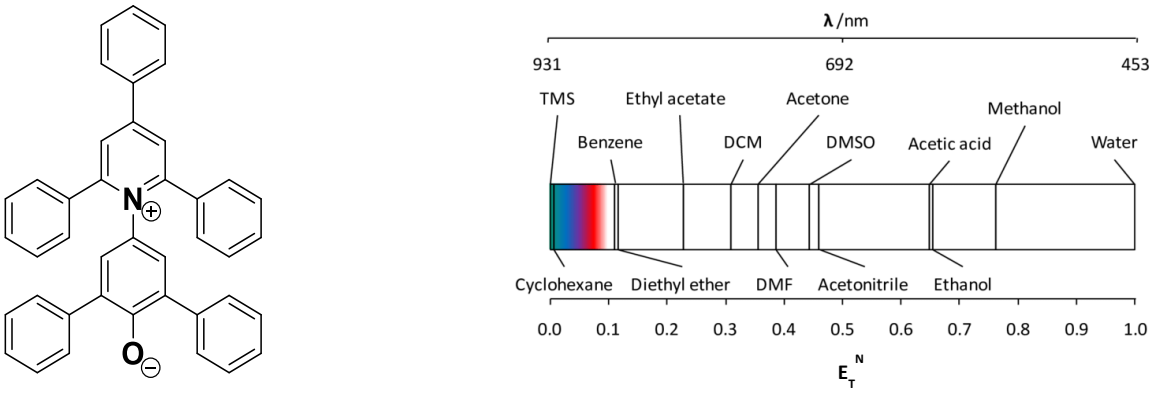
Solvatochromism is a term used to define the noticeable change in position and/or intensity of a UV/Vis absorption band that accompanies a change in the polarity of the medium. It has been established that molecules with π-electrons for which the charge distribution (and consequently the dipole moment) in the electronic ground state is considerably different from that in the excited state exhibit solvatochromism.

Hypsochromic (or blue) shift with increasing solvent polarity is usually known as negative solvatochromism whereas the term positive solvatochromism corresponds to bathochromic (or

red) shift under the same conditions. Solvent effects on electronic transitions, and hence the resulting spectra are dependent on the chromophore, a molecule which is responsible for light absorption under consideration, and the nature of the transition ($\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$, and charge-transfer absorption). The relative polarity of excited and ground states determines whether positive or negative solvatochromism is observed. More polar ground state means higher energy transition in polar solvent and has longer wavelength absorption (blue shift) on increasing solvent polarity. As primarily $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ electronic transitions occur in the UV/Vis region, these electronic transitions and charge-transfer absorption are of particular interest.[13]

One of the most widely used probes is Reichardt's Dye 30, depicted in Fig. 1.1 (a). The common use of this dye as a polarity probe is based upon the solvatochromic shift of its lowest energy transition. For example, in non-polar diphenyl ether λ_{max} = 810 nm while in water λ_{max} = 453 nm. Reichardt's Dye 30 has a highly polar ground state and less polar first excited state leading to negative solvatochromism due to the differential solvation of these two states. There is a considerable charge transfer from phenolate to the pyridinium part of the dye upon excitation.

Due to the zwitterionic nature of Reichardt's dye 30, it is susceptible to the dipolarity of the solvent as there is a large dipole moment. It is the phenolate that makes it susceptible to stabilization by a hydrogen-bond donating solute as $C\text{-}O^-$ is a strong hydrogen bond accepting group.[14]



(a) ***(b)***

Figure 1.1 *(a) Structure of Reichardt's Dye 30, a solvatochromic polarity probe (b) The solvent polarity scale derived from Reichardt's dye*.[15]

The $E_T(30)$ polarity scale was developed by Dimroth and Reichardt and defined as the energy of the lowest energy excitation of Reichardt's dye 30 in kcal mol^{-1}. $E_T(30)$ has since been determined for a wide range of solvents.[16] The E_T (30) polarity scale was subsequently normalised to form the dimensionless E_T^N scale such that water has E_T^N = 1.00 and tetramethylsilane has E_T^N = 0.00, as shown in Figure 1.1 (b) and Table 1.1.

Table 1.1 *Empirical parameters of solvent polarity, $E_T(30)$ and normalised $E_T(30)$ values, derived using Reichardt's dye 30, measured at 25 °C.*[13]

Solvents	**$E_T(30)$/(kcal . mol^{-1})**	**E_T^N**
Tetramethylsilane	30.7	0.000
Benzene	34.3	0.111
Diethyl ether	34.5	0.117
Ethyl acetate	38.1	0.228
Acetone	42.2	0.355
Dimethyl sulfoxide	45.1	0.444
Acetonitrile	45.6	0.460
Ethanol	51.9	0.654
Methanol	55.1	0.762
Water	63.1	1.000

As this scale is derived from solvent-solute interactions it leads to a more realistic measurement of solvent polarity in terms of its effect on chemical processes.[17] By means of UV-Vis spectroscopy and using Reichardt's dye 30, E_T (30) is calculated according to equation (1)[18] :

$$\mathbf{E_T\,(30)\ (kcal\ mol^{-1})} = \boldsymbol{h}_{\mathbf{CUmax}}\ \boldsymbol{N_A}$$

$$= \mathbf{(2.8591 \times 10^{-3})\ U_{max}\ (cm^{-1})}$$

$$= \mathbf{28591\ /\lambda_{max}} \qquad \mathbf{(1)}$$

Converting from the E_T (30) scale to E_T^N is performed using equation (2) :-

$$\mathbf{E_T^N} = \frac{\mathbf{E_T\,(30)_{SOLVENT} - E_T\,(30)_{TMS}}}{\mathbf{E_T\,(30)_{WATER} - E_T\,(30)_{TMS}}} \qquad \mathbf{(2)}$$

Here, TMS stands for tetramethylsilane. Since $E_T(30)_{WATER}$ = 63.1 kcal mol^{-1} and $E_T(30)_{TMS}$ = 30.7 kcal mol^{-1}, this equation simplifies to (3) :-

$$\mathbf{E_T^N = \frac{E_T(30)_{SOLVENT} - 30.7}{32.4}} \quad \textbf{(3)}$$

It has been reported that the Reichardt's Dye 30 is incompatible with acidic solvents such as carboxylic acids due to the protonation of phenolate group.[12] This has led to the development of alternative probes. Reichardt's Dye 33 is an analogue of Reichardt's Dye 30 that can similarly serve as a solvatochromic probe. Reichardt's Dye 33 has two chlorine atoms adjacent to the phenolate, as shown in Fig. 1.2. Introduction of these electron-withdrawing substituents reduces the basicity of the phenolate moiety, which allows the direct determination of $E_T(30)$ values for more acidic solvents.

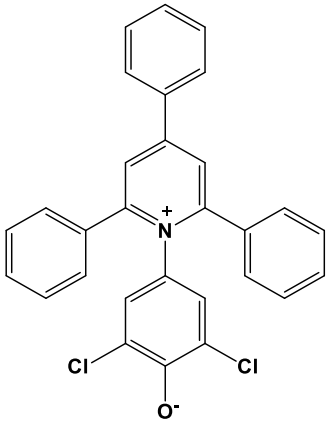


Figure 1.2 *Structure of Reichardt's Dye 33, a solvatochromic polarity probe.*

$E_T(33)$ values are computed using equation (4) using the lowest energy absorption maximum for Reichardt's Dye 33, analogous to the calculation for $E_T(30)$.

$$\boldsymbol{E_T(33) = \frac{28591}{\lambda_{max}}} \quad \textbf{(4)}$$

The empirical relationship between $E_T(30)$ and $E_T(33)$ that has been determined using linear regression is given by equation (5) and enables direct comparison between the two scales,[18]

$$\boldsymbol{E_T(30) = 0.995 \times E_T(33) - 8.113} \quad \textbf{(5)}$$

1.3.2. Multiple parameter measures of solvent polarity: Kamlet-Taft scale

Although many studies have focused on single parameter approaches, there are some limitations to these approaches. Most notably they cannot reflect the diversity of interactions present in solution. Hence, a number of multiple parameter scales have been developed to try and separately quantify the array of possible intermolecular interactions. The most widely used scale is that developed by Kamlet and Taft.

The Kamlet-Taft scale has been developed based on the concept of linear solvation energy relationships (LSERs) of the form given by equation (6).[19] In this relationship: XYZ is the solute property (solubility, reaction rate, equilibrium constant, etc.) in a given solvent, $(XYZ)_0$ is the same solute property in a reference state, α represents the hydrogen-bond donation ability, β represents the hydrogen-bond accepting ability and π^* is the dipolarity and polarizability of the solvent. π^* is derived from the change in energy of the absorption of a dye induced by the local electric field generated by the solvent and α and β from specific H-bond interactions between the solvent and probe. The parameters a, b, and s represent the solvent independent coefficients for the modelled process.[20] The values of α, β and π* were obtained from the electronic spectra of a range of solvatochromic dyes, with the most commonly used molecules Reichardt's Dye 30, *N*,*N*-diethyl-4-nitroaniline and 4-nitroaniline depicted in Figure 1.1(a) and 1.3. The selection of solvatochromic dyes is based on the dominant intermolecular interactions between the solvent and the dye. For example, both 4-nitroaniline and Reichardt's dye are dipolar solutes, however, only the phenolate centre on Reichardt's dye is able to act as a significant hydrogen-bond acceptor site affected by the hydrogen-bond donation ability of the solvent.[20]

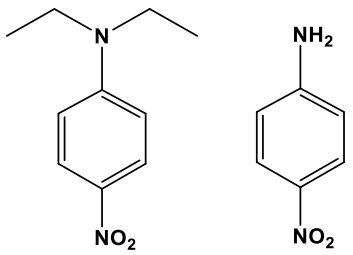


Figure 1.3 *Typical solvatochromic dye molecules used in conjunction with Reichardt's dye for the determination of Kamlet-Taft parameters.*

$$(\boldsymbol{XYZ}) = (\boldsymbol{XYZ})_{\boldsymbol{0}} + \boldsymbol{a\alpha} + \boldsymbol{b\beta} + \boldsymbol{s\pi^*} \qquad \textbf{(6)}$$

The calculation of π^*, β, and α values using the solvatochromic probes 4-nitroaniline, Reichardt's dye 30 and N,N-diethyl-4-nitroaniline can be accomplished by using Equations (7), (8) and (9). E_T (33) can be calculated by using the equations (4) introduced previously. Equation (7), (8) and (9) also require the data obtained from the indicator molecule *N,N*-dimethyl-4-nitroaniline for (7), Reichardt's dye for the E_T (33) for equation (8) and 4-nitroaniline and *N,N*-dimethyl-4-nitroaniline for (9). In these equations, ν is the experimental wave number.[21]

$$\boldsymbol{\pi^* = 0.314(27.52 - \overline{\nu}_{N,N-diethyl-4-nitroaniline})} \quad \textbf{(7)}$$

$$\boldsymbol{\alpha = 0.0649E_T(33) - 2.03 - 0.72\pi^*} \quad \textbf{(8)}$$

$$\boldsymbol{\beta = \frac{1.035\overline{\nu}_{N,N-diethyl-4-nitroaniline} + 2.64 - \overline{\nu}_{4-nitroaniline}}{2.80}} \quad \textbf{(9)}$$

Moreover, the empirical relationship between E_T(33) and E_T(30) that has been determined using linear regression is given by Equation 4 and 5, facilitating the direct determination of α from E_T(33) using Equation 10.

$$\boldsymbol{\alpha = 0.0646E_T(33) - 2.56 - 0.72\pi^*} \quad \textbf{(10)}$$

Table 1.2 represents the Kamlet-Taft solvent polarity parameters of some commonly used organic solvents. The values in Table 1.2 were calculated by using relationships similar to those shown in Equation 7-9 but with the values averaged over several different solvatochromic dye sets. As a result, these solvatochromic parameters are no longer directly based on the solvent effects shown by a single solvatochromic indicator dye which aims to reduce the impact of any effects that are highly specific to the chosen probe.

As discussed above, π^* is a measure of the solvent dipolarity/polarizability that assesses the solvent's capacity to stabilise a charge or a dipole through dielectric effects. Large values of π^* indicates a more dipolar solvent, for example water and dimethyl sulfoxide (Table 1.2). α (hydrogen-bond donating ability) describes the ability of solvent to provide a hydrogen in a solvent-to-solute hydrogen bond. In Table 1.2, the α-scale of organic solvents lies within the range of 0 for non-HBD solvents (e.g. n-Hexane) to 1.12 for acetic acid. Fluorinated alcohols, such as 1,1,1,3,3,3-hexafluoroisopropanol display extremely high α values of up to 1.96. β (hydrogen bond accepting ability), on the other hand, describes the ability of solvent to interact

with a hydrogen in a solvent-to-solute hydrogen bond. The β-scale of solvents (Table 1.2) ranges from 0 for non-HBD solvents (e.g. n-Hexane) to 0.9 for 1-propanol. [13]

Table 1.2 *Kamlet and Taft's solvatochromic parameters α, β, and π^* for selected solvents.*[13]

Solvents	**α**	**β**	**π^***
n-Hexane	0.00	0.00	- 0.11
Dichloromethane	0.13	0.10	0.73
Chloroform	0.20	0.10	0.69
Benzene	0.00	0.10	0.55
Toluene	0.00	0.11	0.49
Diethyl ether	0.00	0.47	0.24
Tetrahydrofuran (THF)	0.00	0.55	0.55
Acetone	0.08	0.48	0.62
Ethyl acetate	0.00	0.45	0.45
Dimethyl sulfoxide	0.00	0.76	1.00
Acetonitrile	0.19	0.40	0.66
Ethanol	0.86	0.75	0.54
Methanol	0.98	0.66	0.60
Acetic acid	1.12	0.45	0.64
2,2,2-Trifluoroethanol	1.15	0.00	0.73
1,1,1,3,3,3-Hexafluoro-2-propanol	1.96	0.00	0.65
1-Propanol	0.84	0.90	0.52
Water	1.17	0.47	1.09

1.3.3. Pyridine-N-oxide probe

Although Reichardt's dye has been widely studied and used for various solvents as probe, there are some problems associated with the use of this dye which makes it unsuitable for the characterisation of polarity for all solvents. As discussed previously, this probe is unable to be used to measure the polarity of acidic solvents since the protonation of the phenolate occurs, which effectively "bleaches" the dye as it changes the HOMO-LUMO gap resulting in the loss of the solvatochromic visible absorbance. To overcome the issue of the bleaching of Reichardt's dye, pyridine-N-oxide (PyO), as shown in Figure 1.4, has been proposed as an alternative probe.

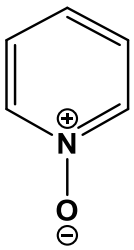


Figure 1.4 *Structure of pyridine-N-oxide for the determination of α value through ^{13}C NMR.*

Pyridine-N-oxide is not a solvatochromic probe; however, the relative position of the ^{13}C NMR signals of carbons 2, 3 and 4 in the ring change with solvent polarity. This probe can be used to characterize the hydrogen-bond acidity of various solvents by measuring the chemical shifts in ^{13}C NMR. The α values of solvents can be determined by the ^{13}C chemical shifts of PyO according to the following equations[22]:

$$\boldsymbol{\alpha_{24} = 2.32 - 0.15 \times d_{24}} \quad \textbf{(11)}$$

$$\boldsymbol{\alpha_{34} = 0.40 - 0.16 \times d_{34}} \quad \textbf{(12)}$$

Where α_{24} and α_{34} are the estimated K-T hydrogen bond donating parameters, d_{24} and d_{34} are the differences (in ppm) of the ^{13}C NMR chemical shifts, δ, of carbons 2 and 4, and 3 and 4, respectively. Empirical linear regression has found that $\alpha_{24} = 0.88\alpha_{RD}$ where α_{RD} is the α value calculated using Equations 8 or 10. Hence dividing α_{24} by 0.88 should yield a comparable value of α to that obtained using Reichardt's Dye 30 or 33 as part of the dye set. This value α_{PyO}, is defined as per Equation (13).[22]

$$\boldsymbol{\alpha_{PyO} = \frac{\alpha_{24}}{0.88}} \quad \textbf{(13)}$$

1.4. Ionic Liquids (ILs)

ILs are salts that have melting temperatures below 100 °C.[23] ILs that are liquid at room temperature are often known as room temperature ionic liquids (RTILs). IL are composed of large organic cations and organic or inorganic anions and often these ions contain a delocalised charge. Many ILs also have ions that are of significantly lower symmetry than conventional salts. The low symmetry of these ions further frustrates packing resulting in reduced melting points of ILs.[24]

Furthermore, ILs are generally considered as "designer" solvents with "tuneable" features that can be adjusted to suit the requirements of a particular process.[22] The chemical and physical

properties of ILs can be altered by the selection of cation and anions. For example, the choice of anion and the alkyl chain length on the cation play an important role on the solubility of ILs in water.[25] Hence, selecting appropriate IL ions is important as they change the properties of the IL which influences their ability to suit certain applications. A more detailed explanation of ILs as designer solvents has been discussed in section 1.4.1. Among the enormous scope of ILs that have been synthesised to date, the most common ILs are based on nitrogen-containing cations combined with different anions. Some examples of these are shown in Figure 1.5

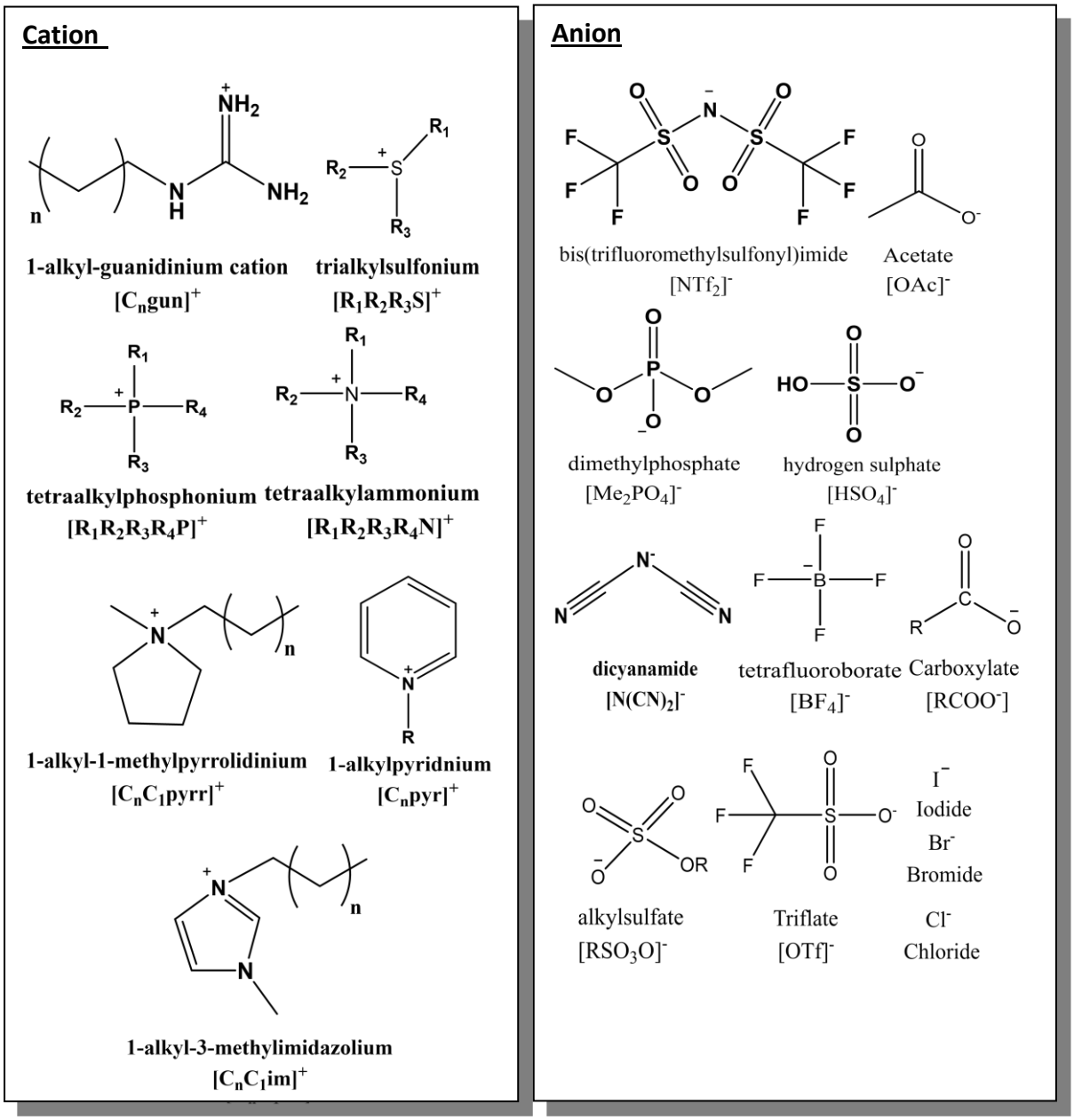


Figure 1.5 *The structures of representative IL cations and anions along with the abbreviations that will be used in this thesis.*

The first known RTIL discovered was ethyl ammonium nitrate ($[C_2N][NO_3]$), which has a melting point 12℃.[25] The next major class of ILs that were developed were based on Lewis acids and involved mixtures of dialkylimidazolium and alkylpyridinium halides with $AlCl_3$ and $AlBr_3$. These were explored as electrolytes in batteries due to their high conductivity.[25] All ILs display unique properties, namely negligible vapour pressure, inherent ionic/electrical conductivity and typically low flammability, while some selected ILs also display properties such as high thermal and chemical stabilities, high solvation ability for organic, inorganic and organometallic compounds and wide liquidus ranges.[23,26] Due to these features, ILs act as a potential alternative to volatile organic compounds (VOCs) for diverse applications. Selected applications of interest for ILs include as electrolytes for batteries and for the electrodeposition of metals, bio-catalysis, and as solvents for organic synthesis, polymerization, separation and extraction processes and for the dissolution of biomolecules and biomaterials.[23,27]

1.4.1. Classification of Ionic liquids (ILs)

ILs can be classified into many categories based on the nature of their ions. Two major categories used to define ILs include the categorisation of ILs as protic ILs or aprotic ILs. These terms are explained below.

1.4.1.1. Protic ILs (PILs)

Protic ILs (PILs) have exchangeable protons attached to the cation as shown in Figure 1.6. PILs tend to be more acidic than aprotic ILs, and are of great interest in fuel cell applications due to their proton transporting ability.[28] As they can be formed by direct reaction of an acid and base they are cheaper to prepare than many aprotic ILs (AILs) but the speciation is more complex as it is affected by the presence of acids/bases.

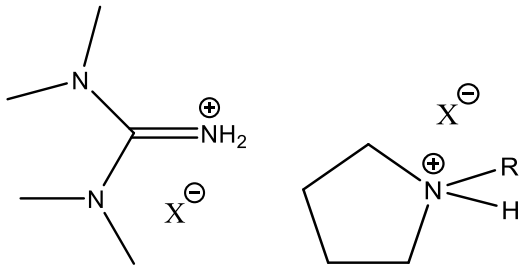


Figure 1.6 *Examples of PILs*

1.4.1.2. Aprotic ILs (AILs)

Aprotic ionic liquids (AILs) have substituents other than a proton (usually an alkyl group) at the position occupied by the labile proton in an analogous PIL.[29] Some examples of AILs are shown in Figure 1.7. Unlike PILs, the speciation of AILs is more straightforward and unaffected by acid/base equilibria but they are often more expensive and complicated to synthesise in comparison.

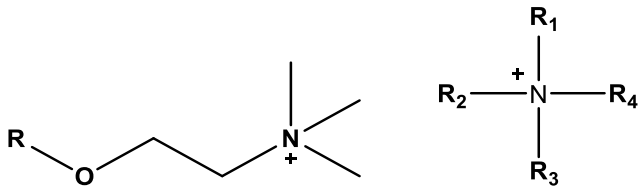


Figure 1.7 *Examples of AILs*

PILs and AILs can be altered with the incorporation of additional functional groups at either the cation or anion to design what is known as a task specific IL. A brief explanation is given below.

1.4.1.3. Task-Specific ILs (TSILs)

ILs that have additional functional groups incorporated are also known as task specific ILs (TSILs). The incorporation of these additional functional groups aims to enhance certain properties of the IL, for example, their ability to interact with catalysts, to make the IL more effective at performing a certain task.[28] The introduction of different functional groups on the imidazolium cation and the general synthesis of TSILs are shown in Figure 1.8 and Scheme 1, respectively.

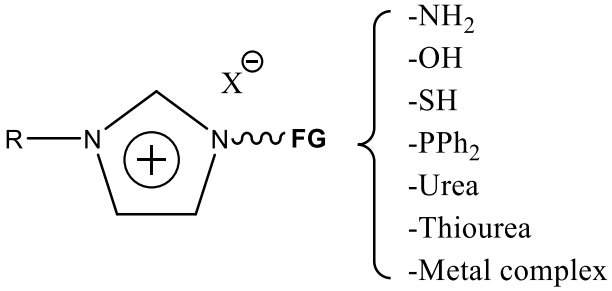
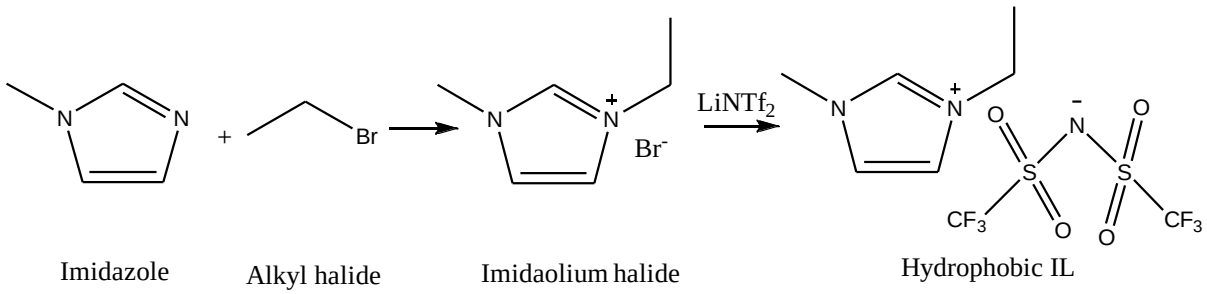


Figure 1.8 *Functionalised imidazolium salts as TSILs*



Scheme 1 *General synthesis of TSILs from 1-alkylimidazoles.*[30]

1.4.2. Synthesis of Ionic liquids (ILs)

The synthesis of AILs can generally be divided into two routes: (i) the formation of the desired cation through alkylation reactions, and (ii) anion exchange through anion metathesis. The synthesis of PILs is more straightforward and typically involves the addition of the appropriate acid to the relevant base (e.g. an alkylamine).

In the alkylation reaction, the most commonly used method for the IL synthesis involves the alkylation of an amine or phosphine, e.g. 1-methylimdazole with a haloalkane.[31] The main advantage of this route is the availability of cheap haloalkanes and the relatively mild temperature employed. The reactivity of the haloalkane increases in the order chloroalkane < bromoalkane < iodoalkane. To obtain the best results and the purest product, all starting reagents and solvents should be thoroughly dried and purified before use.

The properties of the IL can be fine-tuned by anion selection through the metathesis of the anion to afford a new IL. Anion metathesis has become one of the most utilised methods for the synthesis of ILs.[31] The common concept for all anion metathesis procedures is to isolate the desired IL from the by-product salt. Depending on whether the desired IL is hydrophobic or hydrophilic, this process differs. Water can be used as a solvent to remove the unwanted salt if the IL is hydrophobic. In this example, the two salts are combined in an aqueous solution in a stoichiometric quantity. As the desired IL is formed it will separate out of the aqueous phase into a separate IL layer, or precipitate, depending on the IL's melting point. This layer, or precipitate, can then be readily separated, and remaining IL can be extracted by washing the aqueous phase with an appropriate organic solvent such dichloromethane. The procedure for hydrophilic ILs is a little difficult, and it varies based on the properties of the by-product salt. Either a liquid-liquid extraction is utilised, or the counter ion of the salt designed to lead to an

insoluble by-product precipitate. The organic solvent employed to assist liquid-liquid extraction should be water-immiscible, but it should also be able to dissolve the required hydrophilic IL preferentially. Moreover, the produced by-product salt should preferably be an inorganic alkali metal halide salt due to the fact that they have a poor solubility in organic solvents and a high affinity for water, which makes their removal easier. A common method for this is to dissolve a stoichiometric quantity of the two salts in a water-insoluble solvent, such as dichloromethane. As the salt formed, it precipitates out from the liquid phase and may be readily filtered away. To remove the final traces of salt, the filtered solution is washed many times with a small amount of water. The difficulty with aqueous washing is to only extract the by-product salt and not significant amount of the IL. An alternative approach to preparing a hydrophilic IL is the use of salts that form insoluble by-product precipitates, such as silver salts that form insoluble silver halides on metathesis. The challenge with this approach is ensuring there is minimal residual contamination by the added salt. The schematic representation of both alkylation of cation and anion metathesis is shown in Figure 1.9.

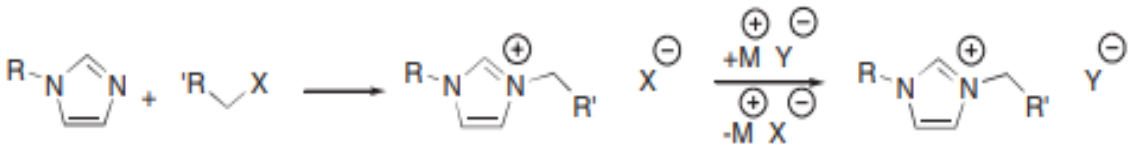


Figure 1.9 *IL synthesised by quarterisation reaction (alkylation) of imidazole (or amine), followed by anion exchange reaction.*[32]

1.5. Deep Eutectic Solvents (DESs)

A DES is generally defined as the mixture of two or more components, which may be solid or liquid, that are capable of self-association, often through hydrogen bond interactions, to form a eutectic mixture with a melting point significantly lower than that of each individual component. In many cases, these components are safe, cheap, renewable, and biodegradable.[33] DESs exhibit eutectic behaviour where the two components are mixed together results in a depression in the freezing temperature of the system, as shown in Figure 1.10. The difference between the freezing temperatures of the eutectic composition of a two-component system compared to the theoretical freezing temperature of the ideal mixture is dependent on the strength of the interaction between the two components and the size and shape of the aggregates formed. Strong interactions and unsymmetrical aggregate structures tend to result in larger

deviations from the ideal freezing temperature, whereas weak interactions result in less pronounced deviations from ideality.

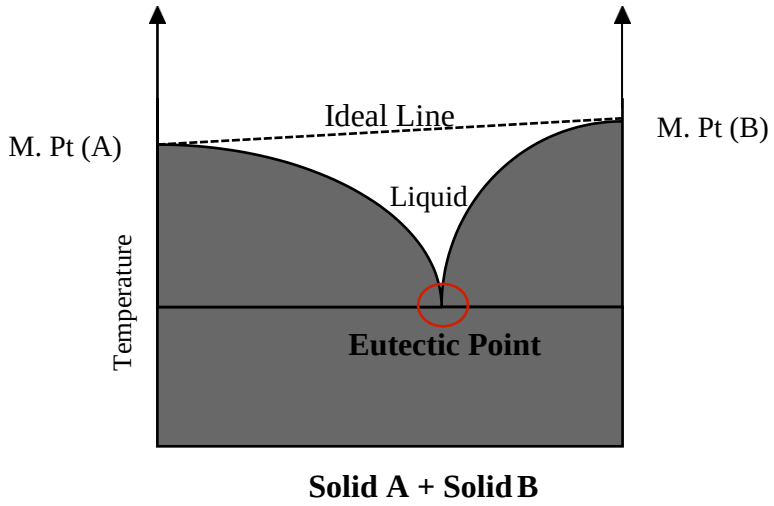


Figure 1.10 *Typical freezing point behaviour of DESs.*[33]

DESs enable the creation of solvents from otherwise high melting components resulting in unique properties such as electrical conductivity and low vapour pressures. DESs can be prepared from cheap, renewably obtained substances and as their preparation simply involves the mixing of these compounds, the energy and resource requirements for their synthesis are typically low.

DESs represent a new family of ionic fluids and display many similarities to ILs. As ILs are often expensive and can be toxic, DESs have been increasingly investigated as alternatives. DESs are obtained by the interaction of a hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA).[18] At ambient temperature, melts consisting of organic compounds can be formed between quaternary ammonium salts as the HBA with HBDs such as amides, amines, carboxylic acids or alcohols.

1.5.1. Classification of deep eutectic solvents (DESs)

DESs are classified based on the nature of the interaction between components and the composition of those components. The following types of DES were originally proposed by Smith et al. in 2014[33], with Type V DESs being added subsequently due to their recent development.[34]

Type I DES = Metal salt + organic salt {e.g., $ZnCl_2$ + ChCl}

Type II DES = Metal salt hydrate + organic salt {e.g., $CoCl_2.6H_2O$ + ChCl}

Type III DES = Organic salt + hydrogen bond donor {e.g., ChCl + urea}

Type IV DES = Metal salt + hydrogen bond donor {e.g. $ZnCl_2$ + urea}

Type V DES = Non-ionic organic mixtures {e.g. menthol and thymol}

Type I include the chloroaluminate/imidazolium salt melts formed with various metal halides such as $FeCl_2$, AgCl, $CuCl_2$ and $SnCl_4$. These DESs are also a subclass of ILs since they consist solely of ions.

Type II eutectics include hydrated metal halides and organic salts such as choline chloride (ChCl) and the relative low cost of many hydrated metal salts coupled with organic salts make their use in large scale industrial processes economically feasible.[35]

Type III DESs contain only organic compounds. The general formula of Type III DESs is Cat^+X^-zY where Cat^+ = Cation, X is a Lewis base (generally a halide anion) and Y is a Lewis acid (z refers to the number of Y molecules that interact with the anion).[36] Cat^+ can be any ammonium, phosphonium or sulfonium cation. There is a complex formed between X^- and either a Lewis or Bronsted acid Y. Type III DES are often simple to prepare, usually unreactive with water, biodegradable and relatively low cost and represent the most widely studied type of DES to date. Due to their ability to solvate a wide range of organic and transition metal species, including chlorides and oxides, this class of DESs has been shown to be particularly versatile. The most common organic salt explored for Type III DES is choline chloride, (ChCl, refer to Figure 1.10). ChCl can be synthesized by extraction from biomass or from fossil fuels. Choline is a nontoxic, low cost and biodegradable cation but ChCl melts over 300°C making it unusable as a solvent in its own right. ChCl is capable of forming many DESs as an HBA. These include with HBDs such as urea, acetamide, glycerol, oxalic acid, and citric acid with many of these combinations being room temperature liquids.[33] A wide range of possible applications has been investigated of Type III DESs including the removal of glycerol from biodiesel, processing of metal oxides, and the derivatisation of cellulose.[33]

Type IV DESs are a combination of type I and type III where a transition metal chloride is incorporated as the salt of the type III DES. For instance, $ZnCl_2$ has been shown to form eutectics with urea, acetamide, ethylene glycol and 1,6-hexanediol.[33,36]

Type V DESs are purely organic mixtures, like type III but do not contain a salt. While many such mixtures have been claimed, relatively few have been shown to be non-ideal beyond the recent discovery of a eutectic mixture of 1:1 thymol and menthol.[37]

Figure 1.11 represents the list of some HBAs and HBDs forming Type III DESs on mixing as these DES are the most widely explored and the DES most relevant to this thesis.

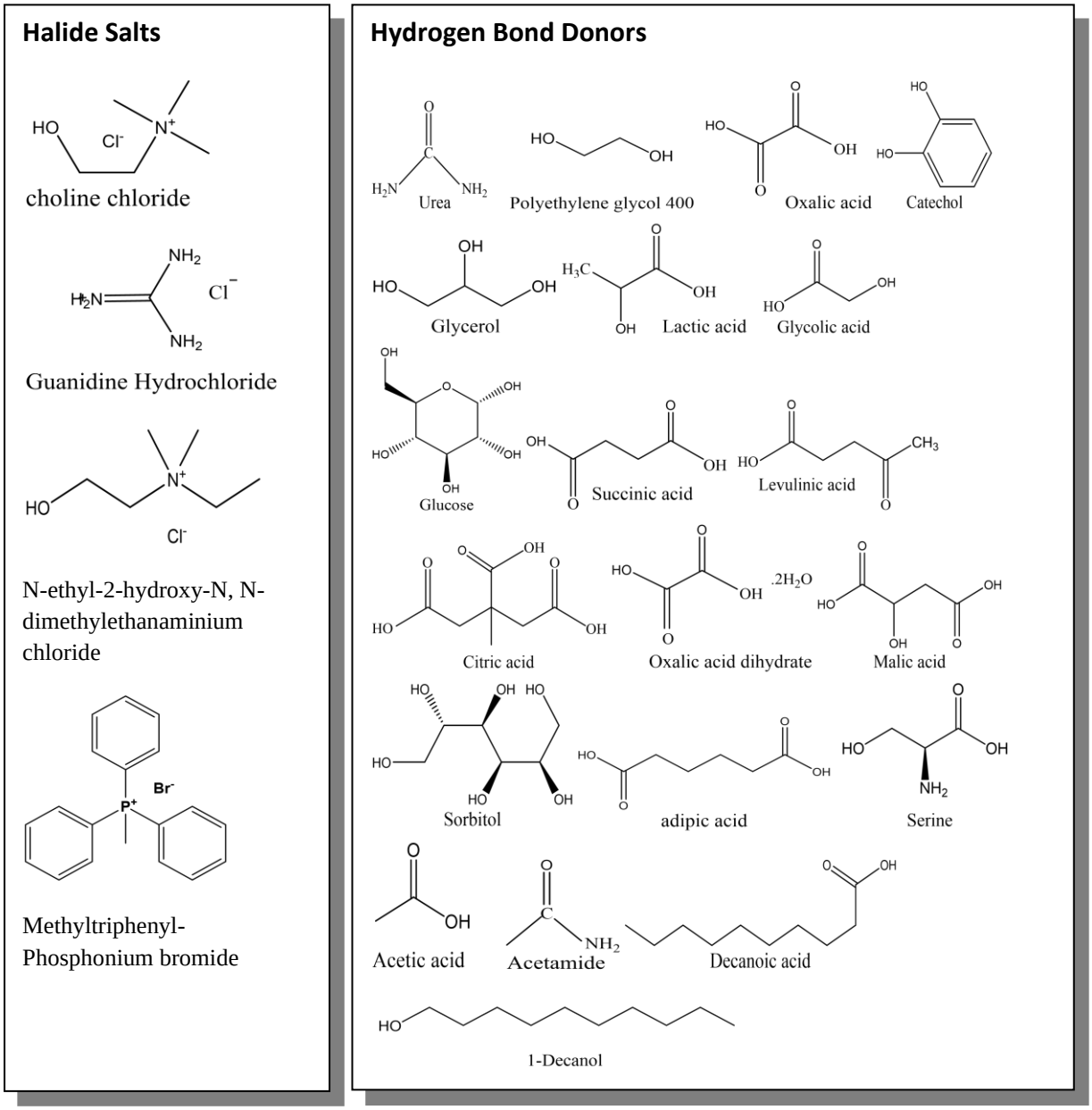


Figure 1.11 *Structures of some halide salts and HBDs used in the formation of Type III DESs.*

1.5.2. Synthesis of deep eutectic solvents (DESs)

DES can be easily synthesised by directly mixing the two or more components and stirring the mixture at either room temperature or at elevated temperature. A mechanism of DES formation for Type III DES has been proposed involving a delocalisation of charge arising from hydrogen bonding between a halide ion and urea.[33] This is responsible for the decrease in the melting point of the mixture relative to the melting point of the individual components as it causes a decrease in lattice energy due to the reduction of the cation-anion interaction energy by the introduction of (weaker) hydrogen bond interactions with the anion, as well as an increase in entropy from the formation of the mixture.[33] One of the most significant deep eutectic phenomena (Type III) is observed for a combination of ChCl and urea in a 1:2 mole ratio, as shown in Fig 1.12.[18] The resulting mixture has a melting point of 12°C, far less than the melting point of pure ChCl (302°C) or urea (133°C).[18] In this DES, ChCl acts as an HBA while urea is an HBD. This significant depression of the freezing point arises from the interactions between urea molecules, the chloride ion and, as has recently been shown, the choline cation as well.[38] A wide range of eutectic mixtures have been prepared from urea and quaternary ammonium salts as well as between ChCl and a range of HBDs.

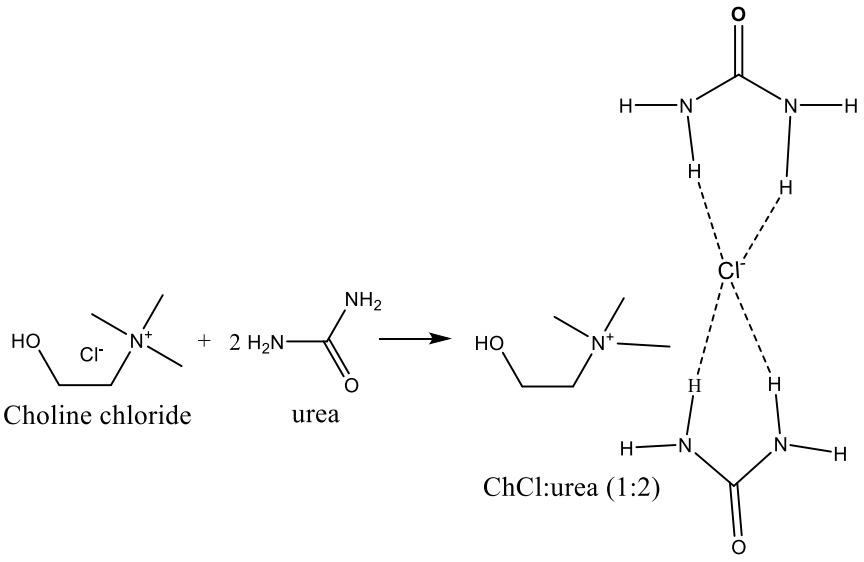


Figure 1.12 *Schematic presentation of one of the proposed ChCl-based DESs hydrogen-bonding interactions.*[39]

1.6. Physicochemical Properties of Ionic liquids (ILs) and Deep eutectic solvents (DESs)

As discussed previously, to qualify as viable green media, ILs and DESs must meet different criteria such as availability, non-toxicity, biodegradability, recyclability, flammability, and affordability among others. Owing to the remarkable properties of ILs and DESs, such as negligible vapour pressure, non-flammability, good solvating properties, and many more, these solvents are of growing interest in many fields of research. The emerging contribution of both solvent classes in catalysis, organic synthesis, dissolution and extraction processes, electrochemistry, CO_2 capture, and material chemistry demonstrates that ILs and DESs have the potential to allow the design of new eco-efficient processes**.** One of the main advantages of ILs is that they are pure compounds which simplifies their recycling but makes their synthesis typically more complex and restricts the salts that can used. Similarly, DESs provide advantages in comparison with pure ILs, such as the fact that DES preparation may be carried out more simply from abundant, cheap and biodegradable precursors and purification is more straightforward, and hence making them more feasible for use in large-scale processes but their reuse is complicated by the potential for changes in the stoichiometry of the mixture or selective leaching of one component. Both solvent classes have unique properties of their own that allow their use for multiple applications.[40,41] A key advantage of both classes of solvents is the ability to control their physicochemical properties through the selection of components. Physical properties important for the use of DESs and ILs including density, viscosity, and conductivity are discussed in the coming sections.

1.6.1. Density

The density is an important physical property for a solvent, particularly when used for liquid-liquid extraction processes. The density of ILs and DESs is strongly influenced by the size and mass of constituent ions and molecules. For ILs this is particularly affected in the case of commonly used fluorinated ions given the much greater mass of fluorine than hydrogen despite their similar volume. For example, ILs containing the bis(methanesulfonyl)amide ($[NMs_2]^-$) anion have lower densities than $[NTf_2]^-$ salts.[42]

In addition to the mass of ions, the packing of ions and molecules affects the densities of both DESs and ILs.[18,36] The packing of DES tends to play a greater role in their density than ILs due to the larger number of components involved which are associated through a variety of

interactions rather than predominantly coulombic interactions. However, the density of both ILs and DESs have been interpreted in terms of the presence of vacancies in the liquid phase as a result of packing effects.[18,36] When the salt of the DES is mixed with the HBD, it has been measured that the average hole radius decreased, resulting in a slight increase of the DES density as compared to that of the individual HBD (e.g. ethylene glycol : 1.11 g cm^{-3}, ChCl: ethylene glycol : 1.12 g cm^{-3}).[18] While an increase in density of ILs can occur in the presence of strong association between the ions of the ILs**.** Moreover, density is a temperature dependent property. It decreases with increasing temperature due to less compact packing and an increase in void spaces within the liquid. Table 1.3 lists the density of some DESs, ILs and molecular solvents at 25 ℃.

Table 1.3 *Density of DESs, ILs and molecular solvents at 298 K.*

Salt	**HBD**	**Salt : HBD**	**Density (g cm^{-3})**	**Ref.**
ChCl	Urea	1 : 2	1.25	[18]
ChCl	Glycerol	1 : 1	1.16	[18]
ChCl	Ethylene glycol	1 : 2	1.12	[18]
ChCl	Malonic acid	1 : 2	1.25	[18]
$[C_4C_1im][Cl]^a$		-	1.08	[43]
$[C_8C_1im][Cl]^a$		-	1.00	[43]
$[C_4C_1im][NTf_2]^a$		-	1.43	[43]
$[C_4C_1im][PF_6]^a$		-	1.36	[43]
	Urea	-	1.32	[44]
	Glycerol	-	1.26	[44]
	Ethylene glycol	-	1.11	[44]
	Malonic acid	-	1.62	[44]
	Water	-	0.897	[44]

{[a] $[C_4C_1im][Cl]$: 1-butyl-3-methylimidazolium chloride; $[C_8C_1im][Cl]$: 1-octyl-3-methylimidazolium chloride; $[C_4C_1im][NTf_2]$: 1-butyl-1-methylimidazolium bis(trifluoromethylsulfonyl)imide; $[C_4C_1im][PF_6]$: 1-butyl-3-methylimidazolium hexafluorophosphate}

It can be seen from the Table 1.3 that the most of the HBDs exhibit higher densities than DESs. It is possible that the reduction in density could be due to less efficient packing in the DES.

However, the density of the ions of the salt also needs to be considered as these may increase or decrease the density of the resultant liquid even with the same extent of packing

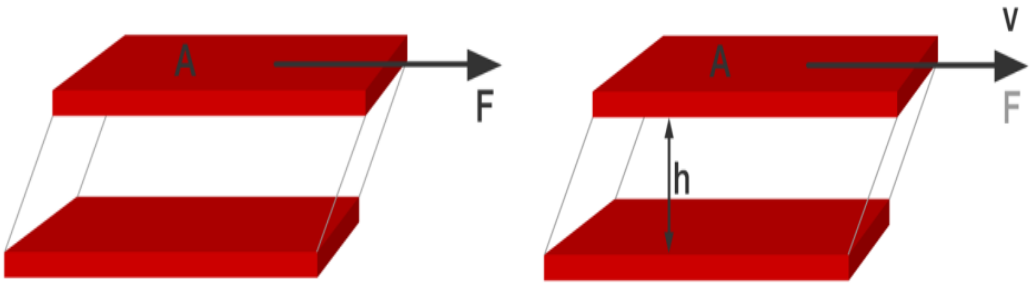
The following trend in density was observed for type III ChCl based DESs: ChCl:urea = ChCl: malonic acid > ChCl: glycerol > ChCl: ethylene glycol. ChCl: urea/malonic acid eutectic mixtures have higher density (1.25 g cm^{-3}) than other eutectic mixtures. It has been concluded that the electron rich oxygen of HBDs (urea and malonic acid) likely interacts with ChCl resulting in tighter intermolecular packing between ChCl and HBD.[18,45] For the remaining two DESs, ChCl: ethylene glycol and ChCl: glycerol, the density decreases due to the less compact nature of intermolecular packing for these systems.

Regarding the density of the ILs, the following trend in density was observed for imidazolium based ILs: $[C_4C_1im][NTf_2]$ > $[C_4C_1im][PF_6]$ > $[C_4C_1im][Cl]$ > $[C_8C_1im][Cl]$.[45] As discussed above, the simple halide salts have lower densities than RTILs that contain larger fluorinated anions. From Table 1.3, it has been determined that the densities of ILs with fluorinated ions are higher than DESs and conventional solvents. The Cl^- ILs had lower densities than the DES which is likely due to the reduced packing density of these ILs compared to the DES. The incorporation of the CH_2 groups to the alkyl chain on the $[C_nC_1im]^+$ cation reduces the density because the hydrophobic CH_2 groups are less densely packed than the charged imidazolium ring. These lower densities are similar to those of common solvents, for example, the density of $[C_8C_1im][Cl]$ is similar to the density of water.[43]

The water content in solvents can also affect the density of the liquid. The addition of water typically decreases the density of the IL or DES due to the slight disruption of the interactions between the ions or other components of the ILs/DESs.[43,46] Understanding the impact of water on the density of these liquids is beneficial in designing many applications such as extraction processes.[46]

1.6.2. Viscosity

Viscosity is of key importance for the design of ILs and DESs for many applications. The viscosity describes the internal resistance of a fluid. This can be visualised using the two-plates model, as shown in Figure 1.14, which provides a mathematical description for viscosity and explains two important terms for understanding the viscosity of fluids, shear rate and shear stress.[47]



(a) **(b)**

Figure 1.14 *Using the two-plates model to calculate the (a) shear stress and (b) shear rate.*[47]

The two-plate model is represented by two plates with fluid placed in between. The shear stress ($N\ m^{-2}$) the liquid is subjected to is the force moving the upper plate divided by the plate area, while the shear rate (s^{-1}) is the velocity of the moving plate divided by the distance between the plates.

There are two classes of viscosity behaviour in fluids with respect to shear rate: Newtonian and non-Newtonian. Newtonian fluids have a constant viscosity coefficient (η) regardless of the imposed shear rate. For example: water, glycerol and acetone. In contrast, non-Newtonian fluids have a variable viscosity coefficient (η) depending on the imposed shear rate and can either shear thicken or shear thin. ILs were initially believed to be Newtonian fluids, but recent results suggest that some, but not all ILs, are non-Newtonian fluids.[48,49] For example, for $[C_nC_1im][BF_4]$ (n = 4, 6, 8) the shear rate produced no difference in the viscosity while for ILs with longer alkyl chains (e.g., $[C_{12}C_1im][BF_4]$) the viscosity was found to decrease with shear rate, displaying non-Newtonian behaviour.

ILs are generally more viscous than molecular solvents, with typical ILs at room temperature have viscosities ranging from 10 cP to over 500 cP, compared to water and ethylene glycol which have viscosities of 0.890 and 16.1 cP, respectively.[42] The viscosities of DESs are often higher and cover an even wider range than what is generally observed with ILs, with viscosities from 36 cP for a eutectic mixture of ChCl and ethylene glycol, to approximately 11000 cP for a eutectic mixture of ChCl, $ZnCl_2$ and urea at 298 K.[18,40]

It has been well established that intermolecular interactions (H-bonding, dispersive and coulombic interactions) play a crucial role in determining the viscosity of ILs and DESs.[50] More attractive interactions typically lead to reduced mobility of ions and constituents, increasing the viscosity of the fluid. For ILs, it was found that increasing the alkyl chain length above

butyl (n > 4) in the $[C_nC_1im]^+$ cation for $[C_nC_1im][PF_6]$ ILs led to an increase in viscosity due to increasing Van der Waals interactions between the aliphatic alkyl chains.[51]

The temperature[50] and additives (halides, water)[46] can also have a significant effect on the viscosity of solvents. Increasing the temperature decreases the strength of intermolecular forces and, therefore, the viscosity of the solvent. Similarly, addition of additives such as water into the solvents weakens the interaction between the components of the solvents and hence, decreases the viscosity.[46]

Another factor which affects the viscosity is the composition of the DESs and ILs. Changes in the size of the molecular constituents and the degree of association between the constituents of the DES affects the ease of movement of the constituents, influencing its viscosity.[50] In the case of ILs, the increased symmetry of the weakly interacting $[PF_6]^-$ and $[BF_4]^-$ anions was found to increase the viscosity of the resultant ILs compared to the less symmetric $[NTf_2]^-$ anion, with some representative data shown in Table 1.4. This is potentially due to the improved ion packing within the symmetric systems leading to stronger interactions between ions and reduced ion mobility. As discussed previously, increasing the length of the alkyl chain attached to the cation tends to increase the viscosity of the IL due to the presence of bulkier ions with stronger Van der Waals interactions between the alkyl chains.[51]

The viscosity data of DESs at 25 °C are also represented in Table 1.4. It can be seen that the viscosity of ChCl-based DESs is strongly influenced on the nature of the HBD.[18] The viscosity of 1:2 ChCl:ethylene glycol, for instance, is relatively low (37 mPa s) compared to the more strongly interacting glycerol (259 mPa s). In contrast, the use of malonic acid as a HBD resulted in DESs with high viscosity (1124 mPa s) as this is a diacid capable of forming more robust intermolecular hydrogen-bond networks.

Table 1.4 *Viscosity of selected DESs and ILs (25 °C).*[18,43]

Salt	HBDs	Salt : HBD	Viscosity/mPa s (η)
ChCl	Ethylene glycol	1 : 2	37
ChCl	Glycerol	1 : 2	259
ChCl	Urea	1 : 2	750
ChCl	Malonic acid	1 : 2	1124
$[C_4C_1im][NTf_2]^a$	-	-	69
$[C_4C_1im][PF_6]^a$	-	-	450
$[C_6C_1im][PF_6]^a$	-	-	585
$[C_8C_1im][PF_6]^a$	-	-	682

{[a] $[C_4C_1im][NTf_2]$: 1-butyl-1-methylimidazolium bis(trifluoromethylsulfonyl)imide; $[C_4C_1im][PF_6]$: 1-butyl-3-methylimidazolium hexafluorophosphate; $[C_4C_1im][PF_6]$: 1-hexyl-3-methylimidazolium hexafluorophosphate $[C_8C_1im][PF_6]$: 1-octyl-3-methylimidazolium hexafluorophosphate }

Hence, the viscosity of most DES and IL systems is a function of three parameters: the composition, the strength of intermolecular interactions and the temperature. Owing to the presence of a strong hydrogen bond network and high degree of association between each component within a DES, the viscosity of DESs are typically higher than ILs, which further results in a lower mobility of ions and molecules within DES.[50,51]

1.6.3. Conductivity

Conductivity is significant for liquids being explored as electrolytes for electrochemical applications. Conductivity is affected by factors such as charge carrier concentration, and carrier mobility. DESs that are prepared from salts and all ILs are conductive indicating that ionic species are dissociated in the liquid and are able to move independently.[52] Similar to viscosity, factors such as temperature and composition play a vital role in changing the conductivity of IL and DES systems.

The electrical conductivity of DESs would be expected to increase with the concentration of charge carriers, i.e. the concentration of the salt in the mixture. Table 1.5 demonstrates that this is not universally true. While the conductivity increases with the composition of salt in ChCl:glycerol DESs, the opposite trend is observed for many DES such as ChCl: ethylene glycol, N, N-diethylethanolammonium chloride : ethylene glycol DESs and phosphonium-

based DESs depicted in Table 1.5. In these DESs, at higher salt concentrations the electrical conductivity decreases due to the strong influence of ionic association and viscosity. Owing to this, the overall mobility and number of effective charge carriers reduces resulting in lower conductivity.[53] Hence, it can be concluded that it is a combination of the mobility as well as the number of charge carriers which control the conductivity of DESs. Increased ionic mobility can be obtained by using smaller and more weakly associated ions in liquids, although unfortunately these effects are often strongly anticorrelated with smaller ions typically being more strongly associated.[18] The conductivities of common DESs and ILs are typically in the range of 0.1 – 14 mS/m.[52]

Table 1.5 *Conductivity of selected DESs at 298 K.*[53]

Salt	**HBD**	**Salt : HBD**	**Conductivity (mS m^{-1})**
ChCl	Ethylene glycol	0.28 : 0.72	8.317 **(DES1)**
	Ethylene glycol	0.36 : 0.64	6.801
	Glycerol	0.25 : 0.75	1.463
	Glycerol	0.34 : 0.66	1.749
N, N – diethyl ethanol ammonium chloride	Ethylene glycol	0.20 : 0.80	5.661 **(DES2)**
	Ethylene glycol	0.28 : 0.72	5.12
	Glycerol	0.25 : 0.75	0.602
	Glycerol	0.34 : 0.66	0.75
Methyl triphenyl phosphonium bromide	Ethylene glycol	0.16 : 0.84	1.942 **(DES3)**
	Ethylene glycol	0.20 : 0.80	1.557
	Glycerol	0.25 : 0.75	0.103
	Glycerol	0.34 : 0.66	0.062

Regarding ILs, the nature of the ions has a strong influence on their conductivity. For example, Yuan et al. have investigated the role of the cation on the conductivities of four dicyanamide ($[N(CN)_2]^-$) anion based ILs as shown in Table 1.6. At 298 K, the increasing order of conductivity for these ILs was as follows: $[C_4C_1im][N(CN)_2]$ > $[C_4C_1C_1^2im][N(CN)_2]$ > $[N_{4442}][N(CN)_2]$ > $[N_{8444}][N(CN)_2]$. The difference of an order of magnitude in the conductivities among these ILs that all share a common counter ion suggests that the

availability of charge carriers is not the only factor influencing the conductivities of the ILs. Ion aggregation/pairing and increased viscosity may reduce the ion mobility resulting in a reduction in accessible charge carriers.[54] Additionally, the order of the viscosities of four dicyanamide ($[N(CN)_2]^-$) ILs was given as follow : $[N_{8444}][N(CN)_2]$ > $[N_{4442}][N(CN)_2]$ $[C_4C_1C_1{}^2im][N(CN)_2]$ > $[C_4C_1im][N(CN)_2]$. The viscosity trend shows that the salts with imidazolium cations are less viscous than the quaternary ammonium-based cations ILs.[54]

Table 1.6 *Abbreviations and conductivity of selected $[N(CN)_2]^-$ ILs at 298 K.*[54]

Ionic Liquids	**Abb.**	**Conductivity (mS cm^{-1})**
1-butyl-3-methylimidazolium dicyanamide	$[C_4C_1im][N(CN)_2]$	10.09
1-butyl-2,3-dimethylimidazolium dicyanamide	$[C_4C_1C_1{}^2im][N(CN)_2]$	2.88
N-ethyl-N,N,N-tributylammonium dicyanamide	$[N_{4442}][N(CN)_2]$	5.08
N-octyl-N,N,N-tributylammonium dicyanamide	$[N_{8444}][N(CN)_2]$	1.08

The conductivity of ILs can also be influenced by the anion. For example, the conductivity of $[C_2C_1im]^+$ salts was found to be dependent on the calculated anion volume and viscosity, with conductivity decreasing in the following anion order : $[BF_4]^-$ (48 Å^3, 13 mS/cm) > $[NTf_2]^-$ (143 Å^3, 8.4 mS/cm) > $[Beti]^-$ (or bis(perfluoroethylsulfonyl)imide) (189 Å^3, 3.6 mS/cm). In general, conductivity is increased by decreasing the ion size for ILs.[55]

Additionally, it has been found that the conductivity of both solvent classes (ILs/DESs) increases exponentially with increasing temperature, due to a decrease of the viscosity, as would be expected. Table 1.7 represents the variation of the conductivity of some of the solvents introduced previously in this section with temperature.[53,56]

Table 1.7 *DES/IL conductivity at different temperatures.* [53, 56]

Temp (K)	Conductivity (mS m^{-1})					
	DES1[53]	**DES2**[53]	**DES3**[53]	**IL1**[56]	**IL2**[56]	**IL3**[56]
298.15	8.317	5.661	1.942	3.58	2.84	1.09
303.15	10.665	6.994	2.57	4.00	3.18	1.21
308.15	12.07	8.245	3.103	4.60	3.56	1.38
313.15	14.133	9.699	3.845	4.95	3.93	1.57
318.15	16.558	11.579	4.437	5.45	4.33	1.77
323.15	17.977	13.408	5.072	5.95	4.73	1.97

1.6.4. Polarity

As discussed in section 1.3.1 and 1.3.2, the polarity of solvents can be evaluated by using empirical polarity scales such as $E_T(33)$, which is the electronic transition energy of Reichardt's dye (33) or α, β and π^*, which are Kamlet-Taft parameters based on the concept of linear solvation energy relationships (LSERs). There have been many publications that have evaluated the polarity of both DES and IL solvents using these scales, although ILs have been more widely characterized by these methods than DESs to date Pandey et al. investigated the polarity of ChCl:ethylene glycol, ChCl:glycerol, and ChCl:urea using Reichardt's dye 33.[14] From measurements using Reichardt's dye, it has been evaluated that the E_T^N of these three DESs are found to be higher than those of short chain alcohols. These DESs also show higher E_T^N values than those observed for the most common pyridinium, imidazolium, and pyrrolidinium ILs and most common molecular solvents, except water, as represented in Table 1.8. According to the response of Reichardt's dye 33 for the three DESs, ChCl:glycerol has the highest E_T^N followed by ChCl:ethylene glycol and then by ChCl:urea although the range of E_T^N values was only 0.03 across all 3 solvents. This could be due to the three hydrogen bond donation sites in glycerol as opposed to two in ethylene glycol and urea. Moreover, the absorbance wavelength maxima obtained from the dye (33) for glycerol and ethylene glycol are similar to those observed for the corresponding DESs, ChCl:glycerol, and ChCl:ethylene glycol, respectively, indicating that the polarity of these DESs is strongly influenced by their H-bond donating components.

In contrast, the E_T^N values for the $[C_4C_1im]^+$ ILs in Table 1.8 are similar to those of short chain alcohols and lower than those of the DESs.

Table 1.8 *UV-vis absorbance of Reichardt's dye 33, $E_T(33)$, $E_T(30)$ and E_T^N values for selected DESs, ILs and molecular solvents.*[14]

Solvent	**$\lambda_{abs\ max}$ of Reichardt's dye 33 (nm)**	**$E_T(33)$ (kcal mol^{-1})**	**$E_T(30)$ (kcal mol^{-1})**	**E_T^N**
ChCl : Ethylene glycol	435	65.7	57.3	0.82
Ethylene glycol	436	65.5	57.1	0.82
ChCl : Glycerol	430	66.4	58.0	0.84
Glycerol	431	66.3	57.9	0.84
ChCl : Urea	437	65.4	57.0	0.81
$[C_4C_1im][BF_4]^a$	464	61.6	53.2	0.69
$[N_{1,4,4,4}][NTf_2]^a$	439	65.1	56.7	0.80
$[C_4C_1im][OTf]^a$	460	62.1	53.7	0.71
$[C_4C_1pyrr][NTf_2]^a$	440	64.9	56.4	0.79
Dimethylsulfoxide	530	53.9	45.5	0.46
Acetonitrile	521	54.8	46.5	0.49
Ethanol	472	60.5	52.1	0.66
Water	399	71.5	63.1	1.00

{^{a} $[C_4C_1im][BF_4]$: 1-butyl-3-methylimidazolium tetrafluoroborate; $[N_{1,4,4,4}][Tf_2N]$: methyl-tributylammonium bistrifluoromethylsulfonyl)imide; $[C_4C_1im][OTf]$: 1-butyl-3-methylimidazolium trifluoromethanesulfonate; $[C_4C_1pyrr][Tf_2N]$: 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide.}

Additionally, the effect of the composition of the ChCl : glycerol DES on the polarity of the DES was investigated using Reichardt's dye 30 and the results are represented in Table 1.9. According to the results, it has been evaluated that by increasing the ChCl/glycerol mole ratio, i.e. increasing the proportion of ChCl, the $E_T(30)$ of the DES and hence its apparent polarity increases up to 1:2 ChCl:glycerol with no significant change seen on increasing this ratio further to 1:1.[18]

The investigation of the solvent polarity using Kamlet-Taft parameters was also performed for these ChCl-based DES.[58] It was observed that the three parameters, E_T (30), π^* and α, generally increase with the increasing concentration of ChCl in a DES as shown in Table 1.9, though only slightly and the E_T(30) and α values do not significantly change between the 1:2 and 1:1 DESs. The β parameter is not affected significantly by the addition of the salt. This latter result is not surprising since Cl^-, the most effective hydrogen bond accepting component of the salt, is likely to be tightly bound by the glycerol such that the OH of the glycerol acts as the most effective acceptor. This suggests that in highly associated high polarity solvent mixtures, solvatochromic indicators are insensitive to compositional changes as the solvent-solvent interactions dominate over solvent-solute interactions.[58]

Table 1.9 *Solvent polarity parameters for various ChCl : glycerol mixtures.*[18,58]

Solvent	**Mole ratio**	**E_T (30) /kcal mol^{-1}**	**E_T^N**	**π^***	**α**	**β**
Glycerol	-	57 .17	0.817	0.956	0.882	0.658
ChCl : Glycerol	1 : 3	57.96	0.841	0.970	0.914	0.657
ChCl : Glycerol	1 : 2	58.28	0.860	0.984	0.937	0.657
ChCl : Glycerol	1 : 1	58.49	0.858	1.003	0.923	0.658

Similar to DESs, a range of ILs and molecular solvents have been investigated using Kamlet-Taft parameters, with some selected examples highlighted in Table 1.10. These data suggest that ILs are indeed polar solvents that can act as both HBD and HBA. It has been found that π^* is typically higher for all ILs studied than DESs and molecular solvents (except water) and while the specific value depends on both the anion and the cation of the IL, values tend to be close to 1 and change only slightly with ion composition.[57,59]

α and β vary more widely with the selection of the ions composing the ILs. α depends mainly on the cation while β is depends mainly on the anion.[57] It is recognised that the cations with hydrogen-bond donating substituents, such as alcohols, tend to have the highest α values. Of the unsubstituted ILs such as those shown in Table 1.10, α values for $[C_nC_1im]^+$ based ILs are generally higher than for the corresponding $[C_4C_1pyrr]^+$ ILs. For ILs with same cation and different anions a secondary but measurable anion effect can be seen. As the anion becomes more basic (increasing β), the hydrogen bond donating ability of the IL decreases. For the β values, it can be seen the basicity of the anions explored for the ILs decreases in the following order : $[OTf]^- > [NTf_2]^- > [BF_4]^-$.[57,59]

Table 1.10 *Kamlet-Taft parameters of DES, ILs and molecular solvents.*[19,20,59]

Solvent	α	β	π^*
Water	1.23	0.49	1.14
Butanoic acid	1.06	0.22	0.47
Hexanoic acid	1.05	0.21	0.43
Octanoic acid	0.94	0.23	0.38
$[C_2C_1im][NTf_2]^a$	0.71	0.23	0.98
$[C_4C_1im][NTf_2]^a$	0.63	0.38	1.05
$[C_6C_1im][NTf_2]$	0.65	0.26	0.97
$[C_4C_1im][BF_4]^a$	0.77	0.39	1.04
$[C_4C_1im][OTf]^a$	0.63	0.46	1.01
$[C_4C_1pyrr][Tf_2N]^a$	0.43	0.25	0.95
$[C_4pyr][NTf_2]^a$	0.64	0.12	0.82
$C_8H_{20}N^+Cl^-$: butanoic acid[b]	0.99	0.76	0.92
$C_{12}H_{28}N^+Cl^-$: hexanoic acid[b]	0.91	0.92	0.85
$C_{16}H_{36}N^+Cl^-$: octanoic acid[b]	0.84	1.19	0.80

{[a] $[C_2C_1im][NTf_2]$: 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide, $[C_6C_1im][NTf_2]$: 1-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide, $[C_4pyr][NTf_2]$: 1-butylpyridinium bis(trifluoromethylsulfonyl)imide [b] $C_{12}H_{28}N^+Cl^-$: tetrapropylammonium chloride, $C_{16}H_{36}N^+Cl^-$: tetrabutylammonium chloride, $C_8H_{20}N^+Cl^-$: tetraethylammonium chloride}

1.7. Introduction to Bark

One of the most significant difficulties that humanity will confront in future decades is the supply of sustainable energy, particularly in light of the need to combat climate change. The replacement of crude oil by more sustainable renewable energy sources brings with it the complication of needing to locate alternative sources for chemical inputs that are currently obtained from these petrochemical sources. Biomass has the potential to make a significant contribution to meet chemical needs in a sustainable manner. It is currently the one of the world's greatest contributors of renewable energy and has substantial potential to develop in the production of transportation fuels.[60] Moreover, the constant depletion of fossil fuels is a

driving force behind research into alternate, renewable alternatives for energy and chemical manufacturing. The use of the biorefinery concept in agroforest-based operations, as well as the requirement to upgrade by-products generated during the processing of agricultural and forestry goods, may help provide solutions to these problems.[61]

Bark is a promising biological resource that can be exploited to produce chemical commodities as a renewable feedstock. The detailed study of the anatomical features and the chemical composition of bark will be discussed in the Section 1.7.1 and 1.7.2.

1.7.1. Anatomical features of bark

It is well known that bark is the outermost layer of the branches, stem and roots of woody plants. Bark is a part of the plant's body that contains all the tissues outside the vascular cambium as shown in Figure 1.13.[62]

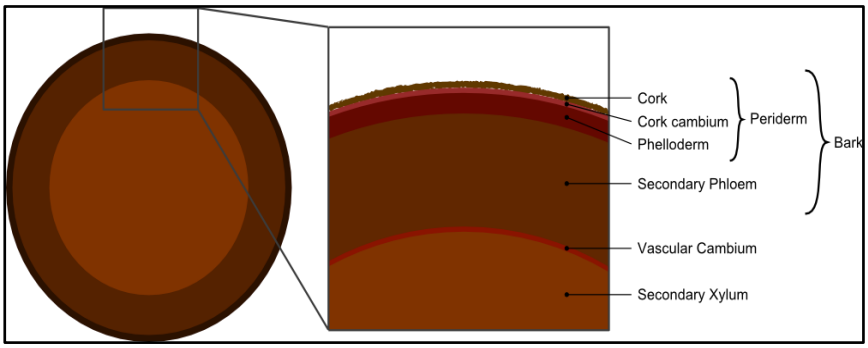


Figure 1.13 *Picture of the anatomical features of bark.*[63]

Bark constitutes on average, about 10-20 % of the volume of a tree. This figure varies depending on the species and age of the tree.[64] There are two layers of the bark - inner bark and outer bark. The inner bark is composed of mostly living cells, which together form the phloem, a system that works primarily for the storage and transportation of carbohydrates.[65] The outer bark consists of the epidermis and periderm in which fat and suberin-rich, dead tissues are present. Underneath the epidermis is the outer skin also known as periderm, which is composed of three types of tissues listed below.[66]

1. <u>Outer phellem (Cork cells)</u> – Outer phellem is responsible for the protective characteristics of bark. The thick layer contains dead cells called cork cells (phellem). This layer is responsible for protecting tree against harsh weather, pests and disease. These cells have a layer of a waterproofing complex called suberin wax. Suberin is a fatty, waxy substance that makes the cells mostly resistant to water and unable to exchange gases and nutrients, hence these cells

quickly die and trap air.[65,67] New phellem cells are being constantly formed in radial rows by meristematic activity, i.e., division and growth, of cells adjacent to the phellogen.[66]

2. <u>Middle phellogen (Cork cambium)</u> – The middle phellogen or the cork cambium is the thin-walled row of cells which is present in the middle of the periderm and separate the phellem from the phelloderm by a thin-walled meristematic cell.[66] When the secondary growth of dicot stem and root occurs i.e., the growth that results from cell division in the cambium and that causes stems and roots to thicken, the epidermal layer of the bark breaks. To protect the stem and root from infections, the cork cells need to be replaced. Therefore, the cork cambium develops from the cortical region and cuts off cells towards both centre and periphery. The cells on the outer side are differentiated into the cork or phellem and the cells on the inside give rise to the secondary cortex or phelloderm. Hence, the phellogen continuously produces new phellem and phelloderm cells.[66]

3. <u>Inner phelloderm (Secondary cortex)</u> – These are the living cells, which can exchange gases and obtain nutrients due to their unthickened and unspecialized cell walls. These are formed from the cork cambium and are forced to the interior of the bark.[67] White phloem fibre bundles may appear in these tissues. The phelloderm cells have precise shapes and are loosely organised. These cells may store starch and undergo photosynthesis.[66]

Moreover, there are small openings in the periderm called lenticels. Lenticels permit the exchange of gases such as oxygen and carbon dioxide with the outer atmosphere and the internal tissue of the stem.[62,65,67]

Inner bark is made up of all the tissues from the vascular cambium through to the innermost periderm. Vascular cambium produces the vertical phloem elements: Sieve cells (conducting elements), phloem parenchyma and phloem fibres which are described in greater detail below.

(i) <u>Sieve cells</u> – Sieve cells are narrow with longitudinal arrangements. The sieve elements are arranged one above the other and form vertical sieve tubes.[67] Sieve cells and sieve tube elements are living cells that have plasmatic content but without a nucleus.[64] The cross section of the sieve tube components have moderately large pores and that is the reason they are called sieve plates. Due to the presence of pores in the sieve tubes, the cell contents and nutrients can easily pass through it.[64]

(ii) Companion cells - The specialized thin-walled parenchyma cells that are associated with sieve elements are called companion cells. They help the sieve tubes transport food material.[67]

(iii) Phloem parenchyma - These cells are associated with phloem and form the main proportion of phloem tissue. These are living cells and are arranged in longitudinal strands.[64] They store starch, resins, tannins and fats. Some parenchyma cells become meristematic and produces phellogen.

(iv) Phloem fibres - The fibres of sclerenchyma associated with phloem are called phloem fibres or bast fibres. They are vertically elongated narrow cells with very thick walls and a small lumen.[62,67] Out of the four types of phloem elements, phloem fibres are the only dead tissue, and are involved in strengthening the stem and supporting conductive cells.[67]

1.8. Chemical composition of bark

The anatomy of the bark and its chemical composition, determine the important properties with respect to its potential use.[64] From a chemical perspective bark consists of several classes of compounds. Specifically, these include biopolymers such as suberin and lignin, carbohydrates and polyphenols including tannins and other extractives. Each component has a different application based on its inherent chemical structure. Table 1.11 listed below represents some of the important components of bark, their properties and examples of their current uses.[67]

Table 1.11 *Current uses and properties of the major classes of chemical compounds in bark.*[67]

Composition of bark	**Properties**	**Application**
Suberin	Resistant to water and heat	In hydrophobic coating
	Waxy and fatty substance	Absorbent for carcinogens
	Mainly found in cork cells	Mutagenesis inhibitor
Lignin	Provides strength	Used in paper and pulp industry
	Natural glue	Adhesive and binder
	Antioxidant	Emulsifier and stabilizer
Carbohydrate	Energy storage	Biofuels and bio-based chemicals

	Strengthening of biomass structure	Biomedical applications
Polyphenols – Phenolic acids, tannins & flavonoids	Antioxidant, antifungal, anti-tumour, anti-carcinogenic, antimicrobial, antidiabetic, anti-inflammatory	Used in food processing and leather industry Used in the manufacture of inks Used in pharmaceuticals and extracts useful against controlling HIV activity

The composition of bark differs from that of wood due to greater abundance of polyphenols, suberin, extractives, ash and lignin with a reduced quantity of carbohydrates.[64,67,68] Extractives that can be dissolved with organic solvents or water without further processing can represent up to 20–40% of bark mass; the rest of the segments are mostly suberin, lignin, and phenolic acids, along with some carbohydrates.[62] Table 1.12 shows the composition of these four classes in bulk wood and bark in softwood and hardwood species.

Table 1.12 *The percent chemical composition between wood and bark in softwood and hardwood.*[69]

	Softwood		**Hardwood**	
	Wood	**Bark**	**Wood**	**Bark**
Lignin	25 – 30 %	40 – 55 %	18 – 25 %	40 – 50 %
Carbohydrates	66 – 72 %	30 – 48 %	74 – 80 %	32 – 45 %
Bark extractives	2 – 9 %	2 – 25 %	2 – 5 %	5 – 10 %
Suberin	Not available	~ 20 %	Not available	~ 20 %

In terms of anatomical differences between the internal (phloem) and external bark (rhytidome), it has been analysed that the content of extractives and polysaccharides decline and the lignin and polyphenolic content increases on moving from the inner to the external bark.[64] Further detail of the chemical composition of the major classes of bark components are given below.

1.8.1. Carbohydrates

The most abundant carbohydrate in bark is cellulose. Cellulose is a linear polymer of glucose linked by β 1-4 glycosidic bonds. The high degree of linearity and rigid structure of cellulose gives rise to a stretched chain conformation as flat sheets due to increased inter and intra molecular hydrogen bonding as shown in Figure 1.14.[70,71] These sheets then interact with each other through Van der Waals forces, packing the cellulose strands into thin crystalline microfibrils and contributing to the high stability of the cellulose.[71]

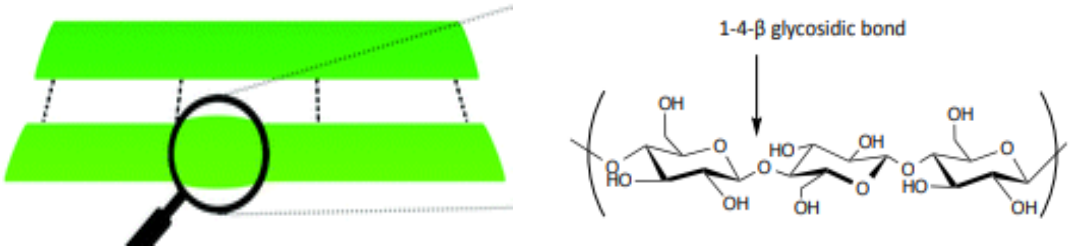
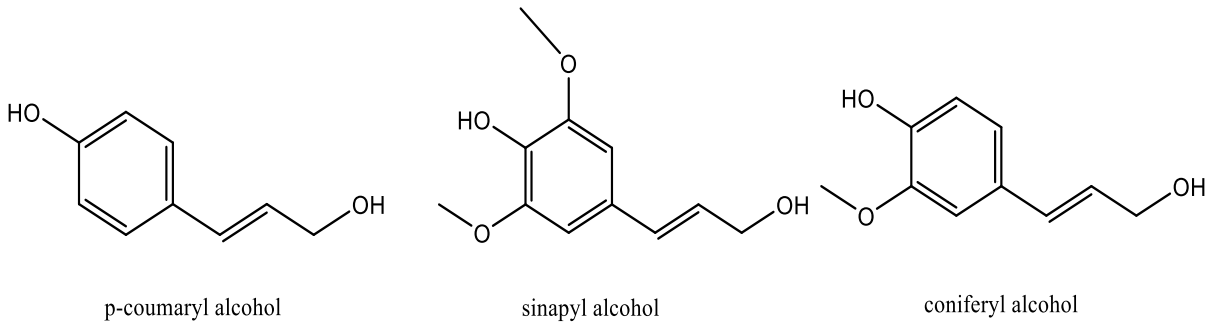


Figure 1.14 *A stretched chain or flat sheet and β 1-4 glycosidic bonds in cellulose.*[70,71]

Moreover, the number of monomer units in one polymer strand can be 10,000 or higher. Although glucose and short oligomers are water-soluble, cellulose is not due to its higher molecular weight and the comparatively low flexibility of cellulose polymer chains.[72] Cellulose is the most important carbohydrate in both bark and wood. Different carbohydrates such as galactose, xylose, and mannose are also present in bark although in small concentrations, with the specific composition varying with the location within the bark and the species of the tree.[62]

1.8.2. Lignin

Lignin is a naturally occurring aromatic cross-linked polymer with a molecular weight of more than 10,000 Dalton. Lignin acts as a linker between cellulose and hemicellulose in lignocellulosic biomass.[28] Lignin provides structural reinforcement to the plants, and prevents water from entering cells due to its hydrophobicity.[71] Therefore, lignin provides an efficient way to transport water and nutrients by efficiently repelling them away from the cells. Hence, transportation cells such as xylem cells have more lignin. Lignin is made up of three main basic monomers collectively known as ‘monolignols’ which include p-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol as shown in Figure 1.15.[28]



p-coumaryl alcohol sinapyl alcohol coniferyl alcohol

(a) **(b)** **(c)**

Figure 1.15 *The structures of lignin monomers*

Depending on the type of plant material, the composition of lignin in terms of the proportion of each lignin monomer varies significantly. For example, softwoods predominantly contain sinapyl alcohol and hardwoods contain more of the coniferyl and p-coumaryl alcohols.

The biological synthesis of lignin involves several key steps. The first step involves the enzymatic oxidation of the monolignols to form resonance stabilised phenoxy radicals, as shown in Figure 1.18.[28] This occurs by H· abstraction by peroxidases from the phenolic group present on the monolignols.[73] These radicals then react with another monolignols to form dimers and this can undergo further reactions on subsequent oxidation, an example is shown in Figure 1.16.

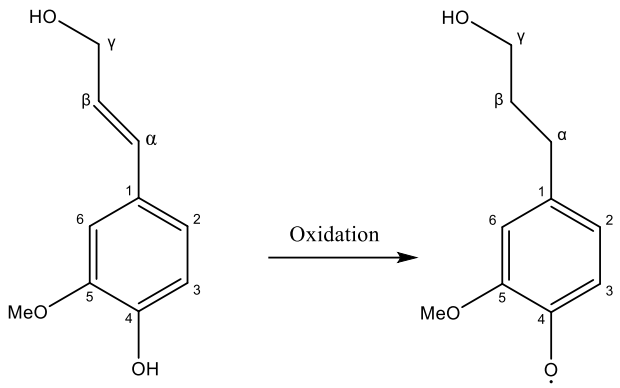
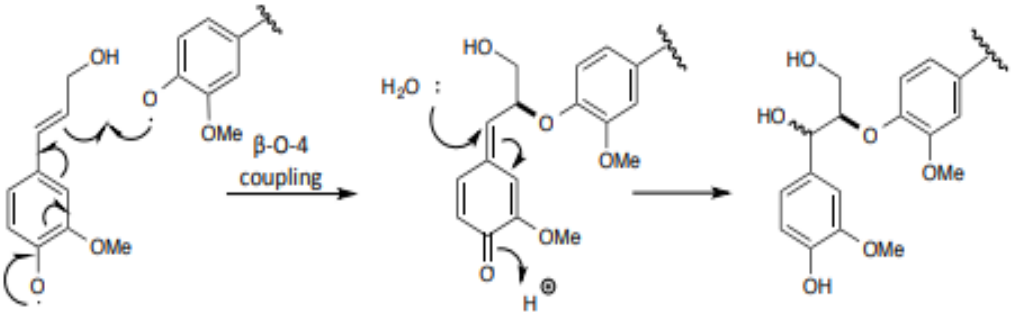


Figure 1.16 *Oxidation of coniferyl alcohol*.[28]

The oxidation and formation of dimers and higher oligomers of these lignin monomers from these radical precursors can lead to a range of different linkages. Common linkages are displayed in Figure 1.17. Out of all the linkages formed, which include β-5, 5-5, 4-O-5, β-β, β-1 and dibenzodioxocin linkages, the β-O-4 linkage (Scheme 2) is the most common among the

three basic units of the lignin.[73] As the most common linkage, there has been work undertaken exploring the ability to cleave this linkage to depolymerise lignin. It has been found that cleaving the β-O-4 bonds gives carboxylic acids and phenol monomers.



Scheme 2 *β-O-4 linkage formation via radical coupling of 2 coniferyl alcohols.*[71]

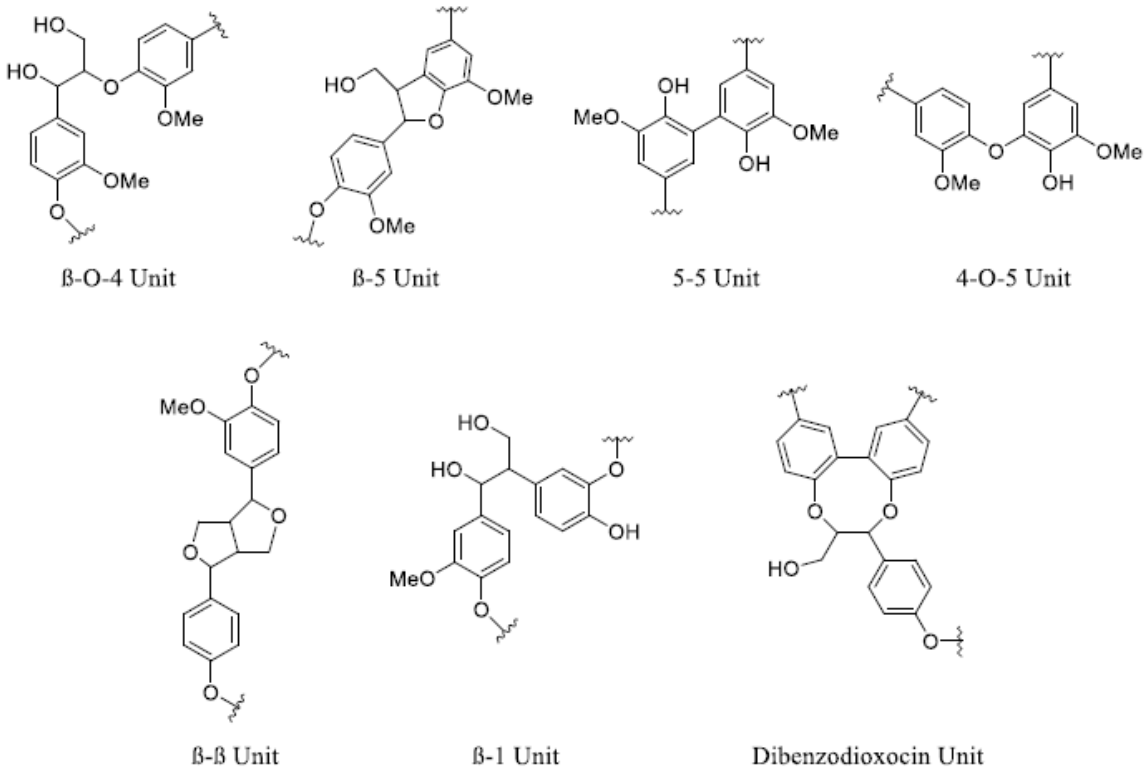


Figure 1.17 *Common linkages in lignin.*[71]

1.8.3. Suberin

Suberin is a hydrophobic, mixed aliphatic and aromatic cross-linked polymer found in outer bark cells. Suberin is another plentiful component of bark, found in the phellem cell wall.[64]

Suberin is a constituent of outer bark. It has a polyester structure made predominantly out of long chain fatty and hydroxy fatty acids with the specific arrangement relying upon the species.[62,64,67] There is no unique repeating chemical structure for suberin because its constitutive moieties vary both in their specific nature and relative abundance with the macromolecular network. Due to its variability and interactions with other cell wall components such as lignin and polysaccharide, the exact chemical structure of suberin has been heavily debated.[74] One proposed model has been put forward by Bernards as shown in Figure 1.18.[75]

Figure 1.18 shows that the aliphatic domains of suberin (in the secondary cell walls) are made up of branched polyester macromolecules mainly composed of long chain hydroxylated fatty acid moieties. Meanwhile there are two distinct aromatic domains in the subernised cell walls, the first lying inside the aliphatic domains which consists mainly of hydroxycinnamates esterified with glycerol or ω-hydroxyfatty acids.[76] The second aromatic domain is a lignin polymer sitting in the primary cell wall and is composed of cross-linked 18-hydroxycinnamic acid moieties, including amides, covalently bound to aliphatic suberin either by ester or ether linkages.[74,77] The extraction and isolation of suberin in its entirety from the cork tissue is challenging due to its poor solubility in water hence, suberin is typically quantified by saponification using alkali treatment.[64]

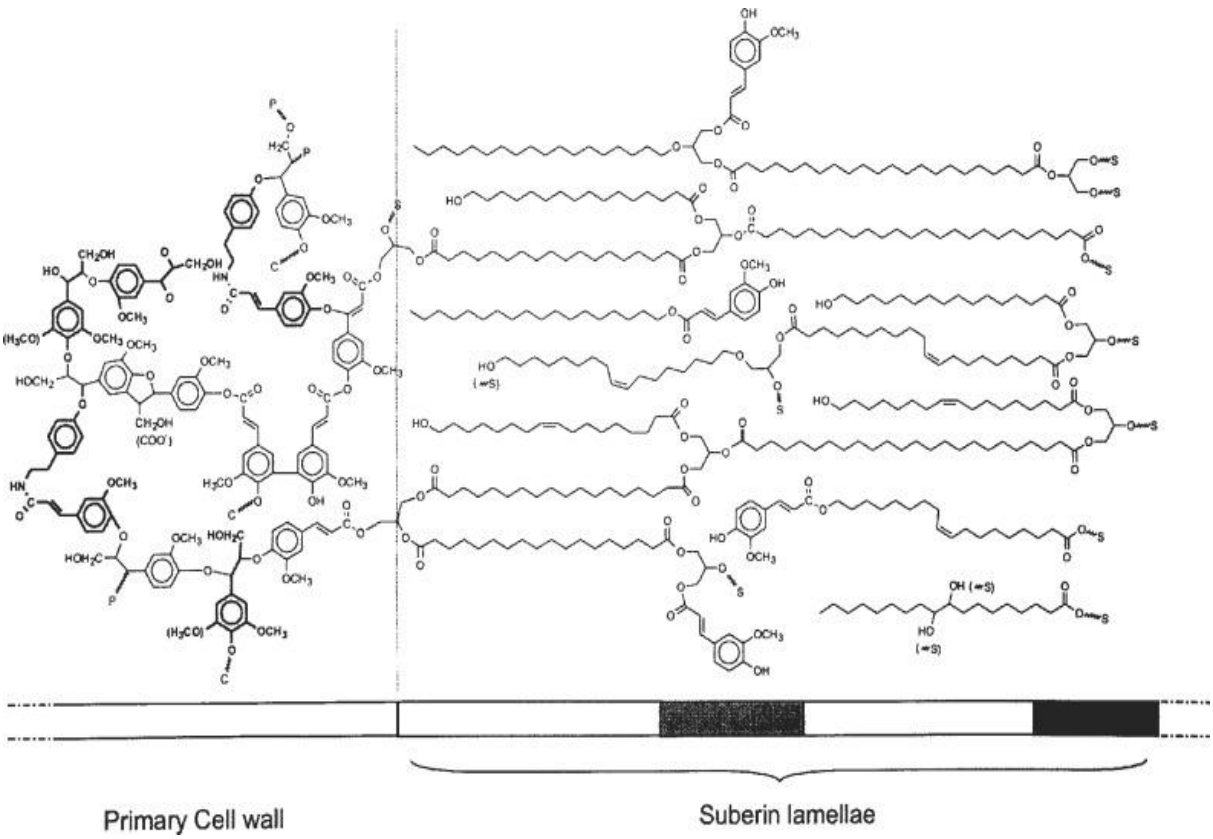


Figure 1.18 *The suberin model proposed by Bernards.*[75]

Suberin can be used chemical precursors in a variety of applications. These macro monomers can be used for the synthesis of novel materials such as renewable hydrophobic polymers. Suberin has also been reported as being an absorbent for carcinogens and a mutagenesis inhibitor.[78]

1.8.4. Extractives

In addition to lignin, suberin and carbohydrates, bark contains relatively high concentrations of low-molecular weight organic compound extractives such as fats, waxes, fatty acids, alcohols, steroids, alkaloids, pigments, terpenes, amino acids, and vitamins.[62,67] These compounds are available in the phelloderm and phellem.[62]

1.8.4.1. Terpenes - These are the components of volatile essential oils, which have low-molecular weights that give characteristic aromas to the barks of many trees (e.g., incense cedar, ponderosa pine, and eucalypts).[62] Many terpenes exhibit biological activities against disease states including cancer, inflammation and diabetes.[23]

1.8.4.2. Alkaloids - Alkaloids are naturally products that contain one or more carbon rings, along with at least one nitrogen atom, for example, morphine.[23] They are found in the

plant, along with some vitamins (e.g., C, B-1). Alkaloids form salts with a range of acids in bark (e.g., citric, tannic). Vitamins are concentrated in some species and tend to fluctuate seasonally.[62]

1.8.4.3. Ash - Extractive ash content, composed of calcium, silica, and phosphorus along with mineral compounds, are multiple times more abundant in bark than in wood. Typically, ash content of the combined inner and outer barks of softwoods ranges from 1 – 2.5% by weight, whereas hardwood barks are 1.5 – 10% ash.[62]

1.8.4.4. Polyphenols – The term polyphenols refer to a various number of related compounds featuring multiple phenol groups primarily derived from flavone derivatives. The polyphenols present in barks are differentiated according to their structure, molecular weight and solubility, with the subclasses discussed below.[64]

(i) **<u>Flavonoids</u>** – Flavonoids are a class of polyphenolic secondary metabolites. They are being used commercially and provide health benefits to humans due to their antioxidant and antitumor activity.[23] Flavonoids are key components of many of the colours found in barks.[62,67] The flavonic core, a basic unit of flavonoids, is shown in Figure 1.19.

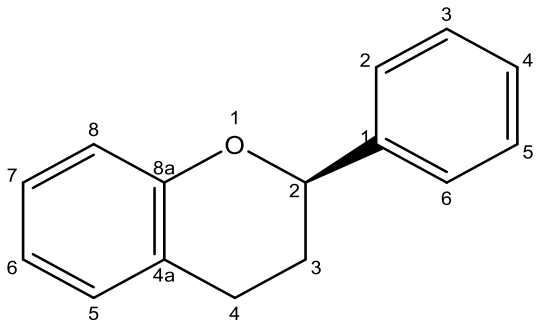


Figure 1.19 *Chemical structure of the flavanic core.*

Flavonoids all contain a common central heterocyclic ring and consist of a range of compounds and compound classes based on their molecular structure. These includes (+)-catechin, and procyanidins. (+)-Catechin, provides human health benefits and can be used for the processing of animal hide into leather. (Figure 1.20). (+)-Catechin represents the monomer unit for the formation of procyanidins. Out of these subclasses, the highest molecular weight compounds are the procyanidins, which mainly consist of tri- and dimeric flavanols. Procyanidins are soluble in hot water[23,64] and display antioxidant properties.[79]

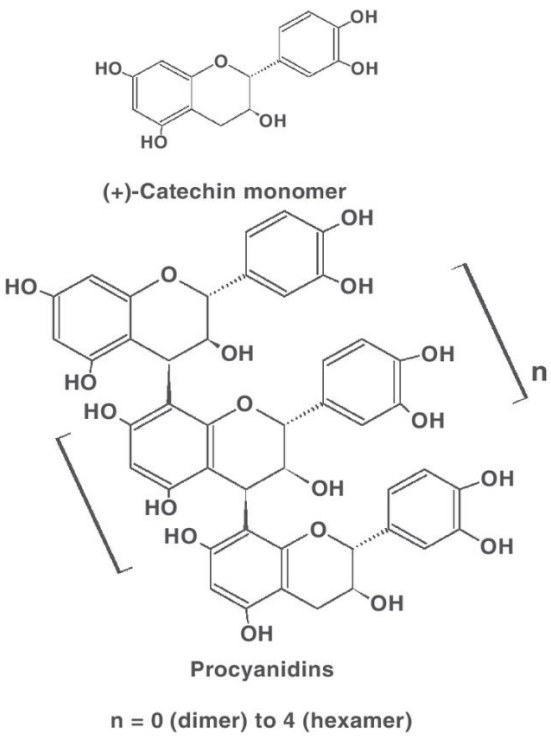


Figure 1.20 *Chemical structure of some flavonoids: (+)-Catechin and procyanidins.*[79]

(ii) **<u>Tannins</u> –** Tannins are high molecular weight nitrogen free phenolic compounds. They are a polymer formed from the condensation of gallic acid and flavan-3-ols and have molecular weights typically between 500 to 3000 Dalton. Tannins are non-crystallizable compounds and are soluble in water, dilute alkali and alcohols including glycerol. They play an important role in the mechanism of hydrogen transfer in plant cells, perhaps due to their high affinity for oxygen.[80,81] Tannins have been extensively used in the fields of pharmaceuticals, food processing and tanning processes.[82] There are two types of tannins known as true tannins and pseudotannins –

(a) True Tannins - They are high molecular weight compounds ranging from 1000 to 5000 Dalton and further sub-classified into 3 groups.

- **Condensed tannins** – The polymers which are formed from the condensation reaction of either flavanols or flavan-3,4-diols are called condensed tannins.[82] Condensed tannins are dimers, oligomers and polymer of catechin, a structural unit of condensed tannin.[83] Proanthocyanidins are considered to be a class of condensed tannin in addition to their classification as flavonoids.[80] Proanthocyanidins could be converted into red insoluble compounds called phlobaphene, on treatment with acids.[64] They are

condensed to form water insoluble precipitates for wood adhesives. The content of condensed tannins in barks varies widely from 5 – 50%.

- **Hydrolysable tannins** – Hydrolysable tannins are polymers which can be hydrolysed in acid to produce a carbohydrate component, such as glucose, as well as gallic acid and ellagic acid. The tannins derived from gallic acid are called gallotannins while from ellagic acid are known as ellagitannins.[82] In ellagitannins, one or more hydroxydiphenol residues are linked to glucose as a diester.[82] The most commonly found gallotannins are glucose and gallic acid polyesters.[84] The chemical structures of these types of tannins are shown in Figure 1.21.

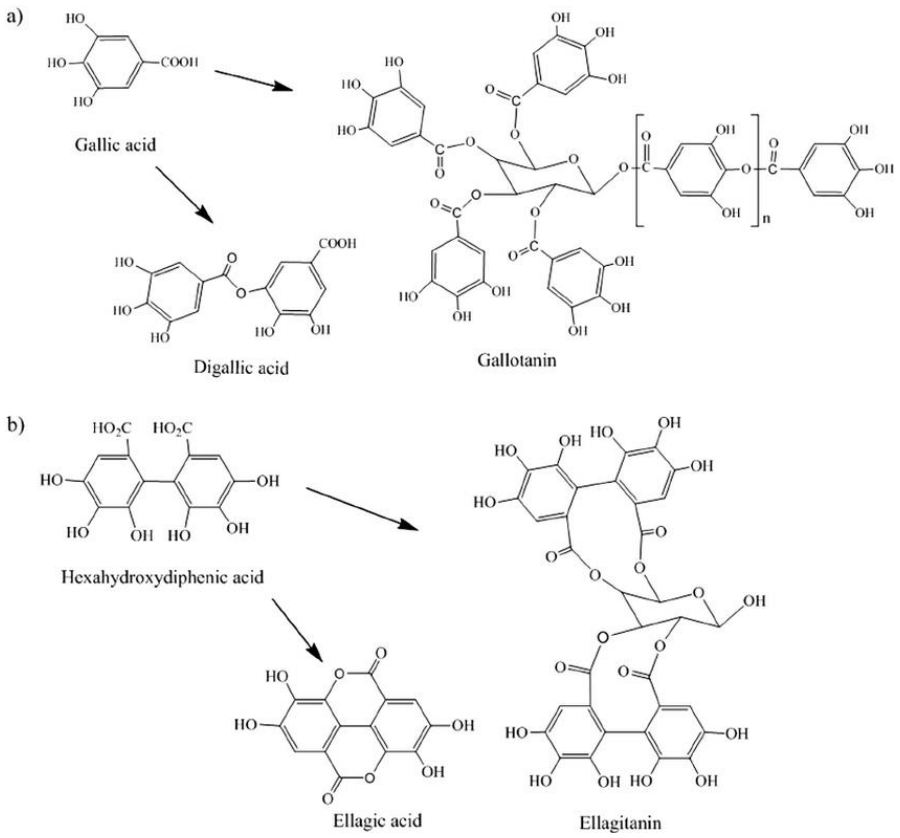
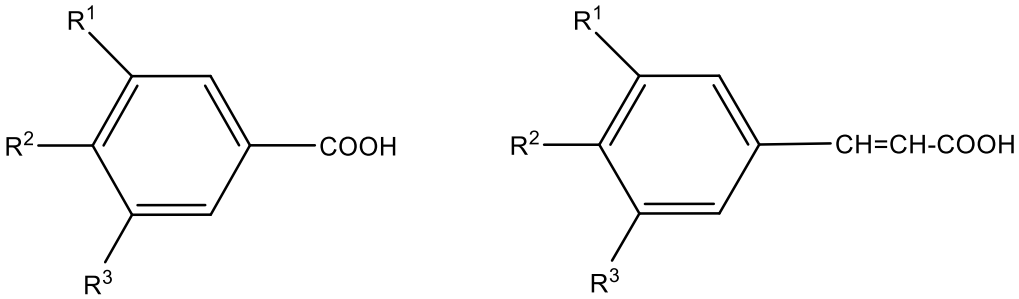


Figure 1.21 *Chemical structure of gallotannins and ellagitannins.*[84]

(b) Pseudo tannins – Pseudotannins are low molecular weight compounds that can be considered tannin monomers, such as gallic acid and catechin.[80,85]

(iii) **<u>Phenolic acids</u>** – Phenolic acids, which are found in outer bark cells, can be subdivided into two major groups: hydroxybenzoic acids (gallic acid) and cinnamic acid derivative (p-coumaric) as shown in Figure 1.22.[86] Pseudo tannins are part of

the phenolic acid class. This group of compounds can be extracted with 1% NaOH at 100 °C. Gallic acid and ellagic acid are some of the examples of phenolic acids which are commonly extracted from the bark along with other compounds from biomass.[23]



Hydroxybenzoic acids **Hydroxycinnamic acids**

Figure 1.22 *Chemical structures of hydroxybenzoic acids and hydroxycinnamic acids, where at least one R group corresponds to an OH group.*[86]

Many polyphenols are beneficial for human health due to their anti-tumour, antiviral, anti-inflammatory, anti-carcinogenic, antimicrobial, antidiabetic, and anti-atherosclerotic properties. Due to the abundance of polyphenols and phenolic acids, bark is more acidic than wood.[64,67]

1.9. Extraction Processes using Ionic Liquids (ILs) and Deep Eutectic Solvent (DESs)

Many commercial extraction processes utilise organic solvents.[25] The issue of using excessive organic solvents is that they are volatile and flammable and may not decompose quickly and thus cause hazards not only to environment but also to the health of living organisms. These adverse effects of conventional solvents have contributed to the development of less hazardous "green extraction" methods. In the development of green extraction methods, the identification of potential green solvents such as ILs and DESs that can replace harmful organic solvents is an important emerging research area. Additionally, green solvents are tuneable which make them distinguish from other solvents and could replace organic solvents in many applications.

The connection between ILs, DESs and green chemistry is clearly related to the solvent properties of ILs and DESs. ILs and DESs act much like conventional organic solvents and can

be developed to dissolve both polar and non-polar species. In many cases, these solvents have been found to perform much better than commonly used solvents.[25] It has recently emerged that ILs and DESs, can be designed to possess unique hydrogen bonding capabilities that enable the solubilisation of traditionally poorly soluble substrates such as drug molecules and biopolymers.[87–89,90] The application of these solvents for extraction processes has been discussed in the sections below.

1.9.1. Extraction of bark components by using ILs and DESs

As has been introduced previously, there are some instances of investigation of the extraction of a few bark species using organic solvents.[64] For example, Fengel et al. observed a higher extractive content, up to 27.5% by extending the extraction procedure to include hexane, benzene, ethyl ether, ethanol, water and 1% NaOH. Howard et al. explored the bark of loblolly pine (Pinus taeda) and determined the total extractives content as 33 %, utilising petroleum ether and toluene.[64,91] Less has been explored on the use of ILs and DESs with bark, although there has been emerging interest in exploring ILs and DESs for the replacement with common organic solvents for the extraction of bio-molecules.[92]

IL properties can be tuned by ion selection to allow optimal solvents to be designed for specific extraction processes. This makes it possible in principle to design ILs that enable the targeted dissolution of specific components of bark.[93] Although considerable work has been done exploring the extraction of wood, relatively little research has been done exploring the extraction of bark using ILs. The higher extractive content and reduced carbohydrate content of bark relative to wood is likely to have a significant impact on the extraction efficiency of these solvents.

The integral components of lignocellulosic biomass such as cellulose and hemicellulose are found in both bark and wood. Due to their supramolecular structures consisting of strong hydrogen bonds, these components are barely soluble in most conventional molecular liquids which has restricted their separation and utilization. In this context, ILs as potential solvents for cellulose have attracted a great deal of interest.[92]

Swatloski et al. has reported the use of ILs for the dissolution of cellulose.[94] Dissolution experiments were carried out by adding cellulose samples (fibrous cellulose, Whatmann cellulose filter papers) to the ILs and heating without agitation either on a hot plate (70 – 100 °C) or in a microwave. The solubility of cellulose was found to be strongly anion dependent

and decreased in the order Cl^- > $[BF_4]^-$ ≈ $[PF_6]^-$. ILs incorporating Cl^- which is a strong hydrogen bond acceptor anion appeared to be the most effective solvents for the dissolution of cellulose, especially when combined with microwave heating. The dissolution of cellulose was attributed to hydrogen-bonding from hydroxyl functionalities of the cellulose to the anions of the solvent, disrupting the intermolecular hydrogen-bond networks of the cellulose.

The extraction of polysaccharides from a woody biomass, Japanese cedar, was investigated using different types of phosphonate ILs (Table 1.13).[93] The extraction was performed by mixing powdered Japanese cedar with each phosphate IL and stirring the mixture at 100 ℃ for 24 h under dry nitrogen.

It was found that the degree of extraction of polysaccharide (the proportion of polysaccharide extracted relative to total mass of biomass) increased with increasing β – value of the IL used (Table 1.13). The degree of the extraction of polysaccharide by $[(C_1OC_3)C_1im][EtPO(OEt)(OH)$ (IL 3) reached 15%, which was three times higher than that of $[C_2C_1im][EtPO(OH)_2]$ (IL1) (5.4%). In other words, polysaccharide extraction was mainly related to the ability of the IL to interact with the hydroxyl groups of polysaccharides and disrupt intermolecular hydrogen-bonding.

Table 1.13 *Solubility of the model cellulose and degree of extraction of polysaccharides from Japanese cedar.*[93]

Ionic liquids (ILs)	**β**	**Dissolution temp /℃**	**Degree of extraction /%**
1-ethyl-3-methylimidazolium ethylphosphonate (IL1)	1.02	25	5.4
1-(3-methoxypropyl)-3-methylimidazolium ethylphosphonate (IL2)	1.12	25	13.3
1-(3-methoxypropyl)-3-methylimidazolium ethyl ethyl phosphonate (IL3)	1.21	Not Given	15.0

Moreover, although the reaction conditions were different to the Swatloski et al. work[94] for the dissolution of cellulose, this has been evaluated that good hydrogen bond acceptor plays an important role in the solubilizing of the cellulose.

Contrasted with cellulose, the dissolution mechanisms for lignin in solution are different as the solubility of cellulose increases with increasing hydrogen bond accepting strength of the IL anion, but this trend does not apply to the dissolution of lignin. The effect of the IL anion was investigated for lignin solubility by Hart et al. using $[C_2C_1im][OTf]$, $[C_2C_1im][BF_4]$, $[C_2C_1im][(O_2P(OCH_2CH_3)_2]$, $[C_2C_1im][CH_3SO_3]$, $[C_2C_1im][NCS]$, $[C_2C_1im][CF_3CO_2]$, and $[C_2C_1im][NTf_2]$ ILs. The effect of the cation was explored using $[C_2C_1im][OTf]$, $[C_4C_1im][OTf]$, $[C_4C_1C_1{}^2im][OTf]$, $[C_4pyr][OTf]$, and $[N_{4448}][OTf]$ ILs.[95] The results indicated that, unlike cellulose, the anion does not have a significant effect on the solubility of lignin although a minimum hydrogen bond basicity (β) is required to enable lignin dissolution. On the other hand, the cation has a significant but subtle effect upon lignin solubility. ILs containing the $[C_4C_1im]^+$, $[C_2C_1im]^+$ and $[C_4pyr]^+$ cations displayed high solubility than those with the $[C_4C_1C_1{}^2im]^+$ and $[N_{4448}]^+$ which suggests that lignin tends to be most highly solubility in ILs that have relatively planar aromatic cations, potentially due to the formation of cation-π interactions.

The extraction of *Cinnamon cassia* bark to obtain both an essential oil enriched in coumarin which has wide range of medicinal properties, and biopolymers (lignin) was investigated utilizing ILs and a mixture of ILs (protic + aprotic). It had been determined that the extraction yields were dramatically improved when a combination of ethylammonium nitrate (EAN) and $[C_4C_1im][OAc]$ were used which was ascribed to a reduction in viscosity by adding EAN to $[C_4C_1im][OAc]$ and improvement in the solvating ability of ILs through the synergistic effect of the IL ions (Table 1.14).[96]

Table 1.14 *Effect of ILs on the yield of biopolymers and essential oil from the extraction of Cinnamon cassia bark.*[96]

S.no.	Mixture of ILs/IL	Temp (°C)	Time (min)	Yield (wt%)	
				Biopolymers	Essential oil
1.	EAN	70	60	4.6	1.2
2.	$[C_4C_1im][OAc]$	70	60	16.7	1.8
3.	EAN + $[C_4C_1im][OAc]$	70	60	22.9	4.7

Ammonium-based ILs have also been used as solvents for the effective extraction of active metabolites from barks of *Cinnamomum cassia,* which can be further used for medicinal applications (Table 1.15).[96] It was reported that that IL (E), which has a $[(HOCH_2CH_2)_3NMe]^+$ cation was the most effective option for extracting alkyl aryl carbonyl compounds such as trans-cinnamaldehyde and coumarin as represented in Table 1.15. The concentration % represents the quantity (%) of the extracts extracted by each ILs. Moreover, ILs (A, B, C) were proportionally better for the extraction of terpenes (caryophyllene and copaene).[97] The rational is not clear although these 3 ILs feature very hydrophobic cations as they all have long alkyl chains so this might be the major factor.

Table 1.15 *GC-MS analysis of extracts from the bark of Cinnamomum cassia by ILs.*[97]

Compounds	Concentration (%)				
	A	B	C	D	E
α-Copaene	2.23	2.24	3.16	0.42	0.14
β-Caryophyllene	1.94	1.19	1.26	0.18	0.13
Trans-cinnamaldehyde	12.5	29.3	50.3	1.19	80.7
Coumarin	0.35	1.31	1.80	0.98	11.8

*{Bezalconium lactate - **A**, Didecyldimethyl lactate - **B**, Bezalconium nitrate, IL – **C**, Didecyldimethyl nitrate - **D**, Tris (2-hydroxyethyl)methylamonium methylsulfate – **E**}*

Suberin, a waxy substance mainly found in cork and bark is difficult to extract from these sources due to its complex hydrophobic structure. Isolation and depolymerisation of suberin is traditionally a laborious process requiring harsh chemical processes involving extensive ester bond cleavage, normally attained by alkaline treatment.[74] While there have not been any studies

performed on the extraction of suberin from *Pinus radiata* bark using ILs, the extraction of suberinic material from cork has been examined.[98]

It had been reported that in order to promote a highly efficient extraction of the suberin domains from cork, cholinium alkanoates ILs namely, cholinium hexanoate, cholinium octanoate or decanoate and one imidazolium IL namely, 1-ethyl-3-methylimidazloium hexanoate, respectively, were selected. It was identified that cholinium hexanoate was the best IL examined and was able to achieve an extraction yield of up to 67.2% (% mass extracted relative to suberin in cork).[98] This might be attributed that the strength of interaction between cation and anion varied in each ILs and hence, the carboxylate moiety of the anion might not be strongly interacted with cholinium cation than imidazolium cation resulting in extraction of suberin from the cork. Moreover, it could also be concluded that slightly smaller non-polar domains actually assisted with the solvation of the suberinic materials, potentially due to the interaction with the ester groups. This suggests that structural features of IL cations and anions are of substantial importance when aiming to extract suberin components.

Like ILs, DESs can be designed to optimize extraction processes by rational selection of the constituents used. For a DES this involves selecting the HBD and HBA as well as varying the molar composition of the constituents. This allows the modification of their properties such as freezing temperature, viscosity, conductivity, polarity, refractive index, density and pH. Owing to the controllable physicochemical properties of DESs and considerable potential benefits such as biodegradability, low cost, and ease of manufacturing, DESs can be potentially useful solvents for the extraction of bark. Additionally, DESs, being multi-component systems, can offer significant advantages over conventional organic solvents. DESs can be chosen as environmentally benign substitutes for organic solvents due to their non-flammability and biodegradability.

DESs have been used as solvents for the effective extraction of bioactive compounds from barks of *Cinnamomum burmannii*. The main compounds in *C. burmannii* bark are cinnamaldehyde and coumarin, which have antidiabetic, anticancer, anti-inflammatory, coagulation inhibition, antifungal, antibacterial and antiviral activity.[99] The extraction of these components was examined using a range of DESs based on ChCl or betaine using an ultrasound assisted extraction (UAE) approach (Table 1.16). It was evaluated that the choline chloride-citric acid DES (ChCl-CA) led to the highest trans-cinnamaldehyde extraction yield (17.67 mg/g) and betaine-lactic acid (B-LA) gave the highest coumarin extraction yield (15.33 mg/g)

of the DESs examined. This was due to the organic acid HBDs driving the solubility through hydrogen bonding interactions with the accepting groups on coumarin and trans-cinnamaldehyde.

Table 1.16 *The effect of DES for the extraction of trans-cinnamaldehyde and coumarin from C. burmannii.*[99]

NADESs	**HBA**	**HBD**	**Molar ratio**	**Yield (mg / g)**	
				trans-cinnamaldehyde	**coumarin**
ChCl-LA	Choline chloride	Lactic Acid	1 : 1	17.19	11.99
ChCl-MA	Choline chloride	Malic Acid	1 : 1	16.52	9.18
ChCl-CA	Choline chloride	Citric Acid	1 : 1	17.67	8.77
B-LA	Betaine	Lactic Acid	1 : 1	13.92	15.33
B-MA	Betaine	Malic Acid	1 : 1	13.56	13.51
B-CA	Betaine	Citric Acid	1 : 1	8.99	12.09

Jablonsky et al. has reported a study on the application of DESs for the extraction of spruce bark, which has a diverse range of antioxidant compounds. The total phenolic content (TPC) and antioxidant activity, in the form of free radical scavenging activity (RSA %), of the polyphenols extracts were measured.

24 solvents based on combination of ChCl, lactic acid (LA), diols and water in different mole ratios were explored. Based on these results, the extracts achieved by a DES made of ChCl, LA, 1,3-butanediol, and water (1: 5 : 1 : 1) displayed the highest antioxidant activity (95%) and phenolic content (596.2 mg gallic acid equivalent (GAE)/100 g dry bark), as shown in Table 1.17. This study displayed that with the addition of a third component i.e., diols, along with the HBA, HBD and water in the DES, the polarity of the extraction system could be further modified resulting in more efficient extraction yields.[100]

Table 1.17 *Prepared DESs, mole ratios of their components, TPC and RSA values for the extraction of spruce bark.*[100]

Sample	Mole Ratio	TPC (mg GAE/100 g Dry bark)	RSA (%)
ChCl:LA:Water	1 : 5 : 0.98	349.4 ± 2.6	92.3
ChCl:LA:1,3-propanediol:Water	1 : 5 : 1 : 0.91	343.8 ± 2.1	94.6
ChCl:LA:1,3-butanediol:Water	1 : 5 : 1 : 1	596.2 ± 7.4	95.0
ChCl:LA:1,4-butanediol:Water	1 : 5 : 1 : 0.91	339.1 ± 1.8	94.1

Skulcova et al. also focused on DES systems and their application in spruce bark extraction.[101] Nine solvents based on combinations of ChCl with organic acids and glycerol were used for extraction. The TPCs and antioxidant activities of the extracts were determined. It was observed that although the greatest extraction yield was obtained from the extraction with ChCl: tartaric acid (27.70%), a higher RSA (%) and TPC was obtained for the extracts from the ChCl:LA DES (16.59% and 463 mg GAE/100 g extract respectively), which was higher than that obtained using other DESs (Table 1.18). A notable finding from these results is that organic acids are the better HBDs in isolating valuable compounds from bark than glycerol.

Table 1.18 *Extraction yield, TPC, an antioxidant activity obtained using different DES systems.*[101]

DESs	Mole ratio	Yield (%)	TPC (mg GAE/100g)	RSA_{30min} %
ChCl: tartaric acid	1 : 1	27.70 ± 1.69	198	15.49
ChCl: lactic acid	1 : 1	22.15 ± 0.68	463	16.59
ChCl: oxalic acid × $2H_2O$	1 : 1	27.08 ± 1.88	191	16.60
ChCl: malonic acid	1 : 1	17.94 ± 0.63	209	15.42
ChCl: glycerol	1 : 2	11.40 ± 0.04	82	15.20

These examples all highlight that DESs and ILs can be valuable solvents in terms of separating extractives from bark. However, the effect of DESs and ILs on *Pinus radiata* bark has not been examined and very few studies involving DES other than those featuring ChCl have been performed.

1.9.2.Drug solubilization

ILs are considered as "designer solvents" and many approaches have been demonstrated for the solubilization of active pharmaceuticals ingredients (APIs) in ILs.[87,102] Kunov-Kruse et al. analysed 18 APIs which were characterised as being primarily HBD or HBA type APIs by using COSMO-RS (representative structures shown in Figure 1.23 and 1.24) and their solubility was tested in three different ILs, $[C_2C_1im][OAc]$, $[C_2C_1im][NTf_2]$ and $[C_4gun][NTf_2]$. HBD APIs were those that contain sites that are significant HBDs such as acetaminophen, albuterol and amoxicillin which all contain phenol, alcohol or carboxylic acid groups. For these HBD APIs, the hydrogen bond acceptor capability of the IL anion is the most important to ensure high solubility. Similarly, HBA APIs are characterised by a lack of accessible hydrogen donating sites and the presence of HBA sites such as fenofibrate, etomidate, ciprofloxacin which contain carbonyl groups and amines. The best ILs for the solubilisation of HBA APIs were predicted to comprise a strong hydrogen bond donating cation and a weakly interacting anion. These predictions were borne out in reality, with $[C_4gun][NTf_2]$ being the most effective solvent for the HBA APIs while $[C_2C_1im][OAc]$ was found to have the best solvating capacity for HBD APIs. The success of $[C_4gun][NTf_2]$ for HBA APIs could be attributed to the strong hydrogen bond donating $[C_4gun]^+$ cation rather than a direct role of the $[NTf_2]^-$ anion, as would be expected, due to the control experiments performed with $[C_2C_1im][NTf_2]$.[87]

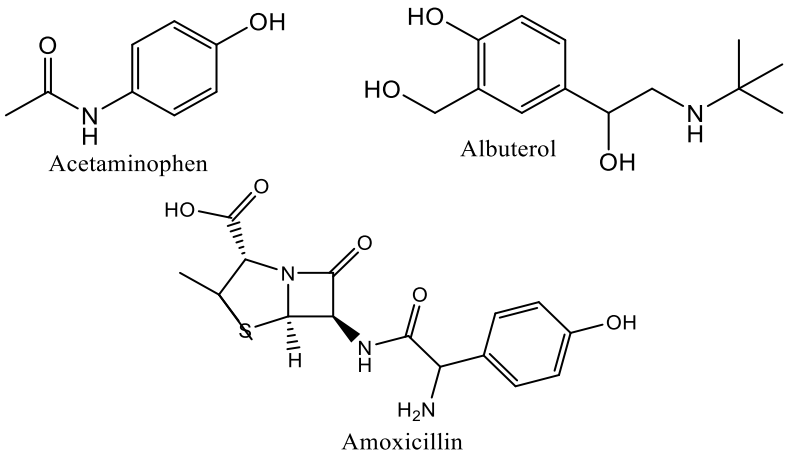


Figure 1.23 *Structures of HBD APIs explored by Kunov-Kruse et al.*[87]

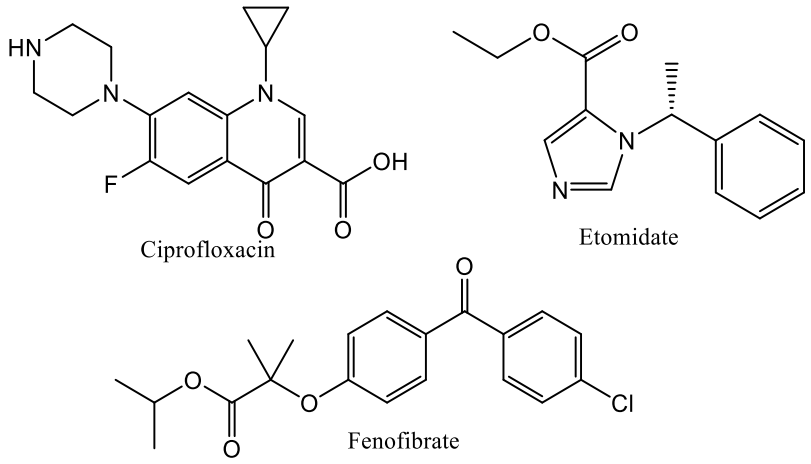


Figure 1.24 *Structures of HBA APIs explored by Kunov-Kruse et al.*[87]

In order to assess the efficacy of the novel IL [C_4gun][NTf_2] as a crystallization solvent for these HBA APIs, fenofibrate (Fig 1.24) was chosen as a model API. The antisolvent, dimethyl sulfoxide (DMSO), which is a good H-bond acceptor, was selected to manipulate the hydrogen bonding in the IL. The hydrogen bond donating sites of the cation could be selectively occupied by the addition of DMSO resulting in the fenofibrate being crystallised in good yield and excellent purity.[87]

Due to their relatively low toxicity and ability to dissolve a wide range of solutes, the use of DES to solubilise drugs has also been explored by Morrison et al.[103] DESs have attracted much attention as some drugs are not soluble in many solvents but can be easily solubilised by DESs. Hence, the solubility of some poorly soluble drugs such as benzoic acid, griseofulvin, danazol and AMG517 (Fig. 1.25) were explored in DESs. Their solubility was measured in pure DESs (ChCl : Urea and ChCl : malonic acid), mixtures of DESs with water, and pure water. AMG517 had a solubility in pure water of <0.0001 mg mL^{-1} compared to 0.01 mg mL^{-1} and 0.4727 mg mL^{-1} in pure ChCl/urea and ChCl/malonic acid, respectively.[18] The high solubility of AMG517 in ChCl/malonic acid was likely to be due to its protonation by the malonic acid which helps the solubility of the drug.[103]

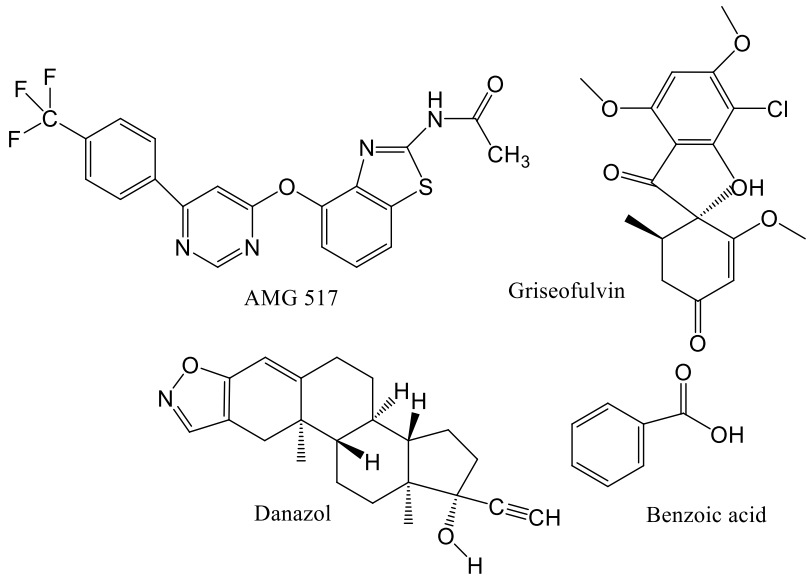


Figure 1.25 *Structure of drugs investigated for their solubility in DESs by Morrison et al.*[103]

A DES based on a 1:1 mixture of ChCl and glycolic acid was synthesised by Zheng et al. to measure the solubility of several common APIs.[90] APIs such as itraconazole, piroxicam, lidocaine and posaconazole were found to have their solubility at room temperature increased by 6700, 430, 28, and 6400-fold respectively, as compared to their aqueous solubility (Fig. 1.26). Probing further, ternary deep eutectic solvent derivatives (DESDs) based on choline chloride, glycolic acid and oxalic acid led to further increases in the solubility of itraconazole with a 53,600–fold increase relative to its aqueous solubility, as represented in Table 1.19. It would be expected that the solubility enhancement of itraconazole would surpass 53,600–fold at even higher mole fractions of oxalic acid due to protonation of the API but the experiment was not feasible due to the high viscosity of such mixtures. As the components of DESs can include biodegradable and biocompatible components that have been used in formulated human products, the potential applicability of suitable DESs to drug delivery could be very attractive.[90]

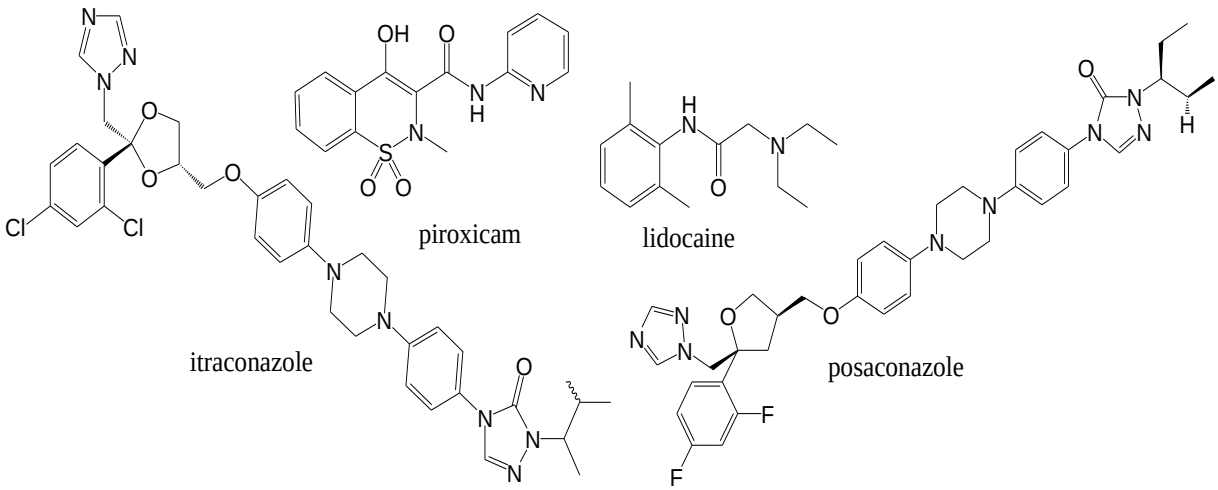


Figure 1.26 *Structure of drugs investigated by Li et al. for their solubility in DESs.*[90]

Table 1.19 *The solubility of four selected drugs in water, DES and DESD.*[90]

Drug	**MW (g/mol)**	**S_w (mg/mL)**	**$S_{1:2}$ (mg/mL)**	**$S_{1:7:0.3}$ (mg/mL)**
Itraconazole	705.6	< 0.001	6.7 ± 0.5	46.4 ± 1.2
Piroxicam	331.3	0.023	9.9 ± 0.9	3.1 ± 0.1
Lidocaine	234.3	3.63	100.6 ± 2.3	295.4 ± 6.8
Posaconazole	700.8	0.012	76.8 ± 1.7	88.4 ± 2.4

{S_w : solubility in water ; $S_{1:2}$: solubility in the DES of ChCl : glycolic acid ; $S_{1:7:0.3}$: solubility in the DESD of ChCl : glycolic acid : oxalic acid.}

Project Aim

The primary focus of the research is to evaluate the use of ILs and DESs for extraction processes. As illustrated by the preceding discussion, ILs and DESs are promising solvents for biomass extraction although very limited work has been performed using these novel solvents on *Pinus radiata* bark.

Pinus radiata is one of the most widely utilised woods in New Zealand. It is a flexible, medium-dense softwood, renowned for fast development and reasonable for a wide variety of uses.[104] From 1900, mass plantings of *Pinus radiata* were common in the Rotorua region of New Zealand and today 89% of the country's plantation forests are this species.[105] Like other barks that have been previously discussed, *Pinus radiata* bark contains five main classes of substances suberin, lignin, carbohydrate, polyphenols, and extractives. Many of these molecules show bioactivity such as antifungal, anti-tumour, anti-carcinogenic, anti-diabetic and anti-inflammatory activity and hence, are gaining attention from researchers.[64,67] *Pinus radiata* bark also represents a significant and growing waste products within New Zealand with limited high value applications at present. Moreover, less work has been conducted examining the separation and isolation of key structural components such as cellulose, lignin and suberin in these barks compared to the work looking at extractives. Therefore, there is a need for further research to explore novel alternatives for the extraction and fractionation of *Pinus radiata* bark with a view to being able to more effectively use the components of this bark as part of future biorefineries.[106]

There has also been very little work performed understanding the polarity of novel DESs and how this may influence their role as extraction solvents. The first major goal of this project will explore the development of new ionic solvents based on the DES concept using strongly hydrogen bond donating cations such as guanidinium and strong hydrogen bond accepting anions. The aim is to develop novel DESs that have greater hydrogen bond donating or accepting capacity than current DESs with the intention of facilitating improved extraction processes. In particular, these solvents have the potential to provide cheaper, more environmentally benign alternatives to ILs that have been recently shown to act as effective solvents for the purification of many difficult to solubilise pharmaceuticals and lignocellulosic biomass. These solvents will then be explored for these applications, both the solubilisation of

pharmaceuticals such as fenofibrate and the extraction of key components from *Pinus radiata* bark.

The second major goal is to understand the relationship between IL and DES structure and the extraction of key components from *Pinus radiata* bark. Relevant IL structural features that will be examined include appropriate IL ions and model the effect of intermolecular interactions on these extraction processes. The length of the alkyl chains on the IL ions will also be modified to provide insight into the effect of the hydrophobic component of the IL on the extraction capacity and selectivity. Therefore, the aim is to enable efficient processing of *Pinus radiata* bark to be designed by utilising DESs and ILs as extractants to understand the relationship between solvent structure and extraction efficacy.

The final aim of this project is to explore the recovery of these extracts. This is an often-overlooked area when using non-volatile solvents such as ILs and DESs but will be essential to any wider application. In this context, the main approach that will be explored will be antisolvent addition. The initial focus will be on the use of antisolvent approaches, with the selection of antisolvents made on the basis of their polarity and the nature of their interaction with the IL.

2. Development of Deep Eutectic solvent (DES) with exceptional hydrogen bonding capabilities

Hydrogen bonding interactions strongly influence the efficacy of solvents in terms of their effect on solubility, reaction kinetics and chemical equilibria. DES provide the opportunity to develop entirely new solvent systems which may display synergistic effects from the constituents. High HBA and HBD abilities could enable improved extraction/separation processes, accelerate chemical reactions, and create opportunities to modify chemical equilibria to change reaction selectivity.

Given the known hydrogen bond donating strength of the guanidinium cation, as discussed in the introduction, it was of interest to explore whether successful DES could be prepared using the guanidinium cation to create solvents with remarkable hydrogen bond donating strength. The design of DESs using a guanidine hydrochloride (GuCl) and various strong HBDs was developed by incorporating a guanidinium salt in the DES. While there have been reports of guanidinium-based DESs,[107] no DESs containing guanidinium salts that are liquid at room temperature have been reported. This emphasises the need to explore alternative HBDs to design guanidinium-based DES that are liquid at room temperature.

The design of hydrogen bond accepting type DESs was also developed by adopting the concept of solvate ILs in order to examine if solvents with unusual hydrogen bonding properties could be prepared as these may have particularly useful properties for chemical extractions.

2.1. Melting points of guanidinium DESs

20 combinations of HBDs with guanidinium chloride were synthesised by directly weighing the HBDs and guanidine hydrochloride in the desired mole ratio i.e., GuCl:HBD (1:9, 1:4, 3:7, 2:3, and 1:1) and then either grinding the mixture at room temperature using mortar and pestle or stirring the mixture at elevated temperature (50 – 80 °C for 1 – 2 h). 10 mol% increments were chosen for the mole ratios. A wide variety of HBDs differing in functionality were screened and the structures of all HBDs are depicted in Figure 2.1. These HBDs include amides (acetamide, urea), carboxylic acids (oxalic acid, lactic acid, levulinic acid, malic acid, acetic acid, citric acid, succinic acid, decanoic acid), sugars or sugar-derived polyols (glucose, sorbitol) and other alcohols (glycerol, 1-decanol) amongst others.

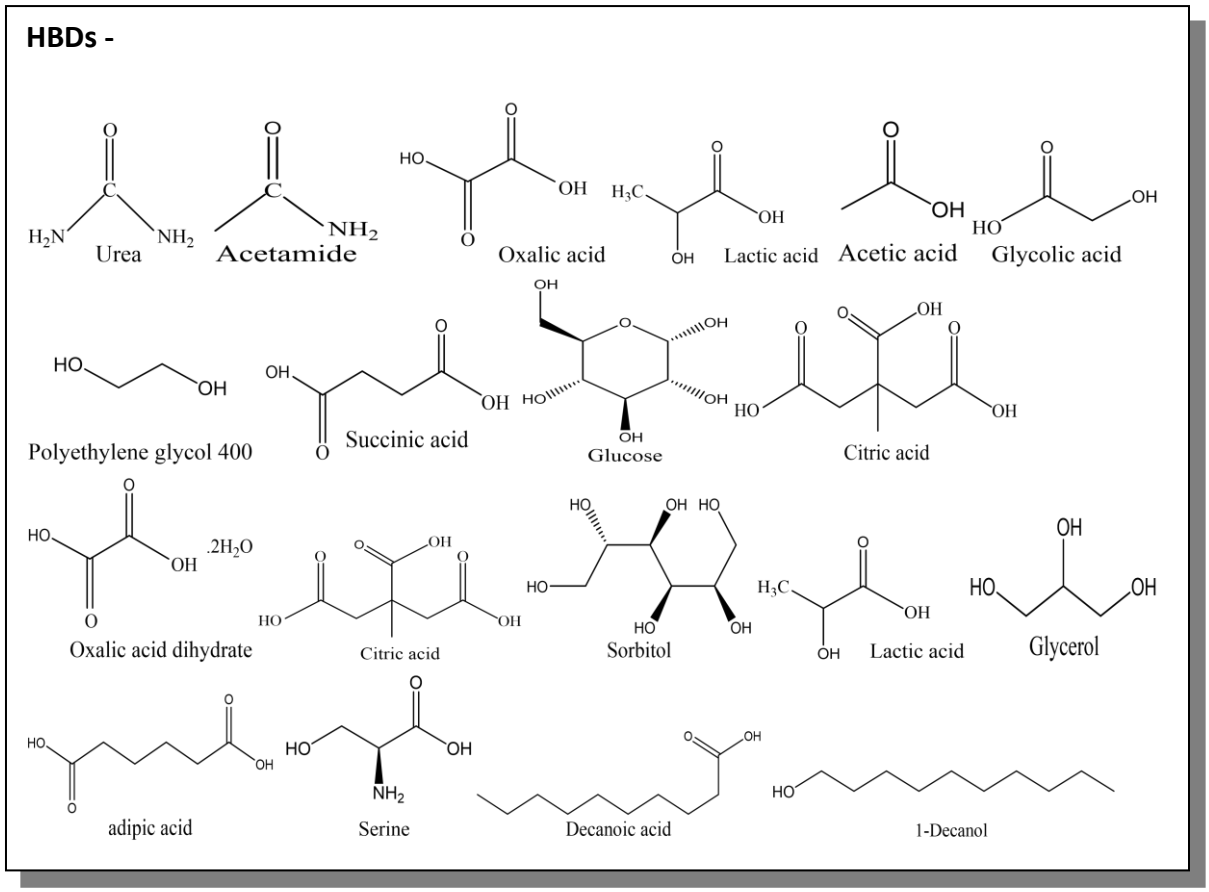


Figure 2.1 Structures of HBDs used to prepare guanidinium DESs

The melting point of all DESs were recorded and these results are provided in Table 2.1. Mixtures that remained liquid after cooling to room temperature are denoted as < 25°C. The reduced ability of HBDs to depress the melting point of mixtures containing GuCl compared to HBAs such as choline chloride (ChCl) is likely to arise from the stronger electrostatic interactions between the guanidinium cation and chloride anion. It is evident from Table 2.1 that only 4 combinations yielded room temperature liquids although several did form eutectic mixtures, with some of the eutectic mixtures leading to significant decreases in melting points (<100°C). Eutectic mixtures with significant melting point decreases (> 75°C) include the combination of GuCl (183°C) and carboxylic acids HBDs such as oxalic acid dihydrate, oxalic acid anhydrous, malic acid, succinic acid, adipic acid and citric acid compared to a linear extrapolation of the pure compounds. This could be due to the availability of hydrogen bond groups on each HBDs for the hydrogen bonding with the Cl^- of the salt. While the following carboxylic acid HBDs such as decanoic acid, and acetic acid did not form the eutectic mixtures resulting in immiscibility of the HBDs in the salt as represented in Table 2.1. The reason behind it is not sure yet. Moreover, even by incorporating a wide range of HBDs in combination with GuCl to produce DESs lower than 100°C only a few numbers of DESs were emerged as liquid

at room temperature such as lactic acid, acetamide, glycerol, and levulinic acid only these type of DESs were further used in measuring polarity parameters, quantification of fenofibrate and viscosity.

Additionally, although the room temperature DESs containing GuCl HBA appeared to be following the similar trend compared to the DESs with ChCl HBA but other combinations of HBA and HBDs are found to be more successful in ChCl DESs than GuCl DESs in terms of lower melting temperature eutectic mixtures.[18]

Table 2.1 *Composition and melting point of lowest melting mixtures of GuCl and each HBD. HBD melting point is provided for reference. < 25°C indicates a room temperature liquid.*

HBD	**HBD m.p.**	**GuCl:HBD**	**GuCl:HBD m.p.**
Oxalic acid dihydrate	101°C	1:1	46°C
Succinic acid	184°C	2:3	125°C
Acetamide	80°C	3:7	<25°C*
Glucose	150°C	1:1	68°C
Malic acid	129°C	1:1	57°C
Urea	134°C	3:7	72°C
Levulinic acid	34°C	1:9	<25°C
Glycolic acid	75°C	1:9	Not miscible
PEG 400	6°C	1:9	Not miscible
Oxalic acid anhydrous	190°C	3:7	160°C
Lactic acid	17°C	3:7	<25°C
Catechol	105°C	1:1	47°C
Decanoic acid	31°C	1:9	Not miscible
Acetic acid	17°C	1:9	Not miscible
Glycerol	19°C	3:7	<25°C
Sorbitol	95°C	1:1	<25°C**
Serine	237°C	1:1	149°C
1-decanol	6°C	1:9	Not miscible
Citric acid	156°C	3:7	98°C
Adipic acid	153°C	1:1	126°C

*Liquid found later to be metastable, **Sticky paste that solidified on standing

2.2. Polarity Parameter Calculations

The polarity of DESs was explored using the solvatochromic dye set developed by Kamlet and Taft.[108] This includes Reichardt's dye 33, 4-nitroaniline and N, N-diethyl-4-nitroaniline[21] or when required, pyridine-N-oxide which had been proposed by Freire and Coutinho as an alternative for acidic solvent systems.[22] The Kamlet-Taft parameters assess the hydrogen bond donating (α), hydrogen bond accepting (β) and dipolarity/polarizability (π*) of a solvent.

2.2.1. Kamlet-Taft Parameters – Solvatochromic Dyes

The structure of a commonly used dye set for the determination of α, β, and π^* parameters for alternative solvents such as ILs and DES involves Reichardt's Dye 30 or Reichardt's Dye 33, N,N-diethyl-4-nitroaniline and 4-nitroaniline are depicted in Figure 2.2.[21,109]

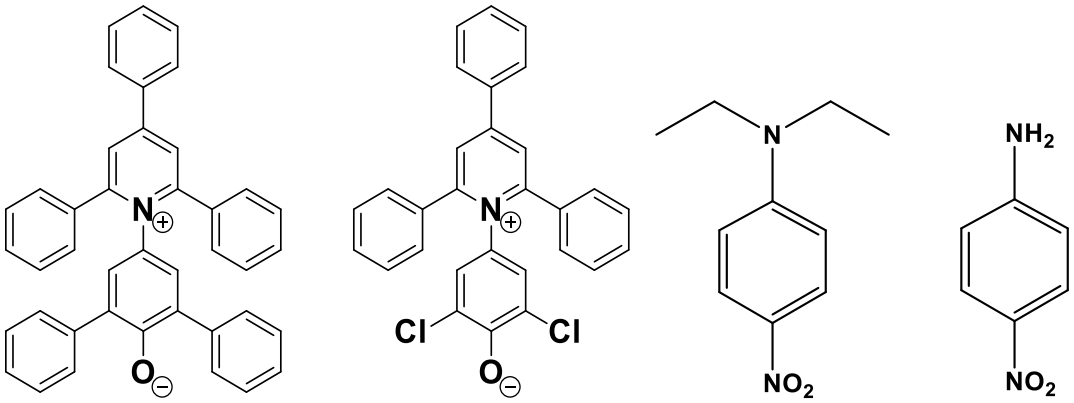


Figure 2.2 *Solvatochromic dye molecules typically used to determine Kamlet-Taft parameters in ionic liquids and deep eutectic solvents. From left: Reichardt's Dye 30, Reichardt's Dye 33, N,N-diethyl-4-nitroaniline, 4-nitroaniline.*

The Kamlet-Taft parameters can be calculated from this dye set using Equations 1-3 below. $\bar{\upsilon}$ is the experimental wave number (cm^{-1}) of the lowest energy absorption maximum with D4NA standing for N,N-diethyl-4-nitroaniline and 4NA for 4-nitroaniline.[21] $E_T(30)$ is determined using Equation 4 based on the lowest energy absorption maximum of Reichardt's Dye 30 (in nm).

$$\pi^* = 0.314(27.52 - \overline{\nu}_{D4NA}) \quad \textbf{(1)}$$

$$\alpha = 0.0649E_T(30) - 2.03 - 0.72\pi^* \quad \textbf{(2)}$$

$$\beta = \frac{1.035\overline{\nu}_{D4NA} + 2.64 - \overline{\nu}_{4NA}}{2.80} \quad \textbf{(3)}$$

$$E_T(30) = \frac{28591}{\lambda_{max}} \quad (4)$$

Reichardt's Dye 33 is a less basic analogue of Reichardt's Dye 30 which makes it more compatible with acidic solvents. $E_T(33)$ values are computed using Equation 5 using the lowest energy absorption maximum for Reichardt's Dye 33, analogous to the calculation for $E_T(30)$. The empirical relationship between $E_T(30)$ and $E_T(33)$ that has been determined using linear regression is given by Equation 6,[12] facilitating the direct determination of α from $E_T(33)$ using Equation 7.

$$E_T(33) = \frac{28591}{\lambda_{max}} \quad \textbf{(5)}$$

$$E_T(30) = 0.9953E_T(33) - 8.1132 \quad \textbf{(6)}$$

$$\alpha = 0.0646E_T(33) - 2.56 - 0.72\pi^* \quad \textbf{(7)}$$

The DES were explored using Reichardt's Dye-33, N,N-diethyl-4-nitroaniline, and 4-nitroaniline although even with the more acid-resistant Reichardt's Dye-33, significant bleaching of the dye could be observed. It was found that Reichardt's Dye-33 was bleached in the presence of carboxylic acid containing solvents such as lactic acid and levulinic acid although it was effective in solvents containing alcohol and amide groups.

2.2.2.Kamlet-Taft Parameters – Pyridine-N-oxide

To overcome the problem of the acid-bleaching of Reichardt's dye, pyridine-N-oxide (Figure 2.3), was used as an alternative probe as it has been shown previously as a viable probe for the determination of α values.[22] Unlike the solvatochromic dyes conventionally used, pyridine-N-oxide uses the ^{13}C chemical shift difference (ppm) between carbon 4 and either carbon 2 (d_{24}) or 3 (d_{34}) to enumerate the α value, denoted α_{24} and α_{34}. A representative spectrum is depicted in Figure 2.3, highlighting the key peaks/differences in the ^{13}C NMR spectrum.

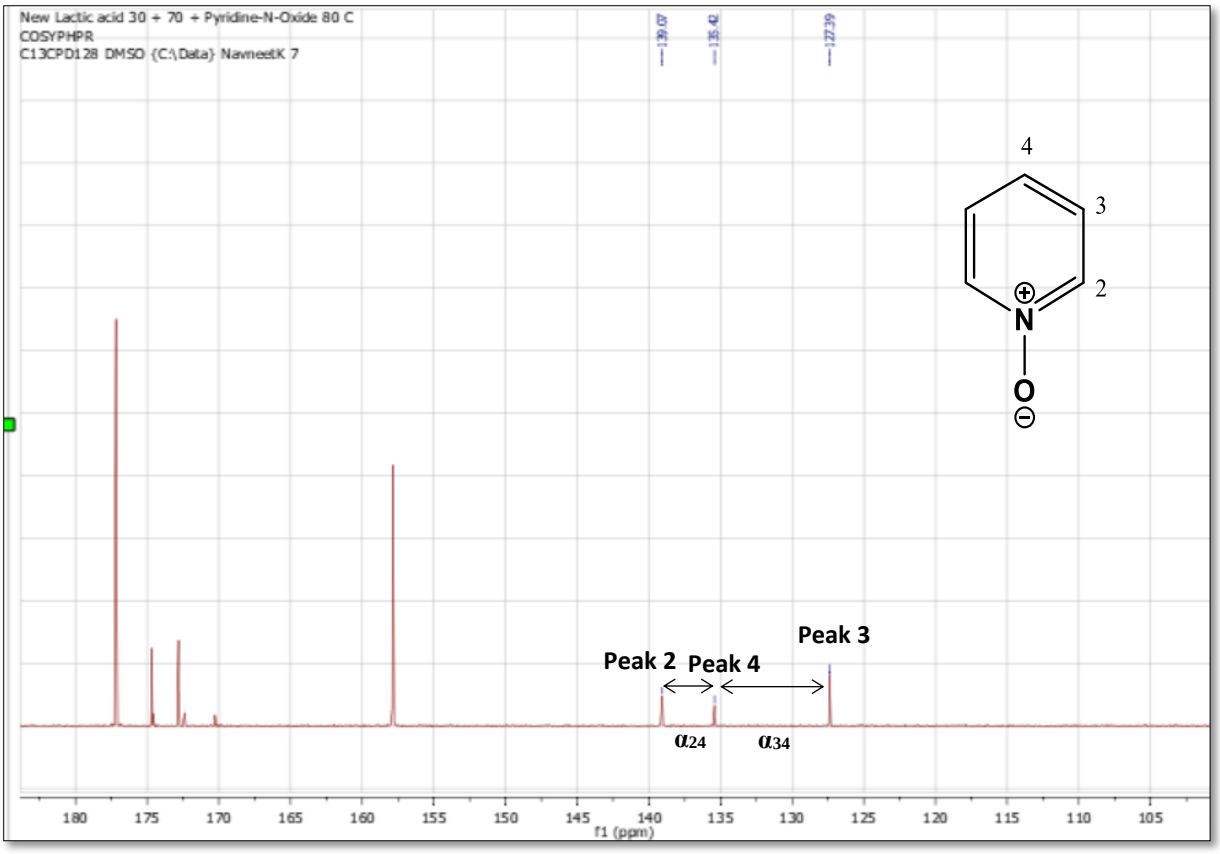


Figure 2.3 *^{13}C NMR spectrum of pyridine N-oxide in GuCl:LA (3:7) highlighting the key peaks and chemical shift differences used to calculate α_{24} and α_{34} along with a structure of pyridine-N-oxide*

α_{24} and α_{34} values are calculated using Equations 8 and 9 respectively.

$$\alpha_{24} = 2.32 - 0.15 \times d_{24} \quad \textbf{(8)}$$

$$\alpha_{34} = 0.40 - 0.16 \times d_{34} \quad \textbf{(9)}$$

Empirical linear regression has been found that $\alpha_{24} = 0.88\alpha_{RD}$ where α_{RD} is the α value calculated using Equations 2 or 7.[22] Hence dividing α_{24} by 0.88 should yield a comparable value of α to that obtained using Reichardt's Dye 30 or 33 as part of the dye set. This value α_{PyO}, defined as per Equation 10.

$$\alpha_{PyO} = \frac{\alpha_{24}}{0.88} \quad \textbf{(10)}$$

2.3. Polarity Parameter Results

2.3.1.Kamlet-Taft Parameters

The Kamlet-Taft parameters were obtained for all of the DESs that were liquid at room temperature at 25°C, 40°C, 60°C and 80°C using both pyridine-N-oxide and Reichardt's dye where possible and are summarised in Table 2.2. These results highlight that the apparent polarity of these solvents, as determined by the solvatochromic dye or pyridine N-oxide measurements, does not vary significantly with temperature. Most changes in the observed parameters were less than ± 0.1 even on heating from 25 to 80°C, which lies within the anticipated experimental error for these measurements.

Table 2.2 Kamlet-Taft parameters obtained for all DES at temperatures from 25-80°C. Parameters are defined in the text. (HFIP = 1,1,1,3,3,3-hexafluoroisopropanol, LA = Lactic acid, Gly = Glycerol, Acet = Acetamide, LevA = Levulinic acid).

DES	GuCl:HBD	Temp (°C)	α_{RD}	α_{PyO}	α_{24}	α_{34}	β	π^*
LA	-	25		1.79	1.58	1.56	0.38	1.15
		40	-	1.76	1.55	1.54	0.41	1.11
		60		1.73	1.53	1.51	0.36	1.09
		80		1.70	1.50	1.48	0.31	1.09
GuCl:LA	1:9	25		2.06	1.81	1.74	0.42	1.17
		40	-	2.05	1.80	1.73	0.40	1.19
		60		2.01	1.77	1.72	0.39	1.17
		80		2.00	1.76	1.70	0.35	1.17
	1:4	25		2.01	1.77	1.70	0.40	1.23
		40	-	2.01	1.77	1.69	0.35	1.25
		60		1.99	1.75	1.68	0.37	1.21
		80		1.97	1.73	1.67	0.38	1.18
	3:7	25		2.07	1.82	1.70	0.39	1.25
		40	-	2.07	1.82	1.70	0.39	1.25
		60		2.07	1.82	1.69	0.29	1.30
		80		2.06	1.81	1.68	0.31	1.27
	2:3	25		1.98	1.74	1.65	0.35	1.30
		40	-	1.98	1.74	1.65	0.33	1.33
		60		1.97	1.73	1.64	0.34	1.30
		80		1.95	1.72	1.63	0.36	1.23
Gly	-	25	0.91	1.34	1.18	1.06	0.53	1.14
		40	0.91	1.30	1.14	1.06	0.48	1.15
		60	0.86	1.26	1.11	1.03	0.49	1.14
		80	0.83	1.22	1.07	1.00	0.49	1.14

GuCl:Gly	1:9	25	1.00	1.34	1.18	1.08	0.50	1.18
		40	0.95	1.31	1.15	1.06	0.44	1.22
		60	0.87	1.28	1.13	1.03	0.47	1.21
		80	0.86	1.24	1.09	1.00	0.44	1.21
	1:4	25	0.97	1.33	1.17	1.07	0.43	1.26
		40	0.92	1.31	1.15	1.05	0.44	1.27
		60	0.86	1.27	1.12	1.03	0.40	1.28
		80	0.86	1.24	1.09	1.00	0.44	1.24
	3:7	25	1.00	1.31	1.15	1.06	0.43	1.27
		40	0.93	1.28	1.13	1.04	0.40	1.30
		60	0.97	1.25	1.10	1.01	0.43	1.27
		80	0.93	1.22	1.07	0.99	0.39	1.27
	2:3	25	0.97	1.28	1.13	1.05	0.43	1.28
		40	0.92	1.26	1.11	1.03	0.38	1.30
		60	0.91	1.24	1.09	1.00	0.37	1.30
		80	0.89	1.20	1.06	0.98	0.36	1.29
LevA	-	25		1.49	1.31	1.31	0.96	0.58
		40	-	1.45	1.28	1.29	0.97	0.56
		60		1.41	1.24	1.26	0.98	0.53
		80		1.27	1.20	1.22	0.99	0.50
GuCl:LevA	1:9	25		1.60	1.41	1.37	0.57	0.96
		40	-	1.58	1.39	1.35	0.53	0.98
		60		1.55	1.36	1.32	0.53	0.97
		80		1.51	1.33	1.30	0.54	0.91
	1:4	25		1.66	1.46	1.39	0.43	1.06
		40	-	1.64	1.44	1.38	0.46	1.04
		60		1.63	1.43	1.36	0.46	1.00
		80		1.60	1.41	1.34	0.47	0.96
GuCl:Acet	3:7	25	0.86	1.00	0.88	0.81	0.52	1.17
		40	0.90	1.00	0.88	0.81	0.54	1.14
		60	0.86	0.95	0.84	0.77	0.48	1.15
		80	0.86	0.92	0.81	0.75	0.52	1.11
	2:3	25	-	1.01	0.89	0.83	-	1.20
		40	0.88	0.83	0.88	0.81	0.49	1.21
		60	0.91	0.99	0.87	0.81	0.49	1.18
		80	0.85	0.97	0.85	0.79	0.47	1.17
HFIP	-	25	1.96[110]	1.80	1.58	1.54	0[110]	0.65[110]

The results in Table 2.2 highlight several key trends. Firstly, the π* values of the DESs tend to increase with increasing GuCl concentration, although most DESs lie within the range 1-1.3 with relatively small changes in π* once at least 10 mol% GuCl is present. There is a small

effect of the HBD on π* where HBDs that have inherently higher π* values (LA, Gly) lead to DES with slightly higher π* values although this becomes less pronounced as the GuCl content increases. This is consistent with trends observed for ILs and other DES where π* is relatively consistent for ionic systems regardless of the nature of the salt in the case of DES.[20,21,109] This is due to the cation-anion charge separation dominating other, weaker, non-specific interactions.

The hydrogen-bond basicity (β), was found to generally decrease as the concentration of GuCl increases, highlighting that the β value is more strongly influenced by the HBD than the presence of the Cl^- anion. This is consistent with results for other DESs where strong hydrogen-bonding interactions between cations and anions reduced the impact of Cl^- ions on β values.[21] In terms of the HBD, the β value increased in the order LA < Gly < LevA < Acet. This trend can be attributed to the functional groups present in the HBDs with the general order of increasing HBA ability being acids < alcohols ≈ ketones < amides.[111]

The trends in α values will be discussed in terms of the α_{PyO} results (qualitatively equivalent to the α_{24} results) as these enable comparison across all solvents explored. α_{PyO} tended to increase in the following order: GuCl:Acet < GuCl:Gly < GuCl:LevA < GuCl:LA. The effect of GuCl on α changed depending on the HBD suggesting that its influence depends on specific interactions between GuCl and the HBD, or between the probe and the HBD. For Gly, the addition of GuCl had no significant effect on the value of α whereas for LA and LevA the addition of GuCl led to an increase in α. However, for LA there was no significant change in α after more than 10 mol% GuCl was present. Notably, the values obtained for the GuCl:LA mixtures were all greater than the α_{PyO} values obtained for the other DES examined and also for 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP) which was characterised under the same conditions. To the best of our knowledge these represent the highest α_{PyO} values observed and highlight that these solvents appear to be exceptional hydrogen bond donors.

Apart from this, there is a slight decrease in observed α_{24} value with heating for many of the DES which is more clearly depicted by Figure 2.4 which may indicate weakening of the hydrogen bond network with increasing thermal energy; however, such an effect is clearly less than the effect of changing the identity of the DES itself. For example, the α_{24} value of GuCl:LA (3:7) at room temperature was observed to be ~1.5 times higher than the α_{24} value of GuCl:Gly at the same temperature conditions and composition whereas the α_{24} value of GuCl:LA (3:7)

did not change with temperature and GuCl:Gly (3:7) only decreased by ~7% on heating from 25-80°C.

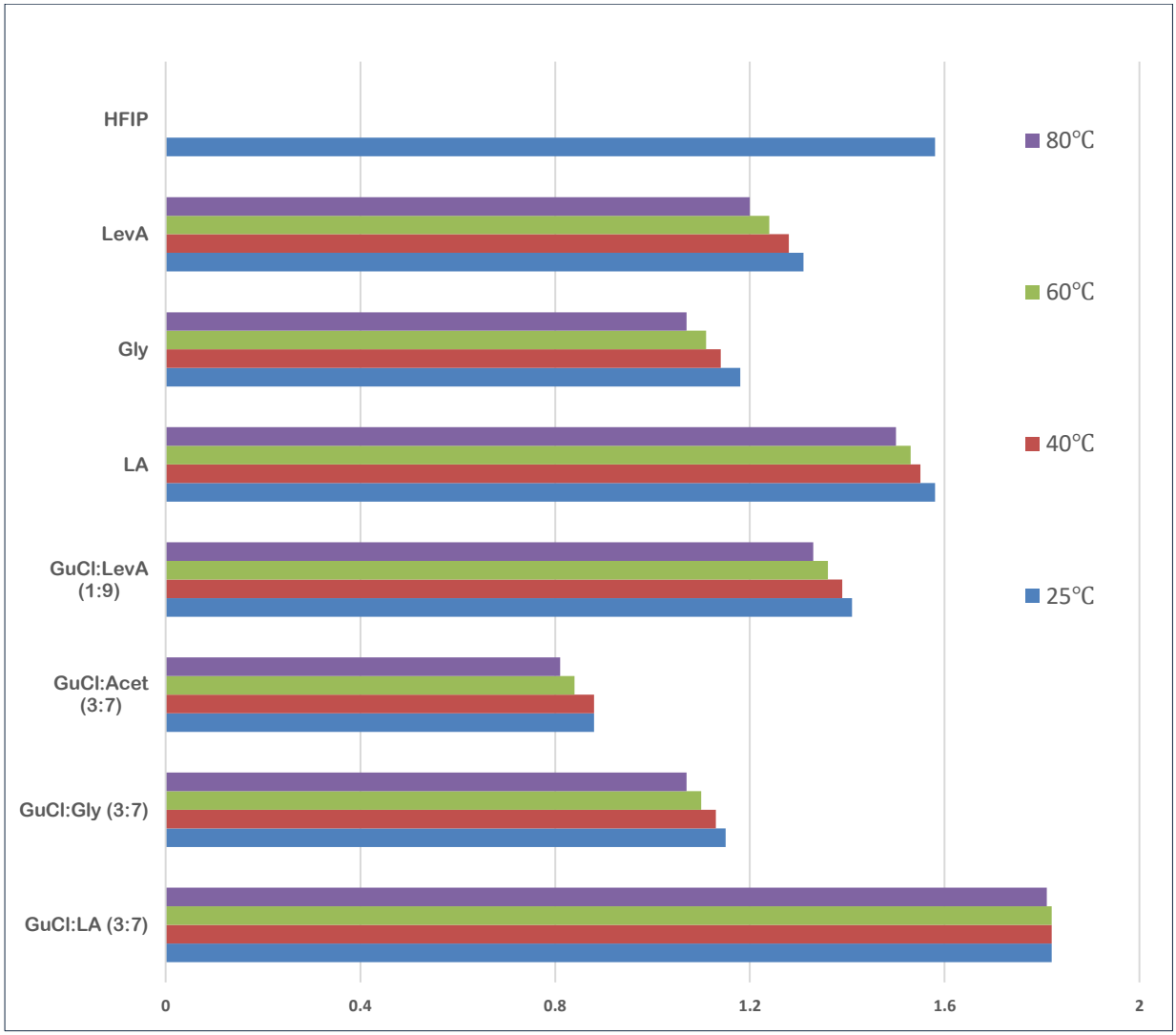


Figure 2.4 *Effect of temperature on the α_{24} parameter for the DES.*

2.4.Viscosity measurement

Viscosity is of key importance for the design of DESs for many applications. Viscosity measures the ease of flow of a liquid. In general, due to the high molecular weights and high degree of association within DESs,[18,33] they have higher viscosities compared with other solvents.

The viscosity describes the internal resistance of a fluid to a shear stress. There are two classes of viscosity behaviour in fluids with respect to shear rate: Newtonian and non-Newtonian.[47] Newtonian fluids have a constant viscosity coefficient (η) regardless of the imposed shear rate. In contrast, non-Newtonian fluids have a variable viscosity coefficient (η) depending on the

imposed shear rate and can either shear thicken, where viscosity increases with applied shear stress, or shear thin, where viscosity decreases with increasing applied shear stress.[47]

In this work, viscosity was measured as a function of shear rate. In all the analysed DESs, the apparent viscosity decreased with shear rate as shown in Figure 2.5 (a), (b), (c) and (d) indicating that most of the DESs are non-Newtonian liquids that experience shear thinning.

The fluid behaviour is also dependent on the components of the system, as it has been well established that intermolecular interactions (H-bonding, dispersive and coulombic interactions) play a crucial role in determining the viscosity of DESs.[112,113] The viscosities of DESs cover a wider range than what is generally observed with ILs and conventional solvents, with viscosities from 36 cP for a eutectic mixture of ChCl and ethylene glycol, to approximately 11000 cP for a eutectic mixture of ChCl, $ZnCl_2$ and urea at 298 K.[18,40] Owing to the presence of a strong hydrogen bond network between each component of a DES, the viscosity of DESs are typically higher than ILs, which further results in a lower mobility of ions and molecules within DES.

Another factor which has a significant effect is the composition of the DES, including its water content. The composition of DES changes the size of the molecular constituents and the degree of complexation between the constituents of the DES, resulting in different degrees of intermolecular interaction, and consequently its viscosity. For instance, it has been found that the eutectic mixture of ChCl and urea showed non-Newtonian fluid properties when pure, but upon the addition of 2.5 % water was found to behave as a Newtonian fluid.[114] The pronounced effect of water on the fluid behaviour of ChCl: urea mixture, was linked to the hydrogen bond donating parameter (α) of the system being lower compared to the other ChCl : HBD systems and pure water, which suggested water was preferentially solvating the chloride anion.[115] While the precise water content of our DES could not be determined due to issues with the Karl Fischer titration, all the analysed four DESs display non-Newtonian fluid behaviour as shown in Figure 2.5.

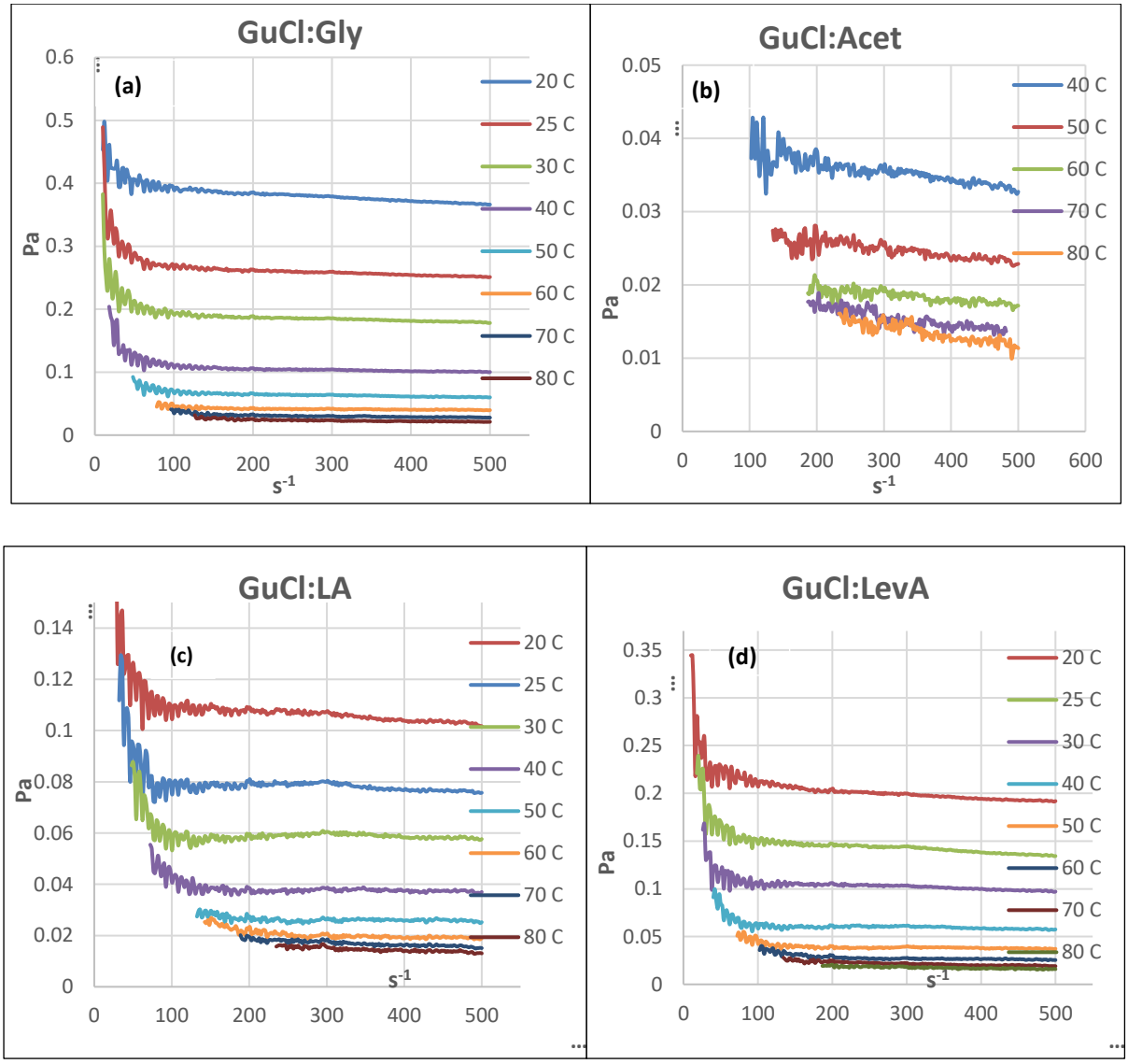


Figure 2.5 *Plots of viscosity v/s shear rate (a) (1:2) GuCl:Gly (b) (1:2) GuCl:Acet (c) (1:2) GuCl:LA (d) (1:4) GuCl:LevA.*

Additionally, increasing the temperature significantly decreases the intensity of these intermolecular forces and, therefore, the viscosity of the solvent. Hence, the viscosity of most DES systems is a function of three parameters; these are the mixture constituents, composition of the mixture and the temperature.[50]

Figure 2.6 (a), (b), (c) and (d) depict the relationship between the viscosity of the four DESs and the temperature at 500 s^{-1} shear rate. It was found that the viscosity of the fluids decreased exponentially with increasing temperature which is in agreement with the published work.[50] Among the four DESs, GuCl:Gly DES was the most viscous DES followed by GuCl:LevA, GuCl:LA and GuCl:Acet.

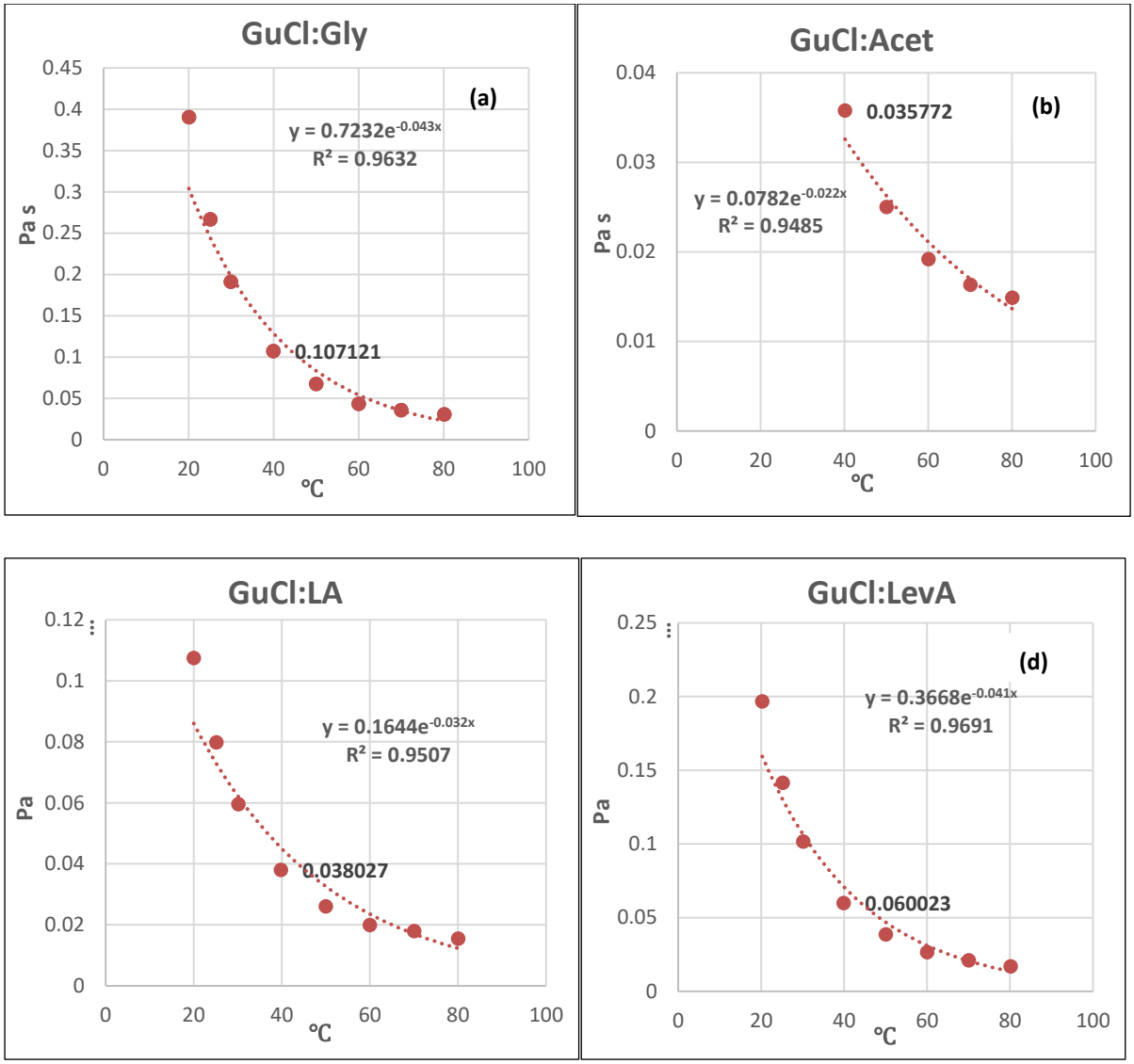


Figure 2.6 *Relationship between viscosity and temperature, viscosity determined at a shear rate of 500 s^{-1} (a) (1:2) GuCl:Gly (b) (1:2) GuCl:Acet (c) (1:2) GuCl:LA (d) (1:4) GuCl:LevA*

The change in viscosity of the DESs with temperature was found to follow the Arrhenius equation (11) and (12) for viscous flow.[33]

$$\eta = Ae^{\frac{-E_a}{RT}} \quad (11)$$

or

$$\ln \eta = \ln A - \frac{E_a}{RT} \quad (12)$$

The activation energy (E_a) and pre-exponential factor (A) of viscosity for each type of DES was determined using the Arrhenius Equation (Table 2.3). This allows these ILs to be compared

with other published DESs. It was reported that the obtained E_a values for the other systems containing acid and alcohol as HBDs such as ChCl:urea (1:2), ChCl: malonic acid (1:1), and ChCl:ethylene glycol are 53.42, 44.36 and 25.43 kJ/mol, respectively,[116] higher than for recent reported DESs resulting in more viscous solvents. From Table 2.3, it had been evaluated that the E_a of GuCl:Gly DES was higher than the remaining analysed DESs indicating the excessive resistance in the fluid's flow and hence, highly viscous solvent.

Table 2.3 *Activation Energy of DES.*

DES	**GuCl:HBD**	**E_a (kJ/mol)**
GuCl:LA	1:2	28.09
GuCl:Gly	1:2	37.41
GuCl:Acet	1:2	20.21
GuCl:LevA	1:4	35.53

The viscosity of these DES was also measured using rolling ball method by Roland Brown and compared with the aforementioned data. The viscosities of the three DESs at 20, 25, 40, 60 and 80°C are provided in Table 2.4 The experimental setup did not permit the measurement of viscosities less than 20 mPa s, hence solvents with a viscosity less than this are simply listed as < 20 mPa s in Table 2.4. Moreover, acetamide-based DES was not analysed by this method due to it being a metastable liquid at room temperature. The measurements recorded at 40℃ using two different techniques are displaying the similar outcome indicating that GuCl:Gly DES has the highest viscosity than the other DESs. Although the viscosity is increased by decreasing the temperature, the trend of viscosity at 40℃ is as follow : GuCl:Gly (253.4 mPa s) > GuCl:LA (50.05 mPa s) > GuCl:LevA (<20 mPa s). The slight variation in the trend could be due to the change in composition of GuCl:LevA (1:9) which is likely to reduce the viscosity of this DES compared to the DES with a higher GuCl content.

Table 2.4 *Viscosity of DES at variable temperatures. Accuracy of viscosity measurements is ±0.5%.*

DES	GuCl:HBD	Temp (°C)	Viscosity (mPa s)
GuCl:LA	1:2	20 25 **40** 60 80	181.4 126.8 **50.05** 20.60 <20
GuCl:Gly	1:2	20 25 **40** 60 80	635.8 411.9 **253.4** 46.45 22.18
GuCl:LevA	1:9	20 25 **40** 60 80	84.32 61.18 **<20** <20 <20

2.5.Quantification of fenofibrate by HPLC-UV

Fenofibrate (FF) was selected as an initial solubility target as it was expected to show significantly enhanced solubility in guanidinium DESs due to the presence of hydrogen bond accepting groups. The structure of FF is depicted in Figure 2.7 It has been observed that the solubilities of HBA active pharmaceutical ingredients (APIs) are higher in $[C_4gun][NTf_2]$ than HBD APIs, which are more soluble in $[C_2C_1im][OAc]$.[87] This can be attributed to the strong hydrogen bond donating $[C_4gun]^+$ cation rather than a direct role of the $[NTf_2]^-$ anion, as would be expected. This would emphasise the possibility of the solubilisation of FF in guanidinium DESs due to the hydrogen bond interaction between the carbonyl group of FF and the guanidinium cation.

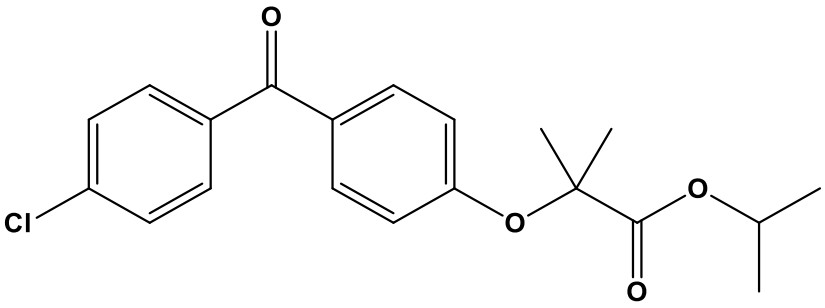


Figure 2.7 *Structure of Fenofibrate (FF)*

2.5.1.Calibration curve for HPLC using fenofibrate

HPLC-UV was used to analyse the solubility of FF in DESs. The calibration curve using standard solutions of FF in acetonitrile is shown in Fig. 2.8.

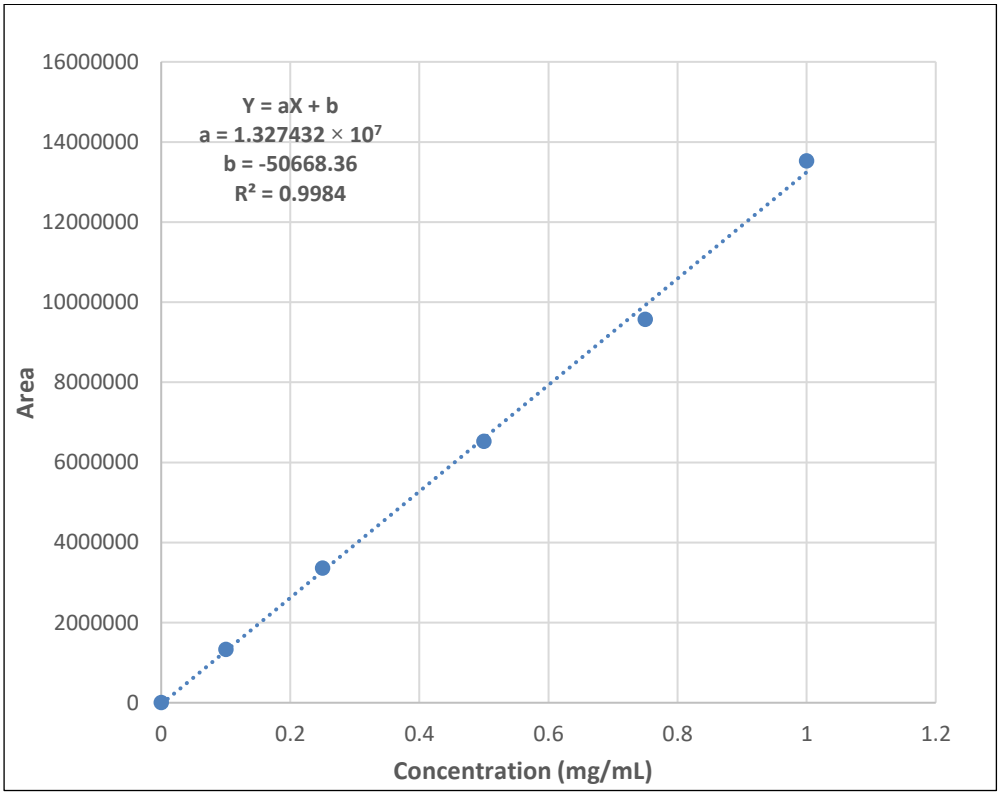


Figure 2.8 *Calibration curve of standard solutions (HPLC-UV)*

The good fit to the calibration curve indicates that HPLC is an appropriate technique for the quantification of fenofibrate.

2.5.2. Effect of DESs structures on the solubility of FF

After having done the calibration, the solubility of fenofibrate and concentration were analysed by HPLC (UV-Vis detector- 280 nm), as shown in Table 2.5. The measurements

were recorded at two different temperatures (25 and 50°C) as not all of the selected DESs, such as GuCl:Acet, were completely liquid at room temperature. It also enabled the effect of temperature on the solubility of FF to be explored. GuCl:Gly was not a good solvent for fenofibrate as it demonstrated no peak in the chromatogram, highlighting that the solubility was below the limit of detection. In addition, out of all DESs, GuCl:LevA emerged as the most suited for the solubilisation of FF; however, the solubility of FF was still very low (1.43 w/w%).

18 APIs which were characterised as being primarily HBD or HBA type APIs by using COSMO-RS (discussed in section 1.9.2) analysed by Kunov-Kruse et al. and their solubility was tested in three different ILs, $[C_2C_1im][OAc]$, $[C_2C_1im][NTf_2]$ and $[C_4gun][NTf_2]$.[87] It has been determined that the solubility of FF was > 30 wt% for the guanidinium IL at 25°C and also solubility in $[C_4C_1im][NTf_2]$ was around 2.5 wt% at 25°C too. This suggests that the 1.4 wt% is actually low relative to ILs similar to these DESs, even ILs not really selected to be optimal solvents.

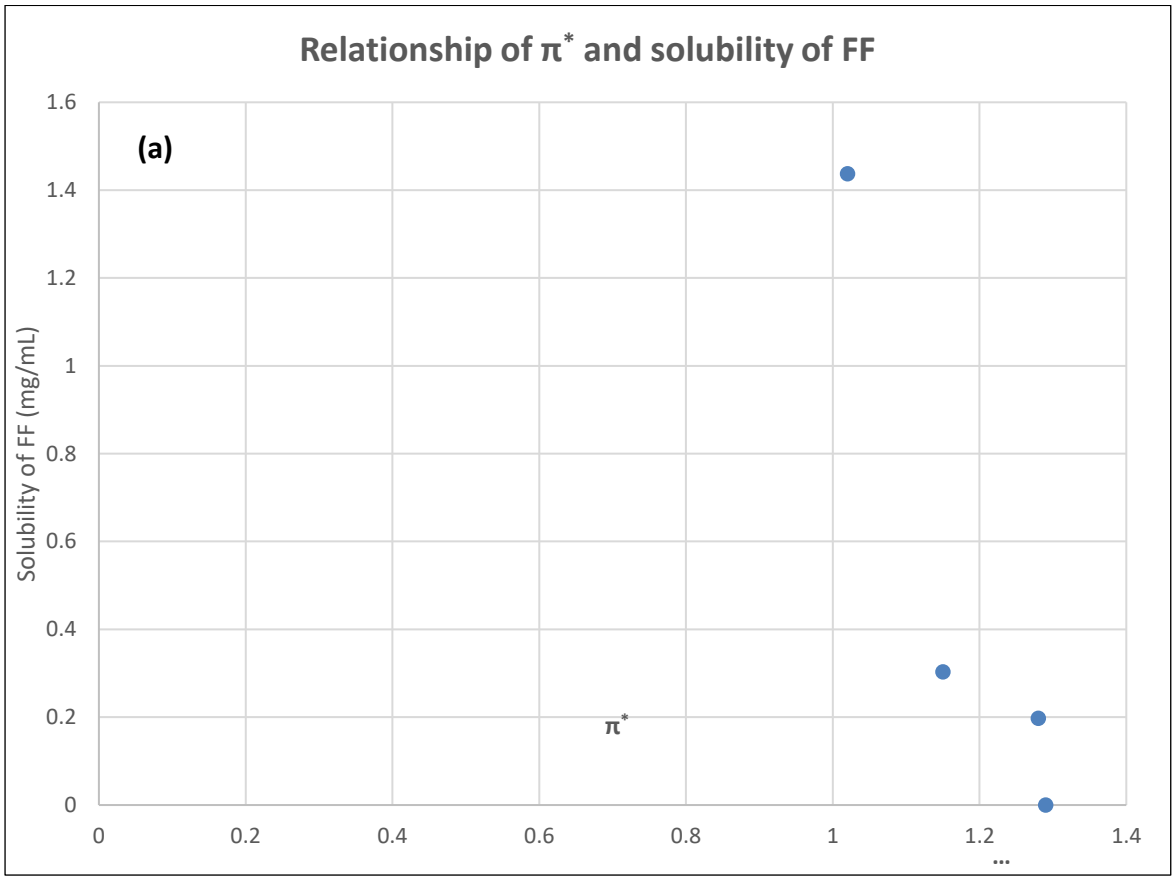
Table 2.5 *Solubility measurement by HPLC-UV and the determined polarity parameters (α, β and π*) of DESs.*

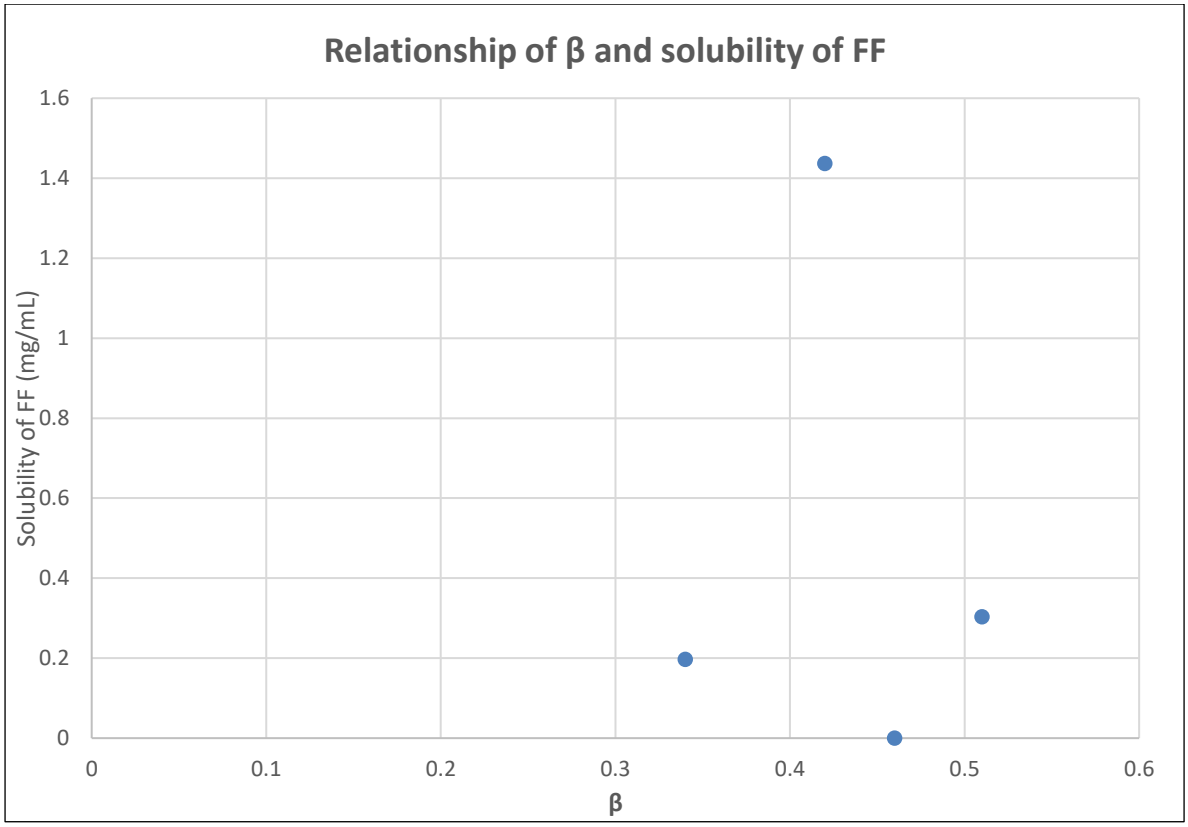
Sample	GuCl:HBD	Temp (°C)	Solubility of FF (w/w %)	α_{24}	β	π^*
GuCl:LA	1 : 2	25	0.164	-	-	-
GuCl:Gly	1 : 2	25	No Peak	-	-	-
GuCl:LevA	1 : 4	50	1.437	1.44	0.42	1.02
GuCl:Acet	1 : 2	50	0.303	0.86	0.51	1.15
GuCl:LA	1 : 2	50	0.197	1.82	0.34	1.28
GuCl:Gly	1 : 2	50	No Peak	1.12	0.46	1.29

(Note : α, β and π value of each DES were calculated by taking the average of 40 and 60 °C)*

Figure 2.9 illustrates the relationship between the α, β and π^* value of four DESs (Table 2.2) and the solubility of FF in DESs. According to the data in Table 2.2, the trend of hydrogen bond donating ability of DESs is as follow : GuCl:LA > GuCl:LevA > GuCl:Gly > GuCl:Acet. While it was originally expected that the solubility would follow this trend, the GuCl:LA DES had the highest α value (1.82) but one of the lowest FF solubilities. In fact, of the Kamlet-Taft parameters, only π* showed a definitive trend with respect to FF solubility, with increasing π* leading to a decrease in solubility. This suggests that the solubility trend

could be due to the strong self-association of the DESs limiting their ability to solvate FF. This would be less of an issue for ILs featuring a weakly interacting anion such as $[NTf_2]^-$ which highlights one of the limitations of the use of these DES as solvents.





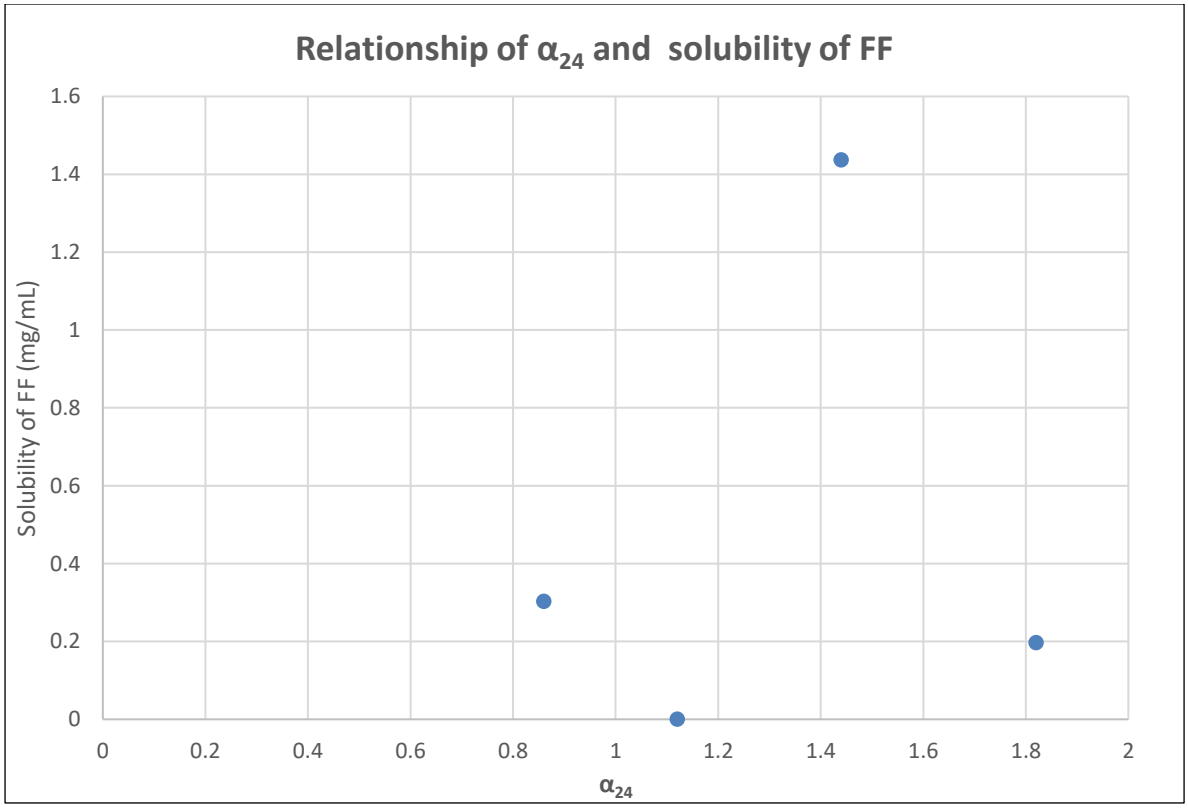
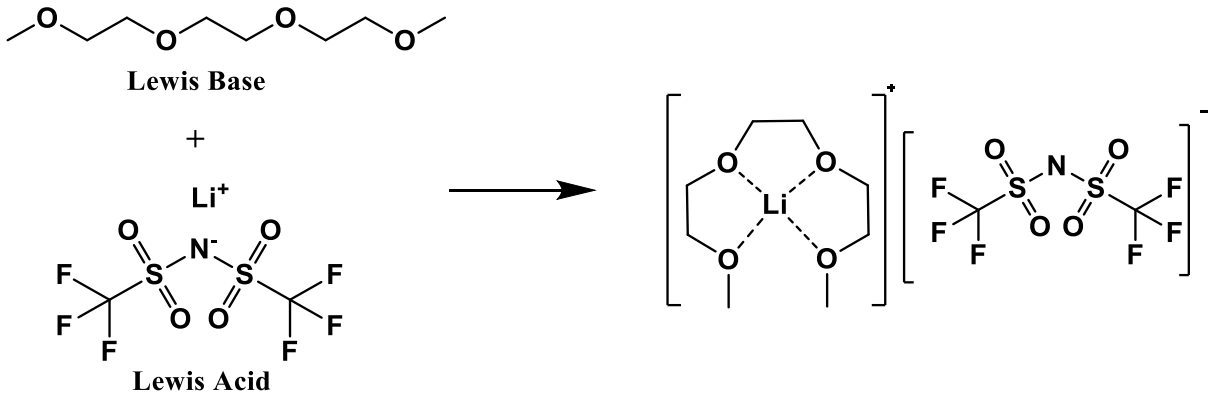


Figure 2.9 *Comparing the solubility of fenofibrate in DESs with the (a) π^* (b) β and (c) α_{24} parameters of the DESs at 50℃.*

2.6. Solvate Ionic Liquids (SILs)

SILs are a relatively new class of ILs composed of at least one complex ion, typically formed by the coordination of a metal salt (usually lithium) with a coordinating solvent (such as glymes) as shown in Scheme 2.1. The coordination occurs due to the polyether being able to chelate the metal by the lone pair of the oxygen atoms of the ether moieties acting as Lewis bases, donating electron to the Lewis acid cation.[117]



Scheme 3 *Addition of metal ion Lewis acid and ether Lewis base to give a SIL.*

The preferred coordination number of lithium ions is 4 or 5 in solution resulting in the greatest effectiveness for appropriate oligoethers that can form such coordination environments, as shown in Figure 2.10. Figure 2.10 illustrates the 4 and 5 centred coordination modes of lithium due to the coordination of lithium in $LiNTf_2$ by either triethylene glycol dimethyl ether or tetraethylene glycol dimethyl ether ($[Li(G_3)][NTf_2]$ (1) or ($[Li(G_4)][NTf_2]$ (2), respectively, as these are amongst the most common SILs reported in the literature.[117,118]

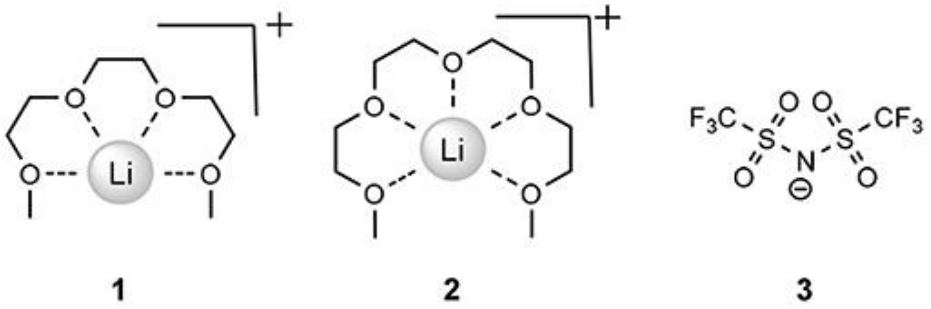


Figure 2.10 *Structure of ($[Li(G_3)]NTf_2$ (1), ($[Li(G_4)]NTf_2$ (2) and NTf_2^- anion (3).*[117]

Additionally, the Lewis basicity of the anion is associated with the relevant physicochemical properties and coordinating structure of the SILs. The coordination ability of the anion

determines whether the SIL is good i.e., strongly interaction of cation with ether or poor i.e., weekly interaction of cation with the ether, respectively.[119] For example, if the anion weakly coordinates to the lithium, the added ether can more readily form complexes with the cation, which produces a good SIL such as $[Li(G_4)][NTf_2]$. By contrast, if the anion is strongly coordinating then there is a significant concentration of free ether and less of the cation is coordinated, producing a poor SIL such as $([Li(G_4)][NO_3]$.[118] Very few SILs containing anions other than $[NTf_2]^-$ anions and cations other than Li have been reported.

The aim of developing various SILs was to design strongly H-bonding accepting SILs using non-conventionally available SILs anions by incorporating to given solvents as shown in Figure 2.11. A wide range of lithium and sodium salts were selected along with oligoethers and hydroxyl ether of differing chain length (Figure 2.11). The selection of the salts was based on the hydrogen bond accepting abilities of the anions chosen such as acetate, nitrate and chloride.

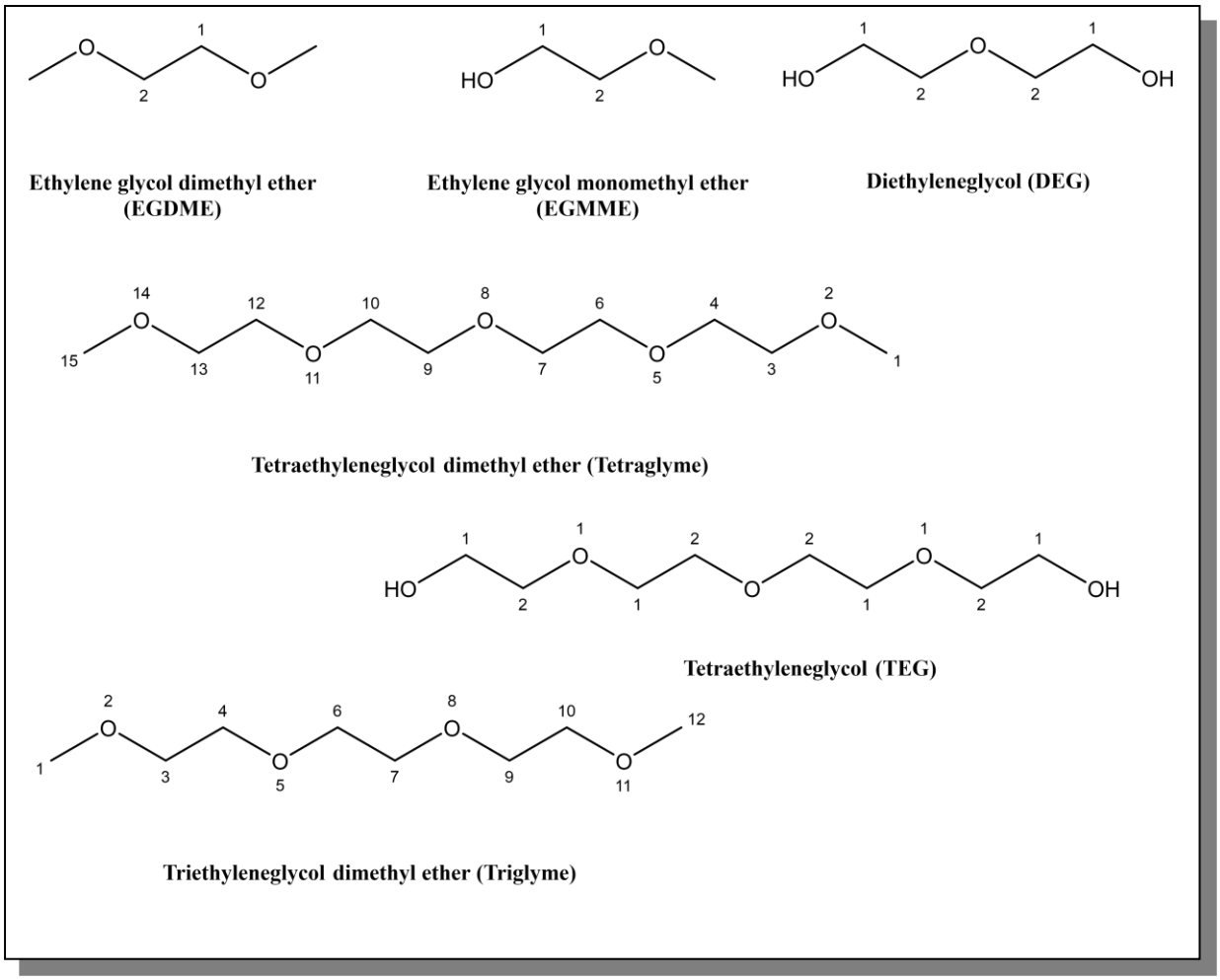


Figure 2.11 *Structure of ethers used to prepare SILs.*

Table 2.6 shows the ability of salts to form homogeneous solutions in different mole ratios and using different ethereal solvents. As shown in Table 2.6, lithium salts are more soluble in the ethereal solvents than sodium. This could be attributed due to lithium being smaller in size than sodium which gives it a higher charge density and stronger interactions with the ethers. Furthermore, being a good hydrogen bond acceptor, acetate anion based SILs appeared to be poor SILs than chloride and nitrate anions based salts as explained in the published work.[118] It has also observed that the hydrates are more soluble, with the exception of LiCl and DEG, than the anhydrous salts. There might be one of the possibilities of that role of the water is assisting in the acetate anion in coordinating with the solvents.

The dissolution of the salts also varies by varying the alkyl chain length of the solvents used. The longer alkyl chain solvents such as tetraglyme and tetraethylene glycol (TEG) do not lead to the solubility of the salts compared to shorter alkyl chain solvents such as ethylene glycol monomethyl ether (EGMME) and diethylene glycol (DEG).

Table 2.6 *Solubility of salts in different coordinating solvents at different molar ratios. Blue shaded regions indicate that salt is soluble.*

Salts	**Salt:Solv**	**EGMME**	**EGDME**	**Triglyme**	**Tetraglyme**	**DEG**	**TEG**
LiOAc. $2H_2O$	1:1						
	3:7						
LiOAc	1:1						
	3:7						
LiCl	1:1						
	3:7						
NaOAc $.3H_2O$	1:1						
	3:7						
NaCl	1:1						
	3:7						
$NaNO_3$	1:1						
	3:7						

2.7. Reverse deep eutectic solvents (RDESs)

Conventional DES are typically composed of an organic molecule as a HBD and a salt as a HBA. Towards the aim of developing solvents with high β values, the opposite approach has been explored which will be termed reverse deep eutectic solvents (RDESs). Rather than a typical HBD-HBA interaction, the purpose was for the HBA to interact with the cation of the salt through effectively ion-dipole type forces. So, this is a bit different concept compared to a typical 'HBD'. RDESs can be developed by exchanging the role of HBD with HBA, i.e., using a salt as the 'HBD' and organic molecule as the 'HBA'. The selection of salts was based on the hydrogen bond accepting ability of anion of the salt. The structures of salts and HBAs are displayed in Figure 2.12.

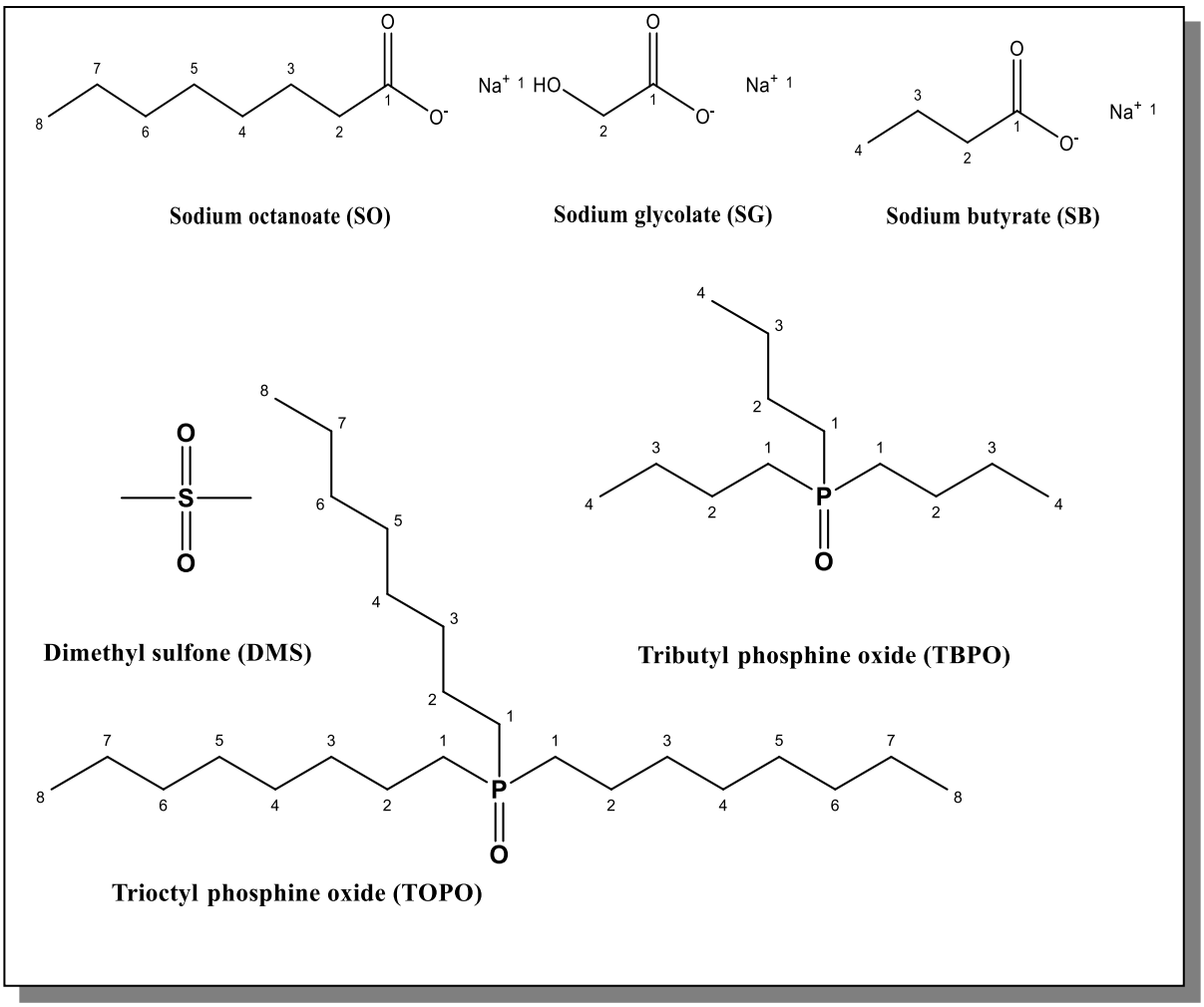


Figure 2.12 *Structure of HBAs and salts used to prepare RDESs.*

The results of forming the RDESs are represented in Table 3.8. According to the data recorded, it had been observed that the resulting melting temperatures of some RDESs were slightly lower than the melting points of the HBDs such as TBPO:SB (1:2) and TBPO:SG

(1:2). This is despite the much higher melting point of the salt component, indicating the formation of a eutectic mixture. For example, a melting point that changes linearly with composition would lead to an expected melting point of 160 ℃ for DMS:SG (1:1) or 180 °C for DMS:SB (1:1) while the actual melting points observed were considerably lower, demonstrating evidence of significant melting point depression.

After all the measurements were analysed, it was concluded that since none of the RDESs were room temperature liquids and there was limited evidence for a significant reduction in melting point, this idea was not pursued any further.

Table 2.7 *Composition and melting point of lowest melting mixtures of salt and each HBD.*

HBA	**Lit. m.p. (℃)**	**Salt**	**Lit. m.p. (℃)**	**Salt : HBA**	**Salt : HBA m.p. (℃)**
TOPO	50 – 52	GuCl	179 – 185	1:1	46.8
TOPO		SO	245	1:2	49.7
TBPO	64 – 69	SB	250 – 253	1:2	47.8
TBPO		SG	210 – 218	1:2	45.6
DMS	107 – 109	SG		1:1	122.7
DMS		SB		1:1	>150

2.8. Summary

In summary, over 20 different HBD combinations were screened with GuCl, with new room temperature liquids based on mixtures with LA, Gly and LevA discovered and characterised. The GuCl:LA mixture was found to possess comparable if not slightly improved hydrogen bond donating abilities to HFIP based on the use of polarity probes. One of these probes involved the use of pyridine-*N*-oxide, a different approach to the traditional method for determining α that was used to avoid complications associated with the possible protonation of the probe molecule. Moreover, in terms of the viscosity measurements of the selected DESs under increasing shear stress, it was concluded that the viscosity of the eutectic mixtures decreased in the order GuCl:Gly (0.107 Pa s) > GuCl:LevA (0.06 Pa s) > GuCl:LA (0.038 Pa s) > GuCl:Acet (0.035 Pa s) DESs with comparable viscosities to which was consistent with the results obtained by using the rolling ball method where direct comparison was possible. These viscosities are lower than the viscosities of ChCl-based DESs following the decreasing

order : ChCl: malonic acid (1.124 Pa s) > ChCl:urea (0.75 Pa s) > ChCl: glycerol (0.26 Pa s) > ChCl:ethylene glycol (0.037 Pa s).[18]

These DESs were further used for the solubilisation of FF under the hypothesis that there could be strong hydrogen bonding interactions between the carbonyl group of FF and the guanidinium cation incorporated in each DESs to help drive its solubility. It was concluded that although the GuCl:LA DES had the highest α value (1.82), it appeared to be one of the least effective solvents, after GuCl:Gly DESs with no solubility, among the other DESs for FF due to the strong self-association of the GuCl:LA DES. In contrast, LevA-based DES displayed higher solubility of FF than other DESs because of the weaker interaction between LevA and the anion resulting in more available to interact with the FF.

Additionally, the synthesis of some other classes of eutectic mixtures were also tried, namely, SILs and RDESs. For SILs, it was determined that hydrated LiOAc emerged as the most suitable in solubilising coordinating solvents (EGMME, EGDME and DEG) than other salts due to smaller size of lithium and potential role of water in helping solubilise the salt. Unfortunately, very few anhydrous salts with strong HBA anions showed the potential to form stable SILs.

For the RDES, several salts based on carboxylate anions and HBA featuring phosphine oxide and sulfone groups were explored. While some RDES displayed slightly reduced melting points, none led to room temperature liquids or exhibited dramatic reductions in melting points so it was concluded that it might not be a good approach for synthesising eutectic mixtures with strong hydrogen bond accepting abilities.

3. Effect of Ionic Liquid (IL) structure on the outcome of extractions of *Pinus Radiata* bark

As discussed in the Introduction, bark consists of a complex mix of extractives, lignin, cellulose and hemicellulose which provide possible opportunities for the use of waste bark as a valuable feedstock for future biorefineries. These applications require the efficient separation and recovery of key bark components. ILs have been widely explored as replacements for common organic solvents, including for the extraction of bio-molecules.[120] ILs have the ability to have their properties tuned based on the structure of the solvent selected to allow optimal solvents to be designed for specific extraction processes. This makes it possible to design ILs that enable the targeted dissolution of specific components of bark.[93] Moreover, ILs with sufficiently long alkyl chains can also exhibit long-range amphiphilic nanostructures comprised of polar and non-polar domains that can be controlled by the choice of the anion and cation.[121] This may provide an additional influence on the fractionation of components of the bark. Despite significant work having been performed exploring the extraction of wood components by ILs, relatively little research has been done exploring the key components of bark. While some work has been reported of the extraction of few species of barks by using organic solvents, [64,91,122,123] extracting and separating the key constituents of bark using ILs has not been widely explored yet and hence the role of the IL ions on the outcomes of these extractions remains relatively unknown.

The aim of this chapter is the investigation of the influence of IL structure on the fractionation of bark to help inform IL design for more targeted bark extractions. The major components of lignocellulosic biomass such as cellulose, hemicellulose and lignin are found in both bark and wood. The components such as cellulose and hemicellulose are barely soluble in most conventional molecular liquids due to their supramolecular structures consisting of strong hydrogen bonds which has further restricted their separation and utilization. Moreover, lignin is typically insoluble using conventional solvents as it is an irregular aromatic polymer. In this context, ILs as potential solvents for disrupting the hydrogen bond network of cellulose have attracted a great deal of interest.[93] The solubility of cellulose tends to increase with increasing hydrogen bond accepting strength of the IL solvent. The effect of the anionic and cationic

components of ILs has also been investigated on lignin solubility.[124] The results indicated that role of the anion is not significant in the overall solubilisation of lignin relative to the cation although a minimum hydrogen bond basicity (β) is required for lignin dissolution in ILs. Meanwhile, the cation has a much more significant effect on lignin solubility with a general trend of increasing solubility with hydrogen bond donating (α) ability of the cation. Other, more subtle, effects may also influence lignin solubility, for example, ion-π interactions between the planar cation $[C_4C_1im]^+$ and the planar aromatic portion of lignin. Suberin is a biopolymer mainly found in cork and bark, rather than wood. Suberin is difficult to extract from these sources due to its complex hydrophobic structure. Isolation and depolymerisation of suberin is traditionally a laborious process requiring harsh chemical processes involving extensive ester bond cleavage, normally attained by alkaline treatment.[74] The extraction of suberinic material from cork using ILs has been examined,[98] although the influence of IL ion selection on the effectiveness of this extraction has not been explored. In addition, there have not been any studies performed on the extraction of suberin from *Pinus radiata* bark using ILs. Hence the influence of IL ions on the extraction efficiency and outcomes of *Pinus radiata* bark will help shed light on the interplay between the composition of this complex substrate and the ability of ILs to be designed to selectively target key bark components.

3.1. Selection of IL ions

Hydrogen-bonding interactions in ILs are likely to a play pivotal role for the fractionation of different components of bark, as discussed above. Moreover, the amphiphilic nanostructure of ILs may have an effect on the segregation of the bark components due to the greater propensity to extract hydrophobic components. To examine the effect of these factors, the nature of the solvents used was systematically varied to explore hydrogen bonding and amphiphilic nanostructure effects both for the extraction of bark and the separation of extracts. For this reason, ILs were selected to contain ions across a range of hydrogen bond donating and hydrogen bond accepting abilities. In particular, cations such as 1-alkyl-3-methylimidazolium $[C_nC_1im]^+$, 1-alkyl-1-methylpyrrolidinium $[C_nC_1pyrr]^+$, and 1-alkylguanidinium $[C_ngun]^+$ were selected on the basis of their increasing hydrogen bond donating strength. Similarly, the anions explored include bis(trifluoromethanesulfonyl)imide $[NTf_2]^-$, dicyanamide $[N(CN)_2]^-$, and dimethylphosphate $[Me_2PO_4]^-$ in increasing order of their hydrogen bond accepting strength. Additionally, different alkyl chain lengths (C_4 and C_8) were used for the cations of the ILs to explore the effect of the alkyl chain length, in particular the role, if any, of amphiphilic

nanostructure on the reaction outcomes. The Kamlet-Taft parameters (α, β, and π^*) of the ILs chosen, where they have been reported, are shown in Table 3.1 and the structures of cations and anions selected are shown in Figure 3.1. Table 3.1 highlights that the α and β trends across the cations and anions are consistent with the anticipated order of hydrogen-bond donating and accepting abilities of the ions chosen. While the K-T parameters of the 1-alkylguanidinium cation has not been studied, the high α values observed for the GuCl DES in Chapter 2 demonstrate that these cations are likely to be the most effective hydrogen bond donors.

Table 3.1 *Kamlet-Taft parameters, using the dye set Reichardt's dye, N,N-diethyl-4-notroaniline and 4-nitroaniline.*

ILs	**Abbreviation**	**α**	**β**	**π^***	**Ref**
1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl) imide	$[C_4C_1im][NTf_2]$	0.72	0.24	0.90	[109]
1-butyl-3-methylimidazolium dicyanamide	$[C_4C_1im][N(CN)_2]$	0.54	0.60	1.05	[88,109]
1-butyl-3-methylimidazolium dimethyl phosphate	$[C_4C_1im][Me_2PO_4]$	0.45	1.13	0.98	[88,109]
1-butyl-1-methylpyrrolidinium bis(trifluoromethanesulfonyl)imide	$[C_4C_1pyrr][NTf_2]$	0.57	0.23	0.87	[109]
1-butyl-1-methylpyrrolidinium dimethyl phosphate	$[C_4C_1pyrr][Me_2PO_4]$	0.24	1.14	1.02	[109]
1-octyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide	$[C_8C_1im][NTf_2]$	0.60	0.29	0.96	[109]
1-octyl-3-methylimidazolium dicyanamide	$[C_8C_1im][N(CN)_2]$	0.43	0.71	0.97	[125]
1-octyl-1-methylpyrrolidinium bis(trifluoromethanesulfonyl)imide	$[C_8C_1pyrr][NTf_2]$	0.80	0.08	0.73	[109]

In addition, triethylammonium hydrogensulfate ([TEA][HSO_4]), was also examined for bark extraction. This IL has been explored for biomass processing and is effective at extracting lignin.[126] This could be due to the strongly hydrogen bond donating cation having an N-H hydrogen bond donor group as well as the acidity of the solvent.

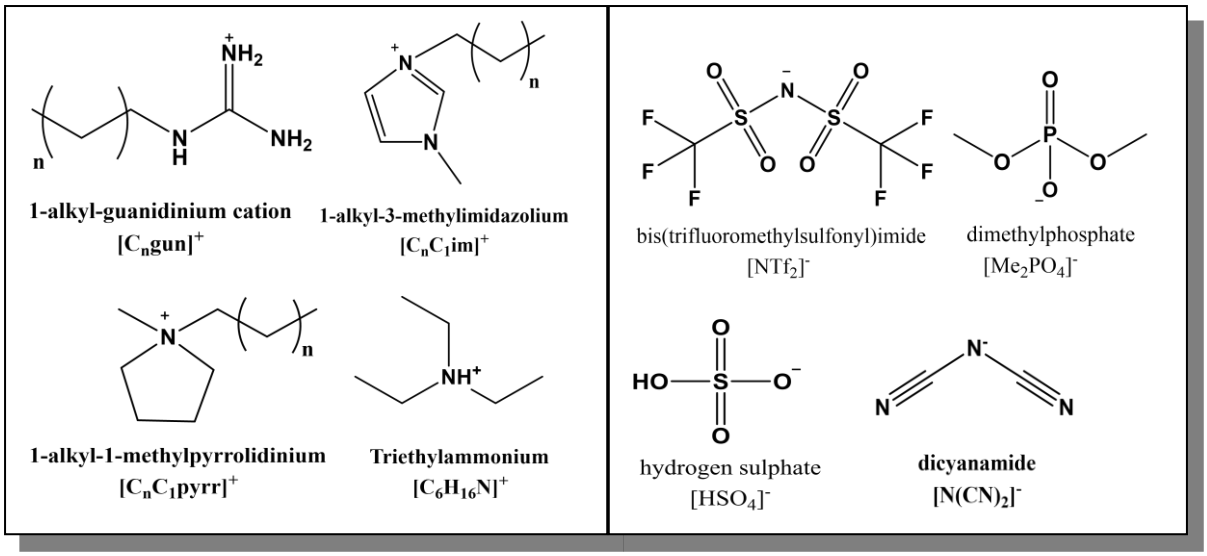


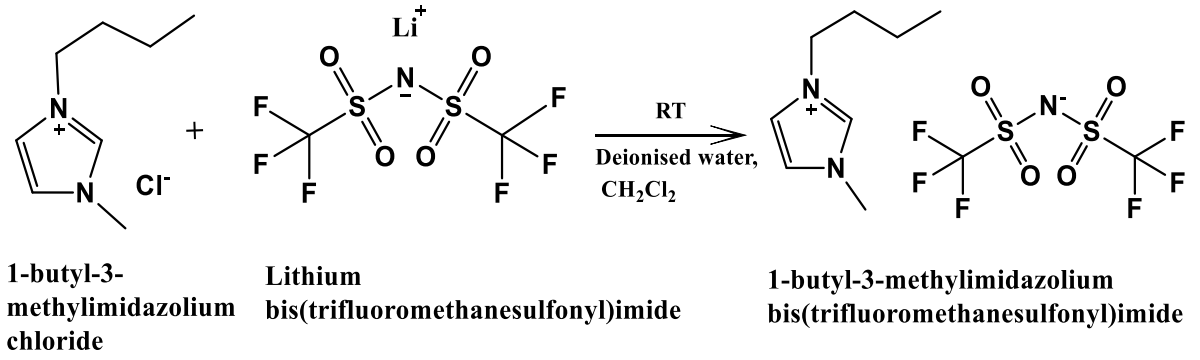
Figure 3.1 *The structures of cations and anions used for the ILs explored for bark extraction*

In terms of the relative hydrogen bond accepting and donating capability of cations and anions, [C_4C_1im]-ILs and [C_8C_1im]-ILs possess intermediate hydrogen-bonding ability while [C_4C_1pyrr]-ILs and [C_8C_1pyrr]-ILs possess relatively weak hydrogen bonding ability. The guanidinium-based cation [C_4gun] and [C_8gun] has strong hydrogen bond acidity. The hydrogen bond accepting ability of anions increases as follows [NTf_2] < [$N(CN)_2$] < [Me_2PO_4].

3.2. Synthesis of ILs

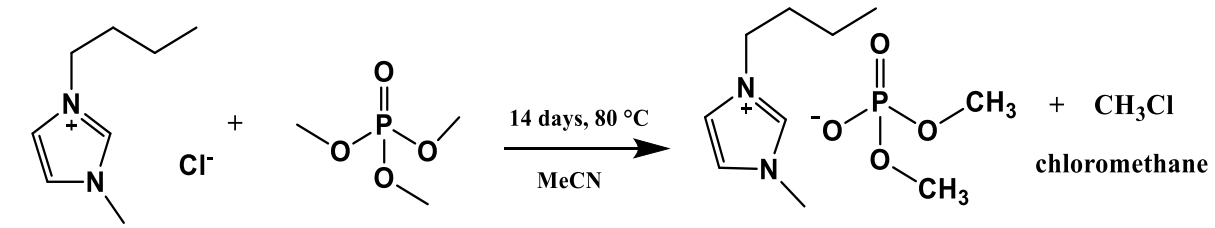
As discussed, a range of ILs were selected to screen cations and anions of differing hydrogen-bonding abilities to develop a model of the effect of ion selection on the extraction of key bark components. A total of 23 ILs or IL intermediates were synthesised with 15 being used for the extraction of molecules from the *Pinus radiata* bark. Some synthesised ILs were not being used for the extraction of molecules due to being high melting temperature salts for example, [C_4gun] ILs were all solid except [NTf_2] anion based ILs. Halide salts were firstly prepared as intermediates for the preparation of ILs based on the $[NTf_2]^-$ and $[N(CN)_2]^-$ anions but they themselves were not used for the extraction of bark as they are typically highly viscous or high melting salts. The same reaction conditions were used for both C_4 and C_8 alkyl chain based ILs for a given cation and anion. The substitution of the anion involved the use of a silver salt to precipitate AgBr in the case of $[N(CN)_2]^-$ or the separation of the undesired LiBr by washing the hydrophobic $[NTf_2]^-$ ILs with water. ILs based on the $[Me_2PO_4]^-$ anion were prepared by reacting trimethyl phosphate with the chloride salt of the appropriate cation. As the reaction

conditions for all ILs bearing the same anion were very similar, representative reaction schemes (4 - 6) of the ILs prepared are shown below.



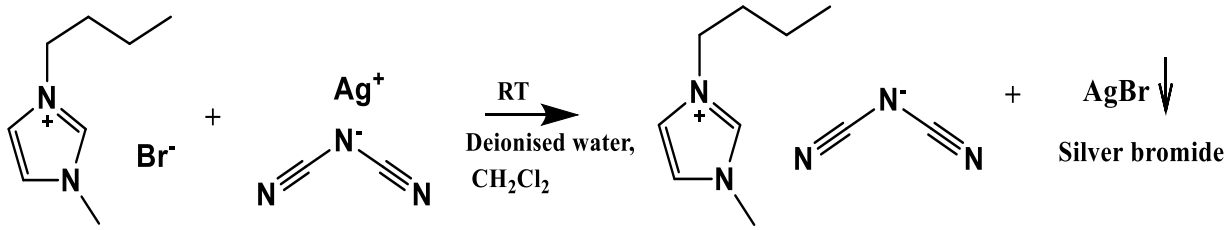
1-butyl-3-methylimidazolium chloride | **Lithium bis(trifluoromethanesulfonyl)imide** | **1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide**

Scheme 4 *The synthesis of [C_4C_1im][NTf_2].*



1-butyl-3-methylimidazolium chloride | **Trimethyl phosphate** | **1-butyl-3-methylimidazolium dimethyl phosphate** | **chloromethane**

Scheme 5 *The synthesis of [C_4C_1im][Me_2PO_4].*



1-butyl-3-methylimidazolium bromide | **Silver dicyanamide** | **1-butyl-3-methylimidazolium dicyanamide** | **Silver bromide**

Scheme 6 *The synthesis of [C_4C_1im][$N(CN)_2$]*

3.3. Effect of pre-drying on the composition of bark

The bark samples obtained can vary significantly in their moisture content which can negatively impact the reproducibility of extractions. For this reason, bark was pre-dried before each attempted extraction to ensure similar moisture contents were observed. This process involved first heating under a flow of nitrogen at 50 °C for 2-3 hours and then in *vacuo* at 50 °C for 16 h. The effect of pre-drying the bark on its composition was determined to examine if this process significantly changed the structure of the bark. Figures 3.2 depicts a comparison of the IR spectra of the bark before and after the drying process, respectively and Table 3.2 lists the literature infrared assignments of the main components of bark.

Figure 3.2 highlights that the drying process reduces the water content of the bark, as would be expected. This is shown by the notable decrease in the peak at approximately 3360 cm^{-1} which corresponds in part to the O-H stretching frequency of water and a sharpening of the peak around 1610 cm^{-1} that includes the O-H bend of water around 1640 cm^{-1}. Outside of these changes there are no significant changes to the IR spectrum of the bark which highlights that its composition is largely unaffected by the drying process.

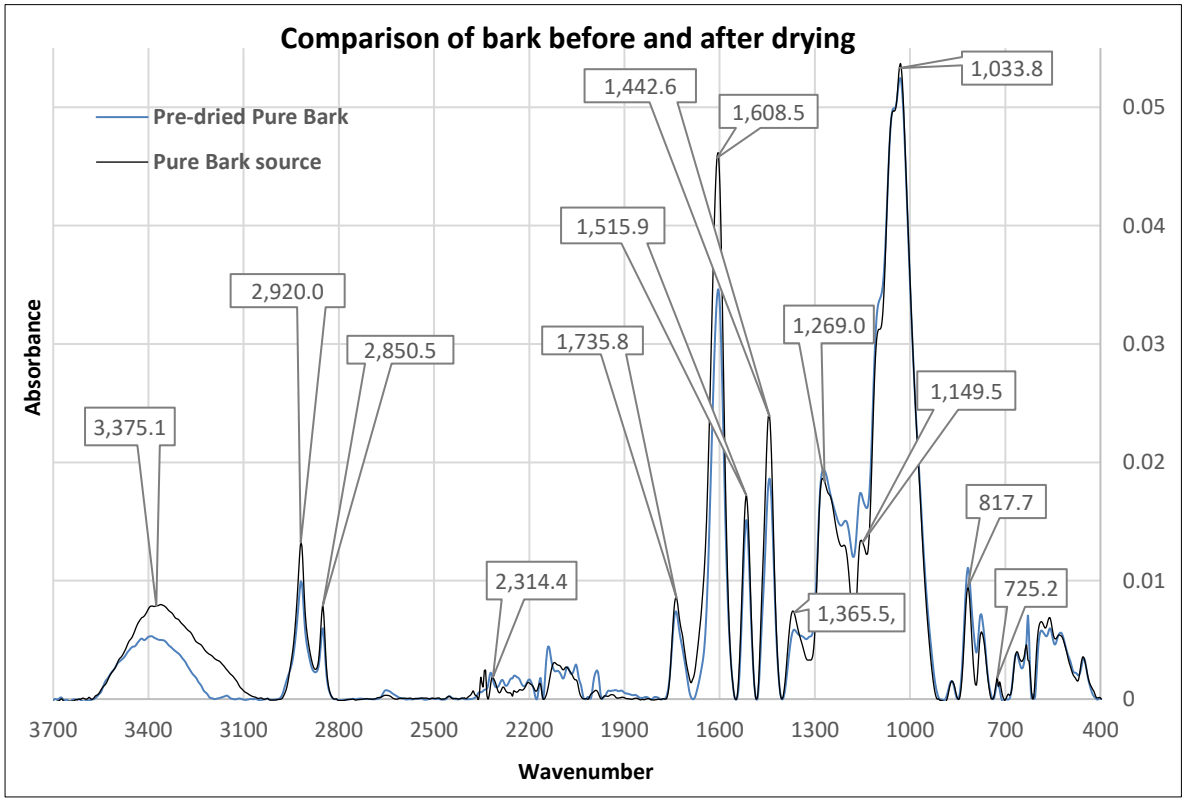


Figure 3.2 *IR of pure bark vs bark after drying under high vacuum at 50 °C overnight.*

Table 3.2 *Infrared assignments of bark components.*

Wavenumber(cm^{-1})	Description	Components
3321 or 3425 or 3464	O-H sym. stretch	Water, alcohol or carbohydrate hydroxyl [127–130]
2933	CH- and -CH_2	Mainly from tannins[131]
2919	CH- aliphatic stretch (C_{16}-C_{30})	Waxes or suberin[74,98,128,129,132]
2850	CH- aliphatic stretch (C_{16}-C_{30})	Waxes or suberin[98,127,129,130,132]
1735	C=O- stretch (ester group)	Mainly from suberin[98,127–130,132]
1600 – 1610	C=C stretch or C=O stretch (aromatic region)	Tannins or lignin[98,127,128,130,133]
1510 – 1515	C=C stretch (aromatic region)	Tannin or lignin[98,127,129,130,133]
1230 – 1260	C-O-C stretch (ester group)	Mainly from suberin[98,127]
1260 – 1270	C-O-C stretch (ester group)	Lignin (guaiacyl type)[127,128,134]
1361 or 1365	CH- sym. Deformation (aliphatic chain)	Mainly from suberin [128,129]
1437 or 1464	CH- asym. deformation (aliphatic chain)	Mainly from suberin [127,129]
1157	CO- asym. stretch (ester group)	Cellulose, hemicellulose or suberin[127,132,134]
1101 or 1036	CH, CO deformation (carbohydrate)	Polysaccharides[127,130]
817 or 858 or 867	C=C (aromatic region)	Lignin (guaiacyl type)[130]
725	$R_1CH=CHR_2$ groups	Suberin[127,132]

3.4. Extraction of bark using ionic liquids (ILs) and characterization of extracts

3.4.1. UV-Vis Data

The UV-Vis was the initial approach adopted in terms of characterising the extracts as it enabled a rapid determination of the extent of extraction and allowed extraction conditions to be developed. For the analyses, a suspension of raw bark pieces in an IL was used with the variation of time and temperature conditions. The bark was not pre-dried for these initial experiments.

3.4.1.1. Time-dependent UV-Vis results

Time-dependent UV-Vis studies were initially performed in $[C_4C_1im][NTf_2]$ at room temperature to assess the rate and extent of extraction, as shown in Figure 3.4 Undried bark pieces (0.1 g) were weighed and added to 1 mL of $[C_4C_1im][NTf_2]$. UV-Vis spectra of the IL were obtained every 5 min for the first 1 h then at longer intervals up to 2 days. The baselines were normalised to the 800 nm absorbance to correct for light scattered by bark chips suspended in the liquid.

However, the rate of increase decreased notably after 1 h which suggested the increase of duration of absorption by the given mixture. Therefore, the absorption course was increased for 2 more days. Due to the increase in absorption with increasing time intervals it was evident that the ILs were extracting conjugated components from the bark over the course of 2 days, although the majority of this extraction appeared to take place in the first hour. Given the intense UV absorptions it was difficult to assign the UV absorption to either lignin or other polyphenolics components as both have the complex structures constituting highly substituted aromatics.

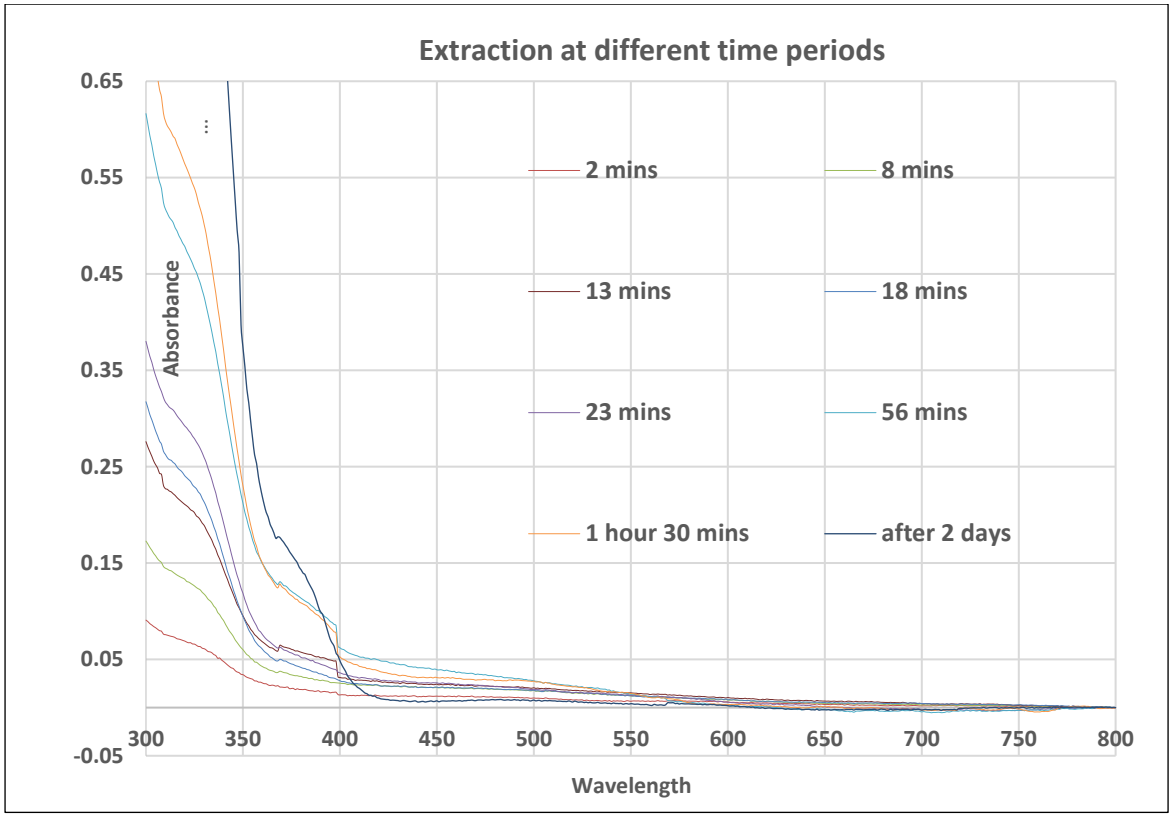


Figure 3.3 *Extraction of bark pieces by NTf_2-IL at different time periods at 25 °C.*

3.4.1.2. Temperature-dependent UV-Vis results

Temperature-dependent UV-Vis studies were performed in $[C_4C_1im][NTf_2]$, $[C_4C_1im][Me_2PO_4]$, and $[C_4C_1im][N(CN)_2]$ to observe the effect of temperature on the extraction of bark.

Figure 3.4 depicts the UV-Vis after 1 h of extraction at 25 ℃ which clearly demonstrates that $[C_4C_1im][NTf_2]$ was less effective at extracting UV-active components from bark as compared to $[C_4C_1im][Me_2PO_4]$ and $[C_4C_1im][N(CN)_2]$ ILs. This was also seen at 50 ℃, as is evident in Figure 3.5. This Figure also demonstrates that there is increased extraction efficiency with elevated temperature as the absorbance in the UV region notably increased. The greater visible absorbance of the $[C_4C_1im][N(CN)_2]$ IL extract might relate to some unusual behaviour of the IL itself which will be explained in a later section.

Moreover, Figure 3.6 illustrates the UV-Vis after 24 h of extraction at 100 ℃ using $[C_4C_1im][Me_2PO_4]$ and $[C_4C_1im][N(CN)_2]$ ILs which demonstrates that they all absorbed light across the spectrum intensely. A more accurate UV-Vis measurement requires dilution of the

sample or the use of smaller path lengths which wasn't explored. It is evident that the higher temperature and longer time conditions (100 ℃ for 24 h) used led to significantly increased extraction of UV-active compounds such as lignin, tannins or other polyphenolics which is likely to indicate that a greater proportion of the bark was extracted overall.

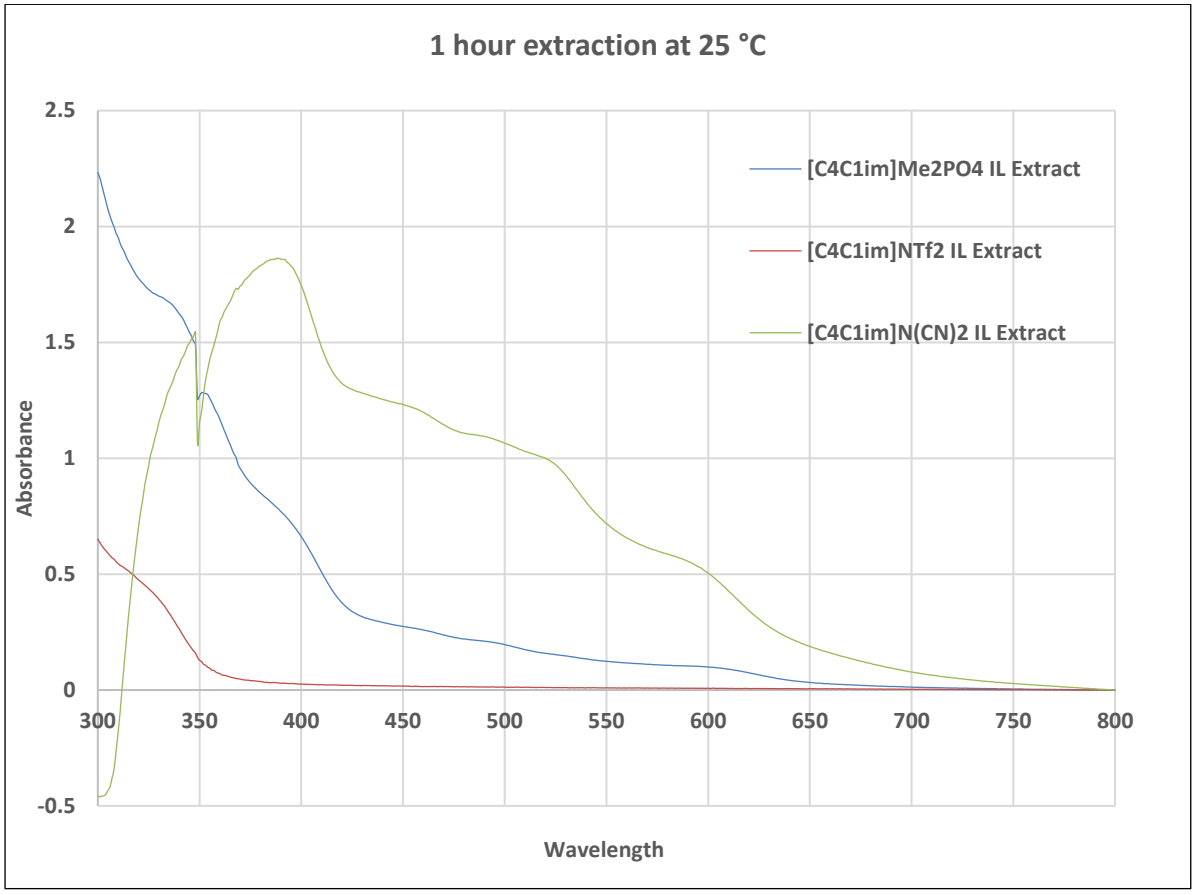


Figure 3.4 *UV-Vis spectrum of given ILs at 25 °C for 1 h.*

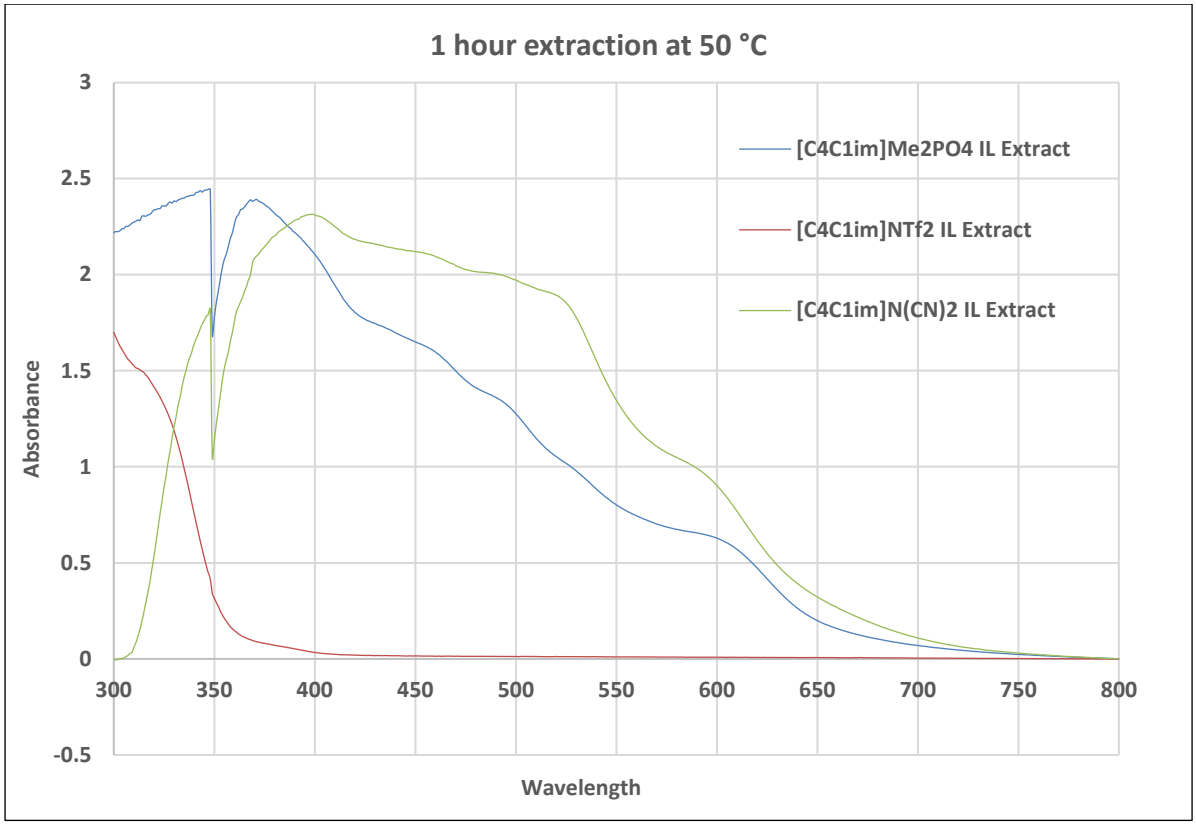


Figure 3.5 *UV-Vis spectrum of given ILs at 50 °C for 1 h.*

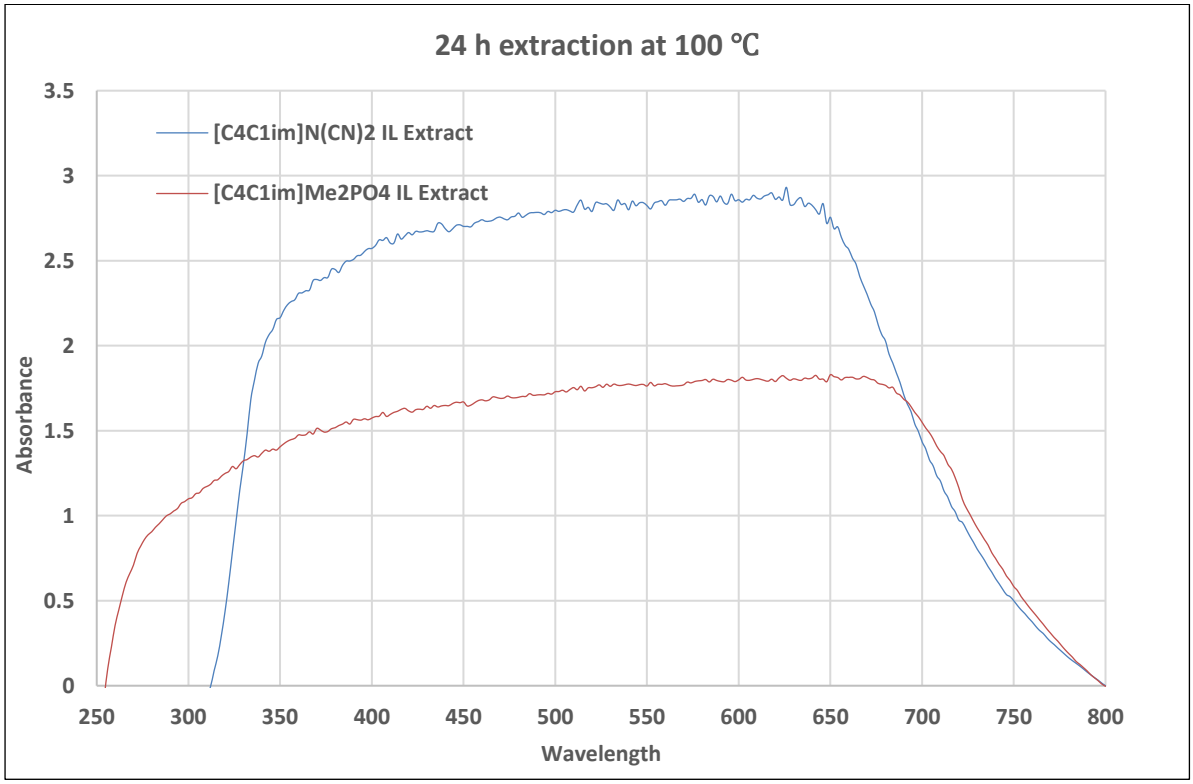


Figure 3.6 *UV-Vis spectrum of $[C_4C_1im][Me_2PO_4]$and $[C_4C_1im][N(CN)_2]$IL extracts at 100 °C for 24 h.*

3.4.2. NMR Analysis of bark extracts

The UV-Vis experiments highlighted that there were bark components being extracted by the $[C_4C_1im]^+$ cation based ILs. To gain insight into the composition of these extract, attempts were made to characterise these bark-ILs mixtures further using solution NMR.

3.4.2.1. Effect of ILs in bark by ^1H and ^{13}C spectrum

The ^1H NMR and ^{13}C NMR of $[C_4C_1im][Me_2PO_4]$ were determined after its use for the extraction of bark at 25 °C and 50 °C for 1 hour. These NMR spectra were compared with the NMR spectra of the pure IL. No extra peaks were present in any region of the spectrum depicting that the concentration of the compounds extracted from bark under these conditions were too low to be reliably detected by NMR. This was true of the other ILs too, as shown in the appendix (Figures A1 and A2).

Similarly, Figures 3.7 and 3.8 represent the zoomed in ^1H spectrum of pure IL and bark-IL at 50 °C for 1 h and at 100 °C for 24 hr as no additional peaks could be seen by scale down spectrum. Again, no extra peaks were present in any region of the spectra even under 100 °C for 24 h condition where a greater proportion of the bark was extracted.

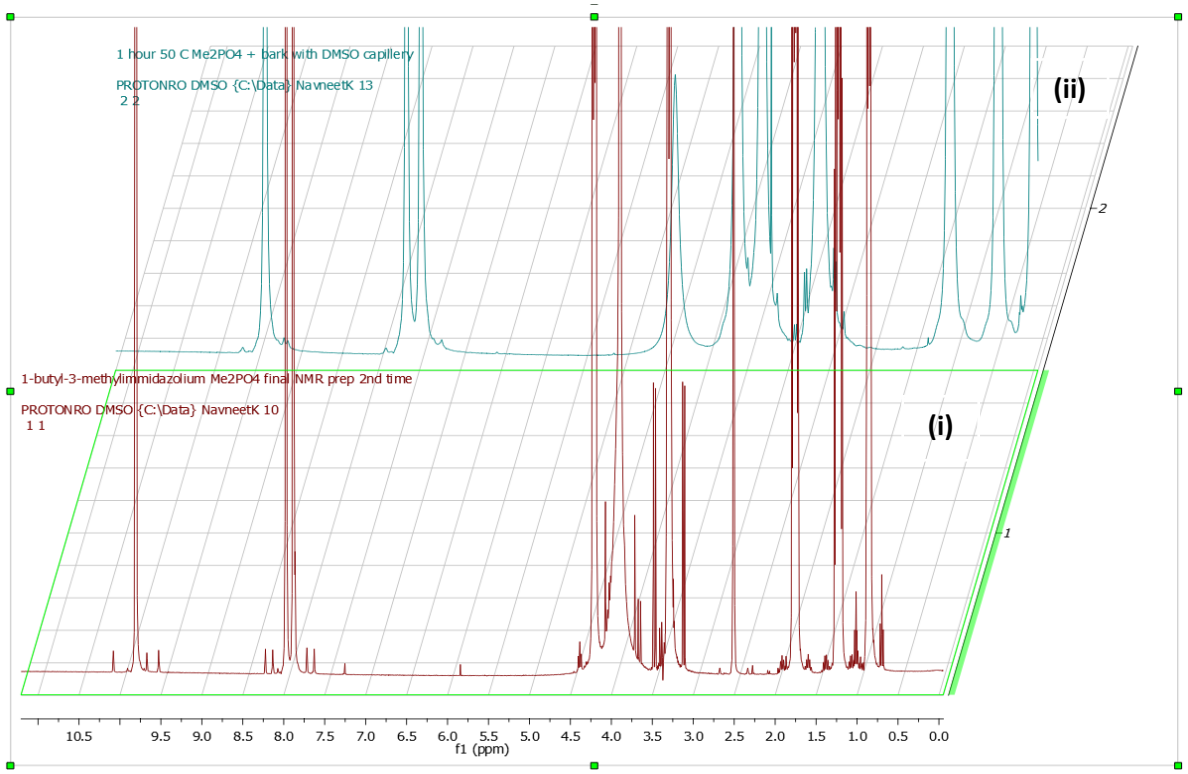


Figure 3.7 *^1H NMR of pure $[C_4C_1im][Me_2PO_4]$ shown in (i) in comparison with bark at 50 °C shown in (ii)*

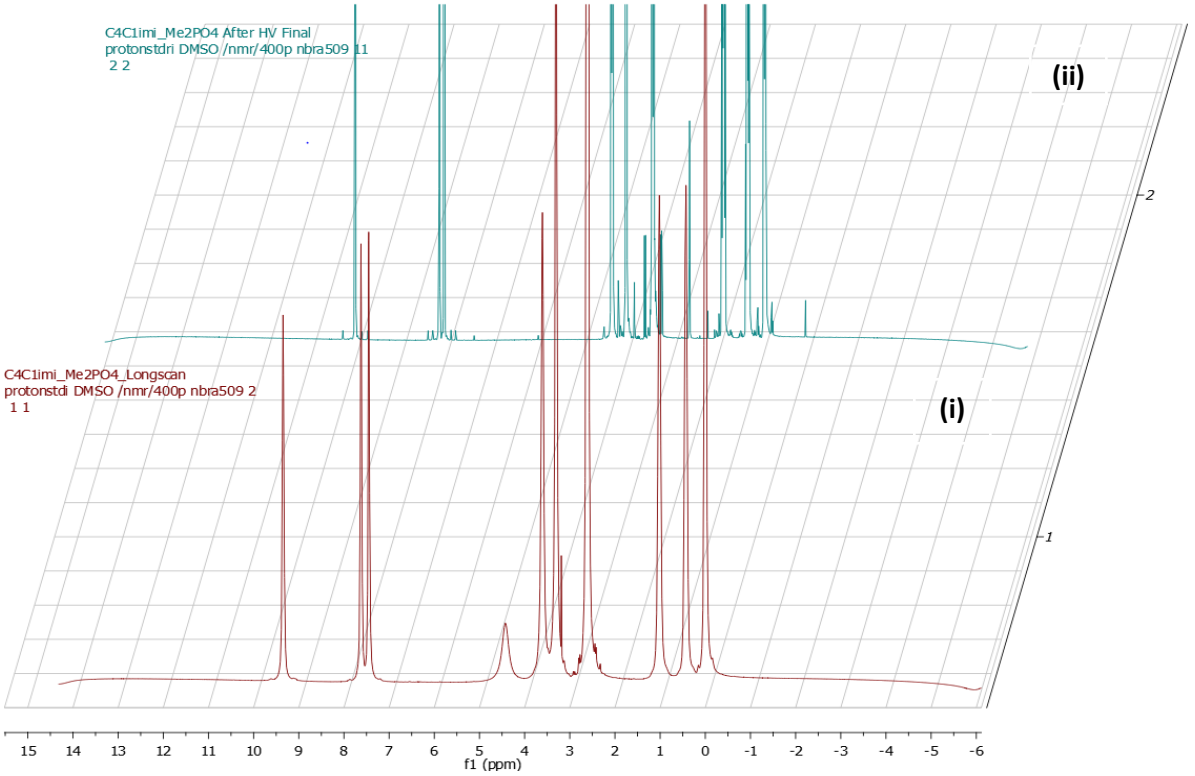


Figure 3.8 *^{1}H NMR of pure $[C_4C_1im][Me_2PO_4]$ shown in (ii) in comparison with bark at 100 °C shown in (i).*

The components observed by UV-Vis under these reaction conditions were not visible by solution NMR (DMSO-d_6 capillary as a lock). The lack of identified peaks may be either due to the low intensity of the NMR peaks or due to key peaks overlapping with the intense IL signals. Given the complexity of these mixtures it is more likely to be due to the low concentration of individual components of the extract as the likelihood that all the relevant signals overlap with the IL is low.

3.4.3.Extraction yield of bark extracts

The next step was to determine the extraction yield of bark. Bark extractions with all 15 ILs and 2 DESs at 100 °C for 24 h had been performed. Pre-dried bark was used with the extraction performed over 24 h at 100 °C as this condition appeared to provide the greatest extent of extraction of those that had been explored. The extractions were scaled up slightly to 0.5 g of pre-dried bark in 10 mL of ILs, to enable more accurate quantification of the proportion of bark extracted and the results are represented in Table 3.3. Both the bark before and after extraction was dried and weighed to ensure a quantitative result. The extraction yield was measured by weighing the amount of dried extract recovered (after extraction process) from the bark and compared with the initial amount of the dry bark (before extraction process).

As some of the ILs such as $[C_4C_1im][NTf_2]$, did not display substantial extraction results therefore, the extraction method was performed only once for such ILs and also the error was not reported. While most ILs displayed fairly reliable extraction results, two solvents provided outcomes that were not readily reproducible, these were $[C_8C_1im][Me_2PO_4]$ and the GuCl:LA DES. The reason behind the inconsistent extraction yields of $[C_8C_1im][Me_2PO_4]$ the IL is likely to be the viscous and gel-like nature of the IL after extraction which makes it difficult to separate from the bark and likely leads to mass transfer limitations. Similarly, for the GuCl:LA DES, the error was again 19, the reason behind is unclear but a possibility is the decomposition of extracts within the solvent as it appeared relatively viscous IL extract compared to some of the others.

Table 3.3 *Extraction yield of bark with ILs under reaction condition of 24 h at 100 °C.*

ILs used	**Yield (%)**
$[C_4C_1im][NTf_2]$	4.2
$[C_4C_1im][N(CN)_2]$	27.6 ± 1.9
$[C_4C_1im][Me_2PO_4]$	33.3 ± 2.4
$[C_4C_1pyrr][N(CN)_2]$	37.4
$[C_4C_1pyrr][Me_2PO_4]$	70 ± 3
$[C_8C_1im][N(CN)_2]$	32.4
$[C_8C_1im][Me_2PO_4]$	48 ± 19
$[C_8C_1pyrr][N(CN)_2]$	9.6
$[C_8C_1pyrr][Me_2PO_4]$	39 ± 4
$[C_4gun][NTf_2]$	14.3
$[C_8gun][NTf_2]$	7.3
$[TEA][HSO_4]$	75 ± 5
GuCl:LA	51 ± 19
GuCl:Gly	38 ± 5

The Table 3.3 highlighted that the extraction is more efficient in $[Me_2PO_4]$-based ILs than $[N(CN)_2]$- and $[NTf_2]$- based ILs, consistent with the other characterisation methods used so far. There were also very significant colour changes observed in the ILs after the overnight extraction with darker solutions observed for those containing good hydrogen bond accepting anions (Me_2PO_4), providing further suggestion of increased extraction efficiencies by these

ILs. Being a strong hydrogen bond acceptor, the $[Me_2PO_4]^-$ anion may interact either with the hydroxyl group of carbohydrate/lignin or extractives respectively, as shown in the Figure 3.9. Moreover, the lowest extraction yields were observed for the $[NTf_2]^-$ based ILs even in the presence of a good hydrogen bond donor cation such as $[C_4gun]^+$. Although some undissolved bark was observed in the $[C_4gun][NTf_2]$ and $[C_8gun][NTf_2]$ IL extracts and colour of the IL extract was changed from colourless to a light brown which could be sign extracting fractions of bark but extraction yield was low for these ILs than others. These results highlight the critical role of the anion on the extraction efficiency.

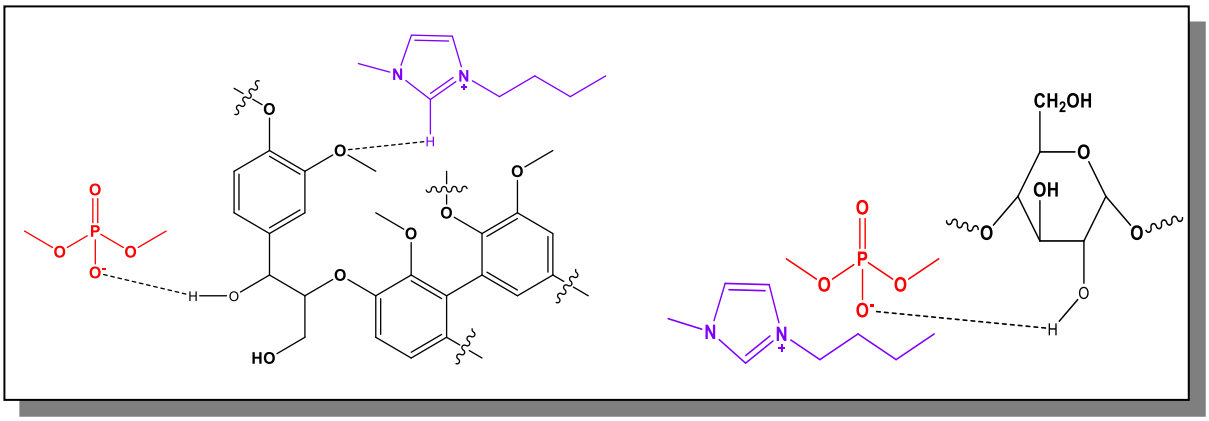


Figure 3.9 *The proposed interaction of cation $[C_4C_1im]^+$ and anion $[Me_2PO_4]^-$ with lignin and carbohydrates.*

The effect of cation on the extraction yield was also examined. Table 3.3 demonstrates that the $[C_nC_1im]^+$ cation tends to reduce the bark extraction yield compared to $[C_nC_1pyrr]^+$ and $[C_ngun]^+$, at least for n = 4, although the relative order of the $[C_nC_1im]^+$ and $[C_nC_1pyrr]^+$ cations changes when n = 8. The proposed interaction of one of the cations ($[C_4C_1im]^+$) interacting with the lignin has been shown in the Figure 3.9. Out of all the cations, pyrrolidinium cation based ILs especially, $[C_4C_1pyrr]$-, displayed the highest extraction yield than the other ILs. This could be attributed that the hydrophobic nature of the cation allows it to extract not only some lignin, carbohydrate and tannin fragments of the bark but also the hydrophobic components of the bark such as suberin or waxes. It may also be due to the reduce cation-anion interactions which will help facilitate stronger anion-solute interactions and reduced solvent association which will improve the solubility of solutes. Moreover, the effect of increasing the alkyl chain of the cations the extraction yield slightly varies although typically leads to a

decrease in extraction yield. This highlights that smaller, more polar ILs tend to be more favourable for extracting bark components at least in terms of quantity (but maybe not based on selectivity) but it is very hard to justify clearly yet given the large error for $[C_8C_1im][Me_2PO_4]$.

Regarding DESs, the extraction yield of the GuCl:LA and GuCl:Gly DESs were comparable to those of the $[C_4C_1im][N(CN)_2]$ and $[C_4C_1im][Me_2PO_4]$ ILs. GuCl:LA DES was found to a stronger hydrogen bond donor than other DESs (Chapter 2) so this DES was expected to potentially extract most of the HBA components of the bark along with some hydrophobic fractions but the extraction yield of GuCl:LA obtained with large error comparison to other yields of ILs. So, it is arguable to state that this DES extracted fractions of bark.

$[TEA][HSO_4]$ showed the highest extraction yield (75%) of the ILs and DESs explored. This is consistent with the reported efficiency of this IL in being an effective solvent for lignin.[126]

3.4.4. Analysis of extracts and residue by FTIR

Given the inability to observe the extracts by NMR and the lack of structural information provided by the UV analysis, FTIR was explored to gain insight into the compounds being extracted. IR spectra of both the solution phase extracts along with the bark and solid residue present after extraction were used to examine the specific nature of the extracted components. As the results of the extraction process of bark by continuous stirring at 100 ℃ after 24 h were quite promising in comparison to other conditions, these conditions were used for the comparison between the different ILs and DESs.

3.4.4.1. Analysis of solution IL and DES extracts

The IL extracts at 25 ℃ or 50 ℃ did not show any significant changes in the IR spectrum after 1 h. This is shown in Figures. 3.10 and 3.11 which compare the IR spectrum of the pure ILs with the IL extracts. Although a change in colour of IL extract was observed, no additional peaks appeared in the spectrum. This highlights that the extent of any extraction under these conditions is relatively low. The other ILs show similar trends and are represented in the appendix (A3 – A6). The analysis of solution DES extracts was not performed under these reaction conditions.

As discussed in the previous sections, by increasing the temperature and extraction duration, it was clear that more notable extraction occurs, with greater extraction typically observed from ILs with good hydrogen-bond accepting anions e.g., [Me_2PO_4] at 100 °C after 24 h as shown in the Figure 3.12 and also in agreement with the results in the Table 3.3. A key new feature was observed in the IR spectrum of these liquid extracts. This feature was a peak at 1600-1660 cm^{-1}, which was identified after extraction for many of these ILs as listed in Table 3.4. The guanidinium based IL and DES extracts did not show this peak. The other ILs and DESs spectra are represented in the appendix (A7 – A21).

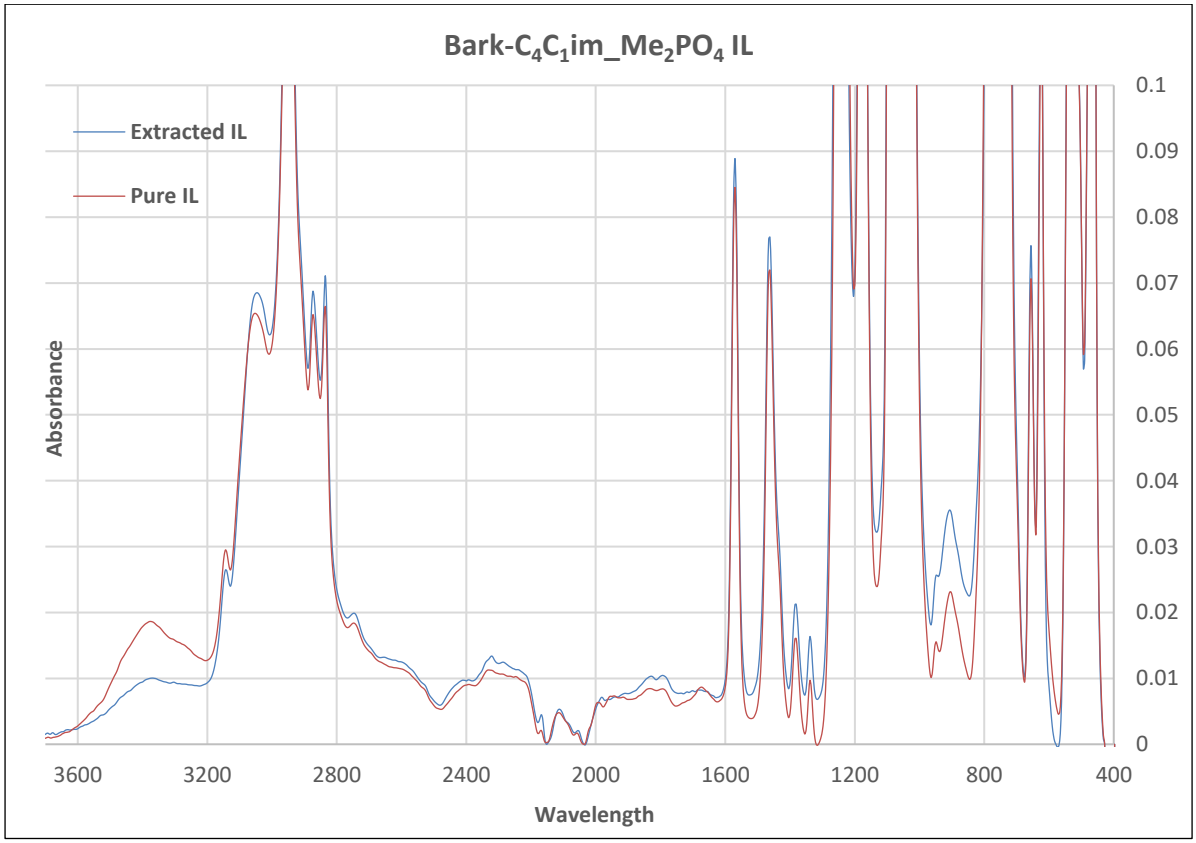


Figure 3.10 *0.1 g of bark in 4 mL of [C_4C_1im][Me_2PO_4] at 25 °C for 1 h*

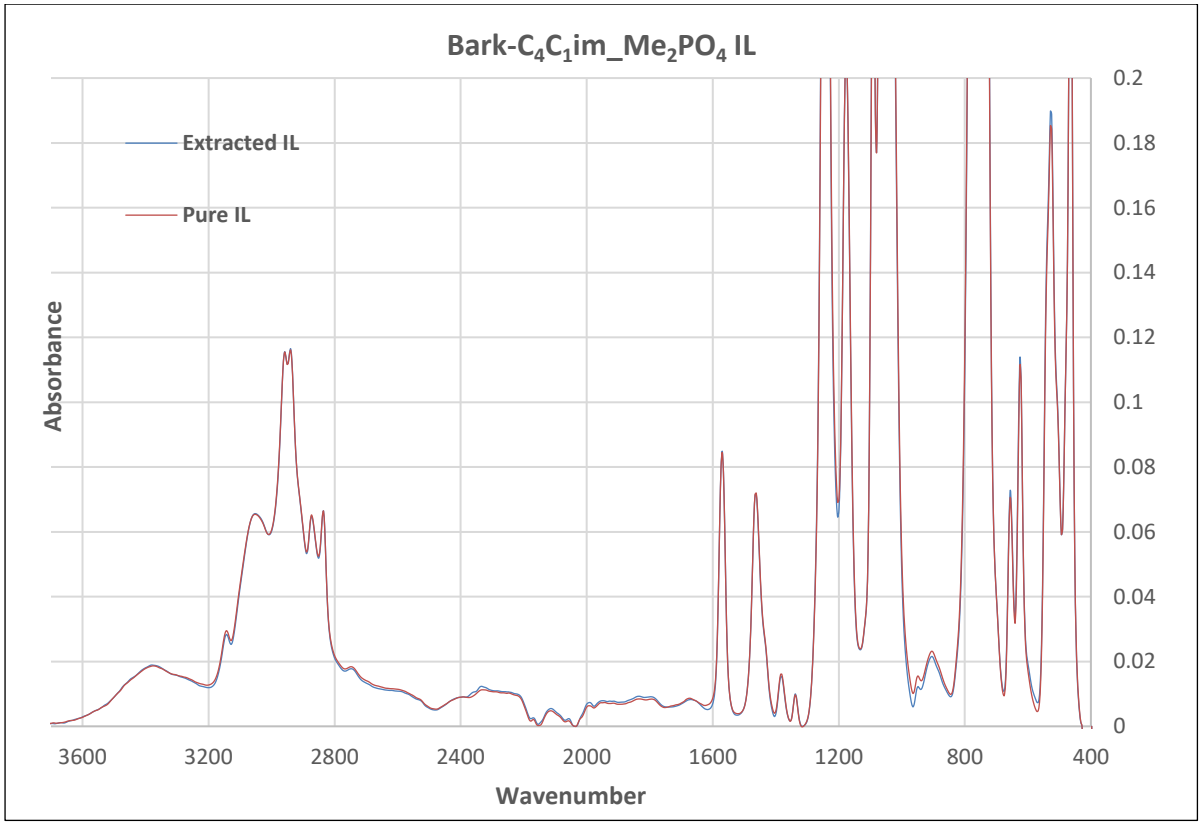


Figure 3.11 *0.1 g of bark in 4 mL of [C_4C_1im][Me_2PO_4] at 50 °C for 1 h.*

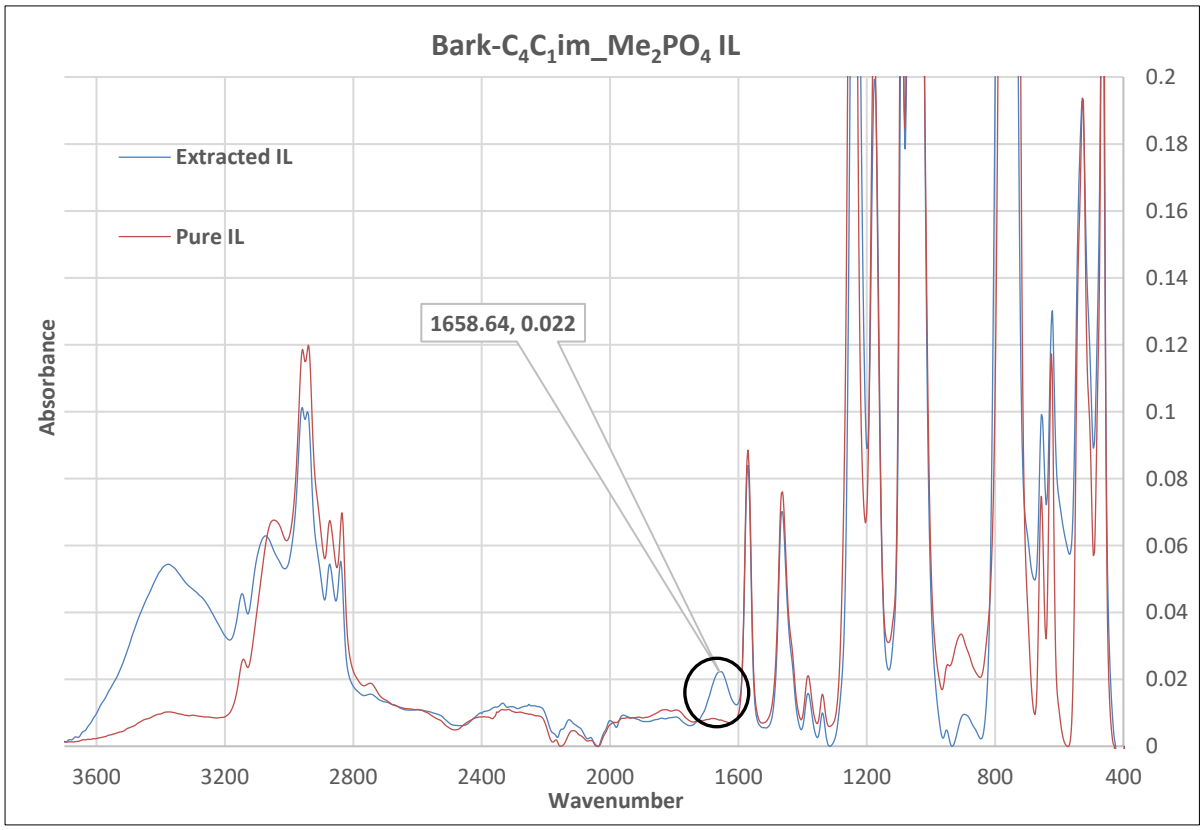


Figure 3.12 *0.1 g of bark in 4 mL of [C_4C_1im][Me_2PO_4] at 100 °C for 24 h.*

Table 3.4 *Intensity and frequency of new peak observed in IR spectrum of IL extracts at 100 ℃ for 24 h.*

ILs	Wavenumber (cm^{-1})	Intensity
$[C_4C_1im][N(CN)_2]$	1639.35	0.0110
$[C_4C_1im][Me_2PO_4]$	1658.54	0.0141
$[C_4C_1im][NTf_2]$	-	-
$[C_4C_1pyrr][N(CN)_2]$	1643.21	0.0075
$[C_4C_1pyrr][Me_2PO_4]$	1604.63	0.0045
$[C_4C_1pyrr][NTf_2]$	-	-
$[C_8C_1im][N(CN)_2]$	1635.49	0.0031
$[C_8C_1im][Me_2PO_4]$	1650.92	0.0138
$[C_8C_1im][NTf_2]$	-	-
$[C_8C_1pyrr][N(CN)_2]$	1647.06	0.0025
$[C_8C_1pyrr][Me_2PO_4]$	1658.63	0.0112
$[C_8C_1pyrr][NTf_2]$	-	-

The IR peak at 1600-1660 cm^{-1} has been assigned in the literature to either extractives or the ring breathing vibrational modes in lignin. Hence, the new peaks in the IR spectrum of the ILs at 1635 cm^{-1} and 1647 cm^{-1} were originally assumed to be either extractives or lignin that have been extracted by ILs with better hydrogen-bond accepting abilities (i.e. those containing the $[Me_2PO_4]^-$ and $[N(CN)_2]^-$ rather than $[NTf_2]^-$ anions). There was also an increase in the intensity of IR peak at around 3300 cm^{-1} which has been assigned in the literature to the stretching bending modes in water. Given the increase in these two peaks appeared to occur together, it was realised that the IR peak at ~1600 cm^{-1} could be due to the extractives but may also be due to water[135]. To examine this, water (0.25 mL) was intentionally added to the IL extract (0.5 mL) and the IR spectrum of the water-IL extract compared with the IR of pure IL and IL extract without water as shown in Fig. 3.13. As the intensity of the peak at 1600 cm^{-1} in the water-IL extract that had been assigned as a possible extractive peak increased significantly in intensity, this demonstrated that the peak was most likely due to the presence of water rather than evidence of extractives in the solution phase. Given the lack of success in characterising the solution phase directly, the solid residue remaining after the extraction was subject to further examination.

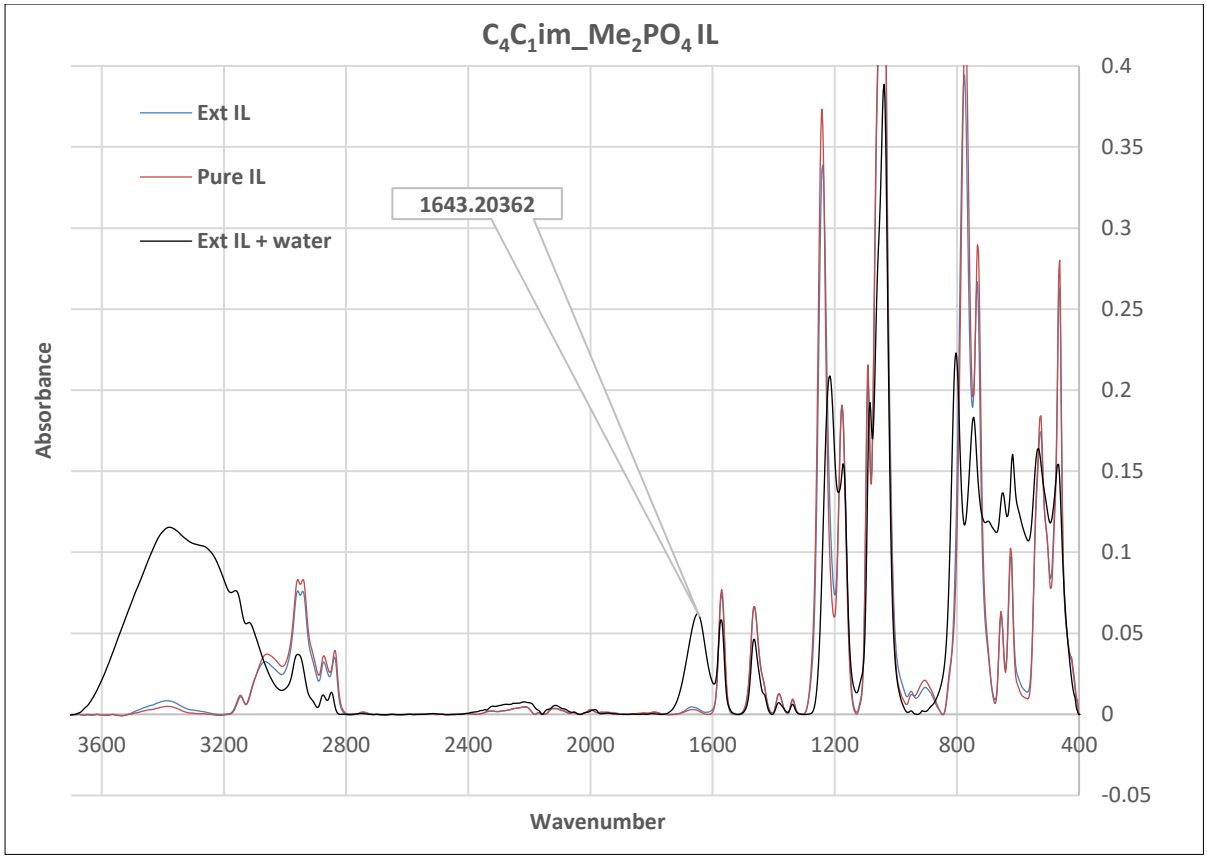


Figure 3.13 *IR of [C4C1im]Me2PO4 IL extract at 100 °C for 24 h with and without water in comparison to pure IL.*

There were some issues with removing residual IL from the solid bark samples during washing. This made the IR more difficult to interpret given the overlap of many of the IL peaks with key peaks in the spectrum. Several solvents (dichloromethane, acetonitrile, and water) were trialled for washing the solid residue after filtration but none were able to successfully remove all of the IL. The better hydrogen bond accepting anions ($[N(CN)_2]^-$ and $[Me_2PO_4]^-$) were more difficult to wash out than ILs containing the $[NTf_2]^-$ anion, being hydrophobic in nature. This might be a sign of the penetration of the IL into the bark samples and suggests that more rigorous washing conditions (e.g. heating and stirring) might be required for the complete removal of the IL.

Ferreira et al. suggested a washing procedure for the removal of residual IL from cork.[98] This involves washing the solid residue with hot water (100 mL at 80 °C) to remove residual IL followed by drying at 50 °C under a nitrogen purge and then high vacuum overnight to remove residual water. Firstly, the washing procedure was done on the pure bark as a control to ensure that no significant changes in bark composition arose from the solvent washing step. It could be observed that almost all the peaks are present even after washing with hot water which

means no notable changes in the composition of bark were observed after washing as shown in Fig. 3.14.

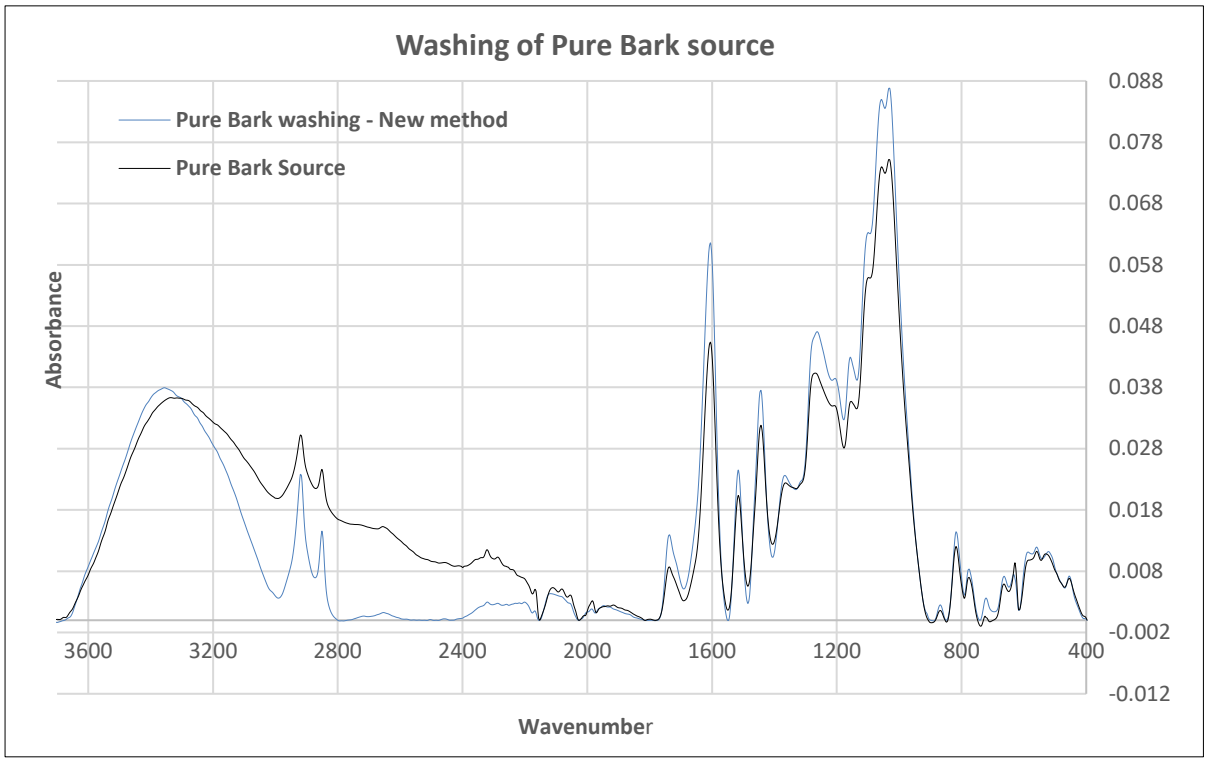


Figure 3.14 *Comparison of composition of bark before and after washing with hot water.*

Moreover, evidence of the reduction in the inclusion of IL in the solid residue could be observed in Fig. 3.15 which represents the extracted bark from $[C_4C_1im][N(CN)_2]$ IL after washing with only small volumes of room temperature water versus the washing of the same solid residue using the new washing method. This appeared to solve the issue of residual IL.

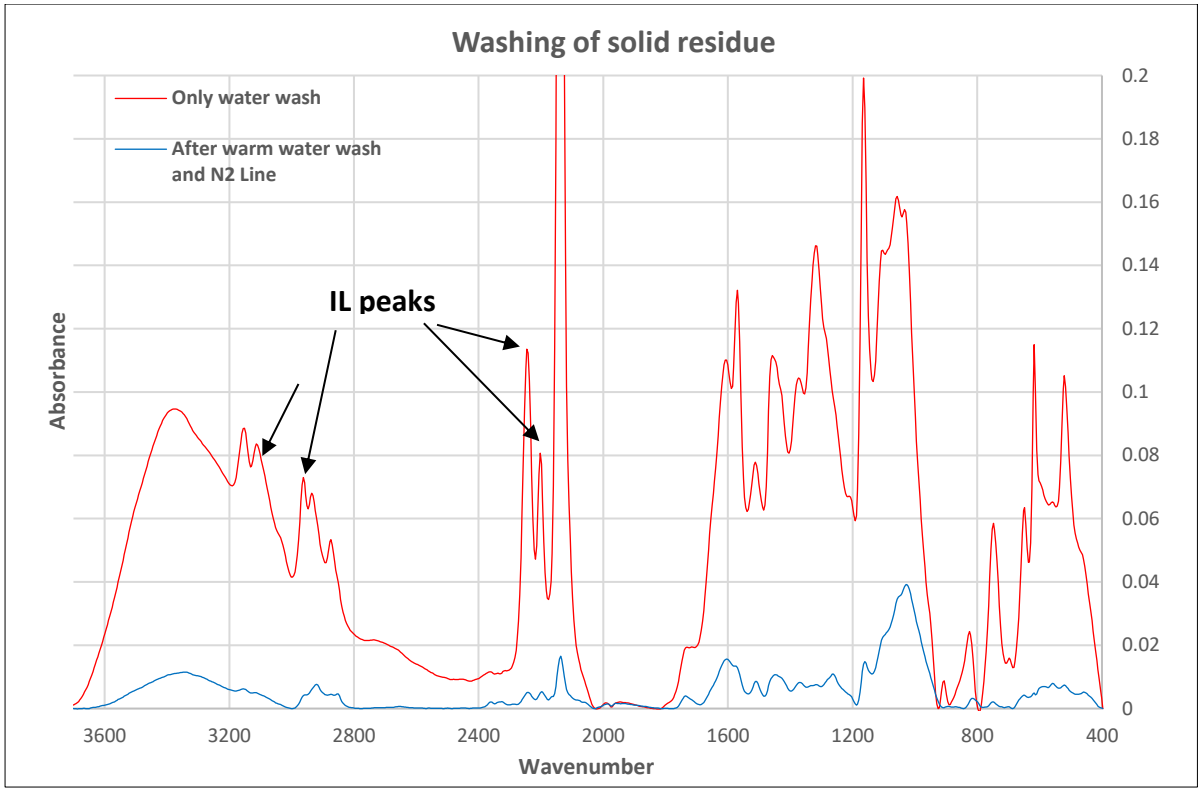


Figure 3.15 *Comparison of washing of extracted bark from $[C_4C_1im][N(CN)_2]$ IL by two different methods.*

3.4.4.2. Analysis of solid residues

Since the IL and DES extracts could not be studied directly, it was of interest to investigate the components extracted by identifying which compounds were no longer present in the bark extract. Like the IL extracts, the bark residue obtained following IL extraction at 25 °C or 50 °C did not show any significant changes in the IR spectrum after 1 h. Evidence of these similarities can be observed in Figs. 3.16 and 3.17 which compare the IR spectrum of the pure bark with the bark extracts. Almost all of the peaks corresponding to pure bark remain present with comparable relative intensities in the bark extracts except some extra peaks coming from residual IL present in the extracted bark. There were some issues with removing residual IL from the solid bark samples during washing which had already been discussed in section 3.4.4.1.

Moreover, even under the given extraction conditions (100 °C, 24 h) where evidence of > 30% extraction had occurred, as described in the previous section, the IR spectrum of the bark residue depicts similar results to the pure bark as shown in Fig. 3.18. This means the > 30% extraction does not appear to be highly selective and the 30% is taken from multiple compounds

of the bark as if it was highly selective one peak or a small group would disappear or significantly reduce in intensity. This is indicating that other techniques might be useful in the analysis of solid residues. All other ILs and DESs IR spectra are represented in the appendix (A22 – A40).

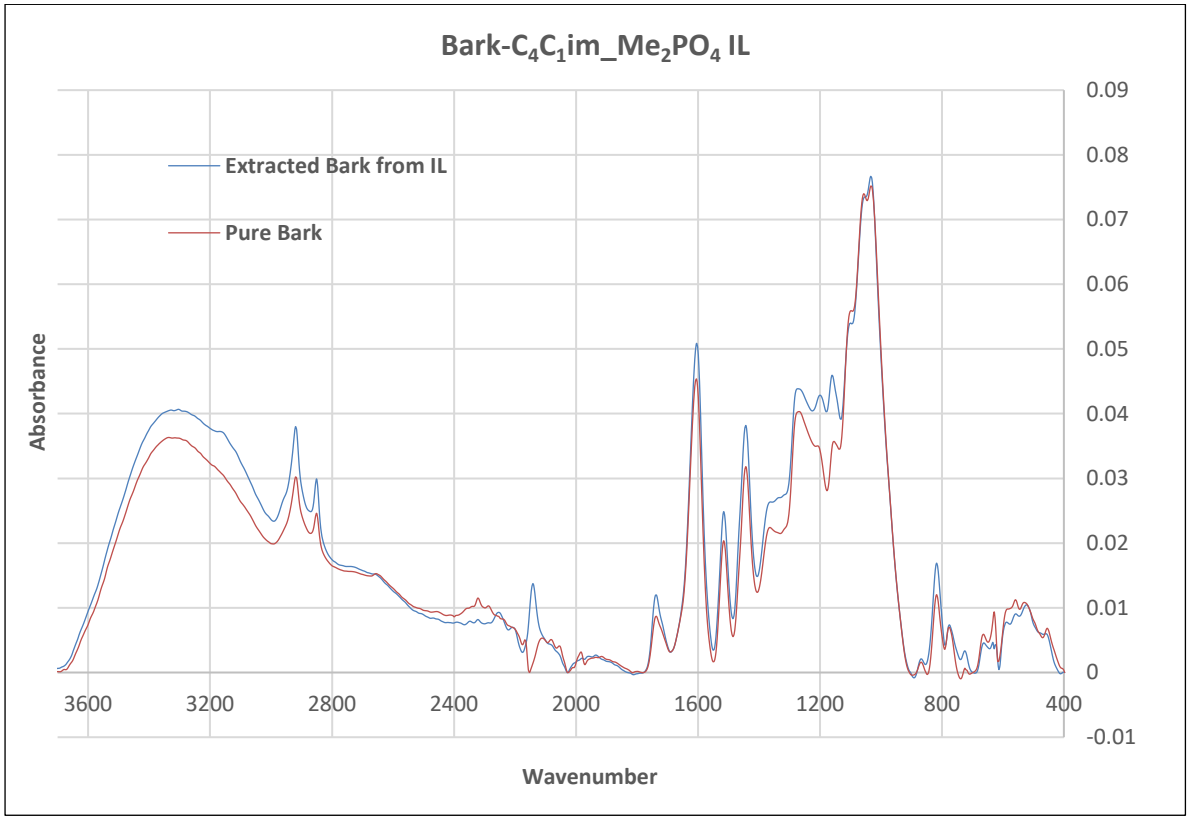


Figure 3.16 *0.1 g of bark in 4 mL of [C_4C_1im][Me_2PO_4] at 25 °C for 1 h.*

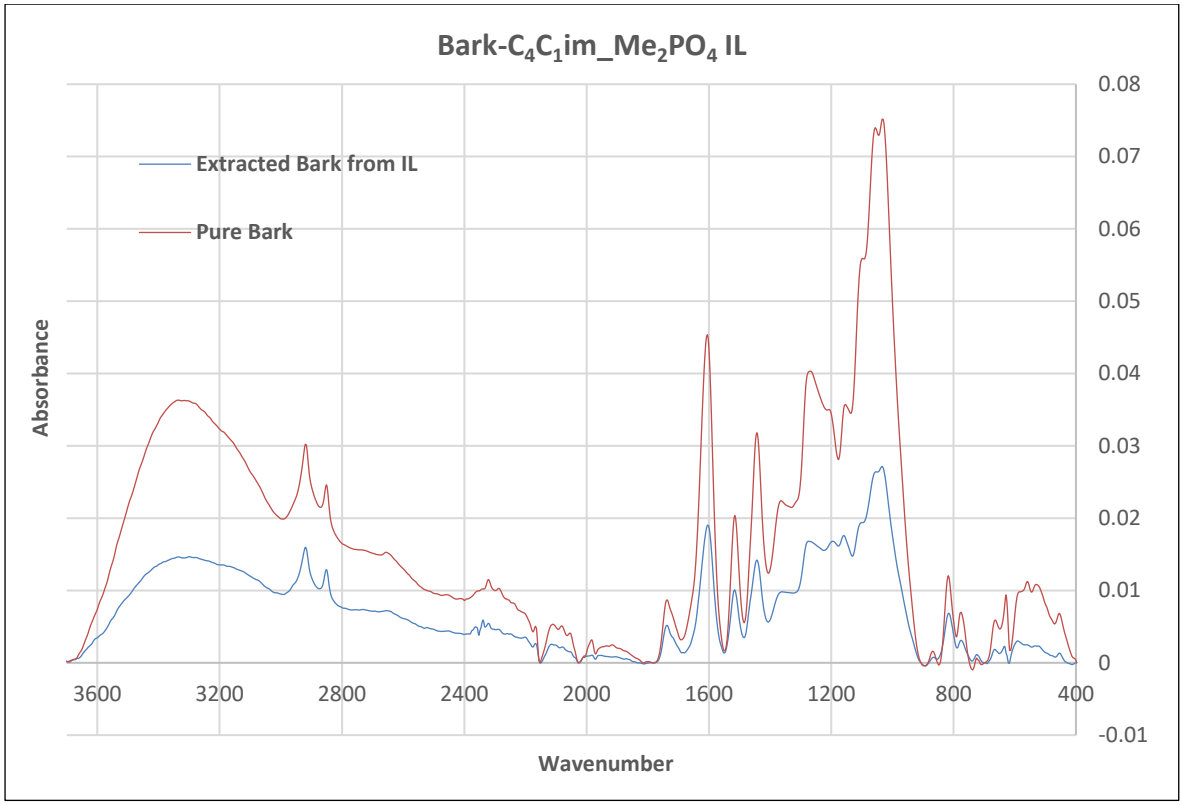


Figure 3.17 *0.1 g of bark in 4 mL of [C4C1im][Me2PO4] at 50 °C for 1 h.*

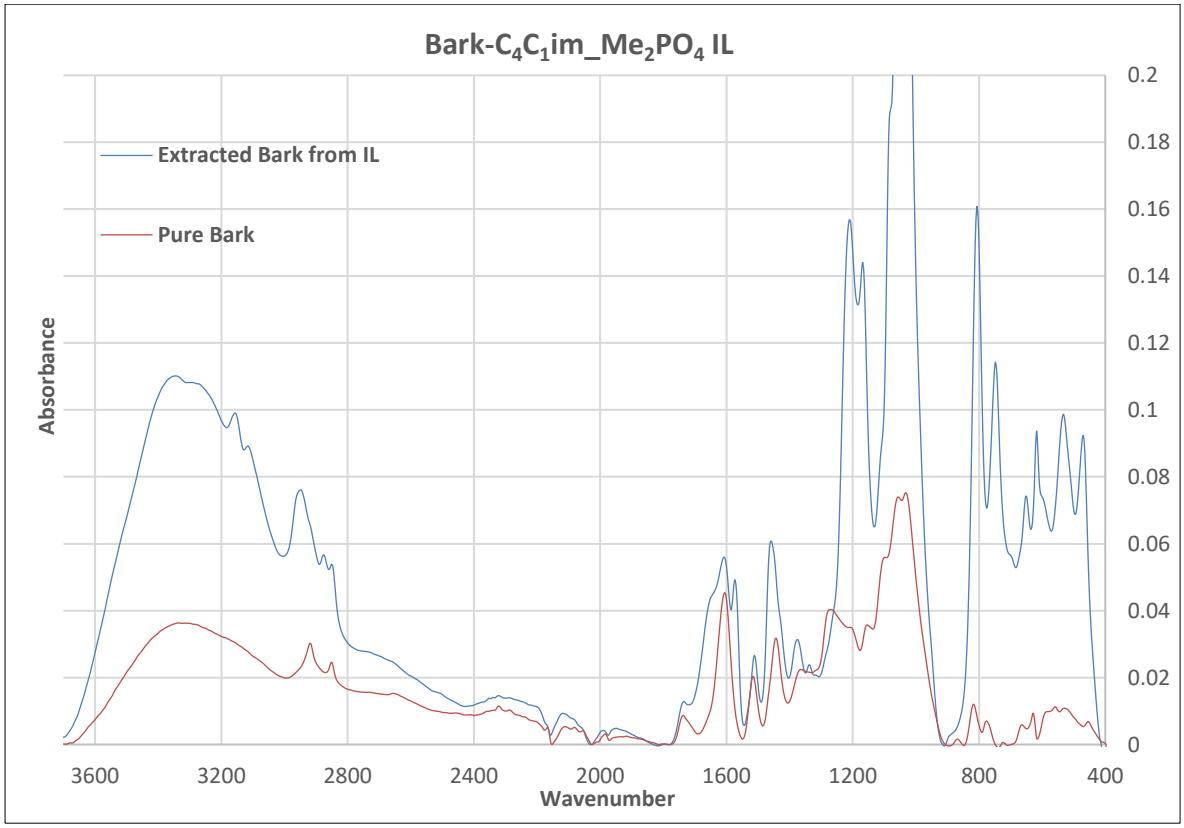


Figure 3.18 *0.1 g of bark in 4 mL of [C4C1im][Me2PO4] at 100 °C for 24 h.*

3.4.5. Analysis of extracts by Headspace solid phase micro extraction - GCMS

As the IR analysis was fairly inconclusive, alternative analytical procedures were employed to investigate the nature of the IL extracts. The head space solid phase micro extraction GCMS (HS-SPME-GCMS) technique is a simple, fast, efficient and solvent-free sample preparation technique which helps in detecting low molecular weight compounds. One of the demerits of this technique is that it doesn't allow the higher molecular weight, non-volatile components of the extract to be easily explored. The use of this technique was explored in order to identify the low molecular weight volatile components extracted from the bark by the different ILs.

SPME is a sampling technique that involves the use of a fibre coated with an extracting phase, that can be a liquid (polymer) or solid (sorbent), which extracts different kinds of volatile analytes (polar and non-polar) from different kinds of media (liquid or gas phase). After adsorption of the analytes, this fibre is exposed to the inlet of GC-MS for a particular period of time for desorption which allows the different analytes present in that sample to be characterised. The quantity of analytes extracted by the fibre is proportional to their concentration in the sample as long as equilibrium is reached.

Pre-dried bark (~100 mg) was run as a control and compared with the solid residue obtained from $[C_4C_1im][Me_2PO_4]$ IL as shown in Figures 3.19 (a). Figure 3.19 (a) shows the chromatograms obtained with the retention time in the range of 2 – 24 min and Figure 3.19 (b) depicts the chromatograms with the next series of retention time (26 – 48 min). These chromatograms depict that there are some peaks that appeared in the solid residue which are not present in the pure bark highlighting that these peaks might either come from residual IL or reactions occurring under the extraction conditions. All the analytes detected from the pure bark sample are listed in the appendix (Table A41) and the blue shaded regions indicating the analytes present in both pure bark and residue following IL extraction.

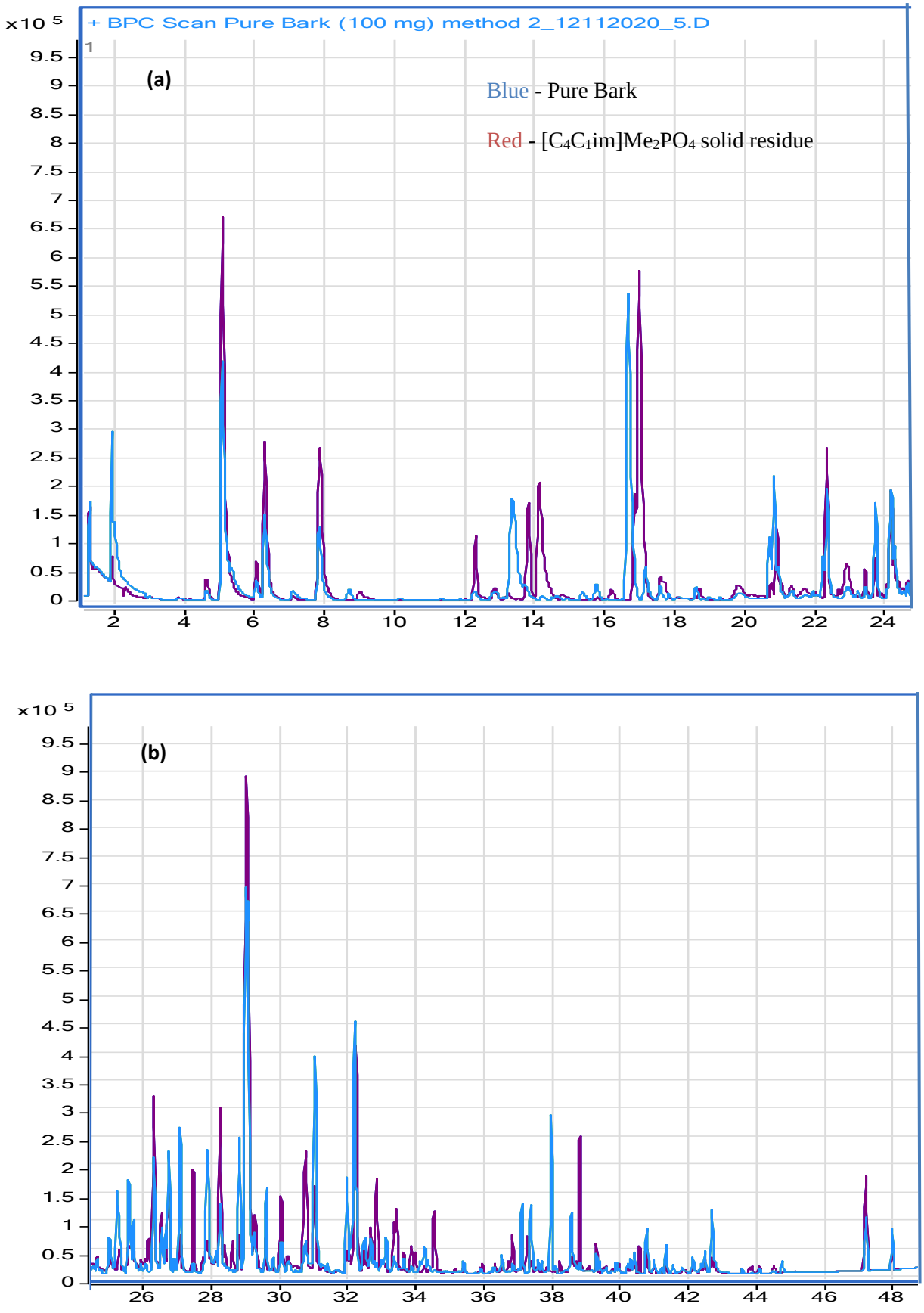
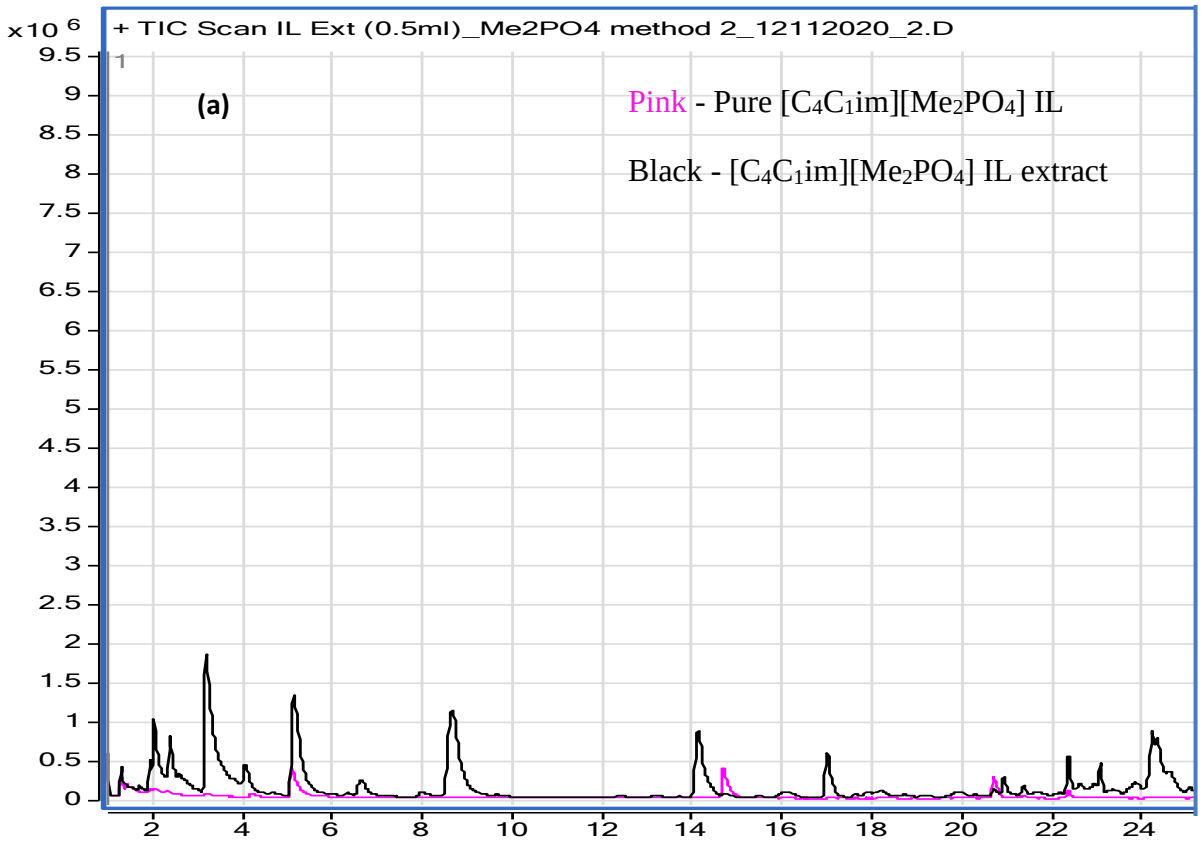


Figure 3.19 *(a) and (b) HS-SPME-GCMS chromatograms of pure bark and [C_4C_1im][Me_2PO_4] IL solid residue.*

IL extracts were also explored using this technique. For this, again the $[C_4C_1im][Me_2PO_4]$ extract was selected. The pure $[C_4C_1im][Me_2PO_4]$ IL was compared with the $[C_4C_1im][Me_2PO_4]$ extract as shown in Figures 3.20 (a) and (b). Figure 3.20 (a) and (b) show the chromatograms obtained with the retention time in the range of 2 – 48 min. Additional peaks could be observed in the chromatograms of the IL extract which are not present in the pure IL, highlighting that these compounds have been extracted from the bark by the IL extract. All the analytes discovered from the IL extract sample excluding the pure IL peaks are represented in the Table 3.5.



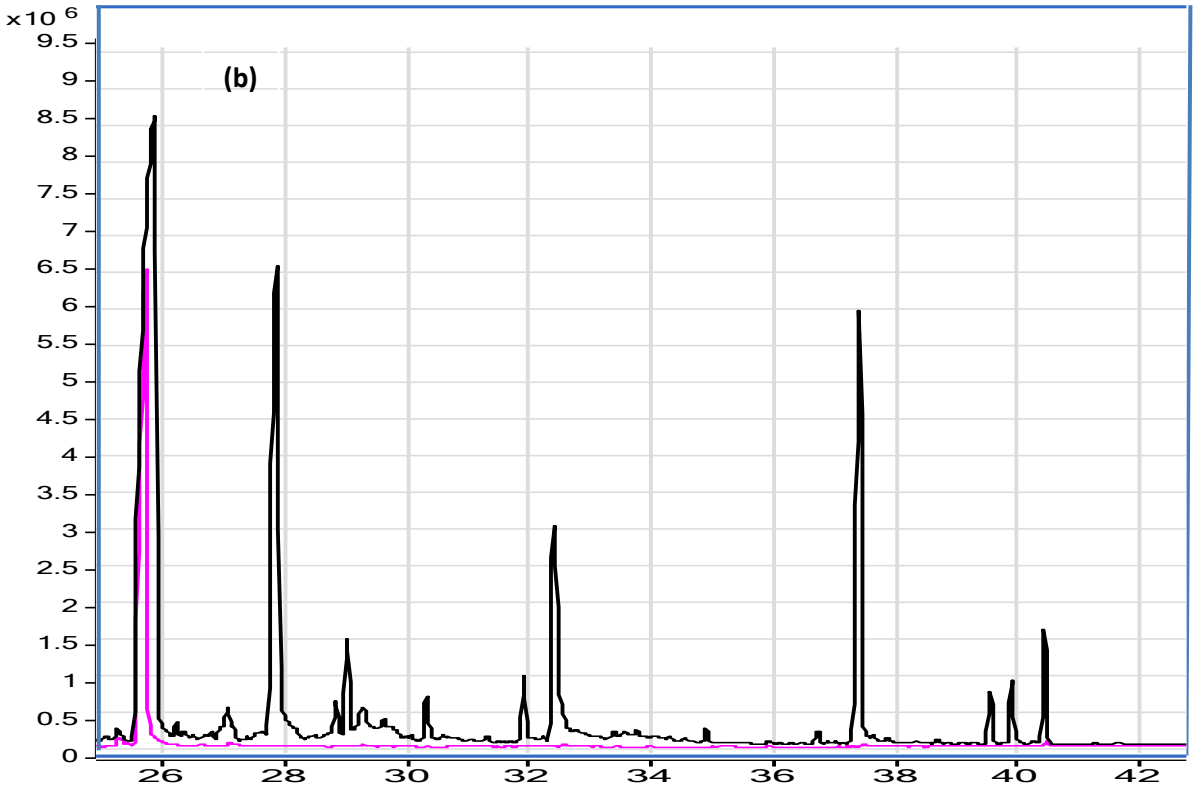


Figure 3.20 *(a) and (b) HS-SPME-GCMS chromatograms of pure [C_4C_1im][Me_2PO_4] and the [C_4C_1im][Me_2PO_4] IL extract obtained at 100°C after 24 h.*

Tables 3.5 represents the compounds present in the [C_4C_1im][Me_2PO_4] IL after extraction that do not arise from the IL itself. Most of these compounds are either terpenes or substituted benzenes. Although most of these compounds are also present in the solid residue except α-pinene, the area of the peaks of these compounds are higher in the IL extract than the solid residue highlighting the extraction of these compounds by the IL. These compounds have been known to represent good pharmacological activities such as anti-inflammatory[136], antidiabetic[137] and can be used as flavouring agents.[138] While this method does provide some insight into some of the extractive components that can be concentrated by the IL, the fact that these are limited only to volatile small molecule components limits the usefulness of this technique in establishing the overall composition of the extracts. Given the likely larger yield of the less volatile polymeric components and their greater utility, this method wasn't explored further for the other ILs.

Table 3.5 *Low molecular weight volatile compounds in the [C_4C_1im][Me_2PO_4] IL extract identified using SPME-GCMS.*

Label	Formula	Score	Base Peak	RT	Width	Area
alpha.-Pinene	$C_{10}H_{16}$	93.73	93.1	5.169	0.883	18511507
Camphene	$C_{10}H_{16}$	95.01	93.05	6.633	0.933	3098765
beta.-Pinene	$C_{10}H_{16}$	97.75	93.1	8.667	1.117	20579066
D-Limonene	$C_{10}H_{16}$	98.47	68.05	14.133	0.7	11919734
Benzene, 1-methyl-2-(1-methylethyl)-	$C_{10}H_{14}$	96.34	119.1	17	0.623	5388080
Benzene, (2-methyl-1-propenyl)-	$C_{10}H_{12}$	91.77	117.1	22.367	0.175	2092807

3.4.6. Solid-State NMR Analysis

In order to gain more insight into the non-volatile components extracted from the bark, ^{13}C solid-state NMR was used to characterise the solid residue formed after the extraction of bark with different ILs and DESs to verify the components that had been removed. The monomer units of lignin, tannin and carbohydrate used for these assignments are shown in Figure 3.21. In addition, suberin, which has a polyester structure made for the most part out of long chain fatty and hydroxy fatty acids, is mainly characterised by the very intense peak at a chemical shift of 33 ppm corresponding to the long aliphatic chains [$(-CH-)_n$] present. Figures 3.22, 3.23, and 3.24 depict an approximate assignment of key solid state NMR peaks to lignin, tannin, suberin and carbohydrate components of the bark, respectively.[128,129,132]

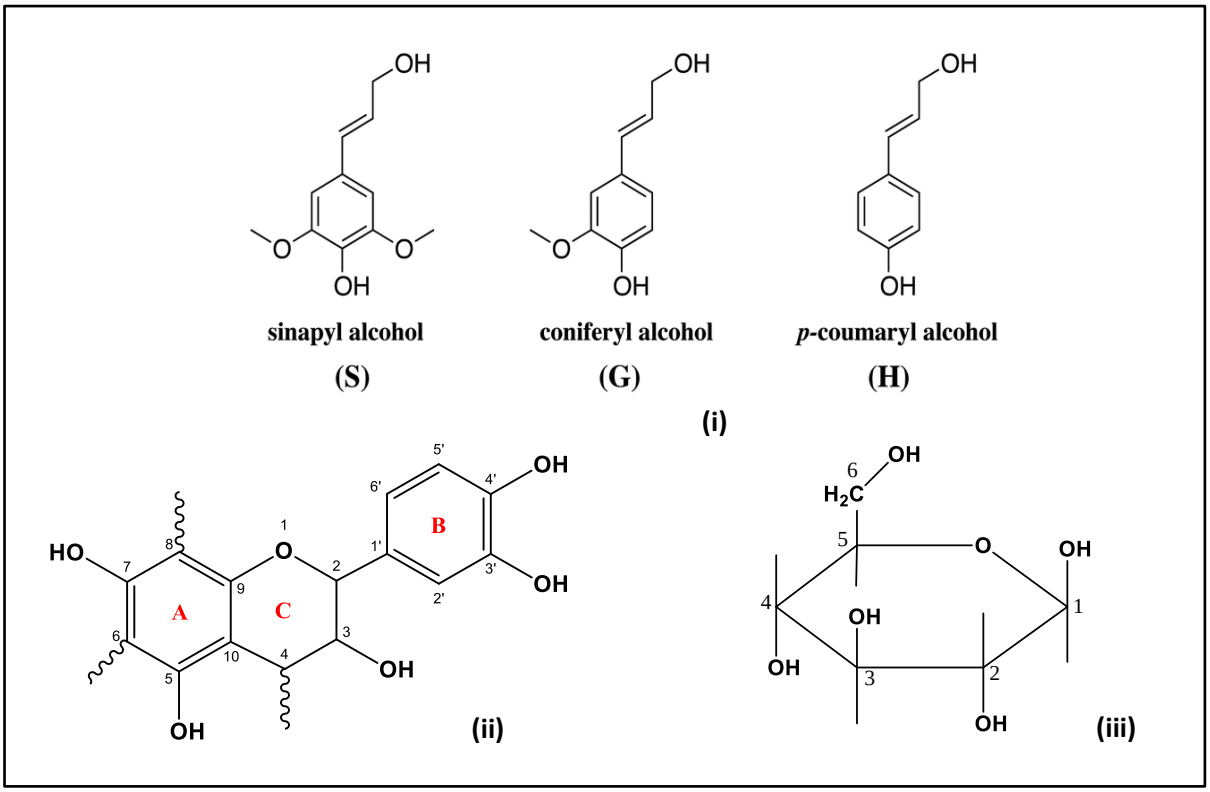


Figure 3.21 *Monomer units of (i) lignin, (ii) tannin, and (iii) polysaccharide.*

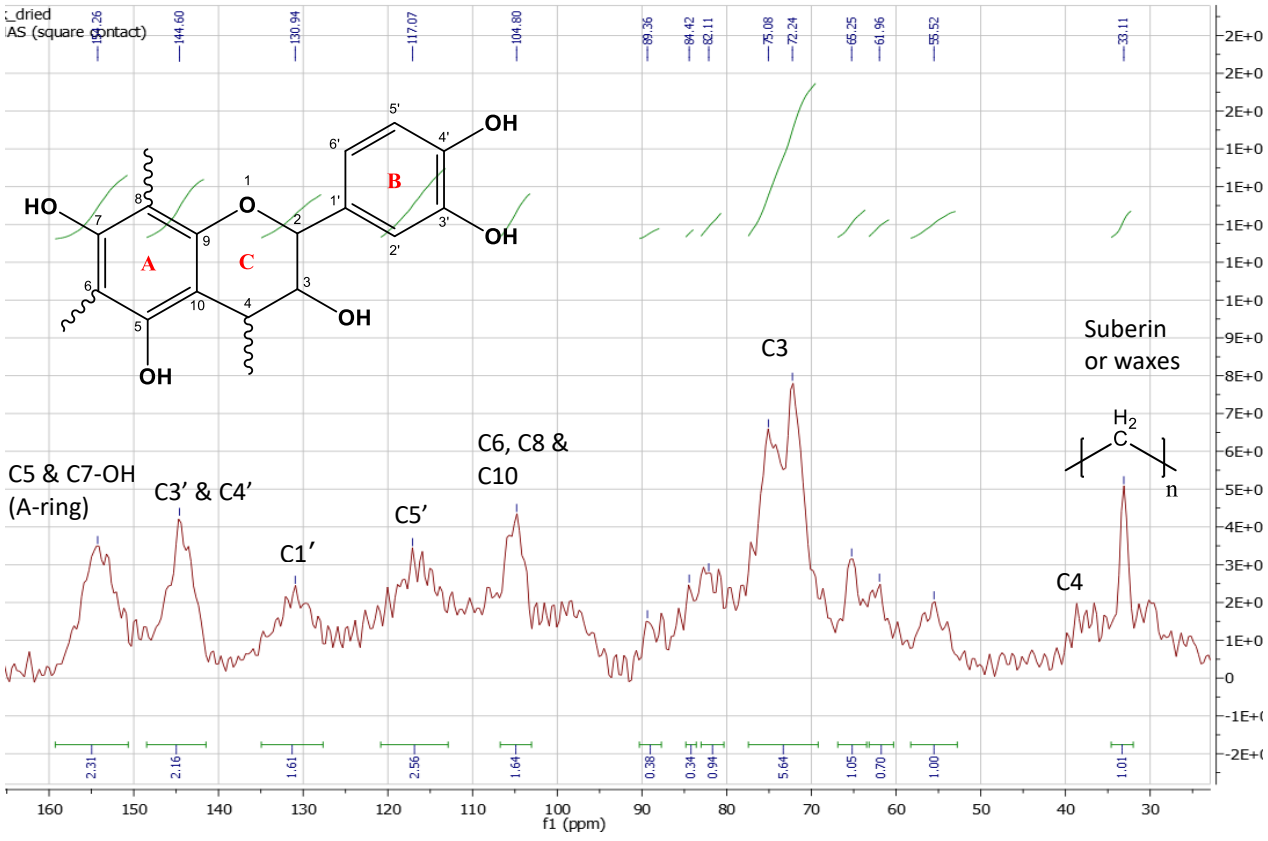


Figure 3.22 *Carbon peaks assigned to tannins (^{13}C solid–state NMR)*

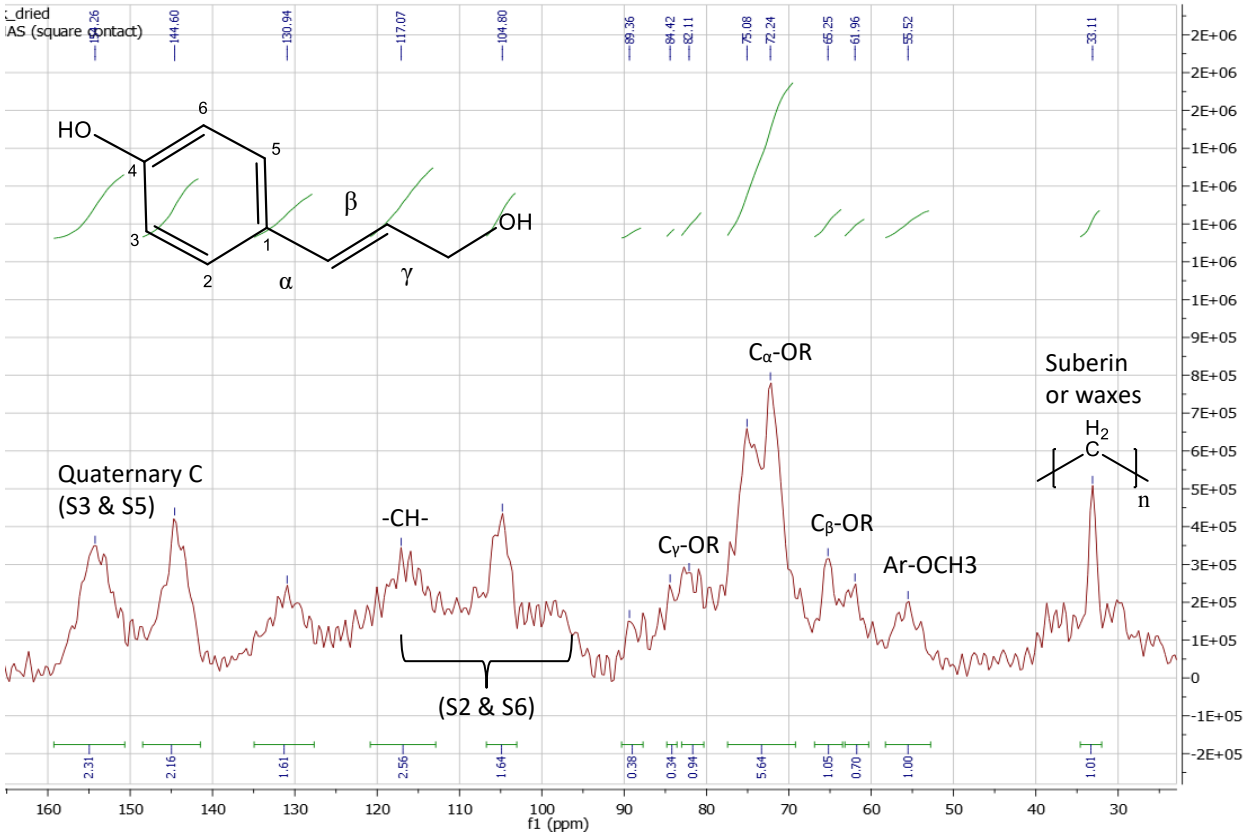


Figure 3.23 *Carbon peaks assigned to lignin (^{13}C solid–state NMR)*

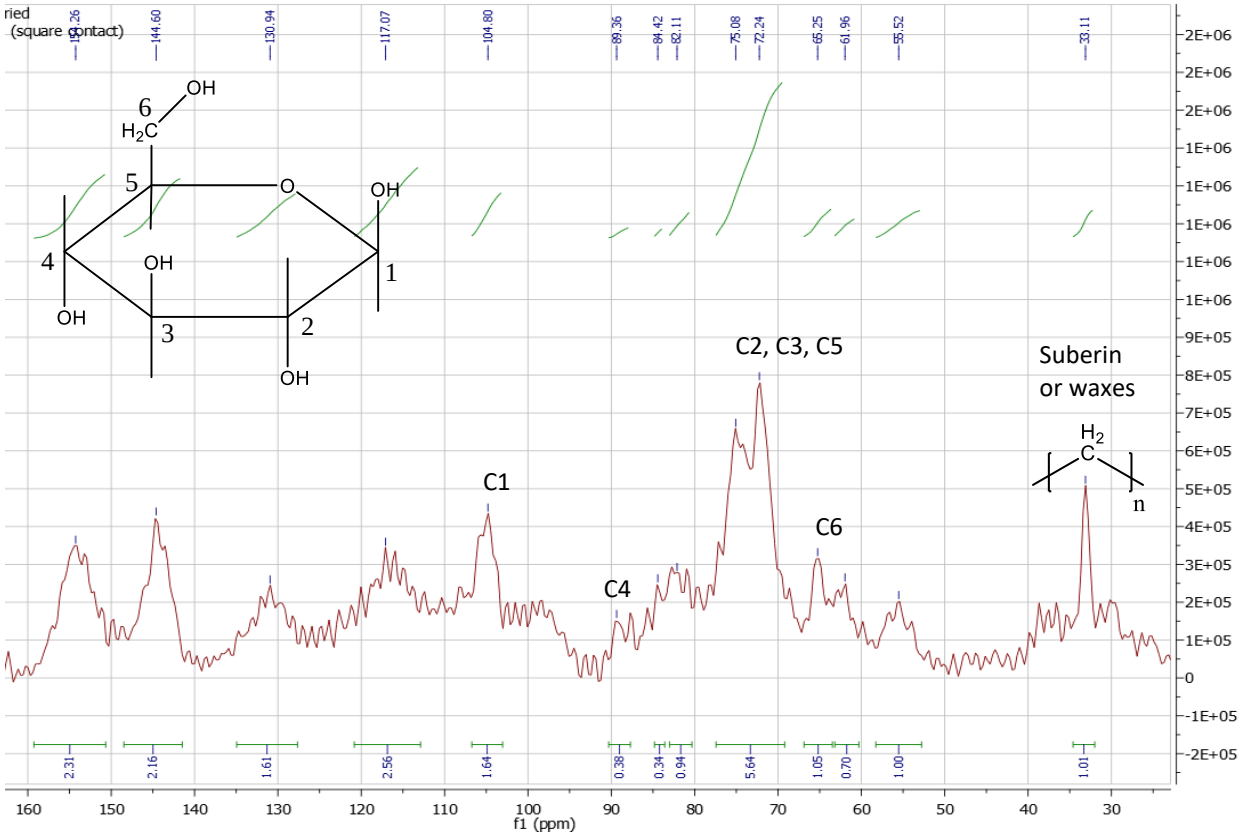


Figure 3.24 *Carbon peaks assigned to carbohydrates (^{13}C solid–state NMR)*

3.4.6.1. Effect of washing the solid residue

The ^{13}C solid-state NMR of extracted bark (solid residue) from the IL and DES extractions were obtained. The residue was washed with hot water for the $[Me_2PO_4]$- and $[N(CN)_2]$-based IL extracts and with dichloromethane for residue obtained from the $[NTf_2]$-based IL extracts. The washing solvent was selected based on the solubility of the ILs. Moreover, pure dried bark was also washed with hot water and dichloromethane to compare with IL-based solid residue to ensure no artifacts were detected arising from the washing step.

Figure 3.25 demonstrates that washing with the hot water slightly reduces the integral of peaks corresponding to aromatic carbons at chemical shifts of 154 ppm and 144 ppm compared to pure dried bark, where the red line represents the spectrum of the pre-dried pure bark source while the green line represents the spectrum of the same bark source after the hot water wash. These peaks are assigned to lignin and tannin and highlights that the water washing step could be extracting a small proportion of the lignin and tannin components. The extraction of polyphenols is also known.[139] This was the only change noticed in the spectrum.

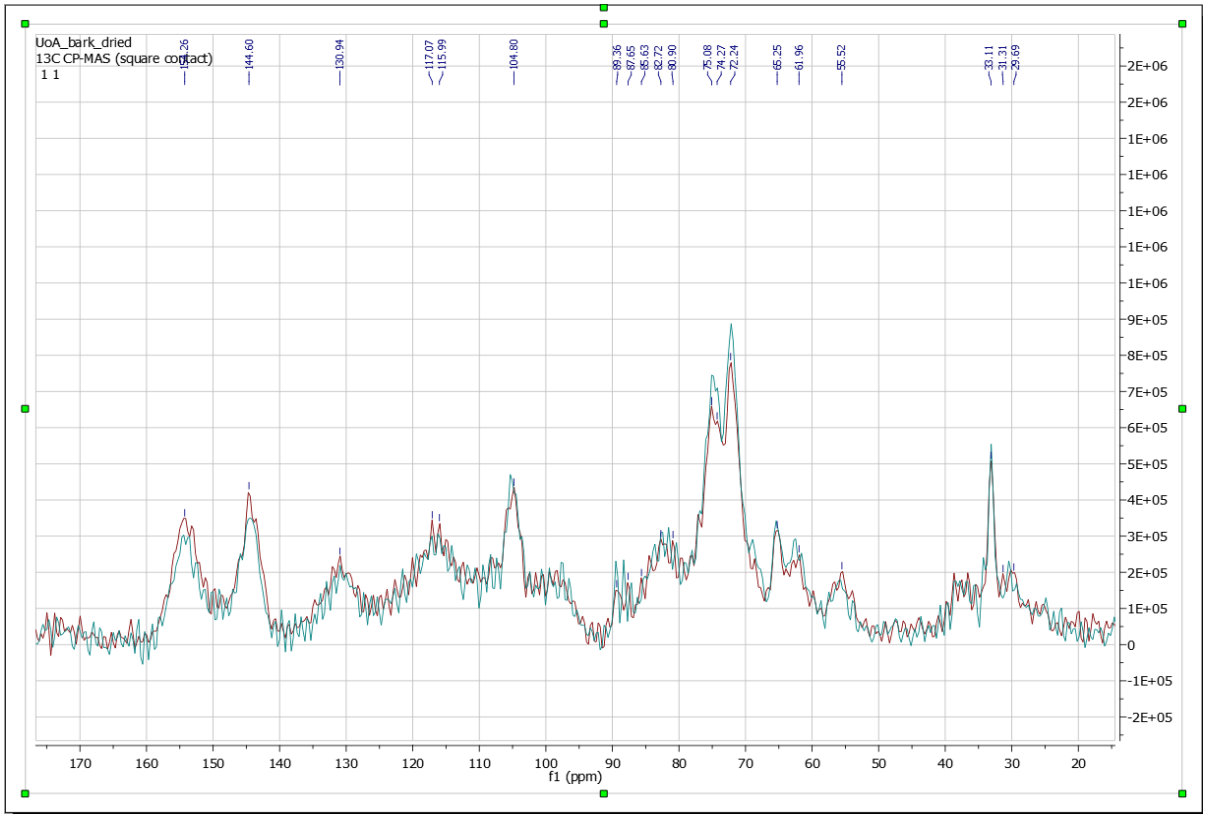


Figure 3.25 *Comparison of the ^{13}C solid-state NMR of water washed bark and pure bark. The red line represents the spectrum of the pre-dried pure bark source while the green line represents the spectrum of the same bark source after the hot water wash.*

The extraction with dichloromethane led to a decrease in the relative integral of the peak at 33 ppm to some extent compared to the unwashed pre-dried pure bark as shown in Fig. 3.26. In this Figure, the red line represents the spectrum of the pre-dried pure bark source while the green line represents the spectrum of the same bark source after washing with dichloromethane. This suggests that the dichloromethane, being non-polar in nature, selectively extracts hydrophobic components (waxes or suberin). This was the only change noticed in the spectrum. The individual integrated spectra of the water washed solid residue and dichloromethane washed solid residue are shown in the appendix (Figures A42 – A43).

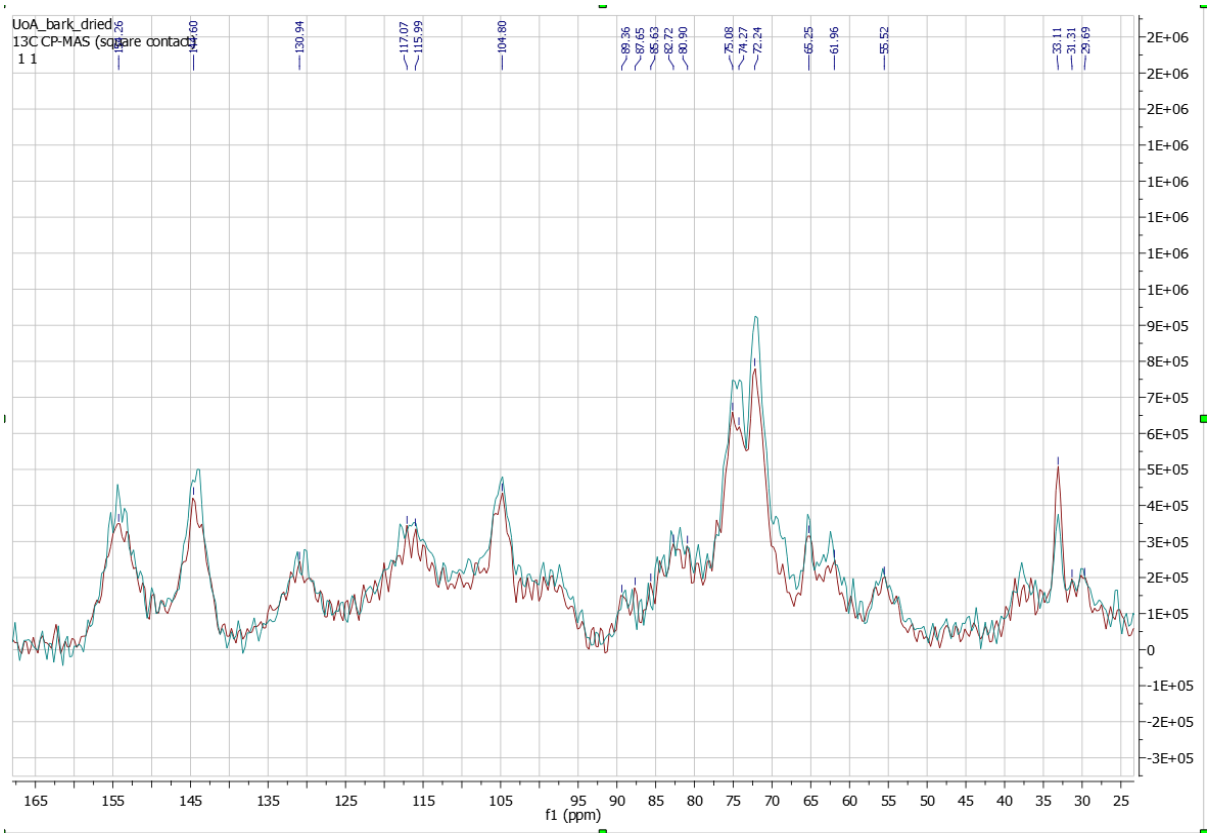


Figure 3.26 *Comparison of the ^{13}C solid-state NMR spectrum of dichloromethane washed bark and pure bark. The red line represents the spectrum of the pre-dried pure bark source while the green line represents the spectrum of the same bark source after washing with dichloromethane.*

3.4.6.2. Solid State NMR analysis of IL bark extraction

The solid-state NMR of the solid residue acquired from bark extractions using the $[Me_2PO_4]$-, $[N(CN)_2]$-, $[NTf_2]$-based ILs and GuCl:LA and GuCl:Gly DESs, were obtained and compared with either the water washed solid residue or dichloromethane washed solid residue depending on the washing regime used to remove residual solvent.

The weakest hydrogen bond acceptor, the $[NTf_2]^-$ anion, was used as a control since the $[NTf_2]$-based ILs did not provide high extraction yields, with the extraction yields of the bark less than 15 % for all of these ILs and less than 10% for the majority. Table 3.6 displays the relative integrals observed in the ^{13}C solid state NMR of the bark samples that have undergone extraction by an $[NTf_2]$-based IL. These integrals were calculated relative the 56 ppm -OCH_3 of lignin which was set to a value of 1. A decrease in the integral relative to the DCM-washed bark highlights that component has been relatively more extracted than lignin whereas an increase suggests the opposite.

The ILs in which the $[NTf_2]^-$, a weak hydrogen bond accepting anion, was combined with the imidazolium and pyrrolidinium cation displayed very less reduction in the integral of peaks corresponding to aromatic carbons of lignin, tannin, carbohydrate and peaks corresponding to aliphatic carbons of suberin at chemical shifts of 154 ppm, 144 ppm, 74 ppm, and 33 ppm, respectively, compared to pure dried bark as represented in the Table 3.6. For instance, the $[C_4C_1im]^+$ and $[C_4C_1pyrr]^+$ cations displayed reduction in the aromatic peaks and a slight increase in carbohydrate which suggest selective aromatic removal. These variations were not observed for the C_8 cations.

Table 3.6 *Comparison of the integrals of the dichloromethane washed bark with $[NTf_2]^-$ IL solid residue.*

ILs	δ (ppm)	Components	Integrals of DCM-washed bark	Integrals of ILs bark residue
$[C_4C_1im][NTf_2]$	154	Lignin / Tannin	2.53	1.83
	144	Lignin / Tannin	2.51	2.05
	72 - 75	Carbohydrate	7.48	7.32
	33	Suberin	0.68	0.34
$[C_4C_1pyrr][NTf_2]$	154	Lignin / Tannin	2.53	1.99
	144	Lignin / Tannin	2.51	2.27
	72 - 75	Carbohydrate	7.48	10.66
	33	Suberin	0.68	0.71
$[C_8C_1im][NTf_2]$	154	Lignin / Tannin	2.53	2.90
	144	Lignin / Tannin	2.51	2.61
	72 - 75	Carbohydrate	7.48	6.59
	33	Suberin	0.68	0.66
$[C_8C_1pyrr][NTf_2]$	154	Lignin / Tannin	2.53	2.98
	144	Lignin / Tannin	2.51	2.80
	72 - 75	Carbohydrate	7.48	7.36
	33	Suberin	0.68	0.62
$[C_4gun][NTf_2]$	154	Lignin / Tannin	2.53	1.92
	144	Lignin / Tannin	2.51	1.58
	72 - 75	Carbohydrate	7.48	5.65
	33	Suberin	0.68	0.33
$[C_8gun][NTf_2]$	154	Lignin / Tannin	2.53	3.33
	144	Lignin / Tannin	2.51	3.04
	72 - 75	Carbohydrate	7.48	9.09
	33	Suberin	0.68	0.41

One of the spectra of $[NTf_2]$-based ILs solid residue, displaying the aforementioned change has been shown in Fig. 3.27 where the red line represents the spectrum of the pure bark after

washing with dichloromethane while the green line represents the spectrum of the $[C_4C_1im][NTf_2]$ IL solid residue after washing with dichloromethane. The other spectra of the $[NTf_2]$-based IL solid residue namely, $[C_4C_1im][NTf_2]$, $[C_4C_1pyrr][NTf_2]$, $[C_8C_1im][NTf_2]$, and $[C_8C_1pyrr][NTf_2]$, are shown in the appendix (A44 – A47).

In addition, the $[NTf_2]^-$ anion combined with the guanidinium cation, a strong hydrogen bond donor, and anticipated that these ILs might interact with the complex structure of the key components of bark via the hydrogen bonding resulting in extracting fractions out of it. The guanidinium cation has a tendency to interact with the ester groups of the suberin/waxes which explains its different behaviour. Figures 3.28 and 3.29 depict the spectra of $[C_4gun][NTf_2]$ and $[C_8gun][NTf_2]$ solid residue. According to these spectra, the intensity of peak assigned to 33 ppm and the integral were lower than the dichloromethane washed bark which highlights that some suberinic components/waxes were extracted by the ILs. This change has also been represented in the Table 3.6. The other spectra of these ILs in terms of the integrals change, are shown in the appendix (A48 – A49).

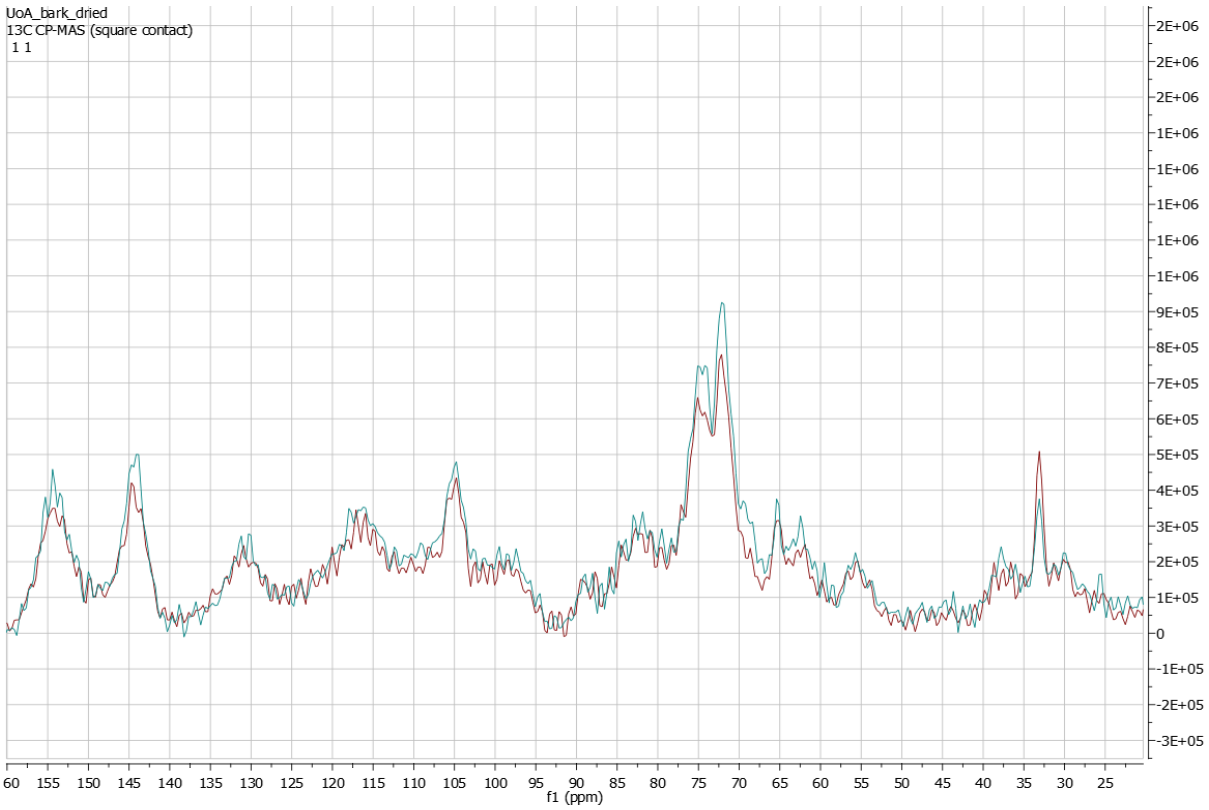


Figure 3.27 *Comparison of the ^{13}C solid-state NMR of dichloromethane washed bark and $[C_4C_1im]NTf_2$ IL solid residue. The red line represents the spectrum of the pure bark after washing with dichloromethane while the green line represents the spectrum of the $[C_4C_1im][NTf_2]$ IL solid residue after washing with dichloromethane.*

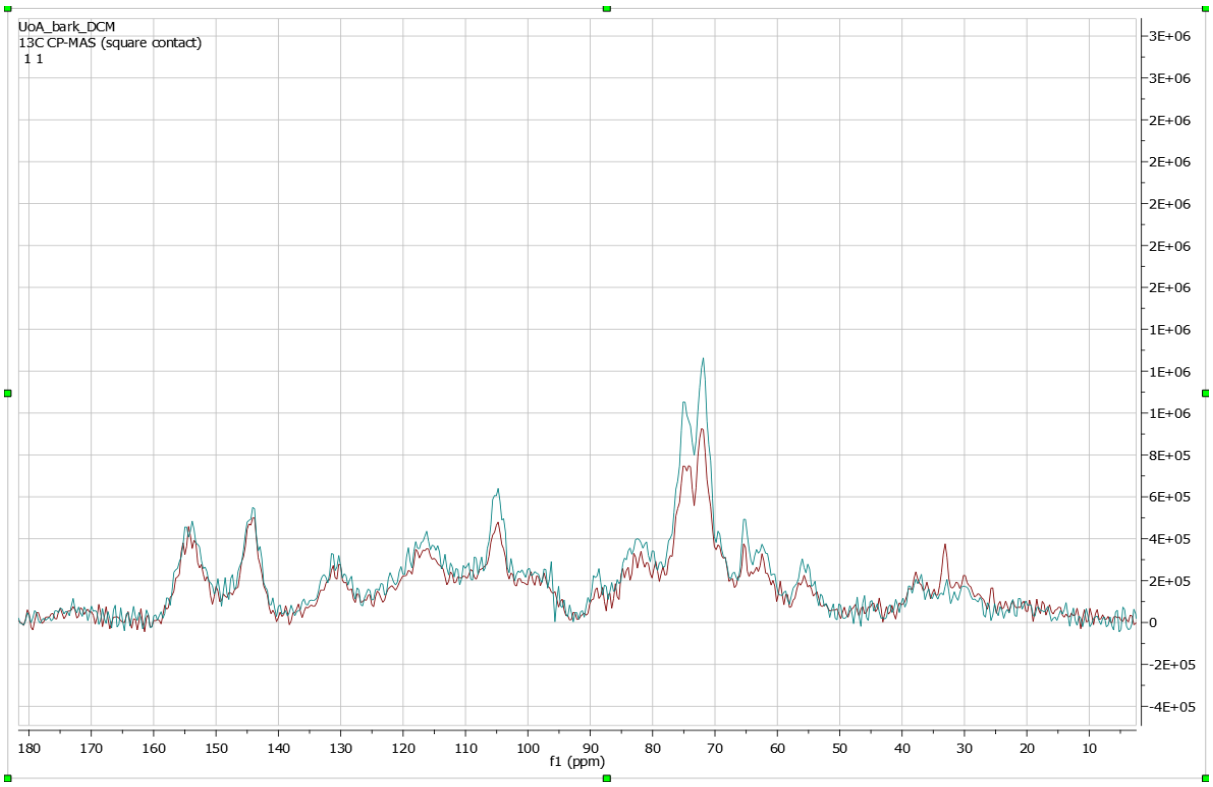


Figure 3.28 *Comparison of the ^{13}C solid-state NMR of dichloromethane washed bark and [C_4gun][NTf_2] IL solid residue. Same description of coloured lines as above.*

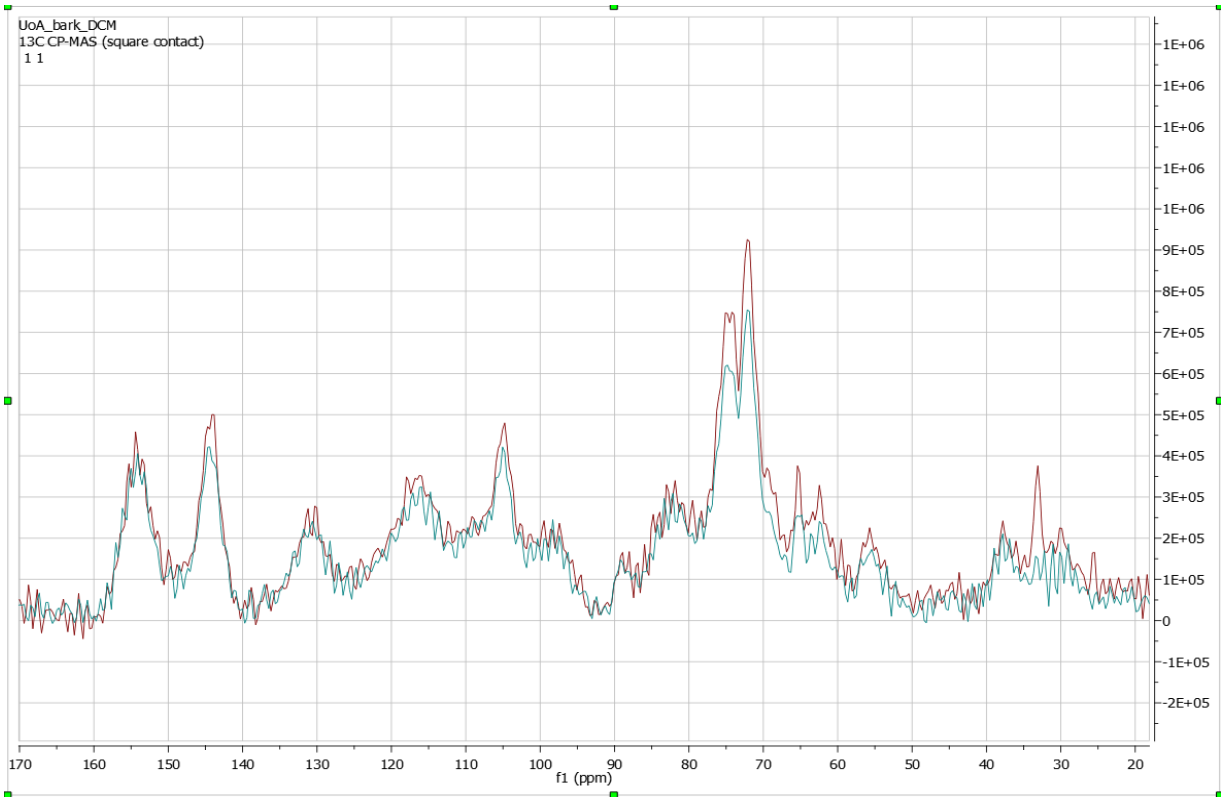


Figure 3.29 *Comparison of the ^{13}C solid-state NMR of dichloromethane washed bark and [C_8gun][NTf_2] IL solid residue. Same description of coloured lines as above.*

The GuCl:LA and GuCl:Gly DES solid state NMR results are summarised in Table 3.7. Unlike the previous ILs discussed, these DESs solid residues were washed with water rather than DCM, hence the values are compared to the integrals obtained for the water washed pure bark. In addition, the colour of the GuCl:LA DES extract was changed from colourless to a reddish which could be due to the incorporation of acidic HBD in the DES resulting in either the side reaction between the bark and the DES[64] or decomposition of lactic acid at high temperature.[140] Although GuCl:LA was found to be a good hydrogen bond donor (Chapter 2), this DES appears to be less capable of interacting with the accepting groups of the bark components such as the ester groups of the waxes or suberinic components than ILs. This could be attributed to the significant self-association within these DESs through the formation of complex hydrogen bonding networks, reducing their ability to act as effective solvents for the extraction of bark. Moreover, both DESs had higher extraction yields than the [NTf_2] ILs. So, some selective components of bark were assumed to be extracted by the DESs. Mixed fractions were obtained which could just be due to the extraction selectivity of DESs was low and so everything was dissolved approximately equally.

Table 3.7 *Comparison of the integrals of the components of water washed bark and DESs solid residues.*

ILs	**δ (ppm)**	**Components**	**Integrals of water-washed bark**	**Integrals of ILs bark residue**
GuCl:LA	154	Lignin / Tannin	2.21	2.22
	144	Lignin / Tannin	2.21	2.78
	72 - 75	Carbohydrate	7.24	6.41
	33	Suberin	1.10	0.98
GuCl:Gly	154	Lignin / Tannin	2.21	2.18
	144	Lignin / Tannin	2.21	2.32
	72 - 75	Carbohydrate	7.24	6.27
	33	Suberin	1.10	1.14

Figures 3.30 and 3.31 depict the change in the integrals of the solid-state NMR of the bark after extraction with these DESs in comparison to the water washed bark. The individual integrated spectra of the solid residue obtained using these DESs as extraction solvents are shown in the appendix (A50 – A51).

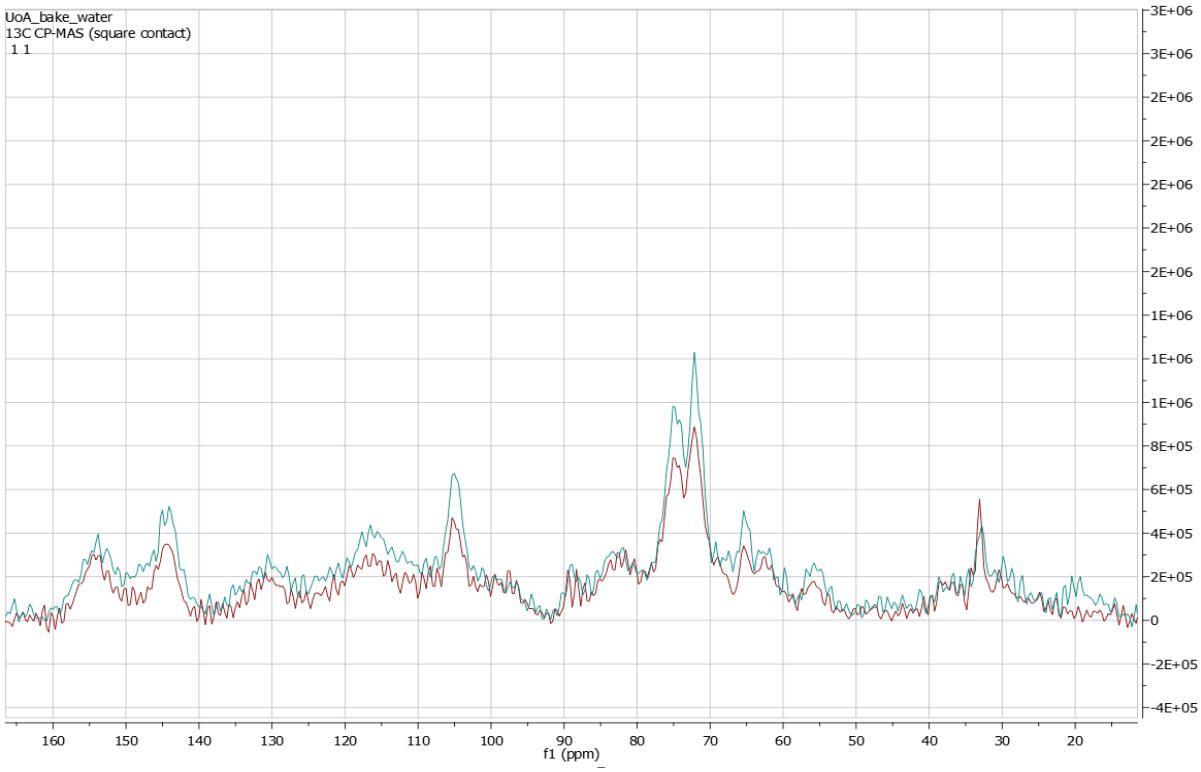


Figure 3.30 *Comparison of the ^{13}C solid-state NMR of water washed bark and GuCl:LA DES solid extract. The red line represents the spectrum of the pure bark after washing with water while the green line represents the spectrum of the GuCl:LA DES solid residue after washing with water.*

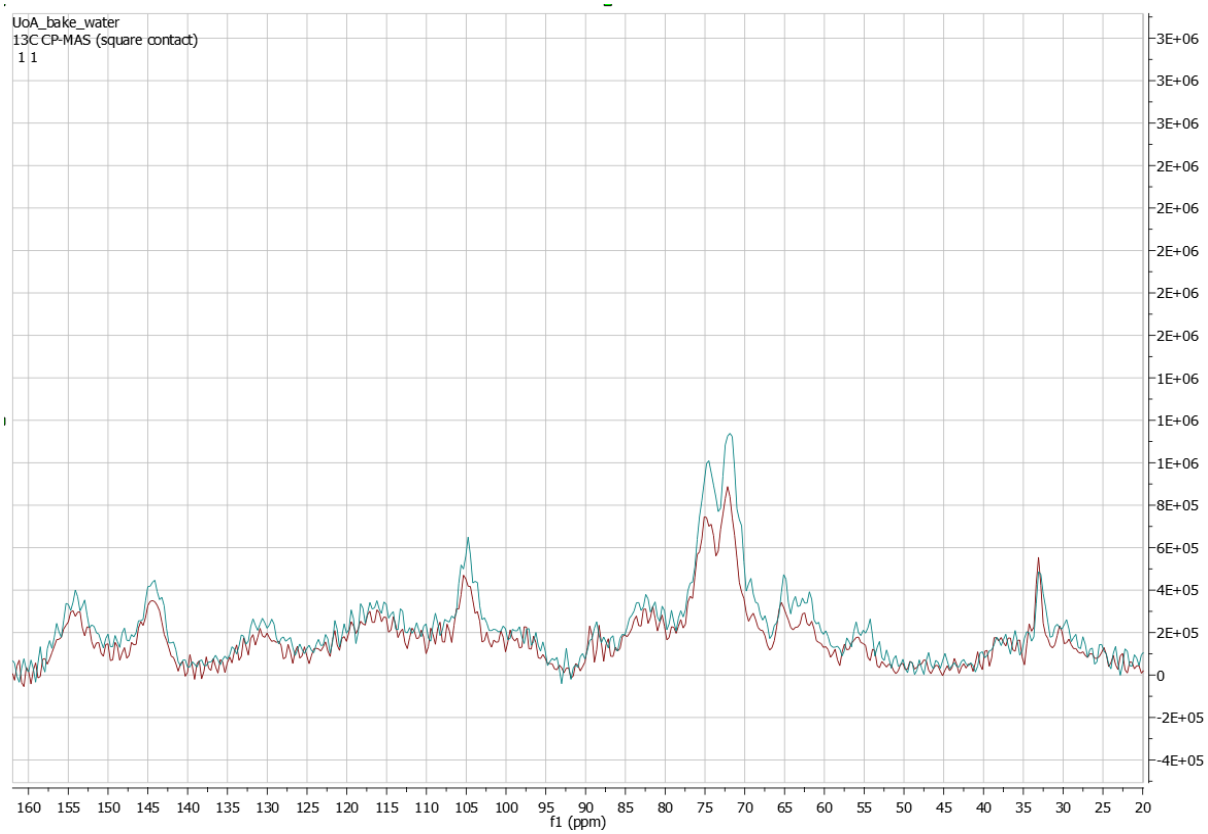


Figure 3.31 *Comparison of the ^{13}C solid-state NMR of water washed bark and GuCl:Gly DES solid extract. Same description of coloured lines as above.*

Table 3.8 shows the relative integrals of the solid-state NMR signals for the $[Me_2PO_4]$-based ILs. With regard to the solid residue obtained following extractions by $[Me_2PO_4]$-based ILs, it was observed the integrals of the peaks assigned to lignin and tannin at chemical shifts of 154 ppm and 144 ppm were notably decreased relative to other residue signals when compared to the water washed bark. This indicates that these components were selectively extracted from the bark which occurred regardless of the cation that was present. This could be due to the strong hydrogen bond accepting nature of anion ($[Me_2PO_4]$-) which interacts with the hydroxyl groups of the lignin and tannins via hydrogen bonding. One of the spectra of $[Me_2PO_4]$-based ILs solid residue, displaying this change is shown in Fig. 3.32, where the red line represents the components of the pure bark after washing with water while the green line represents the components of the $[C_4C_1im][Me_2PO_4]$ IL solid residue after washing with water. The individual integrated NMR spectra for all of the $[Me_2PO_4]$-ILs solid residue are represented in the appendix (A52 – A55).

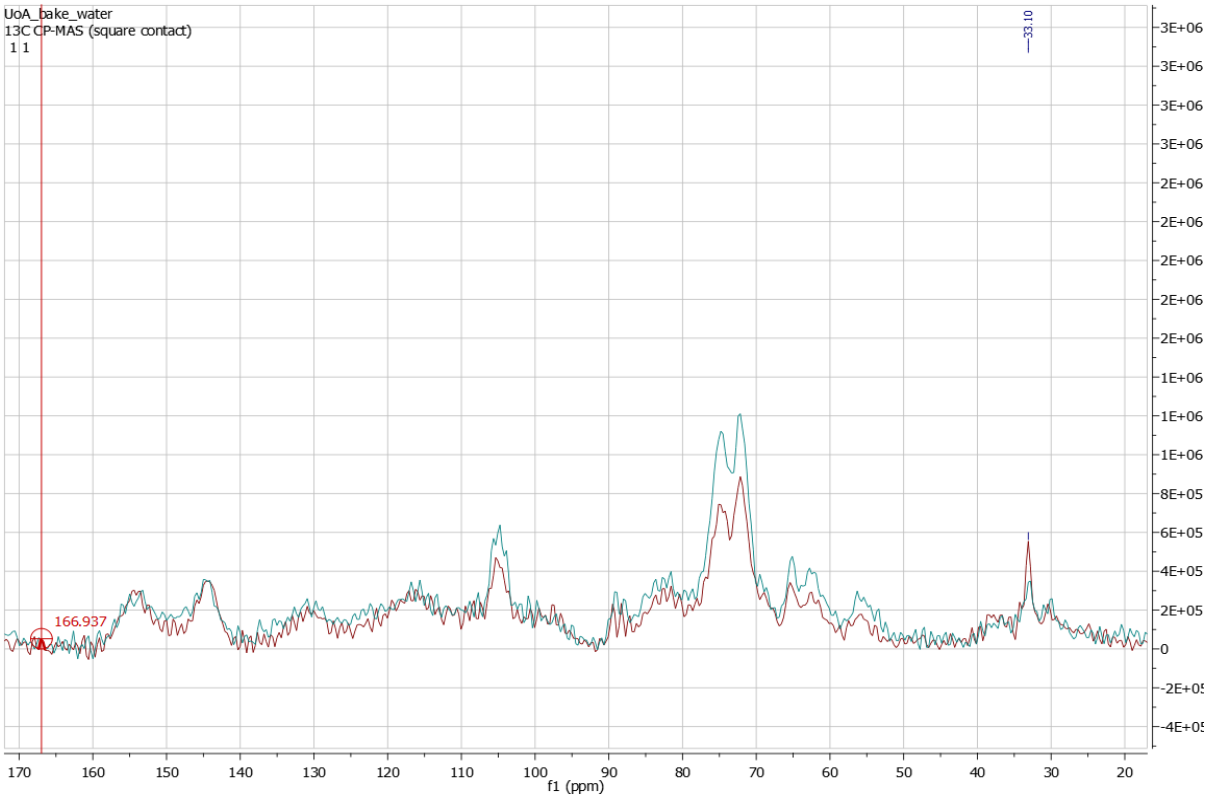


Figure 3.32 *Comparison of the ^{13}C solid-state NMR spectra of water washed bark and the $[C_4C_1im][Me_2PO_4]$ IL solid extract. The red line represents the components of the pure bark after washing with water while the green line represents the components of the $[C_4C_1im][Me_2PO_4]$ IL solid residue after washing with water.*

The change in the relative integral of every component of the solid residue after the extraction by the $[Me_2PO_4]$-based ILs has been represented in Table 3.8. In addition to the apparent decrease of the lignin and tannin peaks, the relative integral of the peak at 33 ppm (suberin or waxes) had declined as well compared to the water washed bark highlighting the potential selective extraction of these components by the IL. Moreover, reducing the hydrophilicity of the IL cation by either using a $[C_nC_1pyrr]^+$ cation or increasing the alkyl chain length to $[C_8C_1im]^+$ was found to increase the proportion of the hydrophobic fragments of the solid residue that were extracted by the IL. This can be seen by the relative integral change of the peak at 33 ppm which is the highest for $[C_4C_1im][Me_2PO_4]$ solid residue (0.51) with similar relative integrals seen for solid residue extracted by ILs with $[C_8C_1im]^+$, $[C_4C_1pyrr]^+$ and $[C_8C_1pyrr]^+$ cations (0.31-0.35). In the all ILs, pyrrolidinium, being a more hydrophobic cation, the hydrophobic components appeared to decrease more than the other components of the bark representing more extraction of waxes or suberin by $[Me_2PO_4]$-based ILs, which was expected.

Although these [Me_2PO_4]-based solvents have often been described as being effective for the dissolution of cellulose[141], it appears they are more effective at extracting hydrophobic compound and lignin since these reduce by a greater proportion than the carbohydrate component.

Table 3.8 *Comparison of the integrals of the components of water washed bark and Me_2PO_4-based ILs solid residue.*

ILs	**δ (ppm)**	**Components**	**Integrals of water-washed bark**	**Integrals of ILs bark residue**
[C_4C_1im][Me_2PO_4]	154	Lignin / Tannin	2.21	1.21
	144	Lignin / Tannin	2.21	1.42
	72 - 75	Carbohydrate	7.24	4.97
	33	Suberin	1.10	0.51
[C_4C_1pyrr][Me_2PO_4]	154	Lignin / Tannin	2.21	0.84
	144	Lignin / Tannin	2.21	1.15
	72 - 75	Carbohydrate	7.24	5.69
	33	Suberin	1.10	0.33
[C_8C_1im][Me_2PO_4]	154	Lignin / Tannin	2.21	1.29
	144	Lignin / Tannin	2.21	1.34
	72 - 75	Carbohydrate	7.24	5.78
	33	Suberin	1.10	0.31
[C_8C_1pyrr][Me_2PO_4]	154	Lignin / Tannin	2.21	1.28
	144	Lignin / Tannin	2.21	1.53
	72 - 75	Carbohydrate	7.24	6.55
	33	Suberin	1.10	0.35

Table 3.9 displays the relative integrals of components observed in the solid residue obtained after extraction by the [$N(CN)_2$]-based ILs with comparison to water-washed bark. The difference in the relative integral change for the bark components is lower for the [$N(CN)_2$]-based solvents than the [Me_2PO_4]-based solvents. From the Figure 3.33 and Table 3.9, it is clear that ILs with the $[N(CN)_2]^-$ anion lead to slight extraction of aromatic components since there was the decrease in the intensity of the peak at the aromatic region (154 ppm and 144

ppm). Moreover, there was also a relative integral change of the peak at 33 ppm which is the highest for $[C_4C_1im]^+$ cation solid residue (1.06) while the least integrals seen for solid residue extracted by ILs $[C_4C_1pyrr]^+$ cation (0.66). This could be attributed that $[N(CN)_2]^-$ anion is a weaker hydrogen bond acceptor than $[Me_2PO_4]$ anion and due to the reduced cation-anion interactions in case of $[N(CN)_2]$-based ILs facilitate effective extraction of bark components. The individual integrated spectra of the $[N(CN)_2]$-based IL solid residue, are shown in the appendix (A56 – A59).

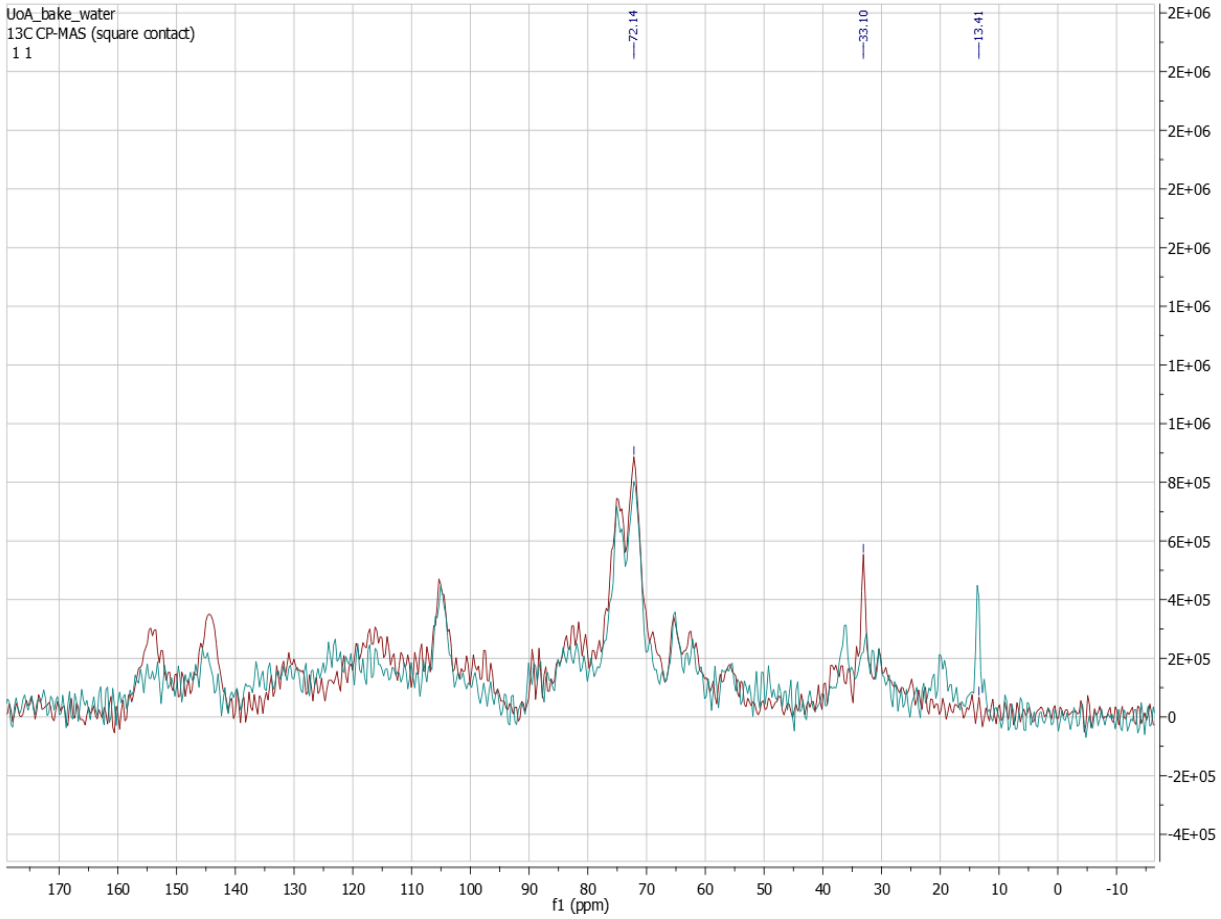


Figure 3.33 *Comparison of the ^{13}C solid-state NMR of water washed bark and $[C_4C_1im]N(CN)_2$ IL solid residue. Same description of coloured lines as above.*

Table 3.9 *Comparison of the integrals of the components of water washed bark and $N(CN)_2$-based ILs solid residue.*

ILs	δ (ppm)	Components	Integrals of water-washed bark	Integrals of ILs bark residue
$[C_4C_1im][N(CN)_2]$	154	Lignin / Tannin	2.21	2.19
	144	Lignin / Tannin	2.21	2.46
	72 - 75	Carbohydrate	7.24	9.60
	33	Suberin	1.10	1.06
$[C_4C_1pyrr][N(CN)_2]$	154	Lignin / Tannin	2.21	1.66
	144	Lignin / Tannin	2.21	1.83
	72 - 75	Carbohydrate	7.24	6.17
	33	Suberin	1.10	0.66
$[C_8C_1im][N(CN)_2]$	154	Lignin / Tannin	2.21	1.87
	144	Lignin / Tannin	2.21	2.14
	72 - 75	Carbohydrate	7.24	7.52
	33	Suberin	1.10	0.91
$[C_8C_1pyrr][N(CN)_2]$	154	Lignin / Tannin	2.21	1.50
	144	Lignin / Tannin	2.21	1.67
	72 - 75	Carbohydrate	7.24	7.35
	33	Suberin	1.10	0.86

3.4.6.2.1. Elemental Analysis on N(CN)2-based solid residue

Unlike the $[Me_2PO_4]$- and $[NTf_2]$-based ILs, however, there were some additional peaks at chemical shifts of 21 ppm and 13 ppm in $[N(CN)_2]$-based ILs. Additionally, there were many peaks present in the IR region 800 – 1600 cm^{-1} as shown in the Figure 3.34 that are not present in the bark itself and in other ILs. The solid residue was also washed with hot water to get rid of IL so that IL not being incorporated in the sample (or at least not much IL) but some additional peaks still appeared in the IR. Hence, IR data were complicated to interpret. This suggests that there is a side reaction occurring between bark and the $[N(CN)_2]$ anion as this was not observed for any other ILs regardless of cation. This could be attributed that

dicyanamide anion might act as a nucleophile for aromatic nucleophilic substitution due to its high polarizability.[142]

Additionally, the colour of the extracted bark residue also changed from brown to black after the extraction, unlike bark extracted by other ILs, which gives more evidence for the occurrence of a side reaction. These unusual observations were figured out using the elemental analysis performed on grounded solid residue.

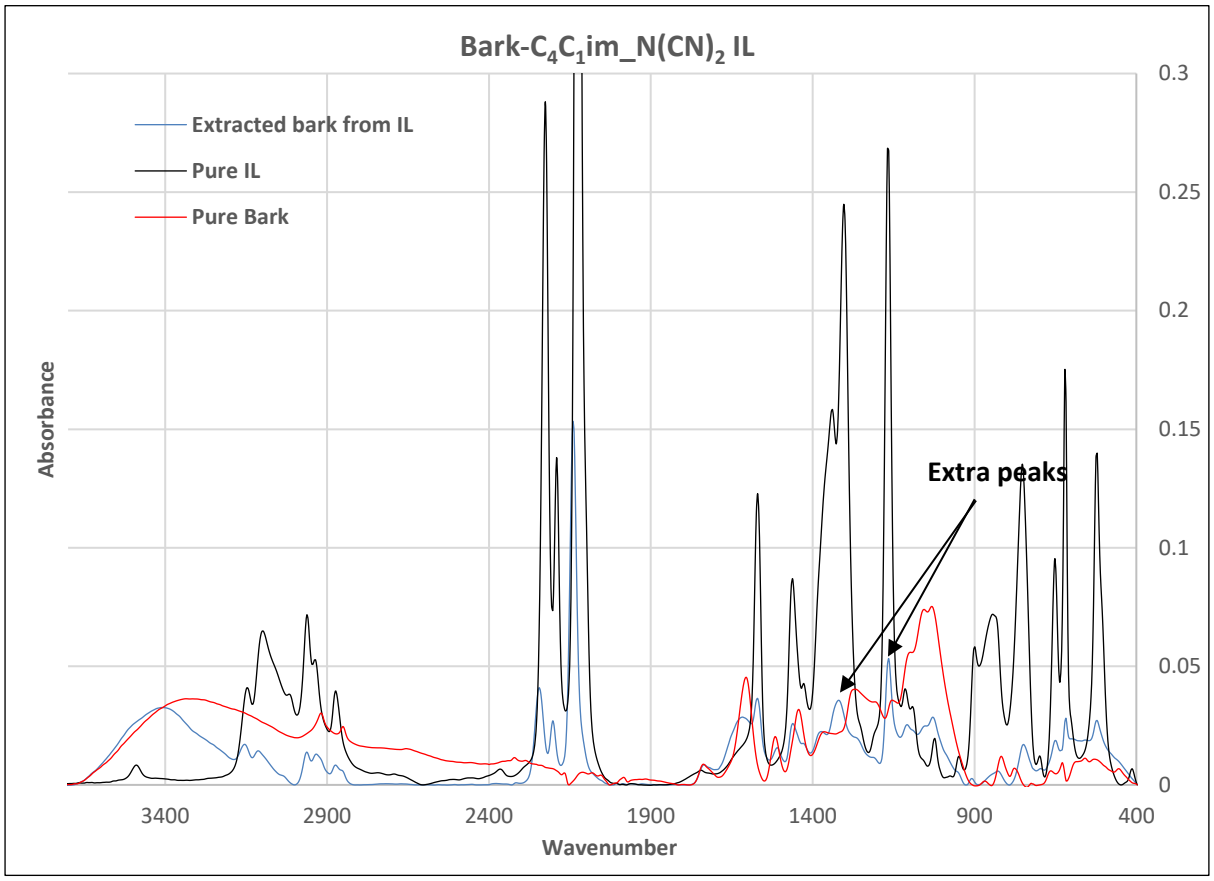


Figure 3.34 *Extra peaks present in [C_4C_1im]$N(CN)_2$ IL extracts.*

The Table 3.10 represents the concentration of carbon, nitrogen and hydrogen present in the solid residue coming from four different ILs and compared with the pure bark source. It had been observed that the pure bark contained only 0.15% N but samples containing of $N(CN)_2$ ILs contained considerably more with up to 6.63% N for [C_4C_1im][$N(CN)_2$] ILs indicating in some sort of side reaction occurring between the of $N(CN)_2$ ILs and bark.

Moreover, in order to confirm further about the side reaction between the ILs and bark, the solid residue from Me_2PO_4-based ILs were also used for elemental analysis. It was observed

that all the Me_2PO_4-based ILs had lower amount of N (0.95 – 1.48%) than $N(CN)_2$-based ILs which further suggests that anion is responsible for the side reaction rather than cation.

Table 3.10 *Elemental analysis results of $N(CN)_2$- and Me_2PO_4- based solid extracts.*

Sample	**N [%]**	**C [%]**	**H [%]**
Pure Bark	0.15	50.3	5.55
$[C_4C_1im][N(CN)_2]$	6.63	52.6	5.92
$[C_4C_1im][Me_2PO_4]$	1.48	49.7	7.42
$[C_4C_1pyrr][N(CN)_2]$	1.11	49.3	5.96
$[C_4C_1pyrr][Me_2PO_4]$	0.97	44.5	6.24
$[C_8C_1im][N(CN)_2]$	3.87	52.4	6.10
$[C_8C_1im][Me_2PO_4]$	1.19	50.0	7.54
$[C_8C_1pyrr][N(CN)_2]$	2.02	51.8	6.35
$[C_8C_1pyrr][Me_2PO_4]$	0.95	50.5	7.68

3.4.7. Pyrolysis-GCMS (Py-GCMS) Analysis

Py-GCMS couples the pyrolysis of a sample with GCMS analysis. Pyrolysis is usually performed at > 400°C and leads to larger molecules being thermally broken up into smaller, more volatile compounds which are then separated by GC and detected by MS.[143] The fragments formed from the pyrolysis are characteristic of their starting materials which allows the non-volatile components of the matrix to be determined. Py-GCMS plays a crucial role in characterising insoluble and complex materials e.g., polymers, cellulose, wood, oil etc. at trace levels often without any sample pre-treatment. The technique helps not only in analysing very small amounts but also providing detailed and unique information about the sample due to the direct sample introduction and the chromatographic separation, which is unobtainable by other analytical techniques.[144] Compared to other techniques, this is actually a more indirect method of obtaining structural information given it depends on the mechanism of degradation of the original components.

The characterisation of the solid residue remaining after bark extraction was performed by pyrolysis GC-MS. The detailed method used for analysis is provided in chapter 6, with

compounds identified by comparison to those in the Agilent MassHunter database. Figure 3.35 displays the % area of the chromatogram corresponding to selected compound. It is important to note that the compounds produced by py-GCMS are not necessarily present in the bark itself but form on exposure to the pyrolysis conditions. In addition, these samples are the solid residue remaining after extraction and so compounds present to a greater extent in these samples represent components that have not been extracted by the IL.

3.4.7.1. Py-GCMS Analysis of the solid extract using ILs

The Py-GCMS of extracted bark (solid residue) from the IL extractions was obtained. The residue was washed with hot water for the $[Me_2PO_4]$- and $[N(CN)_2]$-based IL extracts or with dichloromethane for residue obtained from the $[NTf_2]$-based IL extracts. As mentioned in section 3.4.4, the washing solvent was selected based on the solubility of the ILs. Moreover, pure dried bark was also washed with hot water and dichloromethane to compare with IL-based solid residue to ensure no artifacts were detected arising from the washing step.

Figure 3.35 represents the % area of the furfural, which is an organic compound formed from the pyrolysis of carbohydrates and commonly identified by py-GCMS analysis. This analyte is likely to correlate with the presence of carbohydrates in the solid residue. Although carbohydrates are dissolved due to HBA not HBD ability, the cation might play a role in dissolution of carbohydrate to some extent. In terms of the effect of the cationic structure in the dissolution of carbohydrate, imidazolium based ILs have found to be more capable than the pyrrolidinium based ILs in other studies.[93,124,145] From Figure 3.35, it has been observed that the carbohydrate extracted by the most of the imidazolium based ILs with Me_2PO_4 and $N(CN)_2$ have less proportion of furfural than the pure bark except the $[C_4C_1im][NTf_2]$ (14.2%).

However, according to the Table 3.1, the β value of $[C_4C_1im][Me_2PO_4]$ and $[C_4C_1pyrr][Me_2PO_4]$ ILs is similar, 1.13 and 1.14, along with imidazolium cation, pyrrolidinium-based ILs also appeared to act as a good solvent for carbohydrate such as cellulose which is also consistent with the solid-state NMR results. It appears that the competition with other solutes makes imidazolium based ILs less effective for cellulose/carbohydrate than other components.

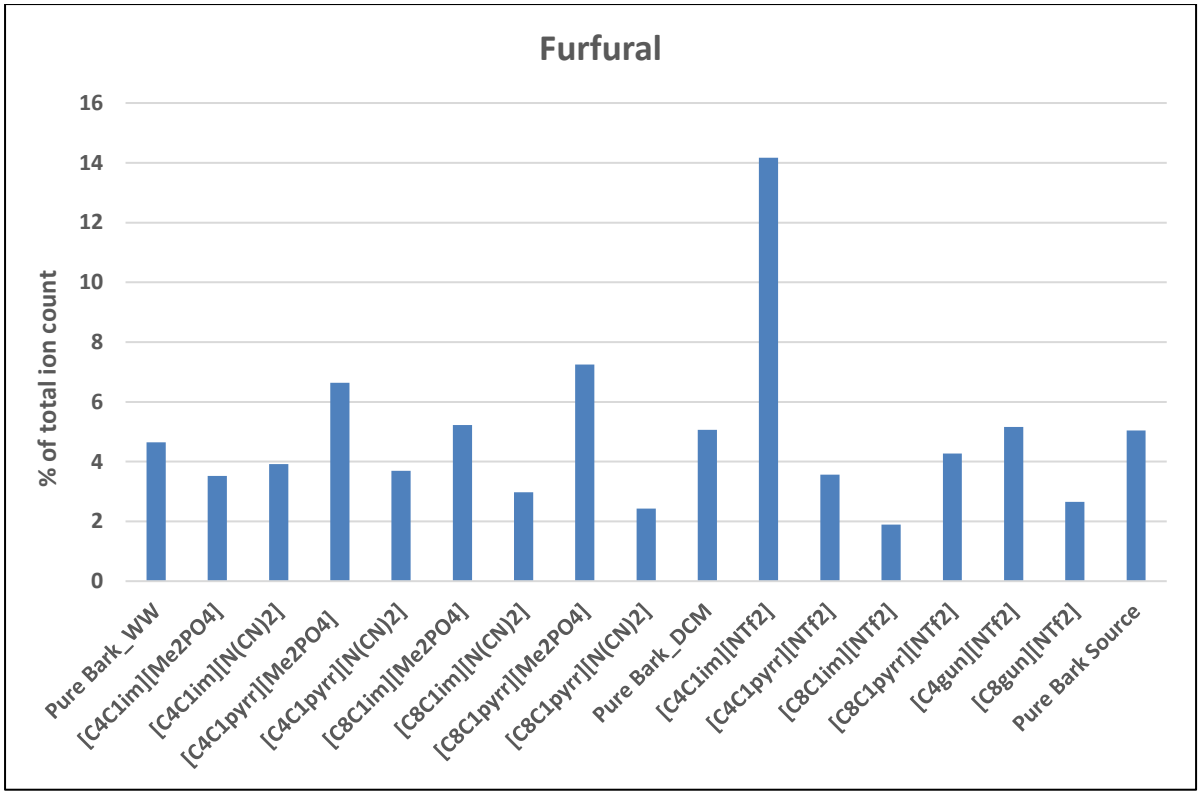


Figure 3.35 *Representing a volatile compound eliminated from solid residue. (Pure bark_WW stands for the bark obtained after water wash and Pure bark_DCM stands for the bark obtained after dichloromethane wash)*

In addition, guanidinium cation is a strong hydrogen bond donor than imidazolium and pyrrolidinium cation but the incorporation of the weakly acceptor anion (NTf_2^-) might be the reason for the less extraction by the $[C_4gun][NTf_2]$ and $[C_8gun][NTf_2]$ ILs of the carbohydrates.

Regarding the effect of the anion, the hydrogen bond accepting ability of the anion has been found to be the most important determinant in the solubility of carbohydrates by ILs.[124] Therefore, due to the strong hydrogen bond accepting ability of the $[Me_2PO_4]^-$ anion, the extraction of the carbohydrate would be expected to be higher in case of the $[Me_2PO_4]$-based ILs than $[N(CN)_2]^-$ and $[NTf_2]^-$ containing ILs. According to Figure 3.35, it was observed that the least % area was shown by the $[C_8C_1pyrr][NTf_2]$ IL and $[C_8C_1pyrr][N(CN)_2]$ representing the highest proportional extraction of the carbohydrate. A valid explanation behind this outcome is not known yet but it is possible that there are other components in the bark that catalyse the formation of this pyrolysis product and those have been extracted by the $[NTf_2]$ and $[N(CN)_2]$- ILs as pyrolysis GC-MS is an indirect technique.

Additionally, it had been evaluated that the solubility of carbohydrates in ILs decreases with an increase of the alkyl chain on the cation[145] which is also in agreement with the observed result as shown in the Table 3.11 except the difference in the % area of the $N(CN)_2$-based ILs. The reason behind this could be the unusual reactivity displayed by the $N(CN)_2$ anion in the extraction process.

Table 3.11 *The variation in the % area of the proportion of the furfural present in the solid residue by increasing the alkyl chain of the cation of the ILs.*

Structural Formula of ILs	**% area**
$[C_4C_1im][Me_2PO_4]$	3.52
$[C_8C_1im][Me_2PO_4]$	5.23
$[C_4C_1im][N(CN)_2]$	3.92
$[C_8C_1im][N(CN)_2]$	2.98
$[C_4C_1pyrr][Me_2PO_4]$	6.64
$[C_8C_1pyrr][Me_2PO_4]$	7.52
$[C_4C_1pyrr][N(CN)_2]$	3.69
$[C_8C_1pyrr][N(CN)_2]$	2.43

It is apparent from the Figure 3.36 that dehydroabietic acid or diterpenoid product was also observed in the Py-GCMS chromatogram from the raw bark (up to 0.16%). This acid is the main component of pine resin obtained from *Pinus Radiata*.[146] Comparing within ILs, dehydroabietic acid, on average, was a major component of moderately acidic cation $[C_4C_1im]^+$ than weekly and strongly acidic cation $[C_4C_1pyrr]^+$ and $[C_4gun]^+$. Moreover, the least amount of dehydroabietic acid, which is 0.01 than original bark (0.09 % with dichloromethane wash and 0.16% with water wash), present in $[C_4C_1pyrr][NTf_2]$ solid residue displaying the efficient extraction capability of a hydrophobic cation present in the IL. It was also observed that with the increase in the after $[C_4C_1pyrr][NTf_2]$ IL, $[C_ngun][NTf_2]$ ILs (0.03 %) appeared to extract the dehydroabietic acid to some extent. This is again due to the strong hydrogen bond donating ability of the guanidinium cation but the incorporation of the weakly acceptor anion (NTf_2^-) might be the reason for the slightly less extraction than $[C_4C_1pyrr][NTf_2]$ IL. Moreover, the highest proportion of the given compound (2.52%) was found in $[C_8C_1im][Me_2PO_4]$ IL this is because it might be more effective at extracting other components than dehydroabietic acid.

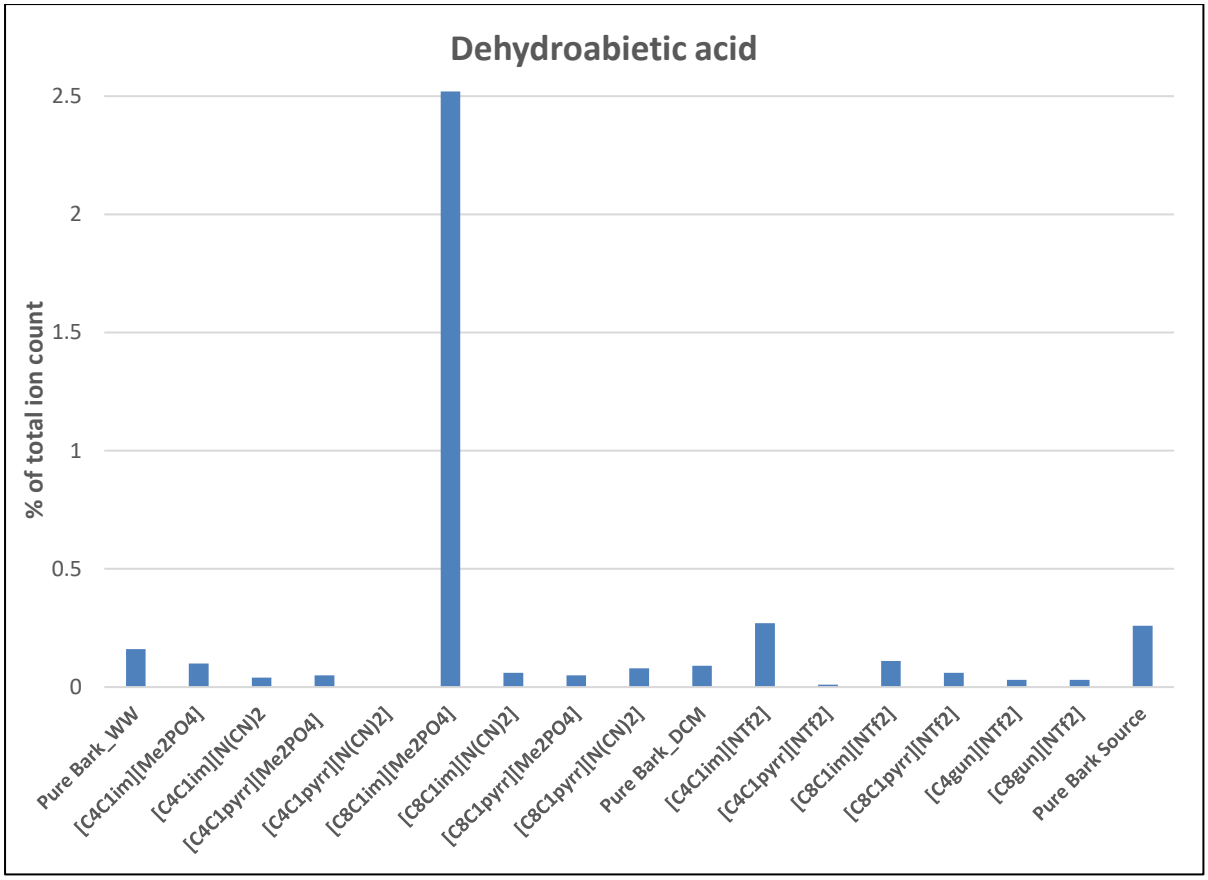


Figure 3.36 *Representing a volatile compound eliminated from solid residue.*

3.4.7.2. Py-GCMS Analysis of the solid extract using DESs

The Py-GCMS of extracted bark (solid residue) from the DES extractions was obtained. The residue was washed with hot water for both of the GuCl:LA and GuCl:Gly DESs and the washing solvent was selected based on the solubility of the DESs. Moreover, pure dried bark was also washed with hot water to compare with DES-based solid residue to ensure no artifacts were detected arising from the washing step.

The Figure 3.37 depicts that the DESs are capable of extracting the products which come from the pyrolysis of the lignocellulosic biomass and long chain fatty acids. It has been evaluated that even though the difference in the % area of the extracted products between these two DESs was less, the GuCl:LA DES appeared to be efficient in the depolymerisation of carbohydrate (furfural (2.69%)), and lignin or tannin (phenol, 4-ethyl-2-methoxy- (3.67%)) and waxes (docosanoic acid (3.28%)) components than GuCl:Gly DES . This could be due to the strong hydrogen bond donating capability of GuCl:LA DES than GuCl:Gly DES (Chapter 2). Additionally, the amount of long chain fatty acid such as docosanoic acid was decreasing

significantly due to the incorporation of the guanidinium cation in the DESs. This can be assumed that being a strong hydrogen bond donor, guanidium cation showing immense effect on the extraction of FAME.

It has been determined that DESs seem to be more effective at removing sugars/polysaccharides than aromatics and waxes which is also consistent with the solid-state NMR results.

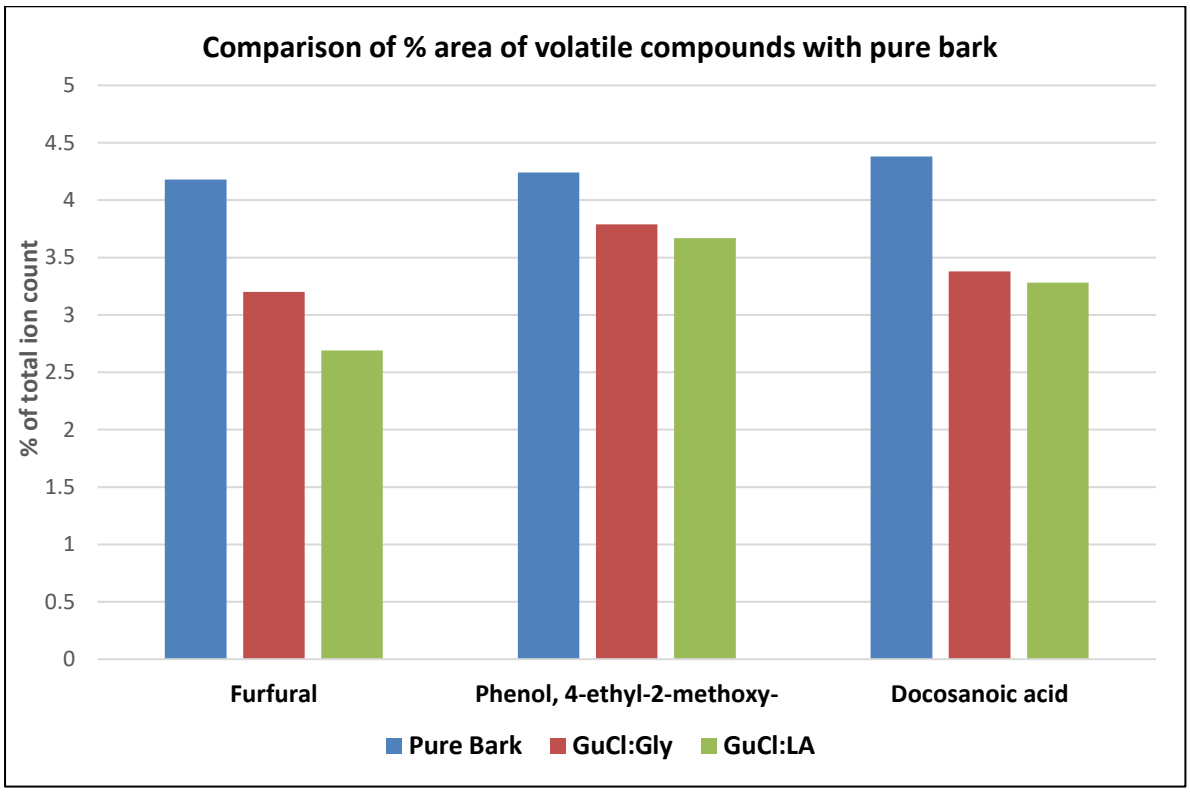


Figure 3.37 *Representing volatile compounds eliminated from solid residue.*

3.4.8. Tetramethylammonium hydroxide Py-GCMS (TMA-OH-Py-GCMS)

Long chain fatty acids are not amenable to GCMS analysis, so in addition to Py-GCMS without derivatisation, samples were also analysed following tetramethylammonium hydroxide (TMA-OH) derivatisation to improve the observation of the fatty acid components. TMA-OH in methanol is a thermochemolysis reagent which acts as a transesterification agent, hydrolysing and methylating fatty acids to generate fatty acid methyl esters (FAME) that are more volatile and readily detectable by GC-MS.[147] This method is often used to characterise long chain fatty acids.

Figure 3.38 represents that the quantity of highly substituted benzenes which most likely arise from lignin or tannin, has decreased in the solid residue from most of the ILs in comparison to original bark source. The proportion of simpler aromatics such as benzene, 1-methoxy-3-methyl- was found to increase in most of the ILs. This can be attributed to the preferential extraction of highly substituted aromatics lignin and tannin components by the ILs or, perhaps more likely, the depolymerisation of these polymers during the extraction process. It is also possible that the IL breaks down the cellular structure of the bark which affects the pyrolysis and hence the distribution of aromatic by-products that form.

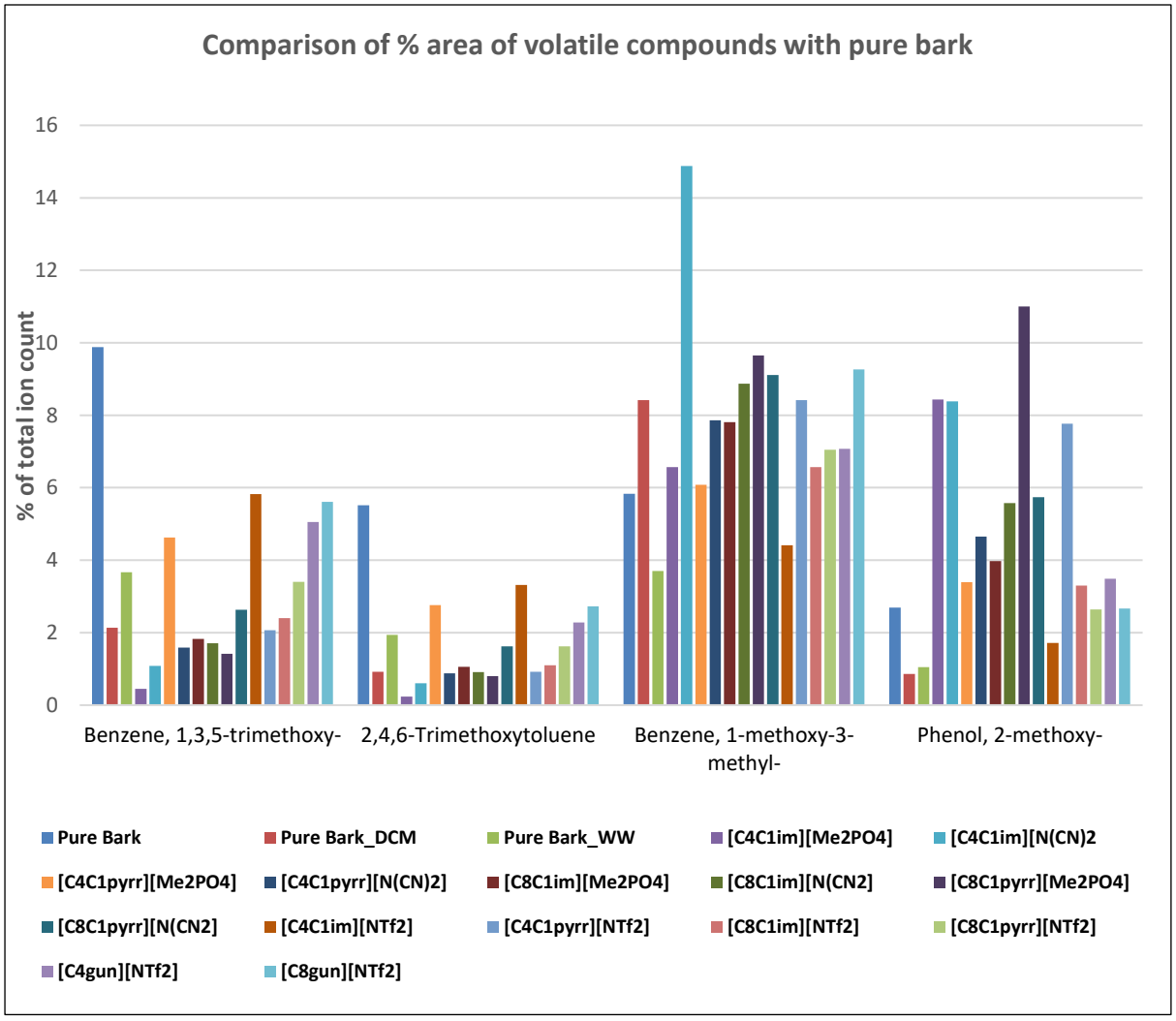


Figure 3.38 *Volatile compounds produced by solid residue under pyrolysis GC-MS conditions using TMA-OH as a derivatising agent.*

Additionally, it has also been determined from the data of TMA-OH-Py-GCMS analysis that among all the ILs [Me_2PO_4]-based ILs are more efficient in the extraction of lignin and tannin fractions due to its strong hydrogen bond accepting ability (refer to Figure 3.38). The % area

of the highly substituted compounds namely, 1,3,5-trimethoxy- benzene and 2,4,6-trimethoxytoluene are less in proportion in the $[C_4C_1im][Me_2PO_4]$ IL while the proportion of simple aromatics 2-methoxyphenol and 1-methoxy-3-methyl- benzene are less in proportion in $[C_4C_1im][NTf_2]$ followed by the $[C_4C_1pyrr][Me_2PO_4]$. In case of $[C_4C_1im][NTf_2]$ IL, the cation $[C_4C_1im]^+$ could play a role in the extraction process but due to the presence of weak hydrogen bond acceptor $[NTf_2]^-$ the reason behind for the extraction is not so sure yet although the results obtained so far are in favour of the solid state NMR results.

Amphiphilic domain of the ILs may play a role in the extraction of polyphenols. It has been observed that by increasing the non-polar domain of the cation, the extraction of the compounds by some ILs decreases.

The extraction of some long chain fatty acids was also displayed by the ILs. The amount of FAME such as hexacosanoic acid, methyl ester and 1,3-benzenedicarboxylic acid, dimethyl ester is decreased significantly in $[C_4C_1pyrr][Me_2PO_4]$ IL as shown in the Figure 3.39. This suggests that the pyrrolidinium cation, being hydrophobic in nature, enables this IL to successfully extract wax/suberin components. This is also in agreement with the solid-state NMR results and the extraction yield obtained which might suggest that this IL is able to extract lignin, tannins along with some suberin fragments.

In addition, regarding guanidinium cation based ILs ($[C_ngun][NTf_2]$), these ILs are not that efficient in extracting long chain fatty acids as expected due to the incorporation of the weak hydrogen bond acceptor $[NTf_2]^-$ with the strong hydrogen bond donor $[C_ngun]^+$.

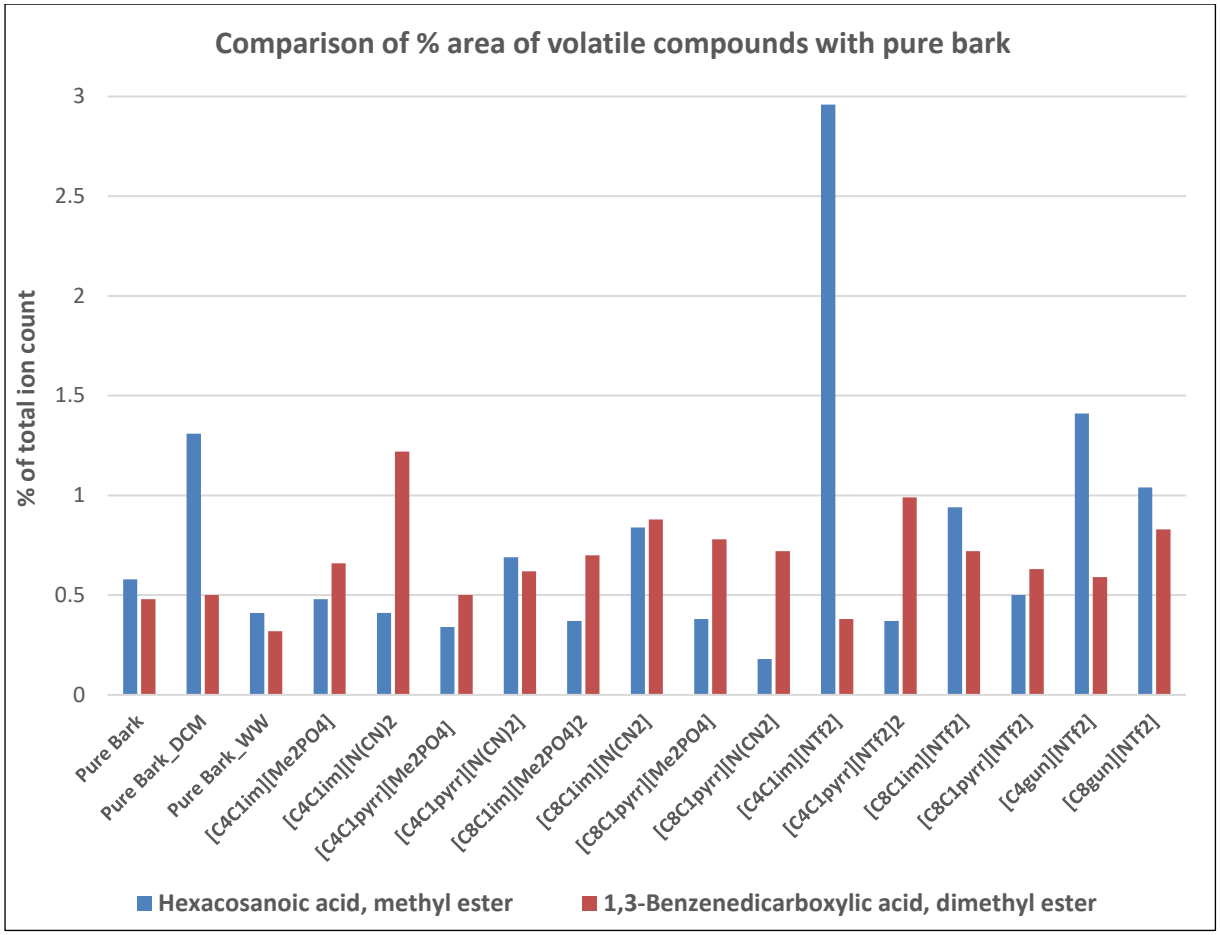


Figure 3.39 *Volatile compounds produced by solid residue using TMA-OH as a derivatising agent.*

3.5. Summary

The extraction of bark by a range of ILs that vary systematically in terms of their hydrogen-bonding capabilities has been explored alongside selected DES based on the work described in Chapter 2. These extracts have been characterised by a range of analytical techniques including UV-Vis, IR, NMR (solid and solution state), HS-SPME-GCMS and pyrolysis GC-MS. It has been evaluated that the anion governs extraction yield, with $[Me_2PO_4]^-$ generally the most effective anion of those explored. The DES systems were found to be competitive with the IL systems for overall extractions efficiency. ILs seem to be more effective at removing aromatics and waxes than sugars/polysaccharides as given in the solid-state NMR and pyrolysis GCMS results. This was even true for $[Me_2PO_4]^-$ ILs which have been described previously as being effective for the solubilisation of polysaccharides, highlighting that the competition arising from components in the complex bark matrix can affect the extraction outcomes.

In terms of the effect of the cation, it has been determined that $[C_nC_1pyrr]^+$ cations facilitate the preferential extraction of the hydrophobic components grafted with strong hydrogen bond accepting anion (Me_2PO_4). In contrast, imidazolium cation based ILs are more efficient in extracting lignin and tannin fractions. The proportion of carbohydrates seem to be extracted by the guanidinium cation based DESs.

In terms of the effect of the alkyl chain grafted on the cations, the solubility of the carbohydrate and lignin or tannin components depend on the alkyl chain length and the dissolution of these fractions decreases with increasing non-polar domain of the cation. The extraction of hydrophobic compounds such as waxes and suberinic material tends to increase with increasing alkyl chain length.

In summary, with respect to the ability to obtain the highest extraction yield and range of components extracted, the $[Me_2PO_4]^-$ anion appears to be the most suitable. For the DESs, GuCl:LA was found to be more efficient than GuCl:Gly. A key next step is how this extraction efficiency can be linked with fractionation and recovery of the resultant extract. This will be explored in the next chapter.

4. Recovery and isolation of *Pinus Radiata* bark extracts

From Chapter 3, it has been determined that among the three IL anions that were explored, the $[Me_2PO_4]^-$ anion appeared to be the most effective for the extraction of a wider range of components than the $[N(CN)_2]^-$ and $[NTf_2]^-$ anions. For the DESs explored, GuCl:LA was found to be more efficient than GuCl:Gly. Additionally, although both ILs and DESs found to be capable of extracting key components of bark, ILs appeared to be more successful at extracting aromatics and waxes while guanidinium cation based DESs appeared to remove carbohydrate (sugars/polysaccharides). It was also evaluated that the pyrrolidinium cation could extract the hydrophobic components significantly, at least when paired to sufficiently strong hydrogen bond accepting anion, while imidazolium cations, on the other hand, are more effective at extracting carbohydrate, lignin and tannin fractions.

To sum up, both cations and anion played a vital role in extracting components of bark, meaning that the selectivity of the extracts obtained changed depending on the solvent used. It is necessary to explore if these fractions, extracted by the solvents, can be selectively recovered and isolated from the solvents. While multiple approaches could be used to isolate the extracts, one of the most feasible to perform on a larger scale is antisolvent addition. There have been many investigations performed exploring the regeneration, purification and separation of compounds from ILs using antisolvents.[87,148,149] Similarly, this approach could also be useful in isolating and recovering readily quantifiable 'classes' of extracts of bark (e.g. suberin, cellulose/hemicellulose, tannins/polyphenols/other polar small molecules extractives, terpenes/other nonpolar small molecules extractives, lignin) which are specific enough to be of interest. Moreover, the recovery process itself could be used to improve the selectivity of the extraction. For example, the choice of antisolvent may influence the nature of the compounds that separate from the IL or DES.

As the $[NTf_2]^-$ ILs have been determined to have negligible efficacy as extraction solvents, these ILs have been excluded from this approach. As an IL currently of commercial interest for biomass extraction processes, $[TEA][HSO_4]$ has been explored alongside the other ILs and DESs discussed in chapter 3 as a comparator. Extracts used for the antisolvent addition have been obtained using the same conditions as those used for the comparison in chapter 3, with the extraction taking place at100°C for 24 h.

4.1. The Antisolvent Addition Approach

An antisolvent is a liquid that causes a solute to separate when added to a solution. In this case the solute represents the compounds extracted from the bark and the solution is the IL extract. The antisolvent reduces the solubility of the solute which causes it to precipitate or separate as an immiscible liquid phase, allowing it to be easily separated and characterized.[150]

Different antisolvents (water, acetonitrile (MeCN) and dimethyl sulfoxide (DMSO)) were used to isolate bark components from IL extracts to identify if this process could lead to their selective separation. These antisolvents were selected on the basis of their polarity and the nature of their interaction with the ILs and DESs. Water was selected due to its polarity and strong hydrogen bond donation ability. DMSO was selected as a polar aprotic solvent which is a strong hydrogen bond acceptor and MeCN was used as a dipolar solvent with less effective specific interactions with the ILs and DESs.

4.1.1. General Method

To test the effect of antisolvent addition, small amounts of antisolvent were added sequentially, with any solid formed isolated and characterised separately to see if the quantity of antisolvent added influenced selectivity. The first step was the addition of 0.5 mL of antisolvent to 1 mL of the IL extract (referred to as Sample 1). Any suspended solids formed in Sample 1 were settled by centrifugation for 5 – 10 min. The Sample 1 supernatant was separated from the solids and the solids were washed with the antisolvent used for their separation. Further antisolvent (0.5 mL) was added to the Sample 1 supernatant after the solids had been removed, which was denoted Sample 2. This process continued with increasing amounts of antisolvent with any solids formed separated and characterised. The solids formed were characterised using IR. Based on the initial IR outcomes, the antisolvent addition process was further refined and scaled up for each solvent to enable its further characterisation using solution-phase NMR and solid-state NMR. The outcome of these antisolvent additions will be explored by IL or DES in the section below.

4.1.2. Recovery/isolation of components

The primary method for the initial characterisation of precipitates was IR. This was performed by using the relative maximum absorbance of the peaks of IR. While relative absorbance does not provide a direct quantitative measure of the amount of each component, it is useful for

determining changes in the proportion of compounds in each extract. These proportions are defined with respect to the reference peak. The calibration data was collected using IR studies with the representative components such as lignin (kraft lignin), tannin (tannic acid), suberin (decanoic acid) and carbohydrates (cellulose microcrystalline). This was done in order to get an idea of the proportion of each component and compared with the relative absorbance obtained from pure bark. From the IR, the specific components that had been isolated could be classified by group with anticipated IR signals for suberin (S), lignin (L), tannins (T) and carbohydrates (C) being present at 2919 cm^{-1}, 2850 cm^{-1}, 1604 cm^{-1}, 1512 cm^{-1} and 1033 cm^{-1}, respectively[98,127] as discussed in Chapter 3, section 3.3. As there appeared to be evidence of the lignin peak at 1605 cm^{-1} in all of the isolated samples this peak was selected as the reference peak for all of the samples analysed. It is important to acknowledge that some extractives will exhibit overlapping absorbances with these structural subclasses. For example, waxes also contain significant C-H stretching absorbances in the same IR region as suberin (2919 cm^{-1}, 2850 cm^{-1}). Other polyphenols such as flavonoids also show aromatic ring-breathing signals[151] although these tends to be at slightly lower wavenumber absorption compared to lignin and tannins (1448 cm^{-1}). Table 4.1 represents the relative absorbance obtained from the bark and mixture of components. Although the bark constituents are complex to interpret due to mixed fractions present in the bark, the underlying ratios could still help give an approximate idea of the proportion of the components. Based on the model compounds that were used in a 1:1:1:1 mass ratio, it appears that the 'suberin' type absorbances are much weaker whereas the carbohydrate signals have a higher extinction coefficient which should be considered in the interpretation of the subsequent results.

Table 4.1 *The relative absorbance maxima of representative components in a mixture of suberin, tannin, carbohydrate and lignin.*

Sample	Absorbance		
	S : L	**C : L**	**T : L**
Mixture of S + L + C + T (1 : 1 : 1 : 1)	0.19, 0.08	1.85	0.55
Pure Bark	0.27, 0.18	1.54	0.45

4.1.2.1. Addition of different antisolvents to the $[C_4C_1im][Me_2PO_4]$ IL extract

The relative absorbance of the IR peaks of isolated solids obtained from the sequential addition of water as an antisolvent to $[C_4C_1im][Me_2PO_4]$ extracts are shown in Table 4.2 using IR data displayed in Figure 4.1. According to data obtained for $[C_4C_1im][Me_2PO_4]$ extract, it was observed that with the addition of 1.5 mL of water in 0.5 mL increments to 1 mL of extract (samples 1, 2, & 3), lignin and carbohydrate components were precipitated along with a small proportion of hydrophobic components (waxes or suberin). This was evident from the intensity of the suberin, lignin and carbohydrate peaks compared with those in the pure bark (Table 4.2 and Figure 4.1). As discussed in the previous section, the ratio of the absorbance in the IR was obtained with respect to lignin (S:L, T:L, and C:L) as the IR peak of aromatic lignin at 1605 cm^{-1} was common in all extracts. In addition, in Table 4.2, there were two intense peaks (2919 cm^{-1}, 2850 cm^{-1}) for hydrophobic components and they both were used for relative absorbance ratio (S : L). It is important to note that these ratios are proportional to be not equivalent to the actual concentration of each component as the extinction coefficients for each of these peaks are likely to be very different.

Although there is a massive baseline artifact for sample 4, adding a further 2 mL water in 1 mL increments to the supernatant (samples 4 & 5) allowed the separation of a good proportion of hydrophobic components as the intensity of the absorbance is very high compared to the pure bark and few other peaks were observed. When more water was added to the filtrates, there was no evidence of the presence of any further wax/suberin components nor carbohydrates. Hence, after sample 5 only the lignin and tannin components were isolated from the extract i.e., Samples 6 – 10. The rest of the IR spectra of the samples are shown in the appendix A60 – A62.

These results demonstrate that carbohydrates tend to be recovered selectively with only small amounts of added antisolvent. The reasonably high initial concentrations of these are due to the ability of the $[Me_2PO_4]^-$ anion for the dissolution of cellulose and other carbohydrates.[141] The poor solubility of these components in other media leads to their relatively rapid precipitation on addition of an antisolvent. The precipitation of hydrophobic compounds (waxes, suberins) on the addition of small amounts of water is due to their poor solubility in aqueous media. Meanwhile, tannins and lignin are more soluble in water than suberin or waxes

which accounts for their more gradual precipitation on the addition of significant amounts of added antisolvent.

Overall, 14.2 mg of solids were recovered from 1 mL of the IL extract which is almost equal to the amount calculated based on the extraction yields reported in Chapter 3 (16.5 mg).

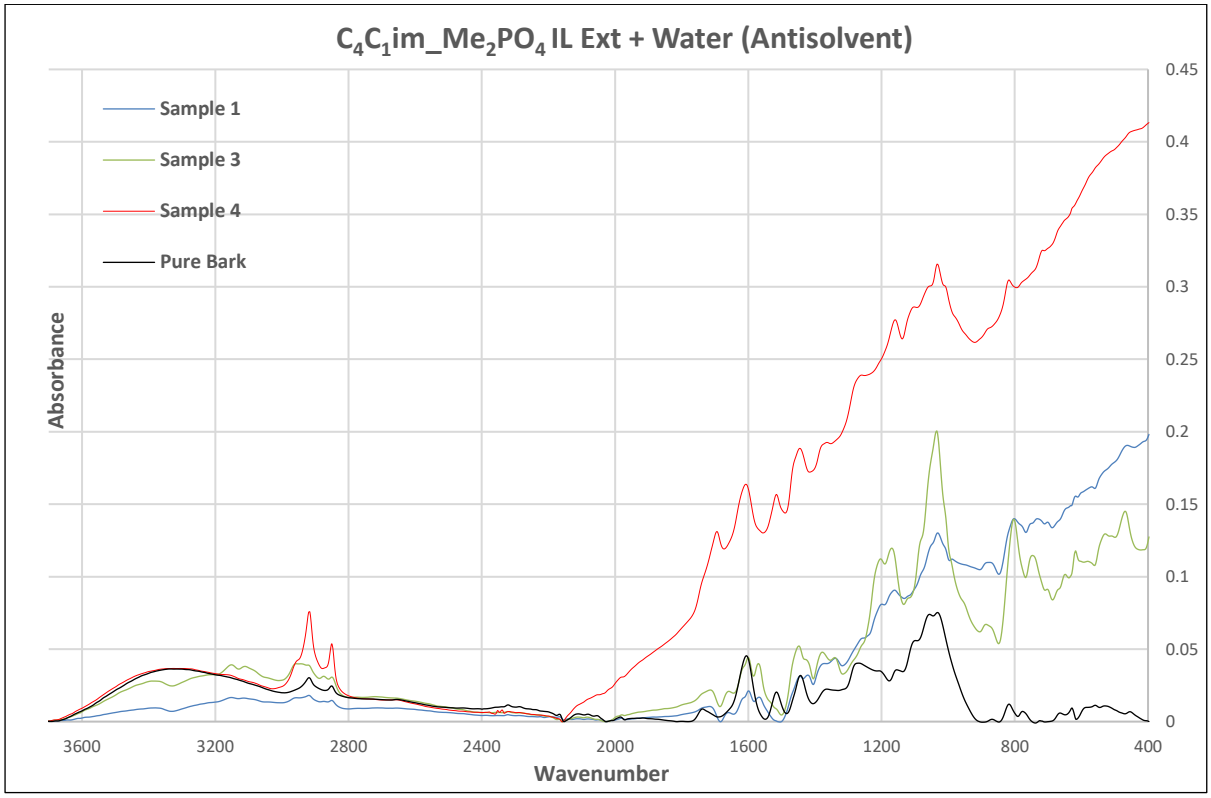


Figure 4.1 *IR spectra of the solids formed from Sample 1, 3 and 4 using 1 mL of [C_4C_1im][Me_2PO_4] IL extract and water as an antisolvent as well as the IR spectrum of pure bark.*

Table 4.2 *The relative absorbance maxima of bark components isolated from 1 mL of $[C_4C_1im][Me_2PO_4]$ IL extract with the addition of water as an antisolvent.*

Sample Labels	**Water (mL)**	**Absorbance**			**Extract (mg)**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27, 0.18	1.54	0.45	-
Sample 1	0.5	0.93, 0.75	6.70	*	3.7
Sample 2	0.5	0.77, 0.56	6.25	*	2.4
Sample 3	0.5	NS	4.55	*	1.0
Sample 4	0.5	0.46, 0.33	*	0.96	1.2
Sample 5	0.5	0.38, 0.27	*	0.96	0.9
Sample 6	0.5	*	*	0.86	0.7
Sample 7	0.5	*	*	0.75	0.3
Sample 8	1	*	*	0.76	1.0
Sample 9	1	*	*	0.90	0.6
Sample 10	1	*	*	0.72	0.2
Sample 11	5	*	*	0.73	1.1
Sample 12	5	**	**	**	0.6
Sample 13	5	**	**	**	0.5
Overall	**21.5**				**14.2 (86.1%)**

*(NS -Not sure i.e., the intensity of the peak is unusual as depicted in Figure 4.1, * the peaks do not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis)*

The characterisation of extracts with the addition of antisolvent was also performed on an increased scale (10 mL of IL extract) as represented in Table 4.3. The reason for the scaling up of the antisolvent addition to the IL extract was to improve the analysis of the extract formed as the scaled-up version would produce larger quantities of each sample produced. To explore in greater depth the different stages of antisolvent addition, the 'samples' were combined based on the relative proportion of each of the components observed to maximise the quantity of recovered solid that could obtain. The first step was the addition of 15 mL of antisolvent to 10 mL of the IL extract (referred to as Sample A). Any suspended solids formed in Sample A were settled by centrifugation for 5 – 10 mins. The Sample A supernatant was separated from the solids and the solids were washed with the antisolvent used for their separation. Further antisolvent (10 mL) was added to the Sample A supernatant after the solids had been removed,

which was denoted as Sample B. This process continued with increasing amounts (75 mL) of antisolvent with any solids formed separated (Sample C). The extracts were characterised using IR, with the IR spectra shown in the appendix (A63 – A65) and the relative absorbances summarised in Table 4.3.

For the scaled-up samples, 160.5 mg of solid was recovered from 10 mL of the IL extract which is almost equal to the amount of the bark that would be expected to be recovered based on the extraction yield (165 mg). This consists of about 64.1 % of mixed fractions including suberin, tannin, lignin and carbohydrates and 35.9 % of pure lignin and tannin fractions.

Table 4.3 *Relative absorbance from the IR spectra of bark components isolated from 10 mL of $[C_4C_1im][Me_2PO_4]$ IL extract with the addition of water as an antisolvent.*

$[C_4C_1im][Me_2PO_4]$ IL extract	**Addition of water (mL)**	**Absorbance**			**Solid recovered (mg)**	**Relative Sample Yields (%)**
		S : L	**C : L**	**T : L**		
Pure Bark	-	0.27,0.18	1.54	0.45	-	
Sample A	15 mL	0.63,0.36	2.67	0.46	102.9	64.11
Sample B	10 mL	*	*	0.56	14.7	9.16
Sample C	75 mL	*	*	0.52	42.9	26.73
Overall	**100 mL**				**160.5 (97.3%)**	**100**

(the peaks do not appear in the IR)*

According to the ^1H NMR spectra of Sample A, B and C (Figures 4.2, 4.3 and 4.4), it has been evaluated that one peak was common in all of the samples at a chemical shift of 1.25 ppm, corresponding to the CH_2 of the aliphatic chain of suberin or wax fragments. The intensity of the absorbance of the peak was higher in case of Sample A than Sample B and started declining with the dilution of the sample and least intense for the Sample C. In addition, broad aromatic signals were observed at a chemical shift of approximately 6.69 ppm for Sample B and C, most likely due to the presence of lignin, tannins and other phenolics. These peaks yielded greater relative intensity for C than B as shown in Figure 4.2 and 4.3. There were also some IL peaks that appeared even after washing the extracts 5-6 times which might be due to mass transfer limitations due to the hydrophobicity of the solid making it difficult for water to penetrate

during the washing steps to remove the residual IL. Moreover, it should be noted that DMSO was used as a solvent to dissolve precipitated bark extracts obtained after antisolvent additions to IL extracts but these solvents were partially soluble in the solvent so ^{1}H NMR data sets are impacted by the solubility issue and hence some IL peaks were observed.

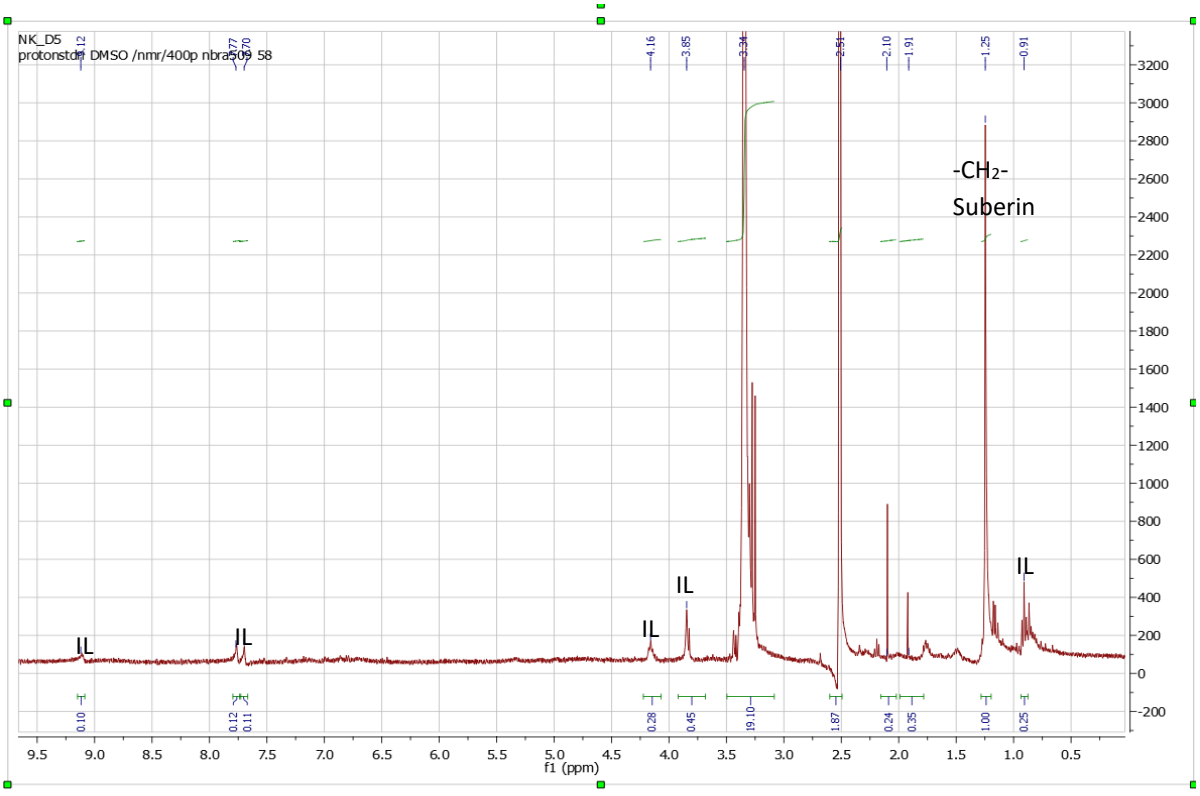


Figure 4.2 *^{1}H NMR of Sample A formed using 10 mL of [C_4C_1im][Me_2PO_4] IL extract and water as an antisolvent.*

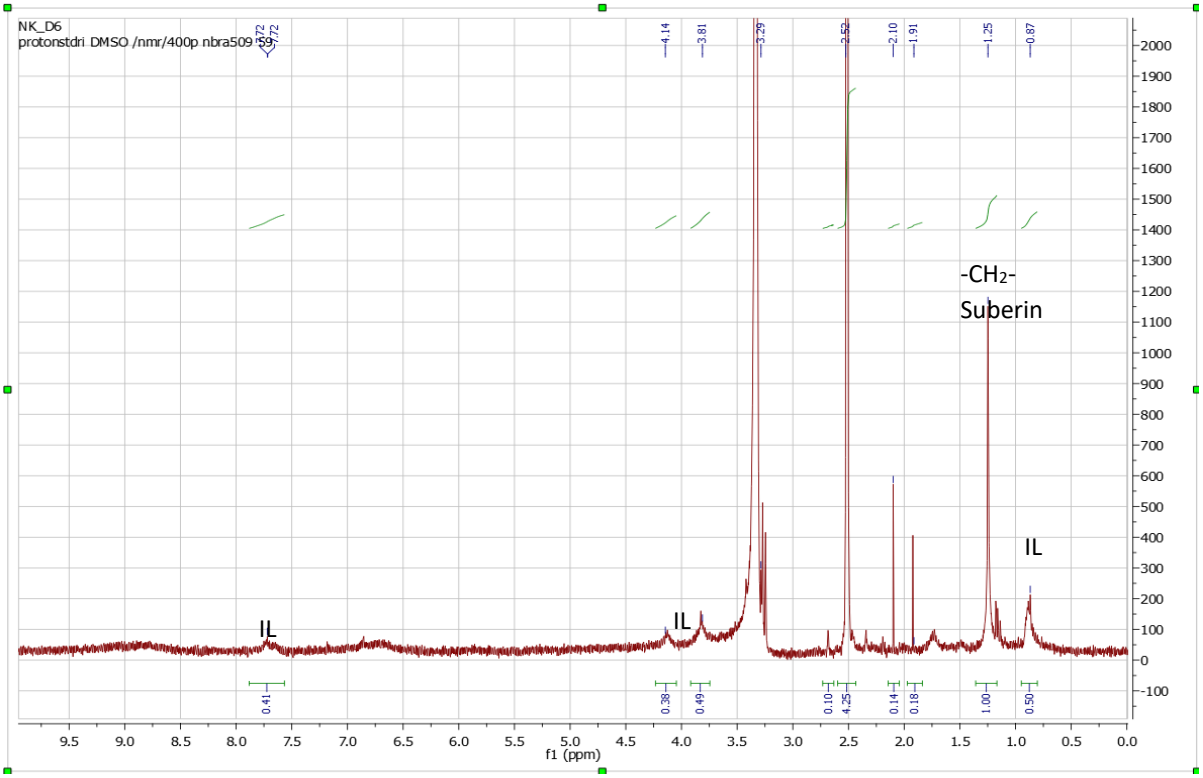


Figure 4.3 *^{1}H NMR of Sample B formed using 10 mL of $[C_4C_1im][Me_2PO_4]$ IL extract and water as an antisolvent*

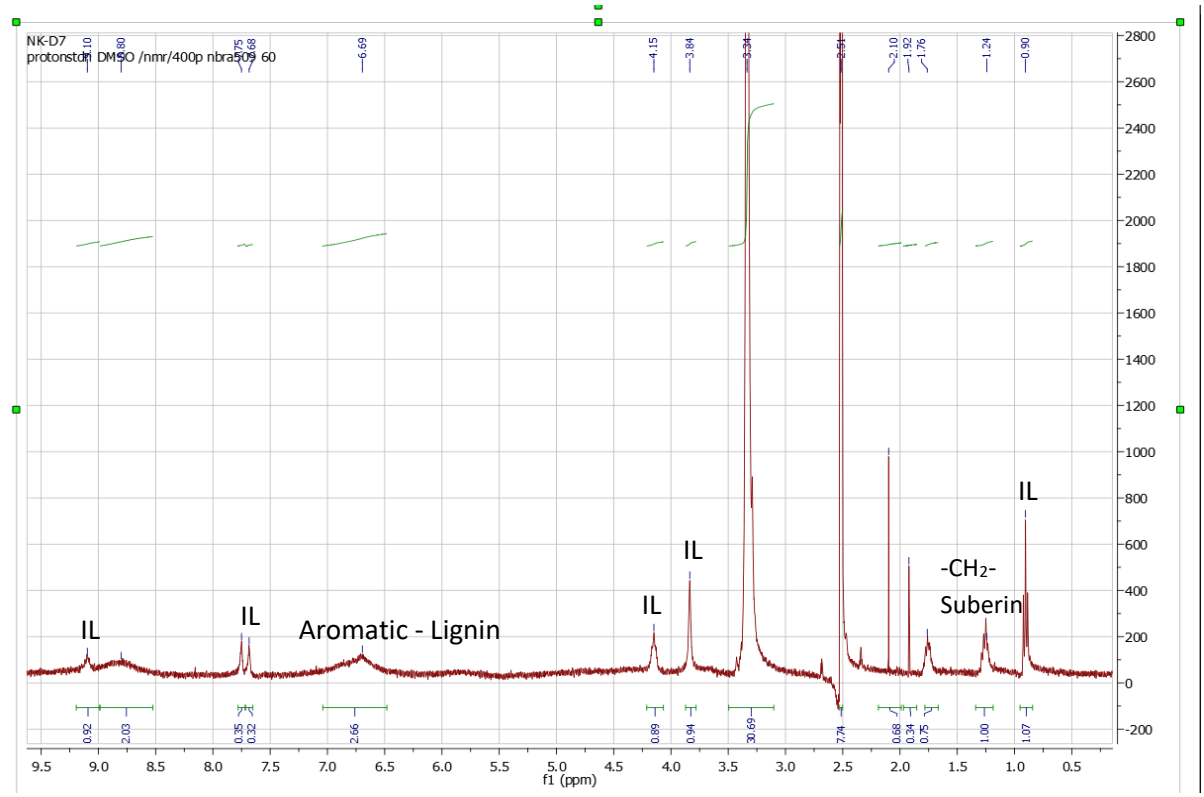


Figure 4.4 *^{1}H NMR of Sample C formed using 10 mL of $[C_4C_1im][Me_2PO_4]$ IL extract and water as an antisolvent*

Some of the samples were also characterised by solid state NMR including the precipitate obtained on adding 100 mL of water to 10 mL of IL extract (i.e. a combined mixture of the recovered bark extracts, Figure 4.5) and Sample A (Figure A80 in the appendix). Due to the small amount of precipitate available for the other samples, they couldn't be used for solid state NMR analysis. The ^{13}C NMR spectrum (Figure 4.5) of the solid recovered from the $[C_4C_1im][Me_2PO_4]$ IL extract depicts that although the aromatic peaks (144.41 ppm) of the lignin and tannins were not as intense as the peaks attributed to suberins/waxes (33 ppm) and carbohydrate (74 ppm), all the peaks of the representative fractions were present. The results are also in agreement with the IR data. The solid-state NMR of solid residue formed using 10 mL of $[C_4C_1im][Me_2PO_4]$ IL extract and 15 mL of water i.e., Sample A is represented in the appendix (Figure A80). The ^{13}C NMR spectrum depicts that the peaks of carbohydrate (74 ppm) were intense as the peaks attributed to suberins/waxes (33 ppm) and the aromatic peaks (144.41 ppm) of the lignin and tannins were not present.

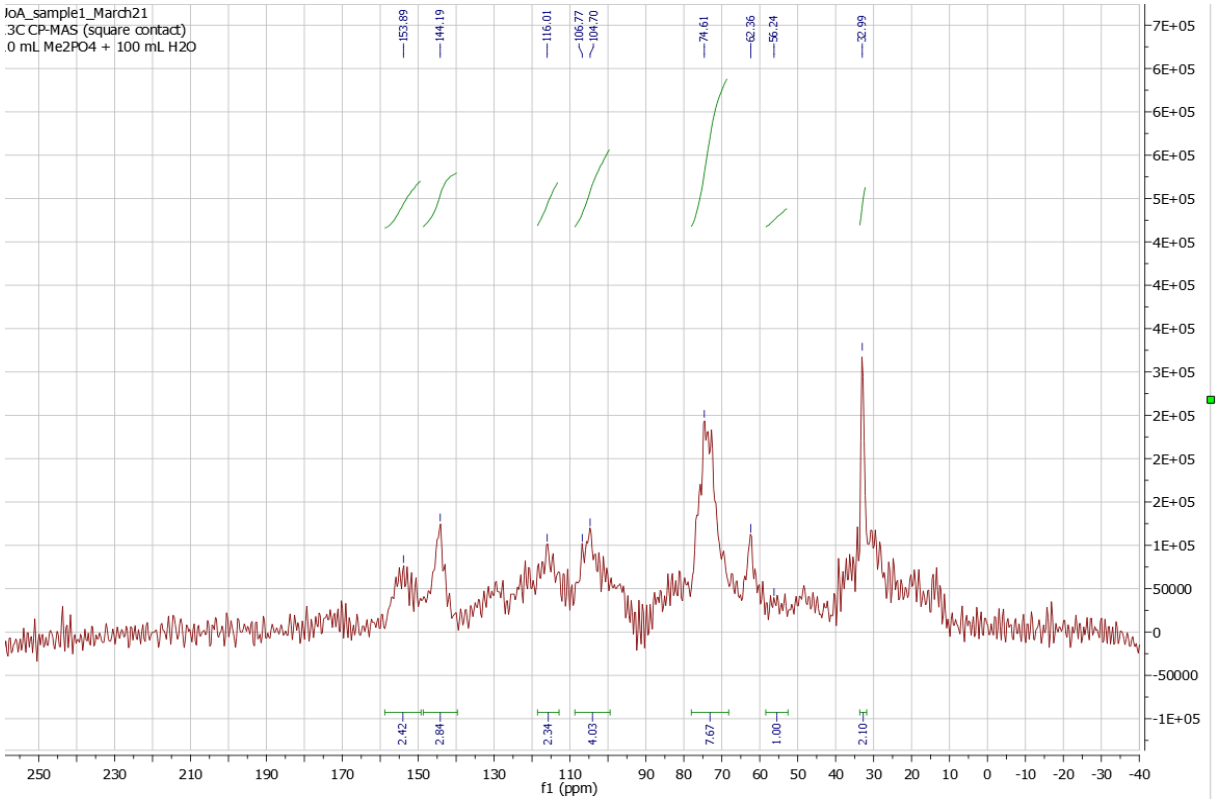


Figure 4.5 *^{13}C solid-state NMR of solid residue formed using 10 mL of $[C_4C_1im][Me_2PO_4]$ IL extract and 100 mL of water as an antisolvent.*

In conclusion, the addition of water to the $[C_4C_1im][Me_2PO_4]$ IL extract initially leads to the separation of a solid residue relatively rich in hydrophobic compounds and some carbohydrates. The proportion of lignin and tannins increased with the addition of more water

(Sample C) while no waxes/suberins or carbohydrates could be observed. Very high solid recovery could be observed from this approach with over 95% of the mass of anticipated solid being recovered on scale-up. This highlights that the quantity of antisolvent addition can assist in improving the selectivity of the extractions even when near quantitative recovery can be observed.

MeCN was the next antisolvent explored for this system. The IR data for these solids are depicted in Figure 4.6 with the summarised relative absorbances shown in Table 4.4. When MeCN was used as an antisolvent, it did not lead to the isolation of the hydrophobic components and led to the formation of mixed fractions that contained lignin, carbohydrate and tannin components (Sample 1 – 8, Table 4.4). In terms of the total extract mass, 11.8 mg of solid was recovered from 1 mL of the IL extract which is less than the amount of the solid expected to be recovered based on the extraction yield (16.5 mg). This indicates that MeCN is not as suitable an antisolvent as water in order to recover the majority of the extract. As the MeCN is less polar than water there is the possibility that it solubilises the hydrophobic components of the dissolved bark in the IL extract rather than facilitating its precipitation, accounting for the lack of formation of any suberin/wax components. Moreover, pure suberin is unlikely to be that soluble in MeCN so this suggests they are more likely to be 'suberin monomers' (i.e. waxes) that were extracted by the IL. The loss of these components in the precipitated extracts may account for the reduced recovery yield.

Table 4.4 *Relative absorbance from the IR spectra of the fractions of bark isolated from 1 mL of [C_4C_1im][Me_2PO_4] extract with the addition of MeCN as an antisolvent.*

Sample labels	**MeCN (mL)**	**Absorbance**			**Extract (mg)**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27,0.18	1.54	0.45	-
Sample 1	0.5	*	3.22	0.74	2.7
Sample 2	0.5	*	2.38	0.72	1.0
Sample 3	0.5	*	3.56	0.71	1.1
Sample 4	0.5	*	5.87	0.68	2.3
Sample 5	0.5	*	3.41	0.69	1.4
Sample 6	0.5	*	3.80	0.75	0.5
Sample 7	0.5	**	**	**	0.7
Sample 8	2	*	2.52	0.47	0.5
Sample 9	2	*	4.64	0.11	1.1
Overall	**7.5**				**11.8 (71.5%)**

(the peaks do not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis)*

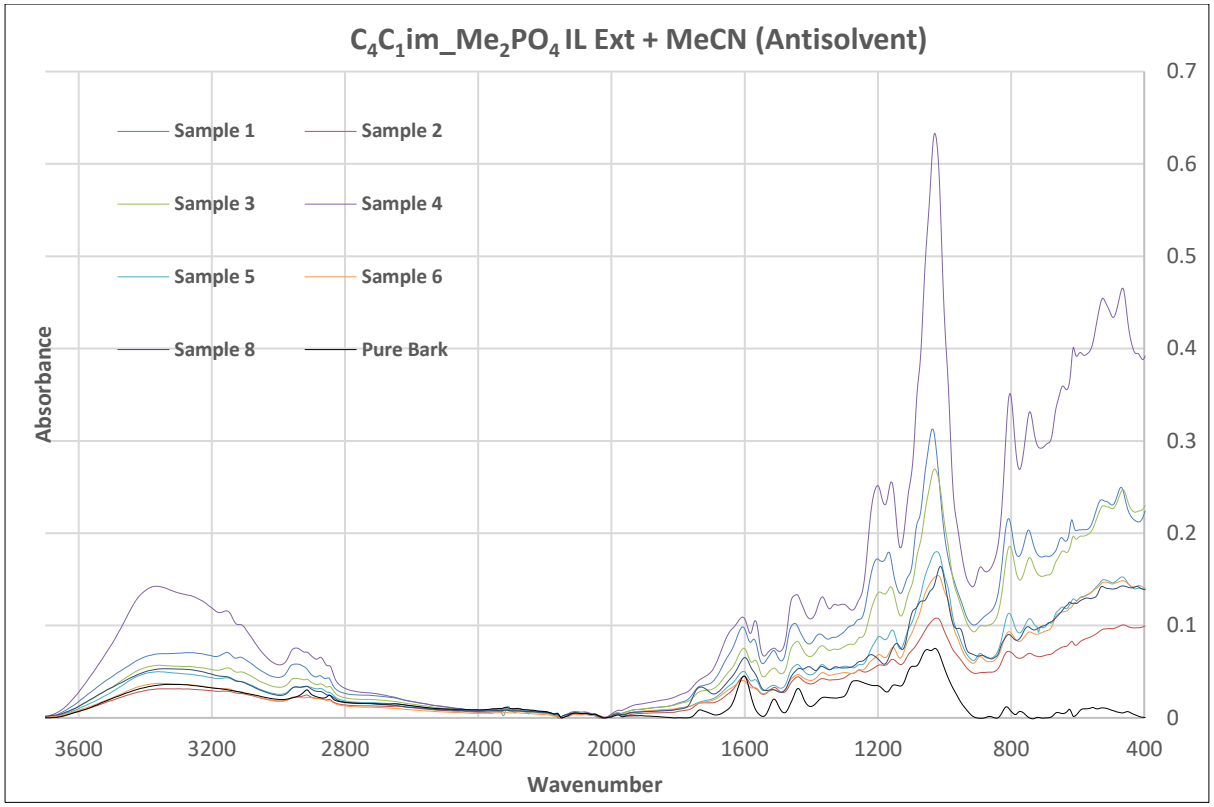


Figure 4.6 *IR spectra of solids formed from Samples 1 – 8 using 1 mL of [C_4C_1im][Me_2PO_4] IL extract and MeCN as an antisolvent.*

The use of DMSO as an antisolvent for the [C_4C_1im][Me_2PO_4] IL extract was also explored with IR data shown in Figure 4.7, with the summarised relative absorbance of key IR peaks shown in Table 4.5. Solids were formed on the addition of up to 1 mL of DMSO to the IL extract; however, no further precipitate was formed after this 1:1 mixture of DMSO: IL extract even on addition of a further 1.5 mL of DMSO. Moreover, the total extract recovered (2.6 mg) is significantly less than the extract recovered using the other two antisolvents and low relative to the theoretical mass that could be recovered based on the extraction efficiency (16.5 mg). The reduced yield (15.8 %) of solid formed relative to other extracts indicates that the majority of the solute remains soluble in the DMSO-IL mixtures. This could be due to ability of DMSO to solubilise key bark components, consistent with the known behaviour of DMSO to act as a cosolvent for the dissolution of carbohydrate and lignin in ILs.[148]

Table 4.5 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 1 mL of [C_4C_1im][Me_2PO_4] IL extract with the addition of DMSO as an antisolvent.*

Sample labels	DMSO (mL)	Absorbance			Extract (mg)
		S : L	C : L	T : L	
Pure Bark	-	0.27,0.18	1.54	0.45	-
Sample 1	0.5	*	1.40	0.28	2.3
Sample 2	0.5	0.88, 0.64	1.76	0.48	0.3
Sample 3	1.5	-	-	-	No ppts
Overall	**2.5**				**2.6 (15.8%)**

(No ppts – no precipitates formed even after adding 1.5 mL of DMSO)

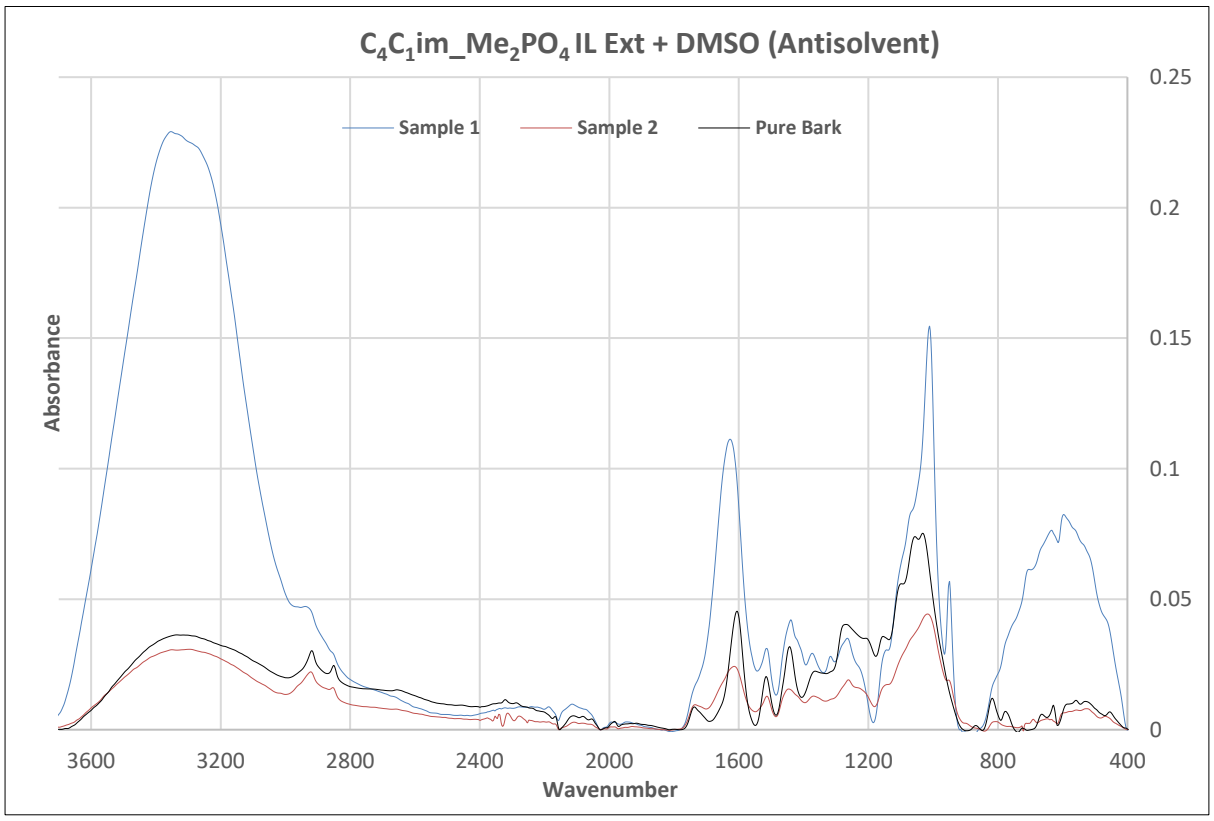


Figure 4.7 *IR spectra of solids formed from Samples 1 - 2 using 1 mL of [C_4C_1im][Me_2PO_4] IL extract and DMSO as an antisolvent*

4.1.2.2. Addition of different antisolvents to $[C_4C_1im][N(CN)_2]$ IL extract

Table 4.6 depicts the data for the addition of water as an antisolvent for the $[C_4C_1im][N(CN)_2]$ IL extract. It was evident that carbohydrates were not isolated from the IL extracts with the addition of water. This is most likely due to the poor extraction of these components by $[C_4C_1im][N(CN)_2]$ as the $[N(CN)_2]^-$ anion is known to be not very effective for the dissolution of cellulose.[88] Hydrophobic components (suberins/waxes) were isolated with the addition of up to 2.5 mL of water to 1 mL of extract (Samples 1 – 5) though the peaks around 2919 cm^{-1} and 2850 cm^{-1} for samples 3 and 4 were unusual, appearing very broad compared to the usual 2 sharp intense peaks (Figure 4.9) seen in samples 1, 2 and 5 as well as in the recovered solids from other IL extracts. Some lignin and tannin fractions were observed in almost all the samples, with the lignin and tannin fractions free of contamination by suberin/wax components obtained after 2.5 mL of water was added. This means that the recovered solid consists of about 79 % of the mixed fractions that consist of suberin and tannin or lignin up to 2.5 mL of added antisolvent and after 2.5 mL addition of water about 21 % of the recovered solid appears to consist of lignin and tannin fractions free of contamination by suberins and waxes. There are also some additional peaks present in the region of 800 – 1600 cm^{-1} that are not present in the bark itself, as shown in Figure 4.8, 4.9 and 4.10. The intensity of the peaks assigned to lignin and tannin are influenced by these additional peaks. In addition, these peaks likely indicate either some sort of side reaction is occurring in between the $[N(CN)_2]^-$ anion and the bark or a significant by-product is being formed from the decomposition of the IL anion. These peaks were not observed for recovered solids from any other ILs.

Moreover, the amount of solid recovered (17.4 mg) is higher than the bark present in the IL extract based on the measured extraction yield (13.8 mg), which could be due to either the presence of residual IL in the solid formed or the side reaction discussed above. These results are consistent with the results of elemental analysis discussed in Chapter 3, section 3.4.6.2.1 that imply a potential reaction between key bark components and the $[N(CN)_2]^-$ anion under the reaction conditions. In conclusion, bark components other than carbohydrates such as suberin, tannin and lignin present in the dissolved bark could be excluded from the IL extract with the addition of water although there was evidence for the potential reaction or decomposition of some of the components recovered.

Table 4.6 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 1 mL of $[C_4C_1im][N(CN)_2]$ IL extract with the addition of water as an antisolvent.*

Sample labels	**Water (mL)**	**Absorbance**			**Solid recovered (mg)**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27,0.18	1.54	0.45	-
Sample 1	0.5	1.17, 0.87	*	1.08	1.0
Sample 2	0.5	0.79, 0.61	*	1.1	1.7
Sample 3	0.5	*	*	1.12	5.9
Sample 4	0.5	*	*	1.03	3.3
Sample 5	0.5	0.92, 0.68	*	0.96	1.9
Sample 6	0.5	*	*	1	0.8
Sample 7	0.5	*	*	0.91	0.8
Sample 8	1	**	**	**	0.9
Sample 9	1	*	*	1.03	0.5
Sample 10	1	***	***	***	0.3
Sample 11	1	***	***	***	0.3
Total	**7.5**				**17.4 (116 %)**

(the peaks did not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis *** amount of the sample was quite low and hence not enough for IR analysis)*

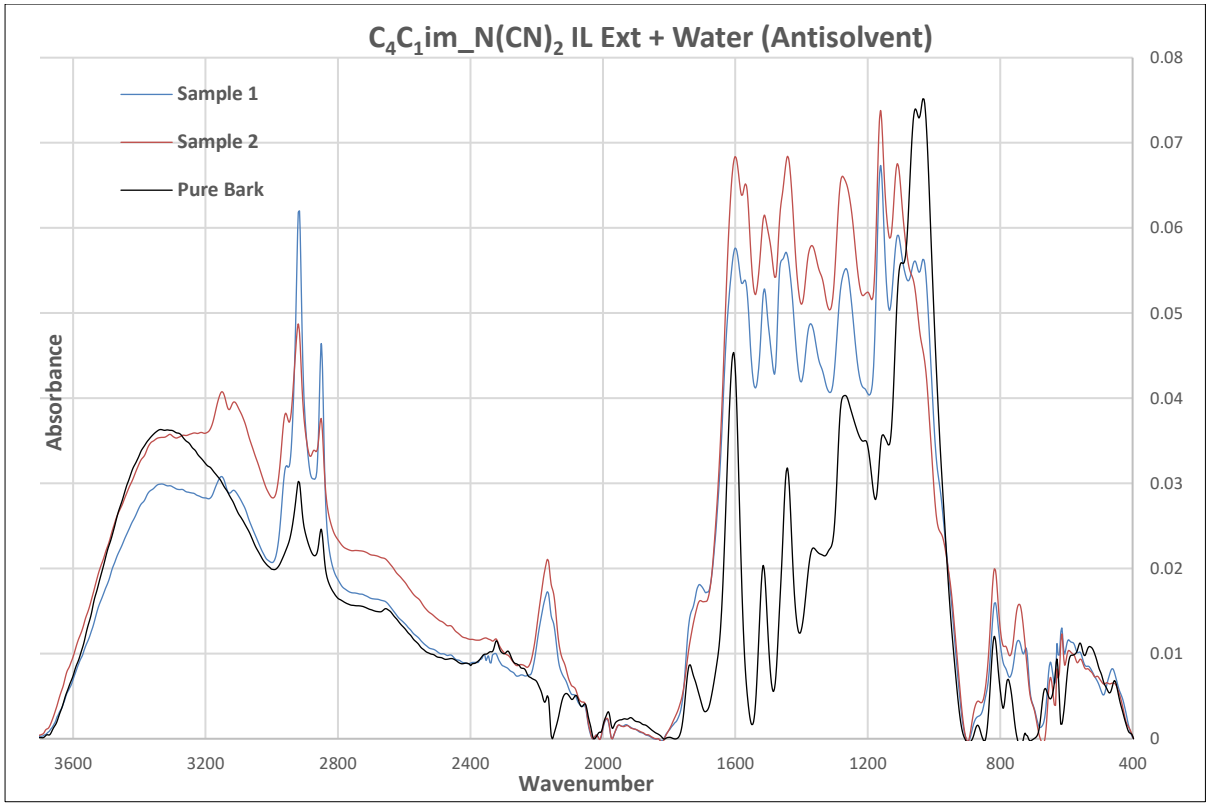


Figure 4.8 *IR spectra of the solids formed from Sample 1 - 2 using 1 mL of [C_4C_1im][$N(CN)_2$] IL extract and water as an antisolvent.*

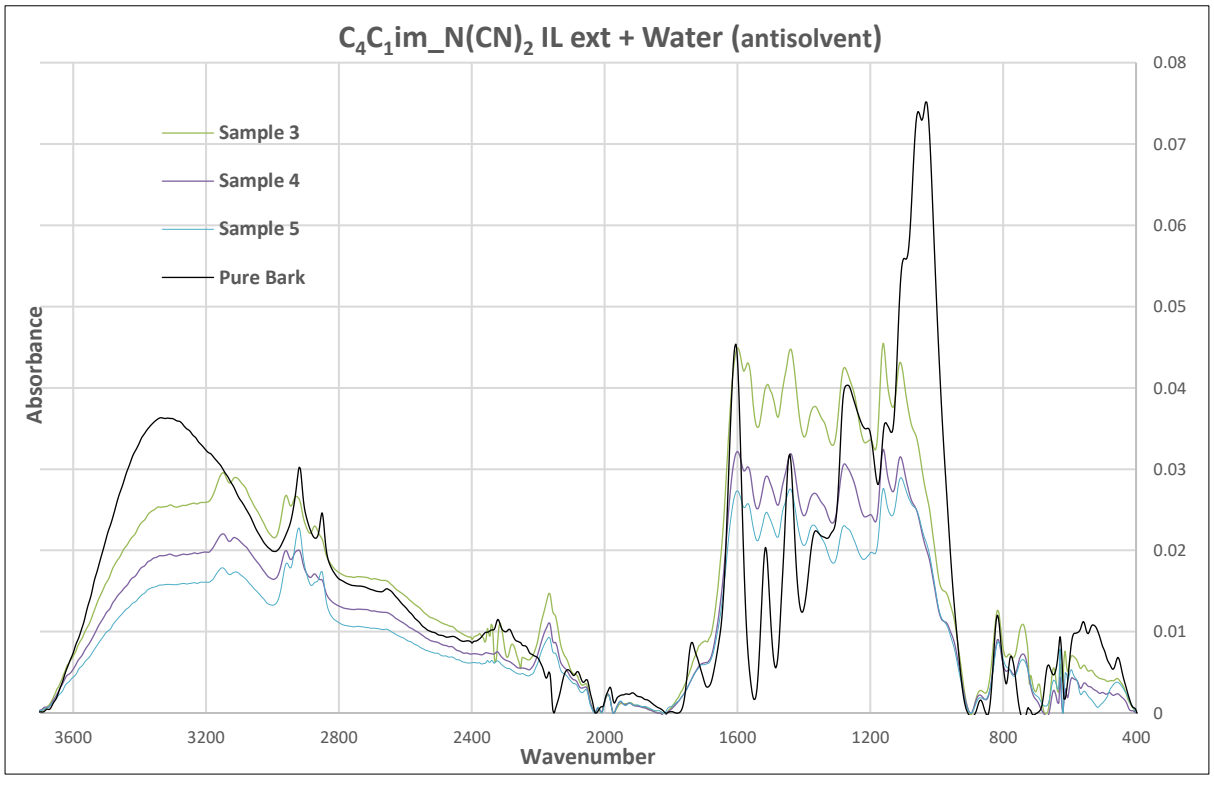


Figure 4.9 *IR spectra of the solids formed from Sample 3 - 4 using 1 mL of [C_4C_1im][$N(CN)_2$] IL extract and water as an antisolvent*

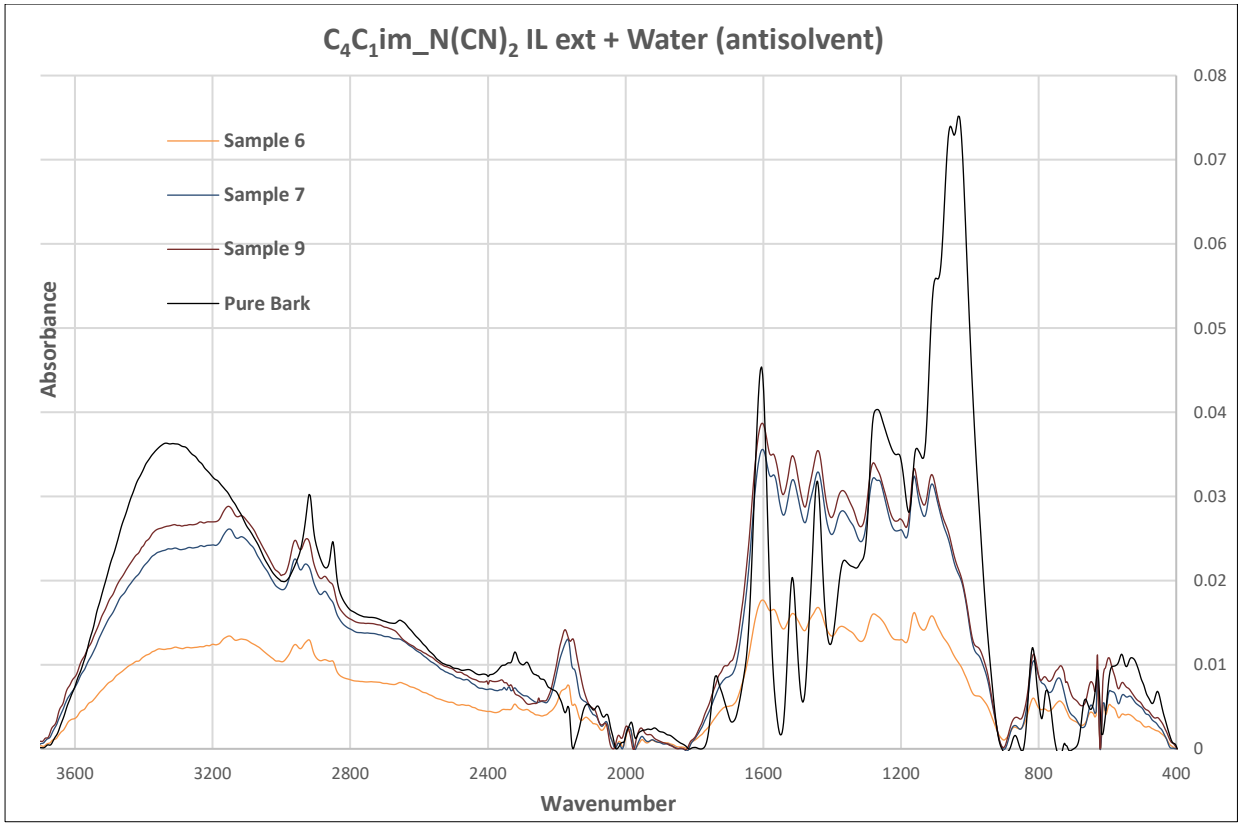


Figure 4.10 *IR spectra of the solids formed from Sample 6 - 8 using 1 mL of [C_4C_1im][$N(CN)_2$] IL extract and water as an antisolvent.*

Similar to the [C_4C_1im][Me_2PO_4] extract, the [C_4C_1im][$N(CN)_2$] IL extract was scaled up (10 mL of IL extract) in order to improve the analysis of the extracts formed. The amount of each extract formed and the relative absorbance of the peaks in the IR spectrum are represented in Table 4.7, with the IR spectra shown in Figures A66 – A67 (Sample D, Sample E and Sample G). Sample F was not characterised due to the small amount recovered. For the scaled-up samples, the maximum solid recoverable was estimated to be 138 mg from 10 mL of the IL extract based on the extraction yield. However, only 54.5 mg of solids were recovered in total from the IL extract. This is only 40% yield and considerably less than was recovered in the smaller scale experiments. This might reflect the greater error in determining the mass at the smaller scale as well as the increased effect and likelihood of included IL in the extracts when dealing with a smaller mass of solid.

Table 4.7 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 10 mL of [C_4C_1im][Me_2PO_4] IL extract with the addition of water as an antisolvent.*

[C_4C_1im] [$N(CN)_2$] IL extract	**Addition of water (mL)**	**Absorbance**			**Solid recovered (mg)**	**Relative Sample Yield (%)**
		S : L	**C : L**	**T : L**		
Pure Bark	-	0.27,0.18	1.54	0.45	-	
Sample D	10 mL	0.3, 0.2	*	0.78	24.3	45
Sample E	10 mL	*	*	0.86	12.9	24
Sample F	5 mL	**	**	**	8.5	15
Sample G	75 mL	*	*	0.89	8.8	16
Overall	**100 mL**				**54.5 (40 %)**	**100**

(the peaks do not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis)*

The scaled-up samples were also characterised by solution phase NMR as shown in Figures 4.11 – 4.14. It has been observed that only the aliphatic CH_2 peak at a chemical shift 1.25 ppm emerged from the extract of all the samples with no strong evidence for aromatic signals or other resonances. This could be due to the low concentration of the fractions or their insolubility.

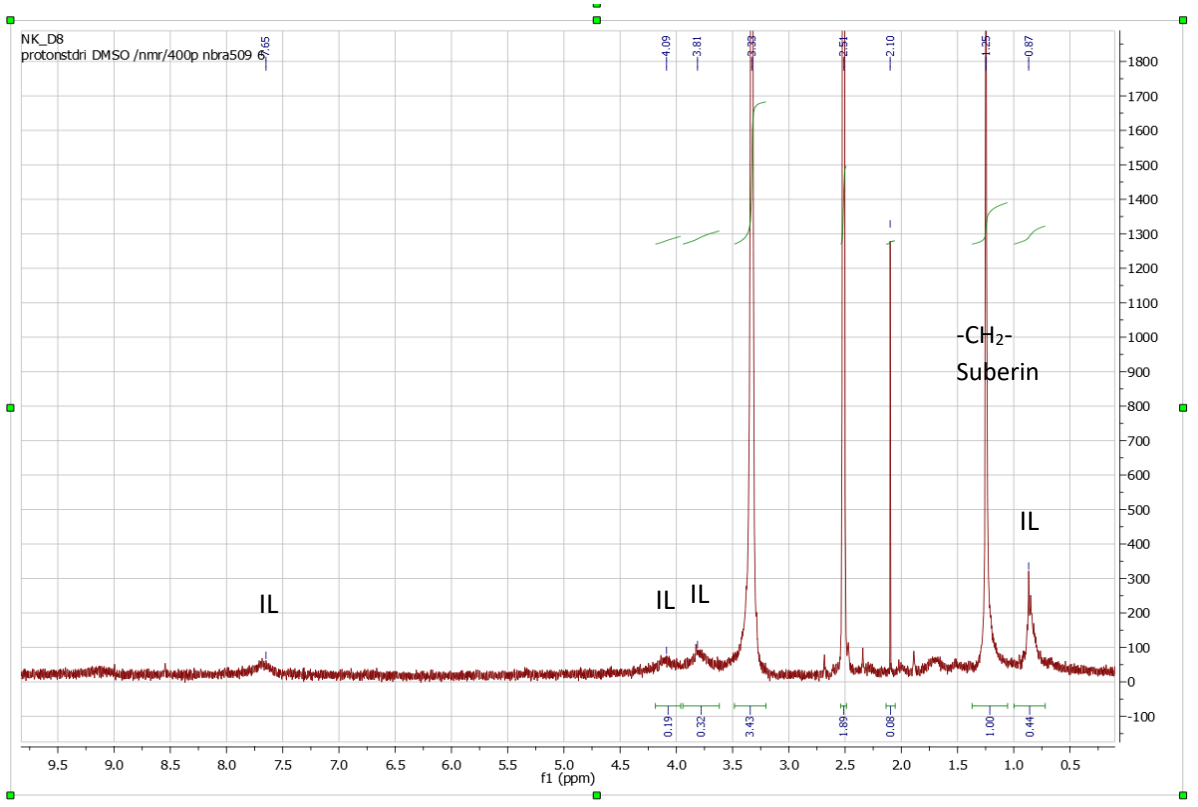


Figure 4.11 *^{1}H NMR of Sample D formed using 10 mL of $[C_4C_1im][N(CN)_2]$ IL extract and water as an antisolvent*

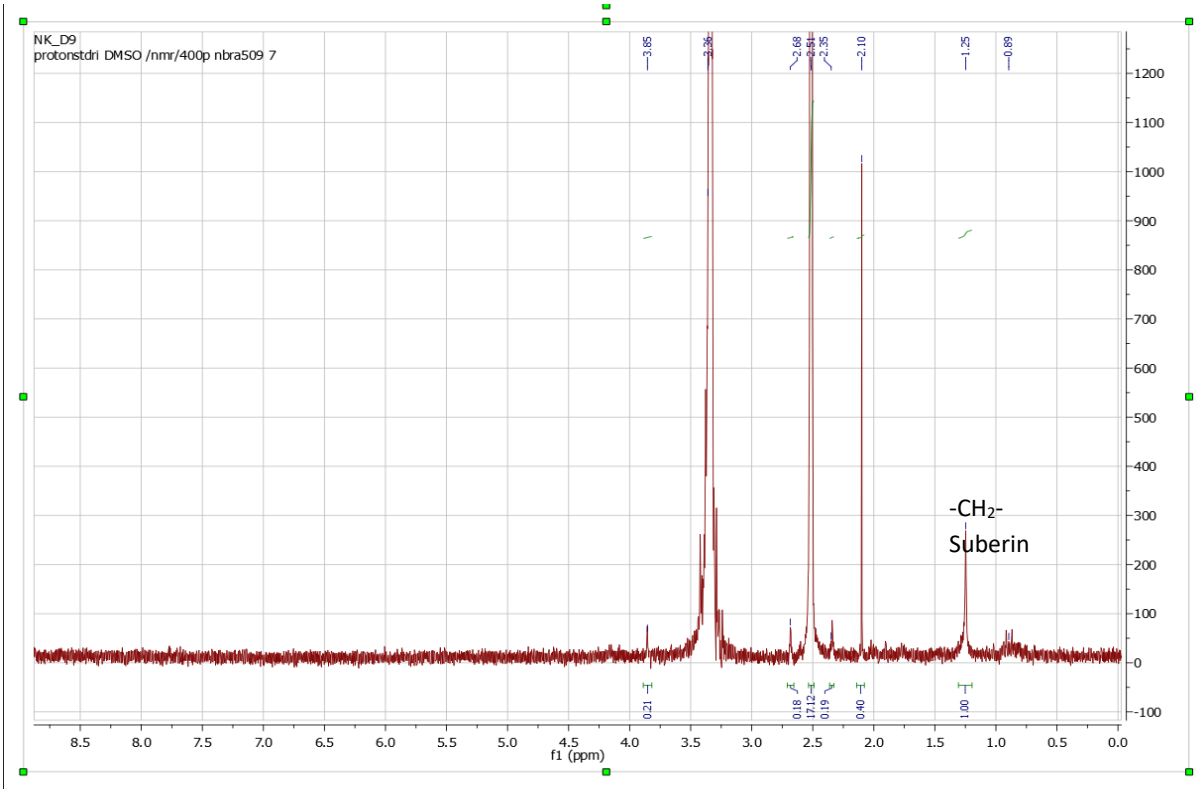


Figure 4.12 *^{1}H NMR of Sample E formed using 10 mL of $[C_4C_1im][N(CN)_2]$ IL extract and water as an antisolvent*

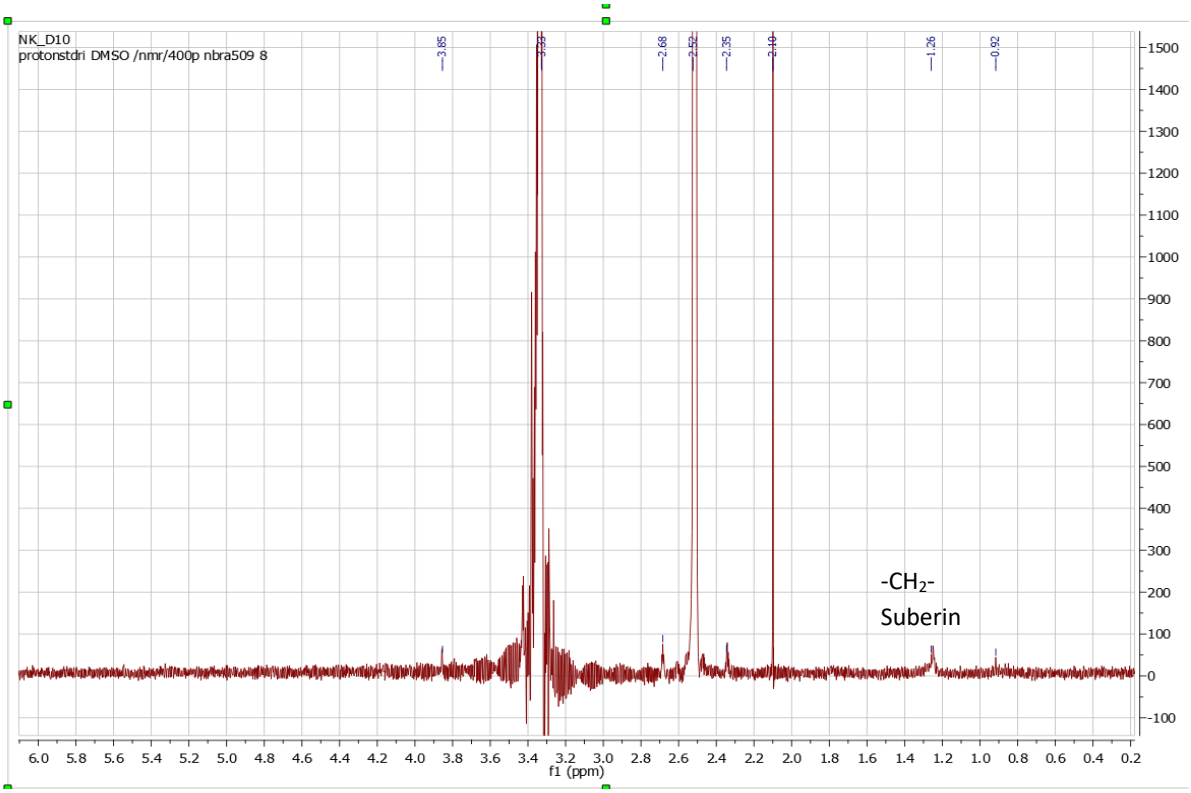


Figure 4.13 *^{1}H NMR of Sample F formed using 10 mL of $[C_4C_1im][N(CN)_2]$ IL extract and* water as an antisolvent

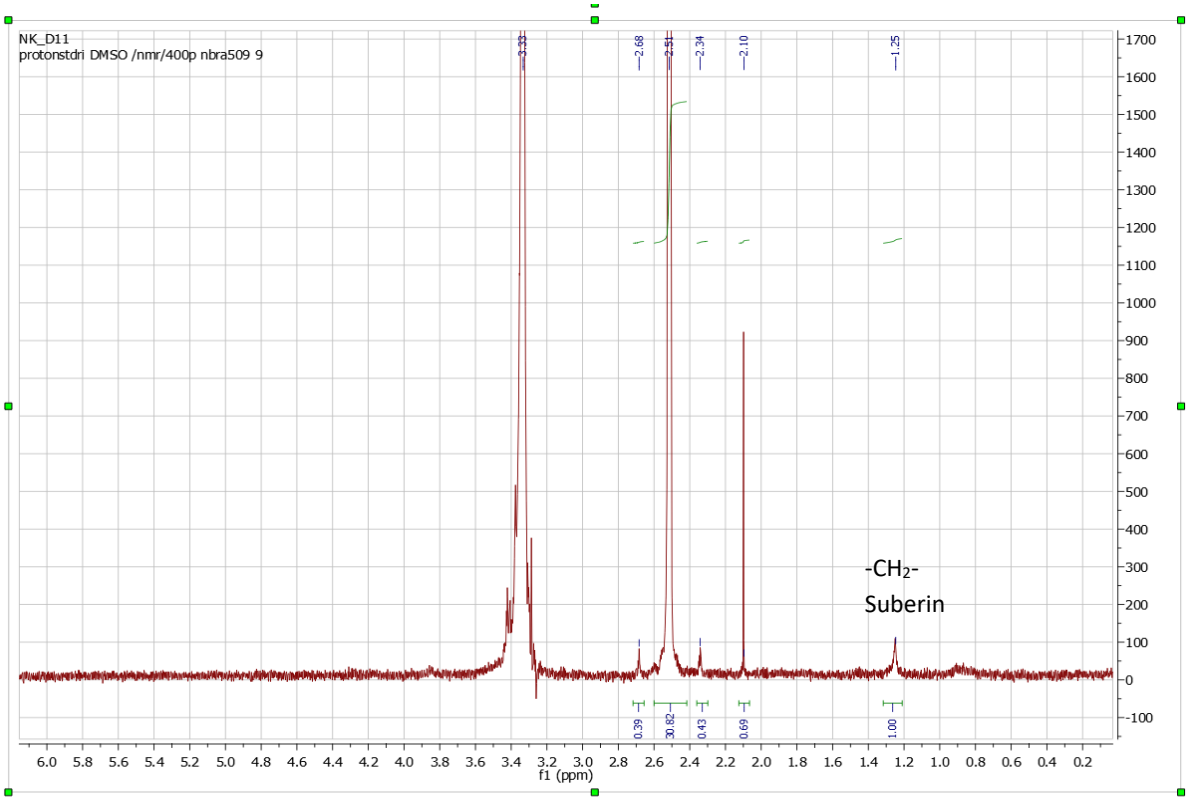


Figure 4.14 *^{1}H NMR of Sample G formed using 10 mL of $[C_4C_1im][N(CN)_2]$ IL extract and water as an antisolvent*

Other antisolvents such as MeCN and DMSO were also used for the $[C_4C_1im][N(CN)_2]$ IL extract but the IR data were complicated to interpret. There were a number of additional peaks were generated in the region 1600 – 800 cm^{-1} due to the side reaction discussed previously. In addition, most of the assigned peaks were less intense (for example – aliphatic region at 2919, 2850 cm^{-1}) than pure bark peaks and interfered with by the additional peaks. These issues are clearly visible in the IR spectra as shown in Figures. 4.15 and 4.16.

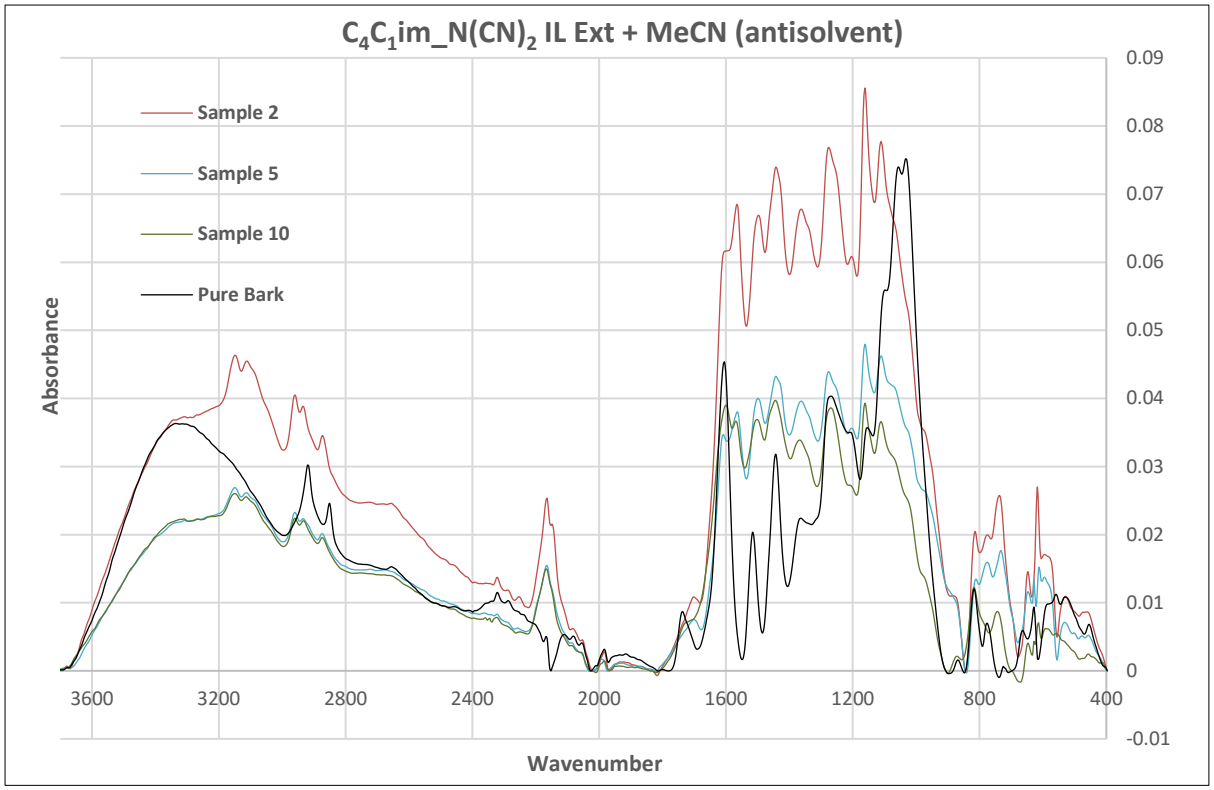


Figure 4.15 *IR spectra of the solids formed from Sample 2, 5, and 10 using 1 mL of* $[C_4C_1im][N(CN)_2]$ *IL extract and MeCN as an antisolvent*

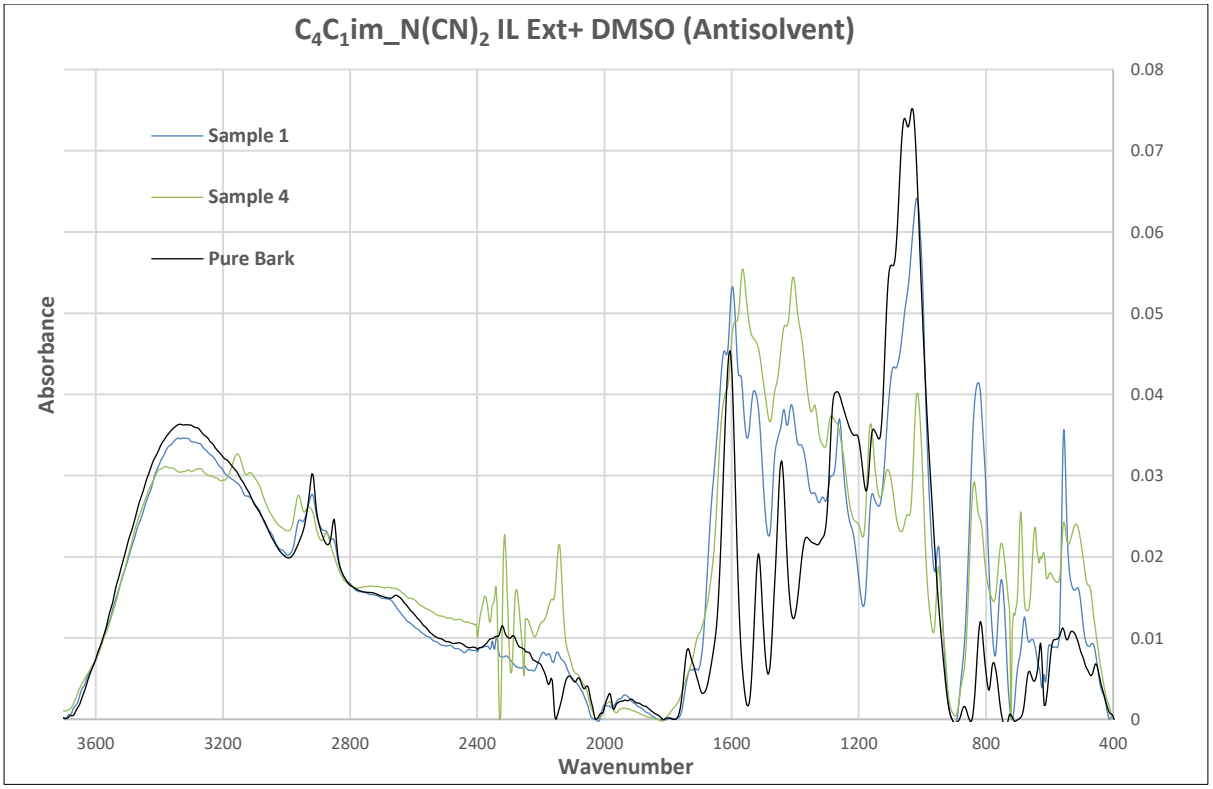


Figure 4.16 *IR spectra of the solids formed from Sample 1 and Sample 4 using 1 mL of [C_4C_1im][$N(CN)_2$] IL extract and DMSO as an antisolvent*

4.1.2.3. Addition of DMSO and MeCN in [C_4gun]- and [C_8gun]-NTf_2 IL extracts

The antisolvents, MeCN and DMSO, were explored for the [C_4gun][NTf_2] and [C_8gun][NTf_2] IL extracts rather than water as these ILs are immiscible in water. Table 4.8 and Figures 4.17 – 4.18 depict the data for the addition of MeCN as an antisolvent for the [C_4gun][NTf_2] and [C_8gun][NTf_2] IL extracts. It has been evaluated that by adding MeCN, only a single mixed fraction could be isolated from the [C_4gun][NTf_2] and [C_8gun][NTf_2] IL extracts, with limited selectivity observed. This occurred after the addition of 0.5 mL of MeCN. No precipitation was observed on further addition of MeCN up to a total of 1.5 mL in either case, so no further addition of MeCN was explored.

Table 4.8 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 1 mL of each [C_4gun][NTf_2] and [C_4gun][NTf_2] IL extracts with the addition of MeCN as an antisolvent.*

IL Extract (1 mL)	**MeCN (mL)**	**Absorbance**			**Extracts (mg) / Yield (%)**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27,0.18	1.54	0.45	-
[C_4gun][NTf_2]	0.5	0.90, 0.66	1.30	0.54	**1.6 (22.1%)**
	1.0	-	-	-	No ppts
[C_8gun][NTf_2]	0.5	0.76, 0.58	0.45	0.66	**1.3 (15 %)**
	1.0	-	-	-	No ppts

(No ppts – no precipitates formed even after adding 1 mL of MeCN)

Overall, 1.6 mg and 1.3 mg of solids containing mixtures of most components, namely, suberin/waxes, cabohydrate, lignin and tannin were recovered from 1 mL of the [C_4gun][NTf_2] and [C_8gun][NTf_2] extracts. This represents only a small proportion of the expected solid recovery for each based on the extraction yield which was anticipated to be 7.25 mg and 8.75 mg for [C_4gun][NTf_2] and [C_8gun][NTf_2], respectively. This indicates that MeCN was ineffective as an antisolvent for these extracts.

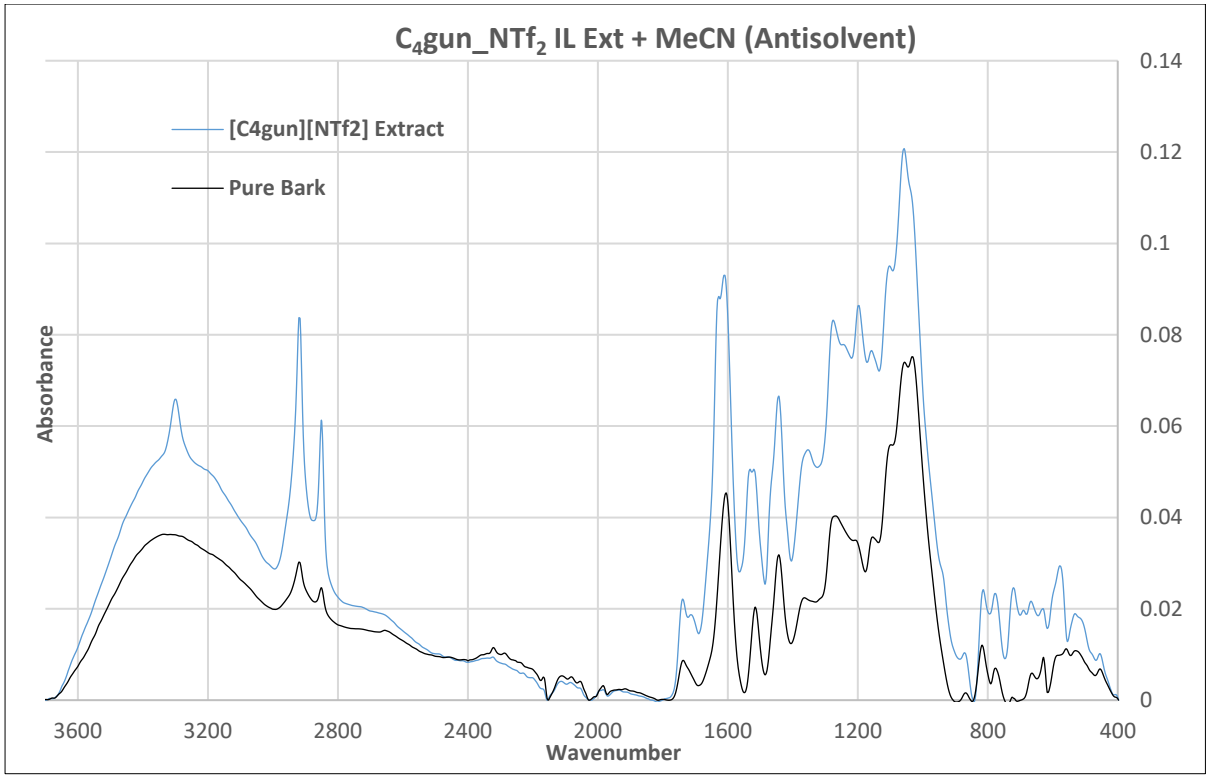


Figure 4.17 *IR spectra of the solids formed from Sample 1 using 1 mL of [C_4gun][NTf_2] IL extract and MeCN as an antisolvent*

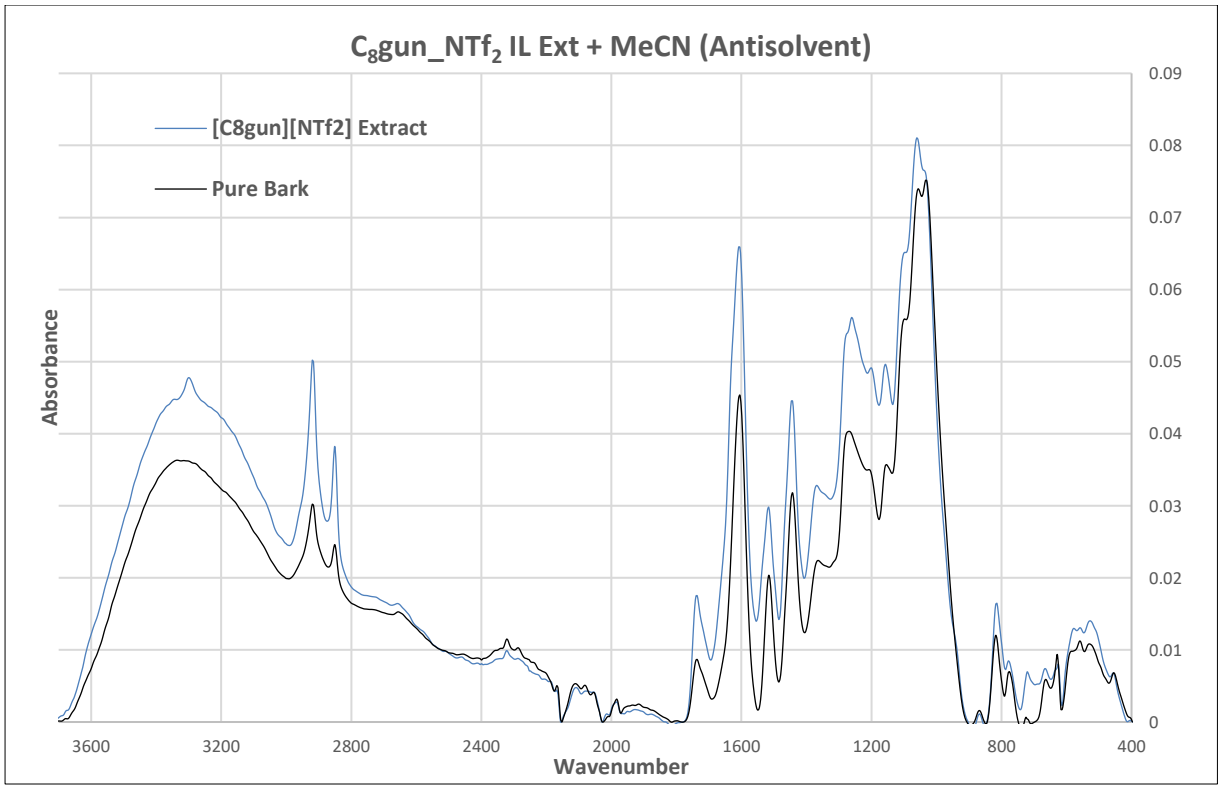


Figure 4.18 *IR spectra of the solids formed from Sample 1 using 1 mL of [C_8gun][NTf_2] IL extract and MeCN as an antisolvent*

The data for the addition of DMSO as an antisolvent in $[C_4gun][NTf_2]$ and $[C_8gun][NTf_2]$extracts is represented in Table 4.9, with Figures 4.19 and 4.20 displaying the IR spectra obtained. It has been determined that only small amount of solid (0.8 mg) could be obtained from the $[C_4gun][NTf_2]$ IL extract relative to the mass expected (7.25 mg). This could be explained on the basis of role of DMSO as lignin or tannin compounds are highly soluble in DMSO so these are less likely to precipitate out. Moreover, on further addition of DMSO (1 mL), no more precipitation occurred. For the $[C_8gun][NTf_2]$ IL extract, DMSO addition led to the formation of a mixed fraction including suberin, lignin, carbohydrate and tannin components. Like MeCN, using DMSO as an antisolvent only led to the formation of a small amount of solid from the $[C_4gun][NTf_2]$ and $[C_8gun][NTf_2]$ IL extracts (0.8 mg and 1.8 mg of solid, respectively) which is considerably less than the expected amount based on the extraction yields of 7.25 mg and 8.75 mg, respectively.

These outcomes revealed that MeCN and DMSO are not the most effective antisolvents for the isolation of bark components from $[Gun][NTf_2]$-based ILs. These results are also in agreement with the earlier results regarding MeCN and DMSO as antisolvents (section 4.1.2.1 and 4.1.2.2), which implies that these solvents retain too high a solubility of key bark components when mixed with ILs to help facilitate their precipitation.

Table 4.9 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components segregated from 1 ml of each $[C_4gun][NTf_2]$ and $[C_4gun][NTf_2]$ IL extracts with the addition of DMSO as an antisolvent.*

IL Extract (1 mL)	**DMSO (mL)**	**Absorbance**			**Extracts (mg) / Yield %**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27, 0.18	1.54	0.45	-
$[C_4gun][NTf_2]$	0.5 1.0	* -	NS -	NS -	**0.8 (11%)** No ppts
$[C_8gun][NTf_2]$	0.5 1.0	0.61, 0.41 -	1.93 -	0.49 -	**1.8 (21%)** No ppts

*((NS - Not sure i.e., the observed peak shapes are unusual as depicted in Figure 4.19, * the peaks do not appear in the IR, No ppts – no precipitates formed even after adding 1 mL of DMSO)*

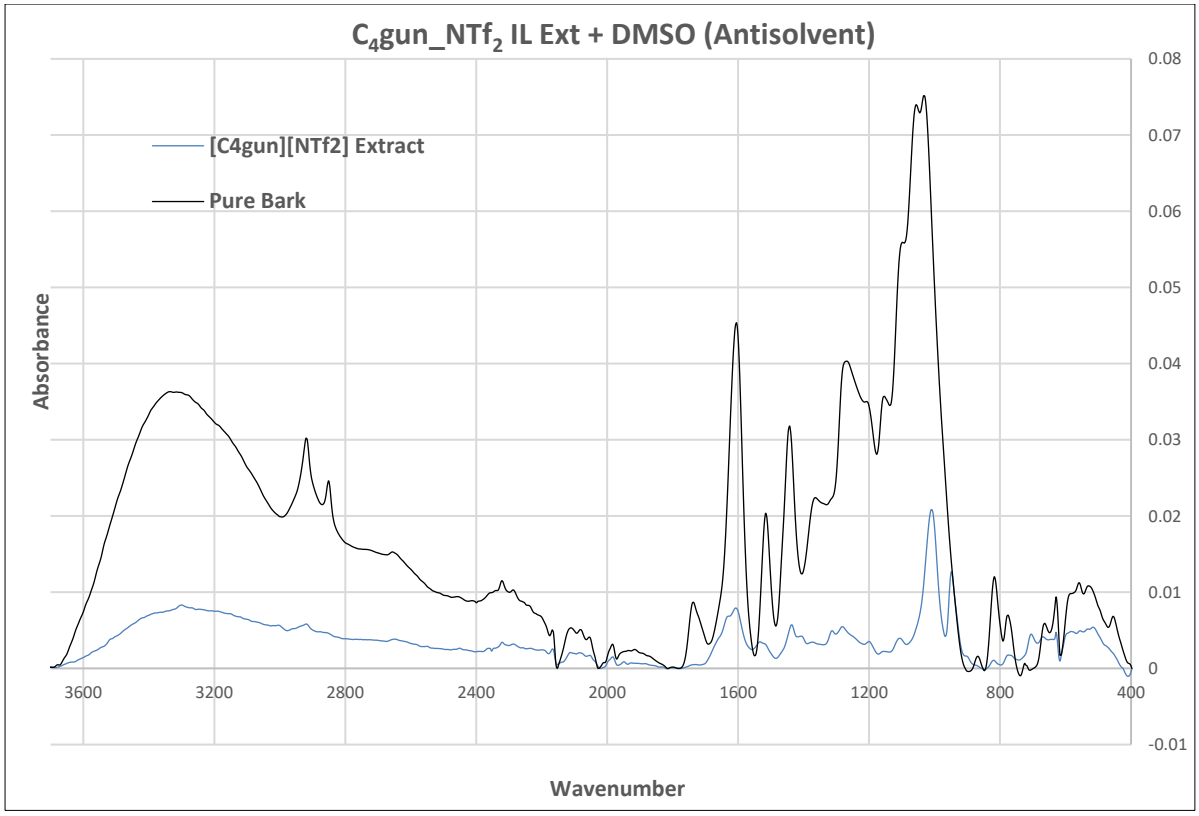


Figure 4.19 *IR spectra of the solids formed from Sample 1 using 1 mL of [C_4gun][NTf_2] IL extract and DMSO as an antisolvent*

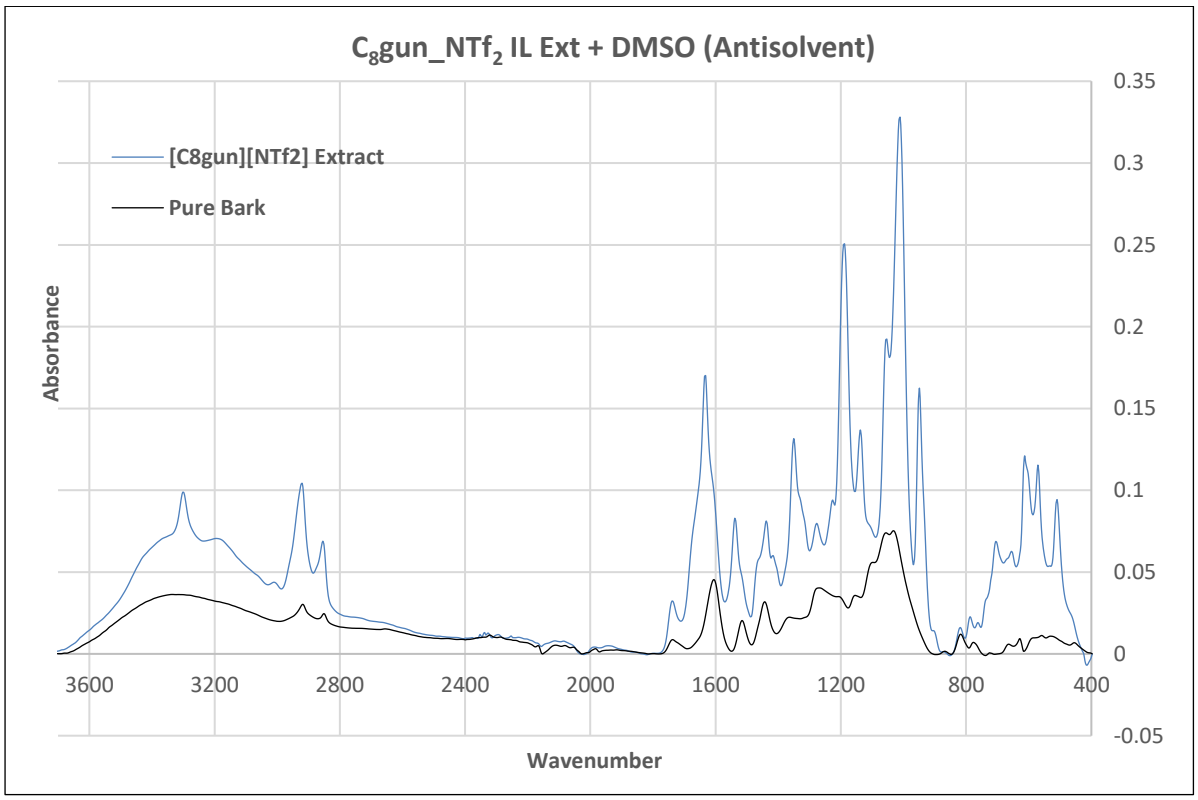


Figure 4.20 *IR spectra of the solids formed from Sample 1 using 1 mL of [C_8gun][NTf_2] IL extract and DMSO as an antisolvent*

4.1.2.4. Addition of water as an antisolvent to the $[C_4C_1pyrr][Me_2PO_4]$ IL extract

Table 4.10 represents the data for the addition of water as an antisolvent for the $[C_4C_1pyrr][Me_2PO_4]$ IL extract based on the IR results displaying in Figures 4.21 – 4.22. Table 4.10 displays that the initial addition of water for samples 1 – 4 led to the precipitation of mixed fractions of lignin, carbohydrates and tannins along with waxes or suberinic components present. The suberin/wax content was particularly high for Samples 2 and 3, due to the strength of signals corresponding to the aliphatic region (2919 – 2850 cm^{-1}). This could be attributed to the more hydrophobic, weakly hydrogen-bond donating pyrrolidinium cation[152] leading to greater extraction of hydrophobic compounds found in bark, as has been discussed in Chapter 3. These could easily be precipitated out by the addition of water due to their hydrophobicity. Along with these hydrophobic compounds, some carbohydrate, lignin and tannins were also recovered in Samples 1 – 4. Moreover, it was observed that following sample 4, only the lignin and tannin fractions were present in the recovered solid. The significant carbohydrate composition detected in the recovered solid obtained from the IL extract is likely to be due to the incorporation of the strongly hydrogen bond accepting anion $[Me_2PO_4]^-$, which is known to assist the dissolution of carbohydrates such as cellulose,[141] despite the presence of pyrrolidinium cation.

On the addition of a total of ~ 15 mL of water only 13.0 mg of solid was able to be obtained less than half of the expected 35.3 mg based on the extraction yield. This could be due to the solubility of some of the extracts in the water itself and a contribution from the experimental error of both the measured extraction yield and solid recovery. Of the solid recovered, 57.7% was present as the tannin/lignin fraction with the remaining 42.3% by mass being the mixed fractions containing waxes and carbohydrates.

Table 4.10 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 1 mL of [C_4C_1pyrr][Me_2PO_4] IL extract with the addition of water as an antisolvent.*

Sample Labels	**Water (mL)**	**Absorbance**			**Extract (mg)**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27,0.18	1.54	0.45	-
Sample 1	0.5	0.53, 0.35	1.81	0.44	1.3
Sample 2	0.5	2.03, 1.36	1.81	0.56	1.1
Sample 3	0.5	2.06 ,1.41	1.41	0.56	0.5
Sample 4	1	0.56 ,0.37	1.04	0.47	2.6
Sample 5	1	*	NS	0.42	2.4
Sample 6	1	*	NS	0.38	1.3
Sample 7	2	*	0.98	0.44	1.6
Sample 8	2	*	0.77	0.33	0.7
Sample 9	2	**	**	**	0.5
Sample 10	5	*	NS	0.50	1.0
Total	**15.5**				**13 (37 %)**

(the peaks do not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis)*

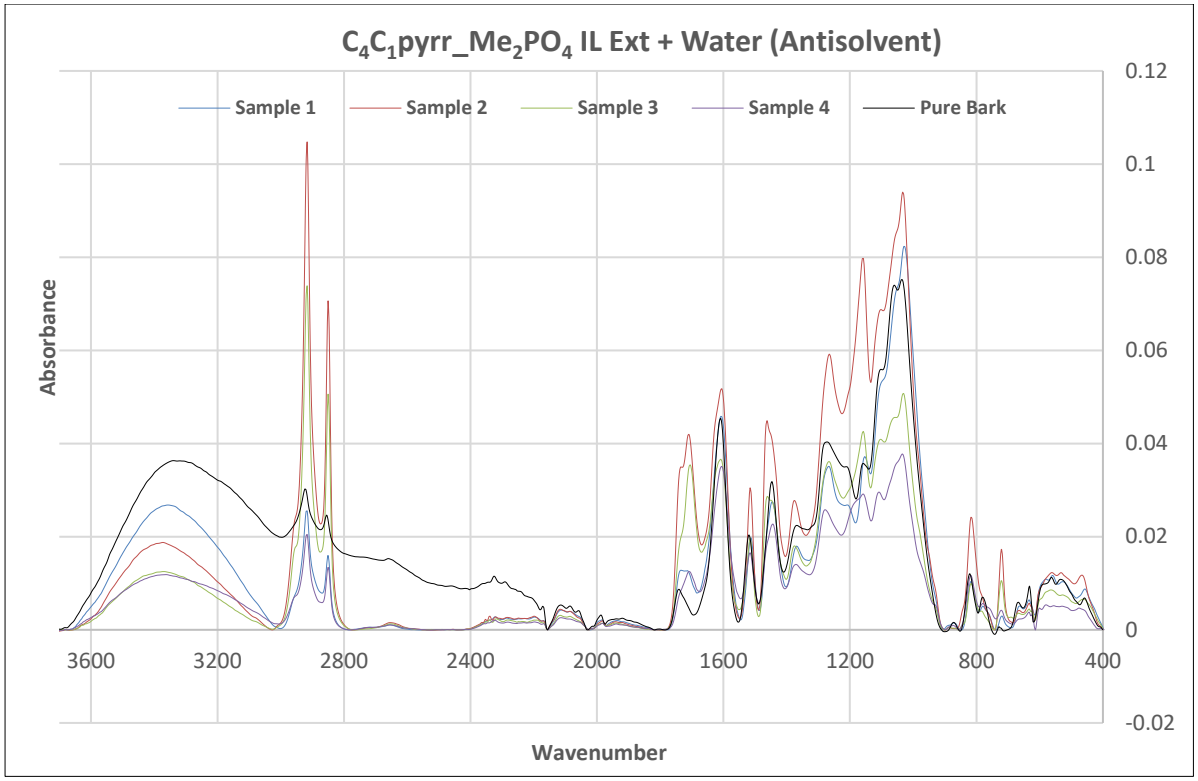


Figure 4.21 *IR spectra of the solids formed from Sample 1 - 4 using 1 mL of [C_4C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

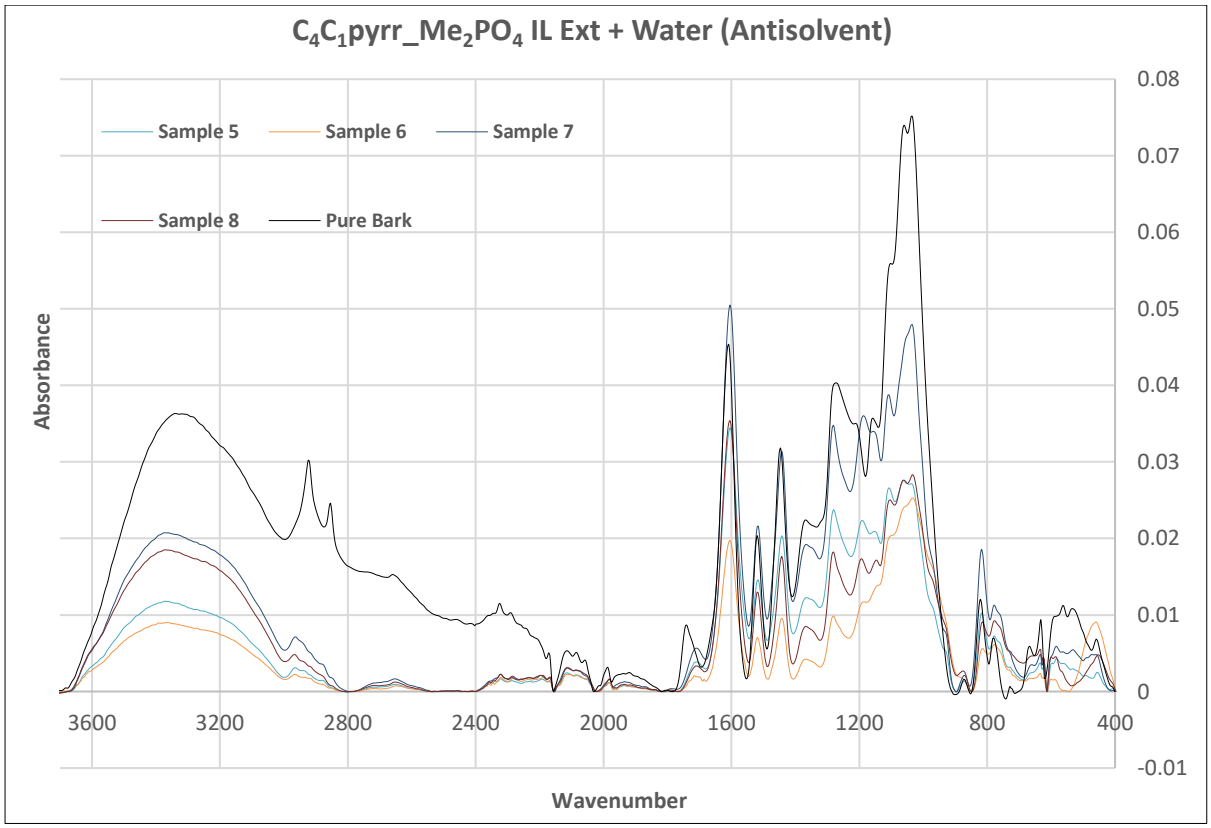


Figure 4.22 *IR spectra of the solids formed from Sample 5 - 8 using 1 mL of [C_4C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

The antisolvent addition of water to the [C_4C_1pyrr][Me_2PO_4] extract was scaled-up, as described previously, although with 5 mL rather than 10 mL IL extract. The outcomes of the addition of water as an antisolvent is represented in Table 4.11, with the IR spectra located in the appendix (Figures A69 – A70). The scaled-up samples (Sample H and Sample I) display the same trend mentioned above, i.e. initially, a mixed fraction formed but after that only polyphenols recovered. In this case, about 61.2 % of mixed fractions including suberin, tannin or lignin and carbohydrates and about 38.8 % of pure lignin and tannin fractions were recovered. These proportions were slightly different to the smaller scale experiments and resulted in reduced overall recovery yield despite the larger relative volume of antisolvent used to collect the lignin and tannin fractions. The clear reason behind it is not known but it might be due to random experimental error.

Table 4.11 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 5 mL of [C_4C_1pyrr][Me_2PO_4] IL extract with the addition of water as an antisolvent.*

[C_4C_1pyrr][Me_2PO_4] IL extract (5 mL used)	**Addition of water (mL)**	**Absorbance**			**Solid recovered (mg)**	**Relative Sample Yields (%)**
		S : L	**C : L**	**T : L**		
Pure Bark	-	0.27, 0.18	1.54	0.45	-	-
Sample H	10	1.09, 0.71	1.54	0.67	24	61.2
Sample I	90	*	*	0.49	15.2	38.8
Total	**100**				**39.2 (22%)**	**100**

(the peaks do not appear in the IR)*

Solution ^{1}H NMR spectra of Sample H and Sample I were obtained in DMSO-d_6 with the spectra displayed in Figures 4.23 and 4.24. These spectra demonstrate the aromatic resonances likely associated with lignin and tannins could be found in Sample I, whereas these peaks were not present in Sample H. This is consistent with Sample I consist primarily of lignin/tannins. Sample H demonstrated a more intense aliphatic CH_2 resonance in the ^{1}H NMR than Sample I, consistent with the greater proportion of waxes in this sample identified by IR in Table 4.11.

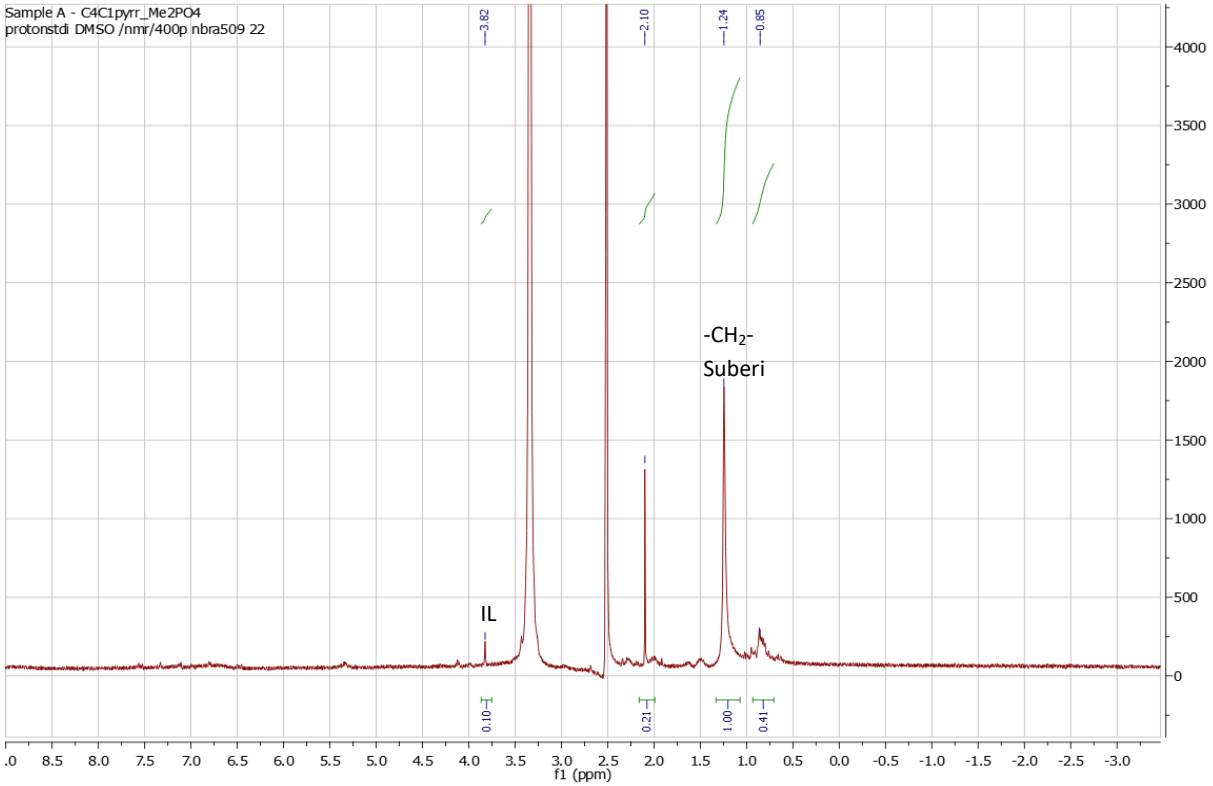


Figure 4.23 *^{1}H NMR of Sample H formed using 5 mL of [C_4C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

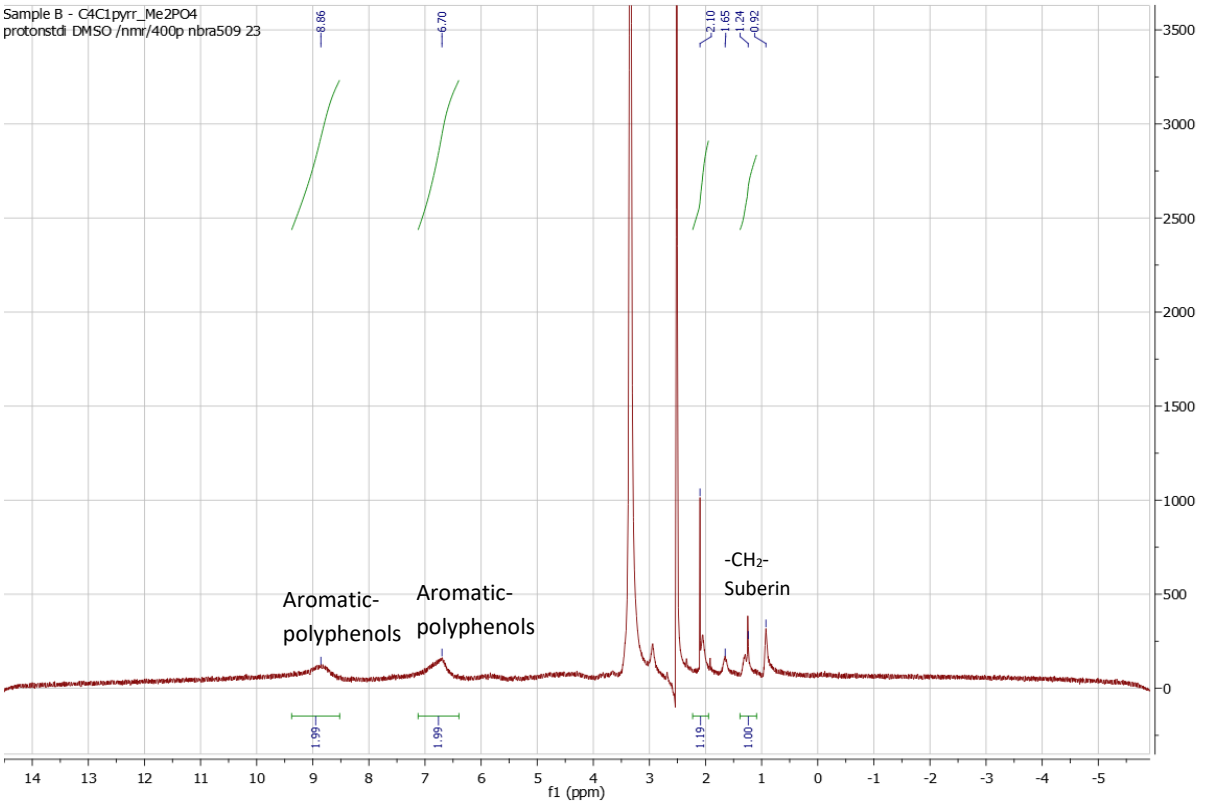


Figure 4.24 *^{1}H NMR of Sample I formed using 5 mL of [C_4C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

The antisolvents, MeCN and DMSO, weren't explored for this IL as these antisolvents didn't display efficient recovery of the extracts from the other ILs, as is evident in the earlier sections (4.1.2.1 and 4.1.2.2).

4.1.2.5. Addition of water as an antisolvent to the $[C_8C_1im][Me_2PO_4]$ IL extract

The $[C_8C_1im][Me_2PO_4]$ IL extract itself was gel-like in appearance after undergoing the extraction process (100 °C, 24 h), which made this IL extract different from the other IL extracts discussed so far. While the pure IL is inherently more viscous than the other ILs discussed, the gel-like appearance is suggestive of the formation of further hydrogen-bonding interactions with solutes following the extraction. The use of water as an antisolvent for this IL extract was explored, with the results summarised in Table 4.12 and the IR spectra displayed in Figures 4.25 – 4.27. Based on the extraction yield, it was anticipated that there should be 24.2 mg of solid able to be recovered from this IL extract. However, only 10.5 mg was obtained in total after addition of 10.5 mL of water, meaning the total recovery of extracted solids was only 43%. This is a significantly lower recovery than was observed for $[C_4C_1im][Me_2PO_4]$ under similar conditions which implies that the longer alkyl chain helps solubilise some of the extracted components.

From Table 4.12, it is apparent that hydrophobic components are emerging even after the addition of 7.5 mL out of the total of 10.5 mL of water. The relatively improved solubility of the hydrophobic components by $[C_8C_1im][Me_2PO_4]$ even on dilution with water is likely due to its ability to retain aspects of its hydrophobic nanostructure on dilution, as has been identified in the literature.[153] This would account for why the solubility of suberin/waxes didn't decrease massively on addition of water leading to their immediate precipitation, as was observed with ILs containing shorter alkyl chains.

Table 4.12 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 1 mL of $[C_8C_1im][Me_2PO_4]$ IL extract with the addition of water as an antisolvent.*

Sample Labels	**Water (mL)**	**Absorbance**			**Extract (mg)**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27,0.18	1.54	0.45	-
Sample 1	0.5	0.8, 0.56	3.44	0.32	2.8
Sample 2	0.5	1.5, 1.03	2.83	0.5	1.2
Sample 3	0.5	1.41, 0.94	2.82	0.44	0.4
Sample 4	1	***	***	***	0.1
Sample 5	1	0.83, 0.63	1.16	0.50	0.8
Sample 6	1	0.32, 0.18	NS	0.48	3.1
Sample 7	1	0.28, 0.14	NS	0.48	1.1
Sample 8	2	0.25, 0.13	NS	0.52	1.0
Sample 9	3	-	-	-	No ppts
Total	**10.5**				**10.5 (43.4 %)**

*(NS – Not sure i.e., the peaks are obscured by overlapping IL peak, *** amount of the sample was quite low and hence not enough for IR analysis, No ppts – no precipitates formed even after adding 3 mL of water)*

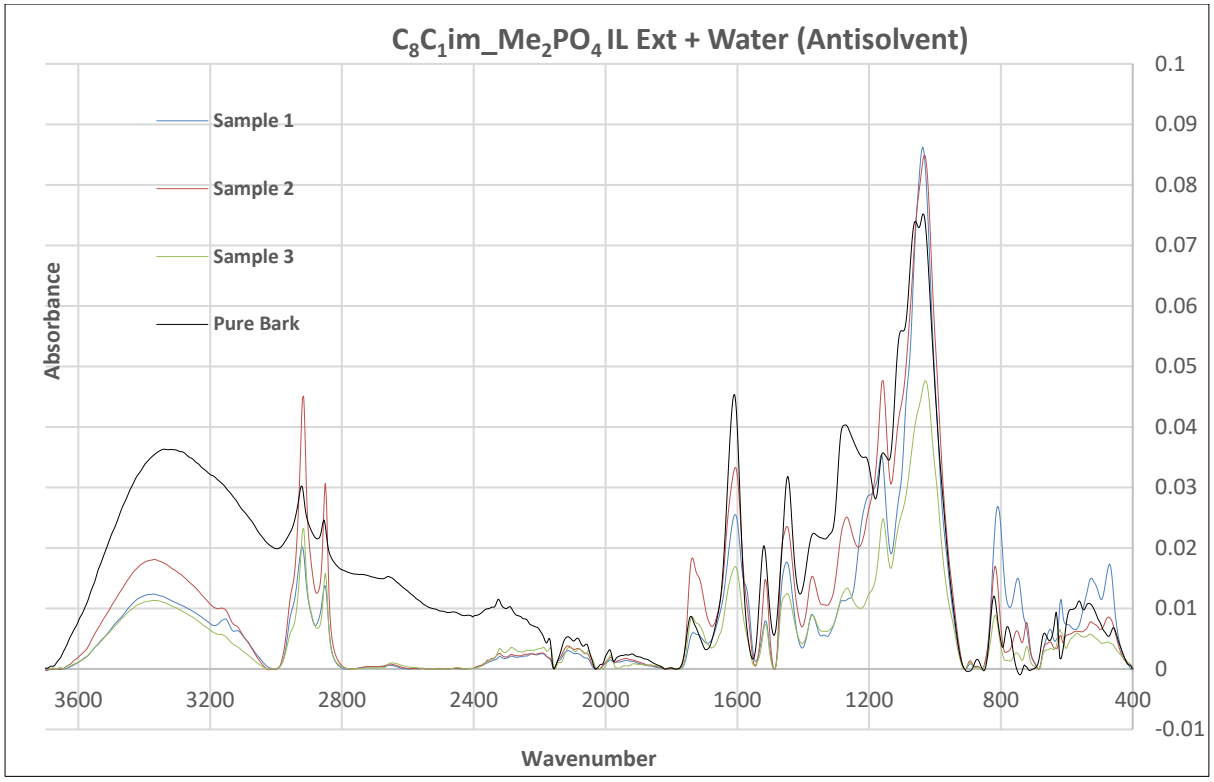


Figure 4.25 *IR spectra of the solids formed from Sample 1 - 3 using 1 mL of [C_8C_1im][Me_2PO_4] IL extract and water as an antisolvent*

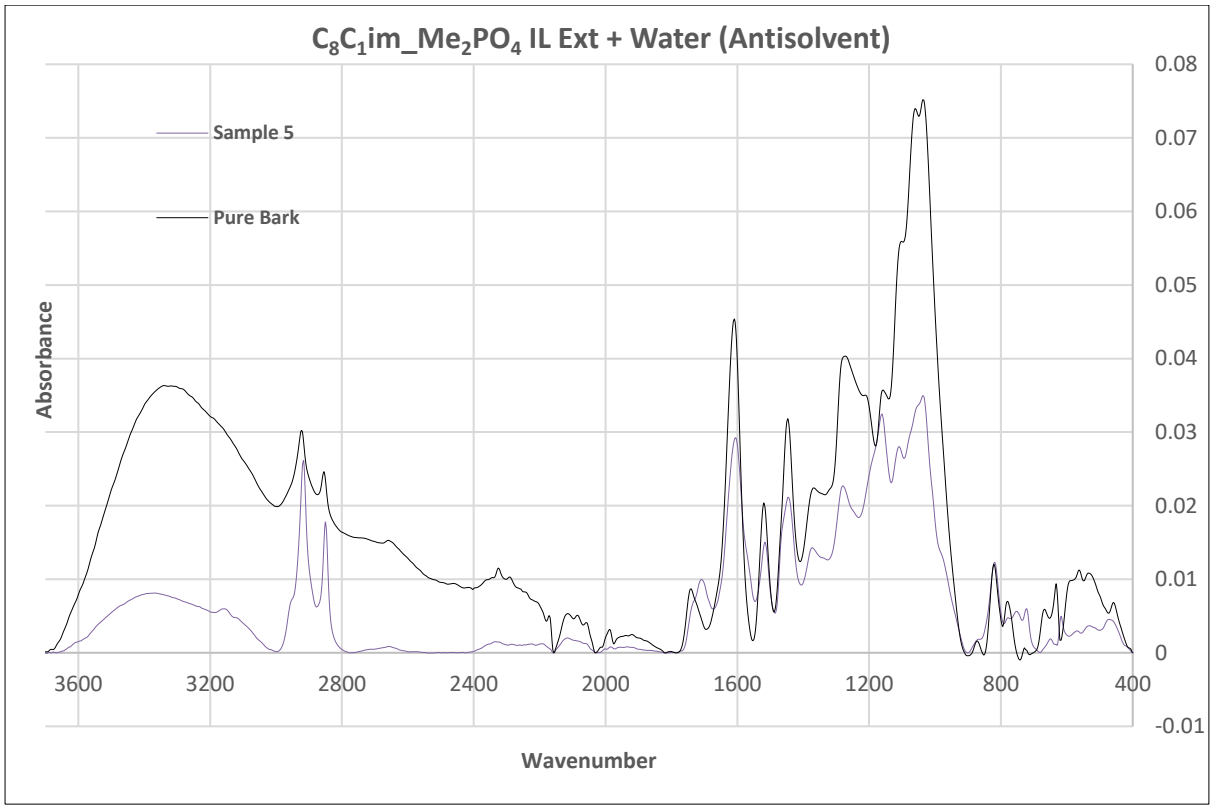


Figure 4.26 *IR spectra of the solids formed from Sample 5 using 1 mL of [C_8C_1im][Me_2PO_4] IL extract and water as an antisolvent*

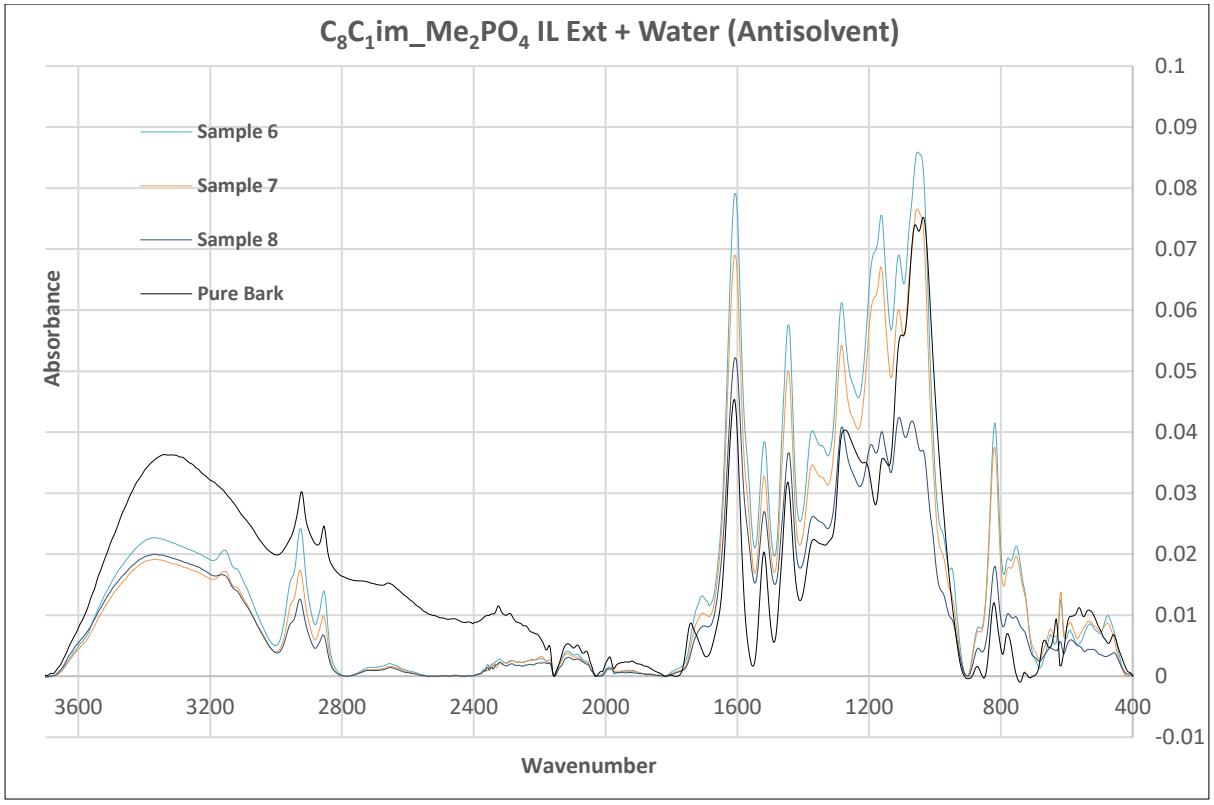


Figure 4.27 *IR spectra of the solids formed from Sample 6 - 8 using 1 mL of [C_8C_1im][Me_2PO_4] IL extract and water as an antisolvent*

Like the other ILs discussed previously, to improve the analysis of the extracts and assist in further understanding the recovery process, the characterisation of extracts recovered with the addition of antisolvent was also scaled-up to use 5 mL of IL extract. These results are represented in Table 4.13, with the IR spectra shown in Figures A71 - A73. These data follow the same trend as given above, in other words, all of the solids recovered are mixed fractions of suberin, carbohydrate, lignin and tannin components. Overall, 88 mg of solid was recovered from 5 mL of the IL extract which is 73% of the expected amount that could be recovered based on the extraction yield (120.8 mg). The recovery yield is higher for the scaled-up experiment compared to the one performed on a smaller scale but the general trend of the results obtained for these samples are similar in that almost all the extracts recovered for each sample appear to be mixed fractions, even with large antisolvent addition volumes.

Table 4.13 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 5 mL of [C_8C_1im][Me_2PO_4] IL extract with the addition of water as an antisolvent.*

[C_8C_1im][Me_2PO_4] IL extract (5 mL used)	**Addition of water (mL)**	**Absorbance**			**Solid recovered (mg)**	**Relative Sample Yields (%)**
		S : L	**C : L**	**T : L**		
Pure Bark	-	0.27,0.18	1.54	0.45	-	-
Sample J	7.5	0.85, 0.54	NS	0.67	44.6	50.68
Sample K	5	0.97, 0.63	3.11	0.60	7.6	8.64
Sample L	87.5	0.56, 0.35	1.40	0.55	35.8	40.68
Total	**100**				**88 (73 %)**	**100**

(NS – Not sure i.e., the peaks are obscured)

The solution ^{1}H NMR data for the recovered solid samples are shown in Figures 4.28 – 4.30. For Sample J and K these spectra display a sharp intense peak at a chemical shift at 8.33 - 8.56. The origin of this peak is unclear given the lack of other features within the spectrum. Given the chemical shift it is likely to correspond to an aldehyde environment although the lack of other notable resonances of similar intensity makes this difficult to assign. This peak was not observed for Sample L but broad signals appeared at a chemical shift of 6.60 ppm which corresponds to the aromatic lignin and tannin and is consistent with the higher relative concentration of these components observed by IR for this sample. This peak has appeared in the other ILs as well as discussed in earlier sections (4.1.2.1 and 4.1.2.4). The aliphatic CH_2 peak (1.25 ppm) associated with extracted waxes and suberin monomers is common in all of the ^{1}H NMR spectra, which corresponds to their presence in all of the fraction isolated for this IL.

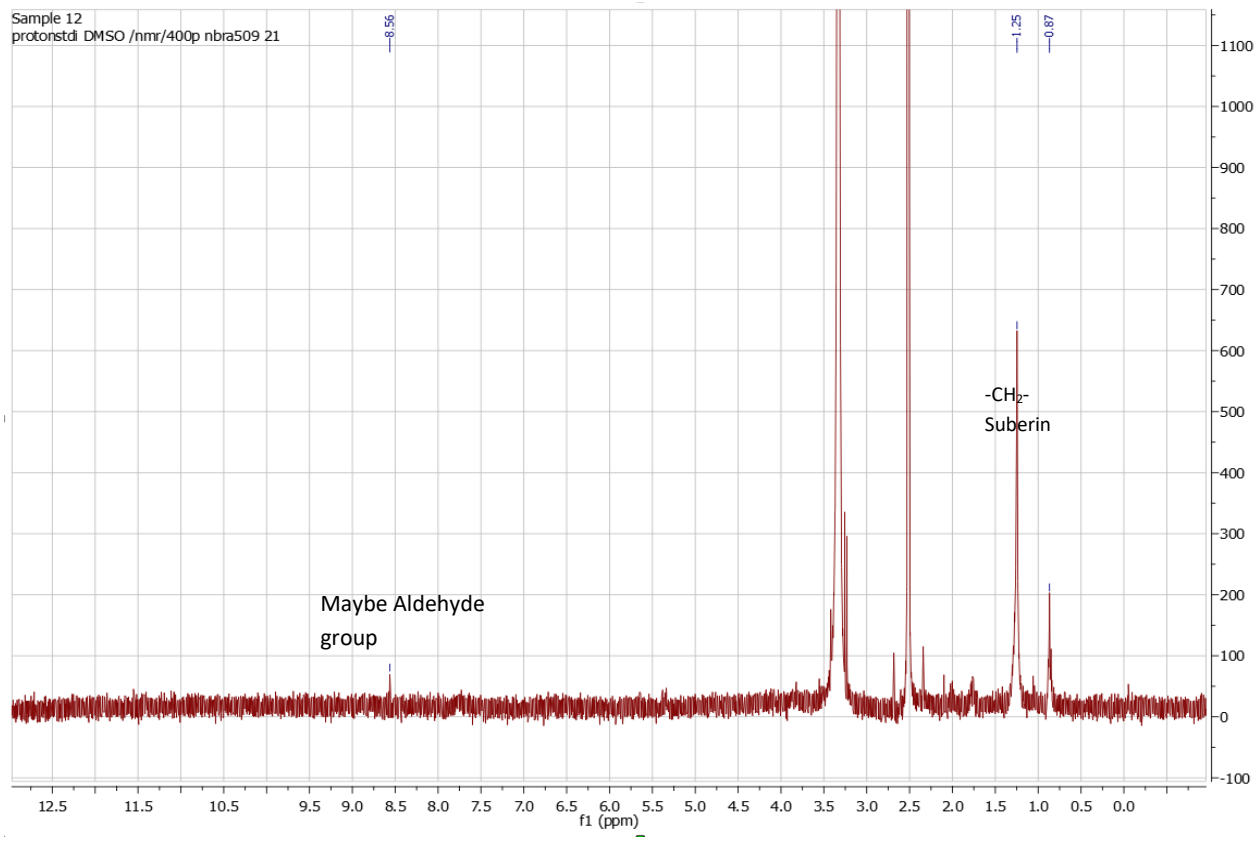


Figure 4.28 *^{1}H NMR of Sample J formed using 5 mL of $[C_8C_1im][Me_2PO_4]$ IL extract and water as an antisolvent*

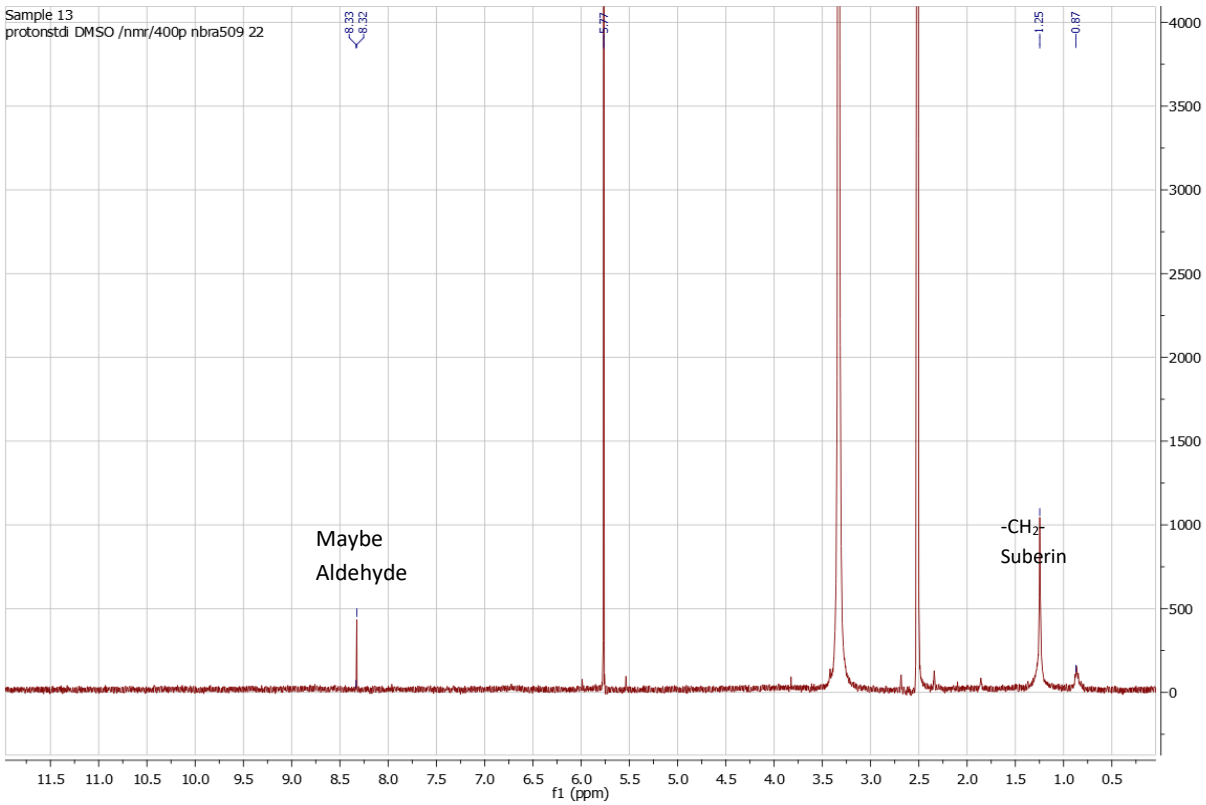


Figure 4.29 *^{1}H NMR of Sample K formed using 5 mL of $[C_8C_1im][Me_2PO_4]$ IL extract and water as an antisolvent*

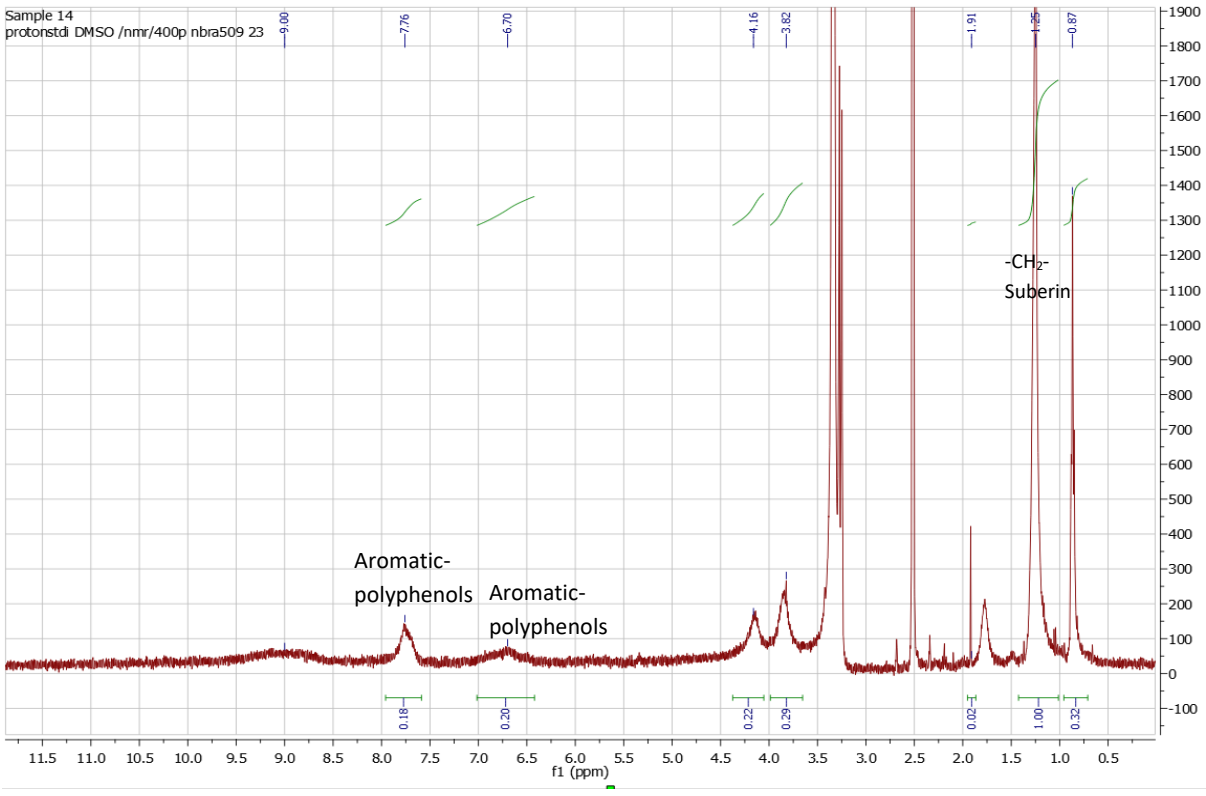


Figure 4.30 *^{1}H NMR of Sample L formed using 5 mL of $[C_8C_1im][Me_2PO_4]$ IL extract and water as an antisolvent*

4.1.2.6. Addition of water as an antisolvent to the $[C_8C_1pyrr][Me_2PO_4]$ IL extract

Table 4.14 represents the results of the $[C_8C_1pyrr][Me_2PO_4]$ IL extracts with the addition of water as an antisolvent, with the IR spectra shown in Figures 4.31 – 4.33. It has been observed that $[C_8C_1pyrr][Me_2PO_4]$ IL extract was able to extract hydrophobic parts to a greater extent than the other fractions if compared with the data of the other fractions recovered. Like $[C_4C_1pyrr][Me_2PO_4]$, suberinic or wax components are present to a significant extent based on the IR peaks detected in the recovered solid (samples 2 and 3 given in Figure 4.32). The solid recovered from the $[C_8C_1pyrr][Me_2PO_4]$ extracts also seems to contain mixed fractions even for the fractions obtained after substantial addition of water as an antisolvent, similar to the behaviour of the solid recovered from the $[C_8C_1im][Me_2PO_4]$ extracts described in the previous section. Hence, this behaviour most likely corresponds with the longer alkyl chain on the cation leading to improved solubility of the hydrophobic extracts which means more water is needed for their precipitation. Moreover, the known tendency of the $[Me_2PO_4]^-$ anion to enable the dissolution of carbohydrates allow the considerable isolation of carbohydrate components compared to ILs containing other anions.[141]

In addition, the maximum theoretical recovery based on the extraction yield (20 mg) was almost 3 times higher than the actual amount recovered (6.7 mg), which might be due to higher hydrophobicity of the IL cation leading to the increased solubility of extracts even on the addition of water. This is similar to $[C_8C_1im][Me_2PO_4]$ which also led to lower recovery, likely for these reasons. There is also evidence that ILs form hydrophobic nanostructures which can be retained on addition of water, which probably helps solubilise some of the more hydrophobic compounds even on further addition of water.[153] Although it has not confirmed yet whether $[C_8C_1pyrr][Me_2PO_4]$ lead to the formation of these nanostructures, it is likely given their existence in $[C_8C_1im][Me_2PO_4]$ so it is possible these play a role in solubilising hydrophobic compounds within $[C_8C_1pyrr][Me_2PO_4]$.

Table 4.14 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 1 mL of [C_8C_1pyrr][Me_2PO_4] IL extract with the addition of water as an antisolvent.*

Sample Labels	Water (mL)	Absorbance			Extract (mg)
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27,0.18	1.54	0.45	-
Sample 1	0.5	0.80, NS	0.98	0.66	0.4
Sample 2	1	2.49, 1.62	1.75	0.66	0.8
Sample 3	1	0.74, 0.44	1.25	0.49	2.2
Sample 4	2	0.27, 0.13	1.00	0.48	2.6
Sample 5	2	**	**	**	0.7
Sample 6	5	-	-	-	No ppts
Overall	**12.5**				**6.7 (34 %)**

*((NS – Not sure i.e., the peaks are obscured, * the peaks do not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis)*

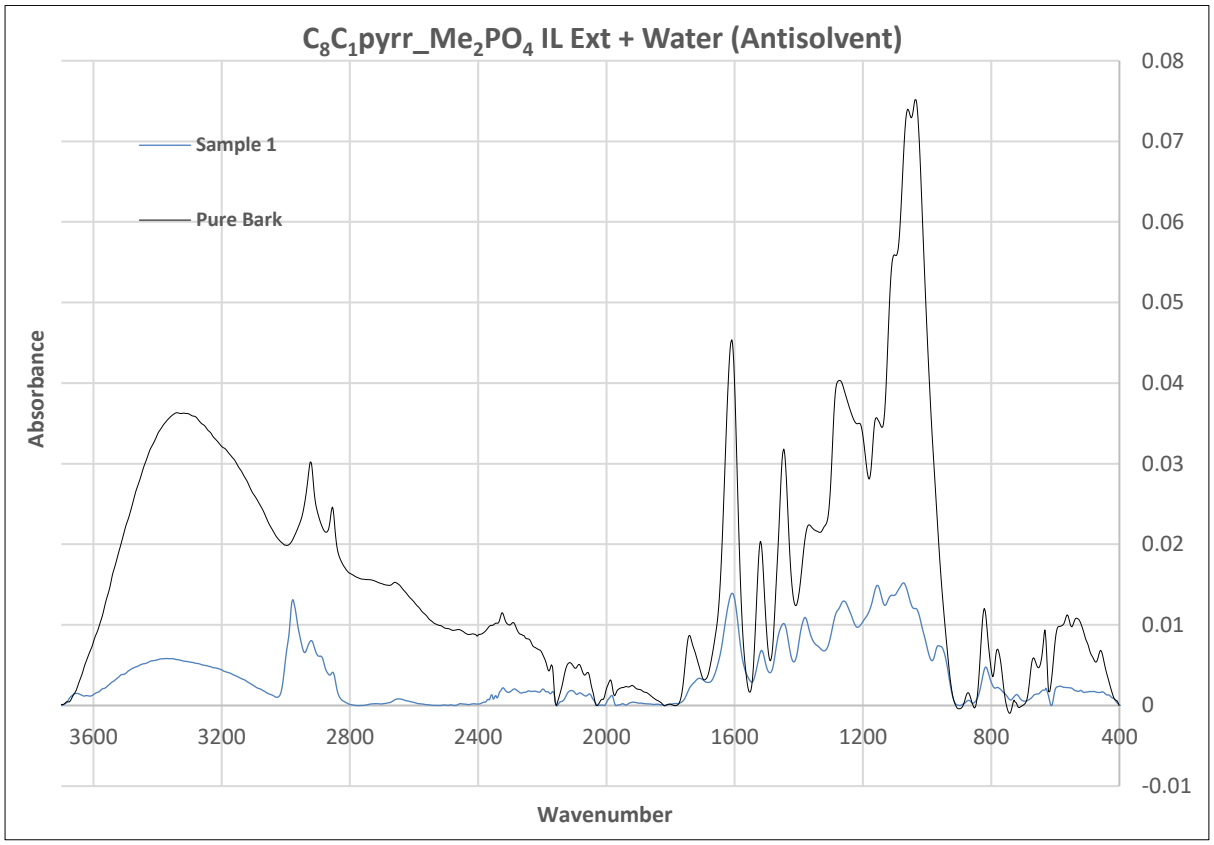


Figure 4.31 *IR spectra of the solids formed from Sample 1 using 1 mL of [C_8C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

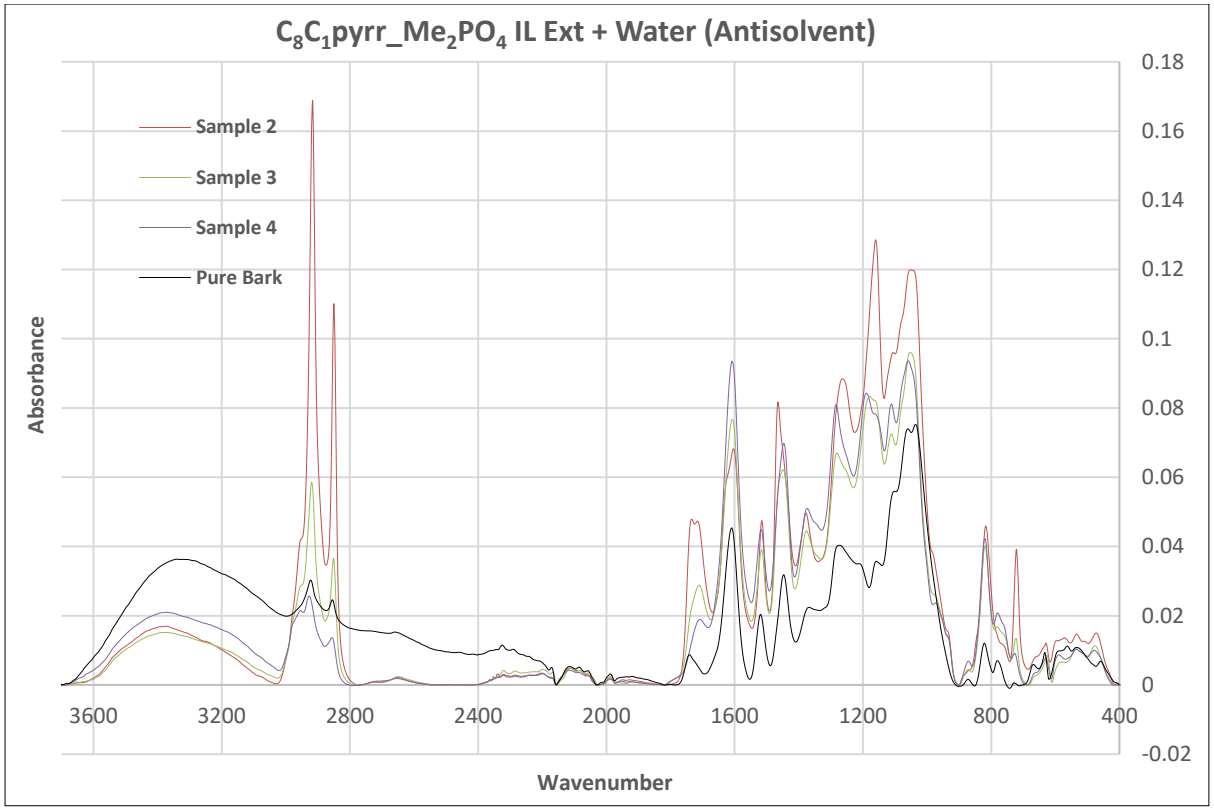


Figure 4.32 *IR spectra of the solids formed from Sample 2 – 4 using 1 mL of [C_8C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

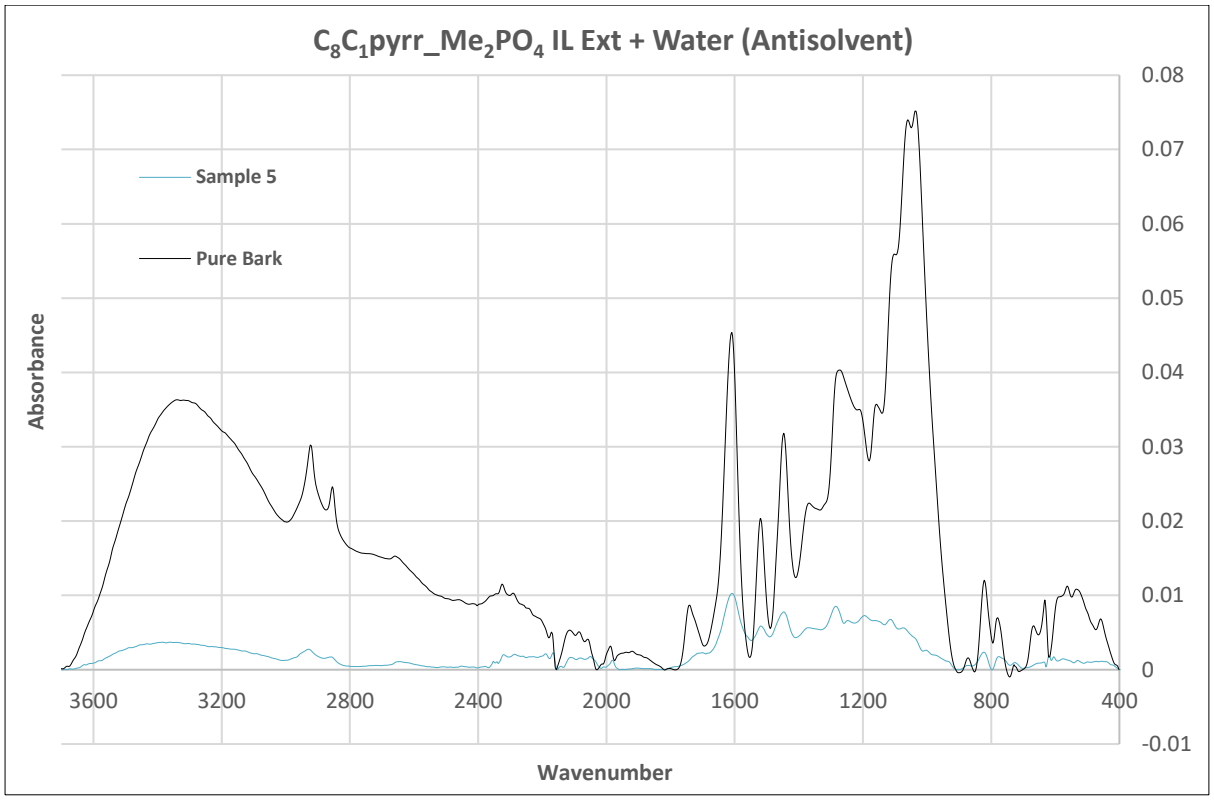


Figure 4.33 *IR spectra of the solids formed from Sample 5 using 1 mL of [C_8C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

As has been performed for previous ILs, the antisolvent addition process was scaled-up to improve the analysis of the extract. Table 4.15 and Figures A74, A75 (in the appendix) depict the data collected for the scaled-up samples (Samples M and O) obtained using 5 mL of the $[C_8C_1pyrr][Me_2PO_4]$ extract. Sample N was not characterised due to the small amount recovered from the round bottom flask as some amounts sticked to the round bottom flask. The outcomes are roughly in agreement with the results for the smaller scale experiments. All of the samples obtained that could be analysed displayed mixed fractions containing both aromatic lignin/tannin components as well as suberins/waxes. Unlike the smaller scale experiments, the final recovered solid (Sample O) contained no evidence of the carbohydrate component. The presence of these mixed fractions is similar to $[C_8C_1im][Me_2PO_4]$, which would correspond with the longer alkyl chain leading to improved solubility of the hydrophobic components which means more water is needed for their precipitation. This also accords with the relatively low recovery yield of around 50% from these ILs as 49.2 mg of solid could be recovered compared to the 100 mg anticipated based on the measured extraction yield.

Table 4.15 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 5 mL of $[C_8C_1pyrr][Me_2PO_4]$ IL extract with the addition of water as an antisolvent.*

$[C_8C_1pyrr][Me_2PO_4]$ IL extract (5 mL used)	**Addition of water (mL)**	**Absorbance**			**Solid recovered (mg)**	**Relative Sample Yields (%)**
		S : L	**C : L**	**T : L**		
Pure Bark	-	0.27,0.18	1.54	0.45	-	-
Sample M	5	0.94, 0.61	3.19	0.36	20.8	42.28
Sample N	30	**	**	**	24	48.78
Sample O	65	0.19, 0.1	*	0.55	4.4	8.94
Overall	**100**				**49.2 (50 %)**	**100**

(the peaks do not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis)*

The ^1H NMR spectra for these fractions are depicted in Figures 4.34 – 4.36. These spectra show an increase in the relative presence of peaks in the aromatic region related to either lignin or tannin fragments with an increase in the addition of water. At the same time the proportion of

the aliphatic CH_2 peak corresponding to wax and suberinic fragments started declining which is consistent with the relative IR absorbances observed for these samples.

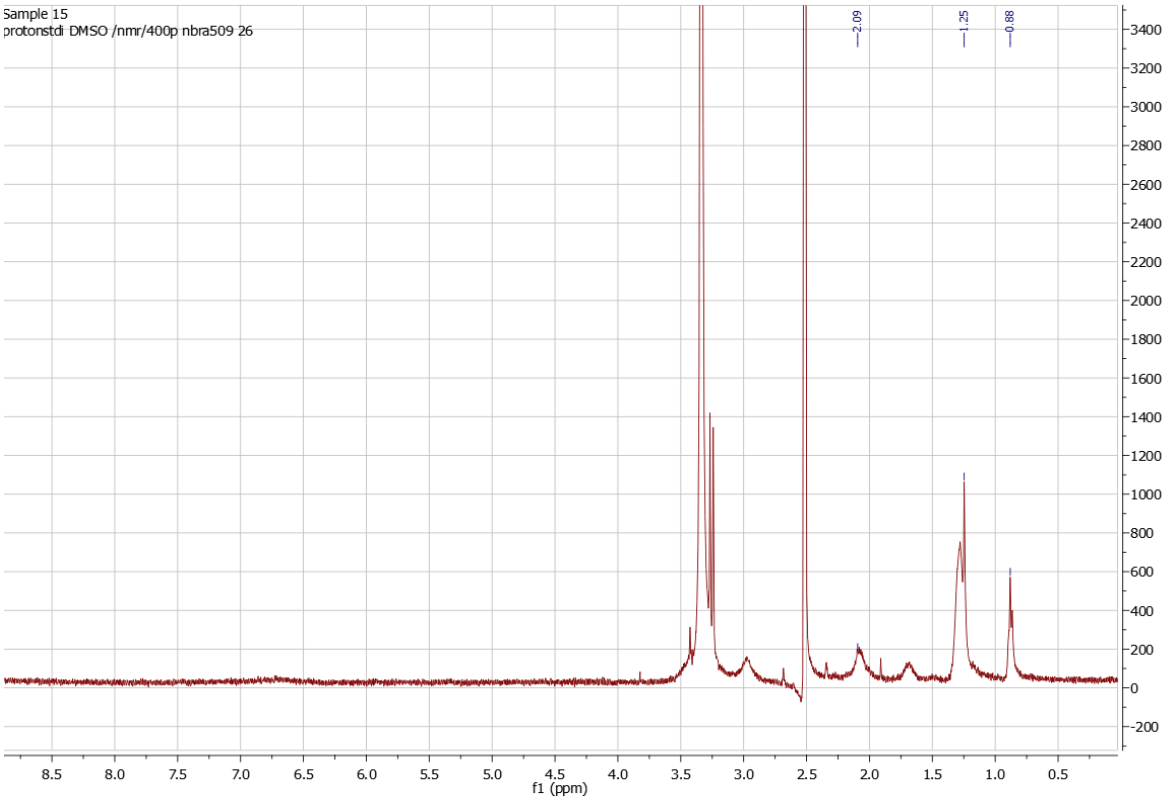


Figure 4.34 *^{1}H NMR of Sample M formed using 5 mL of $[C_8C_1pyrr][Me_2PO_4]$ IL extract and water as an antisolvent*

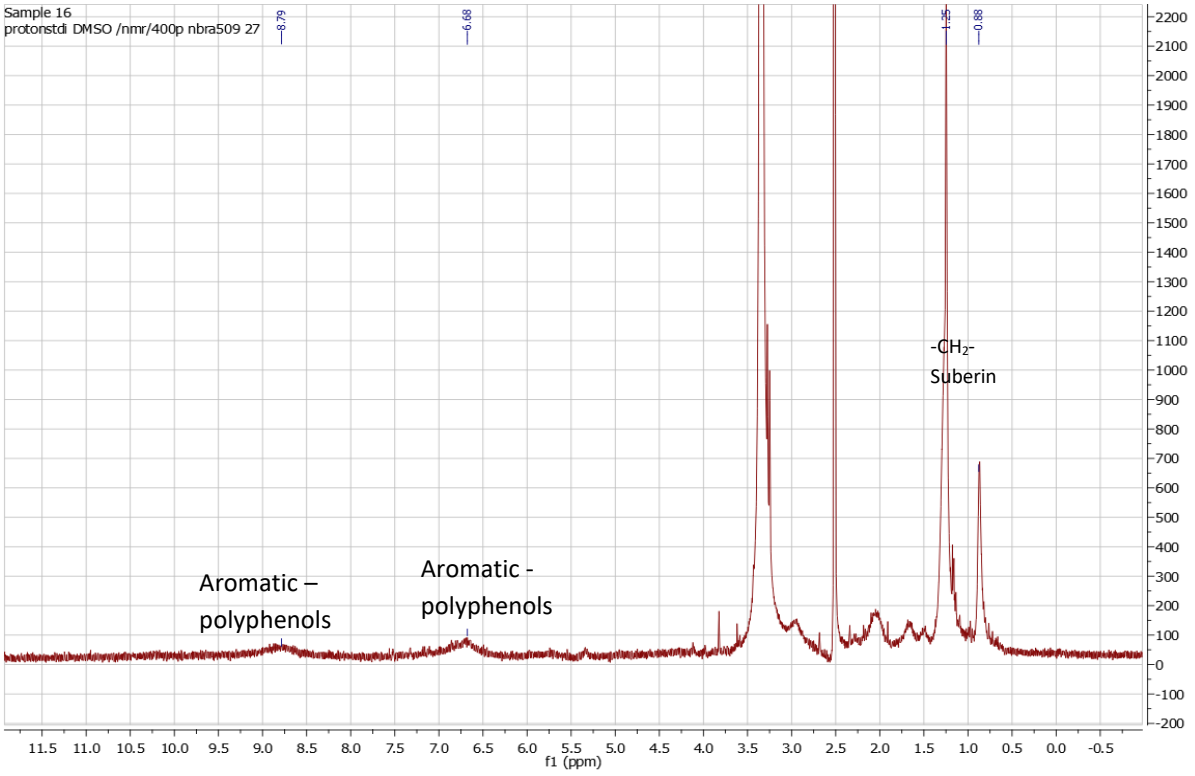


Figure 4.35 *^{1}H NMR of Sample N formed using 5 mL of $[C_8C_1pyrr][Me_2PO_4]$ IL extract and water as an antisolvent*

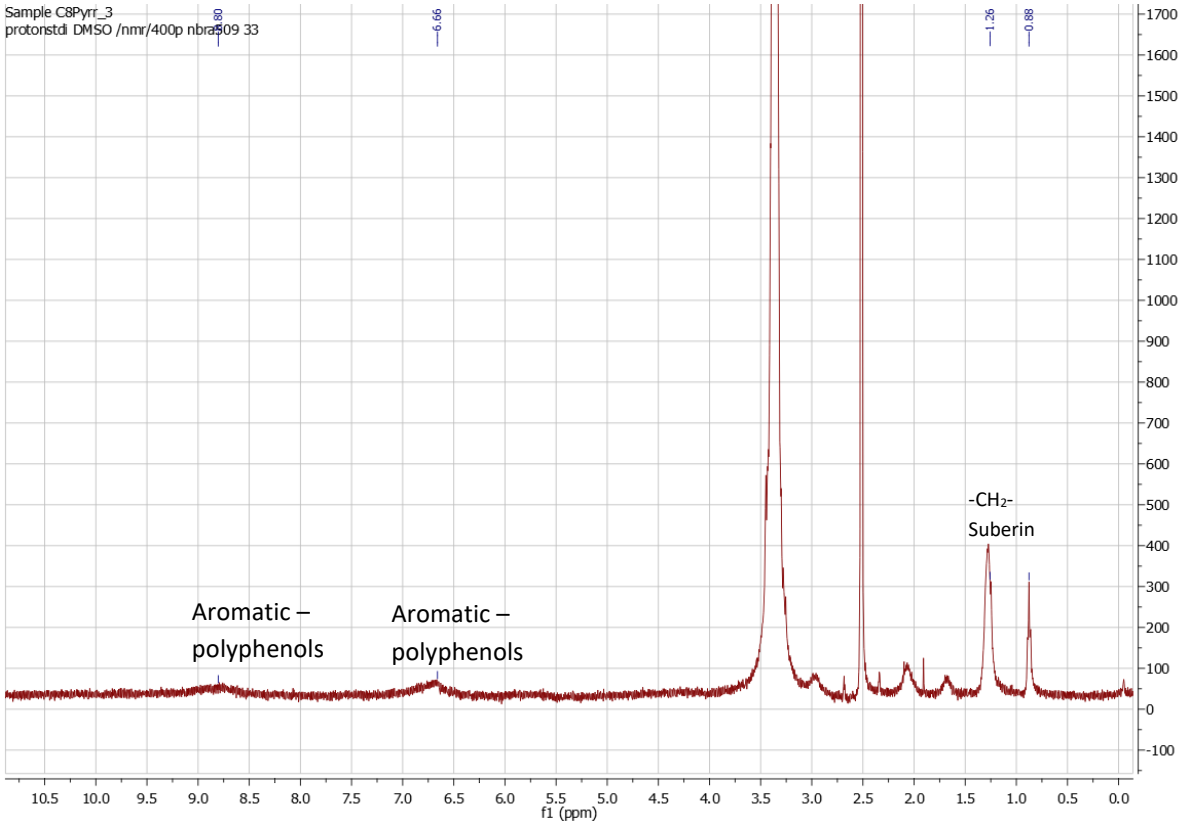


Figure 4.36 *^{1}H NMR of Sample O formed using 5 mL of $[C_8C_1pyrr][Me_2PO_4]$ IL extract and water*

4.1.2.7. Addition of water as an antisolvent to the GuCl : LA DES extract

Out of all the DESs prepared, GuCl:LA DES was selected for extraction of bark due to its high hydrogen bond donating capability and its ability to form a room temperature liquid as mentioned in the DES results and discussion chapter (Chapter 2).

After being used for the extraction of bark (100 ℃, 24 h) the GuCl : LA DES changed colour to form a red coloured solution. This might be due to a reaction of the acidic DES with key bark components not seen for the other solvents used[64] (as discussed in section 3.4.6.2). This could be due to either polysaccharide depolymerisation after undergoing acid catalysed hydrolysis reactions in the presence of acidic HBD or, more likely, the presence of polyphenols from the bark which are pH sensitive and change to red in the presence of the LA containing solvent. The specific origin of this colour was not investigated further due to time limitations.

Table 4.16 represents the results of the solid recovery from the GuCl:LA DES extracts with the addition of water as an antisolvent, with the IR spectra shown in Figures 4.37 – 4.38. It has been determined that even with the addition of only 0.5 mL of water around ~ 15 mg of solid was precipitated (Sample 1). This sample contained suberins/waxes in addition to lignin and tannin fragments. The remaining samples (Sample 2 – 8) appeared to contain only aromatic components. This could be due to the DES polarity since GuCl:LA DES is highly polar (π^* = 1.17 – 1.30, with a high α value, as discussed in Chapter 2) which might exclude the less polar compounds from being extracted in the first place. Hence only the aromatic components could be observed in the recovered solid fractions. This could also account for the relatively high yield for sample 1, as the small amount of these compounds that could be extracted are readily precipitated on addition of water. Overall, 28.6 mg of bark extract could be recovered which compared favourably to the amount of the bark extract expected (~ 30 mg).

The results obtained for recovery of fractions from DES extracts are not favouring the results discussed in Chapter 3 as it was observed previously that DESs were more capable of extracting carbohydrates than other fractions while according to the Table 4.16, there are no evidence of the presence of carbohydrates. This might be due to the acidic hydrolysis of polysaccharides under harsh extraction conditions.

Table 4.16 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 1 mL of GuCl:LA DES extract with the addition of water as an antisolvent.*

Sample Labels	**Water (mL)**	**Absorbance**			**Extract (mg)**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27,0.18	1.54	0.45	-
Sample 1	0.5	0.39, 0.28	*	0.46	15.4
Sample 2	0.5	NS	*	0.53	8.6
Sample 3	0.5	*	*	0.65	1.1
Sample 4	0.5	*	*	0.64	0.8
Sample 5	0.5	*	*	0.69	0.4
Sample 6	0.5	*	*	0.67	0.6
Sample 7	1	*	*	0.69	0.9
Sample 8	1	*	*	0.63	0.4
Sample 9	2	**	**	**	0.4
Total	**7**				**28.6 (95 %)**

*(NS -Not sure i.e., the peak shapes are unusual as depicted in Figure 4.37, * the peaks do not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis)*

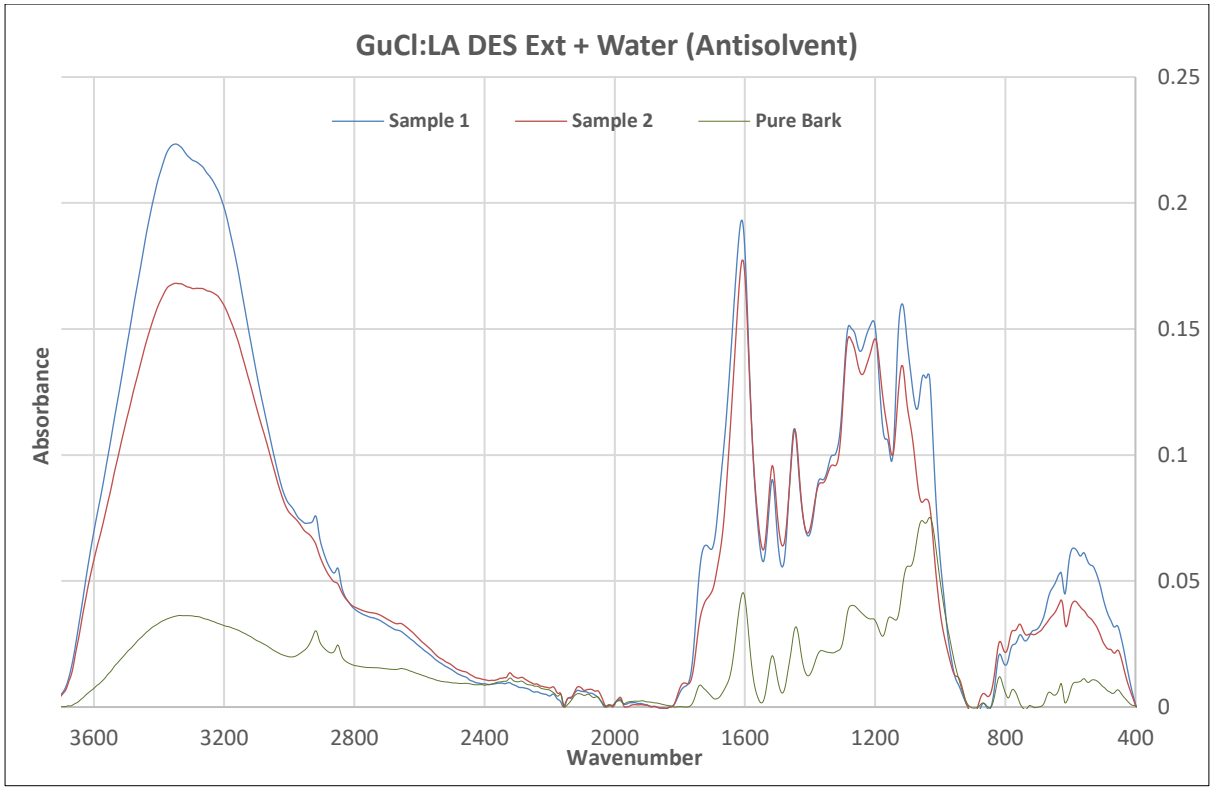


Figure 4.37 *IR spectra of the solids formed from Sample 1 - 2 using 5 mL of GuCl:LA DES extract and water as an antisolvent.*

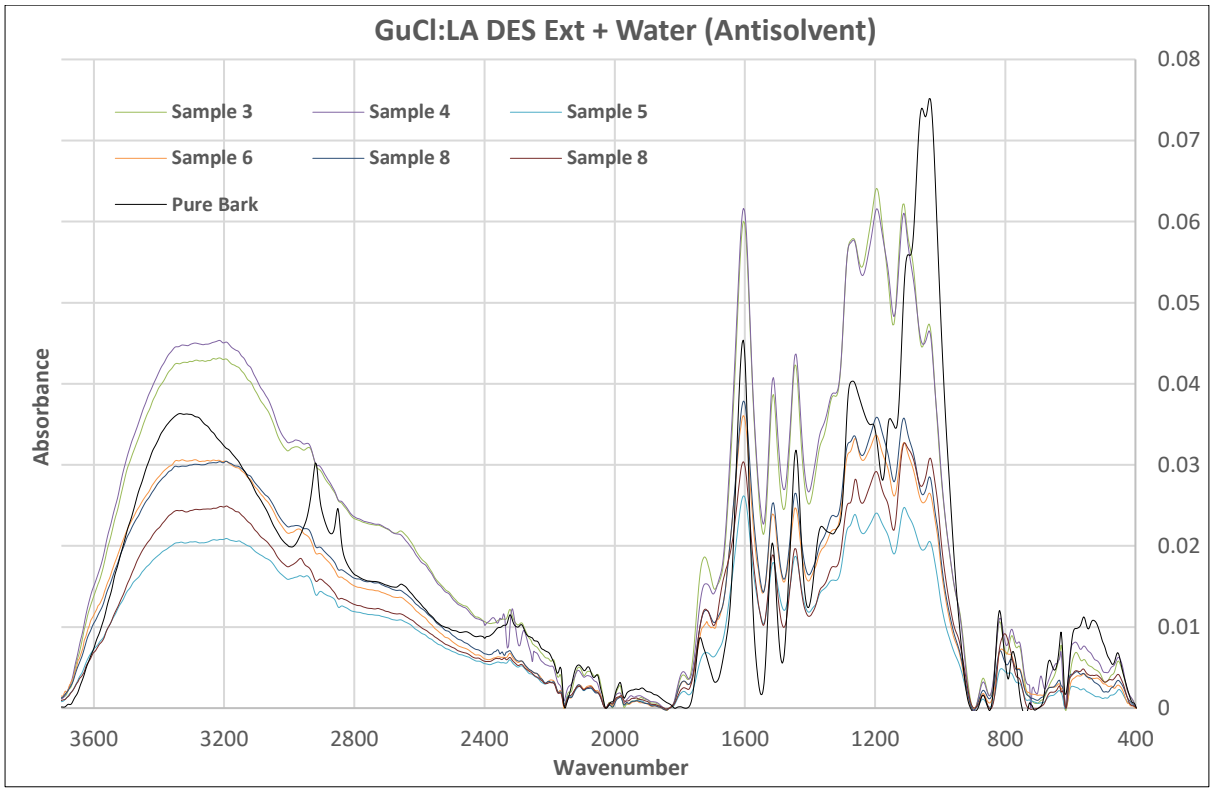


Figure 4.38 *IR spectra of the solids formed from Sample 3 – 9 using 5 mL of GuCl:LA DES extract and water as an antisolvent*

These experiments were scaled-up to 10 mL of the DES extract, as has been discussed for the IL samples previously. Table 4.17 summarises the outcome of these scaled-up antisolvent experiments, with the IR data shown in Figures A76 and A77 in the appendix. These results are consistent with the results given in Table 4.16 as no recovery of carbohydrates could be observed, with a relatively small amounts of suberin/waxes alongside aromatics in the initial fraction and only aromatic components in the later fractions.

Overall, for the scaled-up samples, only 28.6 mg could be recovered from 10 mL of the IL extract, far less than the anticipated 263 mg that would be expected based on the extraction yield. From the recovered fractions in the scaled-up experiment, by mass 84 % was the mixed fractions including suberin and tannin or lignin and about 16 % was the pure lignin and tannin fractions.

Table 4.17 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 10 mL of GuCl:LA DES extract with the addition of water as an antisolvent.*

GuCl:LA DES extract (10 mL used)	**Addition of water (mL)**	**Absorbance**			**Solid recovered (mg)**	**Relative Sample Yields (%)**
		S : L	**C : L**	**T : L**		
Pure Bark	-	0.27,0.18	1.54	0.45	-	-
Sample P	10	0.48, 0.24	*	0.49	24	84
Sample Q	90	*	*	0.57	4.6	16
Overall	**100**				**28.6 (11%)**	**100**

(the peaks do not appear in the IR)*

The ^{1}H NMR of the recovered solid, as shown in Figures 4.39 – 4.40 displays a peak at a chemical shift 6.95 ppm for both samples, which had also seen in some of the other IL extracts. This peak could correspond to the aromatic region of lignin and tannins but as the peak seems to be quite sharp and only present in this DES, so it is argued to declare this peak coming from the protons of aromatic polyphenols (more broader signals). This peak could come also come from probably the guanidinium cation or some other artifact from the solvent.

Along with this peak, there is also an aliphatic CH_2 peak (1.25 ppm) which has been seen in many of the extracts previously described and appears to decrease in relative intensity for the second fraction, consistent with the observed IR results.

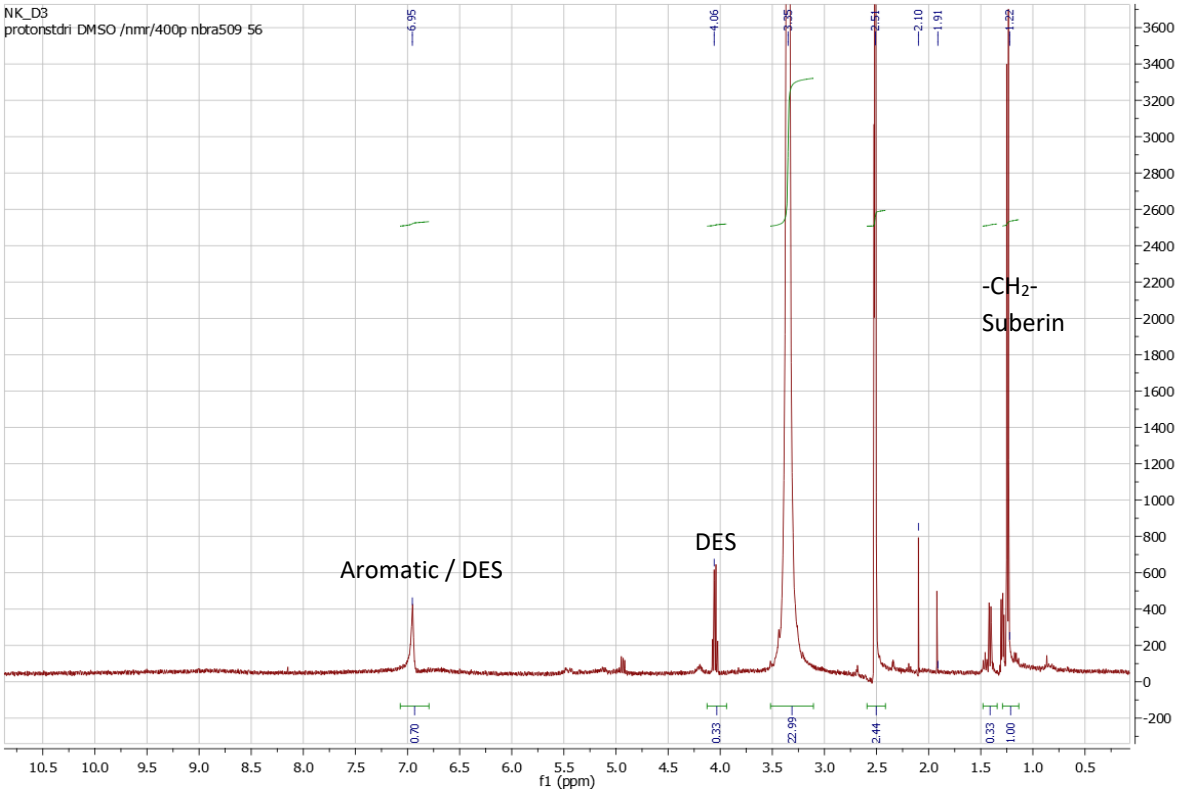


Figure 4.39 *^{1}H NMR of Sample P formed using 10 mL of GuCl:LA DES extract and water as an antisolvent*

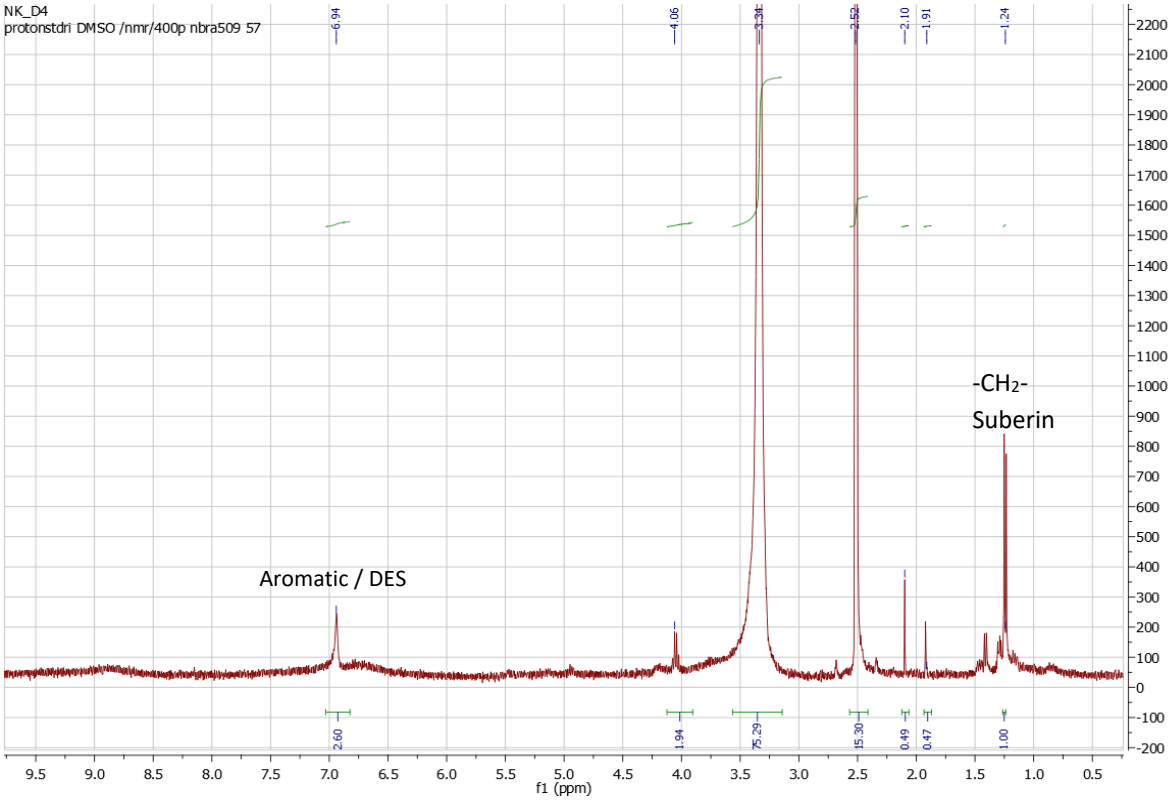


Figure 4.40 *^{1}H NMR of Sample Q formed using 10 mL of GuCl:LA DES extract and water as an antisolvent*

4.1.2.8. Addition of water antisolvent to the [TEA][HSO_4] IL extract

It had been determined that [TEA][HSO_4] potentially can play an important role in biomass processing, particularly for the extraction of lignin.[126] Hence, this IL has been used as a control with which to compare to the other ILs.

Like the GuCl:LA DES extract, the [TEA][HSO_4] IL extract also appeared red in colour (as discussed in section 1.8.4). This supports that this colour comes from the acidity of the solvent as this is the only other acidic solvent that has been explored for these extraction processes.

Table 4.18 represent the results of solid recovery from the [TEA][HSO_4] extract with the addition of water as an antisolvent, with the IR spectra shown in Figures 4.41 – 4.42. It has been observed that the addition of 2 mL of water resulted in the precipitation of mixed fractions containing suberin/waxes alongside the lignin/tannin components (Sample 1 – 4). The first two of these fractions also notably contained carbohydrate fractions alongside the suberins/waxes and tannin/lignin. The later fractions obtained (Sample 5 – 9) from the [TEA][HSO_4] IL extract contained only aromatic constituents (i.e. lignin and tannins).

Moreover, the amount of solid recovered (19.7 mg) was slightly over half of the expected 37.5 mg based on the extraction yield. This difference could be due to the solubility of components

in water or due to reactions between the acidic IL and bark components, such as acid-catalysed hydrolysis reactions.

Table 4.18 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 1 mL of [TEA][HSO_4] IL extract with the addition of water as an antisolvent.*

Sample Labels	**Water (mL)**	**Absorbance**			**Extract (mg)**
		S : L	**C : L**	**T : L**	
Pure Bark	-	0.27,0.18	1.54	0.45	-
Sample 1	0.5	0.41, 0.26	2.06	0.48	4.5
Sample 2	0.5	0.89, 0.60	2.16	0.54	0.3
Sample 3	0.5	1.14, 0.73	*	0.52	0.9
Sample 4	0.5	0.23, 0.10	*	0.38	1.4
Sample 5	0.5	*	*	0.44	4.5
Sample 6	0.5	*	*	0.46	0.5
Sample 7	1	*	*	0.45	2.8
Sample 8	1	*	*	0.50	1.6
Sample 9	2	*	*	0.50	0.7
Sample 10	5	*	*	0.46	1.4
Sample 11	5	*	*	0.47	1.1
Overall	**17**				**19.7 (52.5%)**

(the peaks do not appear in the IR)*

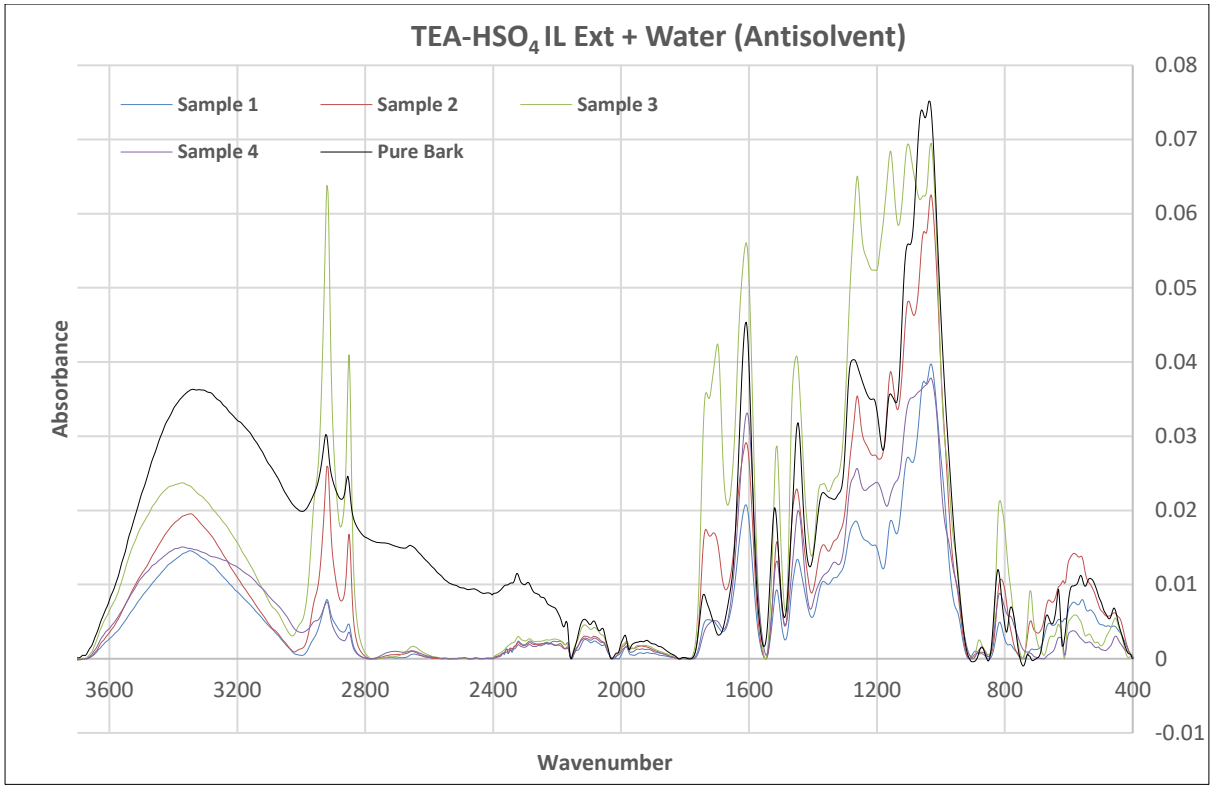


Figure 4.41 *IR spectra of the solids formed from Sample 1 – 4 using 1 mL of [TEA][HSO₄] IL extract and water as an antisolvent*

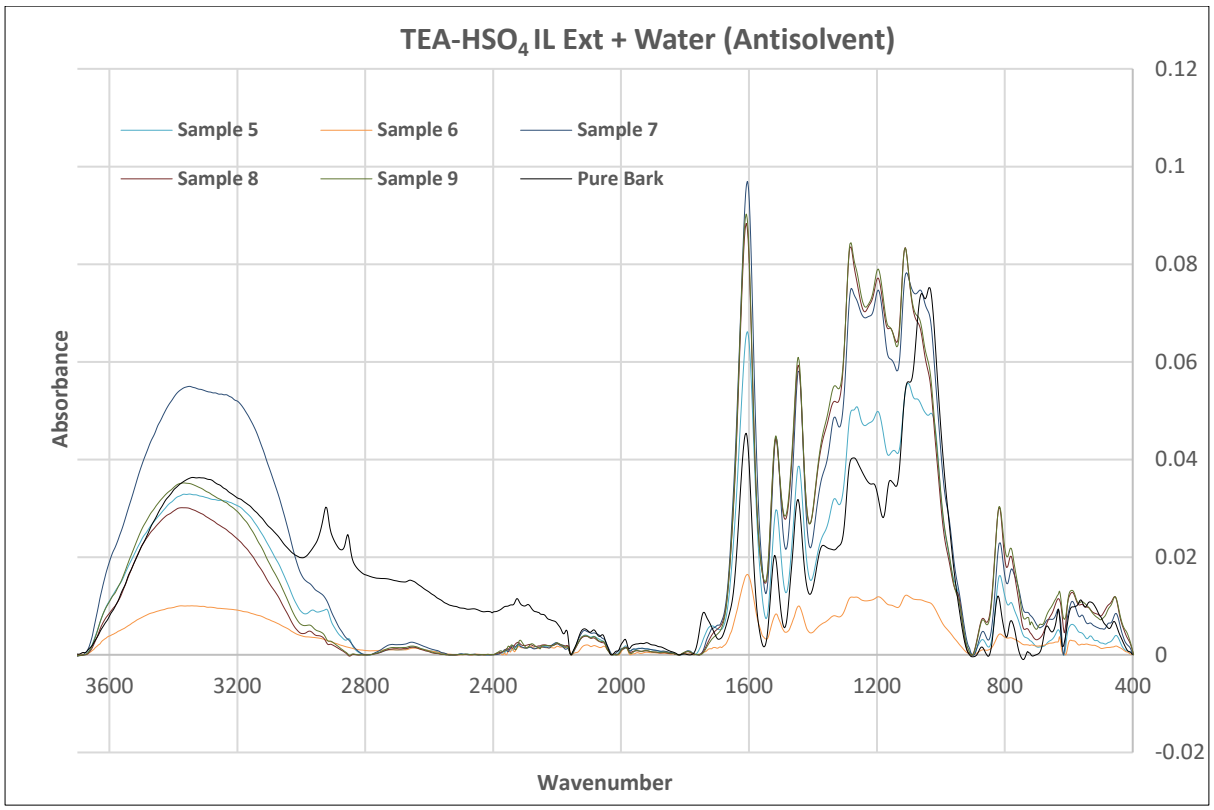


Figure 4.42 *IR spectra of the solids formed from Sample 5 – 9 using 1 mL of [TEA][HSO₄] IL extract and water as an antisolvent*

Similar to other ILs, the recovery of solid fractions from this IL extract was also increased with the antisolvent addition guided by the selectivity identified in Table 4.18. Table 4.19 summarises these results with the IR spectra displayed in Figures A78 and A79 in the appendix. It was evaluated that with 10 mL addition of water to 5 mL of the IL extract about 58 % of mixed fractions obtained including suberin/waxes, tannin, lignin and carbohydrates and further addition of water led to 42 % lignin and tannin fractions recovered which are free of contamination by suberin and carbohydrates. The total recovery yield at 25.7% was very low and appears consistent with many previous observations that the yields of the scaled-up samples tend to be lower than the smaller scale experiments, although this was not universally observed. This might be due to the higher inherent error in measuring small masses for the smaller scale experiments and the larger proportional impact of traces of solvent residue in these extracts.

Table 4.19 *Relative IR absorbance and solid recovery yields observed for the fractions of bark components isolated from 5 mL of [TEA][HSO_4] IL extract with the addition of water as an antisolvent.*

[TEA][HSO_4] IL extract (5 mL used)	**Addition of water (mL)**	**Absorbance**			**Solid recovered (mg)**	**Relative Sample Yields (%)**
		S : L	**C : L**	**T : L**		
Pure Bark	-	0.27,0.18	1.54	0.45	-	-
Sample R	10	0.49, 0.30	1.93	0.50	28	58.09
Sample S	90	*	*	0.50	20.2	41.91
Overall	**100**				**48.2 (25.7 %)**	**100**

(the peaks do not appear in the IR)*

Moreover, the ^{1}H NMR spectra of these samples (Figures 4.43 and 4.44) depicted that for Sample R, obtained from the first addition of water, a significant peak corresponding the aliphatic CH_2 resonance of the wax components (1.25 ppm) was observed. The precipitate formed from the further addition of water led to only the observation of broad aromatic peaks corresponding to lignin and tannin (6.67 – 6.70 ppm). These results are similar to the IR results discussed above in terms of the selective isolation of the fractions.

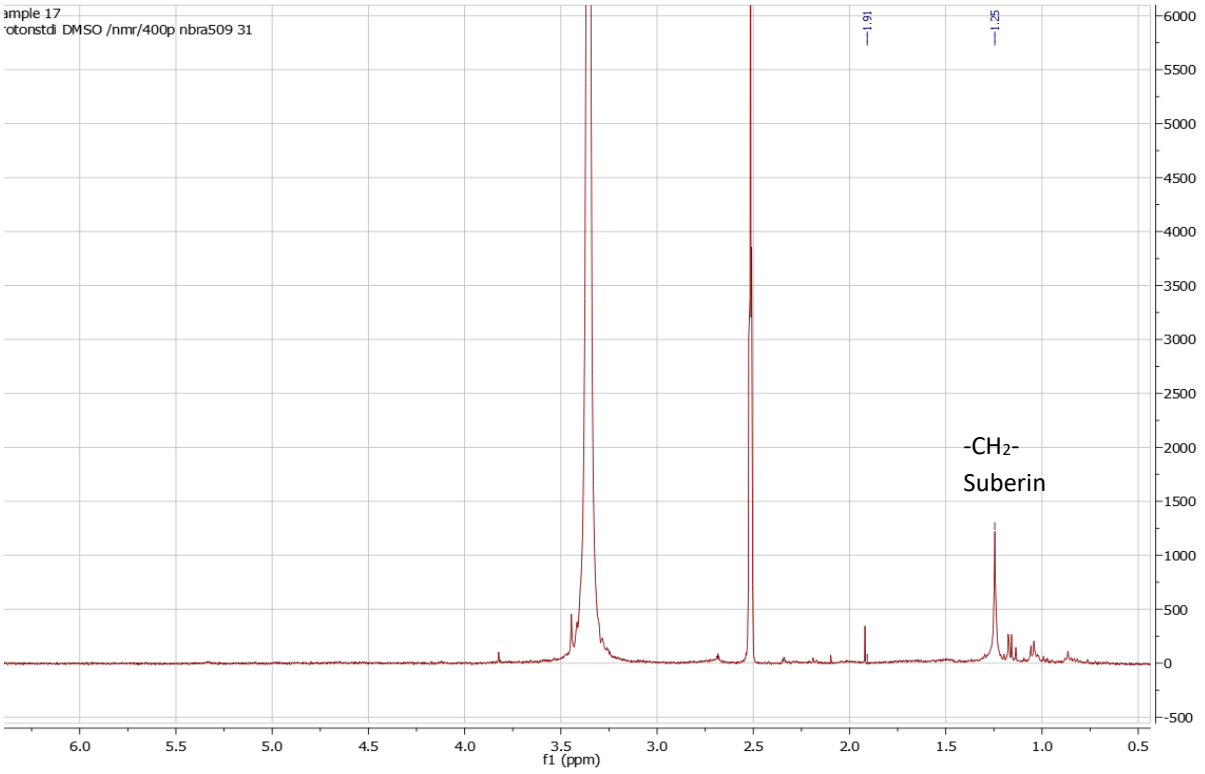


Figure 4.43 *^{1}H NMR of Sample R formed using 5 mL of [TEA][HSO$_4$] IL extract and water as an antisolvent*

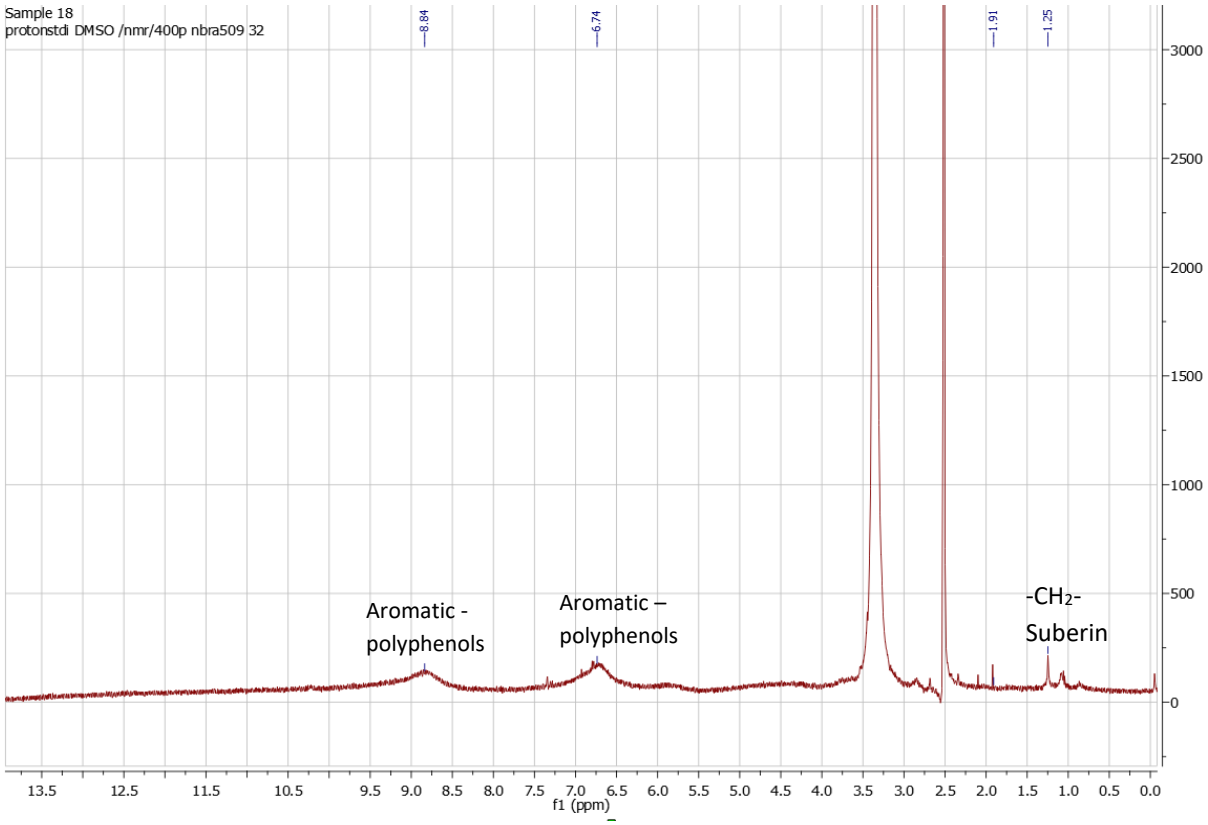


Figure 4.44 *^{1}H NMR of Sample S formed using 5 mL of [TEA][HSO$_4$] IL extract and water as an antisolvent*

4.2. TMA-OH Pyrolysis GC-MS Analysis

To more completely quantify the suberin/wax components observed in the scaled-up solid extracts obtained, TMA-OH pyrolysis GC-MS analysis was used. TMA-OH is a derivatising agent for pyrolysis GC-MS analysis, which causes the hydrolysis of fatty acid esters and the subsequent methylation of the fatty acids to generate fatty acid methyl esters (FAME) that are more volatile and readily detectable by GC-MS (as discussed in Chapter 3).[147] This technique not only determines the long chain fatty acid or suberin/waxes components but also some highly substituted benzenes and fragments of carbohydrates to help corroborate the composition of the sample in terms of carbohydrates, suberins/waxes and lignin/tannins.

Figure 4.45, 4.46, and 4.47 depicts the proportion of different representative volatile compounds that were observed using TMA-OH pyrolysis GC-MS. These compounds were the methyl ester of tetracosanoic acid, 2,4,6-trimethoxytoluene and 3-methylfuran. The methyl ester of tetracosanoic acid is representative of the suberin/wax extracts of these samples, 2,4,6-trimethoxytoluene is representative of the lignin component and 3-methylfuran representative of the carbohydrate components of the different IL extract samples. Additionally, Table 4.20 represents the list of the samples used in TMA-OH pyrolysis GC-MS technique for analysing recovered volatile compounds.

Table 4.20 *List of samples used in determining the volatile compounds obtained from TMA-OH Py-GCMS.*

Label	**Sample**
Sample A	10 mL of $[C_4C_1im][Me_2PO_4]$ IL extract + 15 mL of water as an antisolvent
Sample B	Sample A filtrate + 10 mL of water as an antisolvent
Sample C	(Sample A + Sample B) filtrate + 75 mL of water as an antisolvent
Sample D	10 mL of $[C_4C_1im][N(CN)_2]$ IL extract + 10 mL of water as an antisolvent
Sample G	(Sample D + Sample E + Sample F) filtrate + 75 mL of water as an antisolvent
Sample P	10 mL of GuCl:LA DES extract + 10 mL of water as an antisolvent
Sample Q	Sample P filtrate + 90 mL of water as an antisolvent

According to Figure 4.45, it was observed that the significant proportion of long chain fatty acid (tetracosanoic acid, methyl ester) was present in the Sample A and Sample D i.e., 10 mL of $[C_4C_1im][Me_2PO_4]$ IL extract and 15 mL of water (antisolvent) and 10 mL of

$[C_4C_1im][N(CN)_2]$ IL extract and 10 mL of water (antisolvent), respectively, compared to pure bark and other recovered samples. This is in agreement with the IR and NMR results as in the case of all IL extracts, suberin/wax components and carbohydrates, where they were present, were isolated first with the addition of water followed by the isolation of primarily lignin and tannin components later. Also, being polar in nature, water leads to the precipitation of the less polar components prior to those that are more polar. Moreover, the amount of the FAME observed decreased for samples obtained on further addition of water. This means, for Sample C - $[C_4C_1im][Me_2PO_4]$ IL extract filtrate and 75 mL of water and Sample G - 10 mL of $[C_4C_1im][N(CN)_2]$ IL extract filtrate and 75 mL of water. Several samples display higher tetracosanoic acid, methyl ester proportions on small amounts of water antisolvent addition than pure bark, highlighting that these wax components are concentrated on extraction and recovery from the IL.

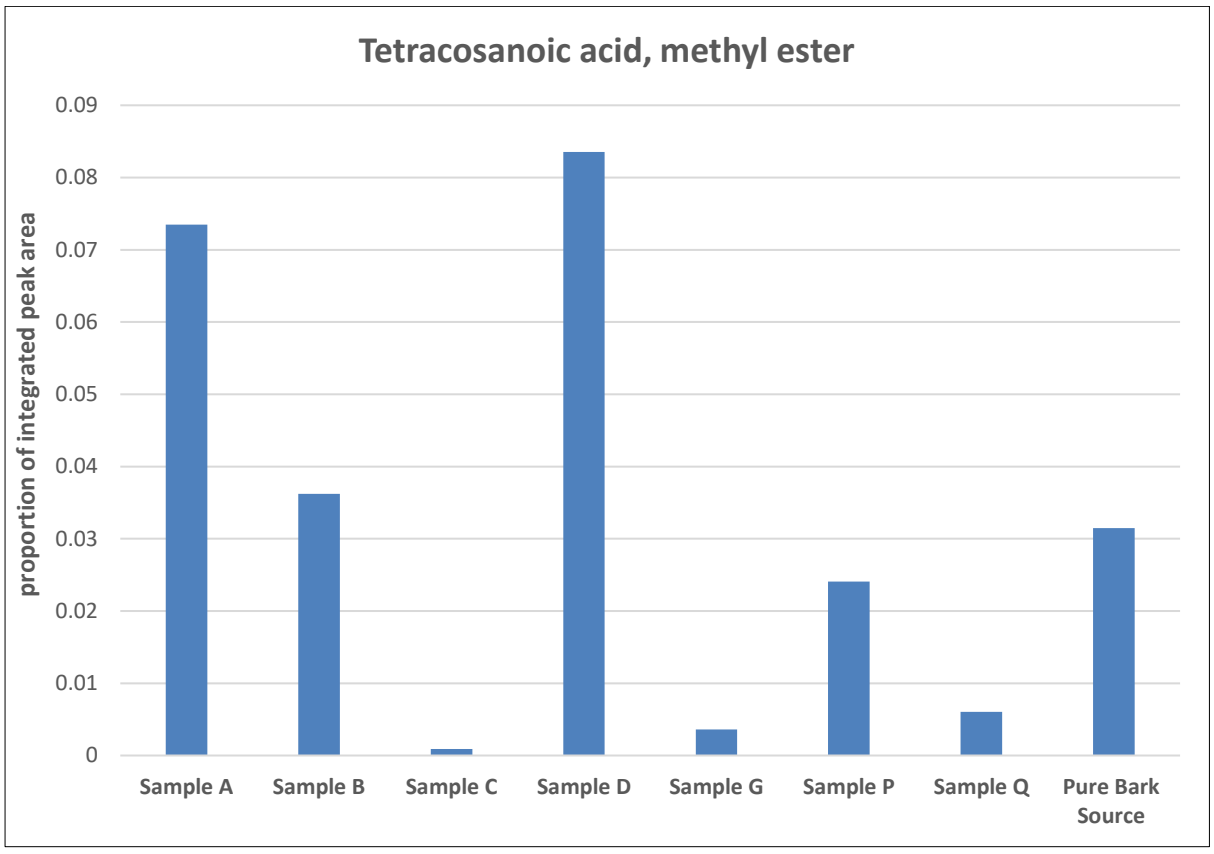


Figure 4.45 *Observed relative TIC areas for tetracosanoic acid, methyl ester following TMA-OH-py-GC-MS analysis for recovered solid fractions from IL extracts. Sample ID is given in Table 4.20.*

From Figure 4.46 it has been determined that that the quantity of 2,4,6-trimethoxytoluene as a representative highly substituted aromatic which most likely arises from lignin or tannin, is

almost equally in the [C_4C_1im][Me_2PO_4] IL and [C_4C_1im][$N(CN)_2$] recovered extract than DES extract. Moreover, the amount of this compound is less for the Sample A and increases with dilution (Sample B and Sample C). Similar trends of smaller aromatic concentration for samples obtained from the addition of small amounts of water as an antisolvent were observed for the other ILs explored. This is consistent with the observation that the proportion of aromatics in these fractions typically increases with increasing antisolvent addition.

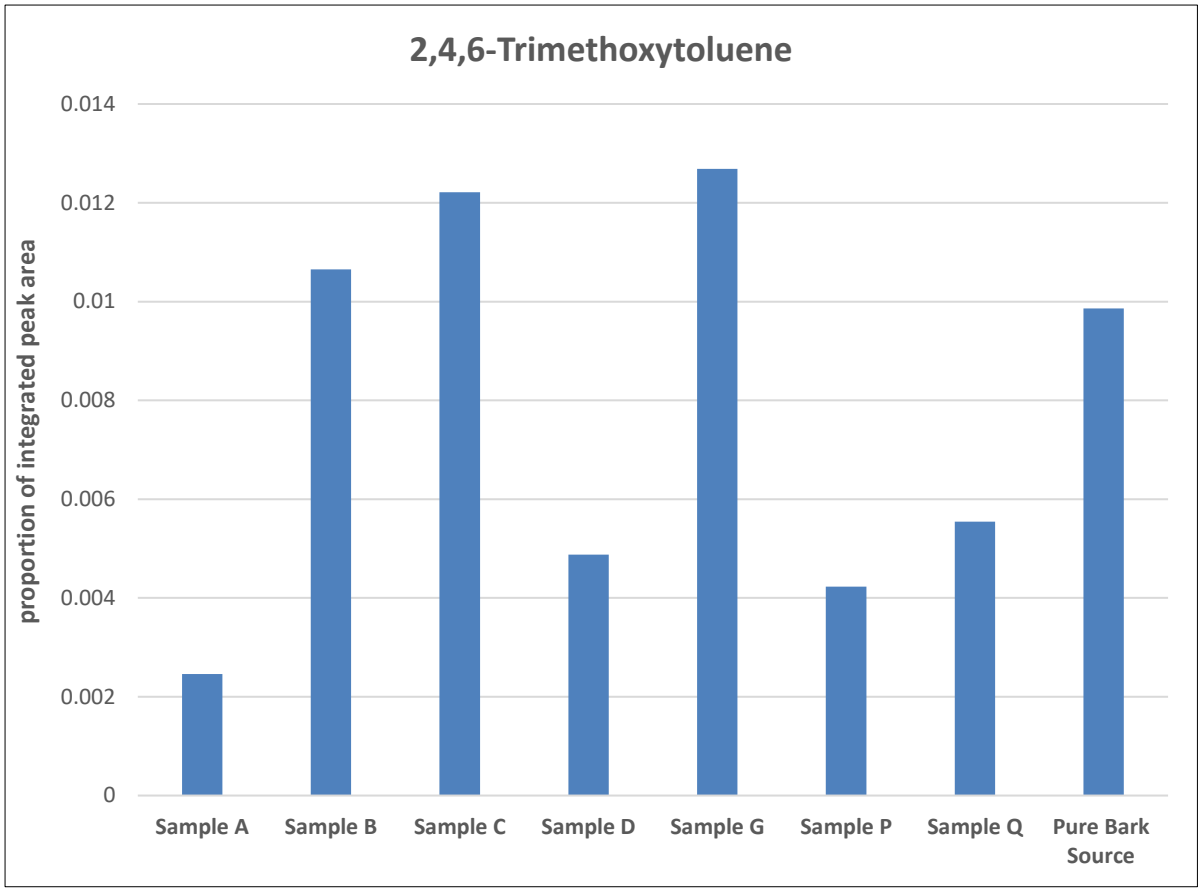


Figure 4.46 *TMA-OH-py-GC-MS results for 2,4,6-trimethoxytoluene obtained for solids recovery by water antisolvent addition to IL bark extracts. Sample IDs are summarised in Table 4.20.*

Lastly, Figure 4.47 broadly represents the proportion of carbohydrates recovered from the different samples. It was evaluated that the carbohydrates recovered in a significant amount from [C_4C_1im][Me_2PO_4] IL extract with the addition of water than other IL extracts and DES extract. It is due to the ability of the [Me_2PO_4] anion[141] and imidazolium cation[154] to aid the dissolution of cellulose and related carbohydrates as discussed in section 4.1. The proportion of the carbohydrate also decreased with further addition of water as an antisolvent, in direct agreement with the trends observed by IR and NMR for each of the ILs. All samples show a

reduction in carbohydrate content relative to pure bark, emphasising that ILs preferentially extract components other than carbohydrates from the bark.

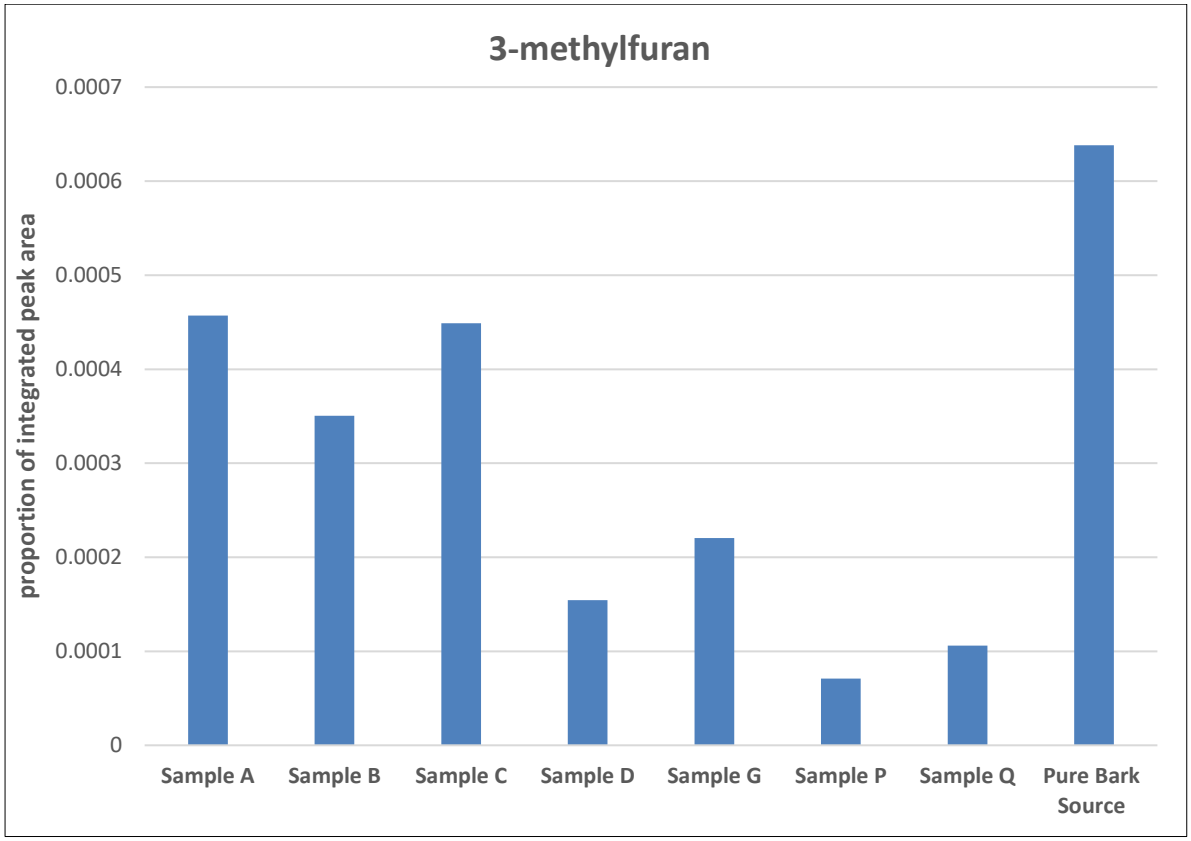


Figure 4.47 *TMA-OH-py-GC-MS results for 3-methylfuran obtained for solids recovery by water antisolvent addition to IL bark extracts. Sample IDs are summarised in Table 4.20.*

4.3. Summary

In conclusion, the addition of several antisolvents have been explored for the recovery of bark extracts from the ILs and DESs explored for the extraction of bark in Chapter 3. The characterisation of the recovered extracts was performed by using IR, NMR and TMA-OH Py-GCMS. It has been determined that out of the antisolvents used for these separations/fractionations, water appeared to be the most suitable in terms of both the selectivity and yield of the fractions that could be obtained of each of the components of bark. The other two antisolvents, MeCN and DMSO, typically led to mixed fractions with limited evidence of preferential precipitation and relatively low yields of solid formation. Although almost all the IL and DES extracts led to the formation of fractions containing suberin/waxes, carbohydrates and lignins/tannins, the quantity and selectivity of these fractions varied significantly between each solvent when water was used as an antisolvent.

It was observed that the $[Me_2PO_4]^-$ IL extracts led to higher yields of carbohydrates than could be recovered from $[N(CN)_2]^-$ and $[NTf_2]^-$ ILs or the DES with the addition of water as an antisolvent. Moreover, the highest proportion of carbohydrate was recovered from $[C_4C_1im][Me_2PO_4]$ due to the presence of the imidazolium cation in combination with strong hydrogen bond acceptor anion. Based on the TMA-OH-Py-GC-MS results, the most carbohydrate enriched fraction obtained from this IL has similar carbohydrate concentrations as pure bark, whereas most of the ILs led to significantly lower carbohydrate concentrations. Additionally, although the highest recovery yield of pure lignin/tannin aromatics (20.2 mg, 41.9%)) was obtained from the $[TEA][HSO_4]$ IL extract, as would be expected given its commercialisation as an IL for delignification, $[C_4C_1pyrr][Me_2PO_4]$ extracts also enabled the recovery of a significant amount of pure lignin/tannin fractions, (15.2 mg, 39%) of the total recovered solid mass, respectively. Regarding the isolation of suberins/waxes as hydrophobic components, $[C_4C_1pyrr][Me_2PO_4]$ IL contributed in extracting and enabling the recovery of the highest proportion of suberin/waxes due to hydrophobicity of the cation. These compounds did co-precipitate with aromatic compounds and carbohydrates, but these precipitates were significantly enriched in these hydrophobic compounds with respect to pure bark. The long chain aliphatic $[Me_2PO_4]$-ILs only gave mixed fractions and limited selectivity of the components was observed, most likely due to the increased solubility of hydrophobic compounds in the presence of water due to the formation of amphiphilic nanostructures within $[C_8C_1im][Me_2PO_4]$ and $[C_8C_1pyrr][Me_2PO_4]$. This also led to relatively low yields of recovered solids.

While $[C_4C_1im][N(CN)_2]$ was the only IL containing the $[N(CN)_2]^-$ anion explored for the recovery of solid fractions, it was determined that this IL did not lead to the recovery of carbohydrates, as would be expected from the poor solubility of these components in this IL and hence their likely poor extraction into the IL. The recovery of the other components such as suberin, tannin and lignin could be obtained although there was evidence of potential side reactions between the $[N(CN)_2]^-$ anion and bark due to the complexity of the IR spectra obtained. Other antisolvents such as MeCN and DMSO were also used for this IL but evidence of similar decomposition products was also determined.

Guanidinium based ILs were also explored although they didn't display significant yield of the recovered solid and only small amounts of mixed fractions could be observed from both $[C_4gun][NTf_2]$ and $[C_8gun][NTf_2]$. This was due to the immiscibility of these ILs with water meaning that only DMSO and MeCN could be used as antisolvents. The GuCl:LA DES led to

the selective recovery of significant amounts of aromatic fractions with limited contamination by suberin/waxes or carbohydrates.

These results indicate that water as an antisolvent can help concentrate hydrophobic components when added in a small amount before leading to the formation of relatively pure aromatic fractions on further antisolvent addition. This controlled antisolvent approach in conjunction with the use of IL solvents for bark extraction highlights a potential method for the fractionation and separation of these extracts. It is also clear that ILs with longer alkyl chains reduce the ability to cleanly fractionate these components by increasing the solubility of wax/suberin fractions on the addition of water. Therefore, in the design of ILs for this antisolvent addition approach it is worth considering those that have higher polar ions that can selectively extract hydrophobic components and aromatic compounds without significant carbohydrate extraction to help facilitate the more efficient fractionation of key structural bark components. The overall picture of characterisation of extracts with the addition of water as an antisolvent performed on an increased scale (10 mL and 5 mL of IL/DES extract) is represented in Table 4.21.

Table 4.21 *The fractions of bark components isolated from 10 mL or 5 mL of IL and DES extract with the addition of water as an antisolvent for scale-up samples. (Total yield % is calculated with respect to extraction yield of each IL/DES)*

IL/DES Extract used (mL)	**Sample**	**Absorbance**			**Solid recovered (mg)**	**Relative Sample Yield (%)**	**Total Yield (mg)**
		S : L	**C : L**	**T : L**			
-	Pure Bark	0.27,0.18	1.54	0.45	-	-	-
$[C_4C_1im][Me_2PO_4]$ (10 mL)	Sample A	0.63,0.36	2.67	0.46	102.9	64.1	160.5 (97%)
	Sample B	*	*	0.56	14.7	9.16	
	Sample C	*	*	0.52	42.9	26.73	
$[C_4C_1im][N(CN)_2]$ (10 mL)	Sample D	0.3, 0.2	*	0.78	24.3	45	54.5 (40%)
	Sample E	*	*	0.86	12.9	24	
	Sample F	**	**	**	8.5	15	
	Sample G	*	*	0.89	8.8	16	
$[C_4C_1pyrr][Me_2PO_4]$ (5 mL)	Sample H	1.09, 0.71	1.54	0.67	24	61.2	39.2 (22%)
	Sample I	*	*	0.49	15.2	38.8	

$[C_8C_1im]$ $[Me_2PO_4]$ (5 mL)	Sample J	0.85, 0.54	NS	0.67	44.6	50.68	88 (73%)
	Sample K	0.97, 0.63	3.11	0.60	7.6	8.64	
	Sample L	0.56, 0.35	1.40	0.55	35.8	40.68	
$[C_8C_1pyrr]$ $[Me_2PO_4]$ (5 mL)	Sample M	0.94, 0.61	3.19	0.36	20.8	42.28	49.2 (50%)
	Sample N	**	**	**	24	48.78	
	Sample O	0.19, 0.1	*	0.55	4.4	8.94	
GuCl:LA (10 mL)	Sample P	0.48, 0.24	*	0.49	24	84	28.6 (11%)
	Sample Q	*	*	0.57	4.6	16	
[TEA] $[HSO_4]$ (5 mL)	Sample R	0.49, 0.30	1.93	0.50	28	58.09	48.2 (26%)
	Sample S	*	*	0.50	20.2	41.91	

*(* the peaks do not appear in the IR, ** sample was too fine to allow its recovery for use for IR analysis)*

Table 4.22 represents the breakdown of the proportions of mixed fractions and pure aromatic fractions relative to the solid recovered from IL and DES extracts. Of the solid recovered from the $[C_4C_1pyrr][Me_2PO_4]$ IL extract, 61.2% was present as the tannin/lignin fraction with the remaining 38.8% by mass being the mixed fractions containing waxes and carbohydrates due to the weakly hydrogen-bond donating pyrrolidinium cation[152] leading to greater extraction of hydrophobic compounds found in bark. Moreover, a significant fraction of 55.4% was also present as the pure aromatic in case of the $[C_4C_1im][N(CN)_2]$ IL extract but not sure about the outcome as a potential reaction between key bark components and the $[N(CN)_2]^-$ anion under the reaction conditions was determined (Chapter 3, elemental analysis in section 3.4.6.2.1).

Table 4.22 *The proportion of mixed fractions and pure aromatic fractions obtained from the solid recovered of IL and DES extracts*

IL/DES Extract	Volume used (mL)	Solid Recovered (mg)	Mixed fractions (%)	Pure aromatic fractions (%)
$[C_4C_1im][Me_2PO_4]$	10	160.5	64.1	35.9
$[C_4C_1im][N(CN)_2]$	10	54.5	44.6	55.4
$[C_4C_1pyrr][Me_2PO_4]$	5	39.2	61.2	38.8
$[C_8C_1im][Me_2PO_4]$	5	88	100	*
$[C_8C_1pyrr][Me_2PO_4]$	5	49.2	100	*
GuCl:LA	10	28.6	84	16
$[TEA][HSO_4]$	5	48.2	58	42

(- only the mixed fractions were obtained)*

5. Conclusion and Future work

5.1. Conclusion

The aim of this research was to develop new solvents with exceptional hydrogen-bonding capability as potential extraction solvents as well as to understand the relationship between IL and DES structure and the extraction of key components from *Pinus radiata* bark. Despite *Pinus radiata* bark being a rich source of chemicals (suberin, lignin, carbohydrate, polyphenols, and extractives) that can be extracted and utilised for various applications, it has little value today due to the inability to extract and separate these compounds effectively. Therefore, there is a need for further research to explore novel alternatives for the extraction and fractionation of *Pinus radiata* bark.

In **Chapter 2**, work has been performed to understand the polarity of novel DESs and how this may influence their role as extraction solvents. The goal was to develop novel DESs that have greater hydrogen bond donating or accepting capacity than current DESs. The intention was to facilitate improved extraction processes and to develop solvents that could act as effective solvents for the purification of many difficult to solubilise pharmaceuticals and lignocellulosic biomass. These solvents were then be explored for these applications, both the solubilisation of pharmaceuticals such as fenofibrate (FF) and the extraction of key components from *Pinus radiata* bark.

In total, 20 different HBD combinations were screened to develop DESs using guanidinium as a strongly hydrogen bond donating cation. The synthesis of these new DES simply involved the direct mixing of the salt and HBD of a desired mole ratio. New room temperature liquids based on mixtures of GuCl with LA, Gly and LevA were discovered and characterised. Firstly, the polarity of DESs was explored using the solvatochromic dye set developed by Kamlet and Taft.[108] This includes Reichardt's dye 33, 4-nitroaniline, N, N-diethyl-4-nitroaniline[21] and pyridine-N-oxide which had been proposed by Freire and Coutinho as an alternative for acidic solvent systems.[22] The Kamlet-Taft parameters assess the hydrogen bond donating (α), hydrogen bond accepting (β) and dipolarity/polarizability (π*) of a solvent.

It was determined that the GuCl:LA mixture was found to possess comparable if not slightly improved hydrogen bond donating abilities compared to existing strong hydrogen bond

donating solvents such as 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP) based on the use of these polarity probes. Moreover, the viscosity of the four room temperature synthesised DESs were also determined under increasing shear stress, it was concluded that the viscosity of the eutectic mixtures decreased in the order GuCl:Gly > GuCl:LevA > GuCl:LA > GuCl:Acet and that all of these DESs displayed shear thinning behaviour.

These room temperatures DESs were further used for the solubilisation of FF under the hypothesis that there could be strong hydrogen bonding interactions between the carbonyl group of FF and the guanidinium cation incorporated in each DESs to help drive its solubility. It was found that LevA-based DES displayed higher solubility of FF than other DESs which was hypothesised to be because of the weaker interaction between LevA and the anion resulting in the LevA being more available to interact with the FF.

Lastly, the synthesis of some other classes of eutectic mixtures based on the SIL and RDES concepts were explored in order to develop strong hydrogen bond accepting solvents. Although hydrated salts emerged as the most suitable for solubilising coordinating solvents (EGMME, EGDME and DEG) in the case of developing SILs, a very few salts with strong HBA anions showed the potential to form stable SILs. The presence of water in these hydrates is also likely to be problematic in their use as strong HBA solvents due to the interaction between water and the other SIL components. For the synthesis of RDES, unfortunately none led to room temperature liquids although some combinations did lead to significant melting point depressions and it was concluded that it might not be a good approach for synthesising eutectic mixtures with strong hydrogen bond accepting abilities.

In **Chapter 3**, the aim was to investigate of the influence of the structure of ILs and DESs on the extraction of *Pinus Radiata* bark. As ILs have the ability to tune their properties based on the structure of the ions selected, the aim was to understand how these structural changes influenced the bark extraction. A range of ILs ions were selected to screen cations and anions of differing hydrogen-bonding abilities to develop a model of their effects on the extraction of key bark components. The length of the alkyl chains on the IL ions was also modified to provide insight into the effect of the hydrophobic component of the IL on the extraction capacity and selectivity.

Overall, 23 ILs or IL intermediates were synthesised with 15 being used for the extraction of *Pinus radiata* bark. Some of the ILs synthesised were not used for the extraction due to their

high melting points. The ILs that were used for the extraction process consisted of nitrogen-based cations that increase in hydrogen-bond donating capability in the order 1-alkyl-1-methylpyrrolidinium ($[C_nC_1pyrr]^+$) < 1-alkyl-3-methylimidazolium ($[C_nC_1im]^+$) < , 1-alkylguanidinium ($[C_ngun]^+$)) with anions that increased in hydrogen-bond accepting capability in the order bis(trifluoromethanesulfonyl)imide ($[NTf_2]^-$) < dicyanamide ($[N(CN)_2]^-$) < dimethyl phosphate ($[Me_2PO_4]^-$). Room temperature DESs such as GuCl:LA and GuCl:Gly, based on the work described in Chapter 2 were also used for the extraction of the bark. Additionally, [TEA][HSO_4] was also examined for bark extraction due to it being reported as an effective solvent for extracting lignin from biomass sources.[126] For the extraction procedure, the IL was added to a pre-weighed dried bark sample and the resultant suspension stirred at various temperatures (25 ℃, 50 ℃, 100 ℃) for different time intervals (1 h or 24 h) although most comparison experiments were performed at 100 ℃ for 24 h as this showed the greatest extent of extraction. The IL and residual bark were separated at the end of the extraction and characterised by a range of analytical techniques including UV-Vis, IR, NMR (solid and solution state), HS-SPME-GCMS and pyrolysis GC-MS. Moreover, the overall extraction yield was also determined.

It has been evaluated that the anion governs extraction yield, with $[Me_2PO_4]^-$ generally the most effective anion of those explored. The DES systems, especially GuCl:LA DES, were found to be competitive with the IL systems for overall extraction efficiency. [TEA][HSO_4] showed the highest extraction yield (75%) of the ILs and DESs explored. This is consistent with the reported efficiency of this IL in being an effective solvent for lignin.[126]

Moreover, ILs seem to be more effective at removing aromatics and waxes than sugars/polysaccharides as demonstrated by the solid-state NMR and pyrolysis GCMS results. This was even true for $[Me_2PO_4]^-$ ILs which have been described as being effective for the solubilisation of carbohydrates,[154] highlighting that the competition arising from components in the complex bark matrix can affect the extraction outcomes. There was evidence of potential side reactions between the $[N(CN)_2]^-$ anion and bark which was reflected in the complexity of the IR spectra obtained of the residual solid as well as from elemental analysis results of the solid obtained.

In terms of the effect of the cation, it has been determined that $[C_nC_1pyrr]^+$ cations facilitate the preferential extraction of the hydrophobic components when combined with a strong hydrogen bond accepting anion ($[Me_2PO_4]^-$). In contrast, $[C_nC_1im]^+$ cation based ILs are more efficient at extracting lignin and tannin fractions.

In terms of the effect of the alkyl chain of the cations, the solubility of the carbohydrate and lignin or tannin components depend on the alkyl chain length and the dissolution of these fractions decreases with increasing non-polar domain of the cation. However, the extraction of hydrophobic compounds such as waxes and suberinic material tended to increase with increasing alkyl chain length.

In **Chapter 4**, the recovery and isolation of components from the bark extracts formed in Chapter 3 from the IL and DESs was examined. The main approach that was explored was antisolvent addition. The selection of antisolvents made on the basis of their polarity and the nature of their interaction with the IL and DESs. Water, acetonitrile (MeCN) and DMSO were selected as the most appropriate antisolvents for the recovery. The characterisation of the recovered extracts was performed by using IR, NMR and TMA-OH Py-GCMS.

It has been determined that out of the antisolvents used for these separations/fractionations, water appeared to be the most suitable in terms of both the selectivity and yield of the fractions that could be obtained of each of the components of bark. The other two antisolvents, MeCN and DMSO, typically led to mixed fractions with limited evidence of preferential precipitation and relatively low yields of solid formation. Although almost all the IL and DES extracts led to the formation of fractions containing suberin/waxes, carbohydrates and lignins/tannins, the quantity and selectivity of these fractions varied significantly between each solvent when water was used as an antisolvent.

It was observed that the $[Me_2PO_4]^-$ IL extracts led to higher yields of carbohydrates than could be recovered from $[N(CN)_2]^-$ or the DES with the addition of water as an antisolvent. Moreover, the highest proportion of carbohydrate was recovered from $[C_4C_1im][Me_2PO_4]$ due to the presence of the imidazolium cation in combination with the strong hydrogen bond acceptor anion. Based on the TMA-OH-Py-GC-MS results, the most carbohydrate enriched fraction obtained from $[C_4C_1im][Me_2PO_4]$ has similar carbohydrate concentrations as pure bark, whereas most of the ILs led to significantly lower carbohydrate concentrations in the recovery components. ILs containing the $[N(CN)_2]^-$ anion lead to relatively low yields of recovered solids, as would be expected from the poor solubility of these components in this IL, potentially further affected by the reactivity of the anion, as discussed in Chapter 3.

Regarding the isolation of suberins/waxes as hydrophobic components, $[C_4C_1pyrr][Me_2PO_4]$ contributed to extracting and enabling the recovery of the highest proportion of suberin/waxes

due to hydrophobicity of the cation. These compounds did co-precipitate with aromatic compounds and carbohydrates, but these precipitates were significantly enriched in these hydrophobic compounds with respect to pure bark. The long chain aliphatic $[Me_2PO_4]^-$ILs only gave mixed fractions and limited selectivity of the components was observed. This was most likely due to the increased solubility of hydrophobic compounds in the presence of the antisolvent water due to the formation of amphiphilic nanostructures within $[C_8C_1im][Me_2PO_4]$ and $[C_8C_1pyrr][Me_2PO_4]$. This also led to relatively low yields of recovered solids. Hence, it is clear that ILs with longer alkyl chains reduce the ability to cleanly fractionate these components by increasing the solubility of wax/suberin fractions on the addition of water.

These results indicate that water as an antisolvent can help concentrate hydrophobic components when added in a small amount before leading to the formation of relatively pure aromatic fractions on further antisolvent addition. This controlled antisolvent approach in conjunction with the use of IL solvents for bark extraction highlights a potential method for the fractionation and separation of these extracts. Therefore, in the design of ILs for this antisolvent addition approach it is worth considering those that have relatively high polarity ions that can selectively extract hydrophobic components and aromatic compounds without significant carbohydrate extraction to help facilitate the more efficient fractionation of key structural bark components.

5.2. Future Work

As the focus of this project was to discover appropriate ILs or DESs that are useful for the extraction of *Pinus radiata* bark, future work would focus on extending the investigations performed here on ILs and DES that share common characteristics with some of the more successful extraction solvents. Based on the relative trends observed for ILs, acetate ($[MeCO_2]$) anion based ILs have been reported which might be better for targeting selective extractions, especially for lignin[155] and cellulose.[156] The strong hydrogen bond basicity of $[C_4C_1im][MeCO_2]$ ($\beta = 1.18$),[109] will make this IL more efficient for the extraction of key components of bark. The first approach will be to utilise $[C_4C_1im][MeCO_2]$ IL for the extraction purpose as reported and investigate the fractionation of bark components from this IL.

Moreover, based on the conclusions above, the aim is to actually minimise cellulose dissolution and maximise suberin/wax extraction which can be fractionated on the addition of small amounts of water. Pyrrolidinium might be effective for this incorporated with acetate anion.

Similarly, in the context of designing a strong HBD guanidinium based DES with even greater HBD ability than GuCl:LA and HFIP, the use of malonic acid (MA) as a HBD incorporated with GuCl could emerge as good HBD DES. It has been reported that the effect of the two acid groups in the malonic acid provides a high α value (1.39) when combined with ChCl salt.[21] Additionally, GuCl:MA DES has not been synthesised so far, so that would be an interesting to see the outcome. If this was successful, then malonic acid-based DES will be useful in dissolution of hydrogen bond accepting group type components present in bark alongside potentially FF and other HBA APIs.

Moreover, ILs have been reported to efficiently absorb microwave energy and microwave-assisted extraction is an attractive and rapid alternative extraction technique.[157] Another approach would be to try using microwave-assisted extraction of the bark[158] rather than prolonged heating at elevated temperatures.

There also are a few characterisation approaches reported that could assist in further understanding the fine detail of the extractions performed, such as thermal gravimetric analysis (TGA), LC-MS analysis and a more thorough compositional analysis of the bark and residues produced. TGA is a convenient method to measure the weight loss with increasing temperature over a definite period of time.[159] The key feature of using TGA would be that most of the components of the bark, especially suberinic material[98], are thermally stable due to their high polymeric structure. LCMS will allow the characterisation of IL extracts to confirm the identity of specific compounds within the separated samples, although relies on the solubility of these components in the LC-MS eluent and their effective separation to enable the identification of key compounds of interest.[82] The gel-permeation chromatography (GPC) could also be add in a list for the characterisation of extracts to get an idea of whether some of the structural biopolymers (e.g. lignin, cellulose etc) are obtained intact or undergo hydrolysis/breakdown under applied conditions.

6. Experimental Procedure

6.1.Materials

2,6,dichloro-4-(2,4,6-triphenyl-N-pyridino)-phenolate - Reichardt's betadine dye 33, 4-nitroaniline, N,N-diethyl-4-nitroaniline, pyridine N-oxide, guanidinium chloride, fenofibrate, 1H-pyrazole-1-carboxamidine hydrochloride, sodium dicyanamide, silver nitrate, and lithium bis(trifluoromethanesulfonyl)imide were used as received.

The following chemicals were purchased from commercial suppliers and were used as received : DL-lactic acid, acetamide, levulinic acid, glycerol, oxalic acid, malic acid, succinic acid, urea, serine, decanoic acid, adipic acid, oxalic acid dihydrate, PEG400, citric acid, sorbitol, glucose, glycolic acid, 1-decanol, catechol, hexafluoro isopropanol, Nile red, potassium hydroxide, lithium acetate dihydrate, lithium acetate, lithium chloride, sodium acetate trihydrate, sodium chloride, sodium nitrate, tributyl phosphine oxide, trioctyl phosphine oxide, dimethyl sulfone, sodium butyrate, sodium octanoate, sodium glycolate, ethylene glycol dimethyl ether, ethylene glycol monomethyl ether, tetraethylene glycol dimethyl ether, diethylene glycol, tetraethylene glycol, and triethylene glycol dimethyl ether.

Acetonitrile and methanol were purified using an LC Technology Solutions Inc SP-1 solvent purification system. Dichloromethane, diethyl ether and ethyl acetate (Laboratory Reagent) were used as received.

1-methylimidazole, 1-methylpyrrolidine, 1-chlorobutane, 1-bromobutane, 1-chlorooctane, 1-bromooctane, trimethyl phosphate, 1-butylamine, and 1-octylamine were distilled under either reduced pressure or a nitrogen atmosphere before use.

6.2.Equipment

6.2.1.Melting point

Melting points of DESs were determined using a digital melting point apparatus (MPS20).

6.2.2.Viscosity

Viscosity of DESs was determined using Rheometer instrument equipped with a MBT 25F-0011 sheath and CCT-25 spindle.

6.2.3.UV-Visible spectroscopy

UV-vis absorption spectra were obtained using a Cary 300 UV-Visible spectrometer equipped with a water recirculating Peltier temperature controller.

6.2.4. HPLC-UV

HPLC-UV spectra were obtained using Shimadzu instrument equipped with SIL-20AC Auto sampler, CBM-20Alite system controller and SPD-20A Detector.

6.2.5. Solution-phase NMR

NMR analysis was performed on a Bruker DRX 400 MHz NMR spectrometer at 300 K. The measurement frequencies for the ^{1}H, ^{13}C, ^{19}F, and ^{31}P NMR were 400.13 MHz, 100.61 MHz, 376 MHz, and 162 MHz respectively.

6.2.6. FT-IR Analysis

FTIR analysis was performed on a Bruker Vertex 70v FT-IR Spectrometer.

6.2.7. Pyrolysis gas chromatography mass spectrometry (Py-GCMS)

The Py-GCMS instrument was composed of a Lab Frontier pyrolysis furnace with an auto-shot sampler (Pyrolysis), an Agilent 7890A gas chromatograph (GC) and Agilent 5973 single quadrupole mass spectrometer (MS).

6.2.8. Headspace solid phase micro extraction (HS-SPME-GCMS)

HS-SPME-GCMS was performed on Shimadzu QP2010 GC-MS instrument with a CTC PAL autosampler.

6.2.9.^{13}C Solid-State NMR

^{13}C solid-state NMR was obtained by using AVIII 200 MHz equipped with a 4 mm CP-MAS probe. Samples were ground in a mortar and pestle to form a fine powder and were inserted into 4 mm ZrO rotors fitted with KeI-F caps.

6.2.10. Elemental Analysis

Elemental analysis of bark extracts was performed using an Elementar Vario EL Cube.

6.2.11 High resolution mass spectral (HRMS)

HRMS data was recorded for ILs using a Bruker microTOF-Q mass spectrometer in both positive and negative ion electrospray ionisation (ESI) modes.

6.3. Deep Eutectic Solvents (DESs)

6.3.1. Synthesis of DESs

DESs were prepared by directly weighing components in the desired mole ratio and then either stirring the mixture at elevated temperature (50-80 ℃ for 1-2 h) or grinding the mixture at room temperature using a mortar and pestle until a homogenous mixture had been formed. Some DESs were dried with activated molecular sieves overnight before use.

6.4.Polarity Parameter

6.4.1. Kamlet-Taft Parameters - Solvatochromic dyes

Solutions containing 0.9 mM 4-nitroaniline and *N, N*-diethyl-4-nitroaniline and 0.4 mM of Reichardt's dye 33 were separately prepared in dichloromethane. 100 µL of 4-nitroaniline, 35 µL *N, N*-diethyl-4-nitroaniline or 100 µL Reichardt's dye 33 were added to a cuvette and the dichloromethane evaporated using reduced vacuum. The precipitated dye was dissolved in the 1 mL of DES in the cuvette and UV-vis absorption spectrum were obtained. The Kamlet Taft parameters α, β and π* were calculated from the wavelength of the lowest energy absorption as described in the polarity parameter calculation given in the results and discussion Chapter 2. Polarity measurements were obtained at 25℃, 40℃, 60℃ and 80℃.

6.4.2. Kamlet-Taft Parameters - Pyridine N-Oxide Probe

Solutions of 0.25 M pyridine N-oxide were prepared in each solvent with a flame-sealed, coaxially inserted capillary containing DMSO-d_6 used as an NMR lock.[22] The ^{13}C NMR of these samples was obtained and used to calculate the α Kamlet-Taft parameter as described in the polarity parameter calculation given in the results and discussion Chapter 2.[22,108] These measurements were obtained at 25 ℃, 40 ℃, 60 ℃ and 80 ℃.

6.5.Viscosity Measurements of DESs

The viscosities of four DESs GuCl:LA (1:2), GuCl:LevA (1:4), GuCl:Gly (1:2), and GuCl:Acet (1:2) were determined in a temperature controlled environment. The solvents (20 mL) were poured up to the inner line in the sheath of the apparatus and the measurements were taken at different temperatures, (20 to 80 ℃) in 10°C increments, against the rotational speed of the spindle attached to the sheath. The other parameters were set up as given in Table 6.1.

Table 6.1 *Parameters to determine the viscosity of DESs.*

Time	Shear rate (start value)	Shear rate (end value)	Measuring point numbers
500 s	0.013 /s	500 /s	250

6.6.Quantification of Fenofibrate by HPLC-UV

Standard solutions of fenofibrate (FF) were prepared in acetonitrile at different concentrations (0.1 mg/mL, 0.25 mg/mL, 0.5 mg/mL, 0.75 mg/mL and 1 mg/mL). Analyte separation took place on a 3.5 micron ZORBAX Eclipse plus C18 column (Agilent) with 4.6 × 100 mm dimensions with a sample injection volume of 5 µL. The analysis was performed for 15 min using an isocratic method with a 70 : 30 mixture of MQ water with 1% formic acid : acetonitrile. The analyte was detected by a UV-Vis detector (SPD – 20A) was set at 280 nm.

The solubility of fenofibrate in the DESs was determined by stirring 1 g of fenofibrate in 10 mL of each DES at room temperature or at 50 °C for 24 h. These mixtures were filtered using a syringe filter. The filtered mixtures (DESs + FF) of ~ 0.7 mL were made up to 10 mL with acetonitrile. These solutions were analysed using the HPLC method described above.

6.7. Reverse deep eutectic solvents (RDESs)

6.7.1. Synthesis of RDESs

RDESs were prepared similarly to the other DESs by directly weighing the two components in the desired mole ratio and grinding the mixture at room temperature using a mortar and pestle until a homogenous mixture had been formed. The melting points of the mixtures were determined. Different combinations of HBAs with salts were explored as represented in Table 6.2.

Table 6.2 *Lists of HBAs and salts in reverse deep eutectic solvents.*

HBA	Salt
Tributyl phosphine oxide (TBPO)	Sodium butyrate (SB)
Trioctyl phosphine oxide (TOPO)	Sodium octanoate (SO)
Dimethyl sulfone (DMS)	Sodium glycolate (SG)
	Guanidine hydrochloride (GuCl)

6.8. Solvate Ionic liquids (SILs)

6.8.1. Synthesis of SILs

All the SILs were prepared by directly weighing components in the desired mole ratio and then stirring the mixture at elevated temperature (40-80℃ for 1-2 h). 12 different combinations of metal salts based on lithium and sodium salts with some ethers and glycol solvents were developed as listed in Table 6.3.

Table 6.3 *List of salts and solvents explored as SILs.*

Salt	Solvents
Lithium acetate dihydrate ($LiOAc.2H_2O$)	Ethylene glycol dimethyl ether (EGDME)
Lithium acetate (LiOAc)	Ethylene glycol monomethyl ether (EGMEE)
Lithium chloride (LiCl)	Tetraethylene glycol dimethyl ether (Triglyme)
Sodium acetate trihydrate ($NaOAc.3H_2O$)	Diethylene glycol (DEG)
Sodium chloride (NaCl)	Tetraethylene glycol (TEG)
Sodium nitrate ($NaNO_3$)	Triethylene glycol dimethyl ether (Tetraglyme)

6.9. Ionic liquids (ILs)

6.9.1. Synthesis of ILs

Synthesis of 1-butyl-3-methylimidazolium chloride[160] ***[C_4C_1im]Cl***

1-Methylimidazole (50 g, 0.609 mol) was dissolved in approximately 100 mL of ethyl acetate in a multinecked round bottom flask fitted with a condenser attached to either a drying test tube (containing $CaCl_2$) or a N_2 line. 1-Chlorobutane (62.02 g, 0.670 mol, 1.1 eq) was added slowly

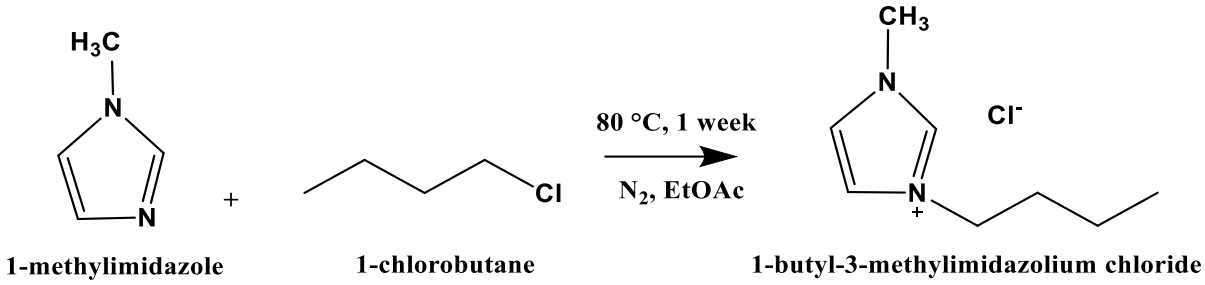
into the 1-methylimidazole solution using a pressure-equalising dropping funnel while the solution was stirred at room temperature. After the addition, the solution was heated at 80 °C for 6 - 7 days. The solution was cooled to room temperature and the resultant liquid was washed thoroughly with ethyl acetate (10 × 50 mL) to give a white solid. Residual solvent was removed by rotary evaporation and the solid was dried *in vacuo* at 50 °C for 16 h to afford [C_4C_1im]Cl (Scheme 7).

White crystalline solid (93.57 g, 0.536 mol, 87.95 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.37 (s, 1H), 7.84 (t, *J* = 1.7 Hz, 1H), 7.76 (t, *J* = 1.7 Hz, 1H), 4.19 (t, *J* = 7.2 Hz, 2H), 3.87 (s, 3H), 1.85 – 1.65 (quint, 2H), 1.35 – 1.13 (sext, 2H), 0.90 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 137.13 (s), 124.06 (s), 122.74 (s), 48.91 (s), 36.19 (s), 31.84 (s), 19.24 (s), 13.75 (s).

HRMS found (ESI): m/z 139.1231 (100%, [$C_4C_1im^+$]), 209.0563 ({[C_4C_1im]Cl_2}$^-$).



1-methylimidazole 1-chlorobutane 1-butyl-3-methylimidazolium chloride

Scheme 7 *The synthesis of [C_4C_1im]Cl IL.*

Synthesis of 1-butyl-3-methylimidazolium bromide*[161] *[C_4C_1im]Br

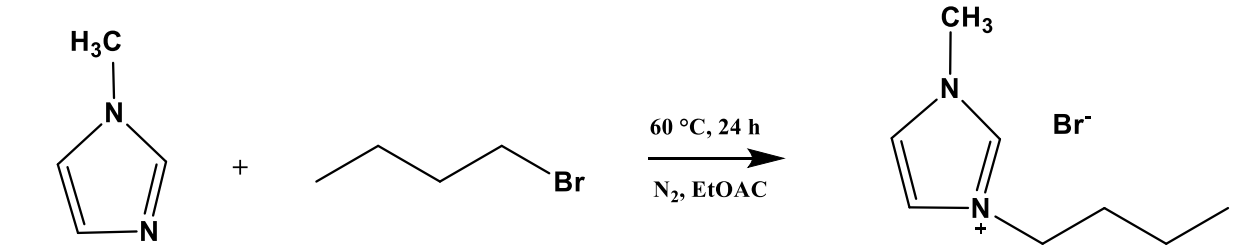
1-Methylimidazole (50 g, 0.609 mol) was dissolved in approximately 100 mL of ethyl acetate in a multinecked round bottom flask fitted with a condenser attached to either a drying test tube (containing $CaCl_2$) or a N_2 line. 1-bromobutane (75.76 g, 0.553 mol, 1.1 eq) was added slowly into the 1-methylimidazole solution using a pressure-equalising dropping funnel while the solution was stirred at room temperature. After the addition, the solution was heated at 60 °C for 24 h. The solution was cooled to room temperature and the resultant liquid was washed thoroughly with ethyl acetate (10 × 50 mL) to give a pale-yellow IL. Residual solvent was removed by rotary evaporation and the solid was dried *in vacuo* at 50 °C for 16 h to afford [C_4C_1im]Br (Scheme 8).

Pale yellow viscous liquid (124.46 g, 0.5678 mol, 93.24%)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.34 (s, 1H), 7.87 (t, 1H), 7.79 (t, 1H), 4.21 (t, 2H), 3.89 (s, 3H), 1.87 – 1.69 (quint, 2H), 1.27 – 1.39 (sext, *J* = 15.1, 7.5 Hz, 2H), 0.90 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 137.01 (s), 124.04 (s), 122.74 (s), 48.92 (s), 36.26 (s), 31.85 (s), 19.22 (s), 13.75 (s).

HRMS found (ESI): m/z 139.1224 (100%, $[C_4C_1im^+]$), 78.9190 (100%, $[Br]^-$), 80.9162 (97%, $[Br]^-$).



1-methylimidazole **1-bromobutane** **1-butyl-3-methylimidazolium bromide**

Scheme 8 *The synthesis of [C₄C₁im]Br IL.*

Synthesis of 1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide[160] ***[C₄C₁im][NTf₂]***

$[C_4C_1im]Cl$ (49.87 g, 0.285 mol) and $LiNTf_2$ (81.27 g, 0.300 mol, 1.05 eq) were separately weighed into conical flasks and dissolved in deionized water. These solutions were combined and stirred, leading to the formation of a biphasic mixture. Dichloromethane (150 mL) was added to increase the volume of the lower phase and the resultant organic phase separated using a separatory funnel. The organic phase was washed with deionized water (~ 150 mL) 3-4 times until Cl^- has been removed, determined by testing the water layer after washing with $AgNO_3$ (0.1 M) solution. Dichloromethane was removed by rotary evaporation and the resultant liquid dried in *vacuo* at 50 °C for 16 h to afford $[C_4C_1im][NTf_2]$. (Scheme 9)

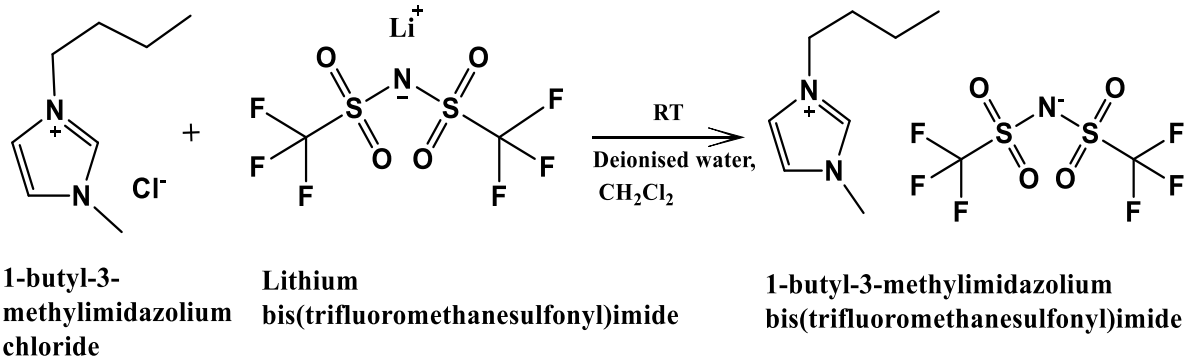
Clear colourless oil (109.82 g, 0.261 mol, 91.76 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.11 (s, 1H), 7.73 (t, *J* = 1.8 Hz, 1H), 7.67 (t, *J* = 1.7 Hz, 1H), 4.18 (t, *J* = 7.2 Hz, 2H), 3.88 (s, 3H), 1.80 (quint, *J* = 12.8, 7.5 Hz, 2H), 1.53 – 1.18 (sext, 2H), 0.92 (t, *J* = 7.4 Hz, 3H)

^{13}C NMR δ (ppm) (101 MHz, DMSO) 136.93 (s), 124.75 (s), 123.96 (s), 122.61 (q, *J* =321.7 Hz), 49.03 (s), 36.01 (s), 31.76 (s), 19.13 (s), 13.36 (s).

^{19}F NMR δ (ppm) (376 MHz, DMSO) -77.17 (s)

HRMS found (ESI): m/z 139.1228 (100%, $[C_4C_1im^+]$), 279.9182 (100 %, $[NTf_2]^-$)



1-butyl-3-methylimidazolium chloride + **Lithium bis(trifluoromethanesulfonyl)imide** → **1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide**

Scheme 9 *The synthesis of $[C_4C_1im]NTf_2$ IL.*

Synthesis of 1-butyl-3-methylimidazolium dimethyl phosphate*[160] *$[C_4C_1im][Me_2PO_4]$

$[C_4C_1im]Cl$ (70 g, 0.151 mol) was dissolved in acetonitrile (150 mL) in a multinecked round bottom flask fitted with a condenser attached to a N_2 line. Distilled trimethyl phosphate (84.20 g, 0.6011 mol, 1.5 eq) was added slowly to the stirred $[C_4C_1im]Cl$ solution using a pressure-equalising dropping funnel. After the addition was complete, the dropping funnel was removed and replaced with a stopper and the solution was heated at 80 °C with stirring for 2 - 3 weeks until a negative test for Cl^- was observed using a $AgNO_3$ (0.1 M) solution. Acetonitrile was removed by rotary evaporation and the resultant liquid was washed with ethyl acetate (10 × 50 mL) to give a pale-yellow viscous liquid. Excess solvent was removed by rotary evaporation and the resultant liquid dried in *vacuo* 50 °C for 16 h to afford $[C_4C_1im][Me_2PO_4]$ (Scheme 10).

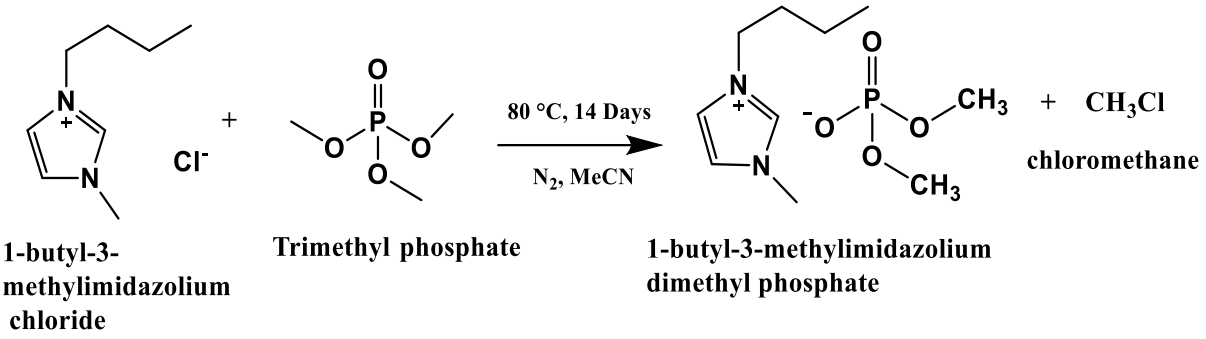
Colourless viscous liquid (100.89 g, 0.3818 mol, 95.27 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.97 (s, 1H), 8.11 (t, *J* = 1.5 Hz, 1H), 8.00 (t, *J* = 1.6 Hz, 1H), 4.28 (t, *J* = 7.2 Hz, 2H), 3.97 (s, 3H), 3.38 (d, *J* = 10.4 Hz, 6H), 1.89 (quint, 2H), 1.35 (sext, 2H), 0.90 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 137.55 (s), 123.02 (d), 51.34 (s), 48.23 (s), 35.34 (s), 31.50 (s), 18.72 (s), 13.08 (s).

^{31}P NMR δ (ppm) (162 MHz, DMSO) 1.45 (hept, *J* = 10.3 Hz).

HRMS found (ESI): m/z 139.1226 (100%, $[C_4C_1im]^+$), 125.0007 (100%, $[Me_2PO_4]^-$)



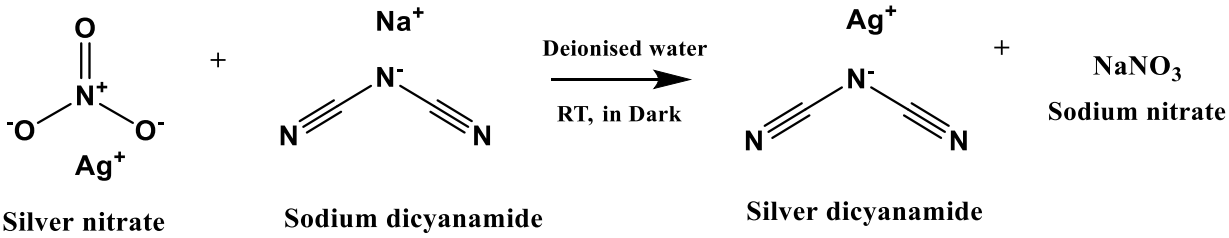
Scheme 10 *The synthesis of $[C_4C_1im]Me_2PO_4$ IL.*

Synthesis of silver dicyanamide*[162] *($AgN(CN)_2$)

$AgNO_3$ (13.122 g, 1 mol) was dissolved in deionised water (~100 mL). $NaN(CN)_2$ (6.878 g, 1 mol) was added to the $AgNO_3$ aqueous solution and the solution stirred for 1 h in the dark. The solid was isolated by filtration and washed with deionised water. The white solid was dried on the suction pump for 2 h then further dried *in vacuo* at 55 °C for 16 h to afford $AgN(CN)_2$ (Scheme 11). $AgN(CN)_2$ was always stored in the dark before use. The infrared spectroscopy data of $AgN(CN)_2$ is shown in the appendix (Fig. A81).

White solid (12.45 g, 0.0716 mol, 92.67 %)

IR (cm^{-1}) – 3598.82, 3081.95, 2302.78, 2248.78, 2175.49 $\nu(C \equiv N)$, 1361.61 $\nu(C - N)$, 952.74, 648.02, 489.87.



Scheme 11 *The synthesis of $Ag[N(CN)_2]$.*

Synthesis of 1-butyl-3-methylimidazolium dicyanamide*[56] *$[C_4C_1im][N(CN)_2]$

$[C_4C_1im]Br$ (22.69 g, 0.104 mol) was dissolved in deionised water (150 mL). $AgN(CN)_2$ (18.0 g, 0.1035 mol, 1.0 eq) was added to this solution and stirred in the dark for 16 h. The resultant slurry was then filtered and the solution checked for the removal of Br^- by testing with saturated

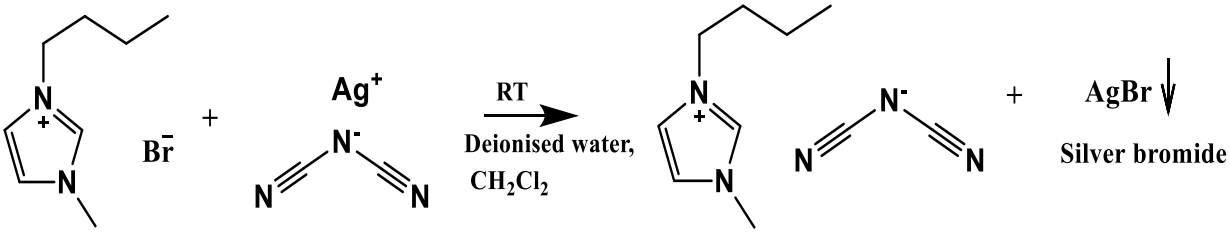
aqueous $AgN(CN)_2$ solution (0.1 g of $AgN(CN)_2$ in 50 mL of water). More $AgN(CN)_2$ was added until no more AgBr precipitate was observed using the $Ag[N(CN)_2]$ test. After the final filtration, water was removed using a rotary evaporator and then dichloromethane (~200 mL) was added to precipitate any remaining dissolved salts. The solution was filtered, solvent removed by rotary evaporation and the resultant colourless liquid dried in *vacuo* at 50 °C for 16 h to afford $[C_4C_1im][N(CN)_2]$ (Scheme 12).

Colourless moderately viscous liquid (15.79 gm, 0.0769 mol, 74.29%)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.15 (s, 1H), 7.77 (t, *J* = 1.7 Hz, 1H), 7.70 (t, *J* = 1.6 Hz, 1H), 4.21 (t, *J* = 7.2 Hz, 2H), 3.90 (s, 3H), 1.81 (quint, *J* = 12.8, 7.5 Hz, 2H), 1.39 (sext, 2H), 0.92 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 136.49 (s), 123.49 (s), 122.16 (s), 119.06 (s), 48.56 (s), 35.66 (s), 31.32 (s), 18.76 (s), 13.12 (s).

HRMS found (ESI): m/z 139.1229 (100%, $[C_4C_1im]^+$), 66.0096 (100%, $[N(CN)_2]^-$)



1-butyl-3-methylimidazolium bromide + **Silver dicyanamide** → **1-butyl-3-methylimidazolium dicyanamide** + **Silver bromide**

Scheme 12 *The synthesis of $[C_4C_1im][N(CN)_2]$ IL.*

Synthesis of 1-butylguanidinium chloride*[163] *$[C_4gun]Cl$

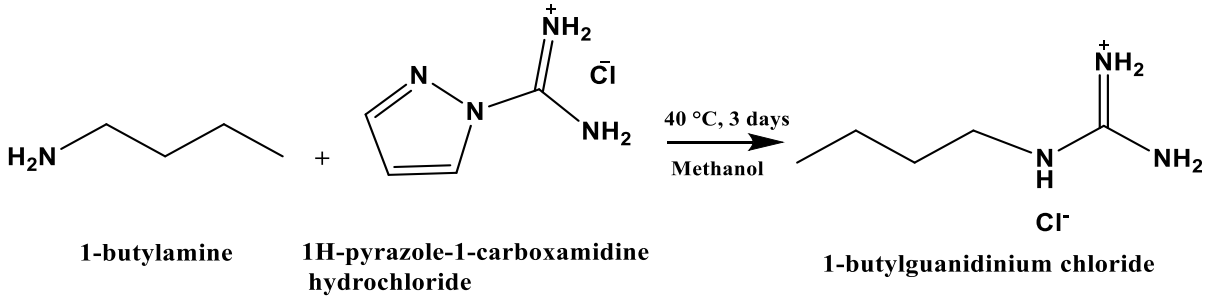
Distilled 1-butylamine (10 g, 0.137 mol) was dissolved in methanol (20 mL) at 40 °C with constant stirring. 1H-pyrazole-1-carboxamidine hydrochloride (20.04 g, 0.1367 mol, 1.0 eq) was added and the solution was stirred at 40 °C for 3 days. Residual solvent was removed by rotary evaporation and the crude product was washed (6 × 100 mL) with diethyl ether before being dried in *vacuo* at 50 °C for 16 h to afford $[C_4gun]Cl$ as a white solid (Scheme 13).

White solid (17.59 g, 0.116 mol, 84.86 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 8.03 – 6.86 (m, 5H), 3.11 (t, *J* = 6.8 Hz, 2H), 1.49 (quint, *J* = 13.8, 6.8 Hz, 2H), 1.32 (sext, 2H), 0.87 (t, *J* = 6.8 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 157.15 (s), 40.11 (s), 30.52 (s), 19.22 (s), 13.48 (s).

HRMS found (ESI): m/z 116.1180 (100%, $[C_4gun]^+$), 186.0579 ({$[C_4gun]Cl_2$}$^-$)



Scheme 13 *The synthesis of [C4gun]Cl IL.*

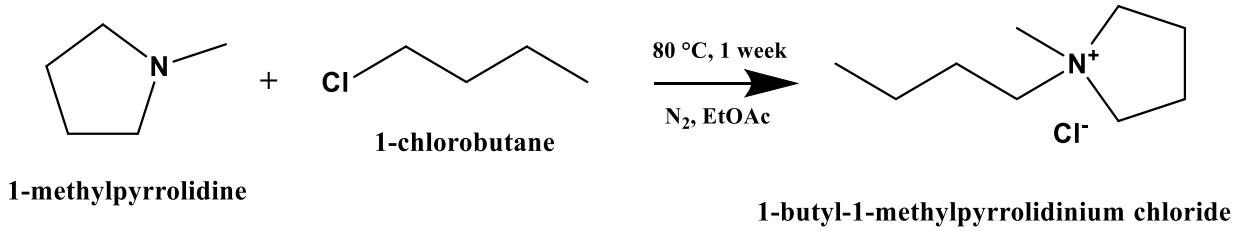
Synthesis of 1-butyl-1-methylpyrrolidinium chloride[160] ***[C4C1pyrr]Cl***

Distilled 1-methylpyrrolidine (100 g, 1.174 mol) was dissolved in ethyl acetate (~100 mL) in a multinecked round bottom flask fitted with a condenser attached to a N_2 line. Distilled 1-chlorobutane (119.59 g, 1.292 mol, 1.1 eq) was added slowly into the stirred 1-methylpyrrolidine solution using a pressure-equalising dropping funnel. The dropping funnel was removed and replaced with a stopper and then the solution was heated at 80 °C for 6 days. The solvent was removed by rotary evaporation and 1-methylpyrrolidine as well as residual solvent were removed *in vacuo* at 50 °C for 16 h to afford $[C_4C_1pyrr]Cl$ as a pale yellow solid. (Scheme 14).

Pale yellow solid (190.36 g, 1.0711 mol, 91.21 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.49 (m, 4H), 3.39 (m, 2H), 3.01 (s, 3H), 2.10 (s, 4H), 1.75 – 1.63 (quint, 2H), 1.38 – 1.27 (sext, 2H), 0.94 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 63.30 (s), 62.75 (s), 47.41 (s), 24.93 (s), 21.04 (s), 19.31 (s), 13.50 (s).



Scheme 14 *The synthesis of [C4C1pyrr]Cl IL.*

Synthesis of 1-butyl-1-methylpyrrolidinium bromide[161] ***[C4C1pyrr]Br***

Distilled 1-methylpyrrolidine (50 g, 0.587 mol) was dissolved in ethyl acetate (~100 mL) in a multinecked round bottom flask fitted with a condenser attached to a N_2 line. Distilled 1-bromobutane (88.50 g, 0.6459 mol, 1.1 eq) was added slowly into the stirred 1-

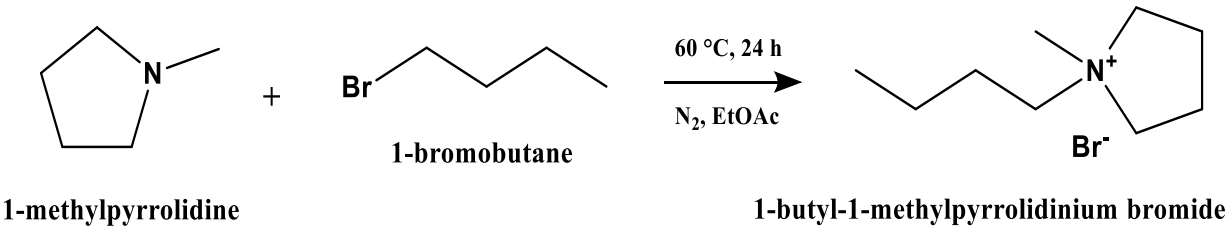
methylpyrrolidine solution using a pressure-equalising dropping funnel. The dropping funnel was removed and replaced with a stopper and then the solution was heated at 60 °C for 24 h. The solvent was removed by rotary evaporation and 1-methylpyrrolidine as well as residual solvent were removed *in vacuo* at 50 °C for 16 h to afford $[C_4C_1pyrr]Br$ as a light yellow solid. (Scheme 15).

Light yellowish solid (124.94 g, 0.5624 mol, 95.97 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.57 (m, 4H), 3.25 (m, 2H), 3.01 (s, 3H), 2.12 (s, 4H), 1.74 – 1.64 (quint, 2H), 1.39 – 1.27 (sext, 2H), 0.95 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) δ 63.36 (s), 62.81 (s), 47.48 (s), 24.92 (s), 21.06 (s), 19.30 (s), 13.49 (s).

HRMS found (ESI): m/z 142.1591 (100%, $[C_4C_1pyrr]^+$), 78.9190 (100%, $[Br]^-$), 80.9162 (97%, $[Br]^-$)



Scheme 15 *The synthesis of [C₄C₁pyrr]Br IL.*

Synthesis of 1-butyl-1-methylpyrrolidinium bis(trifluoromethanesulfonyl)imide[160] ***[C₄C₁Pyrr][NTf₂]***

$[C_4C_1pyrr]Cl$ (60 g, 0.270 mol) and $LiNTf_2$ (81.40 g, 0.284 mol, 1.05 eq) were separately weighed into conical flasks and dissolved in deionized water. These solutions were combined and stirred, leading to the formation of a biphasic mixture. Dichloromethane (150 mL) was added to increase the volume of the lower phase and the resultant organic phase separated using a separatory funnel. The organic phase was washed with deionized water for 3-4 times until Cl^- has been removed, determined by testing the water layer after washing with $AgNO_3$ (0.1 M) solution. Dichloromethane was removed by rotary evaporation and the resultant liquid dried in *vacuo* at 50 °C for 16 h to afford $[C_4C_1pyrr][NTf_2]$ (Scheme 16).

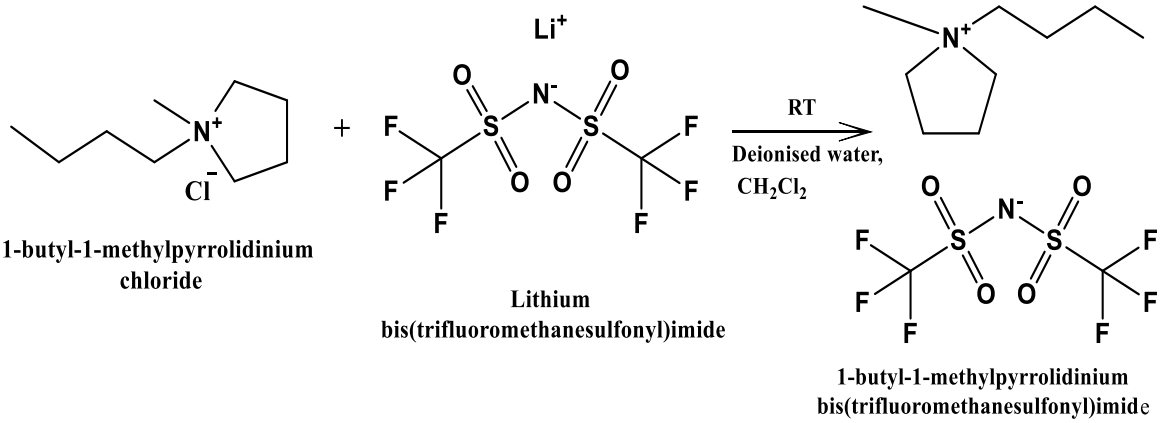
Light pinkish liquid (108.76 gm, 0.2569 moles , 95.34 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.57 (m, 4H), 3.26 (m, 2H), 3.01 (s, 3H), 2.17 (s, 4H), 1.78 – 1.63 (quint, 2H), 1.41 – 1.27 (sext, 2H), 0.96 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 124.28 (s), 119.48 (q, *J* = 321.7 Hz), 63.42 (s), 63.02 (s), 47.27 (s), 24.86 (s), 20.97 (s), 19.17 (s), 13.11 (s).

^{19}F NMR δ (ppm) (376 MHz, DMSO) -78.99 (s).

HRMS found (ESI): m/z 142.1589 (100%, $[C_4C_1pyrr]^+$), 279.9178 (100%, $[NTf_2]^-$)



Scheme 16 *The synthesis of $[C_4C_1pyrr][NTf_2]$ IL.*

Synthesis of 1-butyl-1-methylpyrrolidinium dimethyl phosphate*[160] *$[C_4C_1pyrr][Me_2PO_4]$

$[C_4C_1pyrr]Cl$ (50 g, 0.281 mol) was dissolved in acetonitrile (150 mL) in a multinecked round bottom flask fitted with a condenser attached to a N_2 line. Distilled trimethyl phosphate (59.12 g, 0.4221 mol, 1.5 eq) was added slowly to the stirred $[C_4C_1pyrr]Cl$ solution using a pressure-equalising dropping funnel. After the addition was complete, the dropping funnel was removed and replaced with a stopper and the solution was heated at 80 °C with stirring for 2 - 3 weeks until a negative test for Cl^- was observed using a $AgNO_3$ (0.1 M) solution. Acetonitrile was removed by rotary evaporation and 1-methylpyrrolidine as well as residual solvent were removed *in vacuo* at 50 °C for 16 h to afford $[C_4C_1pyrr][Me_2PO_4]$ (Scheme 17).

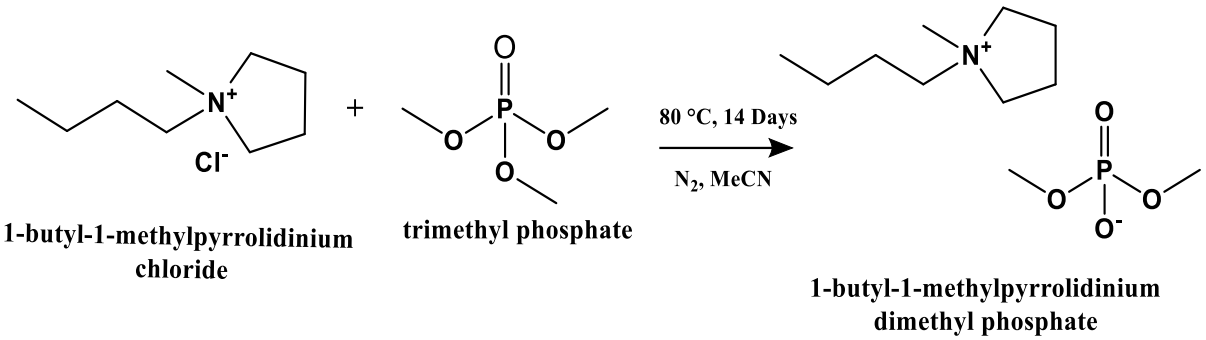
Pale yellow semi-solid (70.99 g, 0.2656 mol, 94.39 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.67 (m, 4H), 3.50 (m, 2H), 3.30 (d, *J* = 10.3 Hz, 6H), 3.07 (s, 3H), 2.09 (s, 4H), 1.77 – 1.61 (quint, 2H), 1.41 – 1.22 (sext, 2H), 0.94 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 62.90 (s), 62.37 (s), 51.22 (d, *J* = 6.0 Hz), 46.90 (s), 25.00 (s), 20.92 (s), 19.29 (s), 13.41 (s).

^{31}P NMR δ (ppm) **(**162 MHz, DMSO) 1.45 (hept, *J* = 10.3 Hz).

HRMS found (ESI): m/z 142.1589 (100%, $[C_4C_1pyrr]^+$), 125.0008 (100%, $[Me_2PO_4]^-$)



Scheme 17 *The synthesis of $[C_4C_1pyrr][Me_2PO_4]$ ILs*

Synthesis of 1-butyl-1-methylpyrrolidinium dicyanamide*[56] *$[C_4C_1pyrr][N(CN)_2]$

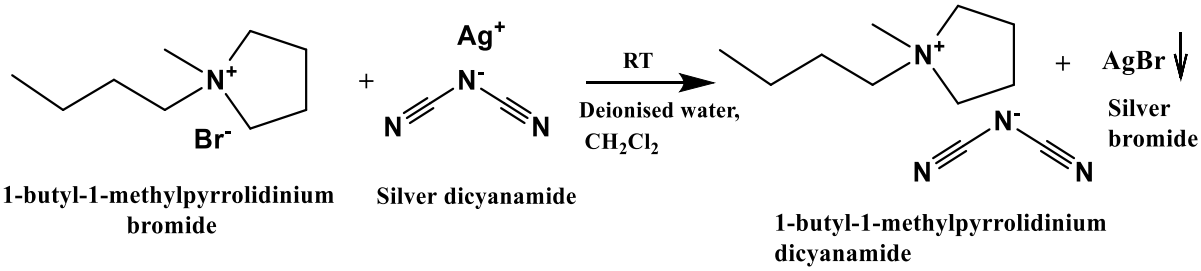
$[C_4C_1pyrr]Br$ (25.55 g, 0.115 mol) was dissolved in deionised water (150 mL). $AgN(CN)_2$ (20.0 g, 0.115 mol, 1.0 eq) was added to this solution and stirred in the dark for 16 h. The resultant slurry was then filtered and the solution checked for the removal of Br^- by testing with saturated aqueous $AgN(CN)_2$ solution (0.1 g of $AgN(CN)_2$ in 50 mL of water). More $AgN(CN)_2$ was added until no more AgBr precipitate was observed using the $AgN(CN)_2$ test. After the final filtration, water was removed using a rotary evaporator and then dichloromethane (~200 mL) was added to precipitate any remaining dissolved salts. The solution was filtered, solvent removed by rotary evaporation and the resultant liquid dried in *vacuo* at 50 °C for 16 h to afford $[C_4C_1pyrr][N(CN)_2]$ (Scheme 18).

Light yellowish liquid (19.7258 gm, 0.09469 moles, 82.34 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) δ 3.57 (m, 4H), 3.28 (m, 2H) 3.02 (s, 3H), 2.12 (s, 4H), 1.77 – 1.64 (quint, 2H), 1.41 – 1.26 (sext, 2H), 0.94 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) δ 119.11 (s), 63.55 (s), 62.96 (s), 47.48 (s), 24.91 (s), 21.13 (s), 19.28 (s), 13.32 (s).

HRMS found (ESI): m/z 142.1589 (100%, $[C_4C_1pyrr]^+$), 66.0098 (100%, $[N(CN)_2]^-$)



1-butyl-1-methylpyrrolidinium bromide **Silver dicyanamide** **1-butyl-1-methylpyrrolidinium dicyanamide**

Scheme 18 *The synthesis of [C_4C_1pyrr][$N(CN)_2$] IL.*

Synthesis of 1-butylguanidinium bis(trifluoromethanesulfonyl)imide[160] ***[C_4gun][NTf_2]***

[C_4gun]Cl (17.59 g, 0.116 mol) and $LiNTf_2$ (34.97 g, 0.122 mol, 1.05 eq) were separately weighed into conical flasks and dissolved in deionized water. These solutions were combined and stirred, leading to the formation of a biphasic mixture. Dichloromethane (150 mL) was added to increase the volume of the lower phase and the resultant organic phase separated using a separatory funnel. The organic phase was washed with deionized water (~150 mL) 3-4 times until Cl^- was removed, determined by testing the water layer after washing with $AgNO_3$ (0.1 M) solution. Dichloromethane was removed by rotary evaporation and the resultant liquid dried in *vacuo* at 50 °C for 16 h to afford [C_4gun][NTf_2] (Scheme 19).

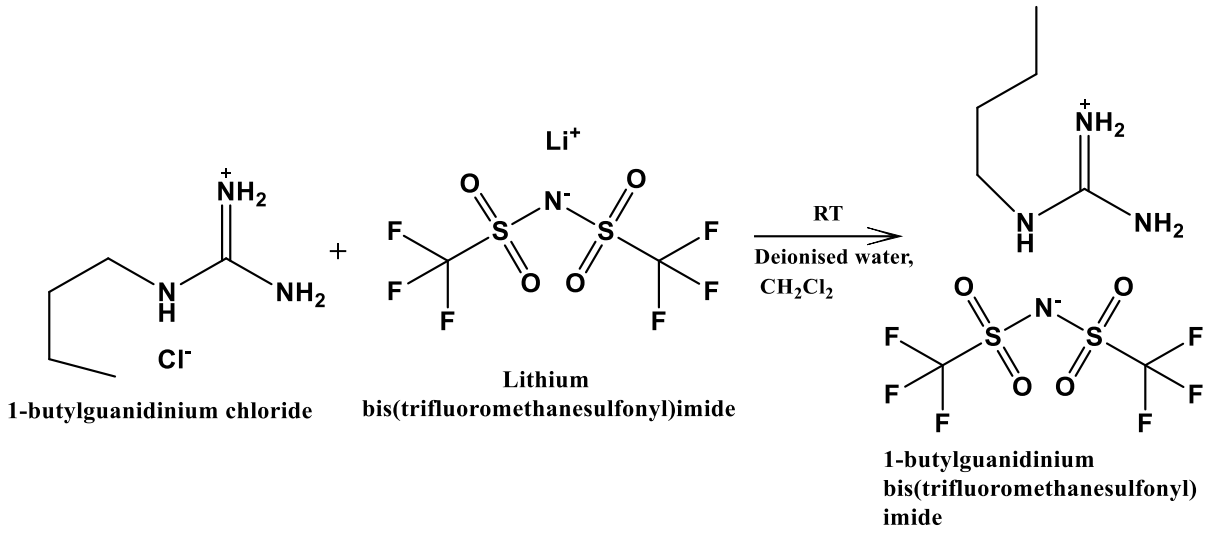
Colourless moderate viscous liquid (35.84 g, 0.0904 mole, 77.96 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 7.50 - 6.31 (m, *J* = 167.0 Hz, 5H), 3.10 (t, *J* = 12.6, 6.9 Hz, 2H), 1.53 (quint, 2H), 1.39 (sext, *J* = 14.2, 7.2 Hz, 2H), 0.90 (t, *J* = 7.3 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 156.69 (s), 119.45 (q, *J* = 321.8 Hz), 40.41 (s), 30.34 (s), 19.09 (s), 13.06 (s).

^{19}F NMR δ (ppm) (376 MHz, DMSO) -79.05 (s).

HRMS found (ESI): m/z 116.1183 (100%, $[C_4gun]^+$), 279.9176 (100%, $[NTf_2]^-$)



Scheme 19 *The synthesis of [C4gun][NTf2] IL.*

Synthesis of 1-octyl-3-methylimidazolium chloride[160] ***[C8C1im]Cl***

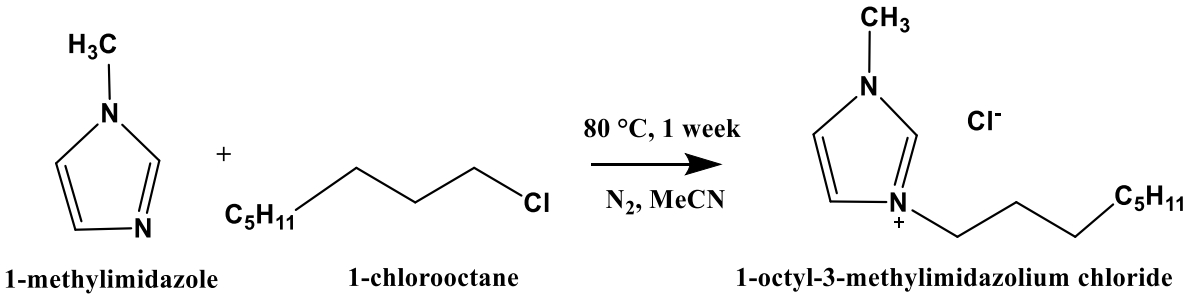
1-Methylimidazole (82.11 g, 1.218 mol) was dissolved in approximately 100 mL of acetonitrile in a multinecked round bottom flask fitted with a condenser attached to either a drying test tube (containing $CaCl_2$) or a N_2 line. 1-Chlorooutane (199.21 g, 1.340 mol, 1.1 eq) was added slowly into the 1-methylimidazole solution using a pressure-equalising dropping funnel while the solution was stirred at room temperature. After the addition, the solution was heated at 80 °C for 6 - 7 days. The solution was cooled to room temperature and the resultant liquid was thoroughly with diethyl ether (10 × 50 mL) to give a white solid. Residual solvent was removed by rotary evaporation and the solid was dried *in vacuo* at 50 °C for 16 h to afford $[C_8C_1im]Cl$ (Scheme 20).

Colourless viscous liquid (230.09 g, 0.9970 mol, 81.86 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.70 (s, 1H), 7.97 (t, *J* = 1.7 Hz, 1H), 7.89 (t, *J* = 1.7 Hz, 1H), 4.24 (t, *J* = 7.2 Hz, 2H), 3.92 (s, 3H), 1.88 – 1.73 (quint, 2H), 1.25 (m, *J* = 4.0 Hz, 10H), 0.86 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 136.78 (s), 123.52 (s), 122.26 (s), 48.61 (s), 35.65 (s), 31.14 (s), 29.45 (s), 28.41 (s), 25.48 (s), 22.02 (s), 13.87 (s).

HRMS found (ESI): m/z 195.1851 (100%, $[C_8C_1im]^+$), 265.1246 ($\{[C_8C_1im]Cl_2\}^-$)



1-methylimidazole **1-chlorooctane** **1-octyl-3-methylimidazolium chloride**

Scheme 20 *The synthesis of [C_8C_1im]Cl IL.*

Synthesis of 1-octyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide[160] ***[C_8C_1im][NTf_2]***

[C_8C_1im]Cl (40.25 g, 0.174 mol) and $LiNTf_2$ (52.57 g, 0.183 mol, 1.05 eq) were separately weighed into conical flasks and dissolved in deionized water. These solutions were combined and stirred, leading to the formation of a biphasic mixture. Dichloromethane (150 mL) was added to increase the volume of the lower phase and the resultant organic phase separated using a separatory funnel. The organic phase was washed with deionized water 3-4 times until Cl^- has been removed, determined by testing the water layer after washing with $AgNO_3$ (0.1 M) solution. Dichloromethane was removed by rotary evaporation and the resultant liquid dried in *vacuo* at 50 °C for 16 h to afford [C_8C_1im][NTf_2] (Scheme 21).

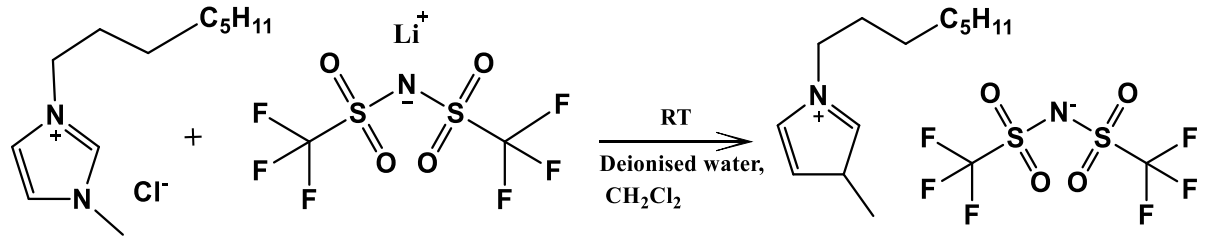
Colourless liquid (79.83 g, 0.168 mol, 96.27 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.11 (s, 1H), 7.74 (t, *J* = 1.8 Hz, 1H), 7.68 (t, *J* = 1.7 Hz, 1H), 4.18 (t, *J* = 7.3 Hz, 2H), 3.88 (s, 3H), 2.06 – 1.63 (quint, 2H), 1.28 (m, *J* = 6.5, 4.0 Hz, 10H), 0.87 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 136.45 (s), 124.28 (s), 123.48 (s), 120.59 (q, *J* = 324), 48.81 (s), 35.55 (s), 31.10 (s), 29.34 (s), 28.33 (d, *J* = 12.7 Hz), 25.43 (s), 21.95 (s), 13.60 (s).

^{19}F NMR δ (ppm) (376 MHz, DMSO) -79.01 (s).

HRMS found (ESI): m/z 195.1856 (100%, $[C_8C_1im]^+$), 279.9177 (100 %, $[NTf_2]^-$)



1-octyl-3-methylimidazolium chloride | **Lithium bis(trifluoromethanesulfonyl)imide** | **1-octyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide**

Scheme 21 *The synthesis of $[C_8C_1im][NTf_2]$ IL.*

Synthesis of 1-octyl-3-methylimidazolium dimethyl phosphate*[160] *$[C_8C_1im][Me_2PO_4]$

$[C_8C_1im]Cl$ (70.25 g, 0.304 mol) was dissolved in acetonitrile (150 mL) in a multinecked round bottom flask fitted with a condenser attached to a N_2 line. Distilled trimethyl phosphate (63.96 g, 0.457 mol, 1.5 eq) was added slowly to the stirred $[C_8C_1im]Cl$ solution using a pressure-equalising dropping funnel. After the addition was complete, the dropping funnel was removed and replaced with a stopper and the solution was heated at 80 °C with stirring for 2 - 3 weeks until a negative test for Cl^- was observed using a $AgNO_3$ (0.1 M) solution. Acetonitrile was removed by rotary evaporation and the resultant liquid was washed with diethyl ether (10 × 50 mL) to give a pale-yellow viscous liquid. Excess solvent was removed by rotary evaporation and the resultant liquid dried in *vacuo* 50 °C for 16 h to afford $[C_8C_1im][Me_2PO_4]$ (Scheme 22).

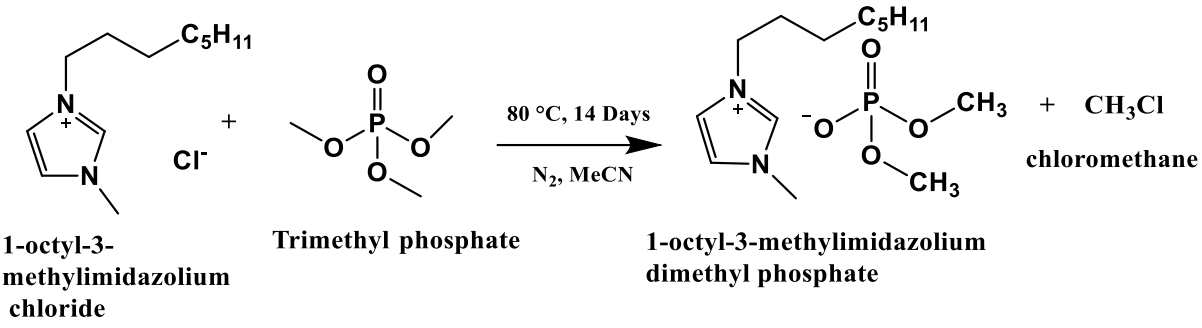
Colourless viscous liquid (52.99 g, 0.1654 mol, 54.54 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.94 (s, 1H), 8.06 (t, *J* = 1.7 Hz, 1H), 7.96 (t, *J* = 1.7 Hz, 1H), 4.23 (t, *J* = 7.3 Hz, 2H), 3.92 (s, 3H), 1.79 (quint, 2H), 1.22 (m, *J* = 11.5 Hz, 10H), 0.81 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 137.57 (s), 123.01 (d), 51.25 (s), 48.51 (s), 35.34 (s), 31.15 (s), 29.62 (s), 28.52 (s), 28.42 (s) 25.53 (s), 22.00 (s), 13.66 (s).

^{31}P NMR δ (ppm) (162 MHz, DMSO) 1.46 (hept, *J* = 10.3 Hz).

HRMS found (ESI): m/z 195.1852 (100%, $[C_8C_1im]^+$), 125.0009 (100 %, $[Me_2PO_4]^-$)



Scheme 22 *The synthesis of [C_8C_1im][Me_2PO_4] IL.*

Synthesis of 1-octyl-3-methylimidazolium dicyanamide[56] ***[C_8C_1im][$N(CN)_2$]***

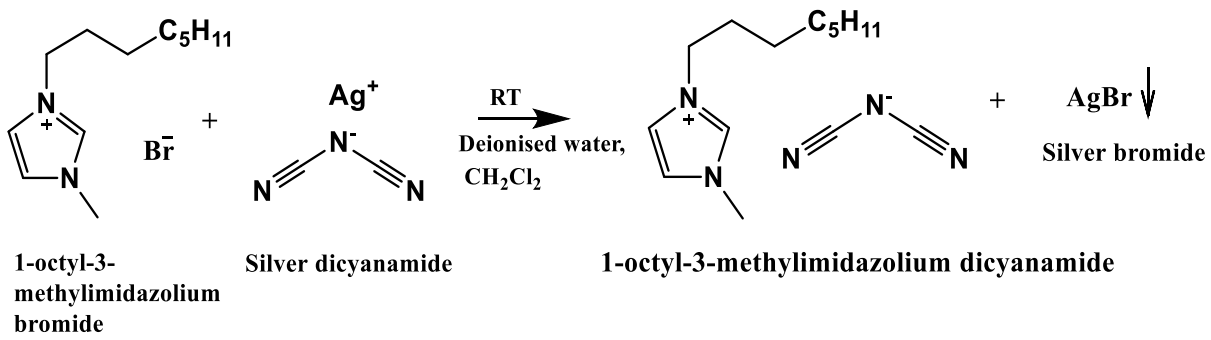
[C_8C_1im]Cl (26.54 g, 0.115 mol) was dissolved in deionised water (150 mL). $AgN(CN)_2$ (20 g, 0.115 mol, 1.0 eq) was added to this solution and stirred in the dark for 16 h. The resultant slurry was then filtered and the solution checked for the removal of Br^- by testing with saturated aqueous $AgN(CN)_2$ solution (0.1 g of $AgN(CN)_2$ in 50 mL of water). More $AgN(CN)_2$ was added until no more AgCl precipitate was observed using the $AgN(CN)_2$ test. After the final filtration, water was removed using a rotary evaporator and then dichloromethane (~200 mL) was added to precipitate any remaining dissolved salts. The solution was filtered, solvent removed by rotary evaporation and the resultant colourless liquid dried in *vacuo* at 50 °C for 16 h to afford [C_8C_1im][$N(CN)_2$] (Scheme 23).

Light yellowish liquid (25.87 g, 0.0989 mole, 86.07 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 9.13 (s, 1H), 7.78 (t, *J* = 1.8 Hz, 1H), 7.71 (t, *J* = 1.8 Hz, 1H), 4.17 (t, *J* = 7.2 Hz, 2H), 3.87 (s, 3H), 1.89 – 1.72 (quint, 2H), 1.35 – 1.07 (m, 10H), 0.86 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 136.49 (s), 123.57 (s), 122.23 (s), 119.08 (s), 48.79 (s), 35.71 (s), 31.14 (s), 29.37 (s), 28.45 (s), 28.31 (s), 25.48 (s), 22.03 (s), 13.88 (s).

HRMS found (ESI): m/z 195.1858 (100%, $[C_8C_1im]^+$), 66.0096 (100 %, $[N(CN)_2]^-$)



Scheme 23 *The synthesis of $[C_8C_1im][N(CN)_2]$ IL.*

Synthesis of 1-octylguanidinium chloride*[163] *$[C_8gun]Cl$

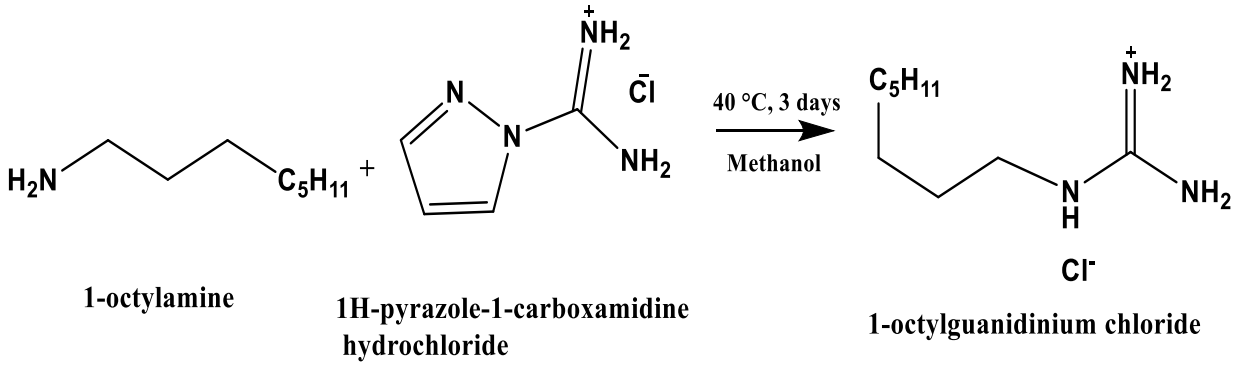
Distilled 1-octylamine (10 g, 0.077 mol) was dissolved in methanol (20 mL) at 40 ℃ with constant stirring. 1H-pyrazole-1-carboxamidine hydrochloride (11.34 g, 0.077 mol, 1.0 eq) was added and the solution was stirred at 40 ℃ for 3 days. Residual solvent was removed by rotary evaporation and the crude product was washed (6 × 100 mL) with diethyl ether before being dried in *vacuo* at 50 °C for 16 h to afford $[C_8gun]Cl$ as a white solid (Scheme 24).

White solid (10.17 g, 0.049 mol, 63.25 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) δ 8.09 – 6.88 (m, 5H), 3.10 (t, *J* = 6.8 Hz, 2H), 1.45 (quint, 2H), 1.27 (m, 10H), 0.87 (t, *J* = 6.8 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 157.12 (s), 40.62 (s), 31.21 (s), 28.60 (s), 28.57 (s), 28.48 (s), 26.04 (s), 22.06 (s), 13.91 (s).

HRMS found (ESI): m/z 172.1808 (100%, $[C_8gun]^+$), 242.1202 ($\{[C_8gun]Cl_2\}^-$)



Scheme 24 *The synthesis of $[C_8gun]Cl$ IL.*

Synthesis of 1-octylguanidinium bis(trifluoromethanesulfonyl)imide*[160] *$[C_8gun][NTf_2]$

[C_8gun]Cl (10.17 g, 0.049 mol) and $LiNTf_2$ (14.75 g, 0.051 mol, 1.05 eq) were separately weighed into conical flasks and dissolved in deionized water. These solutions were combined and stirred, leading to the formation of a biphasic mixture. Dichloromethane (150 mL) was added to increase the volume of the lower phase and the resultant organic phase separated using a separatory funnel. The organic phase was washed with deionized water (~ 200 mL) for 3-4 times until Cl^- has been removed, determined by testing the water layer after washing with $AgNO_3$ (0.1 M) solution. Dichloromethane was removed by rotary evaporation and the resultant liquid dried in *vacuo* at 50 °C for 16 h to afford [C_8gun][NTf_2] (Scheme 25).

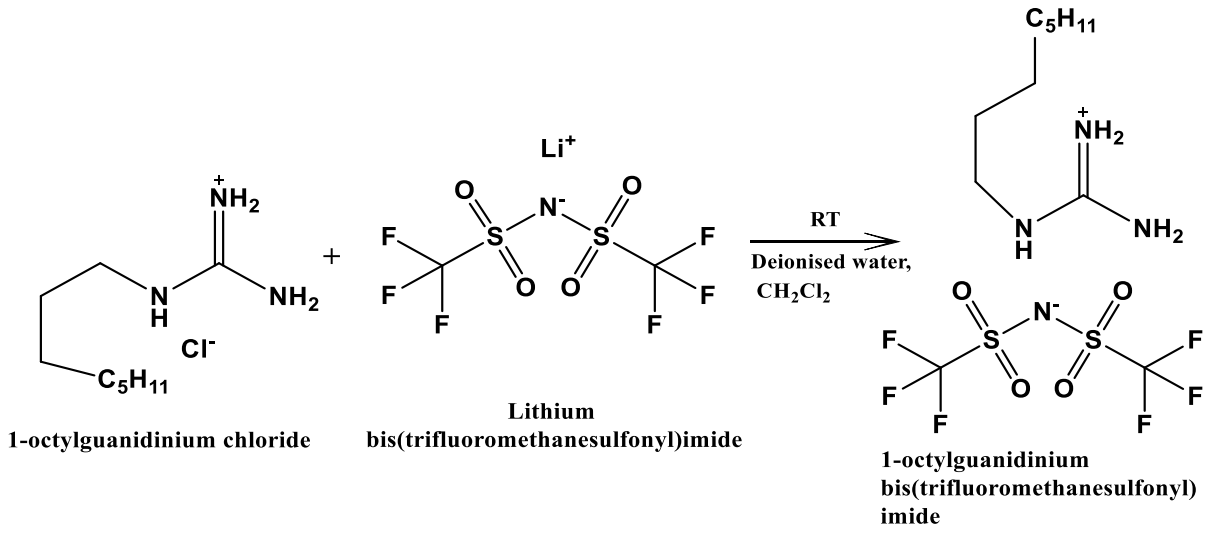
Colourless moderate viscous liquid (18.86 g, 0.0417 mole, 85.19 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 6.92 (m, 5H)**,** 3.10 (t, *J* = 6.3 Hz, 2H), 1.48 (quint, 2H), 1.28 (m, 10H), 0.87 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 156.65 (s), 119.47 (q, *J* = 321.8 Hz), 40.71 (s), 31.14 (s), 28.56 (s), 28.49 (s), 28.36 (s), 25.97 (s), 21.97 (s), 13.63 (s).

^{19}F NMR δ (ppm) (376 MHz, DMSO) -78.97 (s).

HRMS found (ESI): m/z 172.1806 (100%, [C_8gun]$^+$), 279.9179 (100%, [NTf_2]$^-$)



Scheme 25 *The synthesis of [C_8gun][NTf_2] IL.*

Synthesis of 1-octyl-1-methylpyrrolidinium chloride*[160] *[C_8C_1pyrr]Cl

Distilled 1-methylpyrrolidine (50 g, 0.587 mol) was dissolved in acetonitrile (~100 mL) in a multinecked round bottom flask fitted with a condenser attached to a N_2 line. Distilled 1-chlorooctane (96.04 g, 0.646 mol, 1.1 eq) was added slowly into the stirred 1-methylpyrrolidine

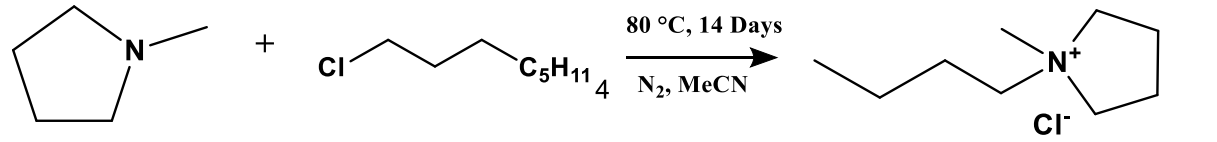
solution using a pressure-equalising dropping funnel. The dropping funnel was removed and replaced with a stopper and then the solution was heated at 80 °C for 6 days. The solvent was removed by rotary evaporation and 1-methylpyrrolidine as well as residual solvent were removed *in vacuo* at 50 °C for 16 h to afford $[C_8C_1pyrr]Cl$ as a colourless solid (Scheme 26).

Colourless solid (133.17 g, 0.5695 mole, 96.99 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.54 (m, 4H), 3.39 (m, 2H), 3.08 (s, 3H), 2.16 (s, 4H), 1.77 (m, 2H), 1.37 (m, 10H), 0.96 (t, *J* = 7.4 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 63.32 (s), 62.98 (s), 47.43 (s), 31.15 (s), 28.46 (s), 25.92 (s), 22.92 (s), 22.03 (s), 21.06 (s), 13.94 (s).

HRMS found (ESI): m/z 198.2210 (100%, $[C_8C_1pyrr]^+$), 268.1610 ($\{[C_8C_1pyrr]_2Cl\}^-$)



1-methylpyrrolidine **1-chlorooctane** **1-butyl-1-methylpyrrolidinium chloride**

Scheme 26 *The synthesis of $[C_8C_1pyrr]Cl$ IL.*

Synthesis of 1-octyl-1-methylpyrrolidinium bromide*[161] *$[C_8C_1pyrr]Br$

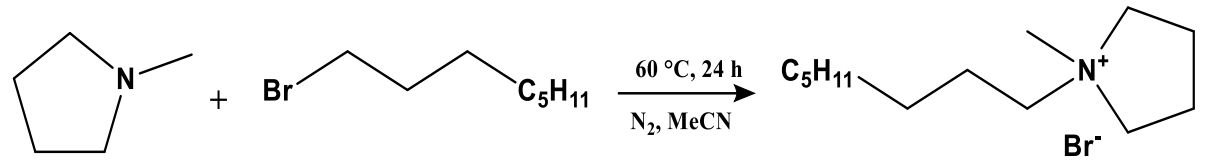
Distilled 1-methylpyrrolidine (70 g, 0.822 mol) was dissolved in ethyl acetate (~100 mL) in a multinecked round bottom flask fitted with a condenser attached to a N_2 line. Distilled 1-bromooctane (174.64 g, 0.904 mol, 1.1 eq) was added slowly into the stirred 1-methylpyrrolidine solution using a pressure-equalising dropping funnel. The dropping funnel was removed and replaced with a stopper and then the solution was heated at 60 °C for 24 h. The solvent was removed by rotary evaporation and 1-methylpyrrolidine as well as residual solvent were removed *in vacuo* at 50 °C for 16 h to afford $[C_8C_1pyrr]Br$ as a light yellow solid. (Scheme 27).

Pale yellow solid (229.59 g, 0.8250 mole, 99.9 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.49 (m, 4H), 3.34 (m, 2H), 3.01 (s, 3H), 2.09 (s, 4H), 1.70 (m, 2H), 1.36 – 1.22 (m, 10H), 0.88 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 63.34 (s), 62.97 (s), 47.47 (s), 31.16 (s), 28.46 (s), 25.91 (s), 22.93 (s), 22.03 (s), 21.06 (s), 13.93 (s).

HRMS found (ESI): m/z 198.2219 (100%, $[C_8C_1pyrr]^+$), 78.9188 (100%, $[Br]^-$), 80.9162 (97%, $[Br]^-$)



1-methylpyrrolidine **1-bromooctane** **1-octyl-1-methylpyrrolidinium bromide**

Scheme 27 *The synthesis of [C_8C_1pyrr]Br IL.*

Synthesis of 1-octyl-1-methylpyrrolidinium bis(trifluoromethanesulfonyl)imide[160] ***[C_8C_1pyrr][NTf_2]***

[C_8C_1pyrr]Br (40 g, 0.144 mol) and $LiNTf_2$ (43.33 g, 0.151 mol, 1.05 eq) were separately weighed into conical flasks and dissolved in deionized water. These solutions were combined and stirred, leading to the formation of a biphasic mixture. Dichloromethane (150 mL) was added to increase the volume of the lower phase and the resultant organic phase separated using a separatory funnel. The organic phase was washed with deionized water (~ 150 mL) for 3-4 times until Cl^- has been removed, determined by testing the water layer after washing with $AgNO_3$ (0.1 M) solution. Dichloromethane was removed by rotary evaporation and the resultant liquid dried in *vacuo* at 50 °C for 16 h to afford [C_8C_1pyrr][NTf_2] (Scheme 28).

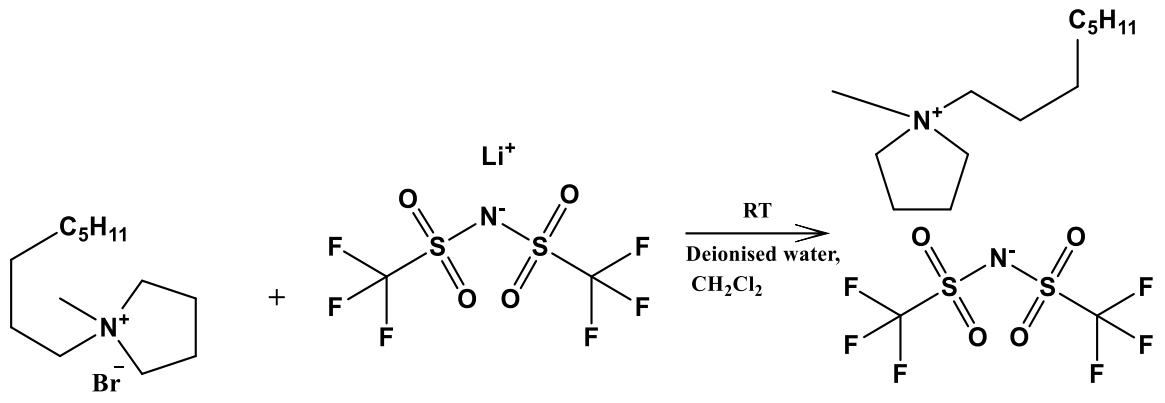
Pale yellow liquid (61.18 g, 0.1279 mol, 88.95 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.50 (m, 4H), 3.39 – 3.27 (m, 2H), 3.03 (s, 3H), 2.15 (s, 4H), 1.81 – 1.70 (m, 2H), 1.33 (m, 10H), 0.91 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 119.49 (q, *J* = 321.7 Hz), 63.43 (s), 47.38 (s), 31.07 (s), 28.33 (d, *J* = 6.1 Hz), 25.72 (s), 22.87 (s), 21.91 (s), 20.90 (s), 13.41 (s).

^{19}F NMR δ (ppm) (376 MHz, DMSO) -79.23 (s).

HRMS found (ESI): m/z 198.2214 (100%, $[C_8C_1pyrr]^+$), 279.9183 (100%, $[NTf_2]^-$)



Scheme 28 *The synthesis of [C_8C_1pyrr][NTf_2] IL.*

Synthesis of 1-octyl-1-methylpyrrolidinium dimethyl phosphate[160] ***[C_8C_1pyrr][Me_2PO_4]***

[C_8C_1pyrr]Cl (70 g, 0.299 mol) was dissolved in acetonitrile (150 mL) in a multinecked round bottom flask fitted with a condenser attached to a N_2 line. Distilled trimethyl phosphate (62.90 g, 0.449 mol, 1.5 eq) was added slowly to the stirred [C_8C_1pyrr]Cl solution using a pressure-equalising dropping funnel. After the addition was complete, the dropping funnel was removed and replaced with a stopper and the solution was heated at 80 °C with stirring for 2 - 3 weeks until a negative test for Cl^- was observed using a $AgNO_3$ (0.1 M) solution. Acetonitrile was removed by rotary evaporation and 1-methylpyrrolidine as well as residual solvent were removed *in vacuo* at 50 °C for 16 h to afford [C_8C_1pyrr][Me_2PO_4] (Scheme 29).

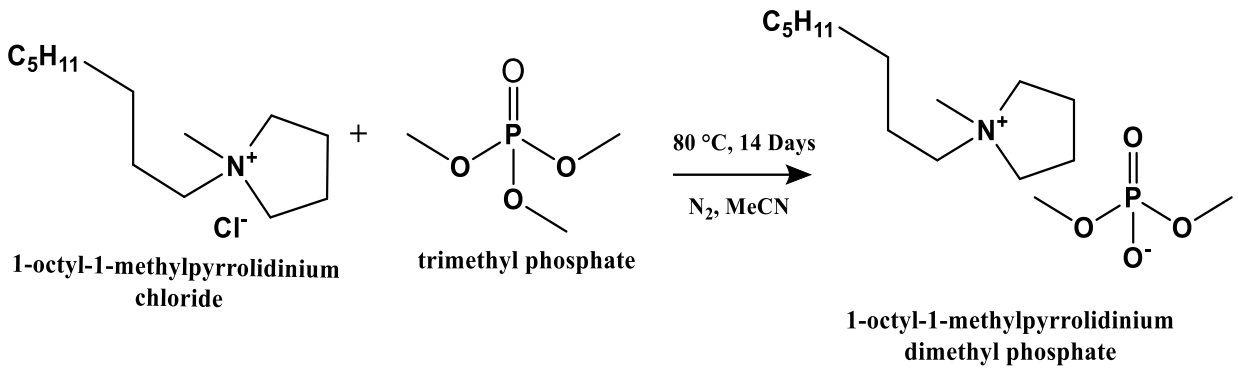
Pale yellow viscous liquid (52.74 g, 0.1693 mol, 54.47 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.50 (m, 4H), 3.39 – 3.31 (m, 2H), 3.29 (d, *J* = 10.3 Hz, 6H), 3.01 (s, 3H), 2.08 (s, 4H), 1.70 (m, 2H), 1.29 (m, 10H), 0.88 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 63.02 (d), 51.30 (s), 47.25 (s), 31.16 (s), 28.49 (s), 25.94 (s), 22.96 (s), 22.03 (s), 21.01 (s), 13.90 (s).

^{31}P NMR δ (ppm) (162 MHz, DMSO) 1.43 (hept, *J* = 10.2 Hz).

HRMS found (ESI): m/z 198.2215 (100%, [C_8C_1pyrr]$^+$), 125.009 (100%, [Me_2PO_4]$^-$)



Scheme 29 *The synthesis of [C_8C_1pyrr][Me_2PO_4] IL.*

Synthesis of 1-octyl-1-methylpyrrolidinium dicyanamide[56] ***[C_8C_1pyrr][$N(CN)_2$]***

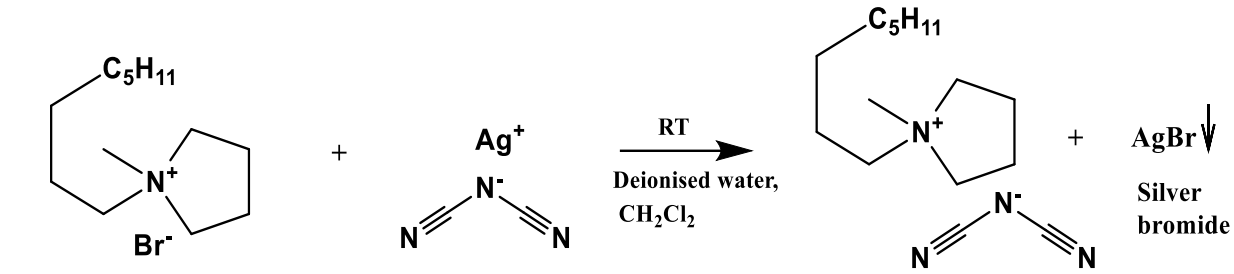
[C_8C_1pyrr]Br (32.00 g, 0.115 mol) was dissolved in deionised water (150 mL). $AgN(CN)_2$ (20.0 g, 0.115 mol, 1.0 eq) was added to this solution and stirred in the dark for 16 h. The resultant slurry was then filtered and the solution checked for the removal of Br^- by testing with saturated aqueous $AgN(CN)_2$ solution (0.1 g of $AgN(CN)_2$ in 50 mL of water). More $AgN(CN)_2$ was added until no more AgBr precipitate was observed using the $AgN(CN)_2$ test. After the final filtration, water was removed using a rotary evaporator and then dichloromethane (~200 mL) was added to precipitate any remaining dissolved salts. The solution was filtered, solvent removed by rotary evaporation and the resultant liquid dried in *vacuo* at 50 °C for 16 h to afford [C_8C_1pyrr][$N(CN)_2$] (Scheme 30).

Pale yellow liquid (23.69 g, 0.089 mole, 77.90 %)

^{1}H NMR δ (ppm) (400 MHz, DMSO) 3.56 – 3.39 (m, 4H), 3.38 – 3.25 (m, 2H), 3.01 (s, 3H), 2.10 (m, *J* = 4.8 Hz, 4H), 1.78 – 1.62 (m, 2H), 1.30 (m, 10H), 0.88 (t, *J* = 6.9 Hz, 3H).

^{13}C NMR δ (ppm) (101 MHz, DMSO) 119.10 (s), 63.27 (s), 63.13 (s), 47.48 (s), 31.16 (s), 28.45 (s), 25.90 (s), 22.93 (s), 22.05 (s), 21.11 (s), 13.86 (s).

HRMS found (ESI): m/z 198.2220 (100%, $[C_8C_1pyrr]^+$), 66.0090 (100%, $[N(CN)_2]^-$)



1-octyl-1-methylpyrrolidinium bromide **Silver dicyanamide** **1-octyl-1-methylpyrrolidinium dicyanamide**

Scheme 30 *The synthesis of [C_8C_1pyrr][$N(CN)_2$] IL.*

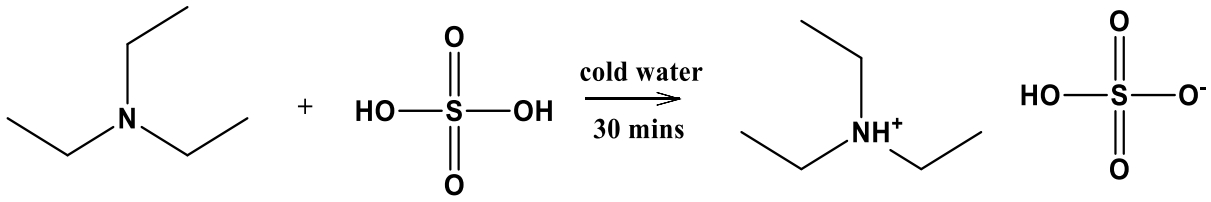
Synthesis of triethylammonium hydrogen sulphate[126] ***[TEA][HSO_4]***

Concentrated H_2SO_4 (98%, 18.4 M, 25.0 g) was added to cold water (200 mL) and this diluted solution was added dropwise to triethylamine (25.39 g, 0.25 mol) that was vigorously stirred in an ice-water bath. The solvent was removed by rotary evaporation and the remaining viscous liquid was dried at 55 ℃ for 1 hour in *vacuo* to afford [TEA][HSO_4] (Scheme 31).

White hygroscopic crystalline solid (50 g, 100%)

^{1}H NMR δ (ppm) (400 MHz, DMSO-d6) 3.08 (q, *J* = 7.3 Hz, 6H), 1.17 (t, *J* = 7.3 Hz, 9H).

^{13}C NMR δ (ppm) (101 MHz, DMSO-d6) 45.88 (s), 8.64 (s).



triethylamine **conc. sulfuric acid** **triethylammonium hydrogen sulphate**

Scheme 31 *The synthesis of [TEA][HSO_4] IL.*

6.10. Extraction of bark by ILs

6.10.1. Pre-drying of bark

Pinus radiata bark was provided by Scion. Prior to the extractions, the bark was pre-dried by heating under a flow of nitrogen at 50 °C for 2-3 hours and then in *vacuo* at 50 °C for 16 h. The drying procedure was performed in 10 g batches and each time resulted in a reduction in mass of 20-23% compared to the undried bark.

6.10.2. General procedure for the extraction of bark

IL was added to a pre-weighed bark sample and the resultant suspension stirred at various temperatures (25 °C, 50 °C, 100 °C) for different time intervals (1 h or 24 h). The IL and residual bark were separated after the extraction using a syringe filter to obtain a sample of the IL and a Buchner funnel to collect the bark. Residual bark was washed using an appropriate solvent (water or dichloromethane) to remove residual IL. The bark residue was analysed using solid-state NMR and pyrolysis GC-MS, while the IL extract was analysed by UV-Vis and solution NMR. Both of the extracts were also characterized by solution NMR, FTIR, and Headspace-SPME GC-MS.

6.10.3. General procedure for antisolvent addition

0.5 mL of antisolvent (water, acetonitrile, dichloromethane and dimethyl sulfoxide) was added to 1 mL of the IL extract (referred to as Sample 1). Any suspended solids formed in Sample 1 were settled by centrifugation for 5 – 10 mins. The Sample 1 filtrate was separated from the solids and the solids were washed with the antisolvent used for their separation. Further antisolvent (0.5 mL) was added to the Sample 1 filtrate after the solids had been removed, which was denoted as Sample 2. This process continued with increasing amounts of antisolvent with any solids formed separated and characterised using IR. The samples were also scaled up (10 mL of IL extract) and further characterised by using IR, solution-phase NMR, and solid-state NMR.

6.11. Analytical Techniques

6.11.1. UV-Visible spectroscopy analysis

UV-vis absorption spectra were obtained of ILs used for the extraction of bark.

6.11.1.1. Time-dependent UV-Vis measurement

Undried bark (0.1 g) was weighed and added to 1 mL of $[C_4C_1im][NTf_2]$. UV-Vis spectra of the IL from 200 – 800 nm were obtained every 5 min for the first 1 h then at longer intervals up to 2.5 days. The baselines were normalised to the 800 nm absorbance to correct for light scattered by bark chips suspended in the liquid.

6.11.1.2. Temperature-dependent UV-Vis measurement

Undried bark (0.3 g) was weighed and added to the IL (3 mL). The mixture was stirred for 1 h at either 25 °C or at 50 °C before the suspension was filtered using a syringe filter to remove suspended bark. The filtered solutions were analysed by UV-Vis spectroscopy from 200 to 800 nm with the baselines normalised to the 800 nm absorbance to account for scattering from residual bark that was not removed by filtration.

To explore the effect of longer extraction times and higher temperatures, 0.1 g of pre-dried bark was weighed and added to 4 mL of each IL. The mixture was stirred for 24 h at 100 °C and filtered using a syringe filter to remove small bark pieces. The solution was analysed by UV-Vis spectroscopy using the same conditions described above.

6.11.2. Solution-phase NMR Analysis

NMR spectra were obtained of the IL extracts and the bark extracts recovered following antisolvent addition. Dimethyl sulfoxide (DMSO-d6) was used as a solvent to dissolve the precipitated bark extracts obtained after antisolvent addition to IL extracts but these extracts were only partially soluble. For the analysis of IL extracts, DMSO solvent was used.

6.11.3. FT-IR Analysis

Both solid and liquid samples were analysed by FTIR. Solid samples were ground into a powder prior to analysis while liquid samples were analysed directly. Data were recorded at room temperature, in the range of 400 – 4000 cm^{-1}, by accumulating 256 scans with a resolution of 8 cm^{-1}. The presence of residual IL was observed in some solid bark residues after the original washing steps. These samples were subjected to an additional washing procedure which had been suggested by Ferreira et al.[98] The solid residue was washed with hot water (3 × 20 mL at 80 °C) for 40 mins to remove residual IL followed by drying at 50 °C under a nitrogen purge and high vacuum overnight to remove residual water. This washing procedure was also done on the pure bark to check its effect on the raw bark residue.

6.11.4. Pyrolysis gas chromatography mass spectroscopy (Py-GCMS) analysis

The sample was weighed directly into pyrolysis cups (0.5 mg ± 0.1). Samples were pyrolyzed at 450 ℃ for 12 seconds before the gas/vapour products were injected into the GC/MS using a 20 :1 split ratio. Analyte separation took place on a 30m HP-5MS column (Agilent) with 0.25 mm inner diameter and 0.25 µm film thickness. The pyrolyzer interface and the GC injection temperature were both set at 250 ℃, the transfer line temperature was 280 ℃ and the ion source temperature was 230℃. Helium was used as the carrier gas at a constant flow rate of 1.2 mL/min. The oven temperature program was set at 2 minutes isothermal heating at 70 ℃, followed by a ramp of 4℃/min to 280 ℃ then held at this temperature for 25.5 minutes. The mass spectra were obtained in electron impact (EI) mode at 70 eV by scanning m/z 35-600 amu at 2.6 scans per sec.[147]

6.11.5. Tetramethylammonium hydroxide pyrolysis gas chromatography mass spectrometry (TMAH-Py-GCMS) analysis

Tetramethylammonium hydroxide 25 wt% solution in methanol was used for the derivatisation. Samples (0.5 mg ± 0.1) were weighed directly into a pyrolysis cup and 10 µL of TMAH solution was added to each to enable their derivatisation during pyrolysis. Samples were pyrolyzed at 500 ℃ for 12 seconds before the gas/vapour products were injected into the GC/MS with a 20:1 split ratio. Analyte separation took place on a 30 m HP-5MS column (Agilent) with 0.25 mm inner diameter and 0.25 µm film thickness. The pyrolyzer interface and the GC injection temperature were both set at 250 ℃, the transfer line temperature was 280 ℃ and the ion source temperature was 230 ℃. The carrier gas was helium at a constant flow rate of 1.0 mL/min. The oven temperature program was set at 1 min isothermal heating at 40 ℃, followed by a ramp of 6 ℃/min to 300 ℃ then held at this temperature for 10 min. The mass spectra were obtained in electron impact (EI) mode at 70 eV by scanning m/z 35 – 800 amu at 2.6 scans/s.[147]

6.11.6. Headspace solid phase micro extraction (HS-SPME-GCMS)

HS-SPME-GCMS was used to determine the analytes present in both the $[C_4C_1im][Me_2PO_4]$ IL extract and the bark residue with the pure IL and pure bark used as controls. Bark residue (100-300 mg) or IL extract (0.5 - 1 mL) were added to a HS-SPME vial. Each vial was incubated at different temperature (60, 80, 100, and 120 ℃) and time intervals (30 minute and 60 minute) to develop the analytical method.

The fibre used in the experiment was a mixture of both polar and nonpolar compounds [divinyl benzene/carboxen/polydimethyl siloxane (DVB/CAR/PDMS)] in order to adsorb both polar and non-polar volatile compounds coming from the sample. The length of the fibre was 2 cm. Volatile compounds were separated on a polar DB wax/RTX wax GC-MS column of 30 m × 0.25 mm × 0.25 µm dimensions. The pre-incubation temperature and time was set at 100 ℃ for 30 min for the sample in the vial to be analysed. 20 mL vials were used, and the fibre penetrated 30 mm into the vial. The extraction time of the fibre was set at 30 min in which fibre was exposed to the headspace of the vial to adsorb the volatile compounds from the sample and desorption time was set at 10 min which allowed the fibre to desorb the volatile compounds into the column. The carrier gas was helium at a constant flow rate of 1.0 mL/min. The oven temperature program was set at 10 min isothermal heating at 35 ℃. The mass spectra were obtained in electron impact (EI) mode by scanning m/z 35 – 650 with a start time at 1 min and end time at 83 min. The source temperature, quad temperature and transfer temperature were set at 180, 230 and 250 ℃, respectively.

6.11.7. ^{13}C Solid-State NMR Analysis

Solid samples were ground into a powder using a mortar and pestle. These samples were packed into 4 mm ZrO rotors fitted with KeI-F caps and analysed. The standard CP-MAS experiments had a 3.5 µs proton preparation pulse following by a 1 ms contact time. A total of 10k transients were collected for each sample, and resulting data were Fourier transformed, applying 25 Hz Gaussian line broadening. Non-quaternary suppressed experiments were collected with a 70 µs (total) dephasing delay.

6.11.8. Elemental Analysis

Carbon, hydrogen and nitrogen of bark residues were determined by accurately weighing ground bark (~ 5 mg) in an aluminium cup. The carrier gas was helium at a constant flow rate of 1.0 mL/min and the combustion time was set for 90 s.

References

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Appendix

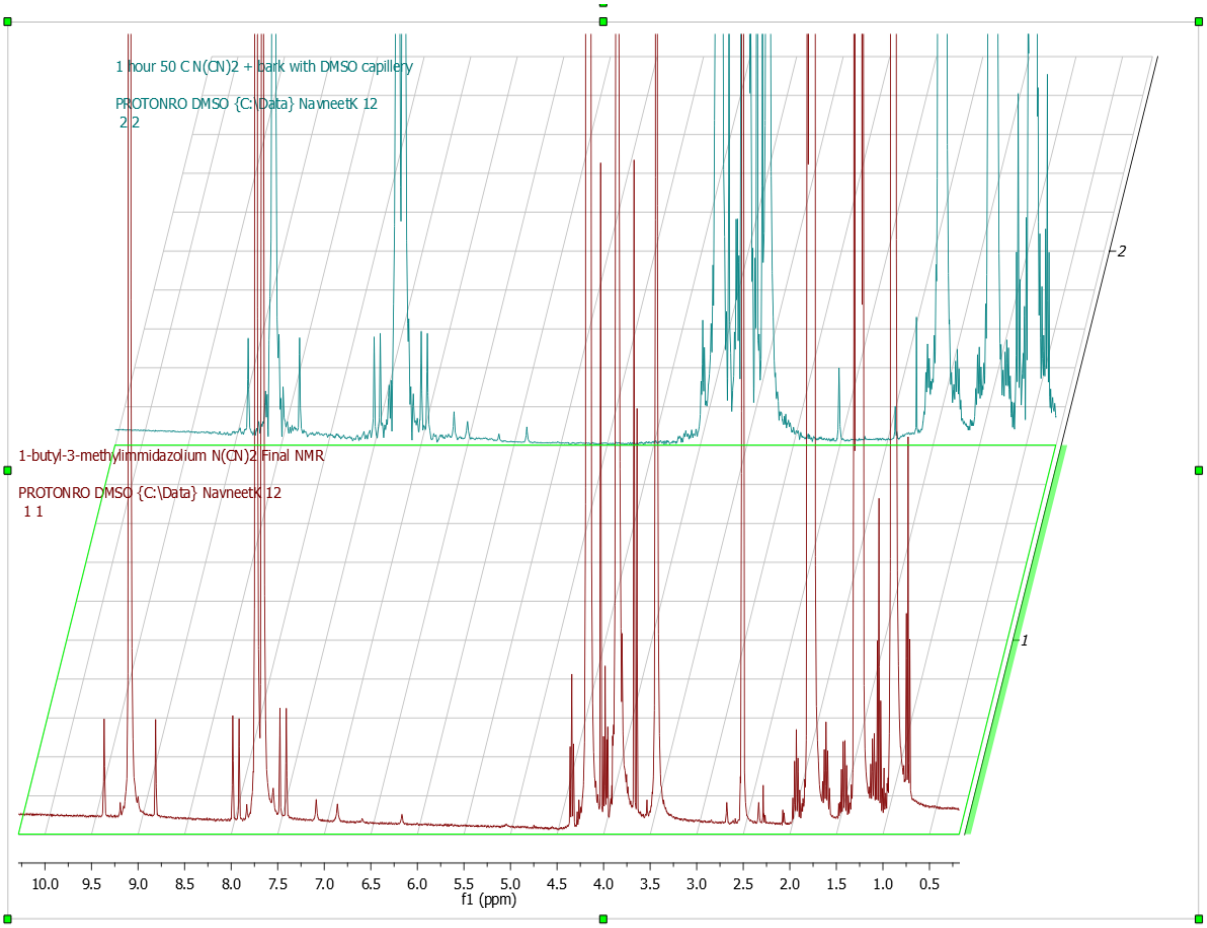


Figure A1 *^{1}H NMR of pure $[C_4C_1im][N(CN)_2]$ shown in (i) in comparison with bark at 50 °C (ii)*

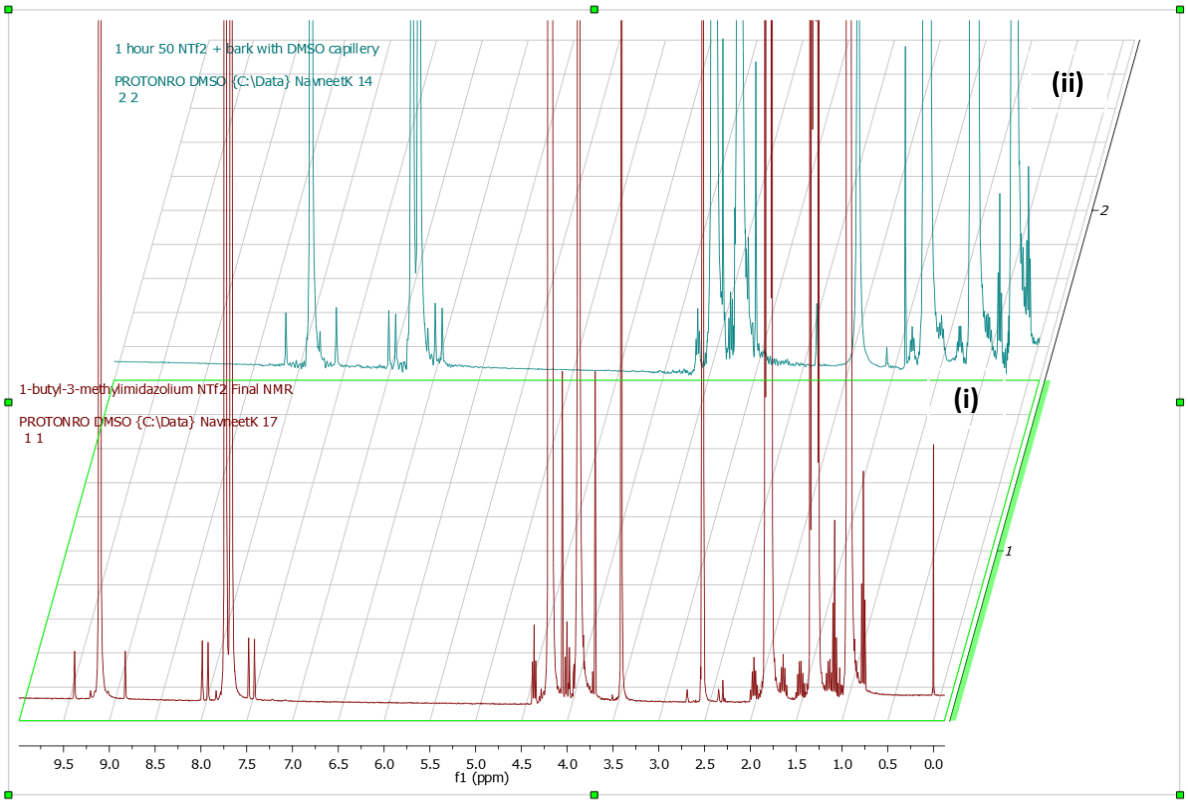


Figure A2 *^{1}H NMR of pure $[C_4C_1im][NTf_2]$ shown (i) in comparison with bark at 50 °C (ii)*

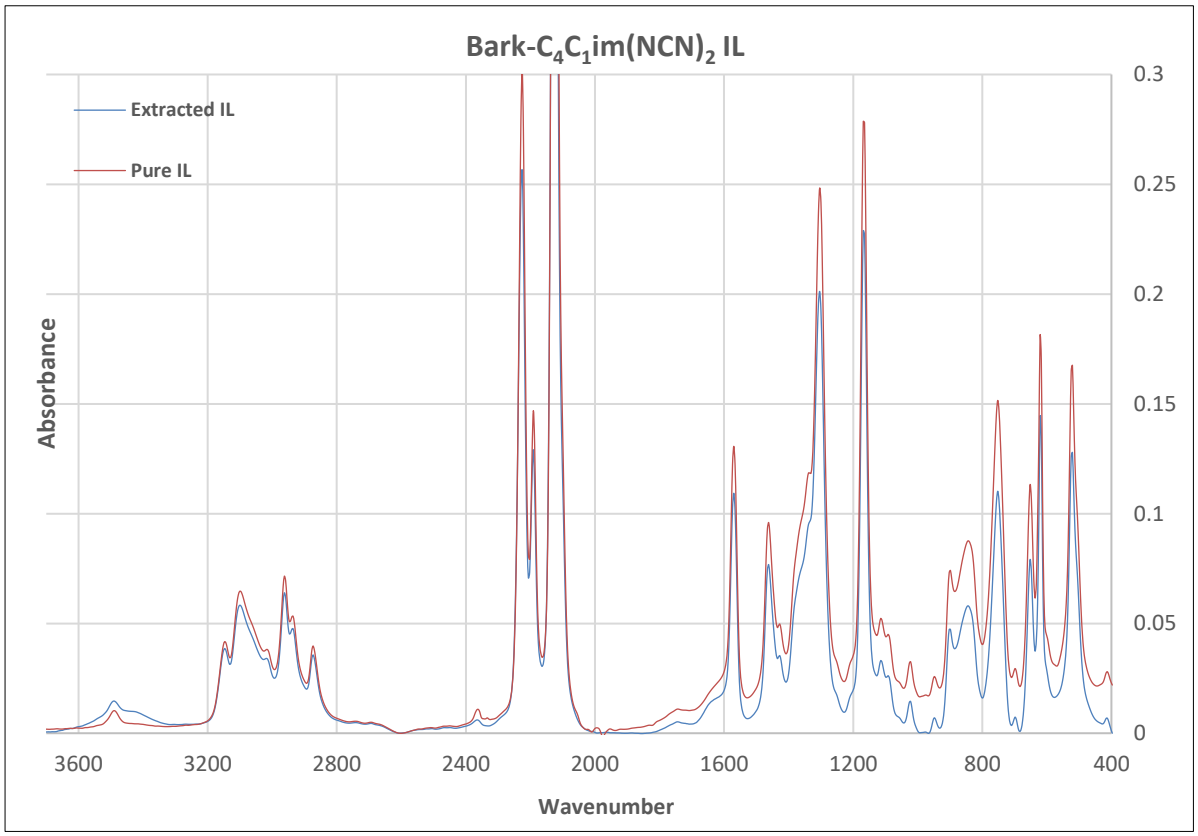


Figure A3. *0.1 g of bark in 4 mL of $[C_4C_1im][N(CN)_2]$ at 25 °C for 1 h.*

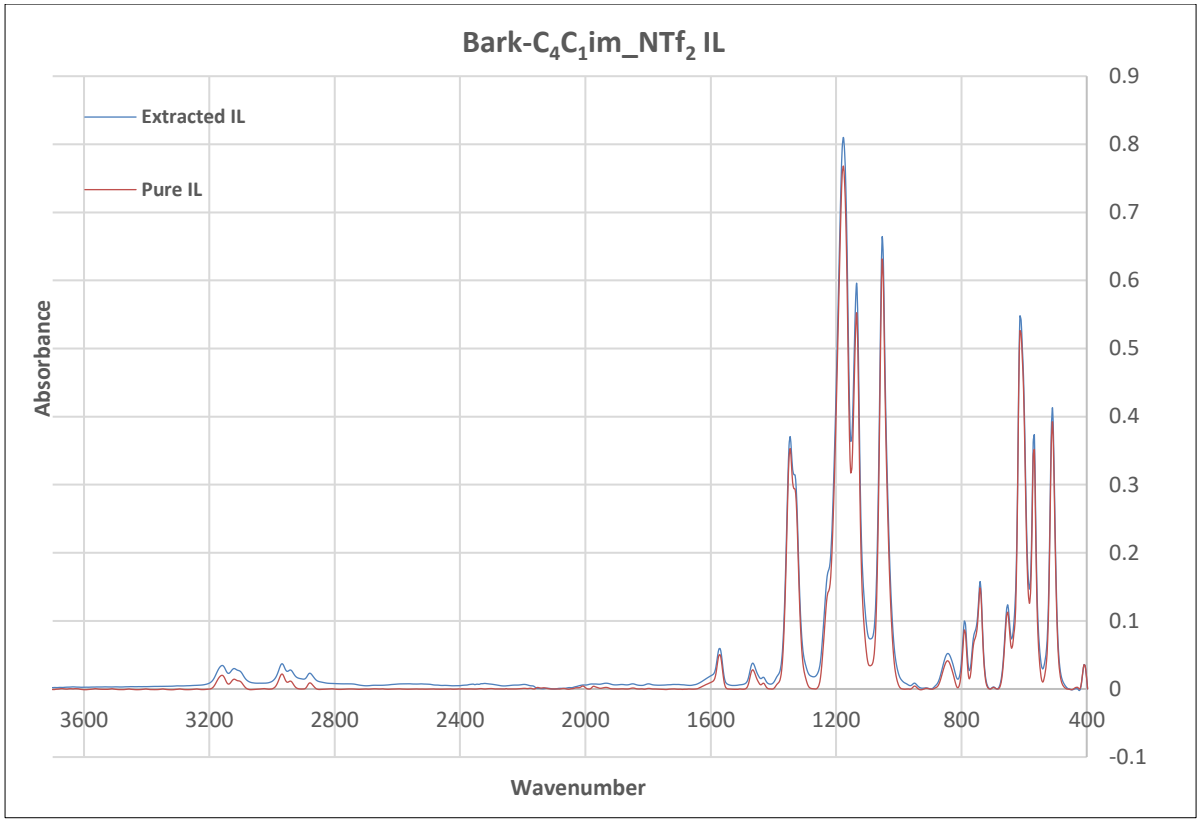


Figure A4. *0.1 g of bark in 4 mL of [C_4C_1im][NTf_2] at 25 ℃ for 1 h.*

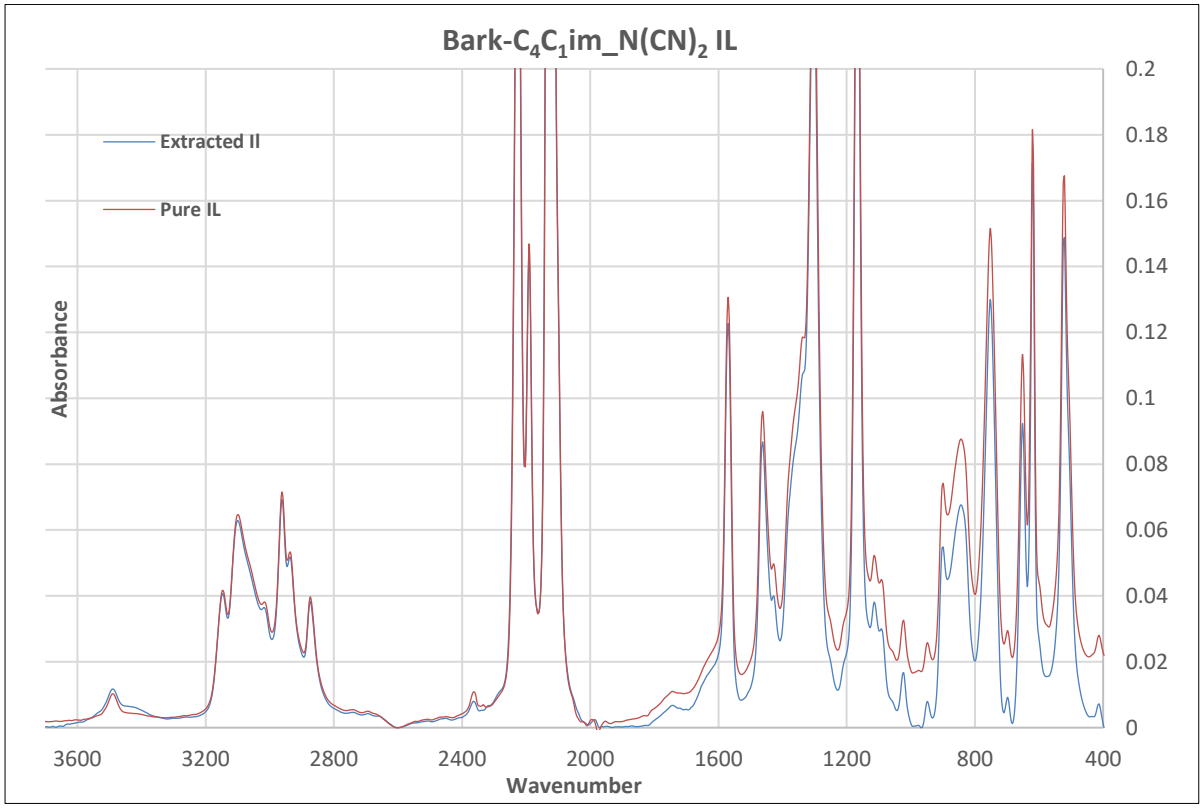


Figure A5. *0.1 g of bark in 4 mL of [C_4C_1im][$N(CN)_2$] at 50 ℃ for 1 h.*

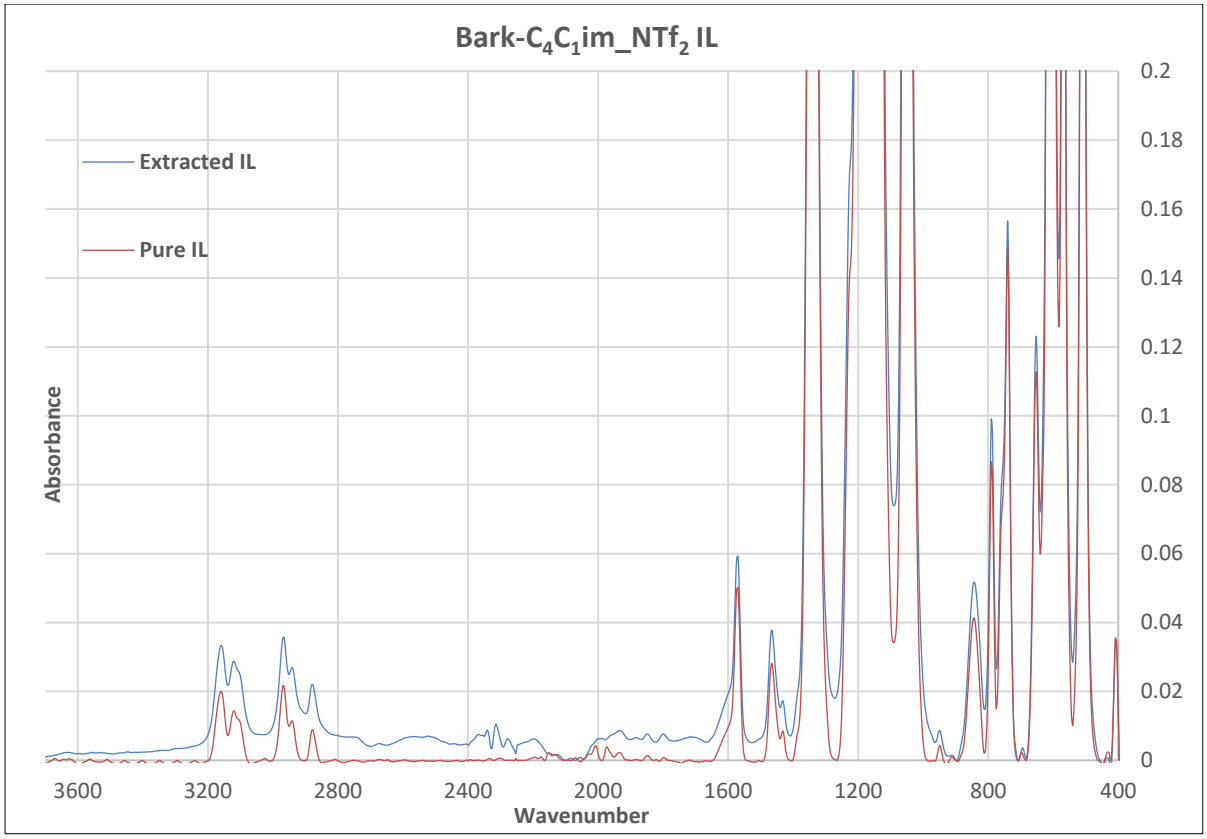


Figure A6. *0.1 g of bark in 4 mL of [C_4C_1im][NTf_2] at 50 °C for 1 h.*

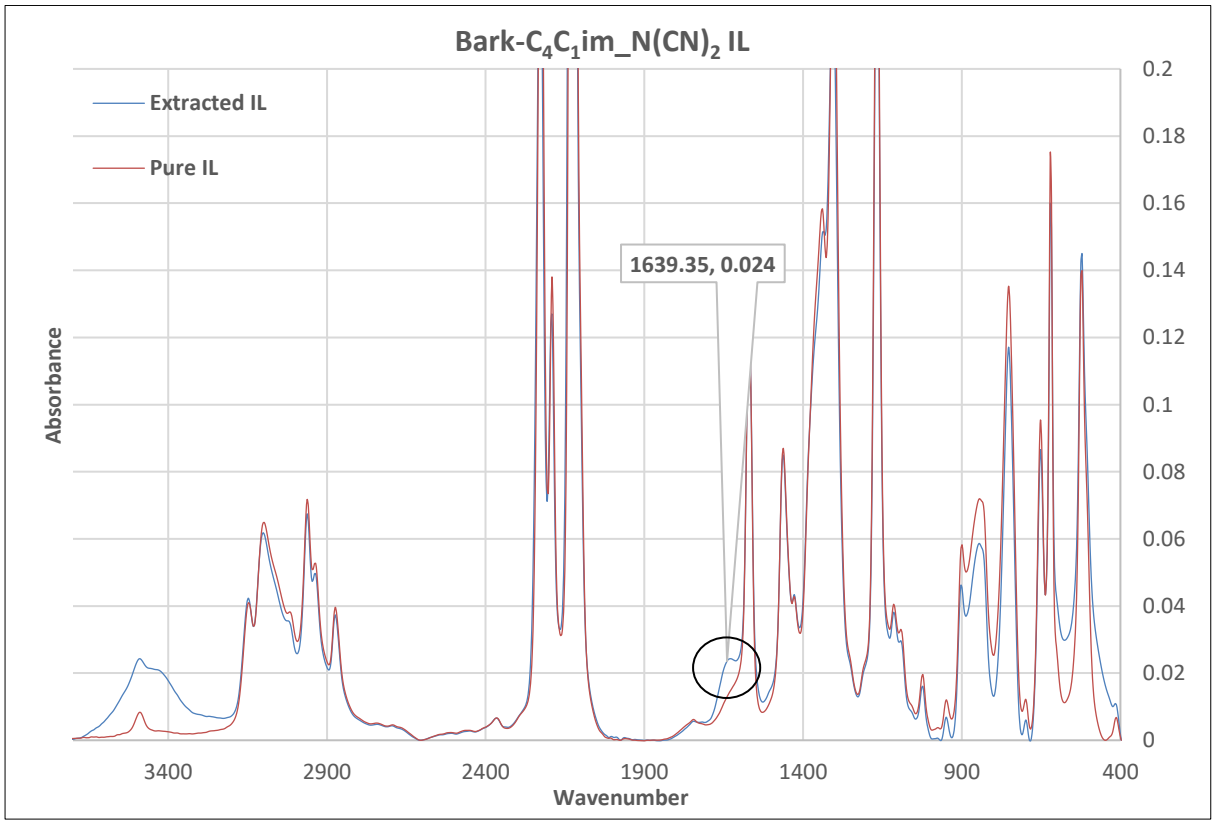


Figure A7. *0.1 g of bark in 4 mL of [C_4C_1im][$N(CN)_2$] at 100 °C for 24 h.*

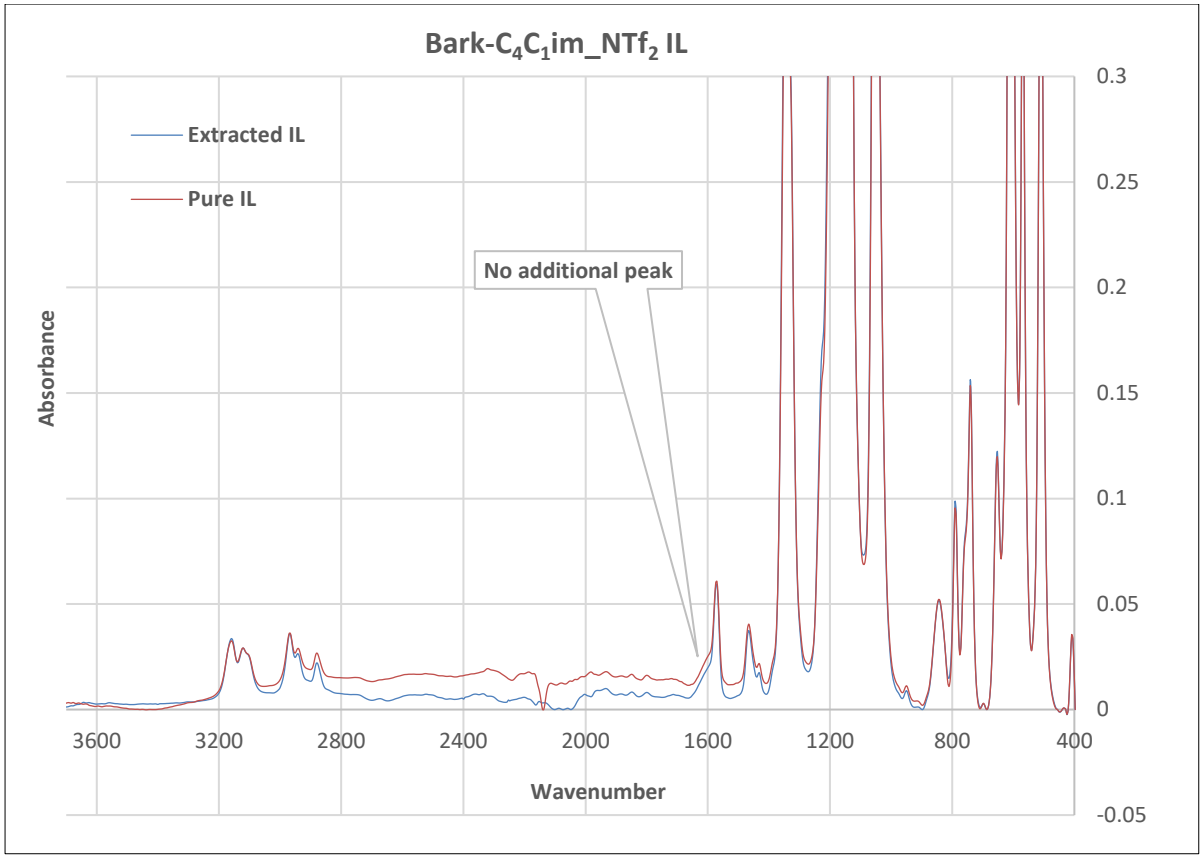


Figure A8. *0.1 g of bark in 4 mL of [C_4C_1im][NTf_2] at 100 °C for 24 h.*

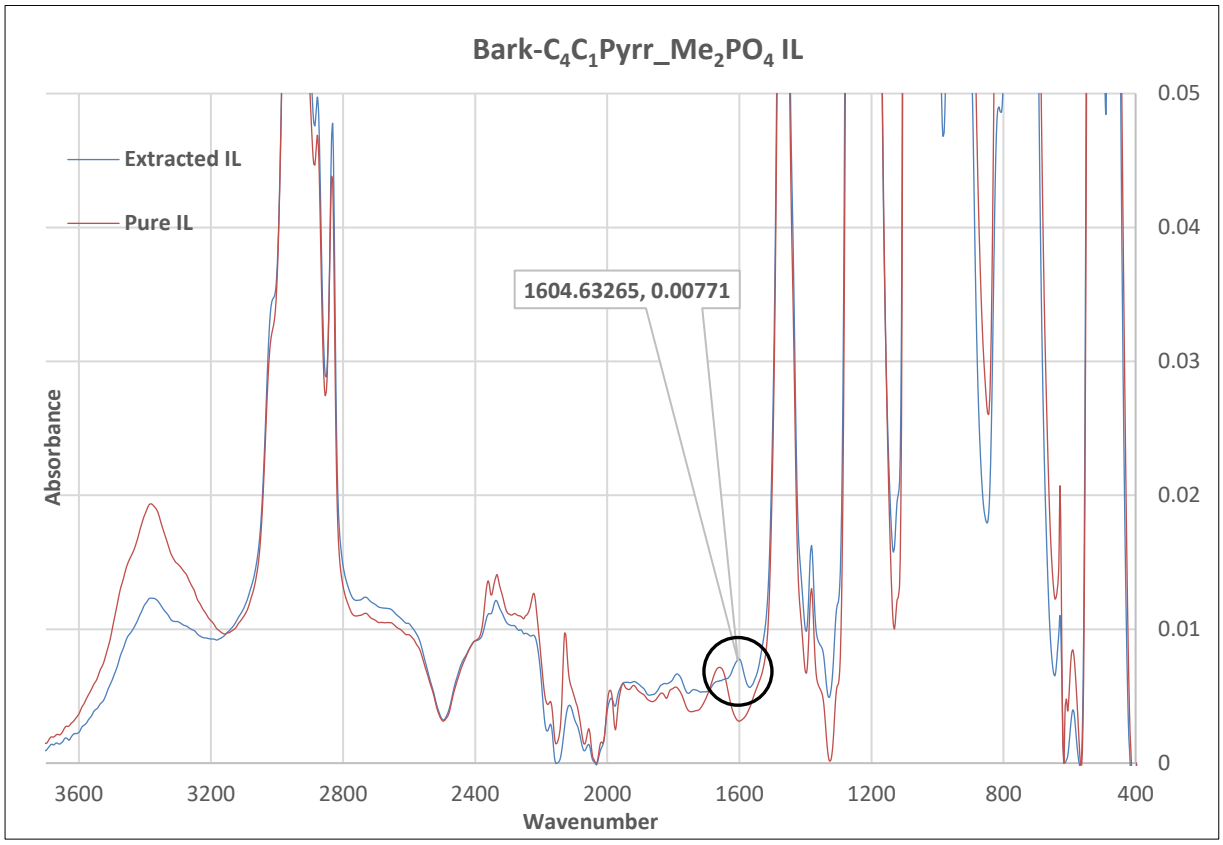


Figure A9. *0.05 g of bark in 2 mL of [C_4C_1pyrr][Me_2PO_4] at 100 °C for 24 h.*

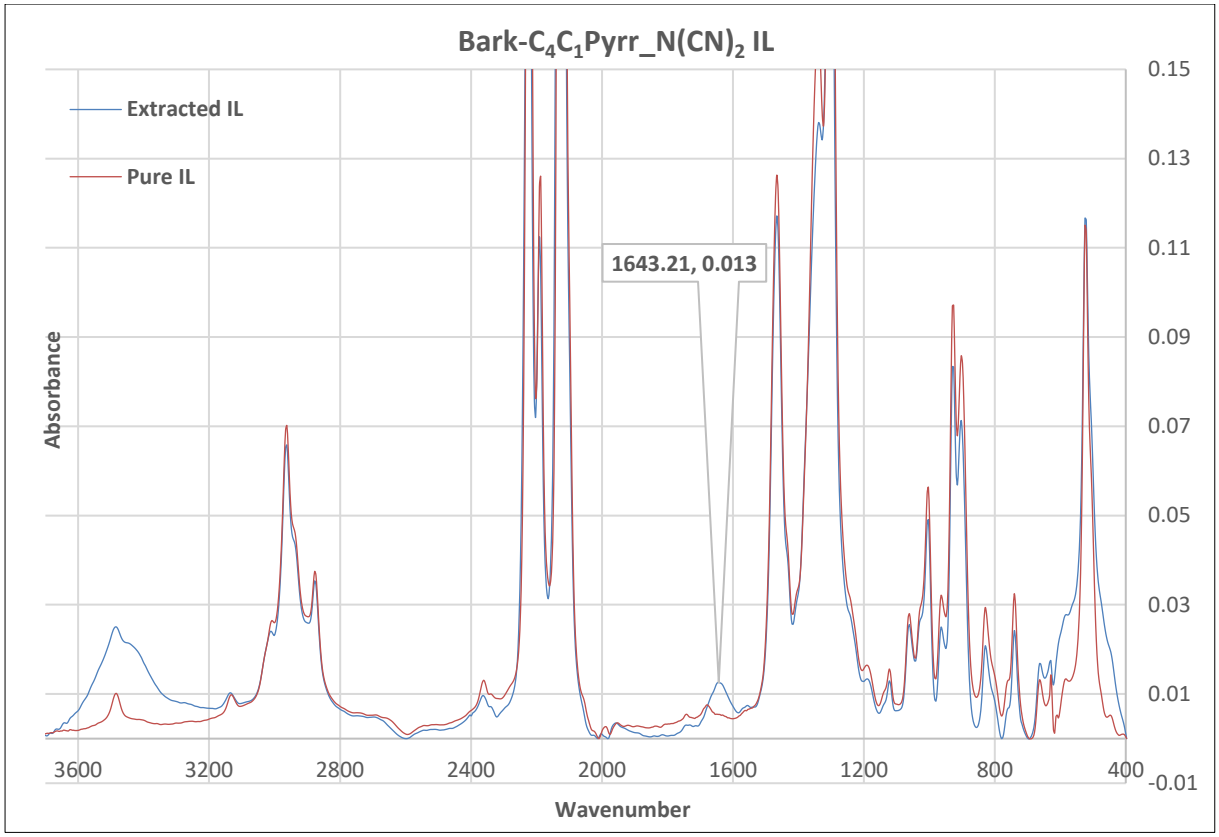


Figure A10. *0.05 g of bark in 2mL of [C_4C_1pyrr][$N(CN)_2$] at 100 ℃ for 24 h.*

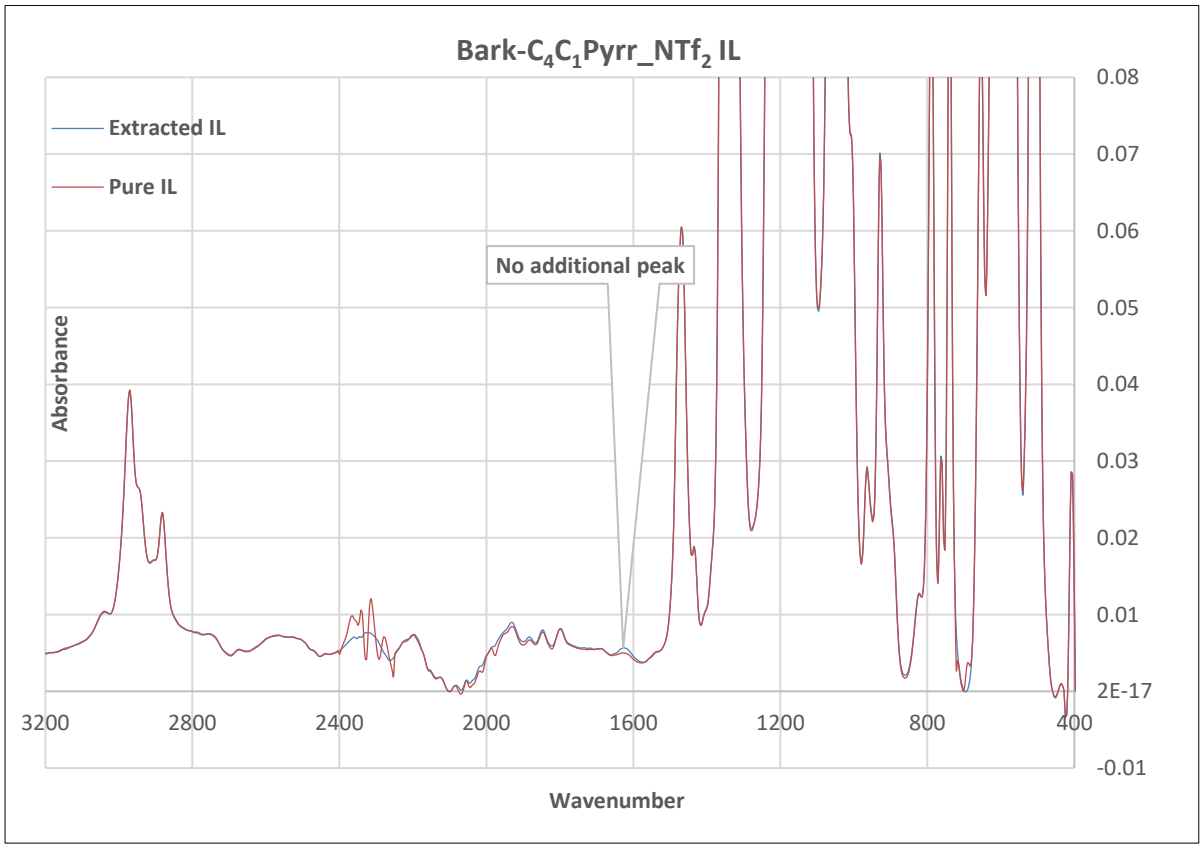


Figure A11. *0.05 g of bark in 2 mL of [C_4C_1pyrr][NTf_2] at 100 ℃ for 24 h.*

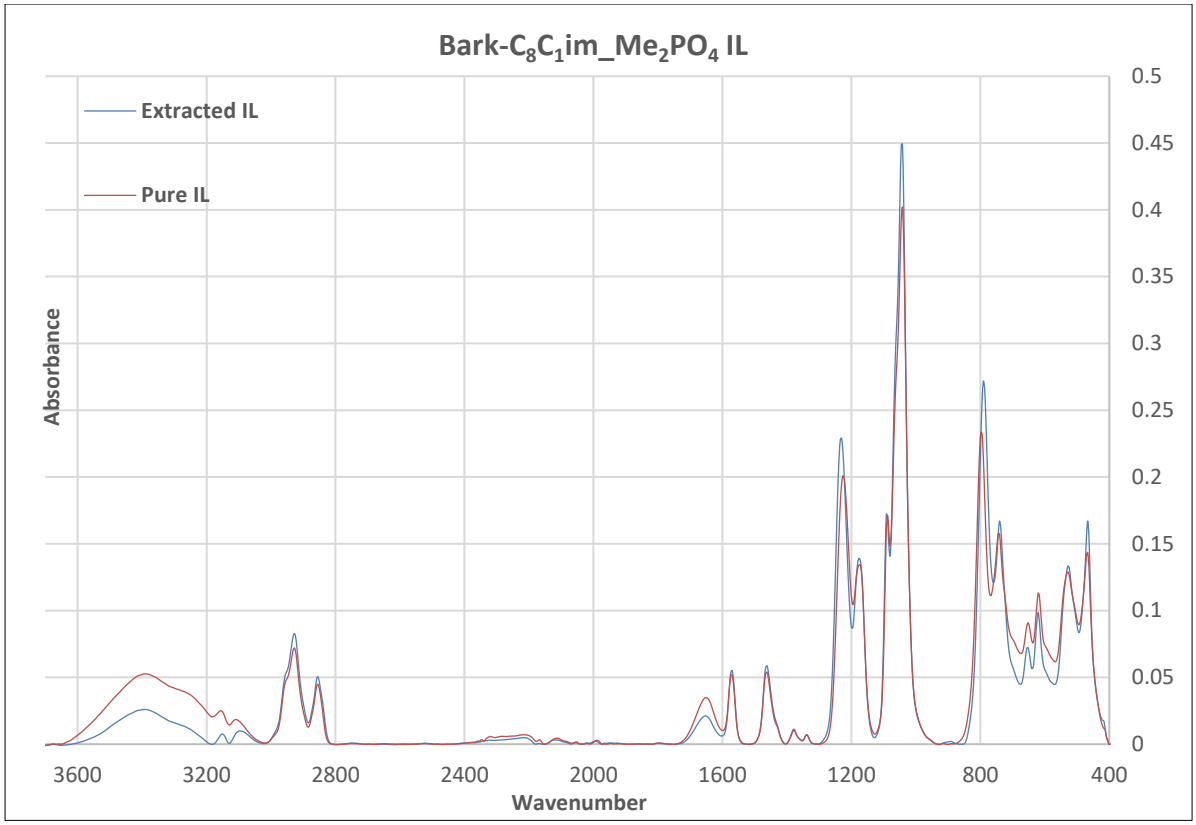


Figure A12. *0.1 g of bark in 4 mL of [C_8C_1im][Me_2PO_4] at 100 °C for 24 h.*

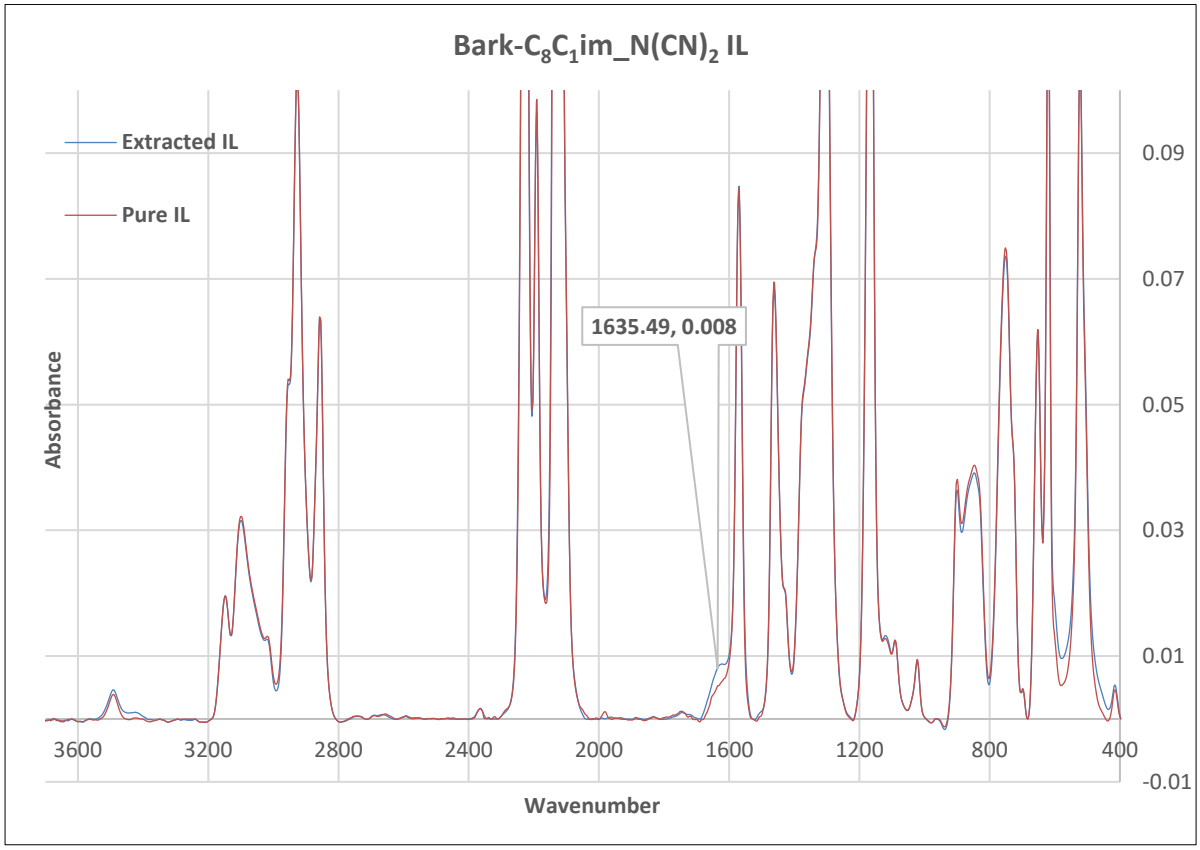


Figure A13. *0.1 g of bark in 4 mL of [C_8C_1im][$N(CN)_2$] at 100 °C for 24 h.*

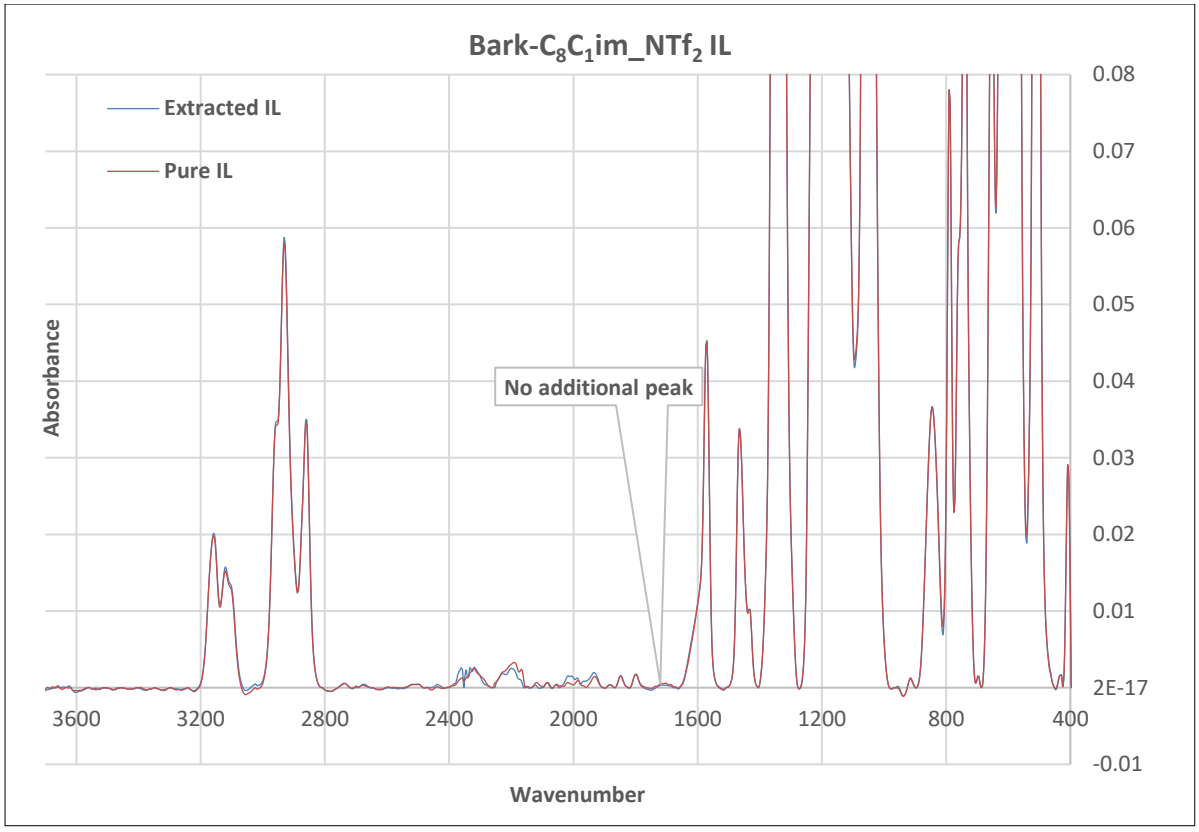


Figure A14. *0.1 g of bark in 4 mL of [C_8C_1im][NTf_2] at 100 °C for 24 h.*

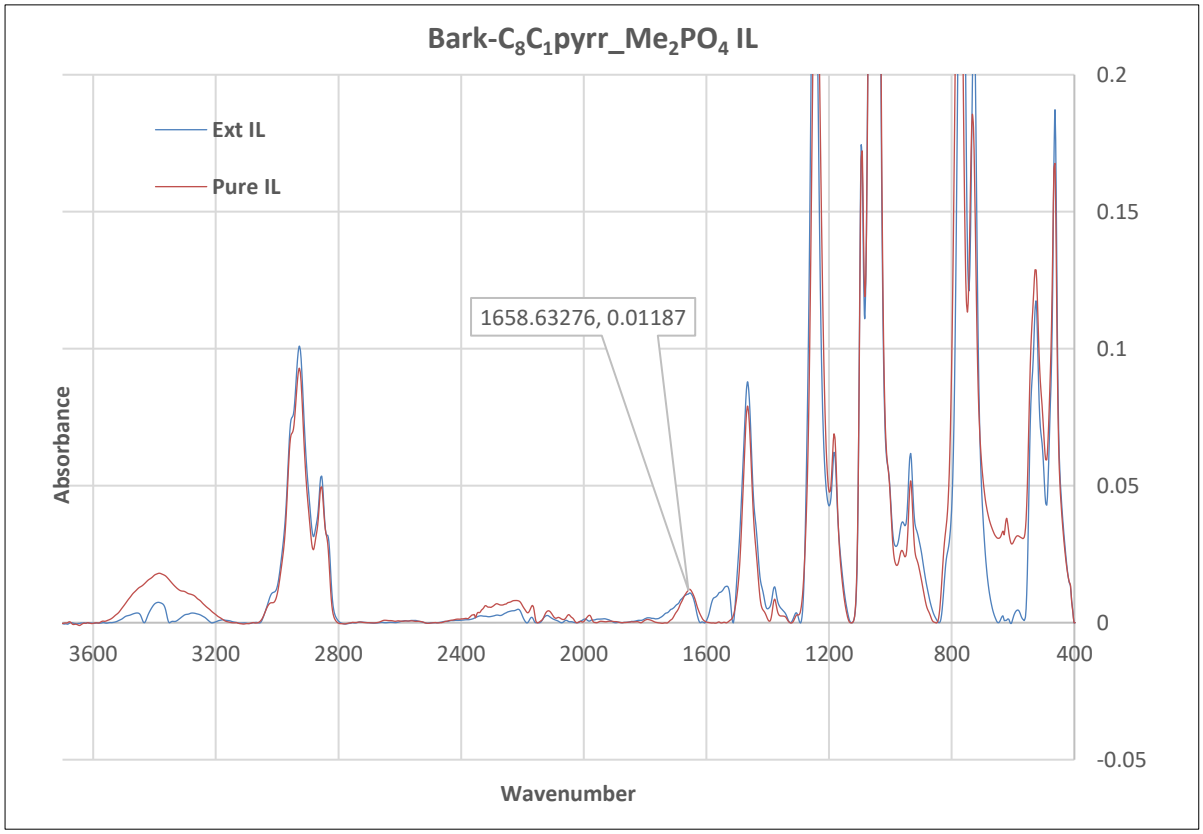


Figure A15. *0.1 g of bark in 4 mL of [C_8C_1pyrr][Me_2PO_4] at 100 °C for 24 h.*

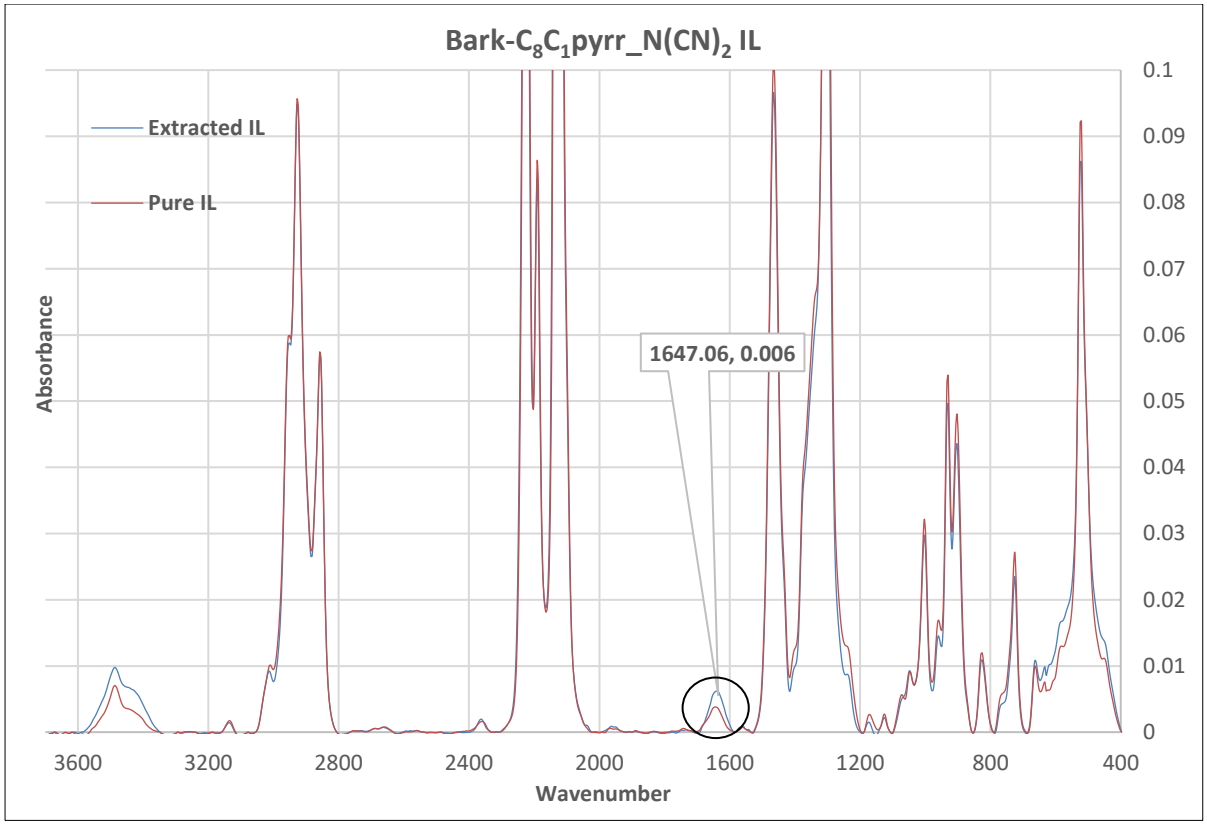


Figure A16. *0.1 g of bark in 4 mL of [C_8C_1pyrr][$N(CN)_2$] at 100 ℃ for 24 h.*

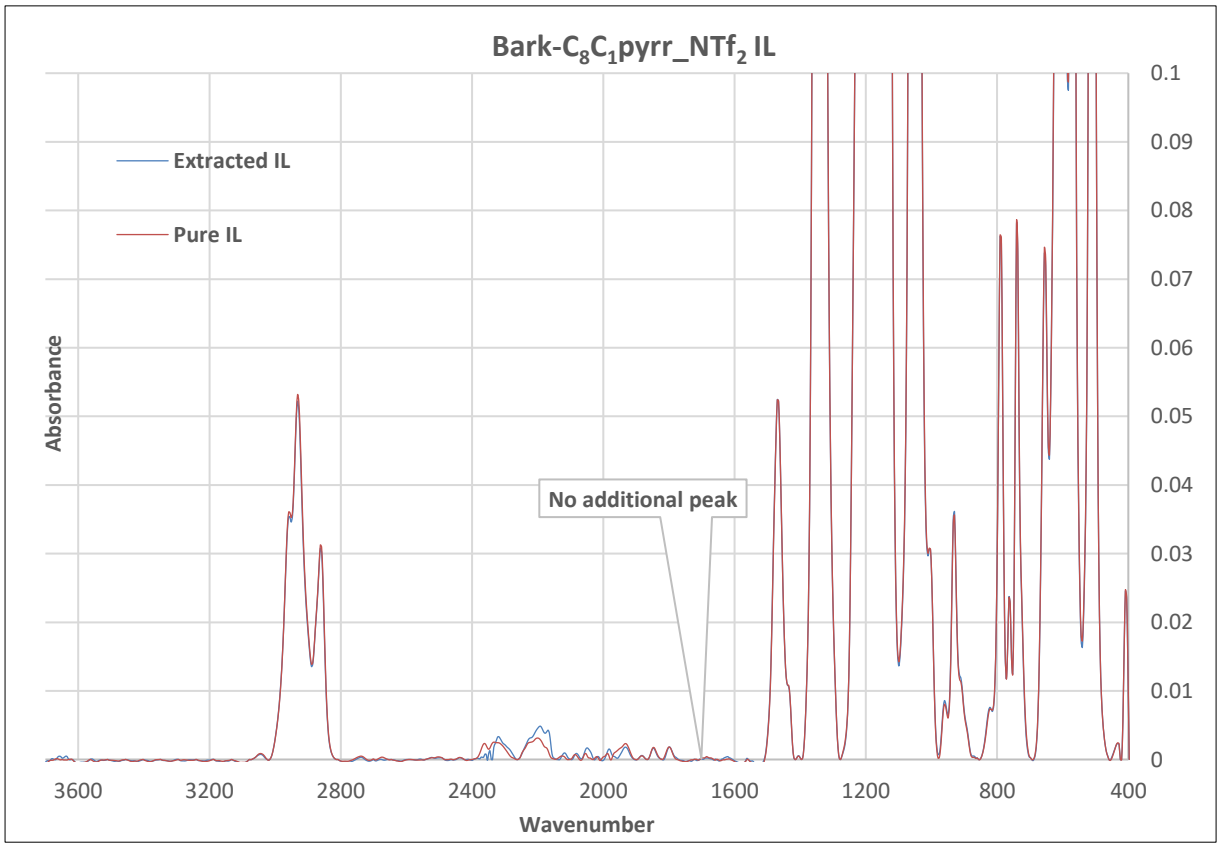


Figure A17. *0.1 g of bark in 4 mL of [C_8C_1pyrr][NTf_2] at 100 ℃ for 24 h.*

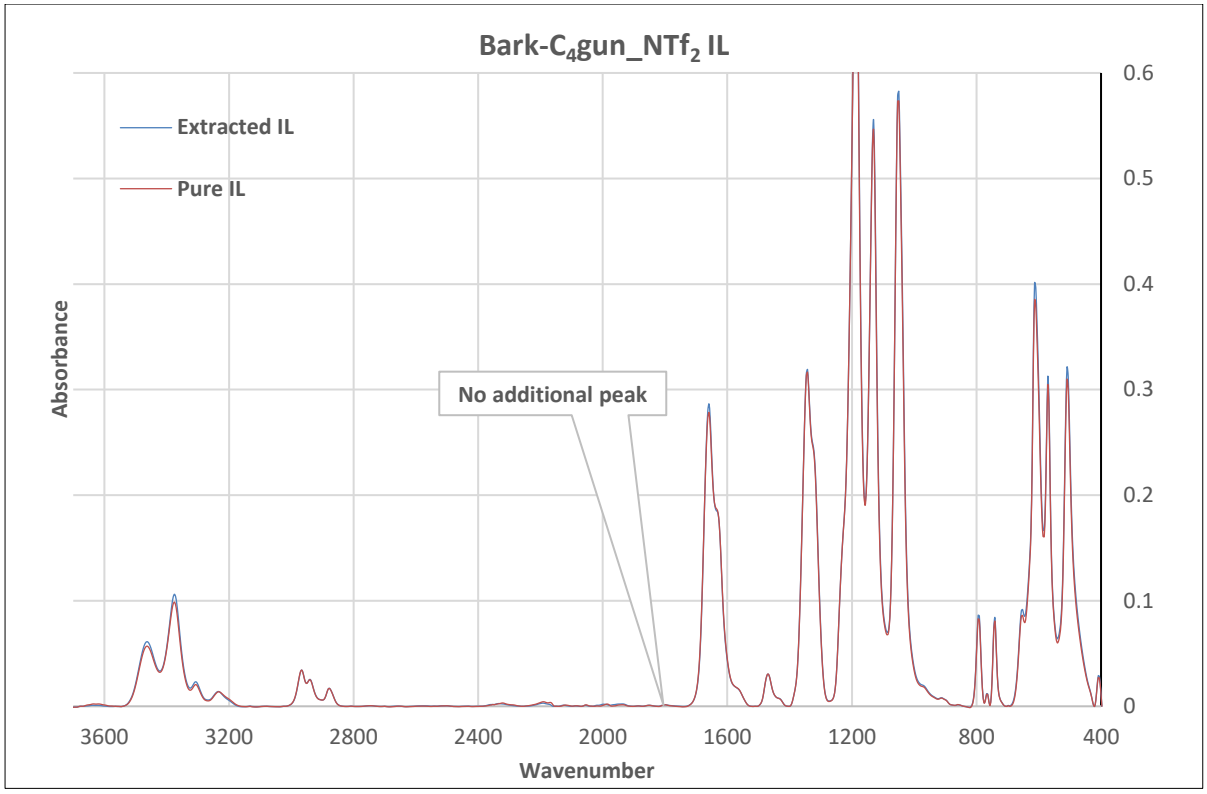


Figure A18. *0.05 g of bark in 2 mL of [C4gun][NTf2] at 100 ℃ for 24 h.*

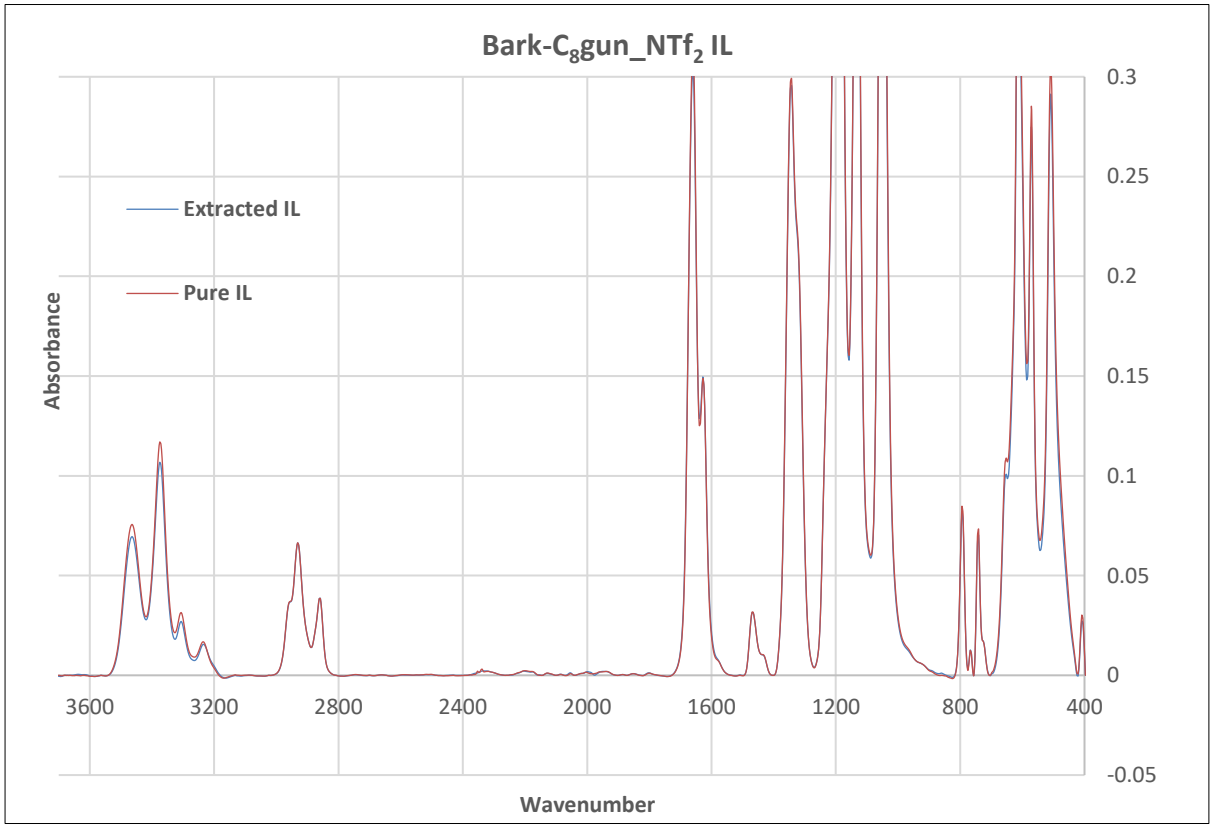


Figure A19. *0.05 g of bark in 2 mL of [C8gun][NTf2] at 100 ℃ for 24 h.*

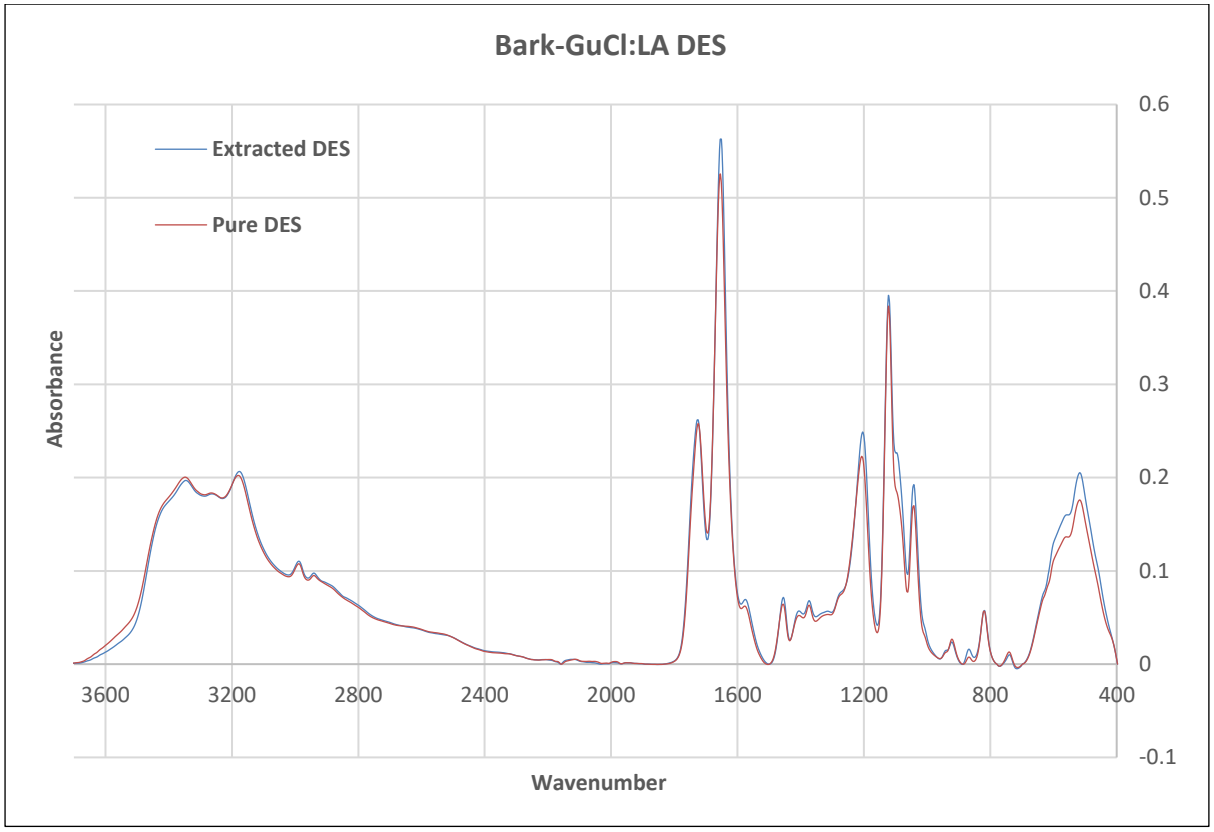


Figure A20. *0.1 g of bark in 4 mL of GuCl:LA at 100 ℃ for 24 h.*

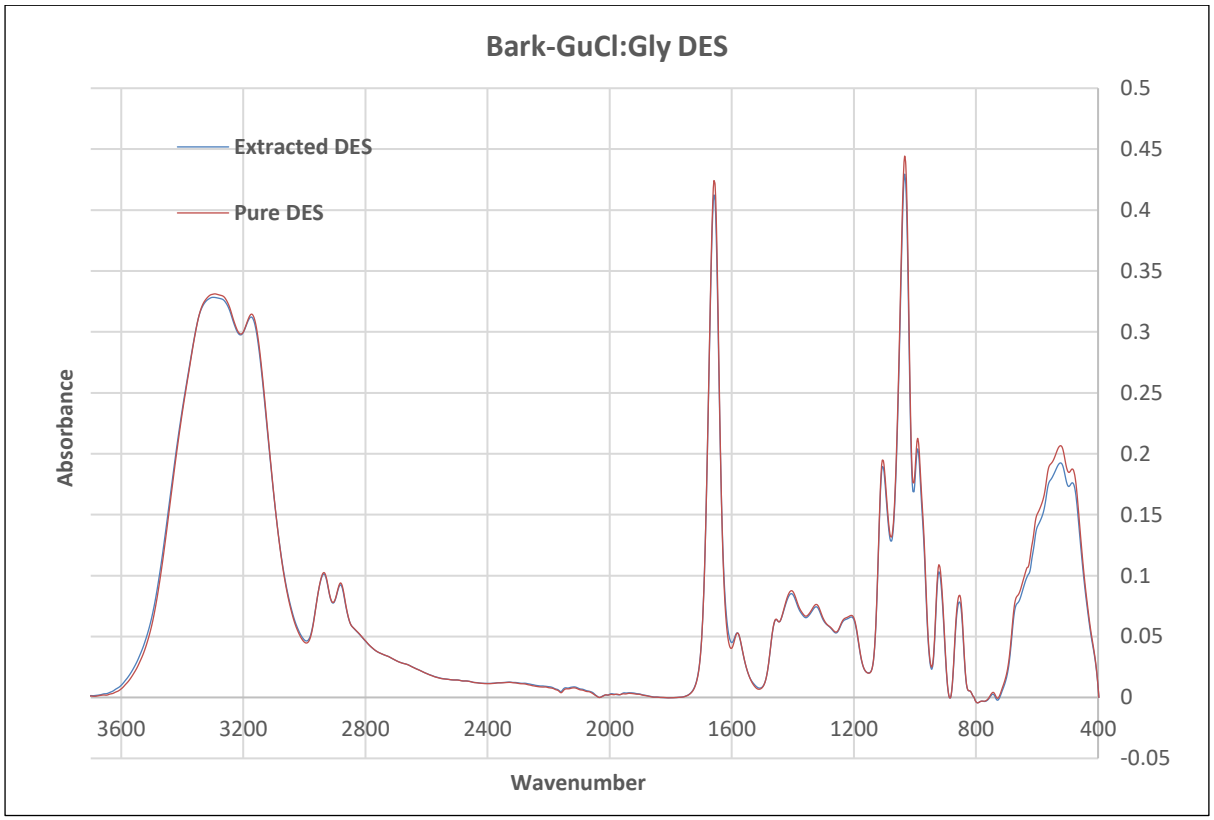


Figure A21. *0.1 g of bark in 4 mL of GuCl:Gly at 100 ℃ for 24 h.*

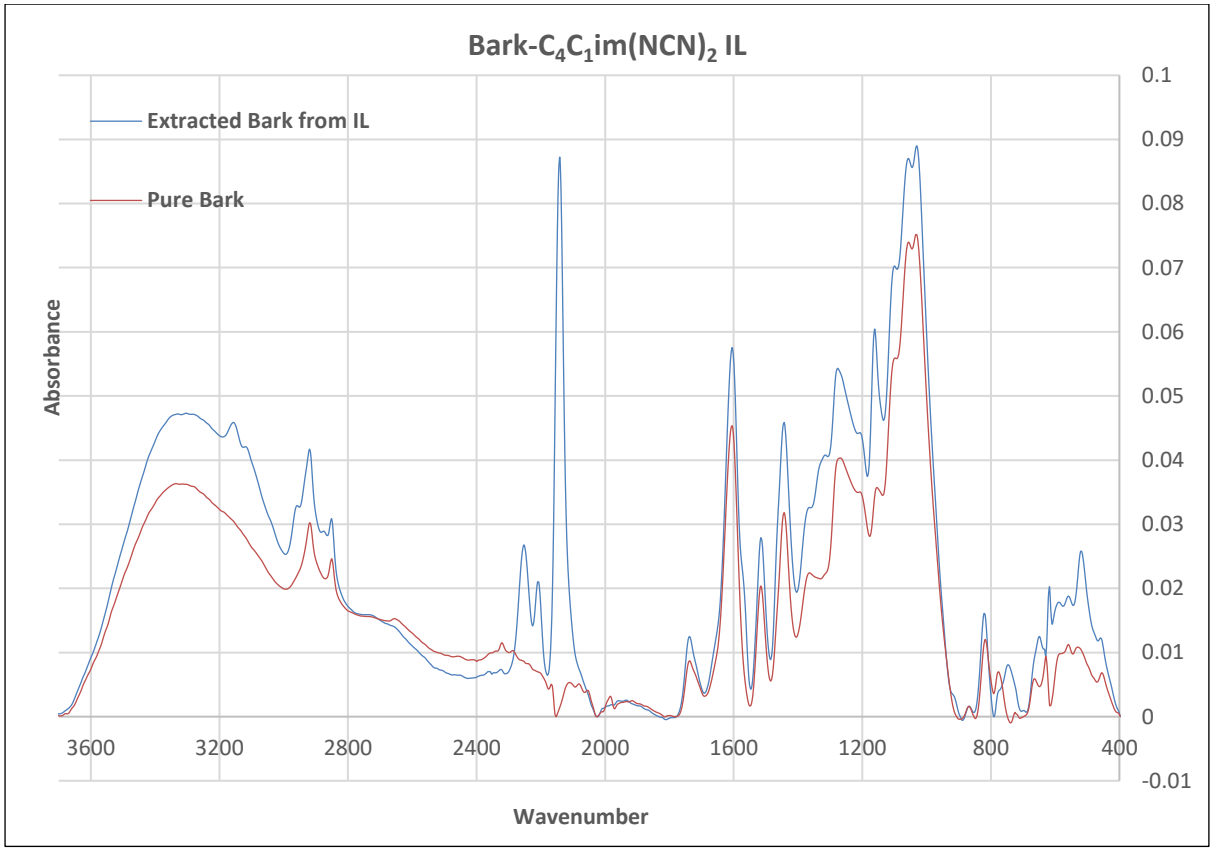


Figure A22. *0.1 g of bark in 4 mL of [C_4C_1im][$N(CN)_2$] at 25 ℃ for 1 h.*

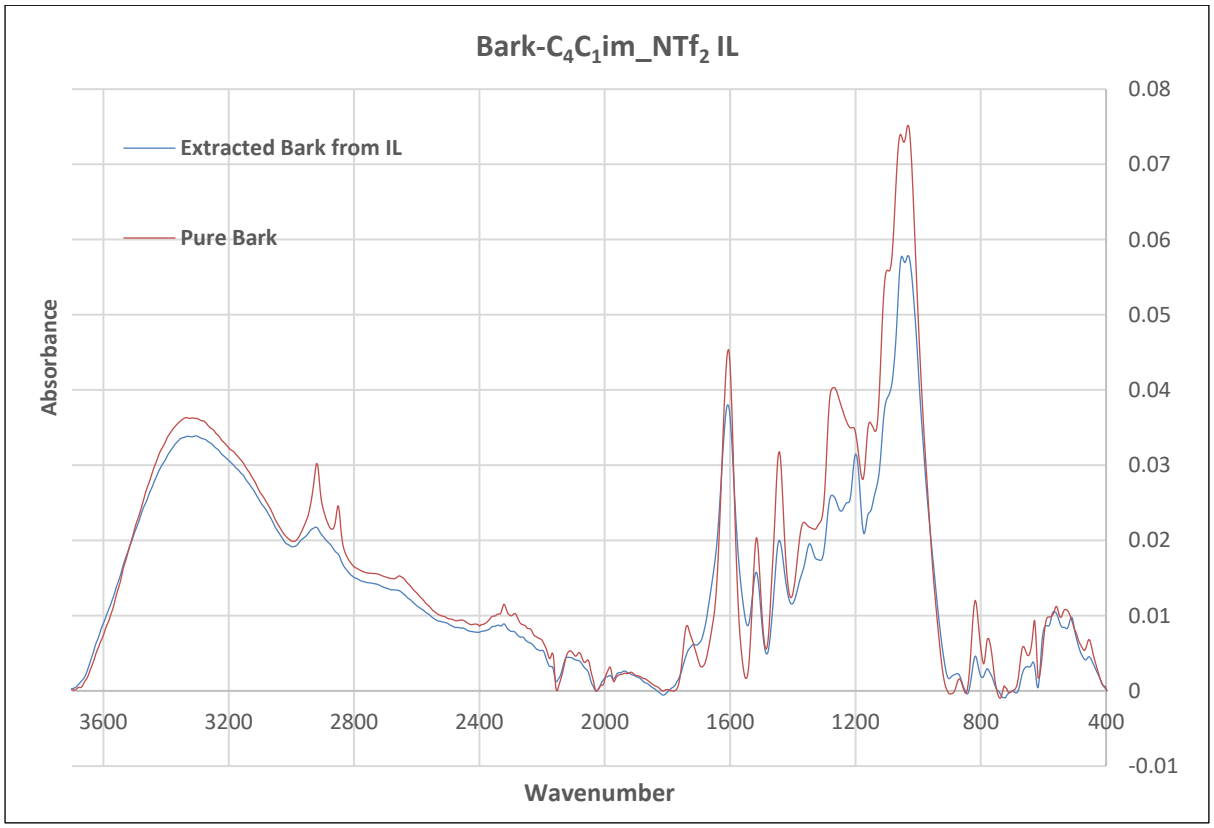


Figure A23. *0.1 g of bark in 4 mL of [C_4C_1im][NTf_2] at 25 ℃ for 1 h.*

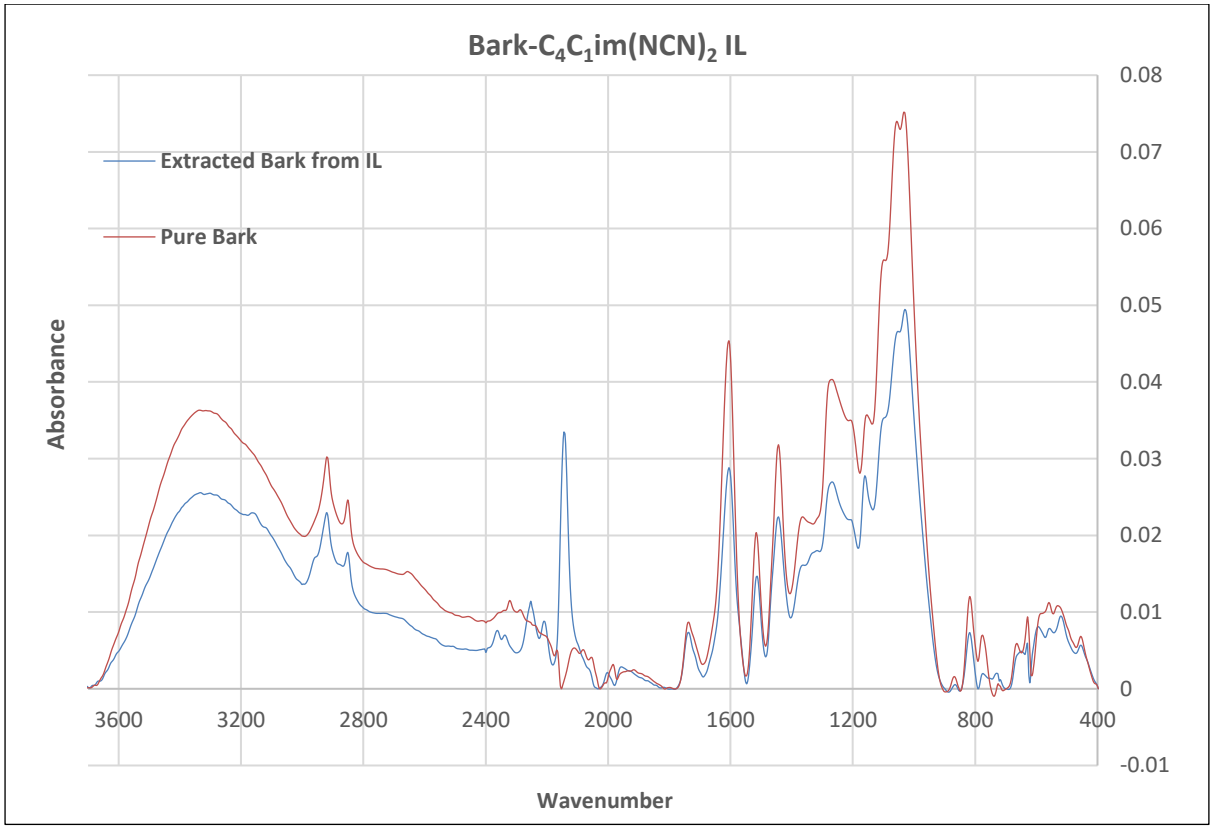


Figure A24. *0.1 g of bark in 4 mL of [C_4C_1im][$N(CN)_2$] at 50 °C for 1 h.*

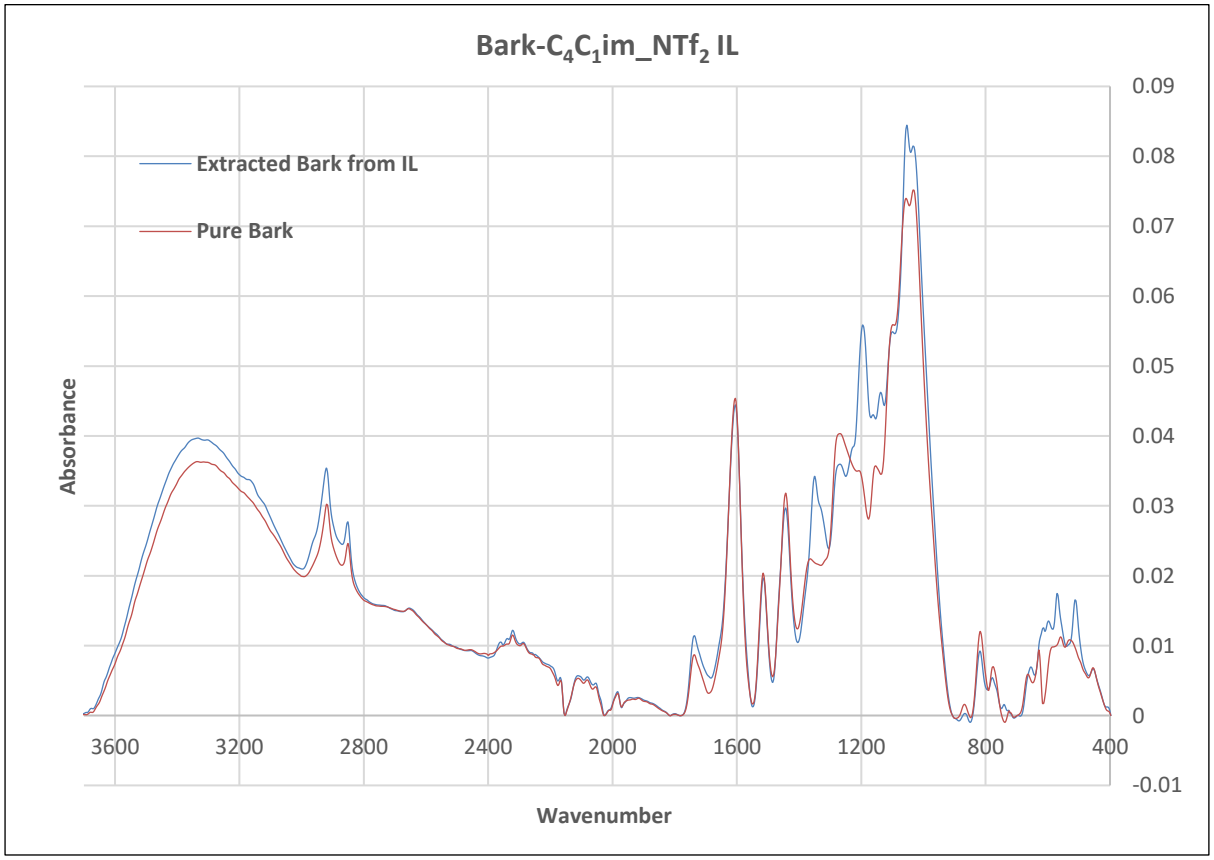


Figure A25. *0.1 g of bark in 4 mL of [C_4C_1im][NTf_2] at 50 °C for 1 h.*

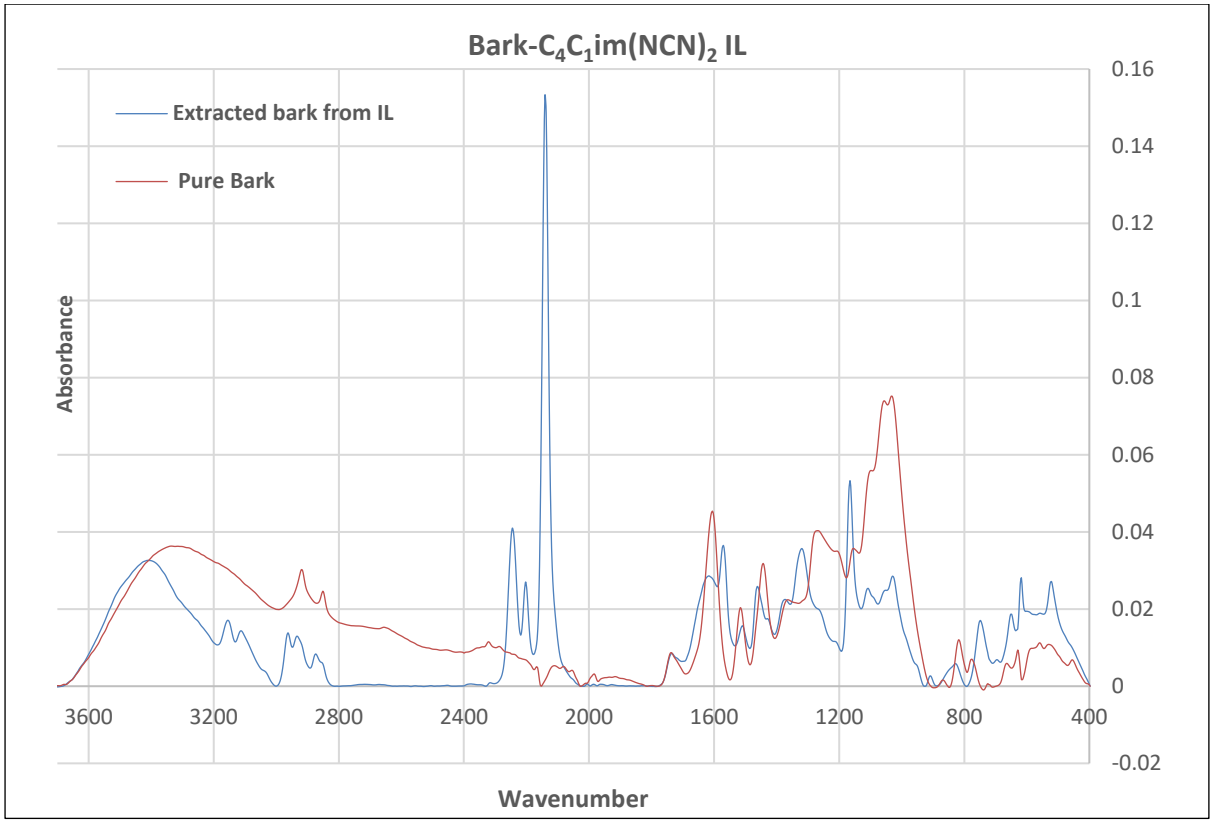


Figure A26. *0.1 g of bark in 4 mL of [C4C1im][N(CN)2] at 100 ℃ for 24 h.*

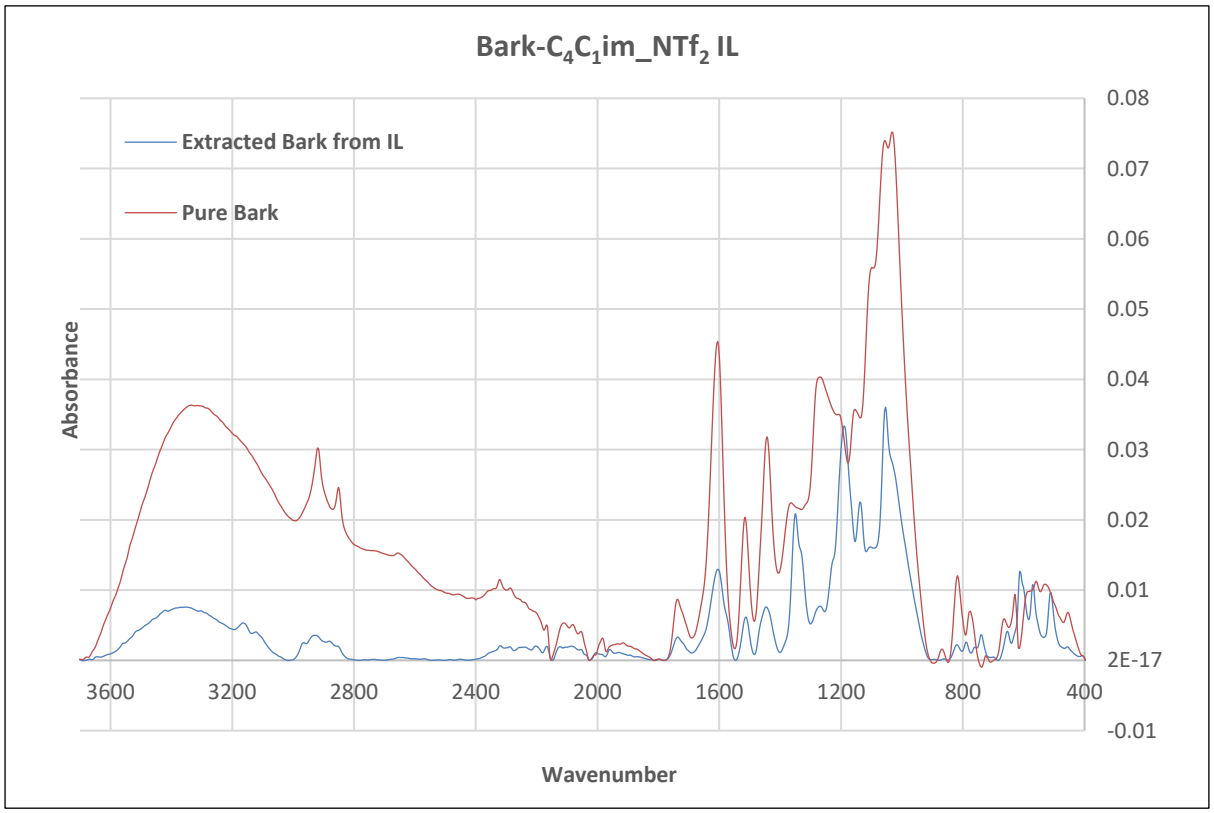


Figure A27. *0.1 g of bark in 4 mL of [C4C1im][NTf2] at 100 ℃ for 24 h.*

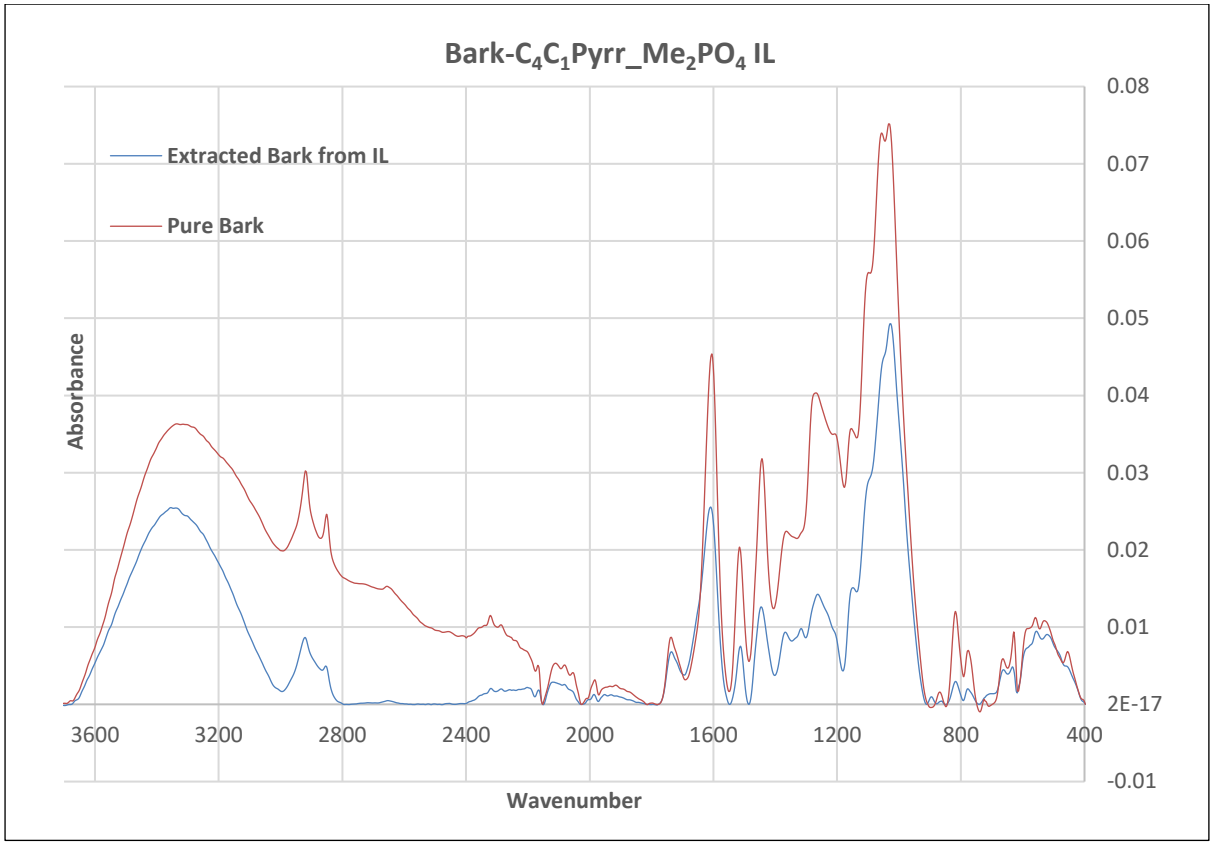


Figure A28. *0.1 g of bark in 4 mL of [C_4C_1pyrr][Me_2PO_4] at 100 ℃ for 24 h.*

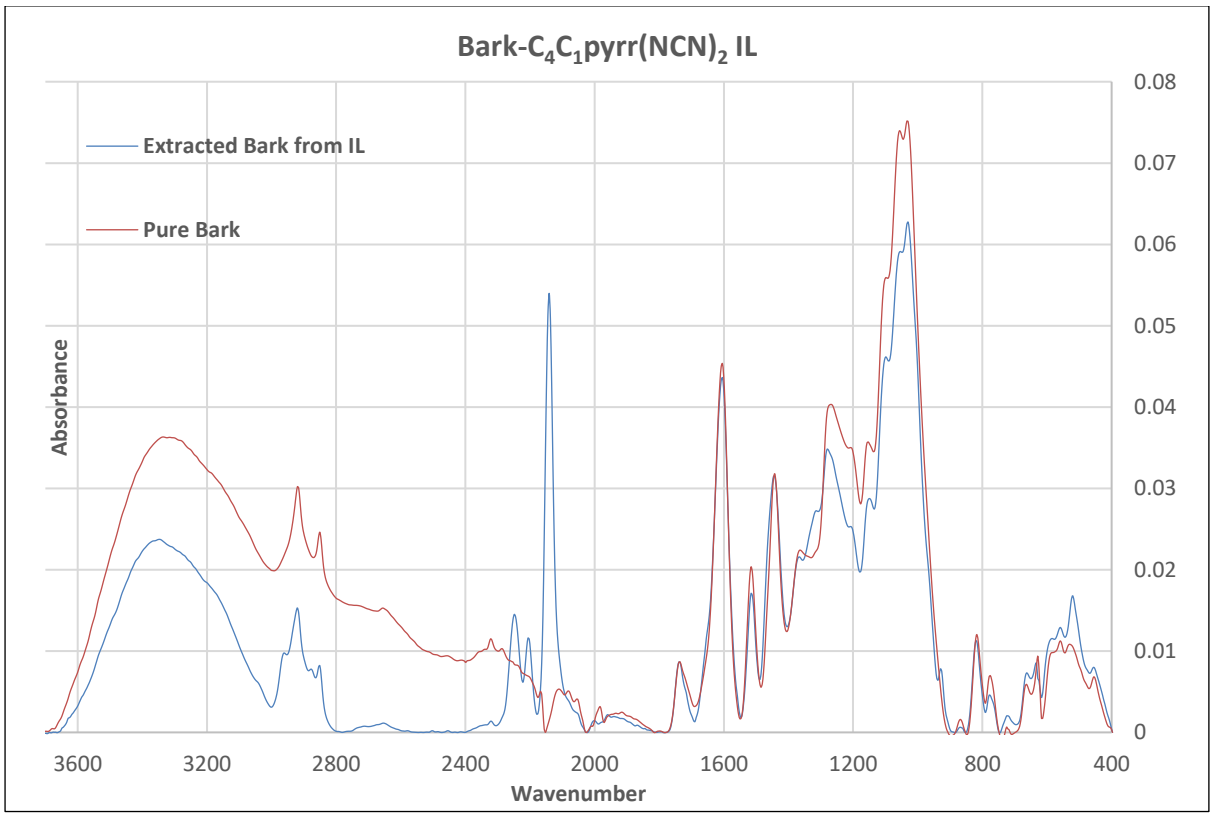


Figure A29. *0.1 g of bark in 4 mL of [C_4C_1pyrr][$N(CN)_2$] at 100 ℃ for 24 h.*

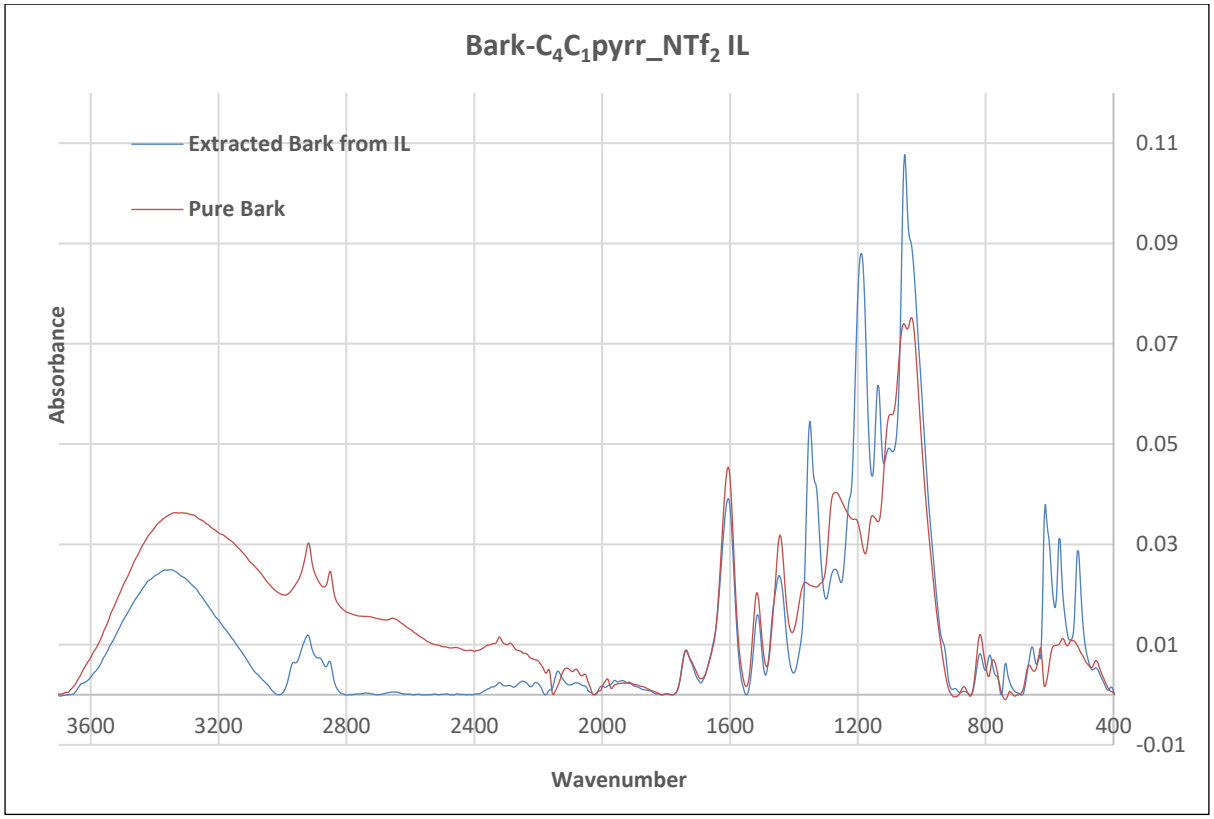


Figure A30. *0.1 g of bark in 4 mL of [C_4C_1pyrr][NTf_2] at 100 °C for 24 h.*

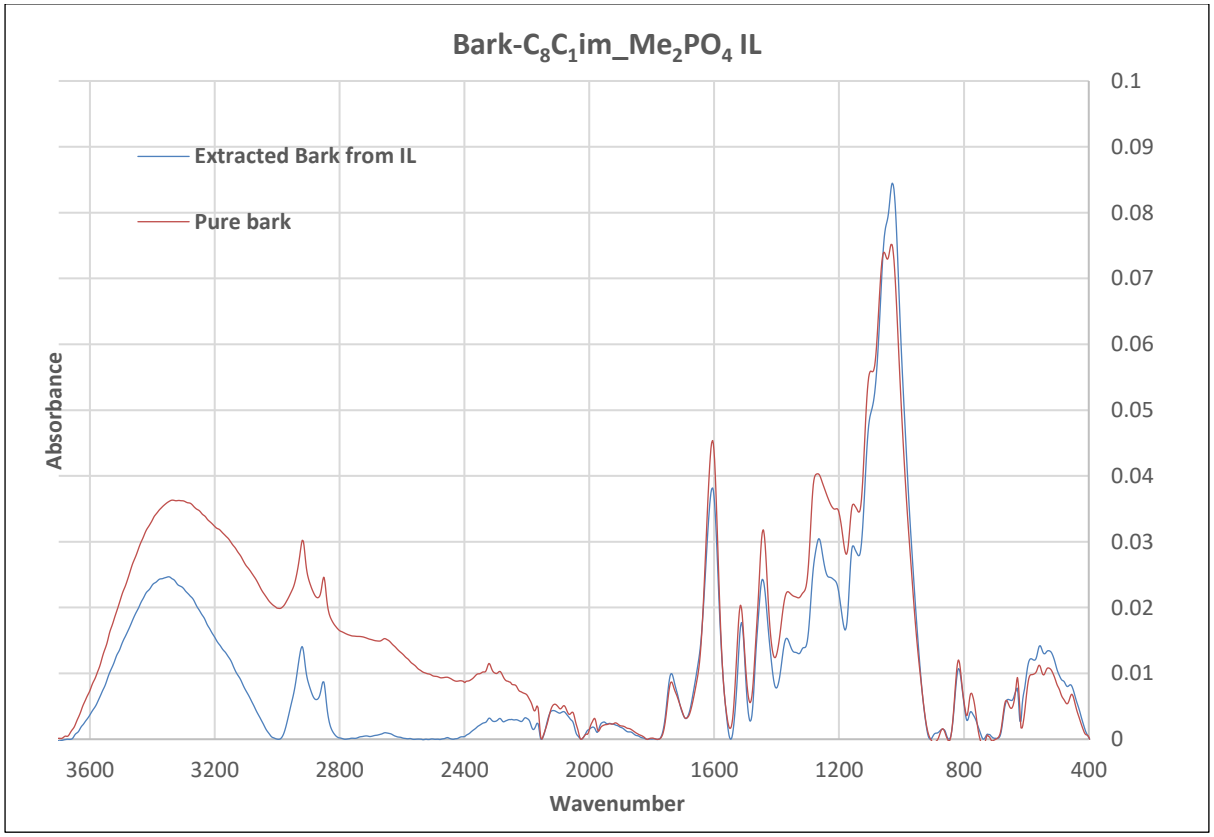


Figure A31. *0.1 g of bark in 4 mL of [C_8C_1im][Me_2PO_4] at 100 °C for 24 h.*

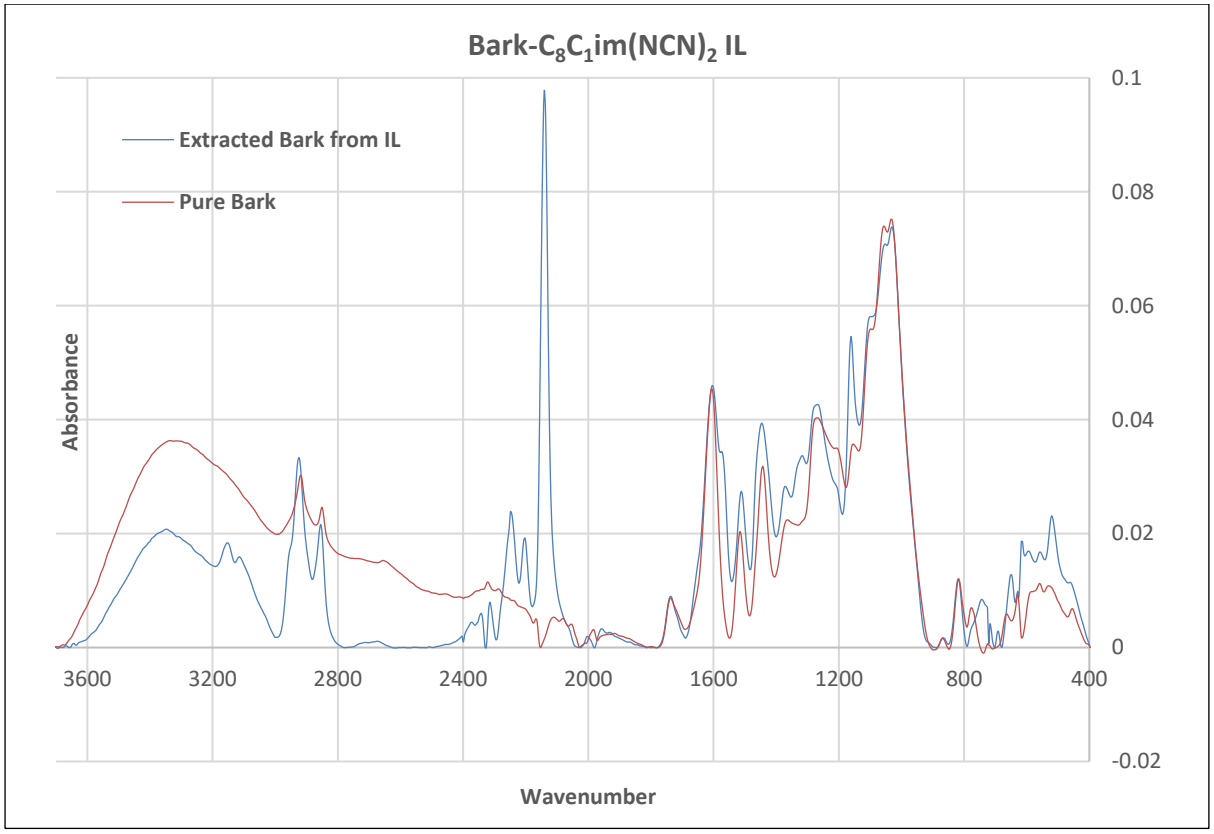


Figure A32. *0.1 g of bark in 4 mL of [C_8C_1im][$N(CN)_2$] at 100 ℃ for 24 h.*

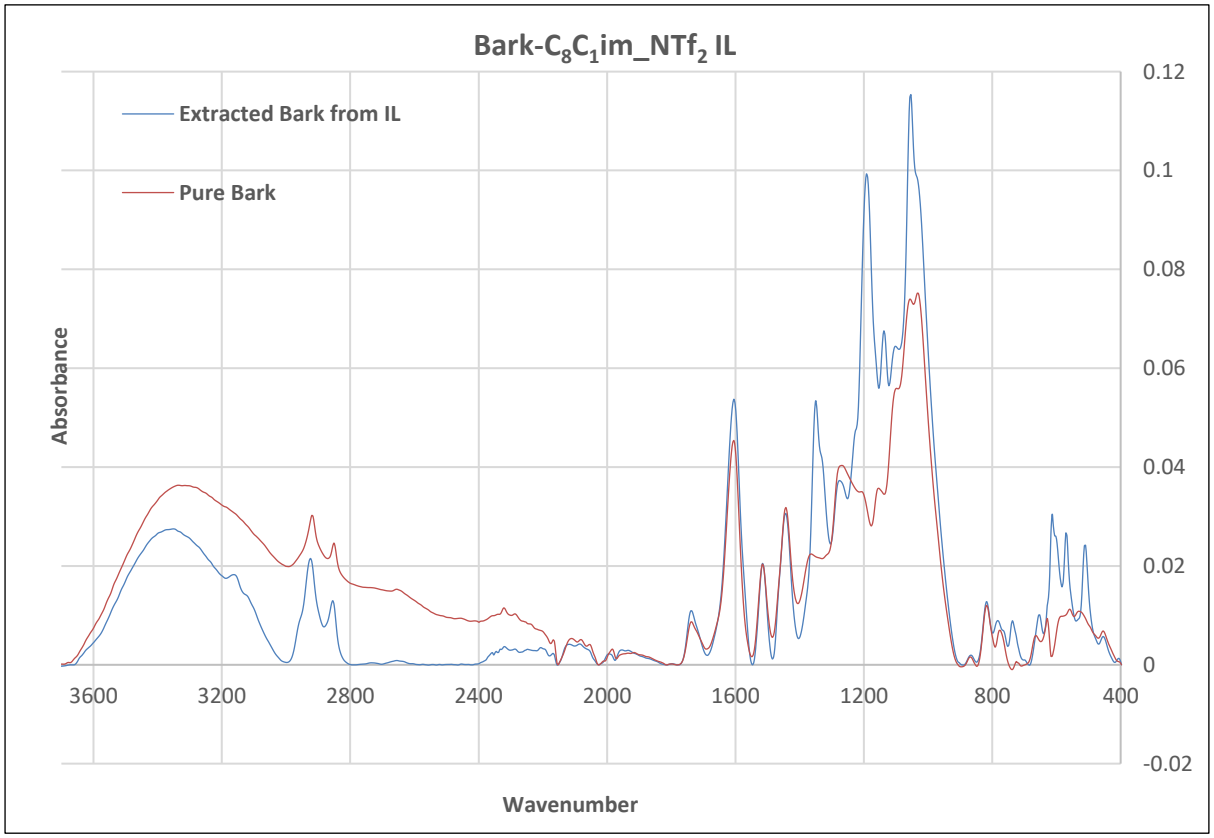


Figure A33. *0.1 g of bark in 4 mL of [C_8C_1im][NTf_2] at 100 ℃ for 24 h.*

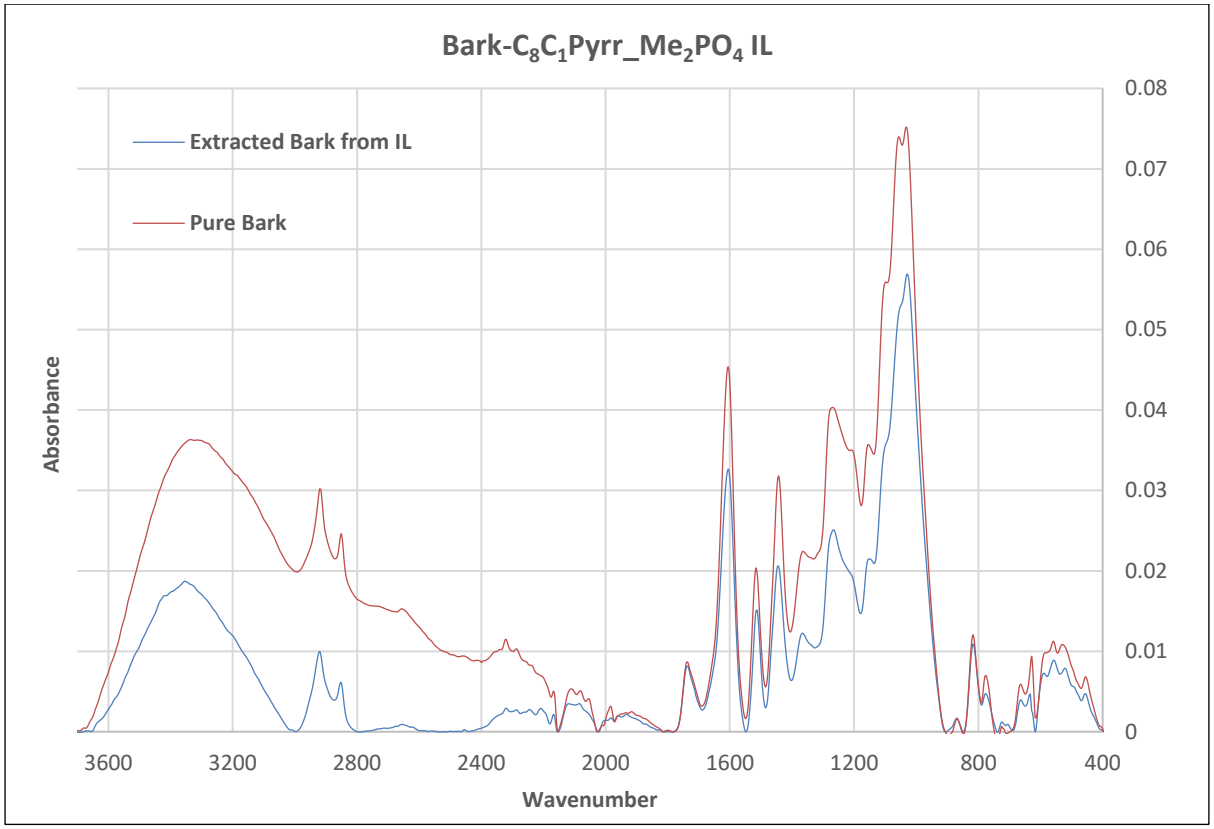


Figure A34. *0.1 g of bark in 4 mL of [C_8C_1pyrr][Me_2PO_4] at 100 ℃ for 24 h.*

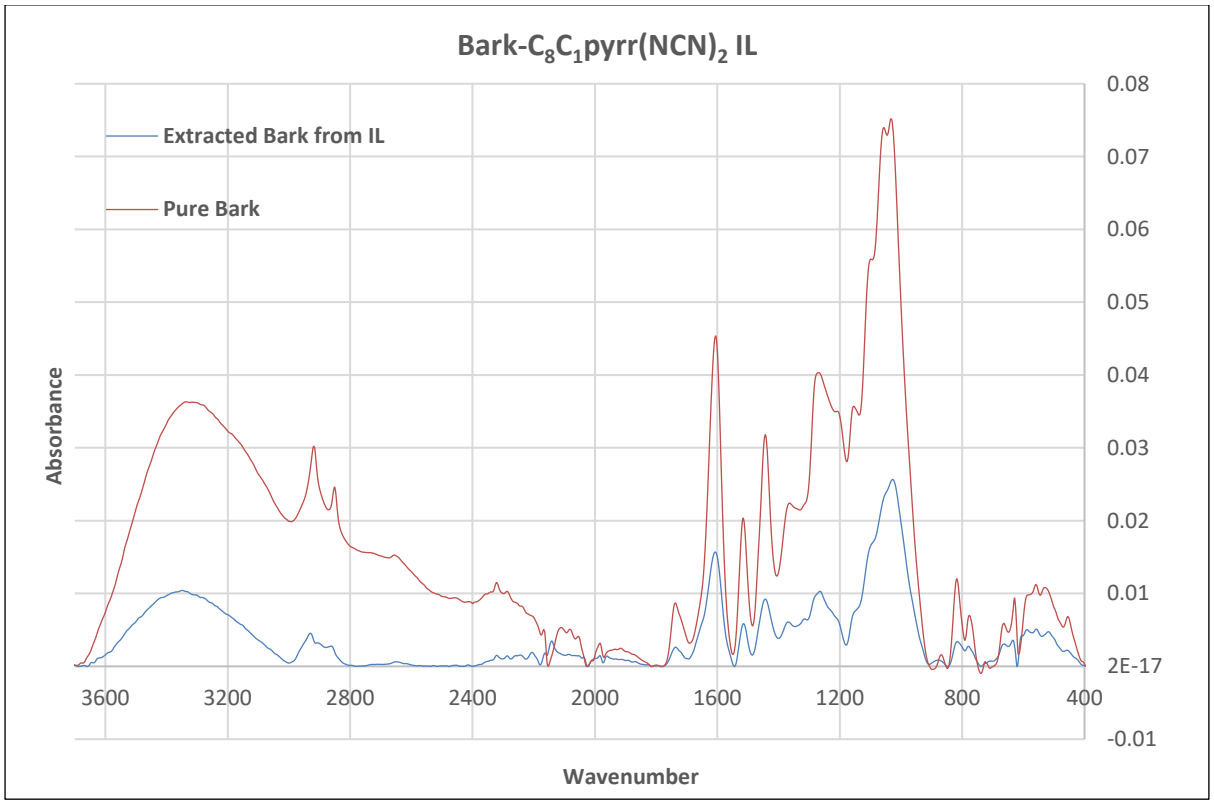


Figure A35. *0.1 g of bark in 4 mL of [C_8C_1pyrr][$N(CN)_2$] at 100 ℃ for 24 h.*

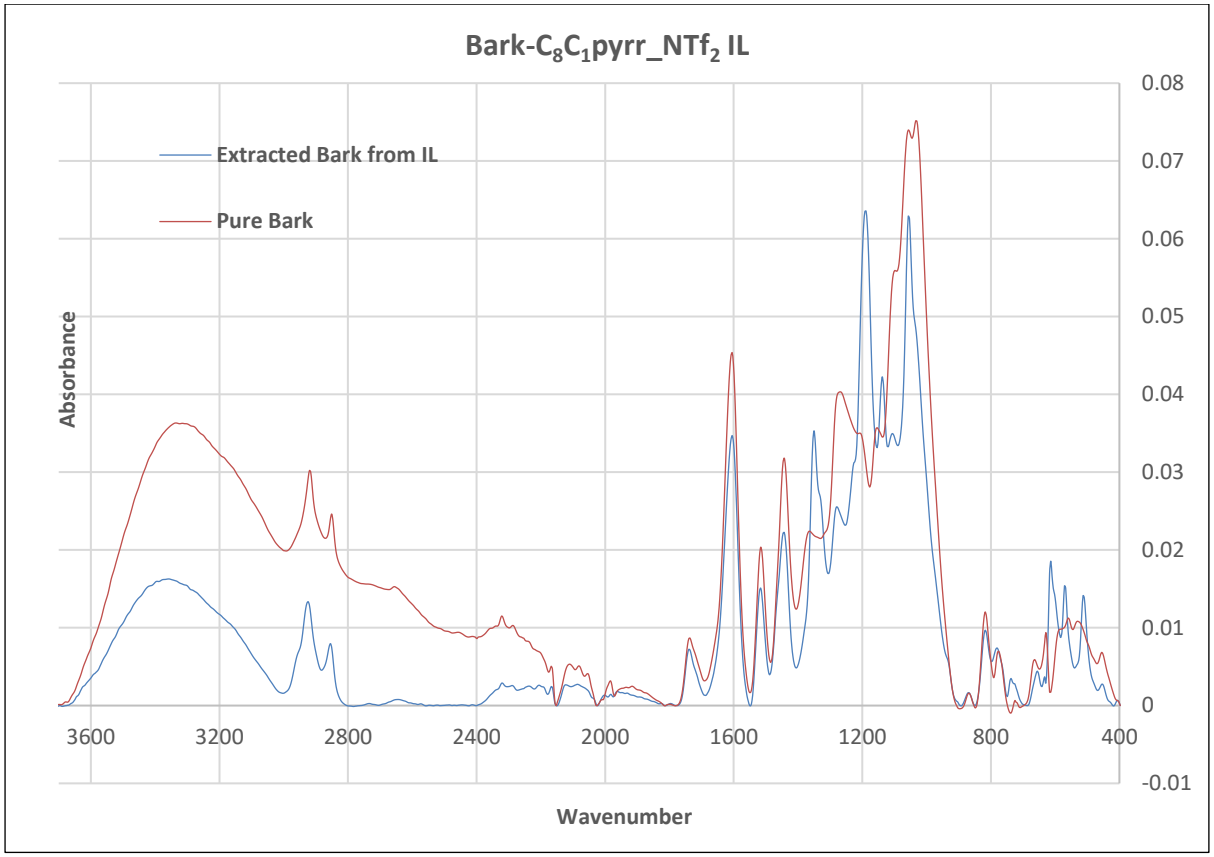


Figure A36. *0.1 g of bark in 4 mL of [C_8C_1pyrr][NTf_2] at 100 °C for 24 h.*

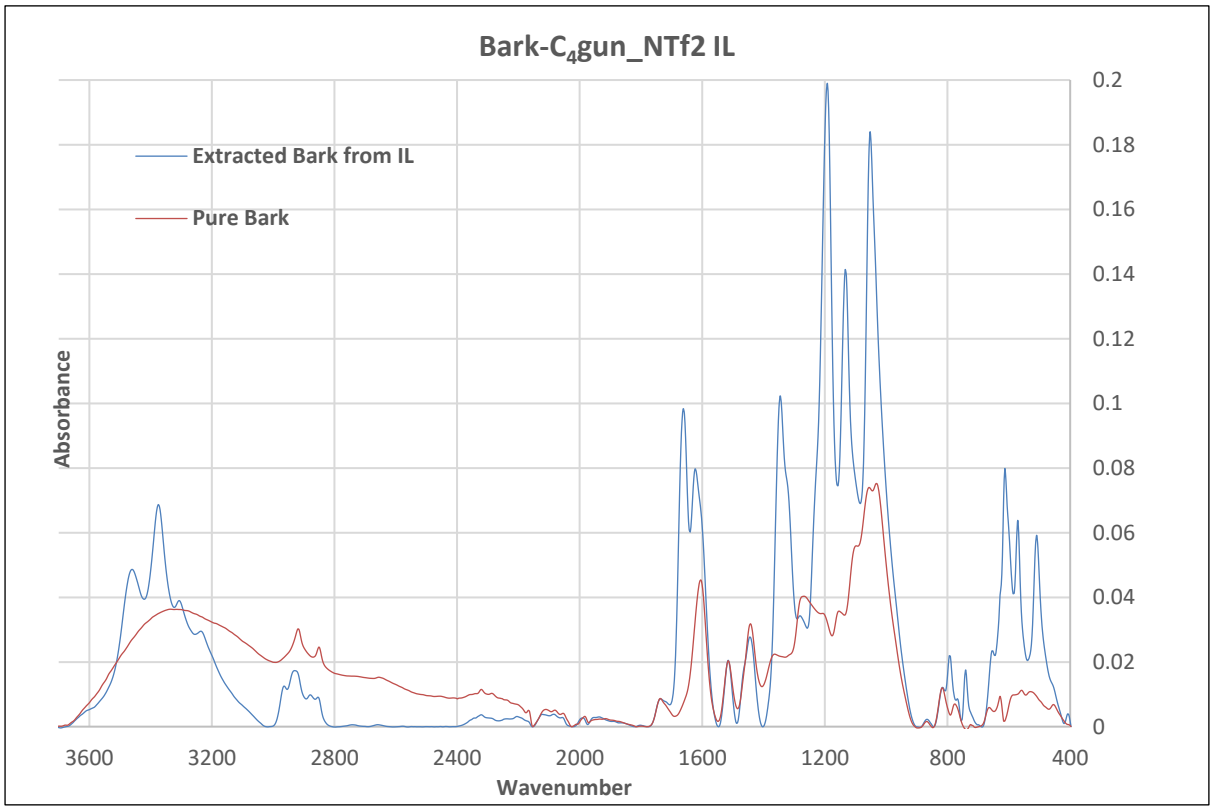


Figure A37. *0.05 g of bark in 2 mL of [C_4gun][NTf_2] at 100 °C for 24 h.*

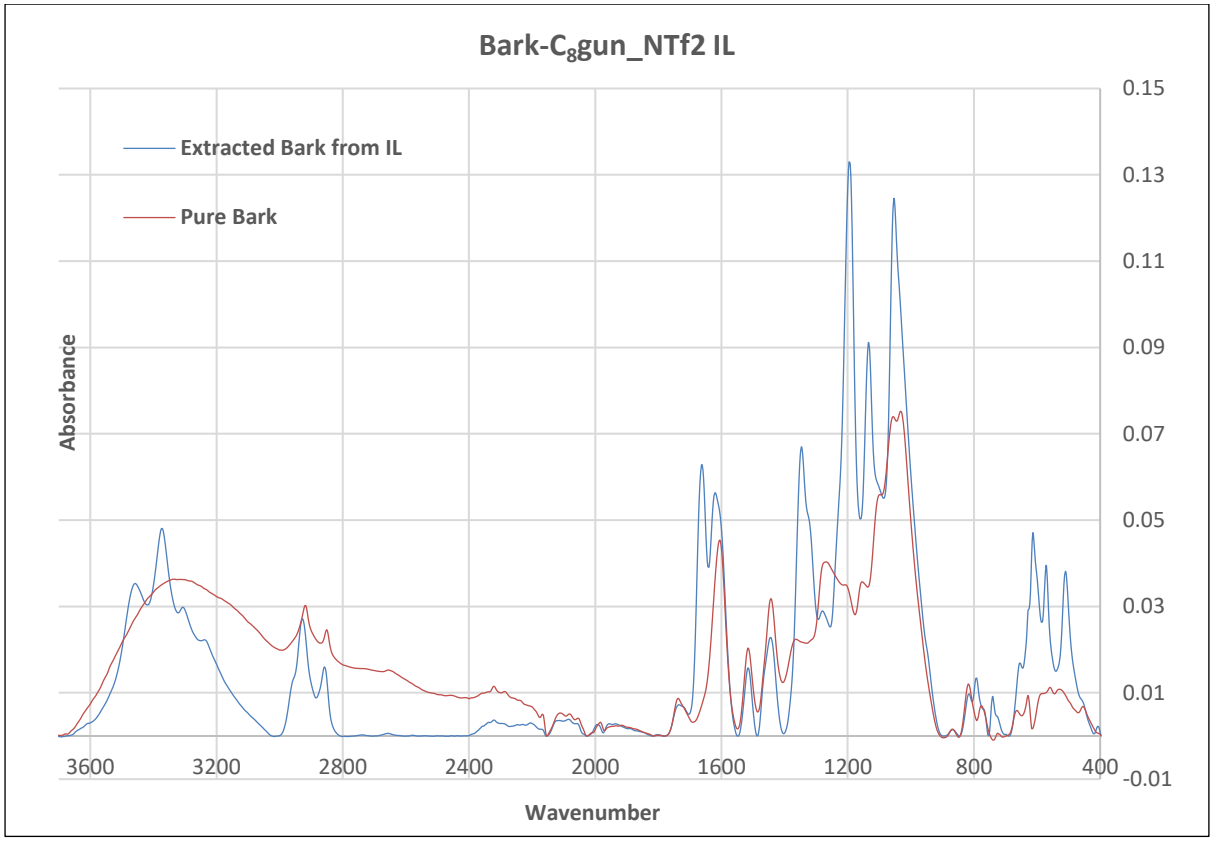


Figure A38. *0.05 g of bark in 2 mL of [C_8gun][NTf_2] at 100 ℃ for 24 h.*

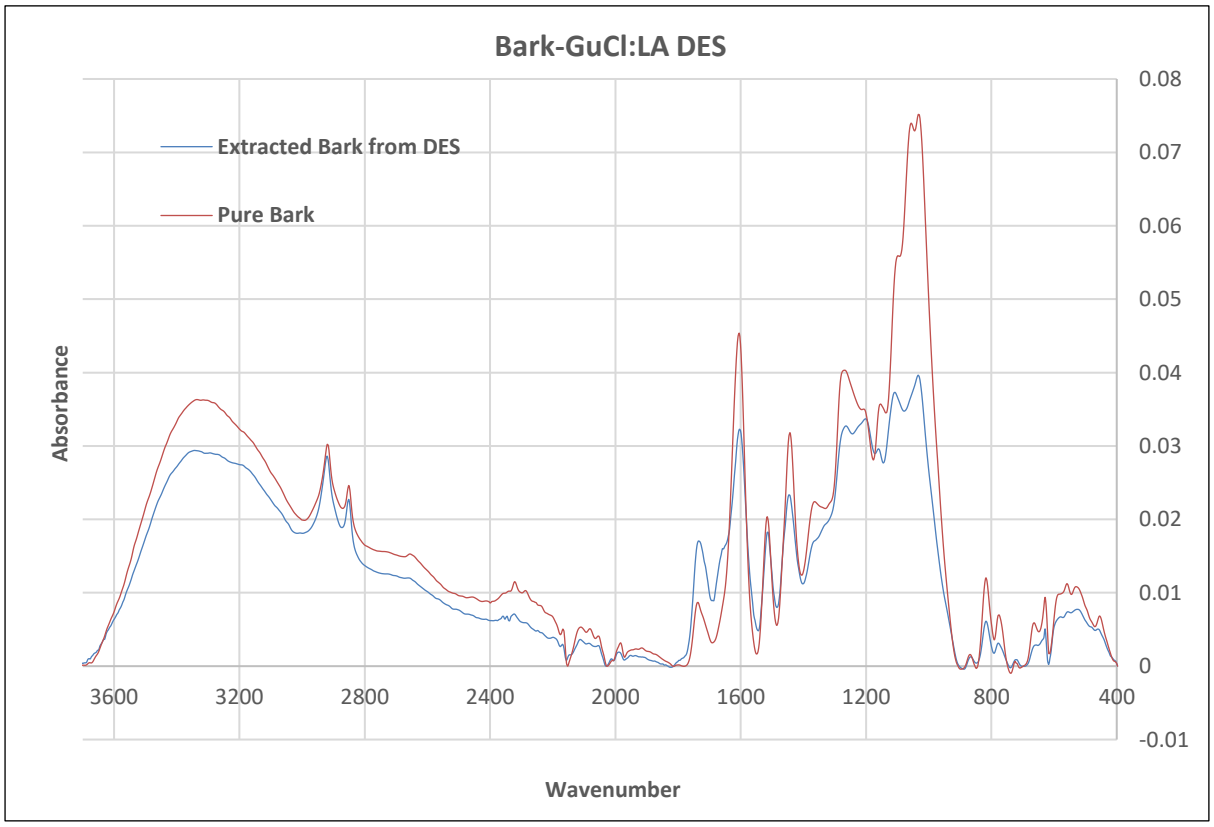


Figure A39. *0.1 g of bark in 4 mL of GuCl:LA at 100 ℃ for 24 h.*

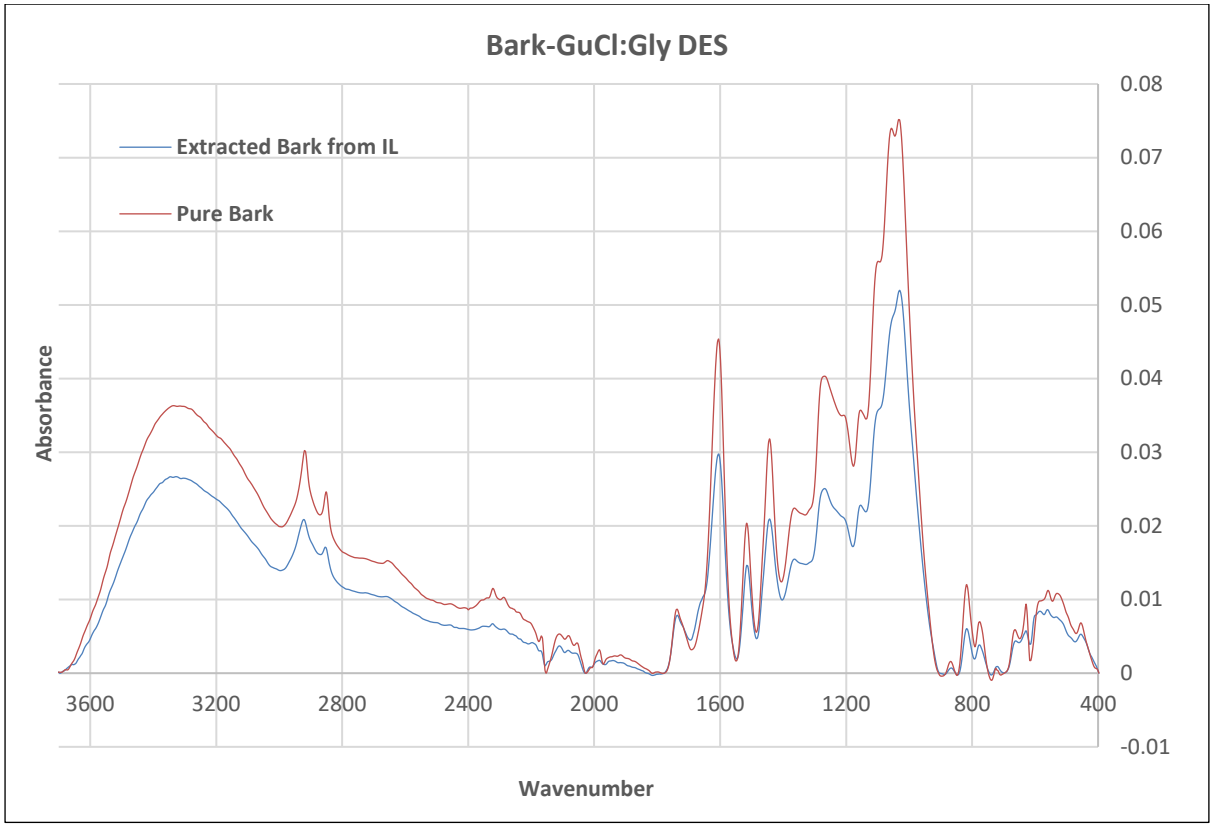


Figure A40. *0.1 g of bark in 4 mL of GuCl:Gly at 100 ℃ for 24 h.*

Table A41. *Low molecular weight volatile compounds in bark identified using SPME-GCMS. (Blue shaded blocks are the analytes present in both pure bark and IL extract)*

Label	Formula	Score	Base Peak	RT	Width	Area
Tricyclo[2.2.1.0(2,6)]heptane, 1,7,7-trimethyl-	C10H16	88.74	93.05	4.617	0.231	384708
.alpha.-Pinene	C10H16	92.75	93.05	5.083	0.597	15906139
Bicyclo[2.2.1]heptane, 7,7-dimethyl-2-methylene-	C10H16	94.68	93.05	6.05	0.233	1489632
Camphene	C10H16	98.5	93.05	6.283	0.617	6902059
.beta.-Pinene	C10H16	96.73	93.05	7.833	0.633	4250038
alpha.-Phellandrene	C10H16	83.5	91.05	8.683	0.529	789354
D-Limonene	C10H16	98.31	68	13.367	0.95	17503728
1,4-Cyclohexadiene, 1-methyl-4-(1-methylethyl)-	C10H16	94.57	93.05	15.75	0.44	1186714
Benzene, 1-methyl-2-(1-methylethyl)-	C10H14	96.65	119.1	16.667	0.447	12752931
trans-3-Caren-2-ol	C10H16O	87.12	119.1	17.167	0.34	1626308
1-Phenyl-1-butene	C10H12	81.49	117.05	17.6	0.531	1310081
3-Cyclohexene-1-methanol, 2-hydroxy-.alpha.,.alpha.,4-trimethyl-	C10H18O2	85.19	79.05	18.6	0.222	725784

1,2-15,16-Diepoxyhexadecane	C16H30O2	87.66	57.05	21.35	0.35	876433
Bicyclo[2.2.1]heptane-2,5-diol, 1,7,7-trimethyl-, (2-endo,5-exo)-	C10H18O2	88.83	108.1	21.933	0.15	584811
Benzene, (2-methyl-1-propenyl)-	C10H12	91.02	117.05	22.233	0.117	1759548
Dodecanoic acid, 3-hydroxy-	C12H24O3	81.61	57.05	22.95	0.367	1067538
3-Cyclopentene-1-acetaldehyde, 2,2,3-trimethyl-	C10H16O	95.1	108.1	23.733	0.38	4986544
Camphor	C10H16O	91.8	95.1	24.183	0.483	13190310
Bicyclo [3.1.1] heptan-3-one, 2, 6, 6-trimethyl-, (1.alpha. 2. alpha., 5.alpha.)-	C10H16O	87.77	55.05	25.017	0.194	3094483
2(10)-Pinen-3-one, (.+/-.)-	C10H14O	90.62	81.05	25.583	0.183	7743764
Bicyclo[3.1.1]heptan-2-one, 6,6-dimethyl-, (1R)-	C9H14O	87.29	83.05	25.717	0.217	3777343
1,2-15,16-Diepoxyhexadecane	C16H30O2	87.65	43	26.083	0.3	1700602
Bicyclo[2.2.1]heptan-2-ol, 1,3,3-trimethyl-	C10H18O	96.56	81.05	26.317	0.267	7819090
Cyclopentanol, 1, 2-dimethyl-3-(1-methylethenyl)-, [1R-(1.alpha, 2.alpha. 3. beta.)]-	C10H18O	83.43	71.05	26.55	0.15	3172243
3-Cyclohexen-1-ol, 4-methyl-1-(1-methylethyl)-	C10H18O	85.2	71.05	26.75	0.25	8044567
(1R)-(-)-Myrtenal	C10H14O	97.75	79.05	27.083	0.417	10117946
Formic acid, 3,7,11-trimethyl-1,6,10-dodecatrien-3-yl ester	C16H26O2	88.19	43	27.483	0.283	1407297
Bicyclo [3.1.1] heptan-3-ol, 6,6-dimethyl-2-methylene-, [1S-(1.alpha.,3.alpha.,5.alpha.)]-	C10H16O	96	92.05	27.883	0.267	12421869
10-Heptadecen-8-ynoic acid, methyl ester, (E)-	C18H30O2	82.44	79.05	28.133	0.15	2608675
Isoborneol	C10H18O	92.67	95.1	28.25	0.15	2971893
Bicyclo[3.1.1]hept-3-en-2-one, 4,6,6-trimethyl-, (1S)-	C10H14O	97.04	107.05	28.833	0.233	11498721
3-Cyclohexene-1-methanol, .alpha.,.alpha.4-trimethyl-	C10H18O	93.19	95.1	29.017	0.267	35492796
6-Nonynoic acid, methyl ester	C10H16O2	80.45	59.05	29.617	0.2	4996742
Epiglobulol	C15H26O	86.39	105.05	29.8	0.2	2310439
alpha.-Muurolene	C15H24	91.09	105.05	30.067	0.25	2803728
Limonen-6-ol, pivalate	C15H24O2	86.43	43	30.483	0.117	409113
Bicyclo[3.1.1]hept-2-ene-2-methanol, 6,6-dimethyl-	C10H16O	94.54	79.05	31.05	0.25	10535462
2-Cyclohexen-1-ol, 2-methyl-5-(1-methylethenyl)-, cis-	C10H16O	95.56	109.05	31.983	0.183	5591173
1,2-15,16-Diepoxyhexadecane	C16H30O2	83.47	69.05	32.517	0.233	6506347
[1,1'-Bicyclopropyl]-2-octanoic acid, 2'-hexyl-, methyl ester	C21H38O2	82.55	43	34.25	0.283	3555883

5,8,11,14-Eicosatetraenoic acid, methyl ester, (all-Z)-	C21H34O2	85.13	43	36.3	0.257	1051923
Z-8-Methyl-9-tetradecenoic acid	C15H28O2	81.49	43	37.583	0.133	933412
Nonanoic acid	C9H18O2	81.2	60	38.567	0.283	5979210
cis-Z-.alpha.-Bisabolene epoxide	C15H24O	82.24	107.05	39.4	0.117	1189433
Bicyclo[2.2.1]heptane-2-carboxylic acid, 3,3-dimethyl-	C10H16O2	83.41	83.05	40.983	0.25	1816664
Phenol, 2,4-bis(1,1-dimethylethyl)-	C14H22O	87.16	191.05	41.35	0.15	770766
1-Heptatriacotanol	C37H76O	86.55	81.05	41.8	0.3	1390678
cis,trans-5,9-Cyclododecadiene-cis-1,2-diol	C12H20O2	81.61	81.05	42.117	0.233	2961074
Cyclopropaneoctanoic acid, 2-[[2-[(2-ethylcyclopropyl)methyl]cyclopropyl]methyl]-, methyl ester	C22H38O2	81.65	81.05	42.383	0.3	2923698
Myrtenoic acid, ethyl ester	C12H18O2	84.85	79.05	42.683	0.433	5839265

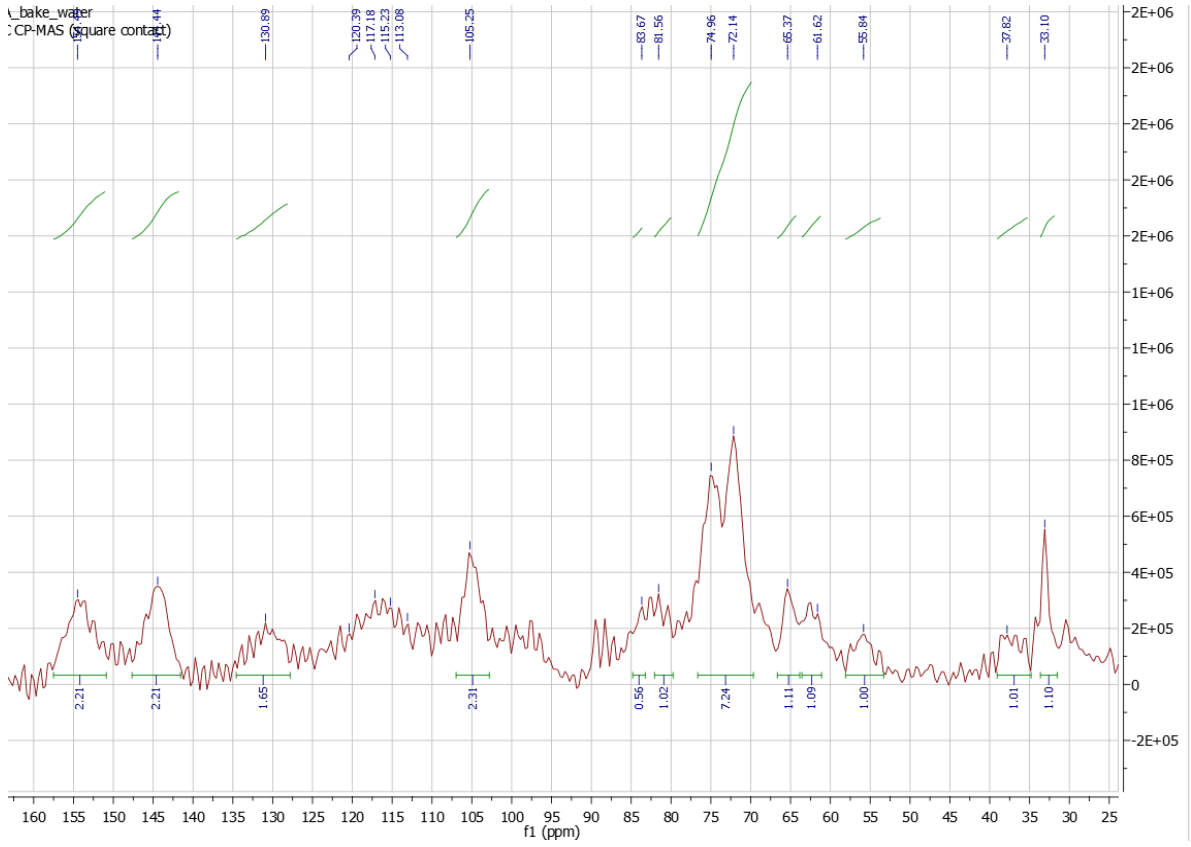


Figure A42. *^{13}C solid-state NMR of components of water washed bark.*

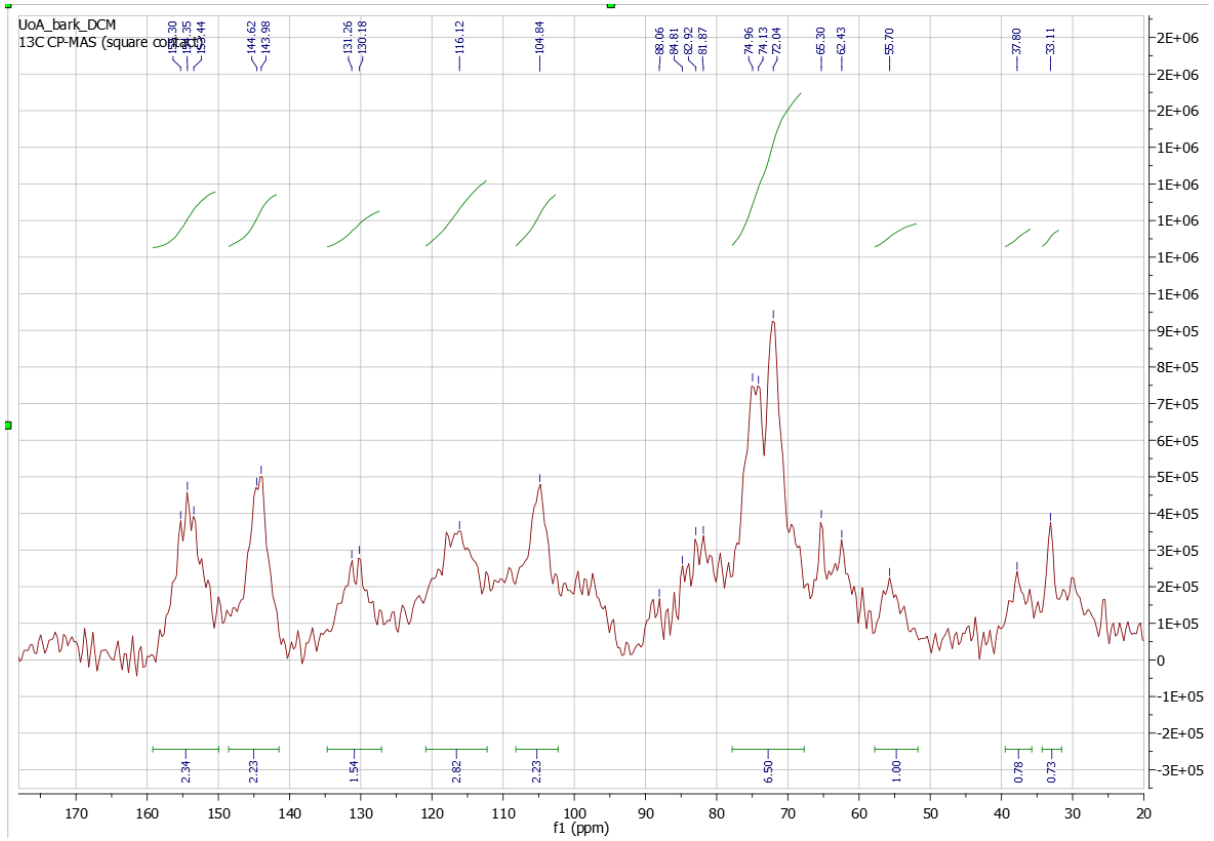


Figure A43. *^{13}C solid-state NMR of components of dichloromethane washed bark.*

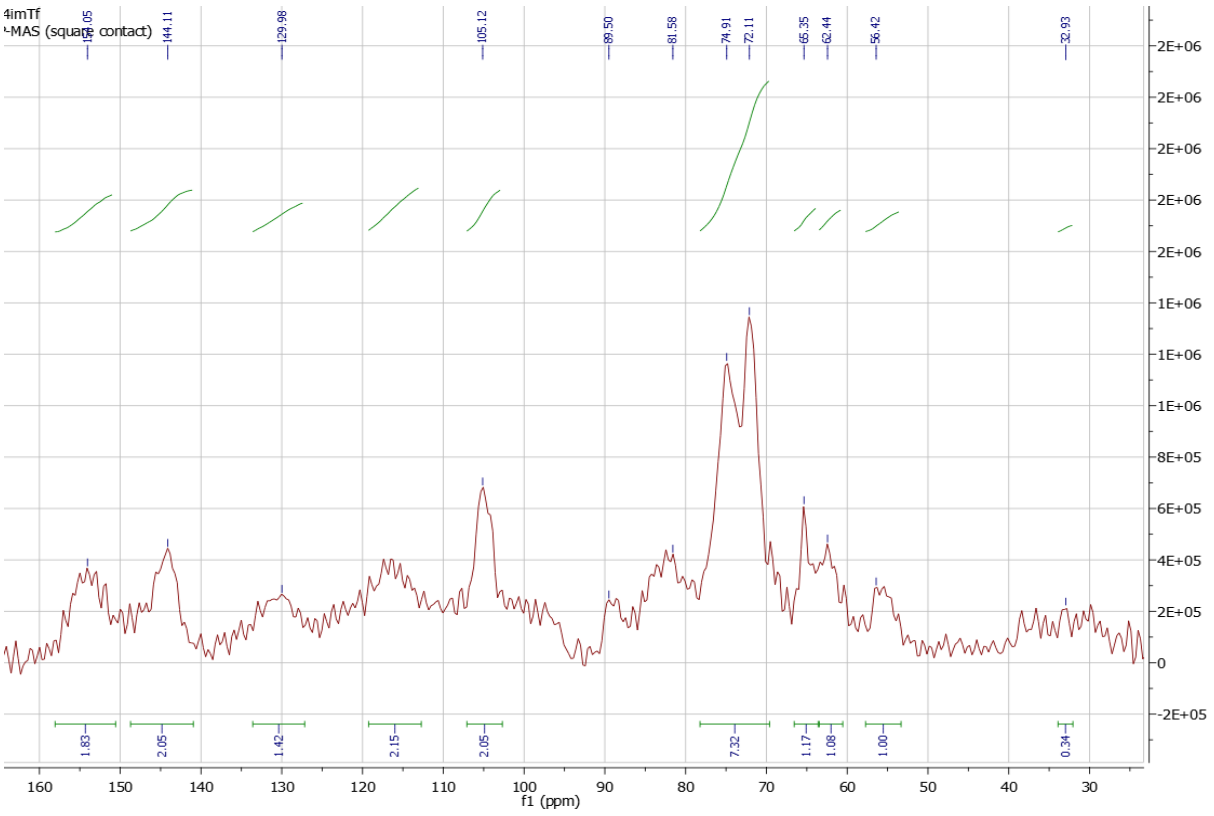


Figure A44. *^{13}C solid-state NMR of components of bark for $[C_4C_1im][NTf_2]$ solid residue*

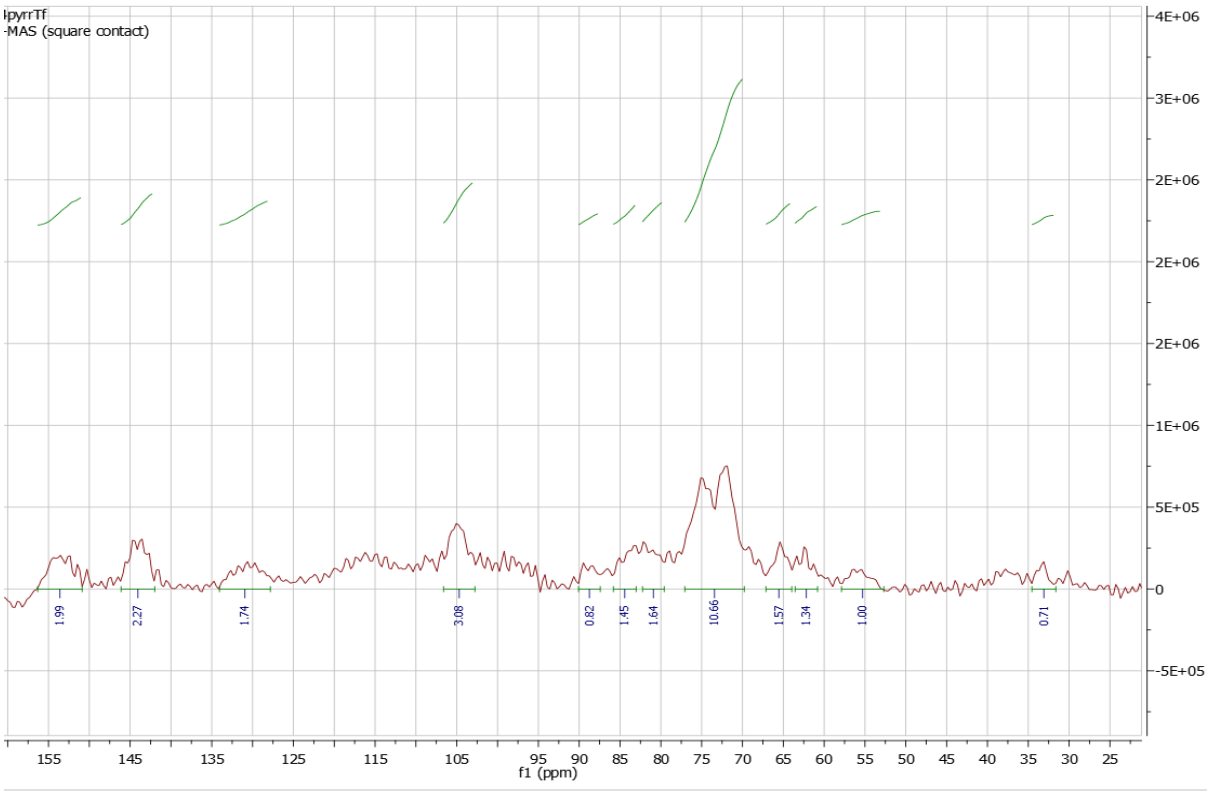


Figure A45. *^{13}C solid-state NMR of components of bark for $[C_4C_1pyr][NTf_2]$ solid residue*

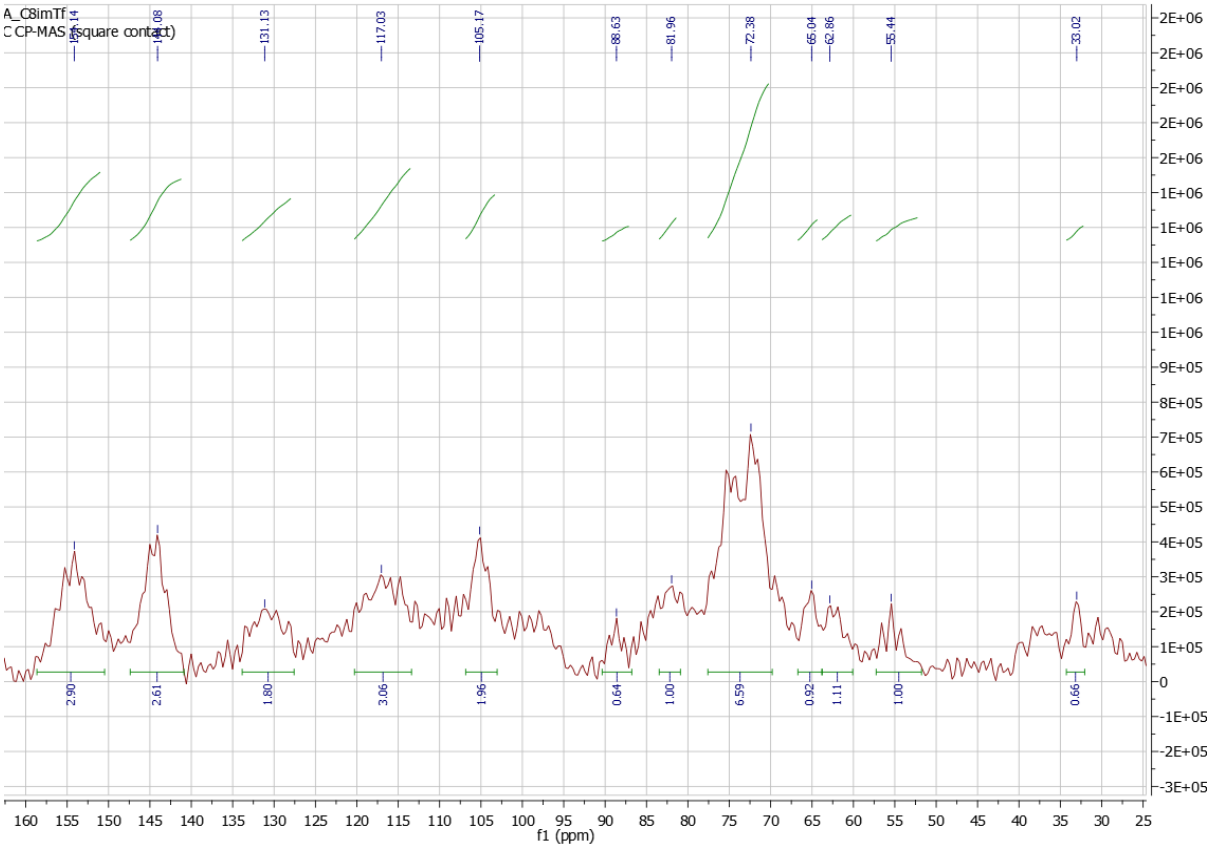


Figure A47. *^{13}C solid-state NMR of components of bark for $[C_8C_1im][NTf_2]$ solid residue*

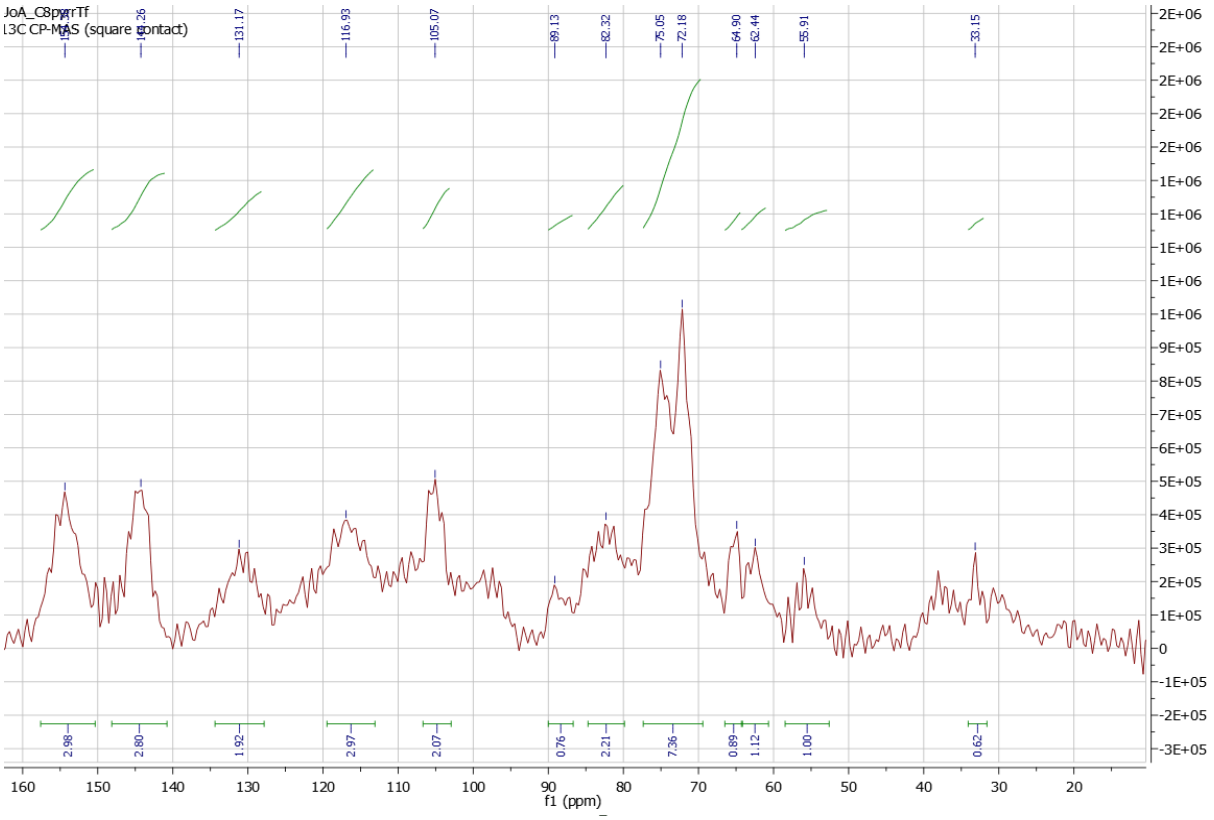


Figure A48. *^{13}C solid-state NMR of components of bark for $[C_8C_1pyr][NTf_2]$ solid residue*

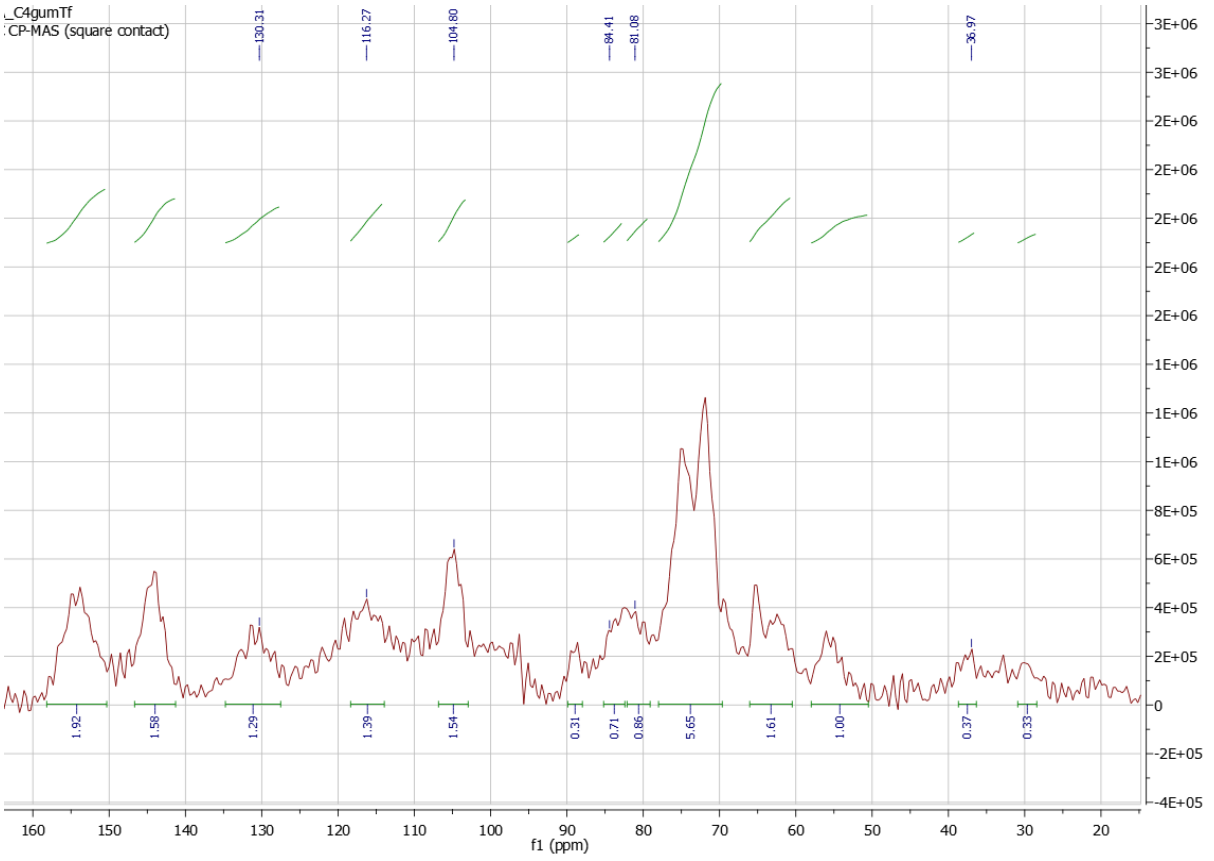


Figure A49. *^{13}C solid-state NMR of components of bark for $[C_4gun][NTf_2]$ solid residue*

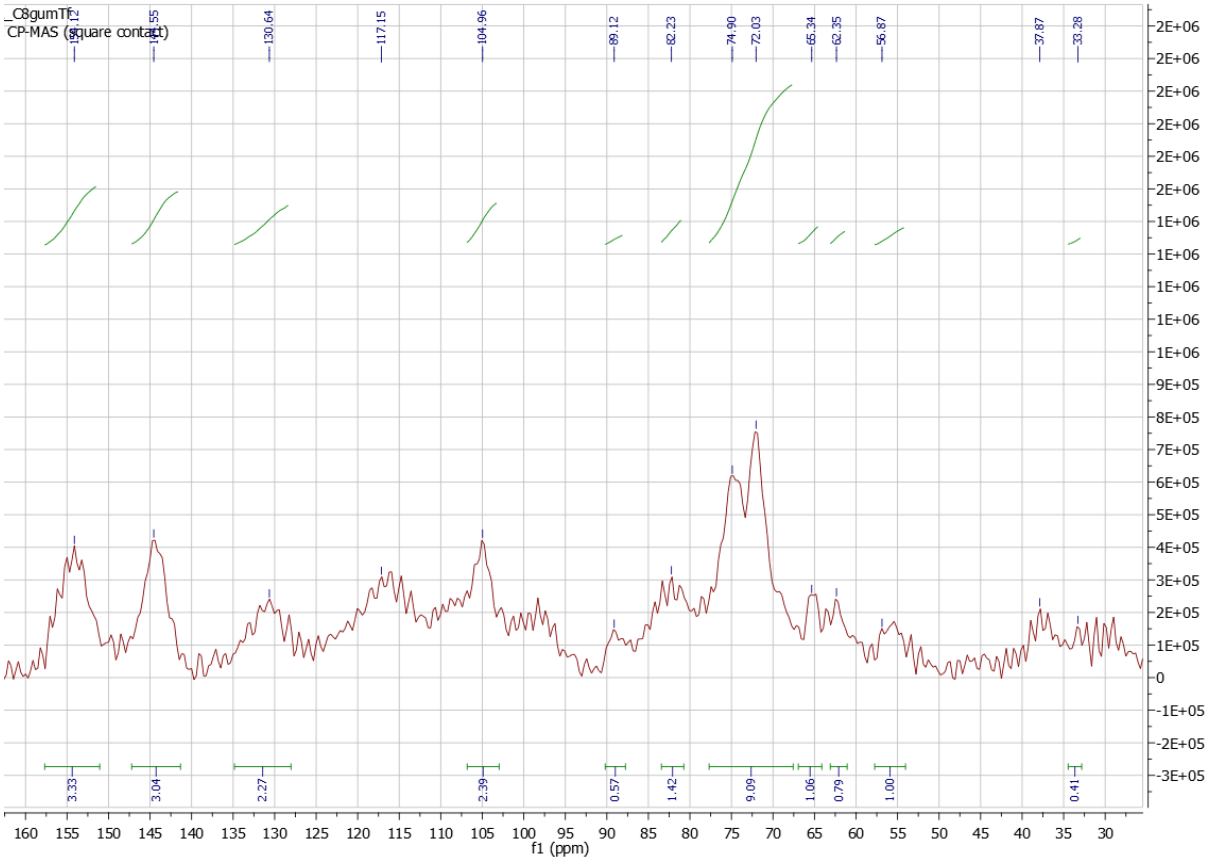


Figure A49. *^{13}C solid-state NMR of components of bark for $[C_8gun][NTf_2]$ solid residue*

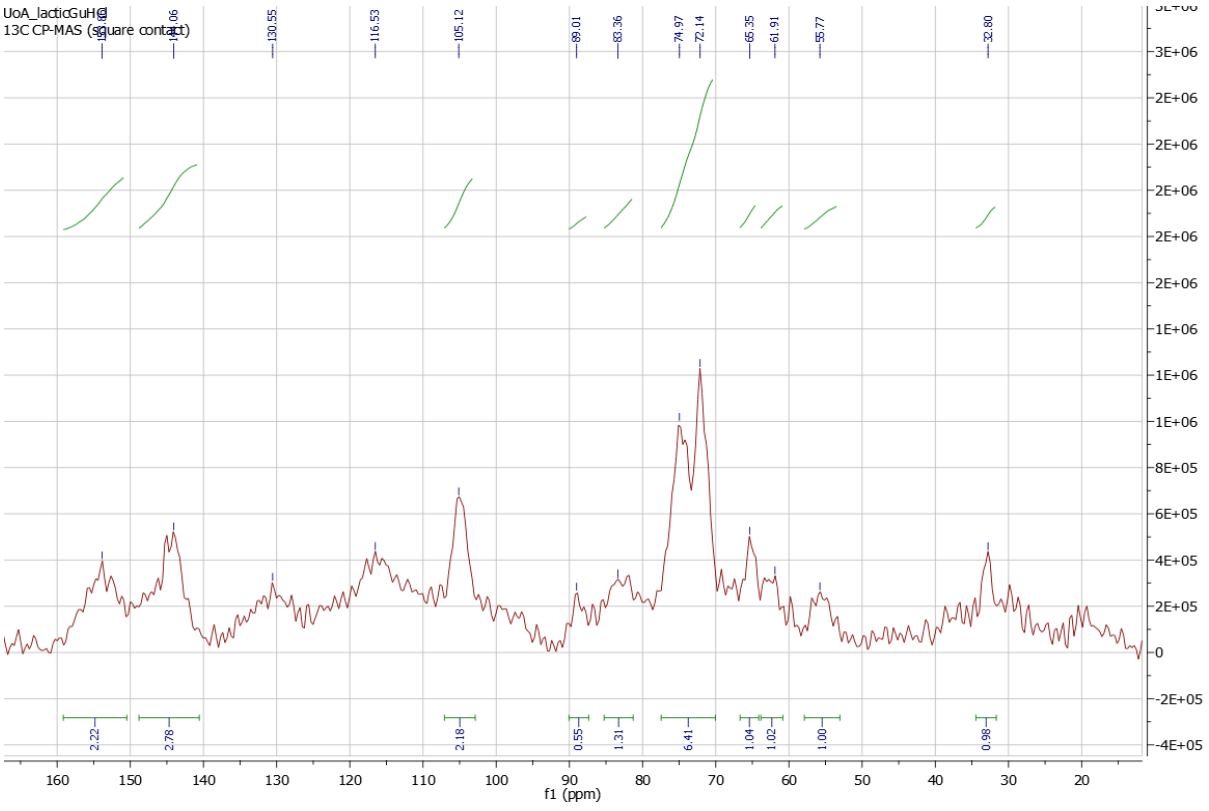


Figure A50. *^{13}C solid-state NMR of components of bark for GuCl:LA solid residue*

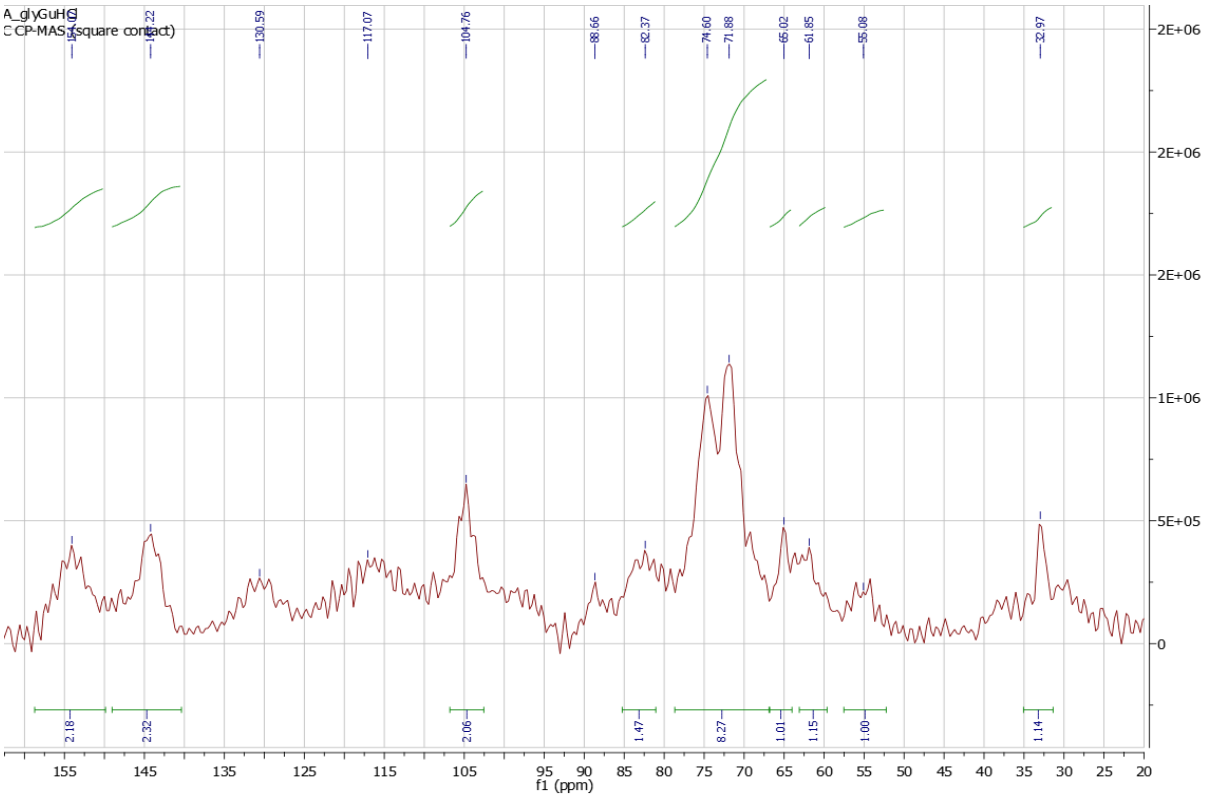


Figure A51. *^{13}C solid-state NMR of components of bark for GuCl:Gly solid residue*

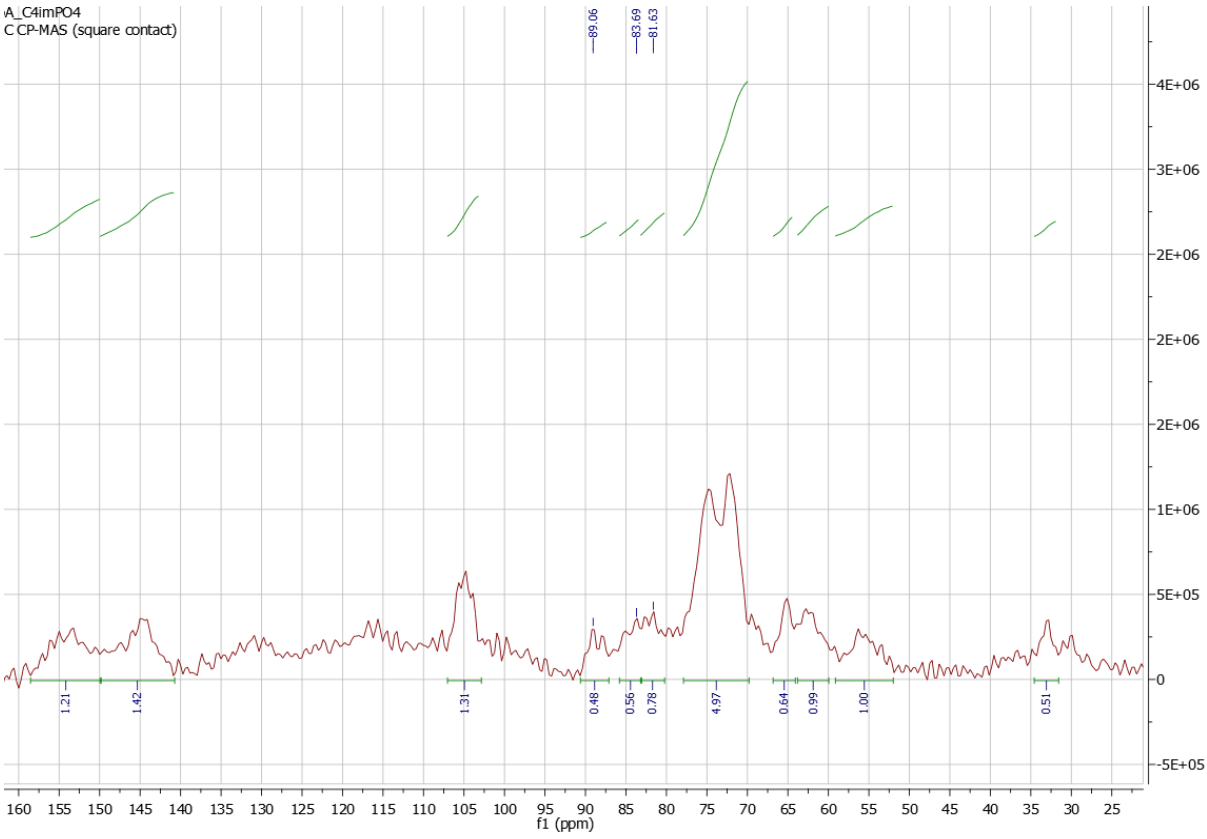


Figure A52. *^{13}C solid-state NMR of components of bark for $[C_4C_1im][Me_2PO_4]$ solid residue*

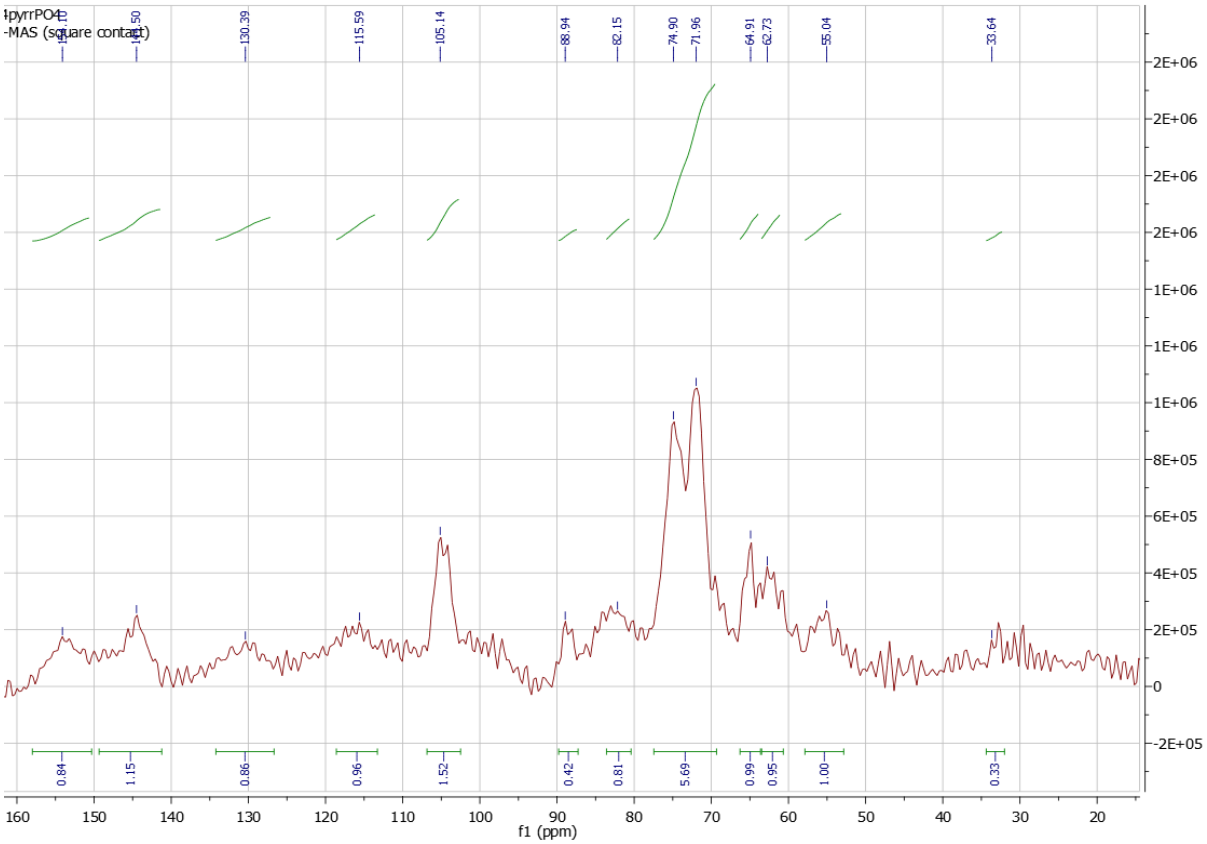


Figure A53. *^{13}C solid-state NMR of components of bark for $[C_4C_1pyr][Me_2PO_4]$ solid residue*

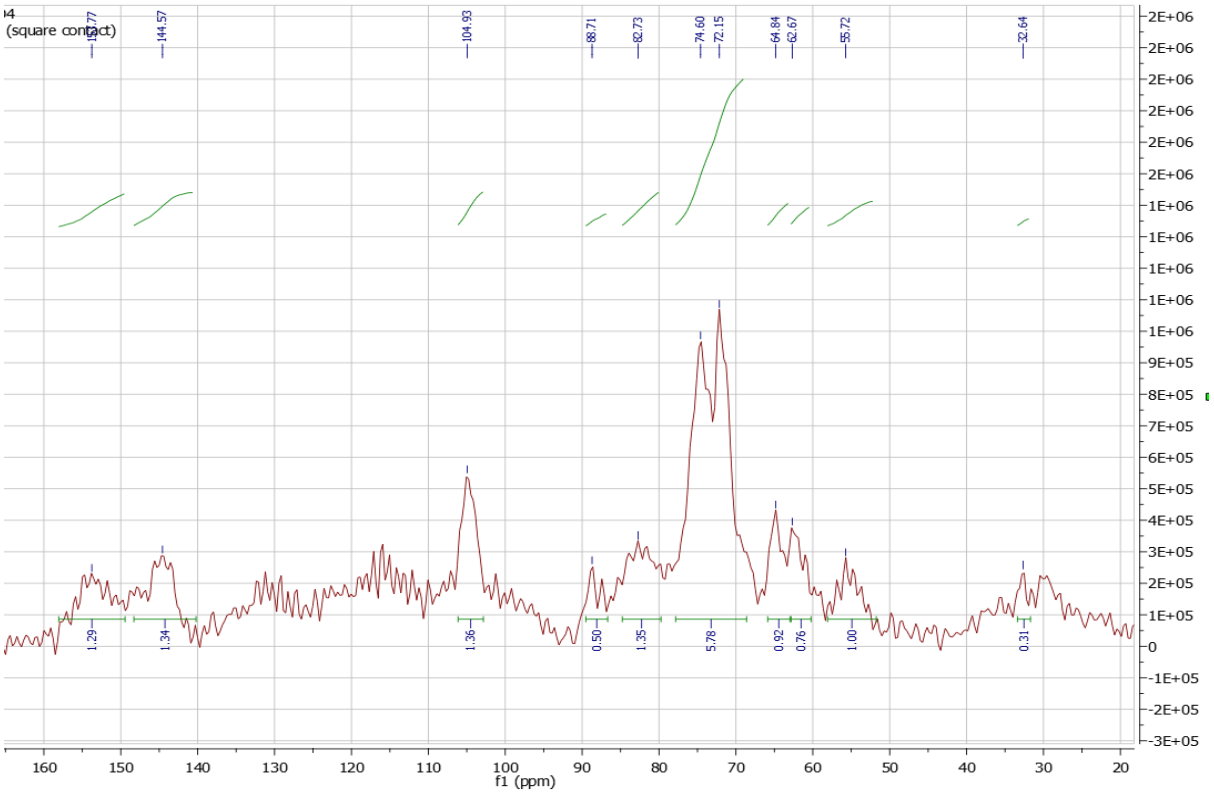


Figure A54. *^{13}C solid-state NMR of components of bark for $[C_8C_1im][Me_2PO_4]$ solid residue*

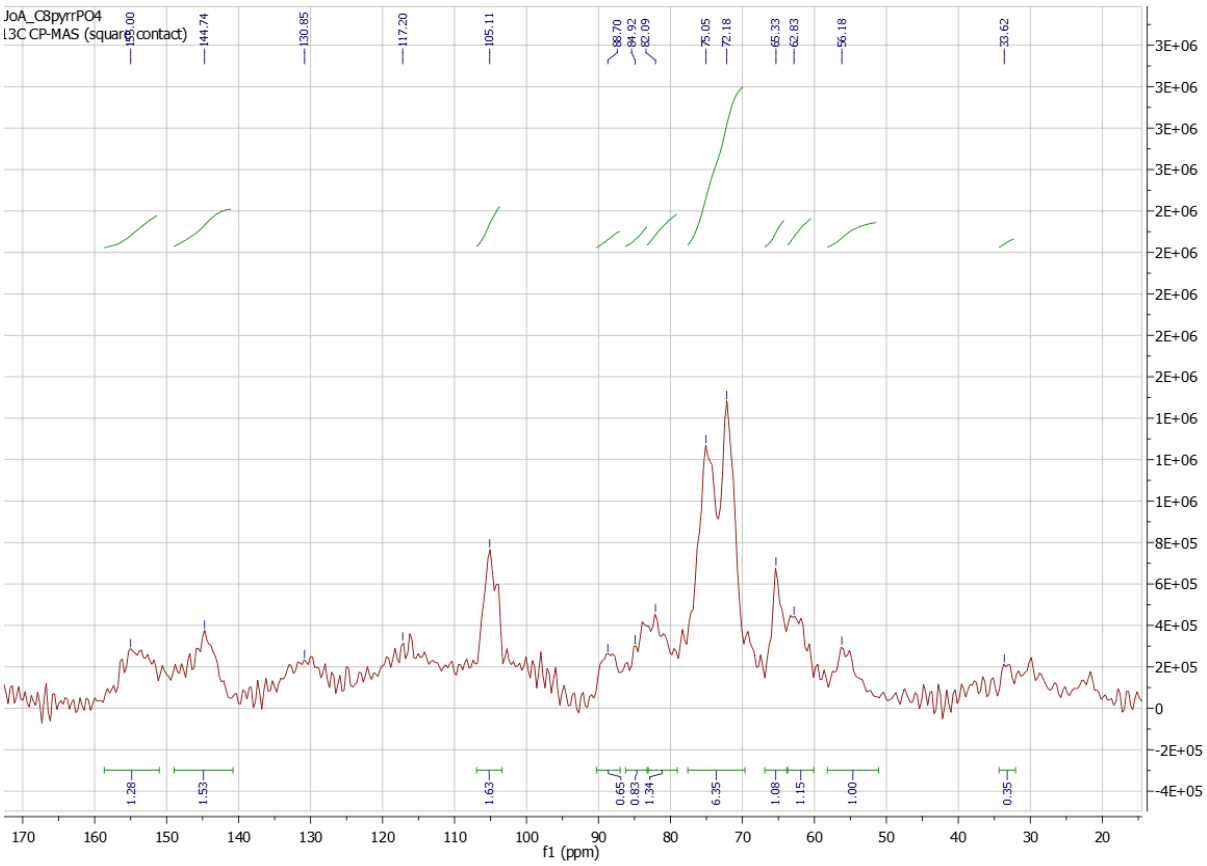


Figure A55. *^{13}C solid-state NMR of components of bark for $[C_8C_1pyr][Me_2PO_4]$ solid residue*

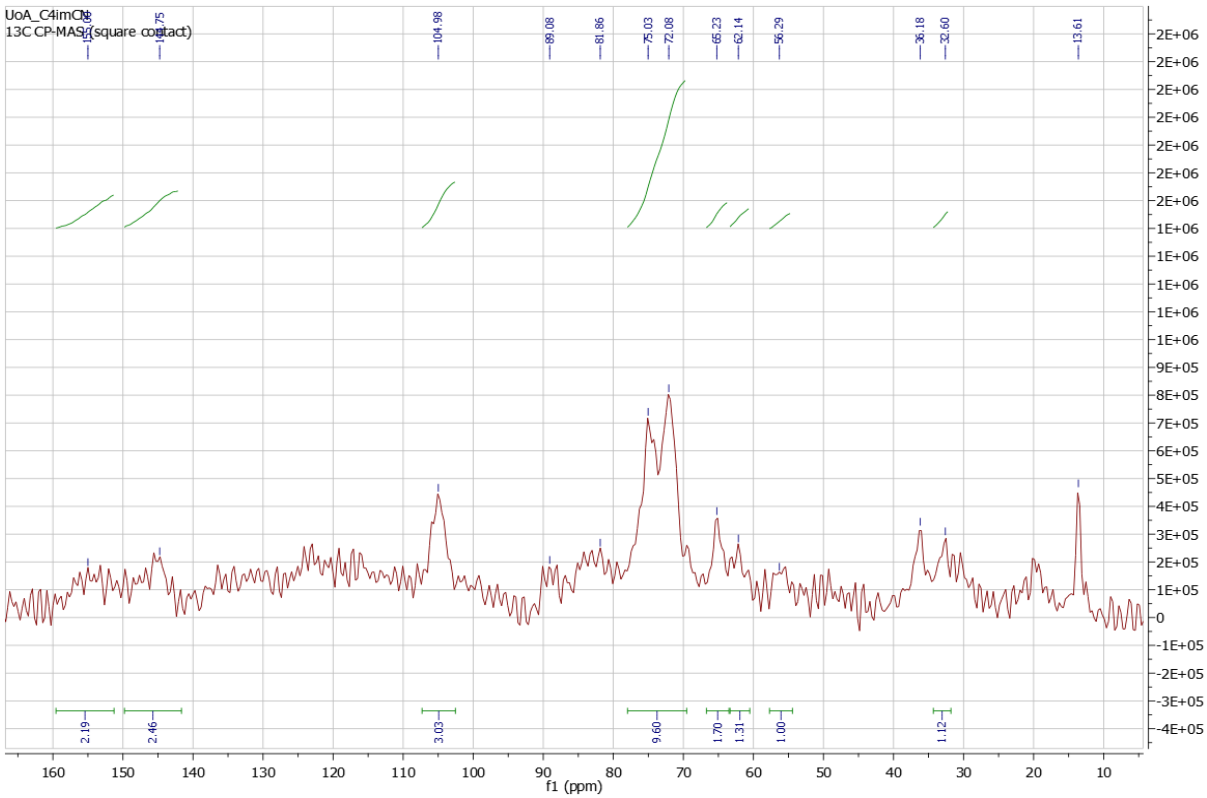


Figure A56. *^{13}C solid-state NMR of components of bark for $[C_4C_1im][N(CN)_2]$ solid residue*

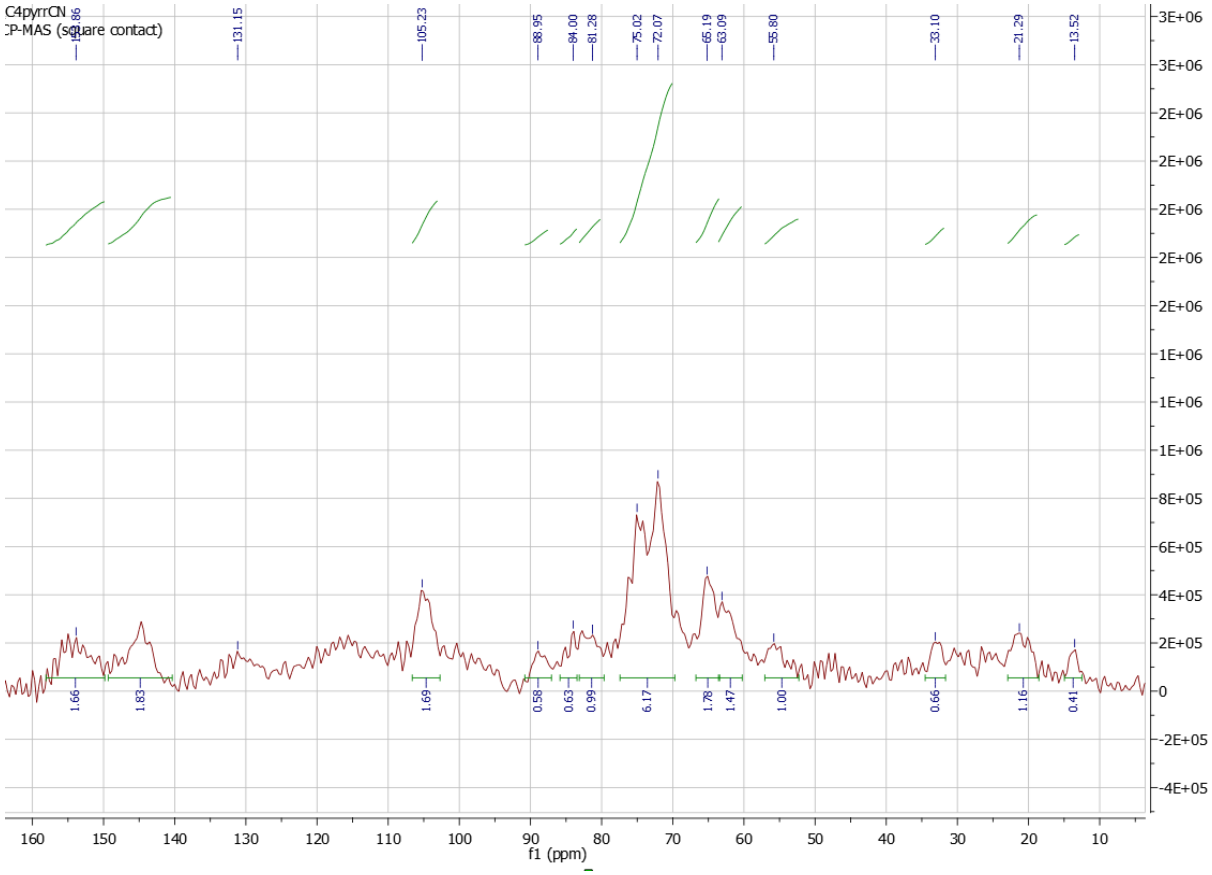


Figure A57. *^{13}C solid-state NMR of components of bark for $[C_4C_1pyr][N(CN)_2]$ solid residue*

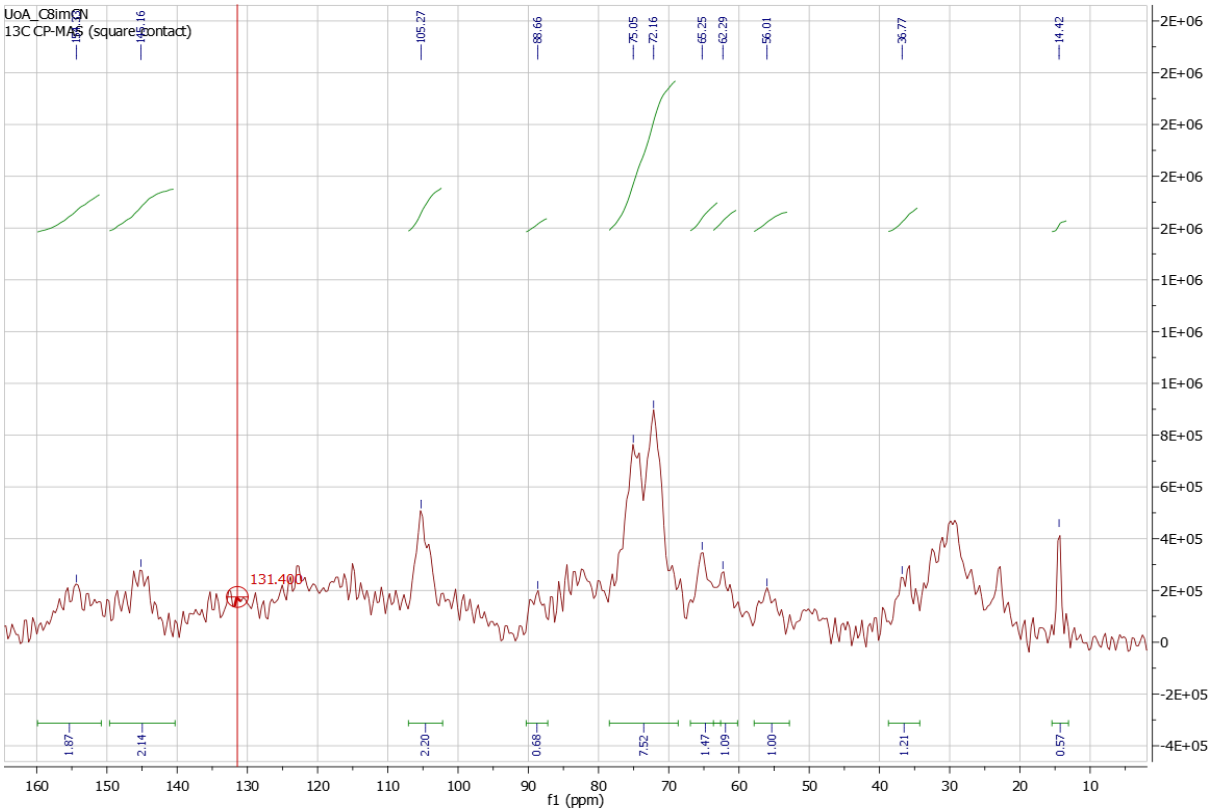


Figure A58. *^{13}C solid-state NMR of components of bark for $[C_8C_1im][N(CN)_2]$ solid residue*

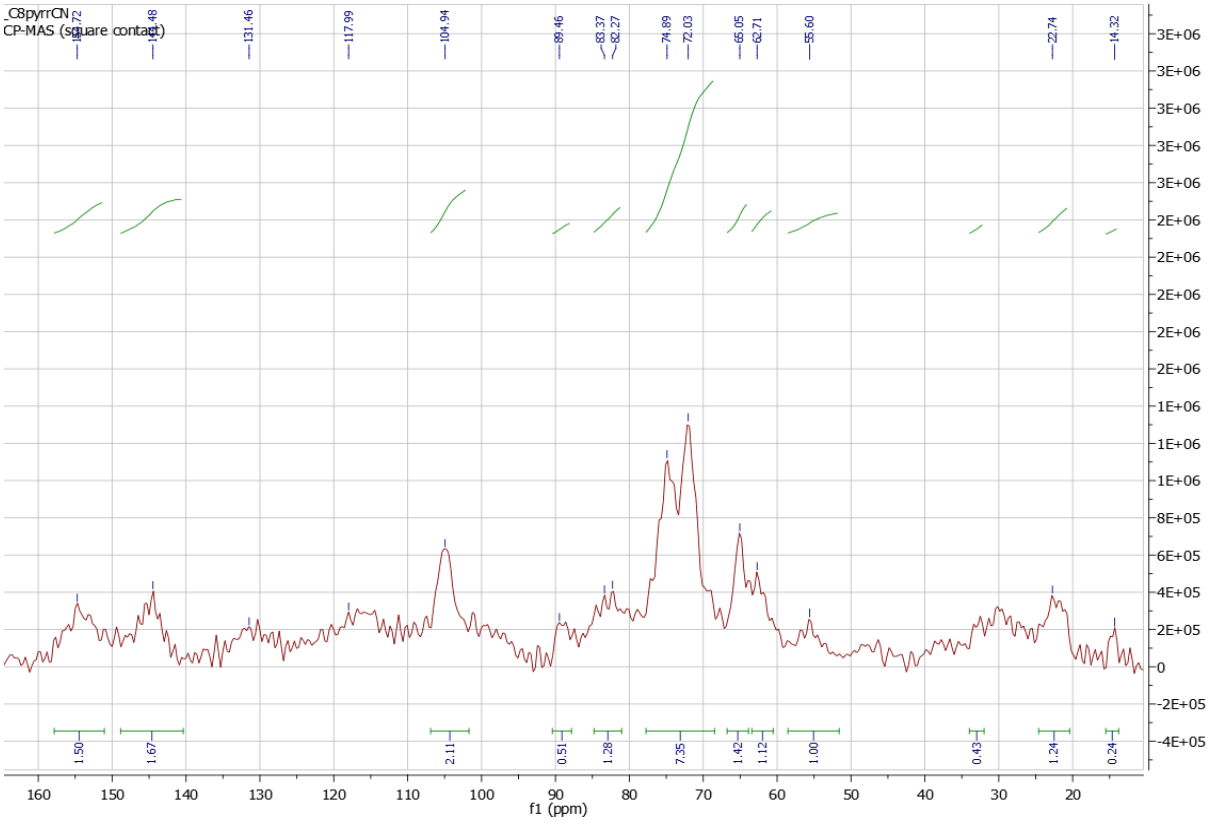


Figure A59. *^{13}C solid-state NMR of components of bark for $[C_8C_1pyr][N(CN)_2]$ solid residue*

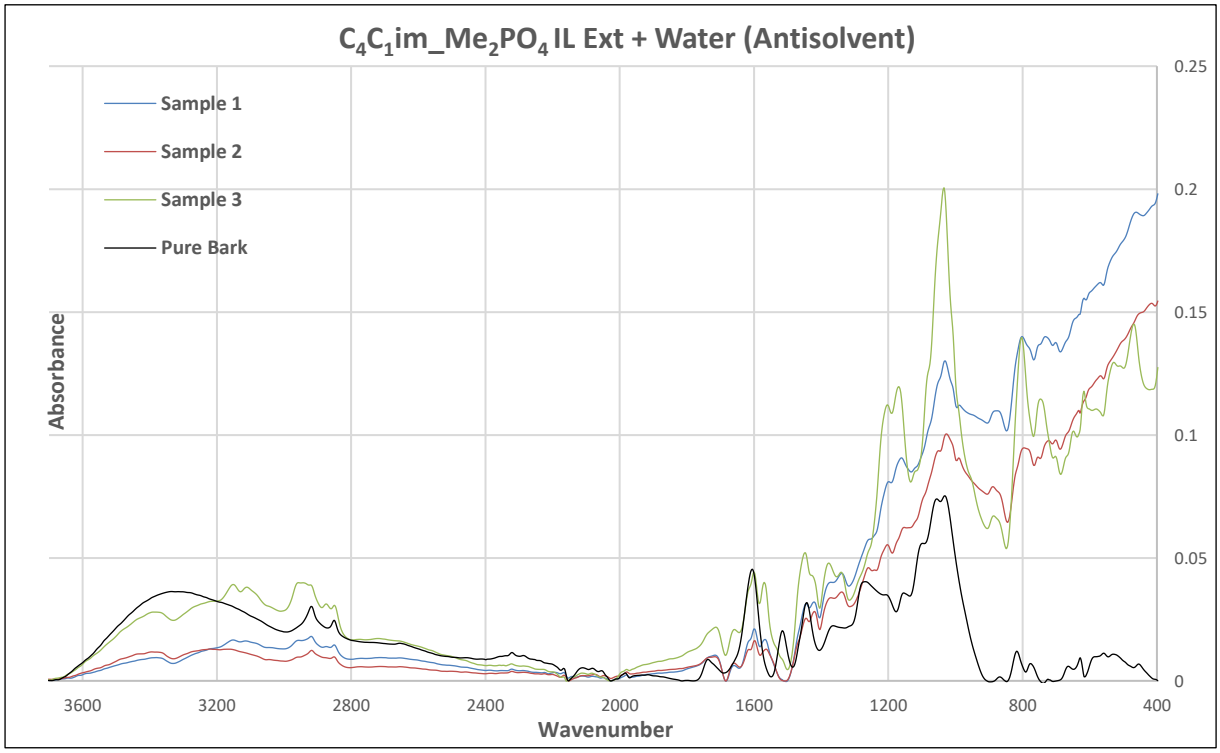


Figure A60. *IR spectra of the solids formed from Sample 1 – 3 using 1 mL of $[C_4C_1im][Me_2PO_4]$ IL extract and water as an antisolvent.*

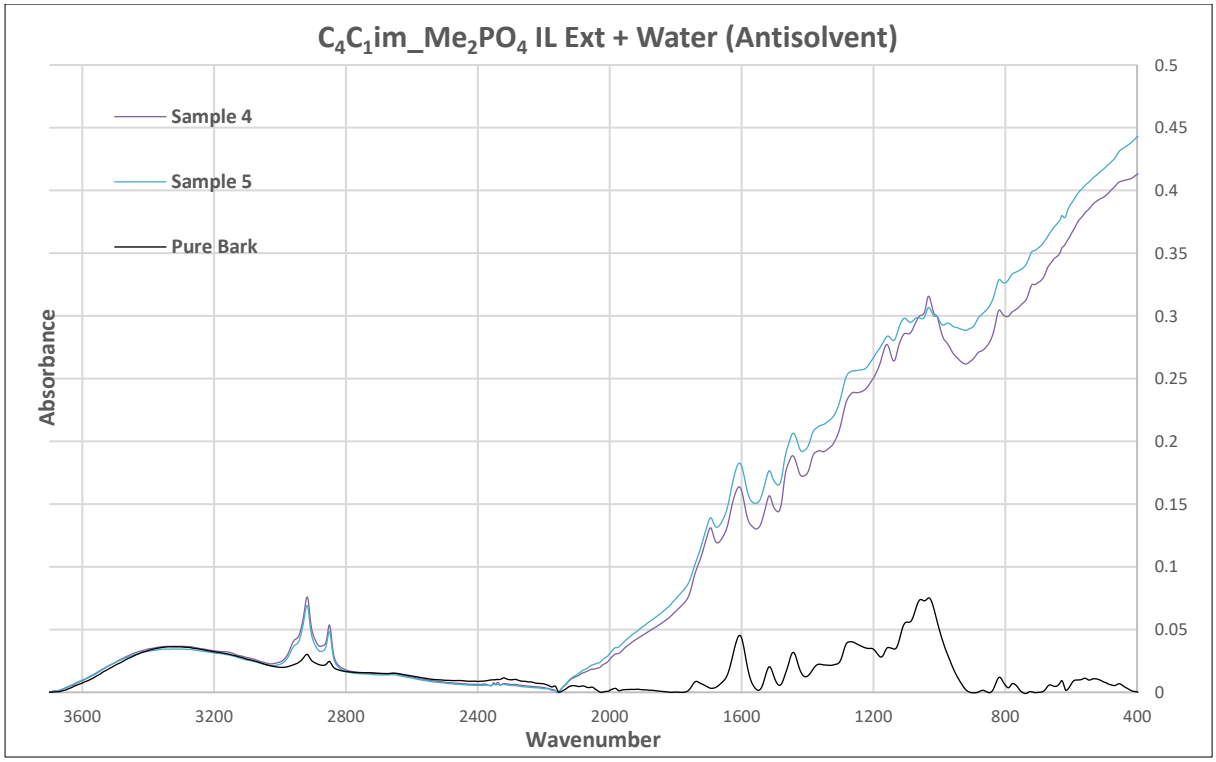


Figure A61. *IR spectra of the solid from Sample 4 - 5 using 1 mL of [C_4C_1im][Me_2PO_4] IL extract and water as an antisolvent.*

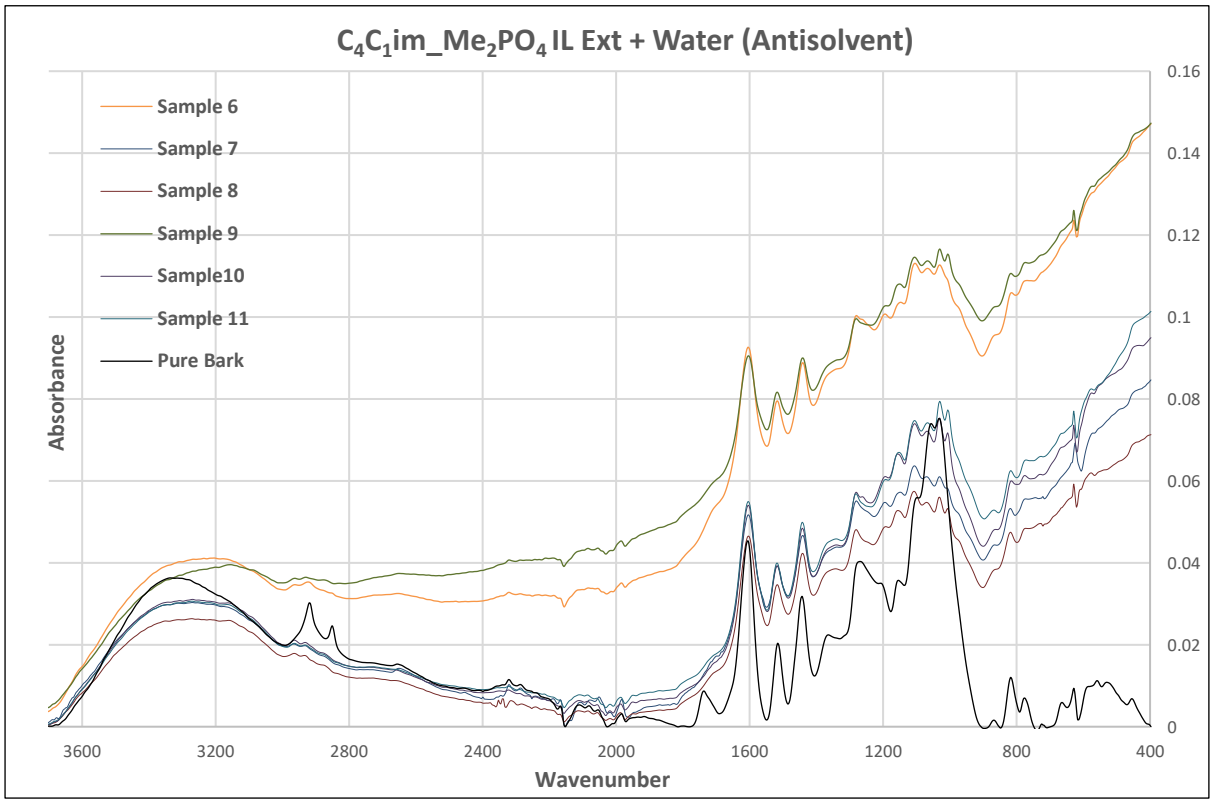


Figure A62. *IR spectra of the solid formed from Sample 6 - 11 using 1 mL of [C_4C_1im][Me_2PO_4] IL extract and water as an antisolvent*

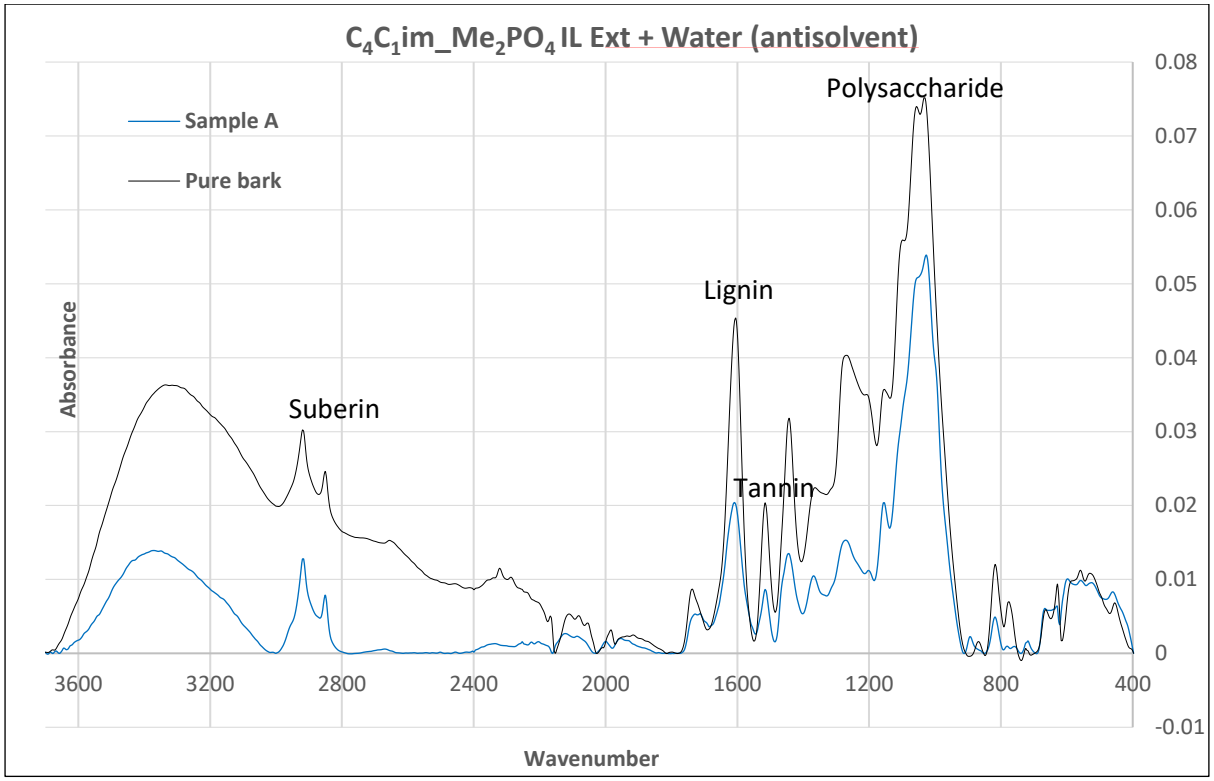


Figure A63. *IR spectra of the solid formed from Sample A using 10 mL of [C_4C_1im][Me_2PO_4] IL extract and water as an antisolvent*

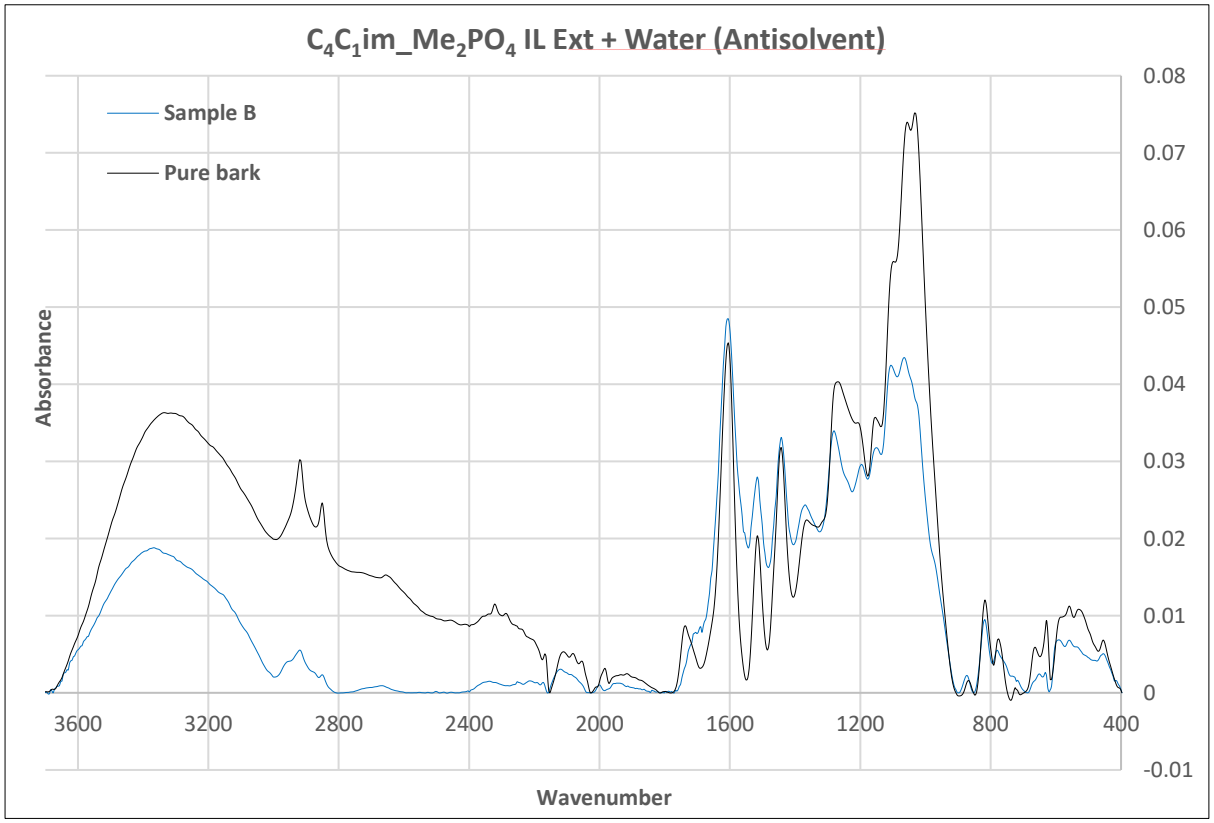


Figure A64. *IR spectra of the solid formed from Sample B using 10 mL of [C_4C_1im][Me_2PO_4] IL extract and water as an antisolvent*

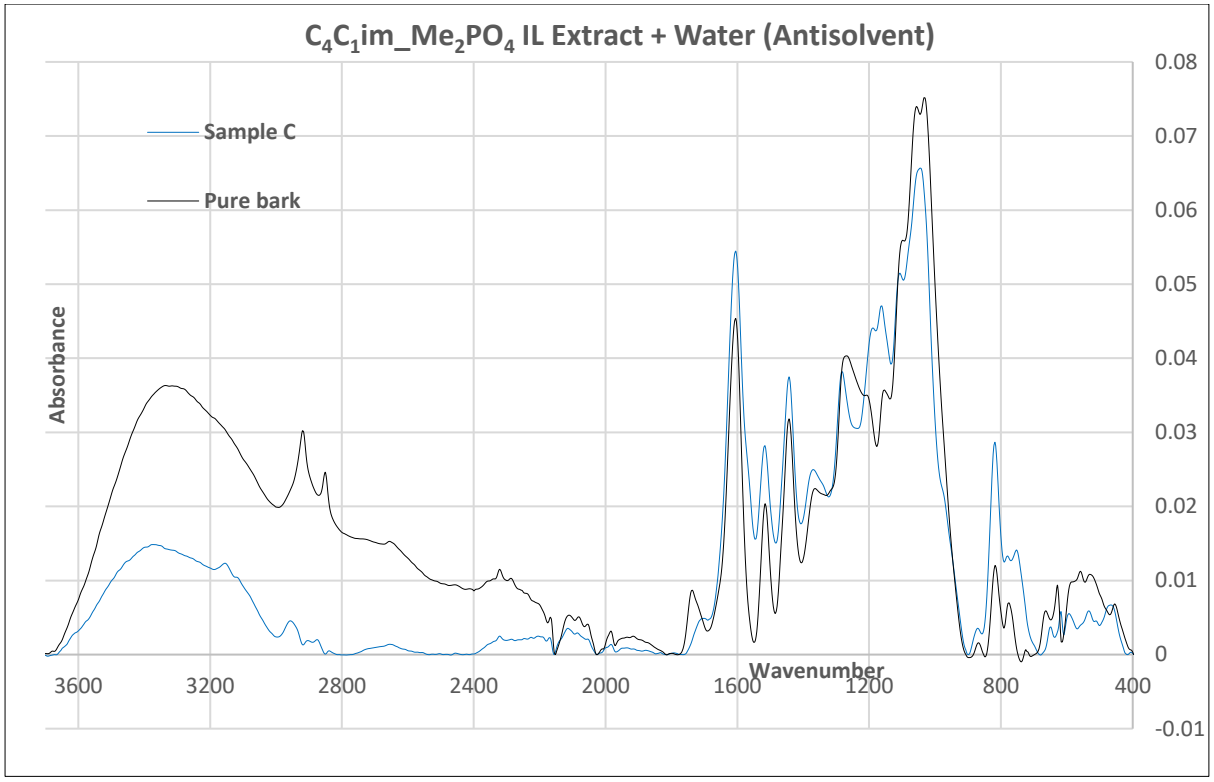


Figure A65. *IR spectra of the solid formed from Sample C using 10 mL of [C_4C_1im][Me_2PO_4] IL extract and water as an antisolvent*

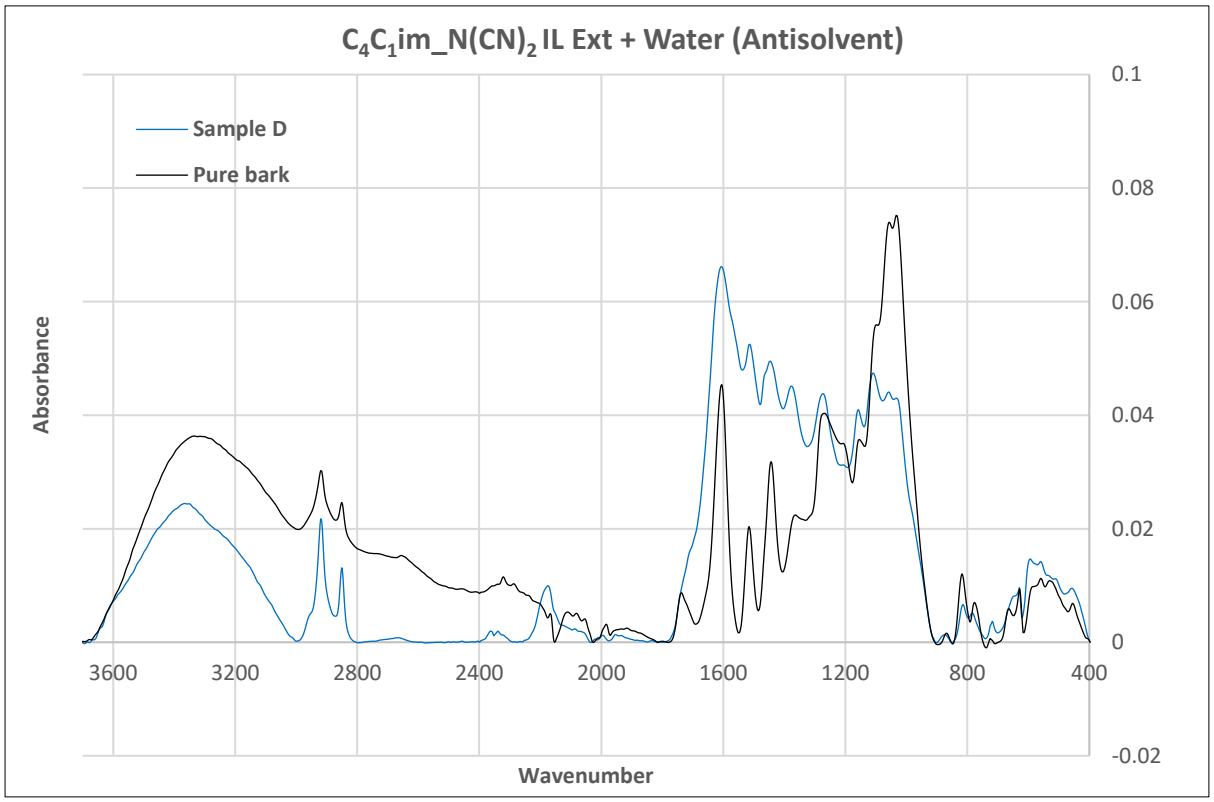


Figure A66. *IR spectra of the solid formed from Sample D using 10 mL of [C_4C_1im][$N(CN)_2$] IL extract and water as an antisolvent*

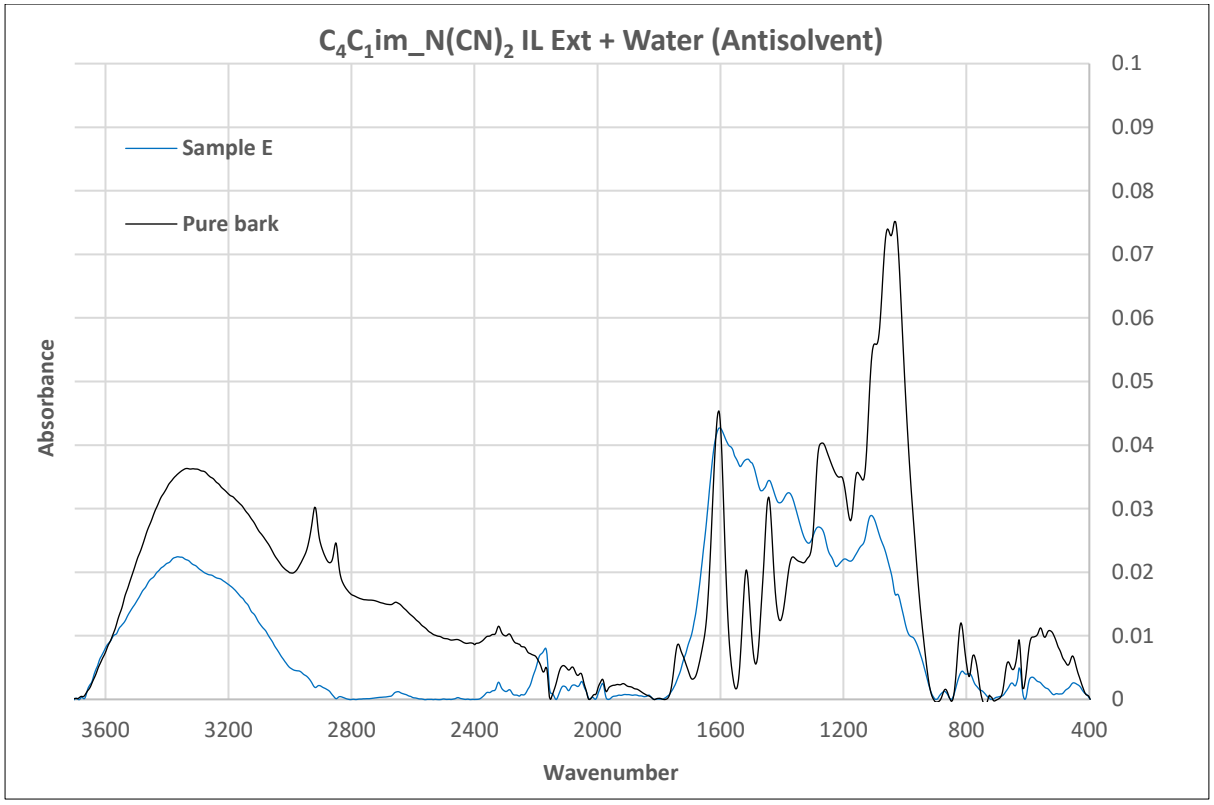


Figure A67. *IR spectra of the solid formed from Sample E using 10 mL of [C_4C_1im][$N(CN)_2$] IL extract and water as an antisolvent.*

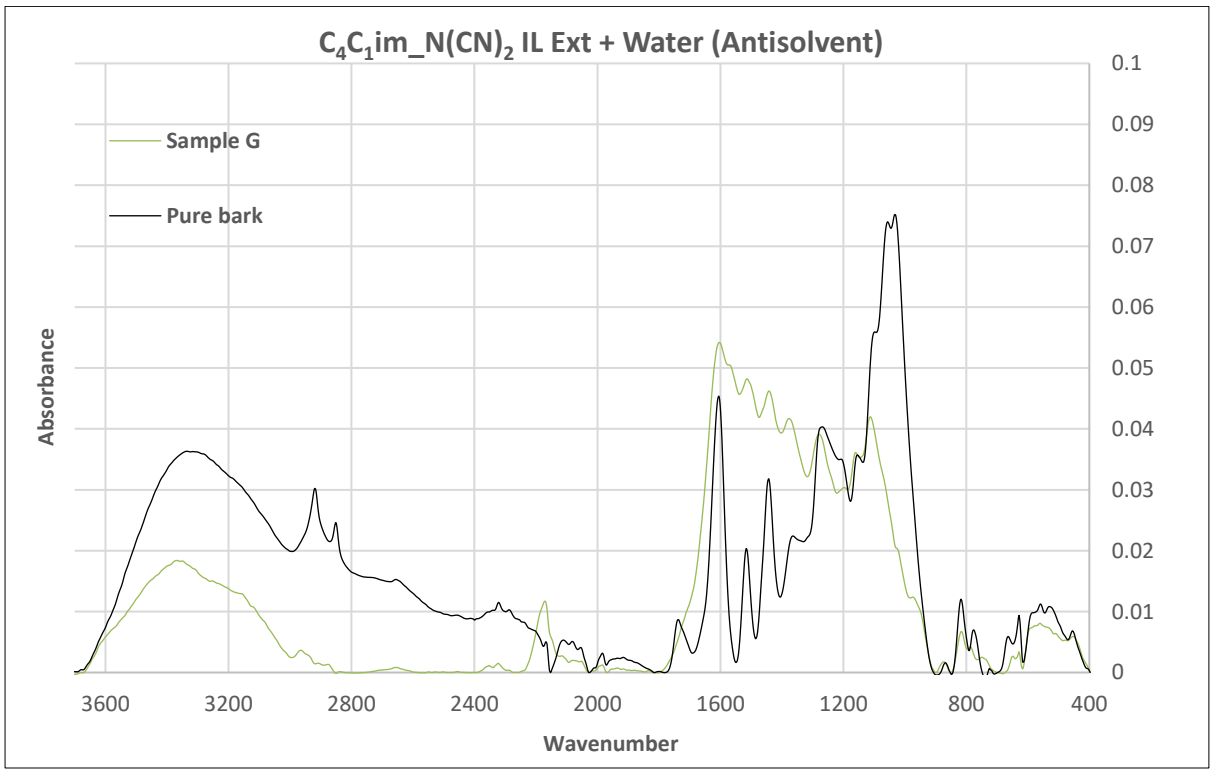


Figure A68. *IR spectra of the solid formed from Sample G using 10 mL of [C_4C_1im][$N(CN)_2$] IL extract and water as an antisolvent.*

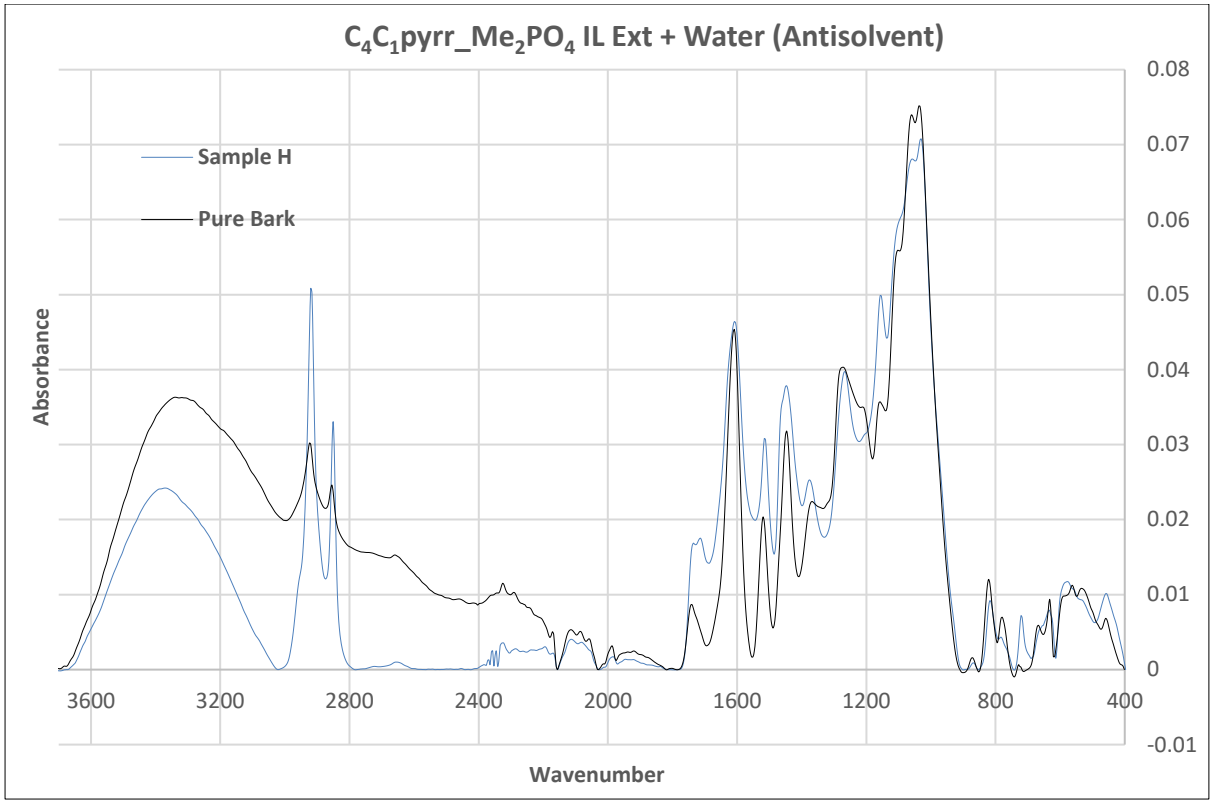


Figure A69. *IR spectra of the solid formed from Sample H using 5 mL of [C_4C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

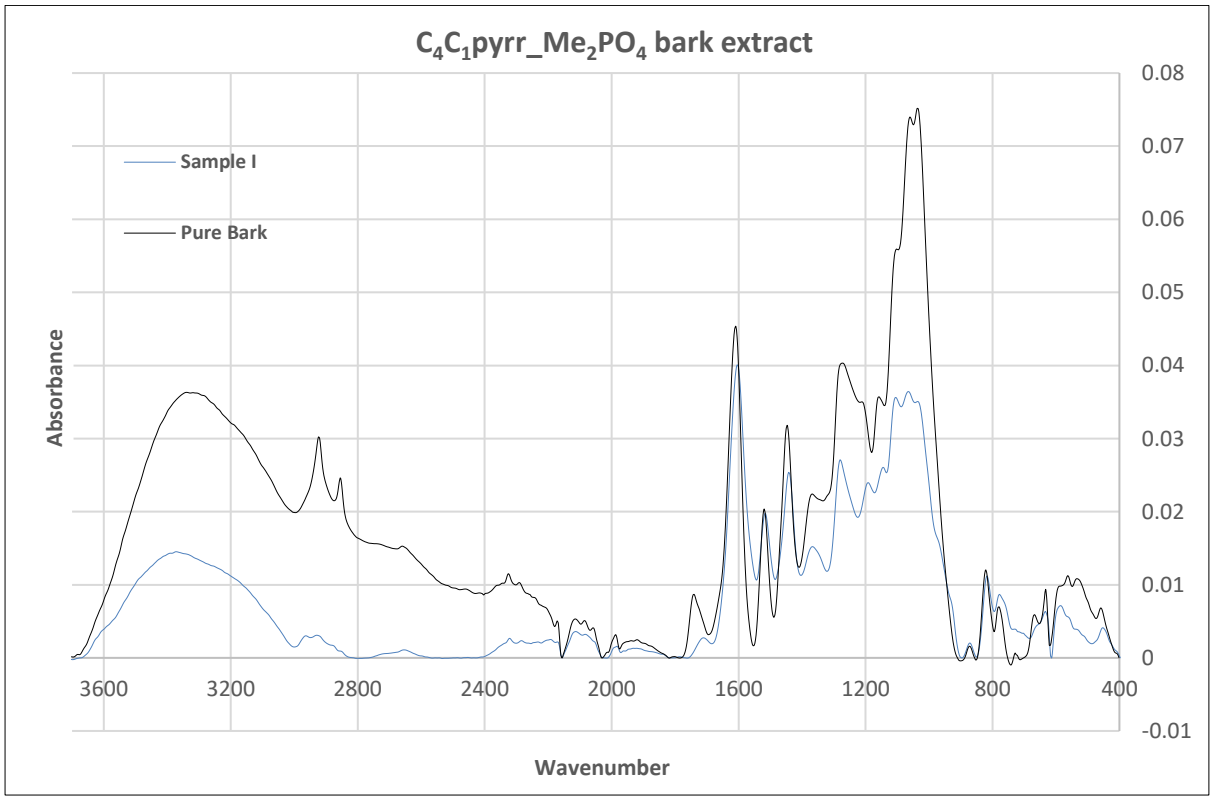


Figure A70. *IR spectra of the solid formed from Sample I using 5 mL of [C_4C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

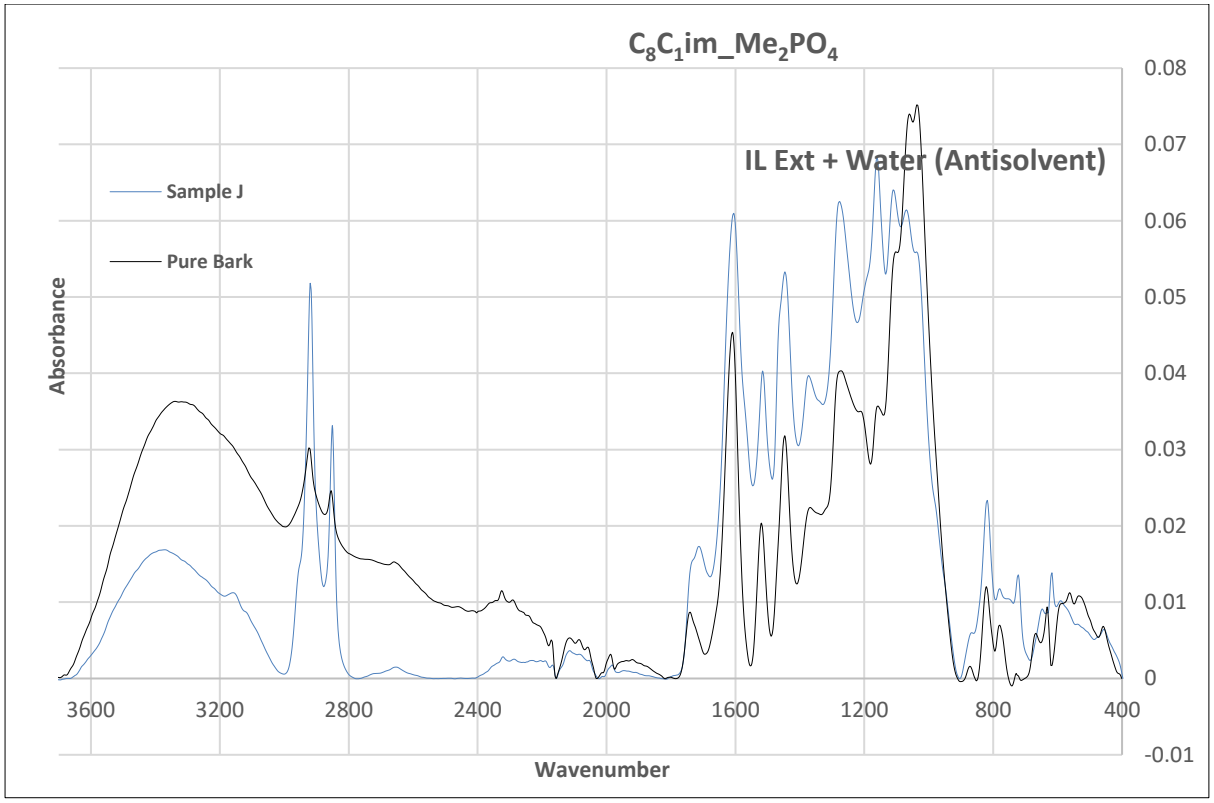


Figure A71. *IR spectra of the solid formed from Sample J using 5 mL of [C_8C_1im][Me_2PO_4] IL extract and water as an antisolvent*

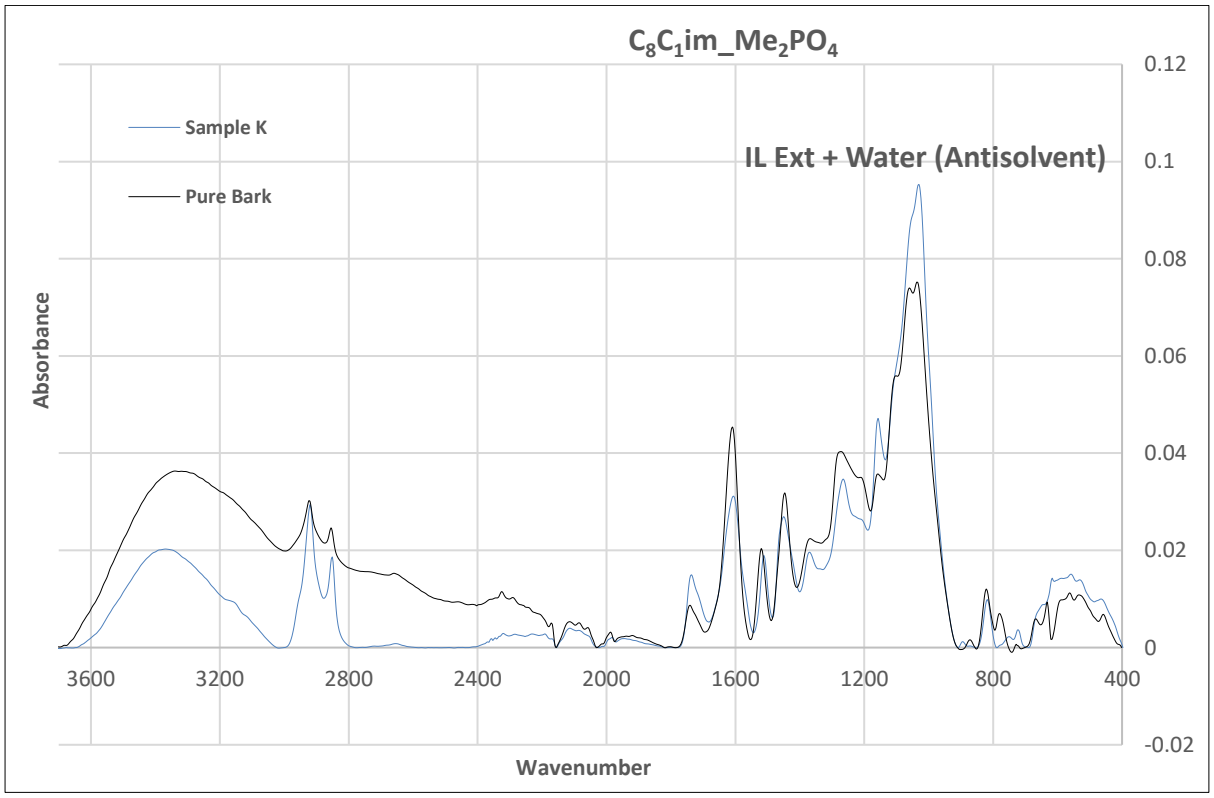


Figure A72. *IR spectra of the solid formed from Sample K using 5 mL of [C_8C_1im][Me_2PO_4] IL extract and water as an antisolvent*

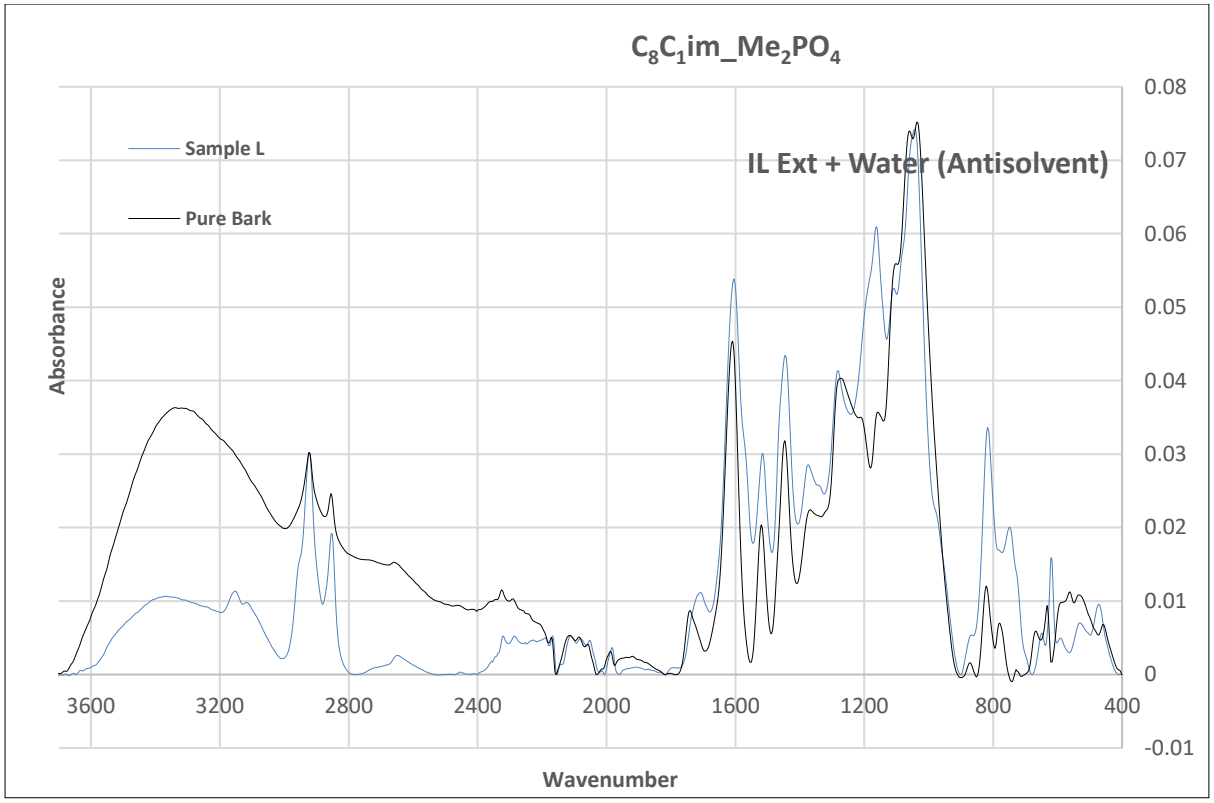


Figure A73. *IR spectra of the solid formed from Sample L using 5 mL of [C_8C_1im][Me_2PO_4] IL extract and water as an antisolvent*

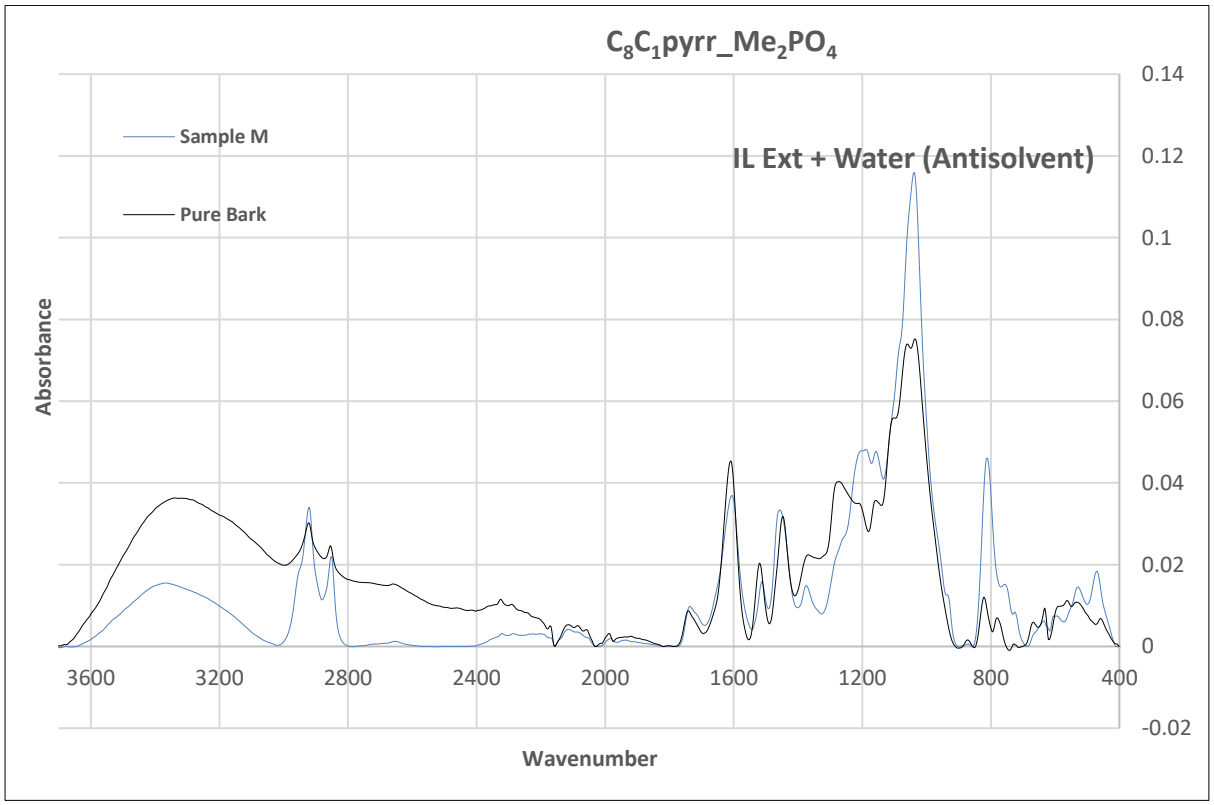


Figure A74. *IR spectra of the solid formed from Sample M using 5 mL of [C_8C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

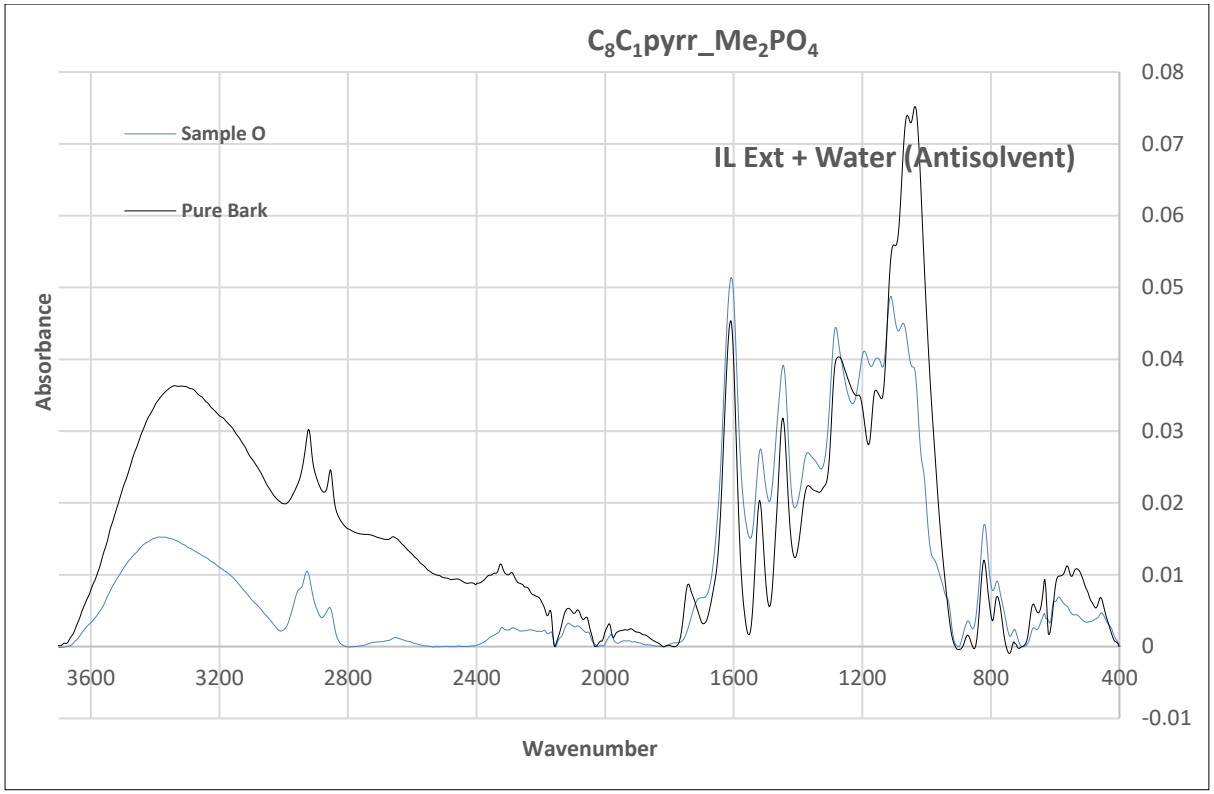


Figure A75. *IR spectra of the solid formed from Sample O using 5 mL of [C_8C_1pyrr][Me_2PO_4] IL extract and water as an antisolvent*

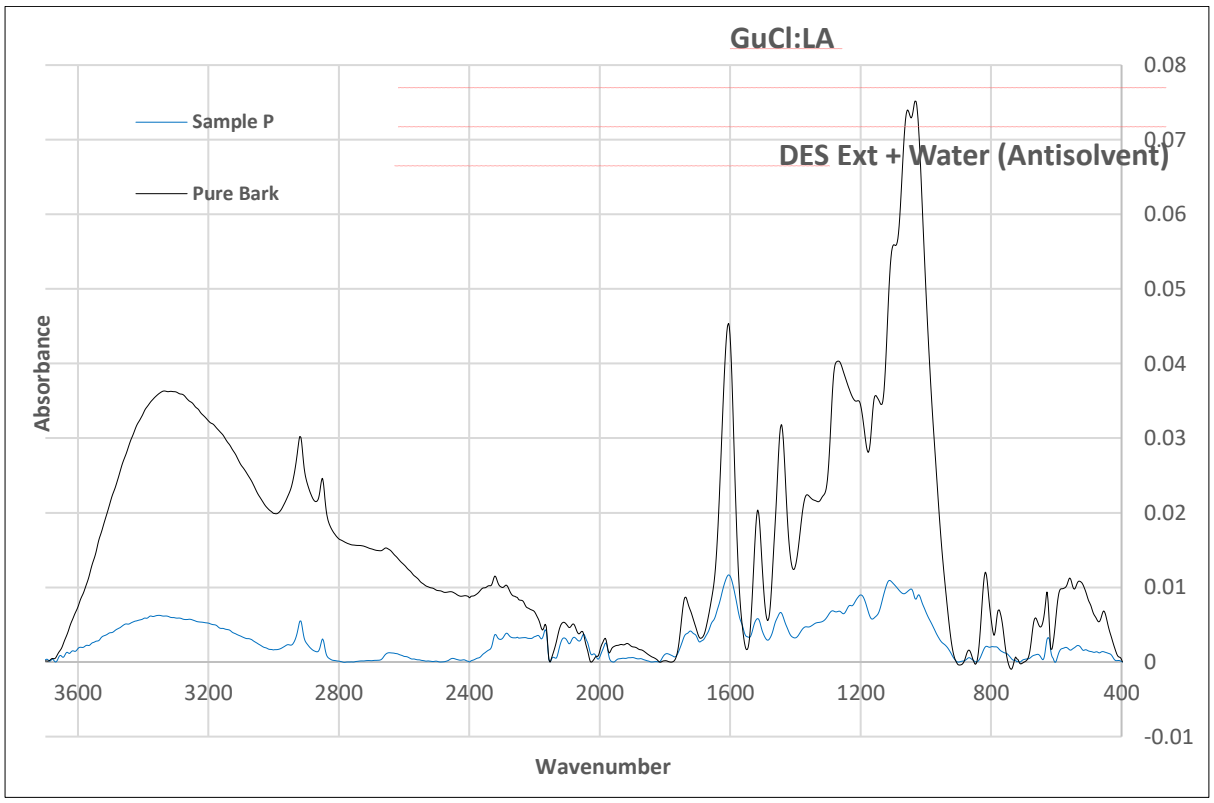


Figure A76. *IR spectra of the solid formed from Sample P using 5 mL of GuCl:LA DES extract and water as an antisolvent*

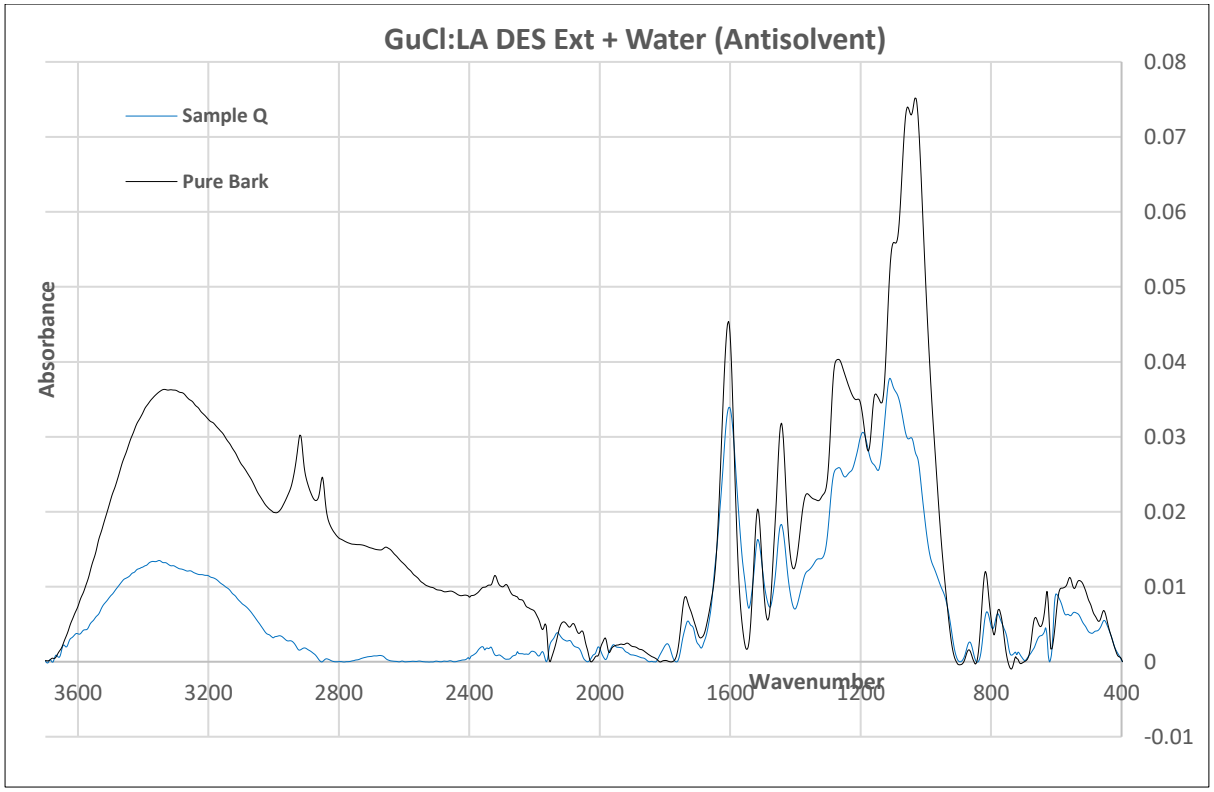


Figure A77. *IR spectra of the solid formed from Sample Q using 5 mL of GuCl:LA DES extract and water as an antisolvent*

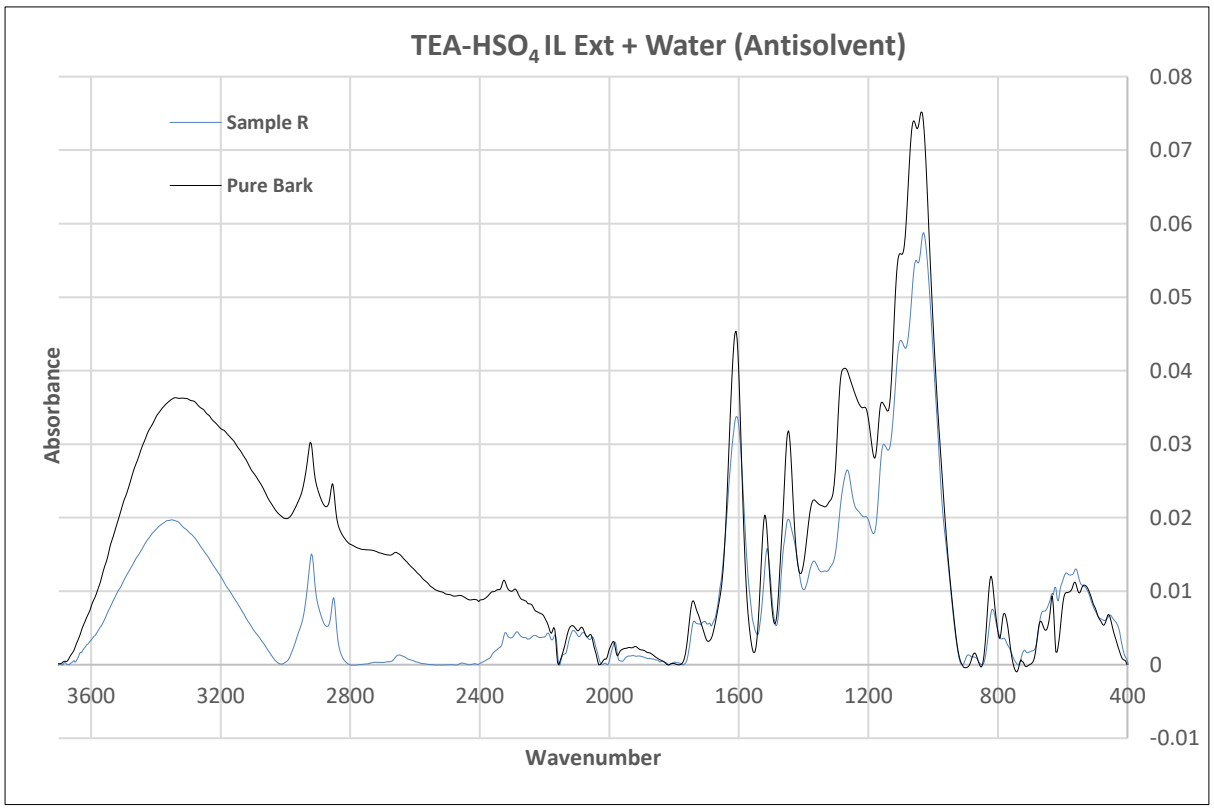


Figure A78. *IR spectra of the solid formed from Sample R using 5 mL of [TEA][HSO$_4$] IL extract and water as an antisolvent*

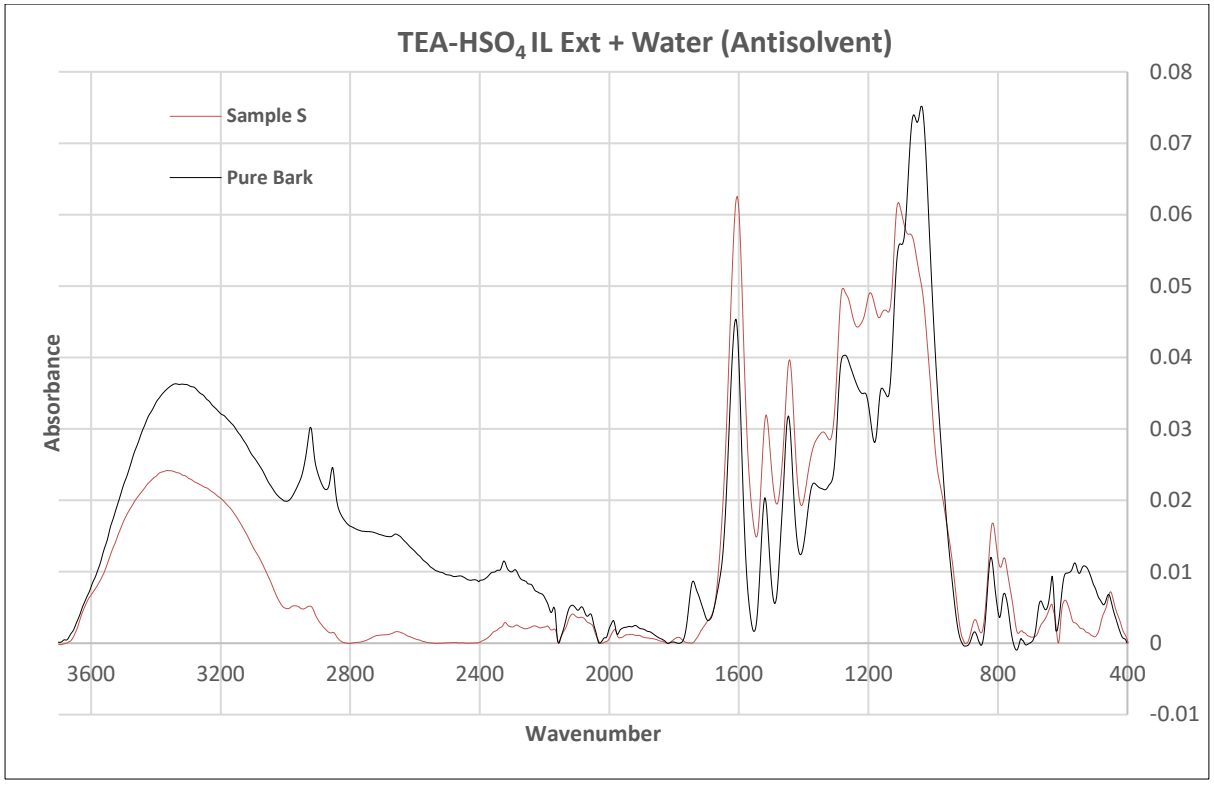


Figure A79. *IR spectra of the solid formed from Sample S using 5 mL of [TEA][HSO_4] IL extract and water as an antisolvent*

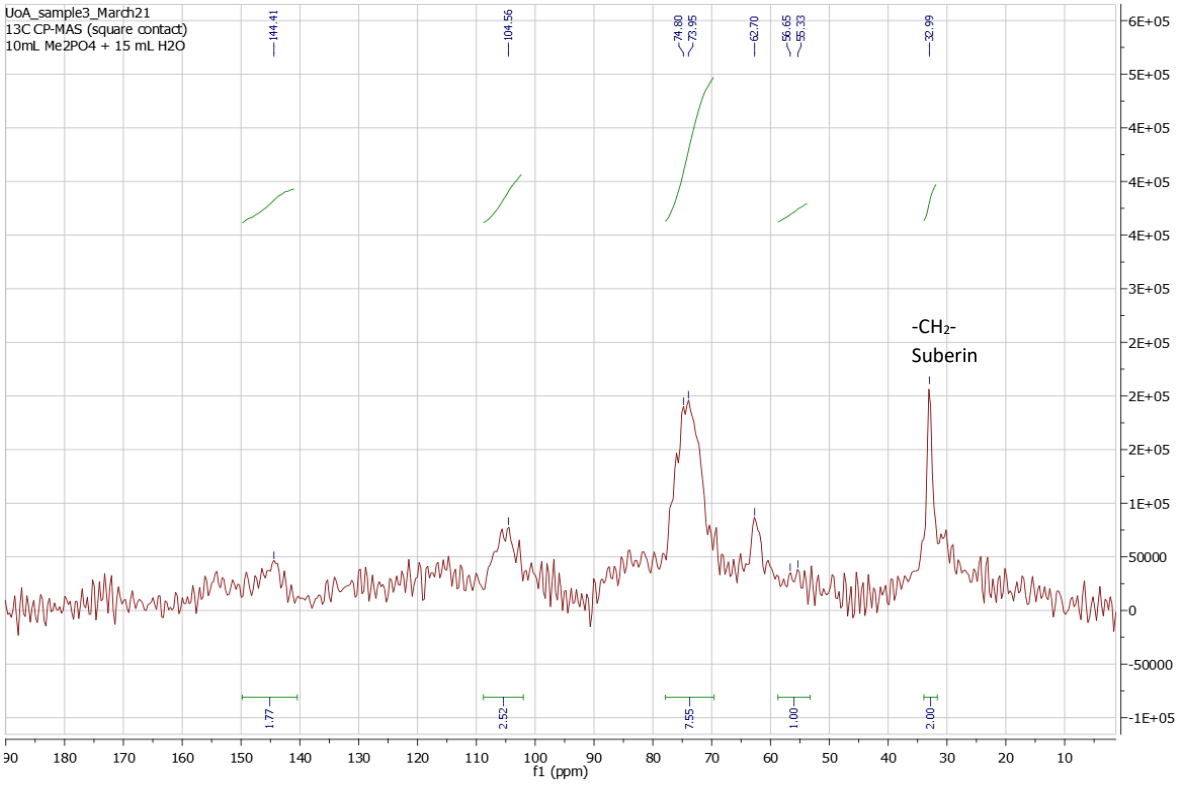


Figure A80. *^{13}C solid-state NMR of solid residue formed using 10 mL of [C_4C_1im][Me_2PO_4] IL extract and 15 mL of water as an antisolvent i.e., Sample A.*

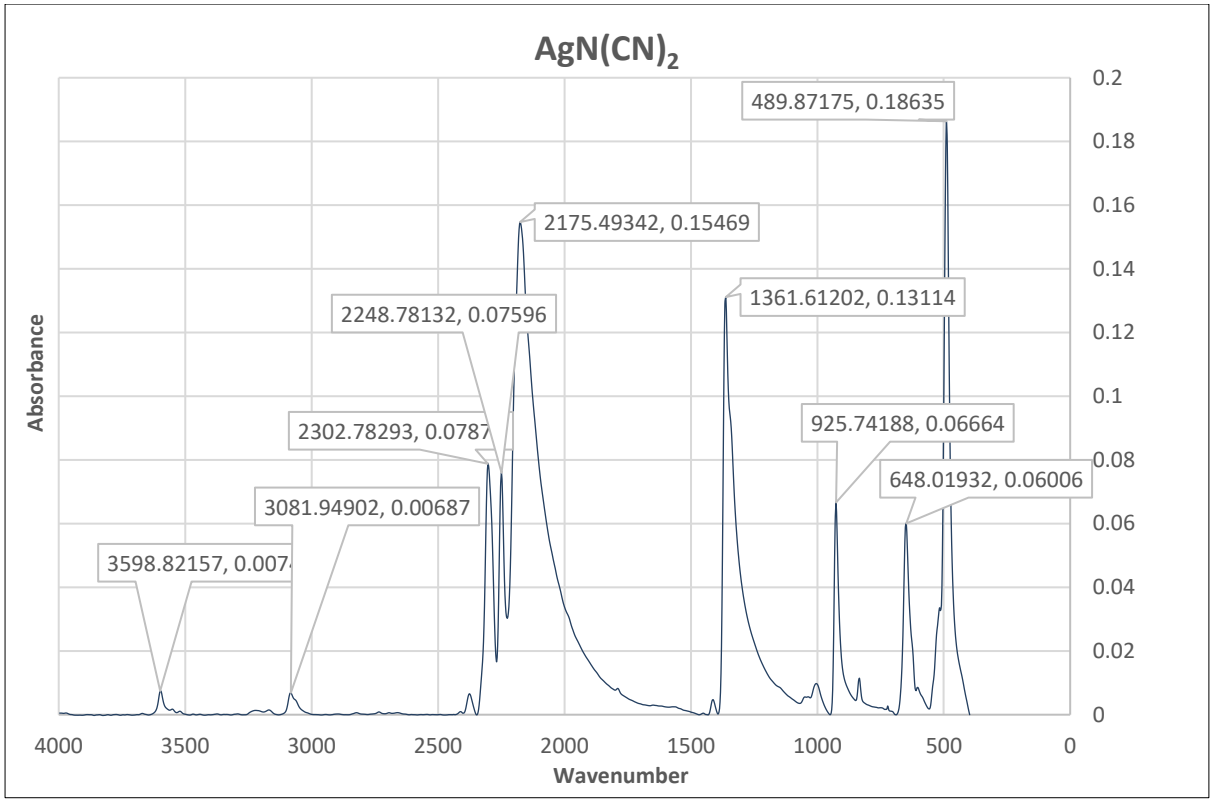


Figure A81 *IR of $AgN(CN)_2$*